

THE MEDICAL AI PROMPT EBOOK

1,000 MEDICAL PROMPTS.

Ready-to-use AI prompts for doctors, residents and medical students, from your first exam to your last publication.

1,000 PROMPTS

3 VOLUMES

THESIS & RESEARCH

SEMINARS & PRESENTATIONS

STUDY, EXAMS & VIVA

Dr. Abhishek J. Benur

Clinical Pulmonologist · Founder, Parabox AI (AI health-tech company)

Every task you will face in your medical career, from exams and thesis to seminars, publications and patient care, has a prompt in this book. **The first and last prompt book you will ever need.**

@abhishekjbenur
paraboxai.com

Find the right prompt in seconds.

You never need to read this book cover to cover. Find the task in front of you today, copy its prompt, fill in your details and paste it into ChatGPT, Claude, Gemini or Copilot.

01

Search

Press **Ctrl + F** (**Cmd + F** on a Mac), or tap the search icon in your phone's PDF reader. Type your task, for example "sample size", "journal club", "viva" or "discharge", or a prompt number such as "#415".

02

Tap the contents

Every section and sub-section in the Contents is a link that takes you straight there. Your PDF reader's bookmarks panel shows the same map.

03

Follow the chain

Most prompts end with → **NEXT**, the prompt to run after it. Tap it to go to the next step. The index at the back links to every prompt by number.

What's inside

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Using a prompt, step by step

1. Copy the full prompt. The length is what makes the answer specific to you.
2. Replace every [SQUARE BRACKET] with your real details. Where a bracket offers options separated by "/", keep the one that applies.
3. Put anything you paste (abstracts, drafts, results, notes) where the prompt shows [PASTE ...].
4. Switch on web search or document upload whenever the prompt asks for evidence.
5. Start a fresh chat for each new task, so old assumptions don't carry over.
6. Push back. If the answer is generic, reply: "Be more specific. Tie every point to my context." Then check every fact before you use it.

Read this **first**.

These prompts are written the way a senior clinician and teacher thinks, not the way a chatbot answers. Many are long on purpose: a vague prompt gets you a vague, confident, wrong answer. Before you paste anything, five non-negotiables.

01 The AI does not know your patients.

Every prompt has a context block. Fill it honestly. If you say you have 200 cases a year when you have 40, the model will design a study you cannot finish.

02 Never accept a citation you have not opened.

Language models fabricate references that look flawless: real journal, real authors, wrong or non-existent paper. Verify every reference on PubMed by PMID or DOI before it enters your work.

03 Never paste identifiable patient data.

No names, hospital IDs, MRNs or scans with burned-in identifiers. De-identify before the data touches a chatbot. This is an ethics committee issue, not a preference.

04 The AI drafts. You decide.

Use it to speed up structure, phrasing and pre-checking. The clinical and scientific judgement, the assumptions and the interpretation stay yours. Your name goes on the work, not the model's.

05 Declare AI use where required.

ICMJE and most journals require disclosure of generative AI use. AI cannot be an author. Check your journal's policy and your university's thesis and exam rules.

Every prompt is built on **P-R-O-M-P-T**.

One formula, six layers. Each layer matches what OpenAI, Anthropic, Google and Microsoft recommend in their own prompting guides.

P Persona Who the AI should be: a senior expert with a trait	R Request One clear task, as numbered steps	O Output Shape, length, tone, tables	M Material Your context and anything you paste	P Parameters Rules, guidelines, accuracy guards	T Tweak Self-critique and the next step
--	---	--	--	---	---

See it in action

BAD	"Tell me about pneumonia."
GOOD	"Explain community-acquired pneumonia management for an MBBS exam."
BETTER	"Act as a pulmonologist. Explain CAP management for an MBBS final-year student as a 6-point answer with empirical antibiotic choices per Indian guidelines. Keep it under 150 words, exam-oriented."

Three kinds of prompt

QUICK 40–80 words One focused output in a few sentences. Each has a TRY IT WITH example.	STANDARD 100–200 words One real task done properly: context, numbered task, rules, output, a pro tip and the next prompt to run.	MASTER 250–450 words A full workflow on its own page, with when to use it, why it exists and how to get more out of it.
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Tags you will see in the outputs

TAG	WHAT TO DO
[CITE] · [CITATION NEEDED]	Find a real source on PubMed and insert it yourself.
[VERIFY] · [ASSUMPTION — VERIFY FROM PUBLISHED DATA]	Replace the number with a sourced figure before use.
[UNVERIFIED BACKGROUND]	Treat as a lead to check, not a fact.
[ASK ME] · [NEED INFO] · [NOT REPORTED]	The model is missing information. Supply it, or leave it out.

Every prompt works in ChatGPT, Claude, Gemini and Copilot. → **NEXT** points to the prompt to run after this one.

What's in the library.

Every volume, section and sub-section below is a link. Each section opens with its own linked list of prompts, and the index at the back of each volume links to every single prompt.

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THESIS, RESEARCH & PUBLICATION.

400 clinician-grade AI prompts for every stage of your thesis and first papers, from finding a feasible topic to the published manuscript, case report and conference poster.

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1.1

Topic selection & research question

Most thesis disasters are topic failures decided in a ten-minute corridor conversation. Build feasibility into the decision from day one.

35

prompts

4

Master

20

Standard

11

Quick

HOW TO USE THIS SECTION

Use it in your first months of residency, before you commit to a thesis topic or write a synopsis.

Start with the Master prompts (**#1–4**) for the full workflow; use the Standard prompts (**#5–24**) for one task at a time; reach for the Quick prompts (**#25–35**) when you have two minutes. Tap any title below to jump to it, and follow → **NEXT** to move to the next step.

#1 Feasibility–first topic shortlist

#2 Five Indian abstracts → six thesis topics

#3 Socratic research–question workshop

#4 Guide–meeting pitch pack for your top 3 topics

#5 India gap finder: 15 topics from scratch

#6 Clinical hunch → PICOT question

#7 FINER stress–test of my research question

#8 Is this topic already done to death?

#9 Turn gaps into Consensus yes/no questions

#10 Topics I can do from existing hospital records

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#12 Topics needing only routine investigations

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#14 Quality–improvement and audit thesis topics

#15 Questionnaire–based topics without the KAP trap

#16 Head–to–head decision matrix for 2–3 topics

#17 Thesis title that passes the committee

#18 What could kill this topic? A risk register

#19 Choose a measurable primary outcome

#20 The questions my guide will ask about my topic

#21 Topics from the gaps in a clinical guideline

#22 Four–week feasibility check before you commit

#23 Five original angles on a crowded topic

#24 Two–month project for an MBBS student (ICMR–STS style)

#25 Ten quick topic seeds in my specialty

#26 Sharpen my research question to one line

#27 PICO or PECO? Frame my question correctly

#28 Check ongoing trials before you commit

#29 The follow–up that surfaces Indian gaps

#30 Identify the variables in my question

#31 My topic is too broad: narrow it three ways

#32 What infrastructure does this topic need?

#33 Does this Western topic apply in India?

#34 Hypothesis–generating or hypothesis–testing?

#35 Summarise my topic on one slide

Feasibility—first topic shortlist

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT (YEAR 1)

USE IT

Months 1 to 3 of your postgraduate course, before you commit to anything.

WHY

Most theses fail at the topic, not the analysis. This prompt checks every idea against your real patient numbers, equipment and time before you fall in love with it.

COPY THE PROMPT BELOW

PERSONA	You are a senior clinical research methodologist with over 20 years of experience supervising postgraduate theses in [YOUR SPECIALTY]. You have sat on an institutional ethics committee and you peer review for specialty journals. You are blunt about feasibility rather than encouraging.
MY CONTEXT	Specialty: [e.g. Respiratory Medicine] Programme and year: [MD / MS / DNB / DM / MCh], year [1 / 2 / 3] Institution type: [government medical college / tertiary private / district hospital] Eligible patients I can realistically access: about [N] per month Months available for recruitment and data collection: [X] Funding: [none / departmental consumables / grant of amount Y] Investigations and equipment I genuinely have: [list] What I do NOT have: [e.g. flow cytometry, PET-CT, research nurse] Statistical support: [none / department statistician / paid] Clinical interests: [2 to 4] Diseases with high burden in my setting: [list]
TASK	Generate 8 candidate research questions I could design, complete, defend and publish inside my timeline. For each, give exactly: 1. Working title as it would appear on a thesis cover page. 2. PICOT: Population, Intervention or Exposure, Comparator, Outcome, Time frame. 3. The gap in two sentences: what is established, what remains unanswered (describe evidence generically). 4. Study design, and one line on why not the obvious alternative. 5. Feasibility score out of 5, justified against my context above. 6. Indicative sample size with every assumption stated. 7. Recruitment timeline in months, assuming 30% of eligible patients are lost. 8. Primary outcome and the exact instrument, scale or cut-off. 9. The three most likely reasons it fails. 10. Honest publication ceiling: local, national indexed, or international.
RULES	Mark every number you need as [ASSUMPTION — VERIFY FROM PUBLISHED DATA] instead of inventing prevalence, effect sizes, trial names or citations, because my guide will ask where each came from. Drop any question needing resources I said I do not have. At least 2 of the 8 must survive if recruitment halves.
OUTPUT	A ranked list from most to least feasible, then your top pick with reasoning.
TWEAK	Finish with AVOID THESE: three topic directions in my field that look attractive to postgraduates but routinely fail, and why.

How to get more out of it

- Run it twice, once with your real patient numbers and once with half. Any topic that survives both is the safe one.
- Take the top three to your guide as a one-page comparison, not a chatbot transcript.
- Read AVOID THESE before you fall in love with an idea.
- → NEXT #4 Guide-meeting pitch pack for your top 3 topics

Five Indian abstracts → six thesis topics

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT

After you have saved 5 recent abstracts from Indian centres in your field (Semantic Scholar or PubMed, last 3 years).

WHY

Topics generated from memory repeat what already exists. Topics generated from real recent Indian abstracts start from what your peers have actually published, and from what they left out.

COPY THE PROMPT BELOW

PERSONA	You are a research mentor at an Indian medical college who has guided over 50 postgraduate theses to completion. You read abstracts like a reviewer: you look for what was not measured, not studied, or not studied here.
MY CONTEXT	Specialty and programme: [e.g. MD Pulmonary Medicine, year 1] Setting: [government medical college / private tertiary / district] Limits: 2-year thesis window, [limited / moderate] lab infrastructure, about [N] eligible patients per month. [PASTE ABSTRACT 1] [PASTE ABSTRACT 2] [PASTE ABSTRACT 3] [PASTE ABSTRACT 4] [PASTE ABSTRACT 5]
TASK	1. For each abstract, write one line on what it studied and one line on what it did not address (population, outcome, setting, follow-up or design). 2. From these, identify 3 research gaps relevant to Indian patients. 3. For each gap, propose 2 thesis topics, each with a PICO question and a measurable clinical outcome. 4. For all 6 topics, give: best study design, rough sample size range, primary outcome measure, the main feasibility challenge in an Indian government hospital, and publication potential in an indexed journal. 5. Rank the 6 and recommend the top 3 to present to my guide.
RULES	Work only from the abstracts I pasted. If a gap needs knowledge beyond them, label it [UNVERIFIED BACKGROUND — CONFIRM ON PUBMED] rather than stating it as fact. Name no additional papers unless you can give a verified PMID or DOI.
OUTPUT	A short table per gap, then the ranked shortlist with one-paragraph reasoning for each of the top 3.
TWEAK	End with the single question I should search on Consensus to check whether my top topic is already saturated.

How to get more out of it

- If an abstract is from a different population (Western, paediatric), say so in the context line. It changes which gaps are real.
- Keep the 5 abstracts. They become the first five studies in your Review of Literature.
- If fewer than 3 real gaps appear, your search was too narrow. Search the gap, not the disease name.
- → **NEXT #8** Is this topic already done to death?

Socratic research-question workshop

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT / MBBS INTERN

USE IT

When you have a vague idea ("something on sepsis in the ICU") but no question.

WHY

A good question is refined, not generated. This prompt makes the AI interview you, one question at a time, until the question is sharp enough to defend.

COPY THE PROMPT BELOW

PERSONA	You are a clinical epidemiologist who runs research-methods workshops for postgraduates. You never hand over answers early. You ask probing questions, one at a time, and you push until the question is specific, measurable and feasible.
MY CONTEXT	My rough idea in one or two sentences: [e.g. something about delirium in elderly ICU patients] Specialty and level: [e.g. MD Anaesthesiology, year 1] Setting: [e.g. 20-bed mixed ICU, government medical college] Time for data collection: [X months]
TASK	Run a workshop with me in rounds: 1. Ask me one question at a time about population, exposure or intervention, comparator, outcome, time frame and setting. Wait for my answer before asking the next. 2. After every 3 answers, show the current version of my research question and score it on FINER (Feasible, Interesting, Novel, Ethical, Relevant), 1 to 5 each, with one line per score. 3. Point out the weakest FINER element and ask the question that fixes it. 4. Stop when every FINER score is 4 or more, or after 5 rounds. 5. Then give the final question in PICOT format, a one-line hypothesis, and a draft working title.
RULES	Do not suggest a finished question before round 3, because the point is to make me think. If an answer I give makes the study unethical or impossible, say so directly and explain why. Keep every question you ask under 30 words.
OUTPUT	Conversational, one question per turn. Use a short FINER scorecard table at each checkpoint.
TWEAK	At the end, list the 3 assumptions in my question that I must verify in the literature before my guide meeting.

How to get more out of it

- › Answer honestly, including "I don't know". The workshop is only as good as your answers.
- › Screenshot the final scorecard. It becomes slide 2 of your topic presentation.
- › Re-run the workshop for your second-choice idea and compare the final scores.
- › → **NEXT #7** FINER stress-test of my research question

Guide-meeting pitch pack for your top 3 topics

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT

The day before you meet your guide or research committee to choose a topic.

WHY

Guides respond far better to a student who arrives with three shortlisted topics, the evidence gap behind each, and an honest feasibility view. This prompt builds that one-page pitch.

COPY THE PROMPT BELOW

PERSONA	You are a head of department who has chaired thesis-topic committees for 15 years. You know which questions guides ask, which topics get approved, and which get sent back.
MY CONTEXT	Specialty and year: [e.g. MS General Surgery, year 1] My guide's known interests: [list, or "unknown"] Department case load relevant to these topics: [numbers per month] Topic 1: [title + PICOT + why it matters] Topic 2: [title + PICOT + why it matters] Topic 3: [title + PICOT + why it matters] Evidence I have for each gap (searches, Consensus result, abstracts): [PASTE NOTES]
TASK	1. Build a one-page comparison table: title, design, sample size needed vs available, time to complete, cost, ethics complexity, novelty, publication ceiling. 2. Give each topic an overall recommendation: strong, possible, or weak, with one reason. 3. For each topic, write the 3 questions my guide is most likely to ask, with a short answer from my notes. 4. Write a 60-second verbal pitch for my first-choice topic. 5. List what I should bring to the meeting (search strings, hit counts, a feasibility estimate).
RULES	Base every feasibility judgement on the numbers I gave, not on general assumptions. Where my notes do not support a claim of novelty, write [GAP NOT YET VERIFIED] instead of asserting it, because a guide will test it. Keep my voice: a first-year resident, not a professor.
OUTPUT	The comparison table first, then the per-topic questions, the pitch script, and a checklist. Fits on two printed pages.
TWEAK	Finish with the one weakness in my first-choice topic that I should raise before my guide does, and how to present it.

How to get more out of it

- › Print the table. A one-page comparison beats scrolling a phone in a guide's office.
- › If your guide rejects all three, ask which constraint mattered most, then re-run topic-feasibility with that constraint added.
- › Practise the 60-second pitch out loud twice.
- › → **NEXT #17** Thesis title that passes the committee

5 India gap finder: 15 topics from scratch

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT (YEAR 1)

USE IT When you have no abstracts or ideas yet and need a starting list today.

PERSONA	You are a senior researcher in [SPECIALTY] at an Indian medical college, known for steering residents towards questions that matter to Indian patients rather than copying Western topics.
MY CONTEXT	I am doing my [MD / MS / DNB] in [SPECIALTY] at a [government medical college / private college] in India. I have limited lab infrastructure and a 2-year thesis window.
TASK	Generate 15 potential thesis topics that: 1. Address conditions with a high burden in India (e.g. TB, diabetes, COPD, anaemia, dengue, snakebite, hypertension) relevant to my specialty. 2. Have limited published data from Indian populations. 3. Can be done as observational, cross-sectional or simple interventional studies. 4. Need no advanced laboratory infrastructure. For each, state the gap it addresses in one line and rate feasibility out of 10.
RULES	Describe gaps generically. Do not name papers or quote prevalence figures from memory; write [VERIFY] where a number would help, because every gap must be checked on PubMed before I pitch it.
OUTPUT	A numbered table: topic, gap, design, feasibility /10. Then the 3 you would pick and why.

PRO TIP Paste your 3 favourites into Consensus as yes/no questions. Keep only the ones with mixed or insufficient evidence.

TRY IT WITH MD General Medicine, government medical college in Bihar.

→ **NEXT #9** Turn gaps into Consensus yes/no questions

6 Clinical hunch → PICOT question

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT / MBBS INTERN

USE IT When you have noticed something on the ward ("these patients seem to do worse") and want to turn it into a question.

PERSONA	You are a clinical research mentor who helps residents turn bedside observations into answerable research questions.
MY CONTEXT	What I noticed on the ward or in OPD: [describe the observation in 2–3 lines] Specialty and setting: [e.g. Paediatrics, tertiary PICU] Rough patient numbers per month: [N]
TASK	1. Restate my observation as a clear clinical question. 2. Convert it into 3 alternative PICOT (or PECO) questions: one descriptive, one comparative, one prognostic. 3. For each, name the study design that answers it and the primary outcome. 4. Say which is most feasible for a postgraduate thesis and why.
RULES	Keep each question to one sentence. If my observation is more likely explained by bias or chance (e.g. referral pattern), say so first, because that changes the question.
OUTPUT	A table of the 3 PICOT versions, then a recommendation in 3 sentences.

PRO TIP Write your observation down the day you notice it. Specific details ("three patients this month on drug X") make much better questions than vague impressions.

TRY IT WITH Elderly patients with hip fracture on our ward seem to develop delirium more often when surgery is delayed beyond 48 hours.

→ **NEXT #7** FINER stress-test of my research question

7 FINER stress-test of my research question

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT Before you submit your topic to the department or research committee.

PERSONA	You are a research committee member at an Indian medical college who scores postgraduate topics every term. You are fair but you do not pass weak questions.
MY CONTEXT	My research question: [paste in PICOT form] Study design: [design] Resources and patient numbers: [brief] Time available: [months]
TASK	1. Score my question on each FINER criterion (Feasible, Interesting, Novel, Ethical, Relevant) from 1 to 5, with two lines of justification each. 2. Identify the single weakest criterion. 3. Rewrite the question twice: once to fix the weakest criterion, once to make it simpler. 4. Say which of the three versions you would approve and why.
RULES	Judge novelty only on what you can reason from the question itself; where novelty depends on the literature, mark it [CHECK ON PUBMED] instead of guessing, because false novelty claims are caught at the viva.
OUTPUT	A FINER scorecard table, the two rewrites, and a verdict of 3 sentences or fewer.

PRO TIP A score of 3 on "Novel" is fine for a thesis. A score below 3 on "Feasible" is not.

TRY IT WITH Is serum vitamin D level associated with exacerbation frequency in COPD patients at a tertiary hospital in north India?

→ **NEXT #8** Is this topic already done to death?

8 Is this topic already done to death?

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT After shortlisting, when you suspect your topic may already be heavily studied.

PERSONA	You are a systematic reviewer who can tell within minutes whether a research question is saturated, contested or open.
MY CONTEXT	My proposed question: [PICOT] Search results I found (titles or abstracts from PubMed, Consensus or Perplexity): [PASTE TITLES / ABSTRACTS HERE]
TASK	1. Sort the studies I pasted by design and setting (Indian vs international). 2. Judge the evidence state: saturated (consistent findings, many studies), contested (mixed findings), or open (few or no studies). 3. If saturated, suggest 3 angles that would still be original (new population, setting, outcome, follow-up, or measurement method). 4. If contested, identify what explains the disagreement and whether my study could resolve it.
RULES	Judge only from what I pasted. If the sample is too small to judge, say so and tell me which search to run next, because a verdict based on six titles is not a verdict.
OUTPUT	One-line verdict, then a short table of studies, then the angles or next search.

PRO TIP Consensus's meter is a quick first pass. This prompt is the second pass on what it returns.

TRY IT WITH Paste 10 PubMed titles for "neutrophil-lymphocyte ratio sepsis mortality".

→ **NEXT #23** Five original angles on a crowded topic

9 Turn gaps into Consensus yes/no questions

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT Between gap-finding (Perplexity) and topic selection, before you check evidence density.

PERSONA	You are a medical librarian who trains residents to query evidence engines such as Consensus effectively.
MY CONTEXT	Research gaps or problem areas I have noted: [PASTE 5–8 GAP AREAS HERE] Population of interest: [e.g. Indian adults with type 2 diabetes]
TASK	1. Convert each gap into one binary research question in the form "Does [factor] affect [outcome] in [population]?" 2. Give a second, narrower version of each question. 3. Tell me how to read the result: what a strong-agreement, mixed or insufficient-evidence result would mean for choosing my thesis.
RULES	Keep each question under 20 words and answerable with yes or no. Use my population exactly, because broad populations return Western data that does not apply to my setting.
OUTPUT	A numbered list: broad question, then narrow question, for each gap. Then a 3-row interpretation guide.

PRO TIP Keep only the questions whose result is mixed or insufficient. Strong agreement means that topic is done.

TRY IT WITH Gaps: sarcopenia in COPD, vitamin D in TB outcomes, early mobilisation in ICU.

→ **NEXT #8** Is this topic already done to death?

10 Topics I can do from existing hospital records

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT / DNB TRAINEE

USE IT When prospective recruitment looks slow, or your thesis window is short.

PERSONA	You are a clinical researcher who has published multiple retrospective and record-based studies from Indian hospitals, and you know their pitfalls.
MY CONTEXT	Specialty: [SPECIALTY] Records available: [e.g. ICU admission registers 2019–2024, discharge summaries, lab information system, OPD registers] How complete and reliable these records are, in my honest view: [brief] Approximate number of records: [N]
TASK	1. Suggest 8 thesis questions answerable from these records. 2. For each, list the variables needed and whether they are routinely recorded. 3. Flag the main bias for each (missing data, misclassification, selection). 4. Say which questions most need an ethics waiver of consent, and what the IEC will want.
RULES	Prefer outcomes that are hard to misrecord (death, readmission, length of stay) over soft outcomes. Mark any question that needs data my records probably lack as [CHECK DATA AVAILABILITY] rather than assuming it is there.
OUTPUT	A table: question, variables, bias risk (low / medium / high), IEC note.

PRO TIP Pull 20 random files before committing. If more than a quarter are missing your key variable, choose another question.

TRY IT WITH Medicine ward records of acute kidney injury, 2020–2024, about 900 admissions.

→ **NEXT #18** What could kill this topic? A risk register

11 Topics that fit a 12-month data window

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · DNB TRAINEE / PG RESIDENT (LATE START)

USE IT When your approved data-collection window is 12 months or less.

PERSONA	You are a thesis coordinator who has rescued many late-starting postgraduate theses by right-sizing the question.
MY CONTEXT	Specialty: [SPECIALTY] Months left for data collection: [X] Eligible patients per month: [N] Current (too ambitious) topic idea, if any: [describe]
TASK	1. Tell me honestly whether my current idea fits the window, with the arithmetic. 2. Suggest 5 alternative or scaled-down questions with outcomes measurable within days or weeks, not months. 3. For each, estimate the patients needed and the months to recruit at my numbers.
RULES	Prefer cross-sectional, diagnostic-accuracy or short-follow-up designs. Show every calculation with its assumptions marked [ASSUMPTION], because I will have to defend the timeline to my guide.
OUTPUT	A verdict paragraph, then a table: question, design, patients needed, months to recruit.

PRO TIP Build in a 2-month buffer. Theses almost never finish recruitment on schedule.

TRY IT WITH DNB Orthopaedics, 10 months left, about 15 eligible patients per month.

→ **NEXT #22** Four-week feasibility check before you commit

12 Topics needing only routine investigations

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT When there is no funding and you can only use tests the hospital already does free or routinely.

PERSONA	You are a researcher who has built a publication record entirely on routine clinical data in low-resource Indian hospitals.
MY CONTEXT	Specialty: [SPECIALTY] Routine tests available at no cost to patients: [e.g. CBC, LFT, RFT, ABG, chest X-ray, ECG, bedside USG] Tests I cannot afford: [list] Patient groups I see often: [list]
TASK	1. Suggest 10 thesis topics that use only the routine tests I listed. 2. For each, give the research question, the key routine variable, the outcome, and why the question still matters clinically. 3. Mark the 3 with the best publication potential.
RULES	Exclude any topic needing a test from my "cannot afford" list, and do not suggest workarounds that shift cost to patients, because the IEC will reject that.
OUTPUT	A numbered table with a one-line clinical-value note for each, then the top 3.

PRO TIP Ratios and indices built from routine tests are popular but crowded. Pair them with a hard outcome to stand out.

TRY IT WITH MD Medicine; routine CBC, RFT, LFT, ABG, ECG, chest X-ray; lots of sepsis, CKD and snakebite.

→ **NEXT #7** FINER stress-test of my research question

13 Validate an existing clinical score in my population

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT When you want a solid, publishable design with a clear outcome, and a well-known score exists in your field.

PERSONA	You are a clinical epidemiologist experienced in external validation of prognostic and diagnostic scores in Indian populations.
MY CONTEXT	Candidate score or rule: [e.g. qSOFA, CURB-65, Alvarado, Wells] Population where it was developed (if known): [describe or "unsure"] My population and setting: [describe] Outcome the score predicts: [e.g. 28-day mortality]
TASK	1. Explain why validating this score in my population is (or is not) a genuine gap. 2. Outline the validation design: inclusion criteria, timing of scoring, outcome ascertainment, blinding of assessors. 3. Name the performance measures to report (discrimination, calibration) and what each means clinically. 4. Give the sample-size logic for validation studies (events needed, not just total patients).
RULES	Do not claim the score has never been validated in India; write [CHECK EXISTING INDIAN VALIDATIONS ON PUBMED], because that search decides whether this topic is novel.
OUTPUT	Four short headed sections, under 400 words in total.

PRO TIP Validation studies are strongest when the score is calculated before the outcome is known. Build that into your protocol.

TRY IT WITH CURB-65 for 30-day mortality in community-acquired pneumonia at a district hospital.

→ **NEXT #19** Choose a measurable primary outcome

14 Quality-improvement and audit thesis topics

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT / FACULTY

USE IT When your department wants a thesis that also improves care, or clinical trials are impractical.

PERSONA	You are a quality-improvement lead trained in PDSA cycles and clinical audit, who has supervised QI projects in Indian hospitals.
MY CONTEXT	Specialty and unit: [e.g. Obstetrics labour room] Care processes that often go wrong or vary: [list 3–5] Existing standards we could measure against: [e.g. WHO, national guideline, hospital protocol, or "unsure"]
TASK	1. Suggest 6 audit or QI topics, each with the standard, the measurable indicator and the target. 2. For the best one, outline a before-and-after design with one PDSA cycle. 3. Explain how to report it (SQUIRE 2.0 headings) and whether it counts as research needing full IEC review.
RULES	Name a standard only if it is a real, named guideline; otherwise write [STANDARD TO BE CONFIRMED], because an audit without a verifiable standard is not an audit.
OUTPUT	A table of 6 topics, then the outline for the best one, under 350 words.

PRO TIP Ask your IEC early whether QI counts as research at your institution. The answer changes consent requirements.

TRY IT WITH Time from admission to first antibiotic in suspected sepsis in the medical emergency.

→ **NEXT #7** FINER stress-test of my research question

15 Questionnaire-based topics without the KAP trap

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT / MBBS INTERN

USE IT When your topic will rely on a questionnaire or scale.

PERSONA	You are a psychometrics-trained researcher who reviews questionnaire studies and is tired of weak knowledge-attitude-practice (KAP) surveys.
MY CONTEXT	Area of interest: [e.g. inhaler adherence, depression in dialysis patients] Population: [describe] Languages my patients speak: [list]
TASK	1. Suggest 5 questionnaire-based questions that go beyond a descriptive KAP survey (e.g. linking a validated score to a clinical outcome). 2. For each, name the kind of validated instrument needed and whether a validated version in my patients' language is likely to exist. 3. Explain what translation and validation steps I need if it does not.
RULES	Do not name a specific instrument's Indian-language validation unless you are sure; write [CHECK FOR VALIDATED TRANSLATION], because using an unvalidated translation is a classic viva question.
OUTPUT	A table of 5 topics with instrument type and translation status, then the validation steps as a numbered list.

PRO TIP A validated instrument plus a hard clinical outcome makes a questionnaire study publishable.

TRY IT WITH Adherence to inhaled therapy and exacerbations in asthma, Hindi-speaking OPD patients.

→ **NEXT #7** FINER stress-test of my research question

16 Head-to-head decision matrix for 2–3 topics

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT When you are stuck choosing between two or three final topics.

PERSONA	You are a pragmatic research supervisor who helps residents decide with criteria, not gut feel.
MY CONTEXT	Topic A: [title + design + patient numbers] Topic B: [title + design + patient numbers] Topic C (optional): [title + design + patient numbers] What matters most to me (rank): [e.g. finishing on time, publication, learning a skill, guide's interest]
TASK	1. Build a decision matrix with my priorities as weighted criteria. 2. Score each topic 1 to 5 per criterion, with one line of reasoning. 3. Calculate weighted totals and give a clear recommendation. 4. State what new information would change the recommendation.
RULES	Use only the facts I gave. Where a score needs information I did not supply, mark it [NEED INFO] rather than guessing, because a guessed matrix decides nothing.
OUTPUT	The matrix table, then a 3-sentence recommendation.

PRO TIP If two topics tie, pick the one with the more reliable primary outcome measurement.

TRY IT WITH A: CPAP adherence in OSA (cohort, 8/month). B: Lung ultrasound vs X-ray for pneumonia (diagnostic, 25/month).

→ **NEXT #4** Guide-meeting pitch pack for your top 3 topics

17 Thesis title that passes the committee

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT When your topic is fixed and you need the formal thesis title.

PERSONA	You are a university thesis committee member who has seen hundreds of titles rejected for being vague, too long or overclaiming.
MY CONTEXT	Research question (PICOT): [paste] Study design: [design] Setting: [hospital, city] My university's title rules, if any: [e.g. word limit, must mention design]
TASK	1. Write 6 title options: descriptive, design-first, PICO-style, question-form, short, and conservative. 2. Check each against the common committee objections (vague population, missing design, causal words, excessive length). 3. Recommend one and explain why.
RULES	Use associational wording (association, correlation, comparison) unless my design is randomised, because causal wording in an observational title is the first thing examiners question.
OUTPUT	A numbered list of 6 titles with word counts, a one-line check result for each, and the recommendation.

PRO TIP Most universities like the design and setting in the title: "A prospective observational study at a tertiary care centre in north India".

TRY IT WITH Association of serum albumin with 28-day mortality in adult sepsis, prospective cohort, AIIMS Rishikesh.

→ **NEXT #19** Choose a measurable primary outcome

18 What could kill this topic? A risk register

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT Right after your topic is approved, before protocol writing.

PERSONA	You are a research operations manager who has seen theses fail for practical reasons: recruitment, equipment breakdowns, lost records and ethics delays.
MY CONTEXT	Approved topic and design: [paste] Recruitment source and expected numbers: [brief] Key equipment or tests: [list] People I depend on (lab, radiology, statistician): [list]
TASK	1. List the 10 most likely risks to completing this thesis on time. 2. For each, rate likelihood and impact (low / medium / high). 3. Give one prevention step and one fallback plan for each. 4. Highlight the 3 risks to discuss with my guide now.
RULES	Be specific to my topic, not generic. Where a risk depends on local facts I did not give, write [CHECK LOCALLY], because generic risk lists get ignored.
OUTPUT	A risk-register table: risk, likelihood, impact, prevention, fallback. Then the top 3.

PRO TIP Review this register every 3 months. Add risks as they appear.

TRY IT WITH Prospective cohort of 150 stroke patients needing MRI within 72 hours.

→ **NEXT #22** Four-week feasibility check before you commit

19 Choose a measurable primary outcome

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT When your topic is agreed but the primary outcome is still vague ("better outcome", "improvement").

PERSONA	You are a trial methodologist who insists every primary outcome is defined precisely enough for two observers to agree.
MY CONTEXT	Research question: [PICOT] Candidate outcomes I am considering: [list] Follow-up I can actually do: [e.g. in-hospital only, 30-day phone call, 3-month OPD visit]
TASK	1. Evaluate each candidate outcome for clinical relevance, measurability, reliability and feasibility with my follow-up. 2. Recommend one primary outcome and up to 3 secondary outcomes. 3. For the primary, write an operational definition: instrument, unit, cut-off and time point.
RULES	Prefer validated scales and objective measures. If an outcome needs a scale, say whether a validated version exists or write [VERIFY VALIDATED TOOL], because unvalidated outcomes weaken the whole thesis.
OUTPUT	An evaluation table, then the recommended outcomes, then the operational definition in 3–4 lines.

PRO TIP Your sample size depends on the primary outcome. Fix this before you calculate anything.

TRY IT WITH Candidates for ARDS: mortality, ventilator-free days, P/F ratio at day 3, length of ICU stay.

→ **NEXT #22** Four-week feasibility check before you commit

20 The questions my guide will ask about my topic

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT Before your first topic discussion with your guide, or before a departmental topic presentation.

PERSONA	You are an experienced thesis guide in [SPECIALTY] who probes every new resident's topic for weaknesses.
MY CONTEXT	My topic and design: [paste] What I know about the background: [brief notes] My expected patient numbers and time: [brief]
TASK	1. List the 12 questions you would ask me about this topic, from basic to tough. 2. For each, show what a weak answer and a strong answer look like. 3. Mark the 3 questions I am least prepared for, based on my notes.
RULES	Ground the model answers in my notes. Where a strong answer needs facts I have not given, write [FIND THIS BEFORE THE MEETING] instead of supplying numbers, because my guide will ask where they came from.
OUTPUT	A numbered list of questions with weak and strong answer pairs, then the 3 danger questions.

PRO TIP Answer the 3 danger questions in writing before the meeting.

TRY IT WITH Topic: point-of-care lung ultrasound vs chest X-ray for pneumonia in children under 5.

→ **NEXT #4** Guide-meeting pitch pack for your top 3 topics

21 Topics from the gaps in a clinical guideline

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT / FACULTY

USE IT When a recent guideline in your field lists recommendations based on weak evidence or states "further research needed".

PERSONA	You are a guideline methodologist familiar with GRADE and with how evidence gaps turn into research agendas.
MY CONTEXT	Guideline document or recommendations section: [PASTE OR UPLOAD THE GUIDELINE / SECTION] My specialty and setting: [brief]
TASK	1. Extract every recommendation graded low or very low certainty, conditional, or expert opinion, and every explicit "research needed" statement. 2. For each, say whether it could become a feasible postgraduate thesis question in my setting. 3. Turn the best 5 into PICOT questions with a suggested design.
RULES	Quote the guideline wording for each gap in quotation marks so I can verify it. Do not describe recommendations that are not in the text I provided, because guideline misquotes are easy to catch.
OUTPUT	A table: quoted recommendation, certainty grade, feasible (yes / no), PICOT. Then the top 5.

PRO TIP Topics built on guideline gaps are easy to justify in the introduction. The gap is already published.

TRY IT WITH Upload the research-gaps section of a recent national TB or sepsis guideline.

→ **NEXT #7** FINER stress-test of my research question

22 Four-week feasibility check before you commit

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT After topic approval, before the full protocol, if recruitment numbers are uncertain.

PERSONA	You are a research methodologist who believes in testing recruitment before promising it.
MY CONTEXT	Topic and design: [paste] Assumed eligible patients per month: [N] Key procedures per patient: [list]
TASK	1. Design a 4-week screening-log exercise: what to record for every potentially eligible patient (screened, eligible, excluded with reason, would consent). 2. Give a simple log template with columns. 3. Define go / amend / stop criteria based on the numbers the log produces. 4. Explain how to present the log to my guide and the IEC.
RULES	A screening log records no patient identifiers; use serial numbers only, because it is collected before formal consent.
OUTPUT	A step list, the log template as a table, and the decision criteria.

PRO TIP A 4-week log is the strongest evidence you can give that your sample size is realistic.

TRY IT WITH Diabetic foot ulcer patients undergoing offloading, assumed 12 per month.

→ **NEXT #18** What could kill this topic? A risk register

23 Five original angles on a crowded topic

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

USE IT When the topic you love turns out to be heavily studied.

PERSONA	You are a researcher skilled at finding original angles within well-studied areas without resorting to trivial changes.
MY CONTEXT	Crowded topic: [describe] What is already well established (from my reading): [brief] My setting's special features: [e.g. rural referral hospital, high TB co-infection, altitude, tribal population]
TASK	Give 5 original angles, one each from: a different population, a different setting, a different outcome, a different measurement method, and longer follow-up. For each, write the new PICOT question and explain why it adds value rather than repeating existing work.
RULES	Ground each angle in my setting's special features. Where novelty depends on the literature, mark it [VERIFY NOVELTY], because the angle only counts if nobody has published it.
OUTPUT	A table of 5 angles: new question, what it adds, feasibility (low / medium / high).

PRO TIP The strongest angles usually come from what is unusual about your hospital's patients.

TRY IT WITH Crowded topic: HbA1c and outcomes in diabetic foot. Setting: tribal district hospital in Jharkhand.

→ **NEXT #7** FINER stress-test of my research question

24 Two-month project for an MBBS student (ICMR-STS style)

STANDARD

1.1 TOPIC SELECTION & RESEARCH QUESTION · MBBS STUDENT

USE IT When you are an undergraduate applying for a short-term studentship or doing your first research project.

PERSONA	You are a medical college research mentor who has guided many MBBS students through short, two-month research projects, including ICMR Short-Term Studentship applications.
MY CONTEXT	My year of MBBS: [e.g. 2nd year] Departments where I have access or a willing mentor: [list] Time available: about 2 months, alongside classes Interests: [list]
TASK	1. Suggest 6 projects completable in 2 months by a student, e.g. a questionnaire study, a record review or a simple cross-sectional study. 2. For each, give the research question, design, a realistic sample size and the data-collection method. 3. Pick the best one and outline a one-page proposal: title, introduction (3 lines), objectives, methods, expected outcome.
RULES	Keep every project within student ethics limits: no invasive procedures and no interventions. Put [CHECK CURRENT STS GUIDELINES] wherever application rules matter, because scheme rules change every year.
OUTPUT	A table of 6 projects, then the one-page proposal.

PRO TIP Mentors say yes faster when you arrive with a one-page proposal instead of "any topic, sir?".

TRY IT WITH 2nd-year MBBS, mentor in Community Medicine, interested in hypertension awareness.

→ **NEXT #26** Sharpen my research question to one line

25 Ten quick topic seeds in my specialty

QUICK

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT (YEAR 1)

Act as a senior [SPECIALTY] faculty member at an Indian medical college. Give me 10 thesis topic seeds for a [government / private] hospital with routine investigations only. For each, give one line on the clinical question and one on why it matters in India. Keep it to a numbered list, and mark any claim that needs checking as [VERIFY].

TRY IT WITH MD Dermatology, government medical college, high burden of fungal infections.

26 Sharpen my research question to one line

QUICK

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT / MBBS INTERN

Act as a research methods tutor. My current research question is: [PASTE QUESTION]. Rewrite it in 3 sharper one-line versions that name the population, exposure or intervention, comparator and outcome. Put the best one first and explain in one sentence why it is better.

TRY IT WITH Does obesity affect asthma?

27 PICO or PECO? Frame my question correctly

QUICK

1.1 TOPIC SELECTION & RESEARCH QUESTION · MBBS STUDENT / PG RESIDENT

Act as a clinical epidemiology teacher. My study idea is: [DESCRIBE IDEA]. Tell me whether it should be framed as PICO (intervention) or PECO (exposure), explain the difference in two lines, then write my question in the correct framework with each element labelled.

TRY IT WITH Comparing outcomes of smokers and non-smokers after coronary bypass surgery.

28 Check ongoing trials before you commit

QUICK

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

Act as a medical librarian. For my topic [TOPIC], give me 5 search-term combinations to check CTRI (ctri.nic.in) and ClinicalTrials.gov for ongoing or recently completed studies, and tell me what to look for in each registry entry to judge whether my thesis would duplicate it.

TRY IT WITH Early enteral nutrition in acute pancreatitis.

29 The follow-up that surfaces Indian gaps

QUICK

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

Act as a research-gap analyst. Based on your previous answer about research gaps in [TOPIC], list what has NOT been studied in Indian or South Asian populations specifically. Give 5 points, each with the population, the missing outcome and why it matters here. Mark claims to verify as [VERIFY].

TRY IT WITH Follow-up after a Perplexity answer on gaps in COPD management.

30 Identify the variables in my question

QUICK

1.1 TOPIC SELECTION & RESEARCH QUESTION · MBBS STUDENT / PG RESIDENT

Act as a biostatistics tutor. For my research question [PASTE QUESTION], identify the dependent variable, the independent variable, likely confounders and effect modifiers. Present them in a 4-row table with the data type and unit of each (continuous, binary, ordinal, categorical).

TRY IT WITH Is anaemia in pregnancy associated with low birth weight at a district hospital?

31 My topic is too broad: narrow it three ways

QUICK

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

Act as a thesis supervisor. My topic [TOPIC] is too broad for a 2-year thesis. Narrow it three different ways: by population, by outcome and by setting. Give each as a one-line research question, then say which is most feasible with about [N] patients per month.

TRY IT WITH Outcomes of diabetes in India.

32 What infrastructure does this topic need?

QUICK

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

Act as a research operations planner. For the study [TITLE + DESIGN], list every piece of equipment, test, trained person, software and consumable needed from recruitment to analysis. Mark each as routine, special or costly, and flag anything a government medical college may not have.

TRY IT WITH Diagnostic accuracy of bedside lung ultrasound for pleural effusion.

33 Does this Western topic apply in India?

QUICK

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

Act as a global health researcher. The topic [TOPIC] is well studied in Western populations. Give 5 reasons its findings may not transfer to Indian patients (disease burden, genetics, nutrition, care access, practice patterns), and turn the strongest reason into an India-specific research question.

TRY IT WITH Statin response and LDL targets in young adults with premature coronary disease.

34 Hypothesis-generating or hypothesis-testing?

QUICK

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

Act as a research methods examiner. For my study [PASTE QUESTION + DESIGN], tell me whether it is hypothesis-generating or hypothesis-testing, why, and what that changes about my sample size, analysis and the wording of my conclusion. Keep it under 150 words.

TRY IT WITH A cross-sectional survey of sleep quality among ICU nurses on night shifts.

35 Summarise my topic on one slide

QUICK

1.1 TOPIC SELECTION & RESEARCH QUESTION · PG RESIDENT

Act as an academic presentation coach. Turn my topic [TITLE + PICOT + GAP] into one slide for a departmental topic presentation: a title, 4 bullets (problem, gap, question, design) and a one-line "why now". Keep each bullet under 15 words.

TRY IT WITH Frailty index and post-operative complications in elderly emergency laparotomy.

1.2

Literature search strategy

A literature review is only as good as the search behind it. A reproducible search string is something examiners and journals now ask to see.

30

prompts

3

Master

17

Standard

10

Quick

HOW TO USE THIS SECTION

Use it before you read anything: build a reproducible search, then screen and store what you find.

Start with the Master prompts (**#36–38**) for the full workflow; use the Standard prompts (**#39–55**) for one task at a time; reach for the Quick prompts (**#56–65**) when you have two minutes. Tap any title below to jump to it, and follow → **NEXT** to move to the next step.

#36 Build a reproducible, multi-database search

#37 Search documentation that survives the viva

#38 Search rescue: iterate until the results are right

#39 MeSH terms and synonyms for one concept

#40 PICO → concept blocks table

#41 Fix the syntax of my search string

#42 Too many or too few hits: fix my search

#43 Translate my PubMed string for other databases

#44 Use PubMed filters without biasing the review

#45 Find studies from Indian centres

#46 Grey literature and trial-registry search plan

#47 Backward and forward citation chasing plan

#48 Map the research landscape in Perplexity

#49 Find the systematic reviews before anything else

#50 Identify the landmark papers in my field

#51 Set up alerts so new papers find you

#52 Reference manager set-up and deduplication

#53 Write the search-methods paragraph

#54 Legal ways to get the full text of a paper

#55 Peer review my search strategy (PRESS checklist)

#56 Boolean operators explained with my topic

#57 Every synonym for my concept

#58 Explode, major topic, subheadings: explained

#59 Adapt my search for Google Scholar

#60 Which databases for my specialty?

#61 Email an author for their paper or data

#62 Has this paper been retracted or corrected?

#63 Where does this study sit in the evidence hierarchy?

#64 Turn my question into Perplexity and Consensus queries

#65 How far back should my search go?

Build a reproducible, multi-database search

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT / FACULTY

USE IT

After your topic is provisionally approved, before you read a single paper in depth.

WHY

"I searched PubMed" is not a method. A reproducible search, with concept blocks, controlled vocabulary and exact strings, is what separates a review from a reading list, and it is what examiners and journals ask to see.

COPY THE PROMPT BELOW

PERSONA	You are a medical librarian and systematic-review methodologist who has co-authored Cochrane reviews and trained postgraduates in reproducible searching. You are precise about search syntax and sceptical of shortcuts.
MY CONTEXT	My full PICOT question: [PASTE QUESTION] Study design I intend to use: [e.g. prospective observational cohort] Databases I can access: [PubMed / Embase / Scopus / Cochrane Library / Web of Science / Google Scholar] Language limits: [English only / English plus others] Date range: [e.g. 2010 to present, and tell me if I should widen it] Reference manager: [Zotero / Mendeley / EndNote / none]
TASK	1. Break my question into concept blocks. For each block, list controlled vocabulary (MeSH for PubMed, Emtree for Embase) and free-text synonyms, including spelling variants, abbreviations and older terms. 2. Write a complete, copy-paste-ready PubMed string with correct field tags, Boolean operators, truncation and explosion. 3. Translate it into Embase and Cochrane Library syntax. 4. Give one deliberately broad and one deliberately narrow version, and say what each will miss. 5. List the steps beyond databases: citation chasing, reference lists, grey literature, trial registries (CTRI, ClinicalTrials.gov) and preprint servers relevant to my field. 6. Give me a search log template: database, date searched, exact string, filters, hits.
RULES	Build strings only from real MeSH and Emtree headings; if you are not certain a heading exists, mark it [CHECK IN MeSH BROWSER], because one invalid heading silently breaks the whole search. Do not report any studies or hit counts from memory.
OUTPUT	Concept-block table, then each string in its own code block, then the broad and narrow versions, the extra steps as a checklist, and the log template.
TWEAK	End by telling me the one concept block most likely to be missing synonyms, and how to test it.

How to get more out of it

- › Run the PubMed string and check the hit count. Under 50 is usually too narrow; over 3,000 is usually too broad.
- › Save the exact string and the date you ran it. Examiners and journals ask for both.
- › Test the string: pick 3 papers you already know are relevant. If any is missing from the results, add its keywords.
- › → **NEXT #42** Too many or too few hits: fix my search

Search documentation that survives the viva

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

USE IT

After you have run your searches, while every string, date and count is still in front of you.

WHY

Examiners and journal reviewers increasingly ask "how exactly did you search?". A documented search (strings, dates, hits, deduplication) turns your ROL from a narrative into a method, and saves you from redoing it months later.

COPY THE PROMPT BELOW

PERSONA	You are a systematic-review methodologist who reviews search reporting for journals and applies the PRISMA 2020 and PRISMA-S reporting standards.
MY CONTEXT	Databases searched, with date of each search: [list] Exact strings used: [PASTE STRINGS HERE] Hits per database: [numbers] Duplicates removed (and how): [number, tool] Records screened, full texts assessed, studies included: [numbers] Reasons for full-text exclusion with counts: [list] Other sources (registries, citation chasing, experts): [list with counts]
TASK	1. Write the search-methods paragraph for my thesis methodology chapter. 2. Produce a PRISMA 2020 flow diagram description with my numbers slotted in, box by box. 3. Check my numbers for internal consistency (screened minus excluded must equal full texts assessed, and so on) and flag any mismatch. 4. List the PRISMA-S items my documentation covers and the ones it is missing. 5. Draft a one-page search appendix table: database, platform, date, string, limits, hits.
RULES	Use only the numbers I gave. Where a number is missing, write [MISSING — FILL IN] rather than estimating, because an inconsistent flow diagram is the first thing a reviewer checks. Keep my exact strings unchanged.
OUTPUT	The methods paragraph (about 150 words), the flow-diagram box list, a consistency check table, the PRISMA-S gap list, and the appendix table.
TWEAK	Finish with the two questions an examiner is most likely to ask about my search, and model answers based only on my documentation. Then tell me which single piece of documentation I should create today because it cannot be reconstructed later (for example, the original hit counts or the date of the first search).

How to get more out of it

- › Draw the final flow diagram in PowerPoint or draw.io from the box list. Don't ask the AI to draw it.
- › Keep screenshots of each database's results page with the date visible.
- › Re-run the search just before submission and report the update date.
- › → **NEXT #53** Write the search-methods paragraph

Search rescue: iterate until the results are right

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

USE IT

When your search returns far too many, far too few, or irrelevant results, and you have already tried tweaking it.

WHY

Searching is iterative. Most residents give up after one bad result, or keep reading 4,000 irrelevant titles. This prompt makes the AI diagnose what went wrong and fix it one change at a time.

COPY THE PROMPT BELOW

PERSONA	You are an expert search consultant at a medical library who debugs failed searches daily. You change one thing at a time and explain the effect of each change.
MY CONTEXT	My research question: [PICOT] Database and current search string: [PASTE STRING HERE] Hit count: [N] What is wrong: [too many / too few / irrelevant results] Five titles from the results that are irrelevant (if any): [PASTE TITLES] Two or three papers I know should appear but do not: [titles or PMIDs]
TASK	1. Diagnose the problem: which concept block, operator, field tag or truncation is causing it. 2. Propose a revised string with one change, and explain what the change should do to the hit count. 3. Ask me to run it and report back the new count and 5 sample titles. 4. Repeat for up to 4 rounds until the known papers appear and the irrelevant titles drop out. 5. Then give the final string with a one-line rationale for each block.
RULES	Change one thing per round, because changing several at once makes it impossible to tell what worked. Do not invent MeSH terms; mark uncertain ones [CHECK IN MeSH BROWSER]. Wait for my results before the next round.
OUTPUT	One round per turn: a short diagnosis, the revised string in a code block, the expected effect on the hit count and the known papers, and exactly what I should report back to you. Keep each round under 200 words so I can act on it quickly.
TWEAK	At the end, summarise the search in 3 sentences suitable for my methods chapter.

How to get more out of it

- › Known papers are the best test of a search. Keep 3 "must-find" papers handy.
- › Irrelevant titles tell you which synonym is too loose. Paste them rather than describing them.
- › If the tweaks never converge, your question is probably too broad. Go back to section 1.1.
- › → **NEXT #37** Search documentation that survives the viva

39 MeSH terms and synonyms for one concept

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT / MBBS STUDENT

USE IT When you are building a search and stuck on one concept.

PERSONA	You are a medical librarian who knows the MeSH hierarchy well and teaches residents to search systematically.
MY CONTEXT	The concept I need to search: [e.g. acute kidney injury] My wider question for context: [one line]
TASK	1. Give the most relevant MeSH heading(s), where each sits in the hierarchy, and whether to explode it. 2. List entry terms and free-text synonyms, including abbreviations, spelling variants (British / American) and outdated terms used in older papers. 3. Combine them into one PubMed block using [MeSH] and [tiab] tags with OR.
RULES	Only give MeSH headings you are confident exist; mark others [CHECK IN MeSH BROWSER], because an invalid heading returns nothing without warning.
OUTPUT	A short table (term, type, tag), then the combined block in a code block.

PRO TIP Check the MeSH introduction year. Papers older than that are indexed under a different term.

TRY IT WITH Concept: ventilator-associated pneumonia.

→ **NEXT #40** PICO → concept blocks table

40 PICO → concept blocks table

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

USE IT The first step of any search, straight after your question is fixed.

PERSONA	You are a search-methods tutor who teaches residents to turn clinical questions into structured searches.
MY CONTEXT	My PICOT question: [PASTE QUESTION] Database I will start with: [PubMed / Embase / Google Scholar]
TASK	1. Split the question into concept blocks and say which PICO elements to search and which to leave out (usually comparator and outcome), with the reason. 2. For each block, give controlled vocabulary and free-text terms. 3. Show how the blocks combine: OR within blocks, AND between blocks. 4. Suggest a validated study-design filter if one fits my question.
RULES	Explain every exclusion decision in one line, because searching every PICO element is the most common reason searches miss relevant papers.
OUTPUT	A concept-block table, then the combined string, then a one-paragraph explanation.

PRO TIP Most good searches use 2 or 3 blocks, not 5.

TRY IT WITH In adults with community-acquired pneumonia, does procalcitonin-guided therapy reduce antibiotic duration compared with standard care?

→ **NEXT #36** Build a reproducible, multi-database search

41 Fix the syntax of my search string

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

USE IT When PubMed gives an error, a strange translation, or zero results.

PERSONA	You are a meticulous search-syntax checker who spots misplaced brackets and wrong field tags at a glance.
MY CONTEXT	My search string: [PASTE STRING HERE] Database: [PubMed / Embase / Scopus / Google Scholar] What PubMed shows under "Details" or "Search translation": [PASTE IF AVAILABLE]
TASK	1. Check brackets, Boolean operators (upper case), field tags, quotation marks, truncation and proximity operators. 2. List every error with its correction. 3. Give the corrected string. 4. Explain how PubMed's automatic term mapping will treat any untagged terms.
RULES	Keep my intended logic unless it is clearly wrong; if you change the logic, say so explicitly, because a silent change alters which papers I find.
OUTPUT	An error table (error, why it matters, fix), then the corrected string in a code block.

PRO TIP Always read PubMed's search translation. It shows what PubMed actually searched, not what you typed.

TRY IT WITH (asthma OR wheez*) AND inhaler adherence AND (child* OR paediatric

→ **NEXT #42** Too many or too few hits: fix my search

42 Too many or too few hits: fix my search

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

USE IT When your search returns thousands of results, or almost none.

PERSONA	You are a search consultant who balances sensitivity and precision for postgraduate literature reviews.
MY CONTEXT	Research question: [PICOT] Current string: [PASTE STRING HERE] Database and hit count: [PubMed, N] Is this for a narrative ROL or a systematic review?: [narrative / systematic]
TASK	1. Diagnose why the count is too high or too low. 2. Suggest 3 targeted changes (e.g. major-topic MeSH, title-only terms, removing a block, adding synonyms, a design filter), ranked by likely effect. 3. Give the revised string for the best option. 4. Say what relevant papers each change risks losing.
RULES	For a systematic review, prefer sensitivity and never restrict by abstract availability or free full text, because those filters introduce bias.
OUTPUT	Diagnosis in 2–3 lines, a ranked table of changes, then the revised string.

PRO TIP For a thesis ROL, 150 to 800 well-targeted hits is a workable range.

TRY IT WITH "sepsis AND lactate" returning 9,000 PubMed results.

→ **NEXT #38** Search rescue: iterate until the results are right

43 Translate my PubMed string for other databases

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT / FACULTY

USE IT When your PubMed search is final and you need Embase, Scopus, Cochrane or Web of Science versions.

PERSONA	You are a medical librarian fluent in the syntax of PubMed, Embase (Ovid and Elsevier), Scopus, Cochrane Library and Web of Science.
MY CONTEXT	Final PubMed string: [PASTE STRING HERE] Target databases and platforms: [e.g. Embase.com, Scopus, Cochrane Library]
TASK	1. Translate the string for each target database, mapping MeSH to Emtree where relevant and converting field tags, truncation and proximity operators. 2. Note any concept that has no equivalent heading, and how you handled it. 3. List any syntax difference that could change the results.
RULES	Mark any Emtree term you are unsure of as [CHECK IN EMTREE], because MeSH and Emtree headings often differ in name and scope.
OUTPUT	One code block per database, with a short notes list under each.

PRO TIP Scopus has no controlled vocabulary. Strengthen the free-text terms there.

TRY IT WITH PubMed string for "high-flow nasal oxygen in acute hypoxaemic respiratory failure".

→ **NEXT #37** Search documentation that survives the viva

44 Use PubMed filters without biasing the review

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

USE IT Before applying filters such as date, language, age group, species or article type.

PERSONA	You are a systematic-review methodologist who warns residents about filters that silently exclude relevant evidence.
MY CONTEXT	My question: [PICOT] Filters I plan to use: [e.g. last 10 years, English, humans, adults, RCT only] Purpose of my search: [thesis ROL / systematic review / quick clinical question]
TASK	1. For each filter, explain what it actually does in PubMed, including how it treats recently added, not-yet-indexed records. 2. Say whether each is appropriate for my purpose, and the bias it could introduce. 3. Suggest safer alternatives (e.g. search-term blocks instead of the "Humans" filter). 4. Give my final string with the recommended filters applied.
RULES	Be explicit about which filters depend on MeSH indexing, because the newest papers are not yet indexed and would be excluded.
OUTPUT	A filter table (filter, what it does, keep or drop, reason), then the final string.

PRO TIP Justify every filter in your methods section. Unexplained filters look like cherry-picking.

TRY IT WITH Filters: humans, English, last 5 years, free full text.

→ **NEXT #36** Build a reproducible, multi-database search

45 Find studies from Indian centres

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

USE IT When you need the Indian evidence for your introduction and ROL, beyond global studies.

PERSONA	You are a research librarian at an Indian medical college who specialises in finding Indian and South Asian clinical studies.
MY CONTEXT	Topic: [TOPIC] Main PubMed string so far: [PASTE STRING HERE]
TASK	1. Show how to add an India block using the [Affiliation] and [tiab] fields (India, Indian, state and major city names, major institutions). 2. Suggest regional and Indian sources to search beyond PubMed, including Indian journals and thesis repositories, and say what each is useful for. 3. Explain how to handle Indian journals that are not indexed in PubMed. 4. Give Google Scholar search tips for Indian studies.
RULES	Name only sources you are confident exist and are currently accessible; mark uncertain ones [VERIFY ACCESS], because Indian indexing services change often.
OUTPUT	The India block in a code block, then a source table (source, what it covers, access), then 4 tips.

PRO TIP Affiliation searches miss studies whose authors list only the city. Add major city names as free text.

TRY IT WITH Topic: scrub typhus in children.

→ **NEXT #46** Grey literature and trial-registry search plan

46 Grey literature and trial-registry search plan

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT / FACULTY

USE IT For any systematic review, and for a thesis ROL where unpublished or ongoing work may matter.

PERSONA	You are an evidence-synthesis specialist who knows that missing grey literature leads to publication bias.
MY CONTEXT	Topic and question: [PICOT] Type of review: [thesis ROL / systematic review / scoping review]
TASK	1. List the grey-literature sources to search: trial registries (CTRI, ClinicalTrials.gov, WHO ICTRP), preprint servers, conference abstracts, theses repositories (including Indian ones), and government or WHO reports. 2. For each, give a simple search approach and what to record. 3. Explain how to handle registry entries with no published results. 4. Give a documentation table template.
RULES	Keep the list relevant to my topic and field, not generic. Mark any source whose current search interface you are unsure of as [CHECK SITE], because registries change their search tools.
OUTPUT	A source table, then the handling notes, then the documentation template.

PRO TIP A trial registered in CTRI years ago with no publication is itself a finding. Note it in your discussion.

TRY IT WITH Systematic review of yoga interventions for hypertension in Indian adults.

→ **NEXT #47** Backward and forward citation chasing plan

47 Backward and forward citation chasing plan

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

USE IT After screening, when you have 3–10 key included papers.

PERSONA	You are a systematic reviewer who uses citation chasing to find the studies that keyword searches miss.
MY CONTEXT	My key papers (title, year, DOI or PMID): [PASTE LIST HERE] Tools I can use: [Google Scholar / Scopus / Web of Science / PubMed "cited by" / Semantic Scholar]
TASK	1. Explain backward chasing (the reference lists) and forward chasing (papers that cite these), and when each adds most. 2. Give step-by-step instructions for each tool I can use. 3. Suggest screening rules to keep chasing manageable. 4. Give a log template to record what each round found.
RULES	Do not list papers that cite mine from memory; tell me how to retrieve them, because any list you produce without live search may be fabricated.
OUTPUT	A numbered method, tool-specific steps, and a log table.

PRO TIP Forward chasing from a landmark paper is the fastest way to find the newest work on your topic.

TRY IT WITH Key paper: a well-known RCT on corticosteroids in severe community-acquired pneumonia.

→ **NEXT #37** Search documentation that survives the viva

48 Map the research landscape in Perplexity

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

USE IT At the very start of a new topic, in Perplexity with Academic mode switched on.

PERSONA	You are a research-landscape analyst who summarises a field in minutes using cited, current sources.
MY CONTEXT	Topic: [TOPIC] Population and setting of interest: [e.g. Indian adults in tertiary care] Time window: [e.g. last 5 years]
TASK	1. Summarise what is well established, with citations. 2. Summarise what is contested or uncertain, and why. 3. Summarise what is unstudied or under-studied in my population and setting. 4. List the 5 most important recent studies or reviews, with year and journal, and a clickable source. 5. Suggest 5 search terms I should use in PubMed next.
RULES	Cite a source for every claim. If you cannot find a source for a statement, leave the statement out, because I will verify every citation before using it.
OUTPUT	Headed sections with short bullets and numbered citations.

PRO TIP Follow up with: "What has NOT been studied in Indian patients specifically?" It surfaces the most useful gaps.

TRY IT WITH Topic: long-term outcomes after dengue shock syndrome.

→ **NEXT #29** The follow-up that surfaces Indian gaps

49 Find the systematic reviews before anything else

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT / FACULTY

USE IT At the start of any ROL or research question, to avoid reinventing existing syntheses.

PERSONA	You are an evidence-based medicine tutor who always starts with the highest level of existing evidence.
MY CONTEXT	My question: [PICOT] My main search string: [PASTE STRING HERE]
TASK	1. Show how to find systematic reviews and meta-analyses: the PubMed systematic-review filter, the Cochrane Library, and PROSPERO for registered reviews in progress. 2. Give the modified strings for each. 3. Explain how to judge whether an existing review already answers my question (scope, date, quality, population). 4. Tell me what to do if a recent high-quality review exists: update it, narrow the angle, or change the question.
RULES	Do not name specific reviews from memory; show me how to retrieve them, because an invented review would mislead my whole plan.
OUTPUT	Numbered steps, the strings in code blocks, and a short decision guide.

PRO TIP A registered PROSPERO review on your exact question means someone is already doing it. Check before you start.

TRY IT WITH Question: effect of early tracheostomy on mortality in mechanically ventilated adults.

→ **NEXT #8** Is this topic already done to death?

50 Identify the landmark papers in my field

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

USE IT When you need the must-cite trials and studies for your introduction, ROL or seminar.

PERSONA	You are a senior academic who knows how to identify landmark literature systematically rather than by memory.
MY CONTEXT	Topic: [TOPIC] Current guidelines I can access: [name them or "none yet"]
TASK	1. Explain 4 reliable ways to identify landmark papers: guideline reference lists, highly cited papers in Scopus or Google Scholar, reviews' "key studies" sections, and editorials. 2. Give the search steps for each. 3. Suggest a table for recording each landmark paper: design, population, intervention, key finding, why it changed practice. 4. List candidate landmark studies you believe are relevant, each marked [VERIFY PMID].
RULES	Every paper you name must carry the [VERIFY PMID] tag, because trial names and years are frequently misremembered by AI. Name fewer papers rather than guessing.
OUTPUT	The 4 methods as a numbered list, the recording table, then the candidate list.

PRO TIP If a paper appears in the reference list of every major guideline, it is a landmark.

TRY IT WITH Topic: fluid resuscitation in septic shock.

→ **NEXT #36** Build a reproducible, multi-database search

51 Set up alerts so new papers find you

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT / FACULTY

USE IT Once your search is final, so you do not miss papers published during your thesis.

PERSONA	You are a research productivity coach who sets up literature alerts for busy clinicians.
MY CONTEXT	Final search string: [PASTE STRING HERE] Tools I use: [PubMed with My NCBI / Google Scholar / journal alerts / RSS] How often I want updates: [weekly / monthly]
TASK	1. Give step-by-step instructions to save the search and create an email alert in each tool I use. 2. Suggest a tighter version of the string so alerts are not flooded. 3. Suggest 5 key journals in my field for table-of-contents alerts. 4. Describe a 15-minute monthly routine to process alerts into my reference manager.
RULES	Keep instructions tool-specific. Where an interface may have changed, say [CHECK CURRENT MENU], because step-by-step menus go out of date.
OUTPUT	Numbered steps per tool, the alert string, the journal list, and the monthly routine.

PRO TIP Re-run your full search before submission anyway. Alerts can miss papers indexed late.

TRY IT WITH Search on "prone positioning in non-intubated COVID-19" with weekly PubMed alerts.

→ **NEXT #52** Reference manager set-up and deduplication

52 Reference manager set-up and deduplication

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

USE IT Before you import your first search results.

PERSONA	You are a research-skills trainer who has set up Zotero and Mendeley for hundreds of thesis students.
MY CONTEXT	Reference manager: [Zotero / Mendeley / EndNote] Word processor: [MS Word / Google Docs] Citation style required by my university: [e.g. Vancouver] Expected number of records: [N]
TASK	1. Give set-up steps: install, browser connector, word-processor plug-in, and the citation style. 2. Explain how to import results from PubMed and other databases. 3. Give a deduplication workflow, including records the tool will miss. 4. Suggest a folder or tag structure for screening (to screen, included, excluded with reason). 5. List the 5 mistakes that corrupt thesis references, and how to avoid each.
RULES	Keep steps specific to my tool. Where menus may have changed, write [CHECK CURRENT VERSION], because outdated steps waste hours.
OUTPUT	Numbered sections, each with short steps.

PRO TIP Never type a reference by hand into your thesis. Insert every citation through the plug-in.

TRY IT WITH Zotero with MS Word, Vancouver style, about 600 records.

→ **NEXT #37** Search documentation that survives the viva

53 Write the search-methods paragraph

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

USE IT When writing the methodology or ROL chapter, after your search is complete.

PERSONA	You are an academic editor who writes concise, reproducible search-methods sections for theses and journals.
MY CONTEXT	Databases and platforms: [list] Dates of search: [dates] Search terms and full strings: [PASTE HERE] Limits applied and why: [list] Additional sources: [registries, reference lists, experts] Screening process: [who screened, how disagreements were resolved]
TASK	1. Write a 120–180 word search-methods paragraph in past tense. 2. Write a one-line pointer to the appendix containing the full strings. 3. List anything a reviewer would say is missing.
RULES	Use only the details I gave. Where something standard is missing (e.g. the date of the last search), insert [ADD DETAIL] instead of inventing it, because search methods must be exactly reproducible.
OUTPUT	The paragraph, the pointer line, and a short missing-items list.

PRO TIP Name the platform as well as the database, e.g. "Embase via Ovid". The results differ.

TRY IT WITH PubMed and Cochrane searched 10 March 2026, English only, two reviewers screening independently.

→ **NEXT #37** Search documentation that survives the viva

54 Legal ways to get the full text of a paper

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT / MBBS STUDENT

USE IT When a key paper is behind a paywall.

PERSONA	You are a medical college librarian who helps students and residents access literature legally.
MY CONTEXT	Paper details: [title, journal, year, DOI] My institution's access, if known: [e.g. institutional subscription, national consortium, none]
TASK	1. List the legal routes in order of speed: institutional access and library, national or consortium access schemes in India, open-access versions (PubMed Central, Unpaywall, author repositories), interlibrary loan, and contacting the author. 2. For each, give the steps. 3. Draft a short, polite email to the corresponding author requesting the PDF for thesis use.
RULES	Recommend only legal routes. Mark any national scheme whose current coverage you are unsure of as [CHECK WITH LIBRARY], because access programmes change.
OUTPUT	A numbered list of routes with steps, then the email (under 100 words).

PRO TIP Authors almost always reply. Most are happy to share their paper with a student.

TRY IT WITH A 2024 RCT in a paywalled critical-care journal.

→ **NEXT #47** Backward and forward citation chasing plan

55 Peer review my search strategy (PRESS checklist)

STANDARD

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT / FACULTY

USE IT Before you run the final search for a systematic review or a thesis ROL your examiners will scrutinise.

PERSONA	You are an information specialist who peer reviews search strategies using the PRESS (Peer Review of Electronic Search Strategies) checklist.
MY CONTEXT	My research question: [PICOT] My full search string and database: [PASTE STRING HERE] Limits applied: [list]
TASK	1. Review the strategy against the PRESS elements: translation of the question, Boolean and proximity operators, subject headings, text-word searching (spelling, synonyms, truncation), spelling and syntax, and limits and filters. 2. For each element, rate it as no revision, revision suggested, or revision required, with a reason. 3. Give the revised string with all required changes applied.
RULES	Explain every required change in one line, because I must be able to justify the final strategy myself. Do not add concepts that are not in my question.
OUTPUT	A PRESS review table, then the revised string in a code block.

PRO TIP Ask a colleague or librarian to run the same review. Two reviews catch more errors than one.

TRY IT WITH A PubMed search on anticoagulation in cancer-associated venous thromboembolism.

→ **NEXT #37** Search documentation that survives the viva

QUICK PROMPTS · 1.2

56 Boolean operators explained with my topic

QUICK

1.2 LITERATURE SEARCH STRATEGY · MBBS STUDENT / PG RESIDENT

Act as a medical librarian teaching a first-time searcher. Explain AND, OR, NOT, brackets, quotation marks and truncation in under 150 words, using my topic [TOPIC] for every example. End with one complete example search string built from my topic.

TRY IT WITH Topic: iron-deficiency anaemia in adolescent girls.

57 Every synonym for my concept

QUICK

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

Act as a medical terminology expert. For the concept [CONCEPT], list all synonyms, abbreviations, British and American spellings, older terms and common misspellings used in paper titles and abstracts. Group them in a table and mark the 5 most important for a search.

TRY IT WITH Concept: chronic obstructive pulmonary disease.

58 Explode, major topic, subheadings: explained

QUICK

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

Act as a PubMed trainer. Using the MeSH term [MeSH TERM], explain in plain language what "explode", "major topic" and subheadings do, and when each helps or hurts a search. Give one example string for each. Keep it under 150 words.

TRY IT WITH MeSH term: Diabetes Mellitus.

59 Adapt my search for Google Scholar

QUICK

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

Act as a search specialist. Rewrite my PubMed string [PASTE STRING] for Google Scholar, which has no MeSH and limited Boolean support. Give the adapted query, explain the 3 main limitations of Google Scholar for a thesis search, and say how to use it as a supplementary source.

TRY IT WITH A PubMed string on tuberculosis preventive therapy in household contacts.

60 Which databases for my specialty?

QUICK

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

Act as a medical librarian. For a thesis in [SPECIALTY] on [TOPIC], tell me which databases to search (core and supplementary), what each adds, and which I can reasonably skip. Present it as a table and mark anything with uncertain access as [CHECK ACCESS].

TRY IT WITH Specialty: Psychiatry. Topic: depression in caregivers of dialysis patients.

61 Email an author for their paper or data

QUICK

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

Act as an academic communication coach. Write a short, polite email to [AUTHOR NAME], author of [PAPER TITLE], asking for [the full text / clarification of a method / supplementary data] for my MD thesis on [TOPIC]. Keep it under 100 words, with a clear subject line.

TRY IT WITH Asking for the supplementary table of a cohort study on sepsis biomarkers.

62 Has this paper been retracted or corrected?

QUICK

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT / FACULTY

Act as a research-integrity officer. Tell me step by step how to check whether a paper [TITLE / DOI] has been retracted, corrected or flagged, using the journal page, PubMed and the Retraction Watch database. Explain how to cite or handle it if it has.

TRY IT WITH A 2020 COVID-19 treatment paper you plan to cite.

63 Where does this study sit in the evidence hierarchy?

QUICK

1.2 LITERATURE SEARCH STRATEGY · MBBS STUDENT / PG RESIDENT

Act as an evidence-based medicine tutor. For the study I describe, [PASTE ABSTRACT], identify the design, its level in the evidence hierarchy, the 2 biggest threats to validity, and how much weight it should carry in my review. Answer in 5 bullet points.

TRY IT WITH A single-centre retrospective study of 60 patients on a new antifibrotic.

64 Turn my question into Perplexity and Consensus queries

QUICK

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

Act as an AI research-tools trainer for doctors. Convert my PICOT question [PASTE QUESTION] into 3 Perplexity Academic queries (broad, focused, India-specific) and 3 Consensus yes/no questions. Explain in one line what I should look for in each tool's results, and how to verify its citations.

TRY IT WITH Does pulmonary rehabilitation reduce readmissions in COPD patients in India?

65 How far back should my search go?

QUICK

1.2 LITERATURE SEARCH STRATEGY · PG RESIDENT

Act as a systematic-review methodologist. For my topic [TOPIC], help me choose a search date range. Consider when key definitions, diagnostic criteria, guidelines or treatments changed. Give a recommended start year, the reason, and one sentence I can use to justify it in my methods.

TRY IT WITH Topic: sepsis outcomes (definitions changed in 2016).

1.3

Writing the Introduction

An introduction fails in two ways: it reads like a textbook chapter, or it inflates the gap with statistics nobody can verify. Keep it tight, sourced and honest.

30

prompts

3

Master

17

Standard

10

Quick

HOW TO USE THIS SECTION

Use it once your literature review is done and you know your gap, to write a tight, sourced introduction.

Start with the Master prompts ([#66–68](#)) for the full workflow; use the Standard prompts ([#69–85](#)) for one task at a time; reach for the Quick prompts ([#86–95](#)) when you have two minutes. Tap any title below to jump to it, and follow → **NEXT** to move to the next step.

[#66](#) The four-movement thesis introduction

[#67](#) Refine my own introduction draft (language only)

[#68](#) Examiner-style review of my introduction

[#69](#) Where to find sourced burden data

[#70](#) Indian-burden paragraph from my sourced figures

[#71](#) Paragraph-by-paragraph outline before you write

[#72](#) Five opening sentences that are not textbook

[#73](#) Write a sharp gap statement

[#74](#) "Need for the study" section

[#75](#) Every claim that needs a citation

[#76](#) Turn a textbook-style intro into a problem-focused one

[#77](#) Cut my introduction to the word limit

[#78](#) Fix the flow between paragraphs

[#79](#) Define key terms by current criteria

[#80](#) Clinical and public-health significance

[#81](#) Does my introduction lead to my objective?

[#82](#) Paraphrase and synthesise sources responsibly

[#83](#) Make my draft sound like me, not like AI

[#84](#) Consistency check: tense, spelling, abbreviations

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The four-movement thesis introduction

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT

After your topic and protocol are approved, when you begin writing the thesis itself.

WHY

Examiners read the introduction to see whether you understand why your question matters. A four-movement structure (global burden, Indian burden, what is known and unknown, the gap and objective) does that in about 500 words, without a single invented statistic.

COPY THE PROMPT BELOW

PERSONA	You are a senior academic and thesis examiner in [YOUR SPECIALTY] who has supervised and marked postgraduate dissertations for over 20 years. You are strict about two things: introductions that read like textbook chapters, and burden figures quoted without a source. You write in formal scientific register.
MY CONTEXT	Specialty: [e.g. Respiratory Medicine] Exact thesis title: [paste] Study design: [cross-sectional / case-control / cohort / diagnostic accuracy / interventional] Setting: [hospital, unit, catchment; tertiary / district / private] Population: [who exactly] Primary objective: [one line] The gap in one sentence, in my own words: [what is not yet answered] Sources I can cite for burden data: [ICMR / NFHS / NCDIR / WHO / Global Burden of Disease / national programme reports / none yet]
TASK	Write a structured introduction of 450–550 words in four movements, each flowing into the next, without sub-headings: 1. Global burden: the scale and clinical importance of the condition. 2. Indian or regional burden: narrow to India or South Asia, point to where national data exists, and state plainly where good Indian data is missing. 3. What is known and what is not: summarise what the literature broadly shows, then state precisely what remains unanswered. 4. The gap and the objective: close on an explicit statement of the gap this study addresses and one clean sentence stating the objective.
RULES	Leave every epidemiological figure as a slot: [CITE] for the source and [VERIFY — INSERT SOURCED FIGURE] for any number I have not supplied, because invented prevalence figures are the most common reason introductions are sent back. Describe the evidence generically unless I supply papers. Frame the question; do not write a literature review. Use associational language unless my design supports causation.
OUTPUT	The 4-paragraph draft within the word limit, then a numbered checklist of every sentence containing a factual claim I must source before submission.
TWEAK	Finish with the one paragraph an examiner is most likely to call "textbook", and a tighter rewrite of it.

How to get more out of it

- › Write your own rough three-movement draft first, then use intro-refine-draft to polish the language only.
- › Take the sourcing checklist to PubMed and your national data portals before you start the ROL.
- › Rewrite the final draft in your own voice. A sudden jump to flawless academic English is a red flag to a guide who knows how you write.
- › → **NEXT #75** Every claim that needs a citation

Refine my own introduction draft (language only)

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT

After you have written a rough introduction yourself.
This is the golden rule: draft first, AI refines.

WHY

Your guide and examiners want your thinking, not the model's. Refining your own draft keeps the ideas, the sources and the voice yours, while fixing structure, flow and academic register, and it keeps you on the right side of your university's AI policy.

COPY THE PROMPT BELOW

PERSONA	You are an academic writing editor for medical theses. You improve clarity, structure and register while strictly preserving the author's content, claims and citations. You never add facts.
MY CONTEXT	My thesis title: [paste] Programme and university: [e.g. MD Pathology, state health university] Target length: [e.g. 400–500 words] Citation style: [Vancouver / other] My rough draft, with citations as I have them: [PASTE DRAFT HERE]
TASK	1. Check the structure against the four movements (background, Indian burden, what is known and unknown, the gap and objective), and tell me which movement is weak or missing. 2. Refine the language paragraph by paragraph: formal academic register, clear topic sentences, smooth transitions, consistent tense. 3. Keep every citation attached to the claim it supports. 4. Mark each place where a claim has no citation with [CITE]. 5. Show a list of the substantive changes you made, so I can accept or reject each one individually.
RULES	Do not add any new fact, statistic, study or reference; improve language and structure only, because anything new would be unsourced and not mine. Keep my meaning exactly, and always flag rather than fix any sentence whose meaning is unclear by writing [UNCLEAR – PLEASE CLARIFY]. Stay within the target length.
OUTPUT	The refined introduction with its word count, then a change log table (original phrase, revised phrase, reason), then the list of [CITE] gaps in the order they appear.
TWEAK	End with 3 sentences that still sound unlike a first-year resident, so I can put them back in my own words.

How to get more out of it

- › Refine one paragraph at a time if your draft is long. You'll keep more control.
- › Accept changes one by one from the change log. Do not paste the whole output blindly.
- › Read the final version aloud. If a sentence is not something you would say in a viva, rewrite it.
- › → **NEXT #75** Every claim that needs a citation

Examiner-style review of my introduction

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT

When your introduction is drafted and you want honest feedback before your guide sees it.

WHY

Your guide may only have ten minutes for your draft. An examiner-style critique with a scoring rubric shows you exactly what to fix first, so the guide's time goes on judgement rather than basic structure.

COPY THE PROMPT BELOW

PERSONA	You are an external examiner for postgraduate medical theses at an Indian health sciences university. You mark introductions against a clear rubric and write direct, specific comments.
MY CONTEXT	Thesis title and design: [paste] Primary objective: [one line] My university's marking emphasis, if known: [e.g. Indian data, references, word limit] What my guide has already commented on: [brief, or "not yet reviewed"] My introduction: [PASTE INTRODUCTION HERE]
TASK	1. Score the introduction out of 10 on each of: relevance of the opening, burden data and sourcing, Indian context, synthesis of what is known, clarity of the gap, link from gap to objective, flow, and academic register. 2. For each score, quote the sentence or passage that drove it. 3. List the 5 highest-priority fixes, ordered by impact on the examiner's impression. 4. Rewrite the weakest paragraph as an example of the fix. 5. Check that the objective at the end matches my stated primary objective exactly, and flag any mismatch.
RULES	Base every comment on the text I pasted; quote rather than paraphrase, because vague feedback cannot be acted on. Do not add facts or references in the example rewrite; use [CITE] and [VERIFY] slots instead.
OUTPUT	A rubric table (criterion, score, quoted evidence, comment) with a total out of 80, then the priority fixes as a numbered list, then the example rewrite with the changed phrases marked in bold so I can see exactly what moved.
TWEAK	Close with a one-line overall verdict (ready for guide, needs one revision, or needs restructuring) and the single change that would raise the total score most.

How to get more out of it

- › Fix the top 2 priorities, then run this prompt again. Scores should rise.
- › Keep each version. Showing your guide the before and after proves the work is yours.
- › If "link from gap to objective" scores low, fix the objective in section 1.5 first.
- › → **NEXT #67** Refine my own introduction draft (language only)

69 Where to find sourced burden data

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT Before writing the burden paragraphs, so every number has a real source.

PERSONA	You are a public-health researcher who knows India's major epidemiological data sources and their limits.
MY CONTEXT	Condition: [e.g. chronic kidney disease] Population of interest: [e.g. Indian adults] Figures I want (prevalence, incidence, mortality, DALYs, trends): [list]
TASK	1. List the most credible sources for each figure: national surveys, ICMR reports, national programme reports, disease registries, Global Burden of Disease, WHO and large Indian cohort studies. 2. For each source, say what it measures, the population it covers, its most recent edition as far as you know, and its main limitation. 3. Explain how to cite a report or dataset correctly in Vancouver style.
RULES	Do not quote any figures. Mark every edition or year as [CHECK LATEST EDITION], because survey rounds and reports are updated often and outdated numbers are easily spotted.
OUTPUT	A table (figure needed, best source, what it covers, limitation), then the citation guidance.

PRO TIP If no Indian data exists for your figure, that absence belongs in your introduction as part of the gap.

TRY IT WITH Condition: gestational diabetes; figures needed: prevalence and regional variation in India.

→ **NEXT #70** Indian-burden paragraph from my sourced figures

70 Indian-burden paragraph from my sourced figures

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT When you have collected real Indian figures with their sources.

PERSONA	You are a public-health academic who writes precise, well-sourced burden paragraphs for medical theses.
MY CONTEXT	My condition and population: [brief] Sourced figures, each with its reference: [PASTE FIGURES WITH CITATIONS HERE] Gaps in Indian data I have noticed: [list]
TASK	1. Write one paragraph (120–180 words) on the Indian burden using only my figures, each followed by its citation number. 2. Draw one meaningful comparison (e.g. regional variation, trend over time, or India versus global) if my figures allow it. 3. End with a sentence on what Indian data is missing.
RULES	Use only the figures I gave, reproduced exactly with their units and years, because a mistyped number looks like fabrication. If two sources conflict, report both and say so.
OUTPUT	The paragraph, then a table of the figures used with their sources.

PRO TIP Always give the year and survey round with a prevalence figure. "12% (NFHS-5, 2019–21)" is stronger than "12%".

TRY IT WITH Anaemia in pregnant women, with NFHS figures for two survey rounds.

→ **NEXT #73** Write a sharp gap statement

71 Paragraph-by-paragraph outline before you write

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT Before drafting, to plan the argument of your introduction.

PERSONA	You are a thesis-writing coach who insists on an outline before any drafting.
MY CONTEXT	Thesis title: [paste] Primary objective: [one line] My key points, in rough notes: [PASTE NOTES HERE]
TASK	1. Arrange my notes into a funnel: broad context → specific problem → Indian context → what is known → the gap → the objective. 2. For each paragraph, give a one-sentence topic sentence and the notes that belong to it. 3. Flag notes that do not fit (they may belong in the ROL or discussion). 4. Point out any step in the logic that is missing.
RULES	Use only my notes. Where a step needs information I have not given, write [ADD POINT], because the outline should expose gaps before I write, not after.
OUTPUT	A numbered outline with topic sentences and assigned notes, then a list of misplaced notes and missing steps.

PRO TIP If a paragraph has no note under it, you need to read more before you write it.

TRY IT WITH Notes on lung ultrasound for pneumonia diagnosis in children in resource-limited settings.

→ **NEXT #66** The four-movement thesis introduction

72 Five opening sentences that are not textbook

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT When your introduction starts with "X is a common disease worldwide..."

PERSONA	You are a journal editor who reads hundreds of introductions and remembers the ones that open with a clear clinical problem.
MY CONTEXT	Topic and population: [brief] The problem my study addresses: [one line] My current opening sentence: [paste]
TASK	1. Explain in 2 lines what is weak about my current opening. 2. Write 5 alternative opening sentences: a clinical-problem opening, a burden opening, a paradox opening, a practice-variation opening, and a patient-outcome opening. 3. Recommend one for a thesis and one for a journal article.
RULES	Leave any number as [VERIFY — INSERT SOURCED FIGURE], because the opening sentence is the one most often quoted back at you. Keep each opening under 35 words.
OUTPUT	The critique, 5 numbered options, and the two recommendations.

PRO TIP The best openings name a problem the reader recognises from their own ward.

TRY IT WITH Topic: delayed diagnosis of pulmonary embolism in emergency departments.

→ **NEXT #71** Paragraph-by-paragraph outline before you write

73 Write a sharp gap statement

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT For the key paragraph of your introduction, just before the objective.

PERSONA	You are a grant reviewer who funds studies only when the knowledge gap is specific and credible.
MY CONTEXT	What is known (my summary): [brief] What is not known: [brief] Why it matters in my setting: [brief] My objective: [one line]
TASK	1. Write the gap statement in 3 sentences: what is known, what is missing, why it matters clinically in my setting. 2. Write the bridging sentence that leads into my objective. 3. Give a weaker and a stronger version, so I can see the difference. 4. Check whether my objective actually addresses the gap as stated.
RULES	Do not overstate the gap with phrases such as "no study has ever examined"; use "few studies" or "limited data" unless I have evidence, because absolute claims invite an examiner to prove you wrong.
OUTPUT	The recommended gap statement and bridge, then the weak versus strong comparison, then the alignment check.

PRO TIP Your gap statement, objective and conclusion should read like three parts of one argument.

TRY IT WITH Known: high-flow nasal oxygen reduces intubation in some adults. Missing: data from Indian district ICUs.

→ **NEXT #81** Does my introduction lead to my objective?

74 "Need for the study" section

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT When your university's thesis format has a separate "Need for the study" or "Rationale" heading.

PERSONA	You are an Indian medical university thesis evaluator who reads many "Need for the study" sections and knows what makes one convincing.
MY CONTEXT	Thesis title and design: [paste] The gap (from my introduction): [paste] Local factors that make this study relevant here: [e.g. high disease burden, unique population, practice variation, cost] Expected benefit of the results: [brief]
TASK	1. Write a "Need for the study" section of 150–200 words that answers: why this question, why now, why here, and who benefits. 2. End with a sentence linking to the aims. 3. List any statement that needs a citation.
RULES	Keep claims of benefit realistic for a single-centre thesis. Mark any statistic as [VERIFY], because overclaimed impact is a common examiner criticism.
OUTPUT	The section, then a citation checklist.

PRO TIP "Why here" is your strongest argument in an Indian setting. Make it specific to your hospital's patients.

TRY IT WITH Rationale for studying post-TB lung disease in a TB-endemic district.

→ **NEXT #81** Does my introduction lead to my objective?

75 Every claim that needs a citation

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT After drafting any section, before you submit it to your guide.

PERSONA	You are a meticulous research-integrity reviewer who checks that every factual claim in a thesis is supported.
MY CONTEXT	My text: [PASTE INTRODUCTION OR SECTION HERE]
TASK	1. List every sentence that contains a factual claim (numbers, associations, guideline statements, "it is well known that..."). 2. For each, say whether it currently has a citation, and whether the citation type fits (primary study, review, guideline, dataset). 3. Flag claims that are too strong for any single source, and suggest softer wording. 4. Group the unsupported claims into a sourcing to-do list.
RULES	Do not supply references yourself; tell me what kind of source each claim needs, because citations you generate may not exist.
OUTPUT	A numbered table (sentence, claim, cited yes or no, source type needed, wording note), then the to-do list.

PRO TIP Work through the to-do list on PubMed before your guide meeting. Every unsupported claim is a viva question waiting to happen.

TRY IT WITH Paste a draft paragraph on vitamin D deficiency prevalence and its link with TB.

→ **NEXT #69** Where to find sourced burden data

76 Turn a textbook-style intro into a problem-focused one

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT When your guide says your introduction reads like a textbook chapter.

PERSONA	You are a thesis supervisor who has rescued many textbook-style introductions by refocusing them on the research problem.
MY CONTEXT	My current introduction: [PASTE INTRODUCTION HERE] My research question and objective: [paste]
TASK	1. Mark every sentence that is textbook content (definitions, pathophysiology, classification, general treatment) not needed to understand the problem. 2. Suggest what to delete, what to move to the ROL, and what to keep. 3. Restructure the remaining content into a problem-focused introduction of no more than [WORD LIMIT] words. 4. Show the new paragraph order with one-line purposes.
RULES	Keep my existing citations with their claims. Add no new content; where the restructured text needs a link, write [ADD LINKING SENTENCE], because new content must come from my own reading.
OUTPUT	A marked-up sentence list (delete / move / keep), then the restructured introduction, then the paragraph map.

PRO TIP If a sentence would appear in any textbook chapter on the disease, it usually doesn't belong in your introduction.

TRY IT WITH An introduction that spends 3 paragraphs on the pathophysiology of preeclampsia before the research question.

→ **NEXT #68** Examiner-style review of my introduction

77 Cut my introduction to the word limit

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT When your section is over the word limit set by your university or journal.

PERSONA	You are a precise academic editor who cuts words without cutting substance.
MY CONTEXT	Current text: [PASTE TEXT HERE] Current word count and target: [e.g. 780 → 500] Parts that must stay: [e.g. the Indian data paragraph]
TASK	1. Identify redundancy, throat-clearing phrases, repeated ideas and content that belongs in another chapter. 2. Produce the trimmed version at or just under the target. 3. List what you removed or merged, so I can restore anything important.
RULES	Keep every citation with its claim and keep all numbers exactly as written, because trimming must never change the evidence. Do not add new content.
OUTPUT	The trimmed text with its word count, then a list of cuts.

PRO TIP Cut whole ideas before cutting words. Trimming every sentence slightly makes the text worse.

TRY IT WITH A 900-word introduction on stroke thrombolysis that needs to be 550 words.

→ **NEXT #78** Fix the flow between paragraphs

78 Fix the flow between paragraphs

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT When each paragraph is fine on its own but the introduction feels disjointed.

PERSONA	You are an academic writing coach who specialises in coherence and argument flow.
MY CONTEXT	My text: [PASTE TEXT HERE]
TASK	1. Summarise the main point of each paragraph in one line, to expose the argument. 2. Identify where the logic jumps, repeats or reverses. 3. Suggest a reordering if needed. 4. Write improved first and last sentences for each paragraph so they connect.
RULES	Change only topic sentences, closing sentences and order; leave the rest of each paragraph as I wrote it, because the content and voice must stay mine.
OUTPUT	The paragraph-point map, a list of problems, then the revised opening and closing sentences for each paragraph.

PRO TIP Read only the first sentence of each paragraph in order. If that doesn't tell the story, the flow is broken.

TRY IT WITH A 5-paragraph introduction that jumps from global burden to treatment options and back to Indian burden.

→ **NEXT #68** Examiner-style review of my introduction

79 Define key terms by current criteria

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT When your introduction or methods needs formal definitions (e.g. sepsis, AKI, ARDS, metabolic syndrome).

PERSONA	You are a clinical academic who insists that every key term in a thesis is defined by a named, current criterion.
MY CONTEXT	Terms I need to define: [list] Specialty and population: [brief]
TASK	1. For each term, name the current standard definition or criteria set (organisation and year) and summarise it. 2. Note any older definition still used in papers I may cite, and how the change affects comparisons. 3. Suggest one sentence for each term suitable for my introduction or methods.
RULES	Mark every criteria name and year as [VERIFY FROM ORIGINAL SOURCE], because definitions are updated and misremembered versions are common. Do not paraphrase numeric thresholds unless you are certain; otherwise write [CHECK THRESHOLD].
OUTPUT	A table (term, current definition, organisation and year, older versions), then the sentences.

PRO TIP Use the same definition throughout your thesis. Mixing criteria between chapters is a classic examiner catch.

TRY IT WITH Terms: sepsis, septic shock, acute kidney injury, delirium.

→ **NEXT #66** The four-movement thesis introduction

80 Clinical and public-health significance

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT For the paragraph that explains who will benefit from your findings.

PERSONA	You are a health-services researcher who writes realistic significance statements for single-centre studies.
MY CONTEXT	Research question: [PICOT] Possible findings and what each would mean: [brief] Stakeholders: [e.g. clinicians, hospital administrators, national programme]
TASK	1. Write a significance paragraph of 100–150 words covering clinical practice, patients, the health system and future research. 2. Give a version for a thesis and a more concise version for a journal article. 3. Point out any benefit that overreaches for a single-centre design.
RULES	Frame benefits conditionally ("if confirmed, these findings could..."), because single-centre results rarely change practice alone, and examiners know it.
OUTPUT	The two versions, then the overreach check.

PRO TIP Name the specific decision your results could change, e.g. "whether to repeat chest X-ray at 48 hours".

TRY IT WITH Study of a bedside frailty score predicting post-operative complications in elderly surgical patients.

→ **NEXT #74** "Need for the study" section

81 Does my introduction lead to my objective?

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT After drafting the introduction and objectives, before submission to your guide.

PERSONA	You are a methodologist who checks the logical chain from problem to gap to objective in research proposals.
MY CONTEXT	My introduction: [PASTE INTRODUCTION HERE] My aims and objectives: [PASTE AIMS AND OBJECTIVES HERE]
TASK	1. Extract the gap as the introduction states it. 2. Compare the gap with each objective: does each objective address it, go beyond it, or ignore it? 3. Check that the population, exposure and outcome wording is identical between the introduction and the objectives. 4. Suggest the minimum edits to align them.
RULES	Point to exact phrases; do not rewrite whole sections, because the fix is usually a few words.
OUTPUT	An alignment table (gap element, objective wording, match yes or no, suggested edit).

PRO TIP Examiners often read the last paragraph of your introduction and your objectives side by side.

TRY IT WITH The introduction says the gap is Indian data; the objective does not mention the setting.

→ **NEXT #138** Primary and secondary objectives in SMART form

82 Paraphrase and synthesise sources responsibly

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT When you have key passages from papers and need to use their ideas in your own words, with citations.

PERSONA	You are an academic integrity adviser who teaches synthesis: combining ideas from several sources in the writer's own words.
MY CONTEXT	Source passages, each with its citation: [PASTE PASSAGES WITH CITATIONS HERE] The point I want to make: [one line]
TASK	1. Identify the shared idea and any disagreement across the passages. 2. Write a synthesis of 2–4 sentences in academic language, combining the sources and citing each claim. 3. Show which part of the synthesis came from which source. 4. Point out any phrase that still sits too close to the original wording.
RULES	Synthesise ideas rather than rewording sentence by sentence, and cite every source used, because the goal is honest scholarship, not getting past a similarity checker. Add no claims beyond the passages.
OUTPUT	The synthesis, then a source-mapping table, then the list of phrases to reword.

PRO TIP Close the source before writing your own version. Then compare. That habit prevents accidental plagiarism.

TRY IT WITH Three passages on the association between air pollution and COPD exacerbations.

→ **NEXT #75** Every claim that needs a citation

83 Make my draft sound like me, not like AI

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT When your text, even though you wrote it, has picked up generic AI-style phrasing after editing.

PERSONA	You are an editor who helps academic writers sound precise and natural rather than generic.
MY CONTEXT	My text: [PASTE TEXT HERE] My natural writing style (a paragraph I wrote without help): [PASTE SAMPLE HERE]
TASK	1. List the phrases that sound generic or AI-like (e.g. "delve", "plays a crucial role", "in today's world", empty transitions). 2. Suggest precise alternatives for each, matching my sample's style. 3. Point out sentences where every sentence has the same length and rhythm, and vary them. 4. Return the revised text.
RULES	Keep all facts, numbers and citations exactly as written. This is style editing of my own work; do not add content, because academic integrity requires that the ideas and sources stay mine.
OUTPUT	A phrase table (original, why it's weak, alternative), then the revised text.

PRO TIP Specific beats impressive. "Mortality was 18%" is better than "mortality remains a significant concern".

TRY IT WITH An introduction paragraph full of "crucial", "pivotal" and "it is important to note".

→ **NEXT #68** Examiner-style review of my introduction

84 Consistency check: tense, spelling, abbreviations

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT Before submitting any chapter, as a final polish.

PERSONA	You are a copy-editor for medical theses who catches inconsistencies authors stop seeing.
MY CONTEXT	My text: [PASTE TEXT HERE] Required conventions: [e.g. British spelling, Vancouver citations, SI units]
TASK	1. Check tense (present for established knowledge, past for specific studies). 2. Check spelling consistency (British or American), abbreviation use (defined at first use, then used consistently), units, number formatting and citation placement. 3. List every inconsistency with its location and correction. 4. Produce a list of the abbreviations used and their expansions.
RULES	Report inconsistencies rather than silently fixing them, because I need to apply the same corrections across all my chapters.
OUTPUT	An issue table (location, issue, correction), then the abbreviation list.

PRO TIP Build one abbreviation list for the whole thesis and check every chapter against it.

TRY IT WITH A methods and introduction section mixing "haemoglobin" and "hemoglobin".

→ **NEXT #77** Cut my introduction to the word limit

85 Short background for a synopsis or protocol

STANDARD

1.3 WRITING THE INTRODUCTION · PG RESIDENT

USE IT When you need a 150–250 word background for a synopsis, IEC form or CTRI registration.

PERSONA	You are a research committee member who reads hundreds of synopses and values brevity.
MY CONTEXT	Thesis title: [paste] Gap and objective: [paste] Key sourced facts I want to include: [PASTE FACTS WITH CITATIONS] Word limit: [e.g. 200 words]
TASK	1. Write a background within the word limit: the problem, Indian relevance, the gap and the objective. 2. Write a 50-word version for form fields with tight limits. 3. List the citations used.
RULES	Use only the facts I gave with their citations; write [CITE] where a needed claim has none, because synopsis claims are checked by the committee.
OUTPUT	The full background, the 50-word version, then the citation list.

PRO TIP Keep the same wording for the gap and objective across synopsis, protocol, IEC form and CTRI. Committees compare them.

TRY IT WITH Synopsis background for a cross-sectional study of depression in post-stroke patients.

→ **NEXT #296** Complete study protocol, IEC-ready

QUICK PROMPTS · 1.3

86 Three hook sentences for my topic

QUICK

1.3 WRITING THE INTRODUCTION · PG RESIDENT

Act as a medical writing editor. Write 3 opening sentences for an introduction on [TOPIC] in [POPULATION]: one clinical-problem hook, one burden hook with a [VERIFY] slot for the number, and one practice-gap hook. Keep each under 30 words and formal.

TRY IT WITH Topic: sepsis-associated acute kidney injury in Indian ICUs.

87 Is my tense right?

QUICK

1.3 WRITING THE INTRODUCTION · PG RESIDENT

Act as an academic copy-editor. Check the tenses in this paragraph, [PASTE PARAGRAPH], using the convention of present tense for established knowledge and past tense for specific studies and my own methods. List each error with its correction, then give the corrected paragraph.

TRY IT WITH A paragraph mixing "Smith found" and "studies show" inconsistently.

88 Turn my bullet notes into a paragraph

QUICK

1.3 WRITING THE INTRODUCTION · PG RESIDENT

Act as an academic writing coach. Turn these notes, [PASTE NOTES WITH CITATIONS], into one coherent paragraph of about 120 words in formal thesis style. Keep every citation with its claim, add no new facts, and mark any missing link as [ADD LINK].

TRY IT WITH Five bullet notes on the burden of snakebite envenoming in rural India.

89 Find the textbook sentences

QUICK

1.3 WRITING THE INTRODUCTION · PG RESIDENT

Act as a strict thesis examiner. In this introduction, [PASTE TEXT], mark every sentence that is general textbook content rather than part of the argument for my research question [QUESTION]. List them with a one-line reason and say whether to delete each one or move it to the ROL.

TRY IT WITH An introduction on thyroid disorders in pregnancy that opens with thyroid physiology.

90 One paragraph on "why now"

QUICK

1.3 WRITING THE INTRODUCTION · PG RESIDENT / FACULTY

Act as a research strategist. In under 100 words, explain why my research question [QUESTION] is timely now: a recent guideline change, a new test or therapy, a policy change, or an emerging burden. Use [CITE] slots for every claim and do not invent dates.

TRY IT WITH Question on point-of-care HbA1c testing in primary health centres.

91 Sentence-level clarity edit

QUICK

1.3 WRITING THE INTRODUCTION · PG RESIDENT

Act as a medical copy-editor. Edit this paragraph, [PASTE PARAGRAPH], for clarity only: shorter sentences, active voice where natural, precise words and no jargon stacking. Keep all facts and citations unchanged, and show the edited version followed by the 3 most important changes.

TRY IT WITH A 90-word sentence describing the pathogenesis and burden of NAFLD.

92 Does my introduction need sub-headings?

QUICK

1.3 WRITING THE INTRODUCTION · PG RESIDENT

Act as an Indian medical university thesis evaluator. My university is [UNIVERSITY] and my introduction is [WORD COUNT] words. Tell me whether to use sub-headings, which headings are conventional in Indian MD/MS theses if so, and how to keep the argument continuous across them. Mark university-specific rules as [CHECK YOUR UNIVERSITY FORMAT].

TRY IT WITH A 1,200-word introduction for an MS Orthopaedics thesis.

93 Check for causal language in my introduction

QUICK

1.3 WRITING THE INTRODUCTION · PG RESIDENT

Act as an epidemiology reviewer. Check this text, [PASTE TEXT], for causal words (causes, leads to, results in, prevents, due to) attached to observational evidence. List each instance with a correct associational rewrite, and leave the causal wording where it is supported by RCT evidence I have cited.

TRY IT WITH A paragraph claiming obesity "causes" poor asthma control, citing cross-sectional studies.

94 Build the abbreviation list

QUICK

1.3 WRITING THE INTRODUCTION · PG RESIDENT

Act as a thesis copy-editor. From this text, [PASTE TEXT], extract every abbreviation, give its expansion, note where it is first used and whether it was expanded there, and flag any abbreviation used only once. Return an alphabetical table ready for my thesis front matter.

TRY IT WITH Introduction and methods of a thesis on COPD exacerbations.

95 Two-sentence background for my abstract

QUICK

1.3 WRITING THE INTRODUCTION · PG RESIDENT

Act as a journal abstract editor. From my introduction, [PASTE INTRODUCTION], write the Background of a structured abstract in exactly two sentences: the problem with its relevance, and the gap leading to the objective. Stay under 50 words, add no new facts, and keep any number exactly as written.

TRY IT WITH An introduction on post-COVID pulmonary fibrosis in Indian patients.

1.4

Review of Literature

A literature review is not a summary of ten papers you happened to find. It is a reproducible search, an evidence table and a defensible gap statement.

40

prompts

5

Master

23

Standard

12

Quick

HOW TO USE THIS SECTION

Use it after your search, to build an evidence table and a gap statement you can defend.

Start with the Master prompts (#96–100) for the full workflow; use the Standard prompts (#101–123) for one task at a time; reach for the Quick prompts (#124–135) when you have two minutes. Tap any title below to jump to it, and follow → NEXT to move to the next step.

#96 Screen, extract and synthesise: the ROL engine

#97 Thematic ROL draft from my evidence table

#98 Evidence table from uploaded PDFs, with quotes

#99 Global and Indian studies: the classic thesis ROL

#100 Examiner-style critique of my ROL chapter

#101 Inclusion and exclusion criteria for screening

#102 Triage a batch of abstracts

#103 Design my data-extraction sheet

#104 Extract one paper into my evidence table

#105 Which risk-of-bias tool for which study?

#106 Apply a risk-of-bias tool to one paper

#107 Find the themes in my evidence table

#108 Explain why studies disagree

#109 One study, one paragraph, done properly

#110 Convert a chronological ROL into a thematic one

#111 Build a literature synthesis matrix

#112 Heading structure for my ROL chapter

#113 The closing paragraph that leads to my study

#114 Summarise a paper, grounded in quotes

#115 Summary table of key studies

#116 Check the numbers in my ROL against the sources

#117 Format and check my reference list

#118 The questions to ask NotebookLM about my papers

#119 Quick critical appraisal of one paper

#120 Read and summarise a meta-analysis

#121 Historical review sub-section without errors

#122 Update my ROL before submission

#123 Register a systematic review on PROSPERO

#124 Pull the PICO out of an abstract

#125 One-line takeaway of a paper

#126 The limitation that matters most

#127 Compare two studies in three sentences

#128 What design is this study really?

#129 Interpret an effect size and its CI

#130 Results in plain language

#131 Does this reference support my sentence?

#132 Format one reference in Vancouver style

#133 How much should this study count?

#134 Customise a NotebookLM audio overview

#135 Is this ROL paragraph a list or a synthesis?

Screen, extract and synthesise: the ROL engine

1.4 REVIEW OF LITERATURE · PG RESIDENT / FACULTY

USE IT

After your search is complete (section 1.2), when you have a pile of records to turn into a review.

WHY

Most thesis ROLs are a string of study summaries. Examiners want to see explicit criteria, a structured evidence table, a quality assessment and a synthesis that ends on the gap. This prompt builds all four from your own papers.

COPY THE PROMPT BELOW

PERSONA	You are a systematic-review methodologist who has co-authored Cochrane reviews and supervised dozens of postgraduate literature reviews. You are sceptical of weak evidence and insist on explicit, reproducible decisions.
MY CONTEXT	My PICOT question: [PASTE QUESTION] Study designs I expect to find: [e.g. cohorts, case-control, a few RCTs] Number of records after deduplication: [N] Papers I have uploaded to this chat: [list, or "none yet"]
TASK	Part A: screen and extract 1. Draft explicit inclusion and exclusion criteria, phrased so a second reviewer could apply them without asking me questions. 2. Design a data-extraction table: author, year, country, setting, design, sample size, population, exposure or intervention, comparator, outcome definition, effect estimate with 95% CI, follow-up, funding, key limitation. 3. Name the right risk-of-bias tool for each design I will encounter, and list its domains. Part B: synthesise 4. Give a skeleton for the review narrative: how to open, how to group studies by theme rather than chronology, how to handle conflicting findings, how to close on the gap. 5. Show how to write a three-sentence gap statement: what is known, what is missing, why it matters in my setting. 6. Draft a PRISMA-style flow description with placeholder numbers.
RULES	Do not give any specific citation, author, journal, year, sample size or result unless I uploaded that paper or you retrieved it through live search with a PMID or DOI, because an invented study in a thesis is academic misconduct. Where you describe what the literature broadly shows, label it [UNVERIFIED BACKGROUND – CONFIRM INDEPENDENTLY]. Never estimate a statistic from memory.
OUTPUT	Criteria as two numbered lists, the extraction table as a column list, a tool table, the narrative skeleton, the gap-statement template, and the flow description.
TWEAK	End with the 3 decisions in this plan an examiner is most likely to question, and how to justify each.

How to get more out of it

- › Upload the actual PDFs before asking for any comparison across papers. That is the difference between synthesis and fabrication.
- › Pilot the extraction table on 3 papers before extracting all of them. Columns always need adjusting.
- › Keep a list of excluded full texts with reasons. It feeds your PRISMA flow diagram.
- › → **NEXT #102** Triage a batch of abstracts

Thematic ROL draft from my evidence table

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT

When your extraction table is complete and you are ready to write the review chapter.

WHY

Writing from an evidence table, rather than from memory or AI knowledge, guarantees every sentence traces back to a real paper you read. Grouping by theme shows examiners you synthesised the literature rather than listed it.

COPY THE PROMPT BELOW

PERSONA	You are a senior academic who writes and examines literature reviews for medical theses. You synthesise across studies, compare findings, and always end each theme with an interpretive sentence.
MY CONTEXT	Thesis title and objective: [paste] Citation style: [Vancouver / Harvard] Target length of the ROL chapter: [e.g. 2,500–3,500 words] My completed extraction table (author, year, design, n, setting, key finding with numbers, limitation, reference number): [PASTE TABLE HERE] Themes I think exist (optional): [list] My guide's expectations for the chapter, if known: [e.g. separate Indian studies section, 25+ references]
TASK	- Propose 4–6 themes that organise the studies, and assign each study to a theme. - For each theme, write 2–4 paragraphs that compare studies: agreement, disagreement and the likely reasons (population, setting, definitions, design, sample size). - Include a short subsection on Indian studies, or say explicitly that Indian data is lacking in a theme. - End each theme with a sentence saying what it means for my study. - Close the chapter with a gap paragraph that leads to my objective. - After each theme, list the reference numbers used, so I can check coverage against my table.
RULES	Use only the studies and numbers in my table, reproduced exactly with their reference numbers, because every figure will be checked against the source. If a theme has a single study, say so rather than generalising. Add no new studies; where support is missing, write [CITATION NEEDED — FIND SOURCE].
OUTPUT	A theme map table first, then the chapter with theme headings, within the target length.
TWEAK	Finish with the 3 places where my evidence is thinnest, so I can search again before submission.

How to get more out of it

- Write one theme per session. Long outputs lose accuracy.
- Spot-check 5 numbers in the draft against your PDFs before you trust the rest.
- Put your own interpretation into the closing sentence of each theme. That is where your voice matters most.
- → **NEXT #116** Check the numbers in my ROL against the sources

Evidence table from uploaded PDFs, with quotes

1.4 REVIEW OF LITERATURE · PG RESIDENT / FACULTY

USE IT

In NotebookLM, Claude or ChatGPT with file upload, once you have 5–15 full-text PDFs.

WHY

Asking an AI to "summarise the literature" invites invented studies. Asking it to extract from PDFs you uploaded, and to quote the sentence each data point came from, lets you verify every cell in minutes.

COPY THE PROMPT BELOW

PERSONA	You are a meticulous research assistant trained in data extraction for systematic reviews. You extract only what is written, you quote your source, and you say "not reported" when information is absent.
MY CONTEXT	My research question: [PICOT] Number of PDFs uploaded: [N] Variables I need from each paper: [e.g. design, country, sample size, population, exposure definition, outcome definition, main effect estimate with 95% CI, follow-up, limitations] Tool I am using: [NotebookLM / Claude / ChatGPT / Gemini] How I will use the table: [thesis ROL / systematic review / journal club]
TASK	1. For each uploaded paper, extract every variable I listed into a table, one row per paper. 2. For each cell, give the supporting quotation from the paper with its section (Methods, Results, Table 2 and so on). 3. Write "NOT REPORTED" where the paper does not state the information. 4. After the table, list any inconsistency within a paper (e.g. abstract and results numbers differ). 5. Summarise in 5 bullets what the set of papers shows collectively, citing papers by first author and year. 6. Flag any paper whose design or population makes it a poor fit for my question.
RULES	Extract only from the uploaded documents; do not supplement from your general knowledge, because I will cite these values in my thesis. Reproduce numbers exactly, including decimals and units. If a PDF is unreadable or partial, say which one.
OUTPUT	The extraction table, then a quotation appendix grouped by paper, then the inconsistency list and the 5-bullet summary.
TWEAK	End by telling me which paper's data you are least confident about, and why.

How to get more out of it

- › In NotebookLM, click the citation numbers to jump to the exact passage. Check 3 per paper.
- › Upload papers in batches of 5 to 8. Very large batches reduce extraction accuracy.
- › Save the quotation appendix. It is your proof if an examiner asks where a number came from.
- › → **NEXT #97** Thematic ROL draft from my evidence table

Global and Indian studies: the classic thesis ROL

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT

When your university expects the traditional Indian thesis ROL format, with global studies, Indian studies and gaps as sub-headings.

WHY

Many Indian universities and guides expect studies described one by one (author, year, sample, finding, limitation) under global and Indian headings. This prompt keeps that format and makes it analytical instead of a list, using only papers you supply.

COPY THE PROMPT BELOW

PERSONA	You are a senior Indian medical academic who has guided and examined theses for 20 years and knows the conventional ROL format. You insist that each study description ends with its relevance to the present study.
MY CONTEXT	Thesis title: [paste] Primary objective: [one line] Global studies, with the extracted details for each: [PASTE STUDY DETAILS HERE] Indian studies, with the extracted details for each: [PASTE STUDY DETAILS HERE] Citation style: [Vancouver] Approximate length expected for the chapter: [e.g. 2,000–3,000 words]
TASK	1. Write a sub-section "Global studies" covering my global studies in chronological order, one short paragraph each: author(s), year, setting, design, sample size, key finding with numbers, one limitation, and one sentence on relevance to my study. 2. Write a sub-section "Indian studies" in the same format. 3. After each sub-section, write a synthesis paragraph comparing the studies: agreement, disagreement and why. 4. Write a sub-section "Gaps leading to the present study" in 3–5 sentences. 5. Add a summary table of all studies.
RULES	Use only the study details I pasted, with numbers reproduced exactly. Where a detail is missing, write [NOT REPORTED] rather than filling it in, because examiners cross-check these paragraphs against the original papers. Write in paragraphs, not bullet points. Use associational language for observational studies, and keep the order of studies consistent between the text and the summary table.
OUTPUT	Three headed sub-sections in formal thesis prose, then the summary table (author, year, country, design, n, key finding, limitation).
TWEAK	Close with a note on any study whose limitation seriously weakens its relevance to my question.

How to get more out of it

- › Most universities expect 15 or more references in the ROL. Check your guide's expectation.
- › The synthesis paragraphs are what raise this from a list to a review. Don't skip them.
- › If your Indian studies are few, say so plainly. That scarcity supports your gap.
- › → **NEXT #113** The closing paragraph that leads to my study

Examiner-style critique of my ROL chapter

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT

When your ROL chapter is drafted and before it goes to your guide.

WHY

An ROL is judged on coverage, synthesis, critical appraisal and how clearly it leads to the gap. An honest examiner-style critique tells you which of these is weak while there is still time to search and rewrite.

COPY THE PROMPT BELOW

PERSONA	You are an external examiner for MD/MS theses who has marked hundreds of Review of Literature chapters. You are specific and fair, and you quote the text you are criticising.
MY CONTEXT	Thesis title and objective: [paste] Number of references in the chapter: [N] Date range of the included studies: [e.g. 2008–2025] Citation style used: [Vancouver / other] What my guide has already commented on: [brief, or "not yet reviewed"] My ROL chapter: [PASTE CHAPTER HERE]
TASK	- Score the chapter out of 10 on each of: coverage, recency, inclusion of Indian evidence, synthesis versus listing, critical appraisal of study quality, handling of conflicting findings, logical flow towards the gap, and accuracy of reporting (numbers and designs described clearly). - Quote the passage behind each score. - List the 6 most important fixes, in priority order. - Identify paragraphs that merely list studies, and rewrite one as a synthesis example. - Flag any claim that is not attributed to a reference. - Check that the studies emphasised at the end of the chapter are the ones that support my gap statement.
RULES	Base all comments on the text I pasted. In the example rewrite, add no new studies or numbers; use [CITE] slots, because the rewrite must stay within my evidence.
OUTPUT	A rubric table with a total out of 80, the priority fixes as a numbered list, the flagged claims, and the example rewrite.
TWEAK	End with a one-line verdict (ready for guide, needs one revision, or needs restructuring) and the single fix with the biggest impact.

How to get more out of it

- › Run this again after revising. Your synthesis score should rise the most.
- › If "recency" scores low, re-run your search for the last 2 years before rewriting.
- › Share the rubric with your guide along with the chapter. It speeds up their review.
- › → **NEXT #97** Thematic ROL draft from my evidence table

101 Inclusion and exclusion criteria for screening

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT Before screening titles and abstracts.

PERSONA	You are a systematic reviewer who writes screening criteria clear enough for two independent reviewers to agree.
MY CONTEXT	My question: [PICOT] Type of review: [thesis ROL / systematic review / scoping review] Designs, languages and dates I intend to include: [brief]
TASK	1. Write inclusion criteria for population, exposure or intervention, comparator, outcomes, study design, setting, language and date. 2. Write exclusion criteria, each with a one-line reason. 3. Give 3 borderline examples and how the criteria would decide each. 4. Suggest a pilot: screen 50 records together and calculate agreement.
RULES	Make each criterion testable from a title and abstract where possible, because criteria that need full text slow screening down badly.
OUTPUT	Two numbered lists, a borderline-examples table, then the pilot plan.

PRO TIP Screen generously at title and abstract stage. Be strict at full text.

TRY IT WITH Question: effect of pulmonary rehabilitation on quality of life in post-TB lung disease.

→ **NEXT #102** Triage a batch of abstracts

102 Triage a batch of abstracts

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT When screening titles and abstracts against your criteria.

PERSONA	You are a careful second reviewer on a systematic review team who applies criteria consistently and explains every decision.
MY CONTEXT	My inclusion and exclusion criteria: [PASTE CRITERIA HERE] A batch of 10–20 titles and abstracts, numbered: [PASTE ABSTRACTS HERE]
TASK	1. Classify each record as include, exclude or unsure. 2. For each exclusion, name the criterion it failed. 3. For each unsure record, state exactly what the full text needs to show. 4. Summarise the counts.
RULES	When in doubt, mark unsure rather than exclude, because wrongly excluded studies cannot be recovered later. Decide only from the text given.
OUTPUT	A table (record number, decision, reason or criterion), then the counts.

PRO TIP You remain the first reviewer. Use this as a second opinion and resolve disagreements yourself.

TRY IT WITH 15 abstracts retrieved on "procalcitonin antibiotic duration ICU".

→ **NEXT #98** Evidence table from uploaded PDFs, with quotes

103 Design my data-extraction sheet

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT Before extracting data from included papers.

PERSONA	You are a data-extraction lead for systematic reviews who designs sheets that capture everything needed and nothing more.
MY CONTEXT	My question: [PICOT] Designs included: [list] Planned analysis: [narrative synthesis / meta-analysis] Tool: [Excel / Google Sheets]
TASK	1. List the extraction columns grouped as study details, population, exposure or intervention, outcomes, results and quality. 2. For each column, give a definition, format (text, number, drop-down) and example entry. 3. Add columns needed if I later want a meta-analysis (e.g. events and totals per group, or means and SDs). 4. Suggest how to pilot and version the sheet.
RULES	Keep outcome columns aligned with my primary and secondary outcomes, because unmatched columns lead to missing data at synthesis.
OUTPUT	A column table (group, column, definition, format, example), then the piloting steps.

PRO TIP Use drop-downs for design and country. They make later summary tables effortless.

TRY IT WITH Diagnostic accuracy studies of lung ultrasound for pneumonia in children.

→ **NEXT #98** Evidence table from uploaded PDFs, with quotes

104 Extract one paper into my evidence table

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT While reading each included paper, to fill your table accurately.

PERSONA	You are a precise research assistant who extracts data only from the text provided and marks everything else as not reported.
MY CONTEXT	My extraction columns: [PASTE COLUMN LIST HERE] The paper (full text or relevant sections): [PASTE OR UPLOAD PAPER HERE]
TASK	1. Fill each column from the paper. 2. Give the section or table each value came from. 3. Mark missing items as NOT REPORTED. 4. Note any discrepancy between the abstract and the results. 5. Write a one-sentence limitation relevant to my question.
RULES	Reproduce numbers exactly, including confidence intervals and units, because a transcription error in the evidence table propagates through the whole ROL.
OUTPUT	A two-column table (column, value with source location), then the discrepancy notes and the limitation sentence.

PRO TIP Extract the primary outcome first. If the paper doesn't report it clearly, reconsider inclusion.

TRY IT WITH A cohort study of NT-proBNP and mortality in acute heart failure.

→ **NEXT #106** Apply a risk-of-bias tool to one paper

105 Which risk-of-bias tool for which study?

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT Before appraising the quality of your included studies.

PERSONA	You are an evidence-synthesis methodologist who knows the major quality-appraisal tools and their correct use.
MY CONTEXT	Designs in my review, with counts: [e.g. 3 RCTs, 8 cohorts, 2 diagnostic accuracy studies] Type of review: [thesis ROL / systematic review]
TASK	1. Match each design to the recommended tool (for example RoB 2 for RCTs, ROBINS-I or Newcastle-Ottawa for non-randomised studies, QUADAS-2 for diagnostic accuracy, AMSTAR 2 for systematic reviews, JBI checklists for others). 2. List each tool's domains and the output (overall judgement). 3. Explain how to present the results in a thesis (table or traffic-light plot). 4. Mention simpler options if a full tool is too heavy for a narrative thesis ROL.
RULES	Name only established tools and describe their domains accurately; if unsure of a detail, write [CHECK TOOL MANUAL], because misapplying a tool is worse than not using one.
OUTPUT	A design-to-tool table with domains, then the presentation guidance.

PRO TIP robvis is a free tool that produces traffic-light plots from your judgements.

TRY IT WITH 4 RCTs and 6 prospective cohorts on early mobilisation in the ICU.

→ **NEXT #106** Apply a risk-of-bias tool to one paper

106 Apply a risk-of-bias tool to one paper

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT When appraising each included study.

PERSONA	You are a trained risk-of-bias assessor who justifies every judgement with a quote from the paper.
MY CONTEXT	Tool: [RoB 2 / ROBINS-I / Newcastle-Ottawa / QUADAS-2 / JBI] The paper (methods and results): [PASTE OR UPLOAD PAPER HERE] My outcome of interest: [outcome]
TASK	1. Assess each domain of the tool for my outcome. 2. Give the judgement for each domain (e.g. low, some concerns, high) with a supporting quote. 3. Give the overall judgement using the tool's own algorithm. 4. Write one sentence I can use in my ROL describing the study's quality.
RULES	Base judgements only on what the paper reports; where information is missing, say so and apply the tool's rule for missing information, because judgements without quotes cannot be defended.
OUTPUT	A domain table (domain, judgement, quote, reason), then the overall judgement and the ROL sentence.

PRO TIP Have a second person assess 2 or 3 papers independently. Report your agreement in the methods.

TRY IT WITH RoB 2 for an RCT of dexamethasone in severe pneumonia, outcome 28-day mortality.

→ **NEXT #107** Find the themes in my evidence table

107 Find the themes in my evidence table

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT After extraction, before writing, to decide the structure of your ROL.

PERSONA	You are a qualitative synthesis expert who groups research findings into meaningful themes.
MY CONTEXT	My research question: [PICOT] My extraction table: [PASTE TABLE HERE]
TASK	1. Propose 4–6 themes, each with a one-line definition. 2. Assign every study to one or more themes. 3. For each theme, summarise in 2 lines what the evidence shows and how consistent it is. 4. Identify studies that don't fit any theme, and what that means. 5. Suggest the order of themes that best leads to my research gap.
RULES	Build themes from the studies in the table only; if a theme rests on a single study, flag it as thin evidence, because thin themes are easily challenged.
OUTPUT	A theme table (theme, definition, studies, consistency), then the suggested order.

PRO TIP Good themes are usually questions: "Does X predict Y?", "In whom?", "How is Y measured?"

TRY IT WITH An extraction table of 18 studies on neutrophil-lymphocyte ratio in sepsis.

→ **NEXT #97** Thematic ROL draft from my evidence table

108 Explain why studies disagree

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT / FACULTY

USE IT When included studies report opposite or very different results.

PERSONA	You are a clinical epidemiologist skilled at explaining heterogeneity between studies.
MY CONTEXT	Studies that disagree, each with design, population, setting, definitions, sample size and result: [PASTE DETAILS HERE]
TASK	1. Compare the studies side by side on population, setting, exposure or intervention definition, outcome definition, measurement method, timing, design and sample size. 2. Identify the most likely explanations for the disagreement, ranked. 3. Say whether chance (small samples, wide confidence intervals) could explain it. 4. Write a paragraph for my ROL that presents the disagreement and the likely reasons. 5. Say what my study could add to resolve it.
RULES	Use only the details I gave; where a comparison needs missing information, write [NOT REPORTED], because speculation presented as fact weakens the ROL.
OUTPUT	A comparison table, the ranked explanations, then the ROL paragraph.

PRO TIP Differences in outcome definitions explain more disagreement than most residents expect.

TRY IT WITH Two cohorts disagree on whether obesity increases mortality in ARDS (the "obesity paradox").

→ **NEXT #97** Thematic ROL draft from my evidence table

109 One study, one paragraph, done properly

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT When writing individual study descriptions in the traditional ROL format.

PERSONA	You are a thesis-writing mentor who writes compact, informative study descriptions.
MY CONTEXT	Study details (author, year, country, design, sample size, population, key finding with numbers, limitation): [PASTE DETAILS HERE] My research question: [one line]
TASK	1. Write a 70–100 word paragraph: who, where, design, sample, key finding with its numbers, one limitation, and one sentence on relevance to my study. 2. Write a 40-word version for a summary table or a shorter ROL.
RULES	Reproduce numbers and terms exactly as given, and use the design's correct language (associated with, not caused, for observational designs), because study paragraphs are cross-checked against the paper.
OUTPUT	The two versions, labelled.

PRO TIP The relevance sentence is what separates a review from a bibliography. Never skip it.

TRY IT WITH A 2021 Indian cross-sectional study of 312 patients on vitamin D and COPD exacerbations.

→ **NEXT #99** Global and Indian studies: the classic thesis ROL

110 Convert a chronological ROL into a thematic one

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT When your guide says your ROL is "just a list of studies by year".

PERSONA	You are an academic editor who restructures list-style literature reviews into synthesised, thematic ones.
MY CONTEXT	My current ROL: [PASTE ROL HERE] My research question: [one line]
TASK	1. Identify the themes present across the studies I have described. 2. Reassign each study paragraph to a theme. 3. Rewrite the ROL thematically: for each theme, an opening sentence, comparison across studies, and an interpretive closing sentence. 4. Keep a short chronological note only where historical development matters.
RULES	Use only the studies and numbers already in my text, with their citations, because restructuring must not introduce new evidence.
OUTPUT	A theme map, then the restructured ROL.

PRO TIP Keep your old version. Some universities still want a brief chronological overview before the thematic review.

TRY IT WITH A 15-paragraph ROL on hypothyroidism in pregnancy ordered from 1995 to 2024.

→ **NEXT #100** Examiner-style critique of my ROL chapter

111 Build a literature synthesis matrix

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT When planning the ROL, to see which studies speak to which sub-questions.

PERSONA	You are a research-methods teacher who uses synthesis matrices to show residents how studies connect.
MY CONTEXT	Sub-questions or themes of my review: [list] Included studies with key findings: [PASTE LIST HERE]
TASK	1. Build a matrix with studies as rows and themes as columns, and a short finding in each relevant cell. 2. Mark empty cells: these are evidence gaps. 3. Summarise each column in one sentence. 4. Identify the column with the least evidence, and relate it to my research gap.
RULES	Fill cells only from the findings I gave; leave the rest blank rather than inferring, because empty cells are the insight.
OUTPUT	The matrix table, column summaries, then the gap note.

PRO TIP Put the matrix in your thesis appendix. Examiners like to see the synthesis visually.

TRY IT WITH 12 studies on AKI biomarkers across themes: diagnosis, prognosis, cost, Indian data.

→ **NEXT #107** Find the themes in my evidence table

112 Heading structure for my ROL chapter

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT Before you write, so the chapter has a logical architecture.

PERSONA	You are a senior thesis guide who has seen what structures examiners find clear.
MY CONTEXT	Thesis title and objectives: [paste] Themes or areas my reading covers: [list] University format requirements, if any: [e.g. must include a historical review]
TASK	1. Propose a heading and sub-heading structure for the chapter (for example: historical perspective, definitions, epidemiology, pathophysiology relevant to the question, themed evidence, Indian studies, gaps). 2. Give a suggested length and number of references for each section. 3. Mark which sections are essential and which are optional for my question. 4. Explain the logic that links the sections.
RULES	Keep pathophysiology and textbook content only where it is needed to understand the research question, because examiners penalise padding.
OUTPUT	A numbered outline with lengths and reference counts, then the linking logic.

PRO TIP If a section cannot be linked to your objective in one sentence, drop it.

TRY IT WITH Thesis: serum magnesium and outcomes in critically ill patients.

→ **NEXT #97** Thematic ROL draft from my evidence table

113 The closing paragraph that leads to my study

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT For the final paragraph of your ROL.

PERSONA	You are a thesis examiner who reads the end of the ROL to judge whether the student understands their own study's purpose.
MY CONTEXT	Summary of what my ROL showed: [brief] Gaps identified: [list] My objectives: [paste]
TASK	1. Write a closing paragraph of 120–180 words that summarises the state of the evidence, names the specific gaps, and states how the present study addresses them. 2. Check that the gaps named here match the gap in my introduction. 3. Suggest one sentence that connects to my methodology chapter.
RULES	Keep the gap claims modest ("limited data", "few Indian studies") unless my ROL demonstrates absence, because absolute claims are easy to disprove.
OUTPUT	The paragraph, then the consistency check, then the connecting sentence.

PRO TIP This paragraph and your introduction's gap paragraph should say the same thing in different words.

TRY IT WITH ROL on steroid use in tuberculous meningitis; gap: outcome data from Indian district hospitals.

→ **NEXT #81** Does my introduction lead to my objective?

114 Summarise a paper, grounded in quotes

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT / MBBS STUDENT

USE IT When you need a reliable summary of a paper you will cite.

PERSONA	You are a critical reader who summarises papers faithfully and never overstates their findings.
MY CONTEXT	The paper: [PASTE OR UPLOAD PAPER HERE] Why I am reading it: [e.g. for my ROL on X]
TASK	1. First, extract 5–8 direct quotes that capture the aim, design, population, main results and the authors' stated limitations. 2. Then write a 120-word summary based only on those quotes. 3. Note what the authors concluded, and whether the results fully support that conclusion.
RULES	Base the summary only on the quotes you extracted, because quote-first summarising prevents misrepresentation of the paper.
OUTPUT	The quotes with section labels, then the summary, then the conclusion check.

PRO TIP If the conclusion claims more than the results show, note it. It's useful for your discussion chapter.

TRY IT WITH A cohort paper on the association of sleep apnoea and resistant hypertension.

→ **NEXT #104** Extract one paper into my evidence table

115 Summary table of key studies

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT For the table that closes or accompanies your ROL.

PERSONA	You are an academic editor who produces clear, journal-quality summary tables.
MY CONTEXT	Key studies with extracted details: [PASTE DETAILS HERE] Columns my guide wants (if specified): [list] Page orientation limit: [portrait / landscape]
TASK	1. Build a summary table: author and year, country, design, sample size, population, key finding with numbers, limitation. 2. Order rows logically (by theme, then year). 3. Write a table title and footnotes explaining abbreviations. 4. Suggest how to split the table if it is too wide.
RULES	Reproduce numbers exactly and write NR where a value is not reported, because table errors are highly visible to examiners.
OUTPUT	The table in Markdown, then the title and footnotes.

PRO TIP Keep each "key finding" cell to one line with a number. Long text cells make tables unreadable.

TRY IT WITH 10 studies on the diagnostic accuracy of GeneXpert for extrapulmonary TB.

→ **NEXT #97** Thematic ROL draft from my evidence table

116 Check the numbers in my ROL against the sources

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT Before submitting your ROL or discussion, to catch transcription errors.

PERSONA	You are a fact-checker for a medical journal who compares every number in a manuscript with its source.
MY CONTEXT	My ROL text: [PASTE TEXT HERE] The source passages or tables for each cited study: [PASTE SOURCE EXTRACTS HERE]
TASK	1. List every number in my text with its citation. 2. Match each against the source passage. 3. Mark each as correct, wrong (give the correct value) or not verifiable from the extracts provided. 4. Flag any rounding, unit or percentage-versus-proportion error.
RULES	Compare only against the extracts I gave; do not "correct" numbers from memory, because the source document is the only authority.
OUTPUT	A table (number, citation, source value, status), then a count of errors.

PRO TIP Check the numbers you copied from abstracts. Abstracts sometimes differ from the results tables.

TRY IT WITH A ROL paragraph quoting prevalence, odds ratios and sample sizes from 6 studies.

→ **NEXT #117** Format and check my reference list

117 Format and check my reference list

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT Before submission, to get the reference list right.

PERSONA	You are a reference-list editor who applies Vancouver (ICMJE / NLM) style precisely.
MY CONTEXT	My reference list as it currently is: [PASTE REFERENCES HERE] Style required: [Vancouver / other]
TASK	1. Reformat every reference to the required style. 2. Flag references missing essential elements (authors, title, journal abbreviation, year, volume, pages, DOI). 3. Flag possible duplicates and numbering problems. 4. List references that should be checked for existence and accuracy on PubMed.
RULES	Do not invent missing details such as volume, pages or DOI; write [MISSING — CHECK PUBMED], because made-up bibliographic details make a real reference look fake.
OUTPUT	The reformatted list, then a problem table (reference number, issue).

PRO TIP Export references from Zotero or Mendeley instead of formatting them by hand wherever possible.

TRY IT WITH 25 references in mixed formats copied from Google Scholar.

→ **NEXT #52** Reference manager set-up and deduplication

118 The questions to ask NotebookLM about my papers

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT After uploading your included papers into a NotebookLM notebook.

PERSONA	You are a research-tools trainer who teaches residents to interrogate their own document collections.
MY CONTEXT	My research question: [PICOT] Number and type of papers uploaded: [e.g. 12 cohorts, 3 RCTs] Sections of my ROL: [list]
TASK	1. Write 15 questions to ask NotebookLM, grouped by: study characteristics, findings, disagreements, methods quality, gaps. 2. For each group, say what a useful answer should contain. 3. Suggest a prompt for NotebookLM's audio overview focused on my question. 4. Explain how to verify answers using NotebookLM's inline citations.
RULES	Phrase every question so the answer must come from the uploaded sources, because NotebookLM is only reliable when it is grounded in them.
OUTPUT	A grouped question list, then the audio-overview prompt, then the verification steps.

PRO TIP Ask "Which papers disagree about X, and how?". It is the fastest way to find material for your synthesis.

TRY IT WITH 15 papers on post-operative delirium in elderly hip fracture patients.

→ **NEXT #98** Evidence table from uploaded PDFs, with quotes

119 Quick critical appraisal of one paper

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT / MBBS STUDENT

USE IT When deciding how much weight a single paper deserves in your review.

PERSONA	You are a journal-club moderator who appraises papers quickly and fairly.
MY CONTEXT	The paper: [PASTE OR UPLOAD PAPER HERE] My research question: [one line]
TASK	1. Identify the design and its natural strengths and weaknesses. 2. Assess selection, measurement, confounding and attrition. 3. Judge whether the results are precise (confidence intervals) and clinically meaningful. 4. Judge applicability to Indian patients and to my question. 5. Give an overall weight for my review: high, moderate or low, with one clear reason.
RULES	Quote the paper when judging a weakness, because unsupported criticism is easy to dismiss.
OUTPUT	Five short headed sections, then a one-line verdict.

PRO TIP Use the verdict line directly in your ROL table's "limitation" column.

TRY IT WITH An observational study claiming statins reduce mortality in sepsis.

→ **NEXT #106** Apply a risk-of-bias tool to one paper

120 Read and summarise a meta-analysis

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT When a key source in your ROL is a systematic review or meta-analysis.

PERSONA	You are a biostatistician who explains forest plots and heterogeneity clearly to clinicians.
MY CONTEXT	The meta-analysis (abstract, results and forest plots, or upload): [PASTE OR UPLOAD HERE] My question: [one line]
TASK	1. State the pooled estimate, its CI and what it means clinically. 2. Explain the heterogeneity (I^2 , τ^2 , prediction interval if reported) and what it implies. 3. Summarise the certainty of evidence (GRADE) if reported. 4. Point out the main limitations (small-study effects, included study quality). 5. Write a 100-word ROL paragraph summarising the meta-analysis accurately.
RULES	Use only the reported numbers; if a statistic is not reported, say so rather than computing it, because an invented pooled estimate misrepresents the review.
OUTPUT	Five short answers, then the paragraph.

PRO TIP A pooled estimate with high heterogeneity is an average of different things. Say so in your ROL.

TRY IT WITH A meta-analysis of prone positioning in ARDS with I^2 of 62%.

→ **NEXT #97** Thematic ROL draft from my evidence table

121 Historical review sub-section without errors

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT When your university's thesis format expects a historical review.

PERSONA	You are a medical historian and academic who insists that every historical claim is dated and sourced.
MY CONTEXT	Topic: [TOPIC] Milestones I have already sourced: [PASTE MILESTONES WITH REFERENCES] Word limit: [e.g. 300 words]
TASK	1. Arrange my milestones into a timeline table: date, milestone, significance. 2. Suggest other milestones I may be missing, each marked [VERIFY DATE AND SOURCE]. 3. Write a historical review of the stated length from my sourced milestones only. 4. End with one sentence linking history to the present research question.
RULES	The prose must use only milestones I sourced; suggested ones stay in the table until I verify them, because dates and "firsts" are frequently misattributed.
OUTPUT	The timeline table, the suggestion list, then the prose.

PRO TIP Keep history short. One well-sourced page beats five pages of unverified dates.

TRY IT WITH Topic: evolution of mechanical ventilation.

→ **NEXT #112** Heading structure for my ROL chapter

122 Update my ROL before submission

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT

USE IT 2–3 months before thesis submission, to add papers published since your first search.

PERSONA	You are a systematic reviewer who updates reviews efficiently and transparently.
MY CONTEXT	Original search date and strings: [PASTE HERE] New records retrieved since then (titles and abstracts): [PASTE HERE] My current ROL themes: [list]
TASK	1. Screen the new records against my criteria. 2. For each included record, say which theme it belongs to and whether it confirms, contradicts or extends existing findings. 3. Suggest exactly where each fits in the text, with a one-sentence insertion. 4. Draft the updated search-methods sentence with the new date.
RULES	Use only the new records I pasted; insertions must use [CITE] slots until I verify the full text, because abstracts alone can mislead.
OUTPUT	A screening table, then an insertion plan (theme, position, sentence), then the updated methods sentence.

PRO TIP If a major new study contradicts your findings, address it in your discussion too.

TRY IT WITH 12 new PubMed records on tele-rehabilitation after stroke since your 2025 search.

→ **NEXT #116** Check the numbers in my ROL against the sources

123 Register a systematic review on PROSPERO

STANDARD

1.4 REVIEW OF LITERATURE · PG RESIDENT / FACULTY

USE IT If your thesis or first paper is a systematic review, before you start screening.

PERSONA	You are a systematic-review methodologist who has registered and peer reviewed many PROSPERO records.
MY CONTEXT	Review question: [PICOT] Planned databases, dates and designs: [brief] Planned synthesis: [narrative / meta-analysis] Review team and roles: [brief]
TASK	1. Draft the main PROSPERO fields: review title, question, searches, condition, participants, intervention or exposure, comparator, outcomes (main and additional), data extraction, risk of bias, synthesis strategy and subgroup analyses. 2. Flag fields where my plan is too vague to register. 3. Explain what must be registered before screening starts and what counts as a deviation.
RULES	Mark field names or rules you are unsure of as [CHECK CURRENT PROSPERO FORM], because the registration form is revised periodically. Do not claim the review is novel; tell me to search PROSPERO first.
OUTPUT	A field-by-field draft, then the vagueness flags, then the rules on timing and deviations.

PRO TIP Registration before screening is what reviewers check. Register early, even if the protocol is short.

TRY IT WITH Systematic review of vitamin D supplementation for TB treatment outcomes.

→ **NEXT #101** Inclusion and exclusion criteria for screening

QUICK PROMPTS · 1.4

124 Pull the PICO out of an abstract

QUICK

1.4 REVIEW OF LITERATURE · MBBS STUDENT / PG RESIDENT

Act as an evidence-based medicine tutor. From this abstract, [PASTE ABSTRACT], extract the population, intervention or exposure, comparator, outcome, design and sample size into a 6-row table. Write NOT REPORTED where missing, and add one line on the main result with its numbers.

TRY IT WITH Abstract of an RCT comparing two regimens for H. pylori eradication.

125 One-line takeaway of a paper

QUICK

1.4 REVIEW OF LITERATURE · PG RESIDENT

Act as a senior academic. Read this abstract or paper, [PASTE TEXT], and give its takeaway in one sentence of no more than 30 words that states the design, population and main finding with its key number, without overstating causation.

TRY IT WITH A cohort study on handgrip strength and mortality in cirrhosis.

126 The limitation that matters most

QUICK

1.4 REVIEW OF LITERATURE · PG RESIDENT

Act as a critical peer reviewer. For this study, [PASTE ABSTRACT OR METHODS], identify the single limitation that most threatens its conclusion, explain in 2 sentences which direction it likely biases the result, and write one sentence I can use in my ROL.

TRY IT WITH A single-centre retrospective study of 80 patients on a new sepsis score.

127 Compare two studies in three sentences

QUICK

1.4 REVIEW OF LITERATURE · PG RESIDENT

Act as a thesis mentor. Compare these two studies, [PASTE DETAILS OF STUDY A AND STUDY B], in exactly three sentences: what each found with numbers, whether they agree, and the most likely reason for any difference. Use associational language for observational designs.

TRY IT WITH Two studies on HbA1c and surgical site infection with opposite results.

128 What design is this study really?

QUICK

1.4 REVIEW OF LITERATURE · MBBS STUDENT / PG RESIDENT

Act as an epidemiology examiner testing a postgraduate. From this methods section, [PASTE METHODS], identify the true study design (authors sometimes mislabel it), justify your answer in 3 short bullets citing the methods, and state what the design can and cannot conclude.

TRY IT WITH A "prospective study" that actually analysed existing hospital records.

129 Interpret an effect size and its CI

QUICK

1.4 REVIEW OF LITERATURE · MBBS STUDENT / PG RESIDENT

Act as a biostatistics tutor. Interpret this result for a clinician: [PASTE RESULT, e.g. OR 1.8, 95% CI 1.1–2.9]. Explain what it means, whether it is statistically and clinically meaningful, and how to phrase it correctly in my ROL in one sentence.

TRY IT WITH Adjusted hazard ratio 0.72 (95% CI 0.55–0.94) for mortality with early antibiotics.

130 Results in plain language

QUICK

1.4 REVIEW OF LITERATURE · MBBS STUDENT / PG RESIDENT

Act as a medical educator teaching interns on the ward. Translate this results passage, [PASTE PASSAGE], into plain language for a junior colleague in under 80 words. Keep every number accurate and explain each statistical term briefly in plain brackets.

TRY IT WITH A results paragraph with sensitivity, specificity, AUC and likelihood ratios.

131 Does this reference support my sentence?

QUICK

1.4 REVIEW OF LITERATURE · PG RESIDENT

Act as a citation auditor. Here is my sentence, [SENTENCE], and the abstract or passage of the reference I cited, [PASTE SOURCE TEXT]. Tell me if the source fully supports, partly supports or does not support my sentence, and rewrite the sentence to match the source exactly.

TRY IT WITH A sentence claiming a drug "reduces mortality", citing a study that reported length of stay only.

132 Format one reference in Vancouver style

QUICK

1.4 REVIEW OF LITERATURE · PG RESIDENT / MBBS STUDENT

Act as a reference editor for medical theses. Format this reference in Vancouver (ICMJE / NLM) style: [PASTE REFERENCE DETAILS]. List any element that is missing rather than inventing it, and show the in-text citation format my thesis should use.

TRY IT WITH Details of a 2023 journal article with 8 authors, copied from Google Scholar.

133 How much should this study count?

QUICK

1.4 REVIEW OF LITERATURE · PG RESIDENT

Act as a systematic reviewer. Given this study's design, size and quality, [PASTE BRIEF DETAILS], tell me how much weight it should carry in my ROL compared with a large RCT or a meta-analysis, and give me one hedged sentence to describe it.

TRY IT WITH A 40-patient pilot study with a striking positive result.

134 Customise a NotebookLM audio overview

QUICK

1.4 REVIEW OF LITERATURE · PG RESIDENT / MBBS STUDENT

Act as a NotebookLM power user. Write a customisation instruction for an Audio Overview of my uploaded papers on [TOPIC] for a [PG resident / exam aspirant]. Focus it on findings, disagreements and gaps, skip history, and keep it under 80 words.

TRY IT WITH 10 uploaded papers on non-invasive ventilation in COPD exacerbations.

135 Is this ROL paragraph a list or a synthesis?

QUICK

1.4 REVIEW OF LITERATURE · PG RESIDENT

Act as an ROL examiner. Read this paragraph, [PASTE PARAGRAPH], and tell me whether it synthesises or merely lists studies, in one line. Then give 3 specific edits that would make it compare and interpret the studies, without adding new evidence.

TRY IT WITH A paragraph describing four studies one after another in date order.

SECTION 1.5

1.5

Aims, objectives & hypotheses

Your objectives are the contract for the whole thesis: every method, every table and every conclusion must trace back to them. Get them precise now.

20

prompts

2

Master

12

Standard

6

Quick

HOW TO USE THIS SECTION

Use it before your synopsis: every method, table and conclusion must trace back to these objectives.

Start with the Master prompts (**#136–137**) for the full workflow; use the Standard prompts (**#138–149**) for one task at a time; reach for the Quick prompts (**#150–155**) when you have two minutes. Tap any title below to jump to it, and follow → **NEXT** to move to the next step.

#136 Aims, objectives and hypotheses, fully aligned

#137 Committee review of my objectives

#138 Primary and secondary objectives in SMART form

#139 Null and alternative hypotheses, correctly stated

#140 Aim versus objective: fix the confusion

#141 Map each objective to outcome, measurement and analysis

#142 Objectives for a descriptive study

#143 Objectives for a diagnostic accuracy study

#144 Too many objectives: trim to what is feasible

#145 Secondary objectives that add value at no extra cost

#146 Fix vague verbs in my objectives

#147 From PICOT question to a full set of objectives

#148 Objectives for an interventional study or RCT

#149 Objectives for qualitative or mixed-methods studies

#150 Write my aim in one sentence

#151 Rewrite one vague objective

#152 Is my hypothesis testable?

#153 One-sided or two-sided test?

#154 Which of my objectives should be primary?

#155 Objectives squeezed into a form field

Aims, objectives and hypotheses, fully aligned

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

USE IT

After your topic is approved and before you write the protocol or synopsis.

WHY

Vague objectives ("to study", "to see") produce vague methods and unanswerable viva questions. Precise objectives, each mapped to an outcome, a measurement and an analysis, make the rest of the thesis almost write itself.

COPY THE PROMPT BELOW

PERSONA	You are a research methodologist and thesis committee member at an Indian medical university who has reviewed hundreds of synopses. You insist that every objective is measurable, feasible and linked to a specific analysis.
MY CONTEXT	Programme and specialty: [e.g. MD Community Medicine] Thesis title: [paste] Research question (PICOT): [paste] Study design: [cross-sectional / case-control / cohort / diagnostic accuracy / interventional] Sample size planned and time available: [N, months] Variables I plan to collect: [list]
TASK	1. Write one overall aim in a single sentence. 2. Write one primary objective and up to 3 secondary objectives in SMART form, each starting with a precise verb (to estimate, to compare, to determine the association between...). 3. Write the null and alternative hypotheses for the primary objective (and for secondary objectives where a comparison is made), and say whether the test is one-sided or two-sided, with the reason. 4. Build an alignment table: objective → outcome variable → how and when it is measured → statistical analysis. 5. Flag any objective my design or sample size cannot support.
RULES	Use my design's language: descriptive designs estimate, analytical designs compare or associate, and only randomised designs test effects, because objectives that claim more than the design can deliver are the most common synopsis rejection. Keep each objective to one sentence and one outcome. Where a timing or measurement detail is missing, write [SPECIFY] instead of assuming it.
OUTPUT	Aim, numbered objectives, hypotheses, then the alignment table, then the flags.
TWEAK	Finish with the 3 questions a synopsis committee is most likely to ask about these objectives, and short answers.

How to get more out of it

- Your conclusion will answer the primary objective in one sentence. Write that sentence now as a test.
- Keep secondary objectives few. Every extra objective adds variables, time and multiple-testing problems.
- Paste the alignment table into your protocol. It shows the committee your thinking in one glance.
- → **NEXT #141** Map each objective to outcome, measurement and analysis

Committee review of my objectives

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

USE IT

When your objectives are drafted, before synopsis submission.

WHY

Synopsis committees return objectives for the same reasons every term: vague verbs, unmeasurable outcomes, too many objectives and a mismatch with the design. This prompt runs that review before the committee does.

COPY THE PROMPT BELOW

PERSONA	You are the member-secretary of a postgraduate research committee at an Indian medical college. You review objectives strictly against the design and feasibility, and you write clear, actionable comments.
MY CONTEXT	Thesis title and design: [paste] Sample size and duration: [brief] My aims, objectives and hypotheses as drafted: [PASTE HERE] My introduction's gap statement: [paste] Comments already received from my guide: [brief, or "none yet"] My university's synopsis format, if it prescribes one: [brief]
TASK	1. Review each objective for: a precise verb, a single measurable outcome, a defined population and time frame, fit with the design, and feasibility with my sample size. 2. Mark each as approved, minor revision or major revision, with a reason. 3. Check that the primary objective directly answers the gap statement. 4. Check the hypotheses: correctly stated, matched to the primary objective, testable with my design. 5. Rewrite every objective that needs revision, keeping as much of my original wording as possible. 6. Recommend whether to cut, merge or demote any objective to "exploratory", and explain the effect on my workload and analysis.
RULES	Comment only on what I wrote, and quote the words you object to, because vague committee feedback is useless to a resident. Do not change my research question; if the objectives cannot be fixed without changing it, say so plainly.
OUTPUT	A review table (objective, verdict, issue, rewrite), then the hypothesis check, then the cut, merge or demote advice.
TWEAK	End with the revised full set of aims, objectives and hypotheses, ready to paste into my synopsis.

How to get more out of it

- › Take the revised set to your guide with the review table. It shows why each change was made.
- › If a secondary objective needs an extra test or visit, weigh it against your timeline before keeping it.
- › Keep the objective wording identical across synopsis, protocol, IEC form and thesis.
- › → **NEXT #138** Primary and secondary objectives in SMART form

138 Primary and secondary objectives in SMART form

STANDARD

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

USE IT When drafting objectives for your synopsis or protocol.

PERSONA	You are a thesis guide who writes objectives that are specific, measurable, achievable, relevant and time-bound.
MY CONTEXT	I am doing [MD / MS / DNB] in [SPECIALTY] in India. My thesis topic is [TOPIC] and my design is [cross-sectional / case-control / cohort / interventional]. Setting: [hospital]. Duration: [months].
TASK	1. Write one primary aim. 2. Write 3 specific secondary objectives. 3. Write one null and one alternative hypothesis for the primary objective. 4. After each objective, name the outcome variable it measures in brackets.
RULES	One sentence each, formal academic language, and a verb that matches my design (estimate, compare, determine the association), because the verb tells the committee what analysis to expect.
OUTPUT	A numbered list: aim, objectives, hypotheses.

PRO TIP If you can't name the outcome variable for an objective, the objective is not ready.

TRY IT WITH MD Paediatrics; topic: vitamin D levels and recurrent wheeze in children aged 1–5; cross-sectional.

→ **NEXT #139** Null and alternative hypotheses, correctly stated

139 Null and alternative hypotheses, correctly stated

STANDARD

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

USE IT When your protocol or synopsis needs formal hypotheses.

PERSONA	You are a biostatistician who teaches residents to state hypotheses that match their analysis.
MY CONTEXT	Primary objective: [paste] Design and comparison groups: [brief] Primary outcome and its type (continuous / binary / time-to-event): [brief]
TASK	1. Write the null (H0) and alternative (H1) hypotheses in words and in parameter form (e.g. $\mu_1 = \mu_2$). 2. Recommend a one-sided or two-sided test, with the reason. 3. Name the test that will evaluate the hypothesis. 4. Say whether a hypothesis is needed at all (purely descriptive objectives may not need one).
RULES	State hypotheses about population parameters, not sample results, because that is the most common error examiners catch. Default to two-sided unless there is a strong, pre-specified reason.
OUTPUT	The hypotheses, the sidedness decision, the test, then the "is it needed" note.

PRO TIP A non-inferiority or equivalence study needs a different hypothesis structure and a margin. Ask separately if yours is one.

TRY IT WITH Comparing mean HbA1c reduction between two oral regimens over 12 weeks.

→ **NEXT #141** Map each objective to outcome, measurement and analysis

140 Aim versus objective: fix the confusion

STANDARD

1.5 AIMS, OBJECTIVES & HYPOTHESES · MBBS STUDENT / PG RESIDENT

USE IT When your aims and objectives say the same thing twice, or mix levels.

PERSONA	You are a research-methods teacher who explains the hierarchy from aim to objectives to hypotheses clearly.
MY CONTEXT	My current aims and objectives: [PASTE HERE] My research question: [paste] My programme: [MD / MS / DNB / MBBS project]
TASK	1. Explain the difference between an aim, an objective and a hypothesis in 3 lines. 2. Classify each of my statements as aim, objective, hypothesis or none. 3. Rewrite them into a clean hierarchy: one aim, then objectives, then hypotheses.
RULES	Keep my research intent unchanged; only restructure and reword, because the content must still match my approved topic.
OUTPUT	The 3-line explanation, a classification table, then the rewritten set.

PRO TIP An aim is the "why"; objectives are the "what exactly"; hypotheses are the "what I expect".

TRY IT WITH Aims: "to study thyroid function in CKD" and "to see if thyroid is affected in CKD patients".

→ **NEXT #138** Primary and secondary objectives in SMART form

141 Map each objective to outcome, measurement and analysis

STANDARD

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

USE IT After finalising objectives, before designing the proforma and the analysis plan.

PERSONA	You are a methodologist who builds the traceability table linking objectives to data and analysis.
MY CONTEXT	My objectives: [PASTE OBJECTIVES HERE] Variables I plan to collect: [list] Software: [SPSS / R / Stata / Jamovi]
TASK	1. For each objective, give the outcome variable, its type, the measurement tool, the time point, and the statistical test or effect measure. 2. Flag any objective with no matching variable, and any variable with no matching objective. 3. Suggest what to add or drop.
RULES	Choose tests that fit the variable type and design, and mark any uncertain choice as [CONFIRM WITH STATISTICIAN], because this table will become part of your analysis plan.
OUTPUT	A mapping table, then the orphan objectives and variables, then the recommendations.

PRO TIP Variables with no objective should usually leave your proforma. They cost time and add nothing.

TRY IT WITH Objectives on prevalence of depression in dialysis patients and its association with serum albumin.

→ **NEXT #196** Pre-specified statistical analysis plan

142 Objectives for a descriptive study

STANDARD

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT / MBBS STUDENT

USE IT When your study describes prevalence, patterns or characteristics rather than testing a comparison.

PERSONA	You are an epidemiologist who writes objectives for cross-sectional and descriptive studies without forcing hypotheses on them.
MY CONTEXT	Topic and population: [brief] What I want to describe: [prevalence / pattern / distribution / characteristics] Any associations I also want to explore: [list]
TASK	1. Write objectives that estimate (with precision) rather than test. 2. Write any exploratory association objectives, labelled as such. 3. Say whether formal hypotheses are needed for each. 4. Suggest how sample size should be based on precision (margin of error) rather than power.
RULES	Use "to estimate" and "to describe" for descriptive aims, because "to prove" or "to test" signals a design error to the committee.
OUTPUT	A numbered list of objectives with their type, then the hypothesis and sample-size notes.

PRO TIP A descriptive study with a clear prevalence estimate and confidence interval is a perfectly good thesis.

TRY IT WITH Prevalence and pattern of dermatological adverse drug reactions in a tertiary hospital.

→ **NEXT #202** Sample size to estimate a proportion (prevalence)

143 Objectives for a diagnostic accuracy study

STANDARD

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

USE IT When your thesis evaluates a test against a reference standard.

PERSONA	You are a diagnostic-research methodologist familiar with the STARD reporting guideline.
MY CONTEXT	Index test: [e.g. bedside lung ultrasound] Reference standard: [e.g. CT chest] Target condition and population: [brief] Clinical role of the test: [replacement / triage / add-on]
TASK	1. Write the primary objective in terms of sensitivity and specificity (with 95% CI) of the index test against the reference standard. 2. Write secondary objectives (predictive values, likelihood ratios, AUC for continuous tests, inter-observer agreement, time or cost). 3. Explain which measures depend on prevalence and how to phrase them. 4. Note what "role of the test" means for the objectives.
RULES	Make the reference standard explicit in every accuracy objective, because accuracy without a named reference standard is meaningless.
OUTPUT	A numbered objective list, then the prevalence note, then the role note.

PRO TIP If your reference standard is imperfect, say so in your objectives' limitation note, not only in the discussion.

TRY IT WITH Point-of-care ultrasound versus chest X-ray for pneumothorax in trauma patients.

→ **NEXT #205** Sample size for a diagnostic accuracy study

144 Too many objectives: trim to what is feasible

STANDARD

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

USE IT When your list of objectives is long and your guide says "focus".

PERSONA	You are a pragmatic thesis supervisor who protects residents from over-ambitious protocols.
MY CONTEXT	My objectives: [PASTE OBJECTIVES HERE] Sample size and duration: [brief] Tests or visits each objective requires: [brief] Who pays for each test: [hospital / patient / project]
TASK	1. Estimate the workload, cost and statistical burden of each objective. 2. Rank the objectives by importance to the research question. 3. Recommend which to keep, merge, demote to exploratory, or drop. 4. Present the trimmed set.
RULES	Keep the primary objective unchanged; explain every cut in one line, because I'll need to justify it to my guide.
OUTPUT	A ranking table (objective, burden, importance, recommendation), then the trimmed set.

PRO TIP Three objectives done well beat eight done badly.

TRY IT WITH Eight objectives for a 2-year study of 100 patients with liver cirrhosis.

→ **NEXT #136** Aims, objectives and hypotheses, fully aligned

145 Secondary objectives that add value at no extra cost

STANDARD

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

USE IT When your primary objective is set and you want useful secondary questions from data you already collect.

PERSONA	You are an efficient clinical researcher who makes every data point count.
MY CONTEXT	Primary objective: [paste] Data I will already collect: [list] Design and sample size: [brief] Specialty and setting: [brief]
TASK	1. Suggest 5 secondary objectives answerable with the data I already collect. 2. For each, give the outcome, the analysis and what it adds clinically. 3. Warn which would be underpowered, and how to label them in my protocol.
RULES	Suggest no objective that needs new tests, visits or consent changes, because the point is zero extra cost. Label underpowered analyses exploratory.
OUTPUT	A table (objective, data used, analysis, added value, power warning).

PRO TIP Subgroup objectives are almost always underpowered. Call them exploratory from the start.

TRY IT WITH Primary: 30-day mortality in sepsis; data: vitals, CBC, lactate, SOFA, comorbidities.

→ **NEXT #141** Map each objective to outcome, measurement and analysis

146 Fix vague verbs in my objectives

STANDARD

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT / MBBS STUDENT

USE IT When your objectives use verbs such as "to study", "to see", "to know" or "to evaluate".

PERSONA	You are an academic editor for research proposals who replaces vague verbs with precise ones.
MY CONTEXT	My objectives: [PASTE OBJECTIVES HERE] Design: [design] Primary outcome: [outcome]
TASK	1. Mark each vague verb. 2. Replace it with a precise verb that matches the design and the intended analysis (estimate, compare, determine the association, assess the accuracy, evaluate the effect). 3. Add any missing population, outcome, setting or time frame. 4. Give the final wording for each objective.
RULES	Match the verb to the design, because "to evaluate the effect" in an observational study promises causation you cannot deliver.
OUTPUT	A before-and-after table, then the final list.

PRO TIP Read each objective and ask, "What single number will answer this?". If you can't say, it's still vague.

TRY IT WITH "To study the role of CRP in appendicitis."

→ **NEXT #138** Primary and secondary objectives in SMART form

147 From PICOT question to a full set of objectives

STANDARD

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

USE IT When your question is fixed and you need everything that follows from it.

PERSONA	You are a research mentor who derives objectives systematically from the research question.
MY CONTEXT	My PICOT question: [PASTE QUESTION] Design: [design] Sample size and follow-up I can manage: [brief]
TASK	1. Derive the primary objective directly from the PICOT elements. 2. Suggest 2–3 secondary objectives that follow naturally (other outcomes, time points or subgroups). 3. Write the hypotheses. 4. Check that the population, comparison and outcome wording is identical across question, objectives and hypotheses.
RULES	Do not introduce new populations or outcomes absent from my PICOT without flagging them as additions, because the objectives must stay within my approved question.
OUTPUT	The derived set, then the consistency check.

PRO TIP Copy exact phrases from your PICOT. Paraphrasing the population in the objectives creates confusion later.

TRY IT WITH In adults with COPD exacerbation (P), does NIV started in the ward (I) versus in the ICU (C) change 30-day mortality (O) over 30 days (T)?

→ **NEXT #139** Null and alternative hypotheses, correctly stated

148 Objectives for an interventional study or RCT

STANDARD

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

USE IT When your thesis compares an intervention with a control, randomised or not.

PERSONA	You are a clinical trials methodologist who writes trial objectives that follow CONSORT and SPIRIT conventions.
MY CONTEXT	Intervention and comparator: [brief] Population and setting: [brief] Primary endpoint and time point: [e.g. treatment failure at day 7] Randomised?: [yes / no] Is the aim superiority, non-inferiority or equivalence?: [state or "unsure"]
TASK	1. Write the primary objective with the endpoint, time point and comparison. 2. Write secondary objectives, including safety and adverse events. 3. Write hypotheses matching the aim (superiority, non-inferiority with margin, or equivalence). 4. Explain what changes if the study is not randomised.
RULES	For a non-inferiority or equivalence aim, include a margin slot marked [JUSTIFY MARGIN FROM LITERATURE], because an unjustified margin is the first thing reviewers attack.
OUTPUT	Objectives, hypotheses, then the notes on randomisation and aim type.

PRO TIP Always include a safety objective in an interventional thesis. The IEC will ask for it.

TRY IT WITH Nebulised hypertonic saline versus normal saline in infants with bronchiolitis, length of stay.

→ **NEXT #139** Null and alternative hypotheses, correctly stated

149 Objectives for qualitative or mixed-methods studies

STANDARD

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT / FACULTY

USE IT When part or all of your study uses interviews, focus groups or open-ended data.

PERSONA	You are a mixed-methods researcher in health services who writes objectives fit for qualitative inquiry.
MY CONTEXT	Topic: [TOPIC] Quantitative component (if any): [brief] Qualitative component: [e.g. in-depth interviews with 20 caregivers] Sequence of the components: [parallel / quantitative first / qualitative first]
TASK	1. Write the overall aim. 2. Write quantitative objectives in measurable form. 3. Write qualitative objectives using exploratory verbs (to explore, to understand, to describe experiences of). 4. Explain how the two components integrate (explanatory, exploratory or convergent design).
RULES	Don't give qualitative objectives numeric targets or hypotheses, because that misunderstands qualitative research and reviewers notice.
OUTPUT	Aim, quantitative objectives, qualitative objectives, then the integration note.

PRO TIP Name your mixed-methods design explicitly. Reviewers look for it.

TRY IT WITH Adherence to antiretroviral therapy: a survey of 200 patients plus interviews with 15 non-adherent patients.

→ **NEXT #136** Aims, objectives and hypotheses, fully aligned

150 Write my aim in one sentence

QUICK

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT / MBBS STUDENT

Act as a research-methods tutor for postgraduates. From my topic [TOPIC] and design [DESIGN], write a single-sentence study aim that names the population, the main variable and the setting. Give 3 versions and mark the most precise one, with a one-line reason.

TRY IT WITH Topic: sleep quality in resident doctors; design: cross-sectional survey.

151 Rewrite one vague objective

QUICK

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

Act as a strict thesis committee reviewer. Rewrite this objective, [OBJECTIVE], so it is specific, measurable and matched to my design [DESIGN]. Give the rewrite, name the outcome variable and its unit, and explain the change in one short line.

TRY IT WITH "To assess the outcome of patients with dengue."

152 Is my hypothesis testable?

QUICK

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

Act as a biostatistician who reviews thesis synopses. Tell me whether this hypothesis, [HYPOTHESIS], is testable with my design [DESIGN] and data [PRIMARY OUTCOME TYPE]. If not, explain why in 2 lines and give a testable version with its null hypothesis.

TRY IT WITH "Yoga improves the quality of life of asthmatic patients."

153 One-sided or two-sided test?

QUICK

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

Act as a biostatistics examiner. For my hypothesis, [HYPOTHESIS], tell me whether a one-sided or two-sided test is appropriate, justify it in 3 lines, and explain how the choice affects my sample size and how examiners usually view one-sided tests.

TRY IT WITH A new dressing is expected to reduce wound infection rates compared with standard dressing.

154 Which of my objectives should be primary?

QUICK

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

Act as a thesis guide. From these objectives, [PASTE OBJECTIVES], tell me which should be the primary objective and why (clinical importance, measurability, feasibility, sample size basis). Rank the rest as secondary objectives in order of importance, with one line each.

TRY IT WITH Three objectives on mortality, ICU stay and ventilator days in ARDS.

155 Objectives squeezed into a form field

QUICK

1.5 AIMS, OBJECTIVES & HYPOTHESES · PG RESIDENT

Act as a research-forms expert. Condense my objectives, [PASTE OBJECTIVES], to fit a [WORD OR CHARACTER LIMIT] field on the IEC or CTRI form, keeping the primary objective complete and the secondary objectives recognisable. Show the character count of the final text.

TRY IT WITH Four objectives condensed to 500 characters for a CTRI registration.

1.6

Study design & methodology

The methods chapter is the document you will be held to for three years. Every ambiguity here becomes an unanswerable viva question later.

40

prompts

5

Master

23

Standard

12

Quick

HOW TO USE THIS SECTION

Use it while writing the protocol and methods, to answer the questions your guide, IEC and examiners will ask.

Start with the Master prompts (**#156–160**) for the full workflow; use the Standard prompts (**#161–183**) for one task at a time; reach for the Quick prompts (**#184–195**) when you have two minutes. Tap any title below to jump to it, and follow → **NEXT** to move to the next step.

#156 Choose and defend the right study design

#157 Write the full methodology chapter

#158 Operational definitions for every variable

#159 Case record form (proforma) with a data-entry codebook

#160 Bias and confounding audit of my design

#161 PICO framework and inclusion/exclusion criteria

#162 Make eligibility criteria bedside-proof

#163 Choose and describe my sampling method

#164 Randomisation, allocation concealment and blinding

#165 Choosing and matching controls in a case-control study

#166 Follow-up plan that keeps loss to follow-up low

#167 Tighten a cross-sectional study design

#168 Design a diagnostic accuracy study (STARD)

#169 Record-review (retrospective) methodology

#170 Identify confounders and draw my causal diagram

#171 Inter-observer agreement and quality-control plan

#172 Translate and validate a questionnaire

#173 Study procedure as a patient journey

#174 Blinding when full blinding is impossible

#175 Before-after and quasi-experimental studies done right

#176 Plan a pilot study for my thesis

#177 Data management plan for my thesis

#178 Laboratory sample handling and pre-analytical control

#179 Standardised imaging acquisition and reading protocol

#180 Which reporting guideline applies to my study?

#181 Describe the intervention so others can replicate it (TIDieR)

#182 Designing studies in patients who cannot consent

#183 Convert my protocol into thesis methods

#184 Name my study design correctly

#185 Prospective or retrospective: what's the difference for me?

#186 Strengths and weaknesses of my design

#187 Operationally define one variable

#188 Draw a simple random sample in Excel

#189 Allocation concealment in my trial

#190 Can I apply this criterion at the bedside?

#191 List the likely confounders

#192 Cohort or case-control? Decide now

#193 Could observer bias affect my results?

#194 Write the "study setting" paragraph

#195 My design in one line for the title page

Choose and defend the right study design

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT

After your research question is fixed and before you write the protocol.

WHY

The design decides what you can conclude. Choosing it by habit ("everyone does cross-sectional") leads to objectives your data cannot answer. This prompt compares designs against your actual question and resources, and gives you the defence for the viva.

COPY THE PROMPT BELOW

PERSONA	You are a clinical epidemiologist who teaches research methodology to postgraduates and sits on thesis committees. You are practical about what can be done in a 2-year MD programme at an Indian government medical college, and precise about what each design can and cannot conclude.
MY CONTEXT	Programme and specialty: [e.g. MD General Medicine] Thesis topic: [TOPIC] Research question: [what I want to find out, in PICOT form if possible] Eligible patients per month and months available: [N, X] Follow-up I can realistically do: [none / in-hospital / 30 days / 6 months] Resources: [routine tests only / some special tests / funded] Can I assign the intervention myself?: [yes / no / not applicable]
TASK	1. List the designs that could answer my question (cross-sectional, case-control, prospective or retrospective cohort, diagnostic accuracy, before-after, randomised trial). 2. For each, state what it could conclude, its main bias, its sample-size and time implications, and its feasibility in my setting. 3. Recommend the single most appropriate design and explain why it fits my question better than the obvious alternative. 4. State its key limitation and how to reduce it in the protocol. 5. Say whether a simpler design could answer a slightly narrower version of my question, and what I would lose. 6. Write a 100-word design justification paragraph for my methods chapter.
RULES	Match conclusions to the design: association for observational designs and effect for randomised ones, because overclaiming the design is a classic viva trap. Where feasibility depends on local facts I have not given, write [CHECK LOCALLY] rather than assuming.
OUTPUT	A comparison table (design, can conclude, main bias, time, feasibility), then the recommendation, the limitation plan, the simpler-design note, and the justification paragraph.
TWEAK	Finish with the 3 viva questions about my design choice that examiners ask most often, with model answers.

How to get more out of it

- › If you are unsure between two designs, ask a follow-up: "Compare [design A] and [design B] for my topic on feasibility and bias."
- › The justification paragraph goes straight into your methods chapter. Edit it into your own words.
- › Take the comparison table to your guide. It shows you considered alternatives.
- › → NEXT #161 PICO framework and inclusion/exclusion criteria

Write the full methodology chapter

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT

When writing the thesis after data collection (past tense), or the protocol before it (future tense).

WHY

The methodology chapter must be complete enough for someone else to repeat your study exactly. Writing it from a structured list of your real decisions, rather than from memory, catches gaps before the examiner does.

COPY THE PROMPT BELOW

PERSONA	You are a senior academic editor who writes and reviews methodology chapters for MD/MS theses and journal articles. You write in formal scientific prose with enough detail for exact replication, and you follow the relevant EQUATOR reporting guideline.
MY CONTEXT	Tense needed: [future (protocol) / past (thesis)] Design and reporting guideline: [e.g. prospective cohort, STROBE] Setting and study period: [hospital, department, dates] Population, inclusion and exclusion criteria: [paste] Sampling method and sample size with its calculation: [paste] Procedure, step by step: [paste] Variables with operational definitions: [paste] Data collection tools and who collected data: [paste] Ethics approval number, consent process and registration: [paste or "pending"] Statistical analysis plan: [paste] University format or heading order, if prescribed: [paste or "none"]
TASK	Write the methodology chapter with these headings: study design; study setting; study period; study population; inclusion criteria; exclusion criteria; sample size; sampling technique; study procedure; operational definitions; data collection; quality control; statistical analysis; ethical considerations. Under each heading, write only what my inputs support. Then produce a reporting-guideline checklist showing which item each heading satisfies, and flag missing items. Finally, list every place where the text depends on a decision I have not yet documented.
RULES	Use only the details I gave, in the tense I chose. Where information is missing, write [MISSING: describe what is needed] instead of inventing it, because invented methods are research misconduct. Keep numbers, dates and definitions exactly as supplied.
OUTPUT	The chapter with headings in formal prose, then the checklist table (guideline item, where addressed, status).
TWEAK	End with the 5 sentences an examiner is most likely to question, and what detail would make each defensible.

How to get more out of it

- Generate the chapter one heading group at a time if your inputs are long.
- Keep protocol and thesis versions in sync. Examiners compare the synopsis with the final methods.
- Any deviation from the approved protocol must be stated and explained, not hidden.
- → **NEXT #183** Convert my protocol into thesis methods

Operational definitions for every variable

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT

While writing your protocol, before you design the proforma.

WHY

Most protocols are weakest at operational definitions. "Anaemia", "smoker" and "improvement" mean different things to different observers. Defining each variable with its instrument, cut-off, unit and time point removes judgement calls at the bedside and questions at the viva.

COPY THE PROMPT BELOW

PERSONA	You are a clinical research methodologist who has reviewed several hundred postgraduate protocols. You insist that every variable is defined so that two independent observers would record the same value.
MY CONTEXT	Thesis title and design: [paste] Primary and secondary outcomes: [list] All variables I plan to collect: [list, grouped as demographic, clinical, laboratory, imaging, outcome] Guidelines or criteria I follow for key definitions: [e.g. WHO for anaemia, KDIGO for AKI, or "unsure"] Instruments and equipment available: [list]
TASK	1. For each variable, write an operational definition: exactly what is measured, the instrument or method, unit, cut-off or categories, time point, and who measures it. 2. For categorical variables, list the categories and their coding for data entry. 3. For outcomes, name the criterion or scale they are based on and whether a validated version exists for my population. 4. Flag variables that are subjective, and suggest how to standardise them (training, scripts, blinding, two observers). 5. Flag any variable whose definition differs between common guidelines, and ask me to choose.
RULES	Name a criterion or scale only if you are confident it exists; otherwise write [VERIFY CRITERION / SCALE AND VERSION], because a misnamed criterion is easily caught. Do not fill in cut-offs from memory where they vary between guidelines; write [CHOOSE AND CITE CUT-OFF].
OUTPUT	A variable dictionary table (variable, definition, instrument, unit or categories, coding, time point, measured by), then the subjectivity and conflict flags.
TWEAK	End with the 3 definitions most likely to be challenged by the examiner and how to strengthen each.

How to get more out of it

- This table becomes your data dictionary and your proforma. Build it once, use it everywhere.
- Pilot the definitions on 5 patients with a colleague. Where you disagree, the definition needs work.
- Keep the definitions identical in the thesis, the proforma and the analysis.
- → **NEXT #159** Case record form (proforma) with a data-entry codebook

Case record form (proforma) with a data-entry codebook

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT

After your variables are defined, before recruiting the first patient.

WHY

A badly designed proforma creates missing data, free-text chaos and weeks of cleaning. A well-coded proforma, built to mirror your spreadsheet columns, makes data entry and analysis almost mechanical.

COPY THE PROMPT BELOW

PERSONA	You are a clinical data manager who designs case record forms for investigator-initiated studies in Indian hospitals. You design for fast bedside use, minimal free text and clean data entry.
MY CONTEXT	Thesis title and design: [paste] Variable dictionary (variable, definition, unit, categories): [PASTE TABLE HERE] Time points of data collection: [e.g. admission, day 3, discharge, 30 days] Format: [printed form / Google Forms / KoboToolbox / Epi Info / Excel] Who fills it: [me / resident colleague / nurse] Where it will be filled: [bedside / OPD desk / from records]
TASK	1. Design the proforma section by section: identification (study ID only), eligibility checklist, consent confirmation, baseline, each follow-up time point, outcome, and investigator signature. 2. For each item, give the question wording, response format (tick box, number with unit, date), allowed range and skip logic. 3. Build the matching codebook: column name, variable label, type, codes, missing-value code, valid range. 4. Suggest data-quality checks at entry (ranges, required fields, logical checks). 5. Suggest how to keep identifiers separate from study data. 6. Estimate how long one form takes to complete, and suggest cuts if it exceeds 10 minutes.
RULES	Use a study ID instead of the name or hospital number on the form, because identifiable data on proformas is an ethics breach. Keep free text to a minimum. Use the same variable names in the proforma and the codebook.
OUTPUT	The proforma as a structured outline with item wording, then the codebook table, then the quality checks and the identifier-separation plan.
TWEAK	Finish with a list of items that will probably be missed at the bedside, and how to prevent that.

How to get more out of it

- Fill the proforma for 3 real patients before finalising. Every awkward question shows up immediately.
- Keep a separate, locked linking log (study ID → hospital number).
- Enter data weekly. Monthly entry means months of backlog.
- → **NEXT #177** Data management plan for my thesis

Bias and confounding audit of my design

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT / FACULTY

USE IT

When your protocol is drafted, before IEC submission.

WHY

Every design has predictable biases. Finding them before data collection lets you fix them in the protocol. After data collection, all you can do is list them as limitations.

COPY THE PROMPT BELOW

PERSONA	You are an epidemiologist who reviews study protocols specifically for bias and confounding, and who proposes practical fixes that fit a thesis budget.
MY CONTEXT	Design and setting: [paste] Population and how patients are identified and recruited: [paste] Exposure or intervention and how it is measured: [paste] Outcome and how it is measured, by whom, and whether blinded: [paste] Follow-up plan: [paste] Variables I plan to adjust for: [list] Expected sample size and number of outcome events: [numbers]
TASK	1. Assess selection bias: referral pattern, non-response, consecutive versus convenience sampling, loss to follow-up. 2. Assess information bias: measurement error, recall bias, observer bias, misclassification (and whether it is differential). 3. Assess confounding: list likely confounders for my exposure-outcome relationship, whether I measure them, and how to handle them (restriction, matching, stratification, adjustment). 4. For each bias, rate its likely impact (low, moderate, high) and the direction it could push my result. 5. Propose protocol changes that reduce the highest-impact biases within my resources. 6. Tell me whether my sample size and event count can support the adjustment I plan.
RULES	Be specific to my design and setting rather than listing textbook biases, because generic lists do not improve a protocol. Where a fix needs resources I did not mention, label it [NEEDS RESOURCE] so I can decide.
OUTPUT	A bias table (bias, where it enters, impact, direction, fix), then a confounder list, then the recommended protocol changes in priority order.
TWEAK	End with the two limitations I will still have to acknowledge in the discussion, pre-written as honest sentences.

How to get more out of it

- Draw a simple causal diagram (exposure → outcome, with confounders) to show your guide.
- Blinding outcome assessors is often possible even when blinding patients isn't. Check it.
- Record reasons for non-participation. They let you describe selection bias with numbers.
- → **NEXT #170** Identify confounders and draw my causal diagram

161 PICO framework and inclusion/exclusion criteria

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When drafting the population section of your protocol.

PERSONA	You are a research methodologist at an Indian medical college who writes precise, practical eligibility criteria.
MY CONTEXT	My thesis topic is [TOPIC]. My design is [DESIGN]. I am conducting this study at a [government medical college / tertiary hospital] in India. My population: [brief].
TASK	1. Give a structured PICO (or PECO): population, intervention or exposure, comparison, outcome. 2. Give 5 inclusion criteria. 3. Give 5 exclusion criteria, each with the reason for exclusion. 4. Flag any criterion that could make recruitment too slow.
RULES	Keep all criteria specific, measurable and practical for an Indian hospital; include age limits and consent ability, because vague criteria create inconsistent recruitment.
OUTPUT	The PICO table, then the numbered criteria lists, then the recruitment flags.

PRO TIP Every exclusion shrinks your eligible pool and your generalisability. Keep only those with a real reason.

TRY IT WITH Topic: serum procalcitonin in differentiating bacterial from viral pneumonia in adults; design: cross-sectional.

→ **NEXT #162** Make eligibility criteria bedside-proof

162 Make eligibility criteria bedside-proof

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT After drafting criteria, to remove judgement calls at recruitment.

PERSONA	You are a trial coordinator who has seen recruitment drift when criteria were left open to interpretation.
MY CONTEXT	My inclusion and exclusion criteria: [PASTE CRITERIA HERE] Who will screen patients: [me / colleagues / nurses]
TASK	1. Rewrite each criterion so it can be applied at the bedside without judgement: name the test, cut-off, time window and source document. 2. Mark criteria that need information unavailable at the time of screening, and suggest a fix. 3. Create a one-page eligibility checklist with yes/no boxes.
RULES	Keep my intent; if tightening a criterion changes who is eligible, say so, because the committee approved the original population.
OUTPUT	A before-and-after table, then the checklist.

PRO TIP Put the checklist at the top of your proforma and tick it for every screened patient.

TRY IT WITH "Patients with severe sepsis", "patients with significant comorbidities".

→ **NEXT #163** Choose and describe my sampling method

163 Choose and describe my sampling method

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When writing the sampling section of your protocol.

PERSONA	You are a biostatistician who explains sampling honestly, including when convenience sampling is the realistic choice.
MY CONTEXT	Design: [design] How patients reach me: [OPD / ward / ICU / community] Expected eligible patients per month versus the sample I need: [numbers] Can I obtain a list of all eligible people in advance?: [yes / no]
TASK	1. Recommend the most appropriate sampling method (consecutive, simple random, systematic, stratified, cluster or convenience) with the reason. 2. Describe exactly how it will be implemented day to day. 3. Explain the bias risk and how to reduce it. 4. Write the sampling paragraph for my protocol.
RULES	Call convenience sampling what it is if that is what I will do, because mislabelling it as random is a classic examiner catch. Consecutive sampling needs a defined start date and no skipped days.
OUTPUT	The recommendation, the implementation steps, the bias note, then the paragraph.

PRO TIP Consecutive sampling of all eligible patients is usually the most defensible choice for a hospital thesis.

TRY IT WITH Diabetic patients attending a medicine OPD three days a week, about 40 eligible per week.

→ **NEXT #202** Sample size to estimate a proportion (prevalence)

164 Randomisation, allocation concealment and blinding

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When your thesis is a randomised trial.

PERSONA	You are a clinical trials methodologist who describes randomisation procedures precisely enough for a CONSORT checklist.
MY CONTEXT	Arms and allocation ratio: [e.g. 2 arms, 1:1] Sample size: [N] Stratification factors, if any: [list] Who enrolls patients and who assigns the intervention: [brief] Can patients, clinicians or assessors be blinded?: [brief]
TASK	1. Recommend a randomisation method (simple, block, stratified block) and how to generate the sequence. 2. Describe allocation concealment operationally (e.g. sequentially numbered, opaque, sealed envelopes, or central allocation). 3. Describe who is blinded and how, and what to do if blinding is impossible. 4. Write the CONSORT-style paragraph for my methods.
RULES	Keep the person who generates the sequence separate from the person who enrolls, and say so, because failure of concealment is the main source of bias in trials.
OUTPUT	Numbered procedures, then the paragraph.

PRO TIP Keep the randomisation list with someone outside the study team, such as the department statistician.

TRY IT WITH RCT of two analgesic regimens after caesarean section, 1:1, 120 patients, stratified by parity.

→ **NEXT #174** Blinding when full blinding is impossible

165 Choosing and matching controls in a case-control study

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When your thesis compares cases with a condition to controls without it.

PERSONA	You are an epidemiologist who has designed many hospital-based case-control studies.
MY CONTEXT	Case definition: [paste] Where cases come from: [brief] Exposure of interest: [brief] Candidate control sources: [e.g. hospital patients with other conditions, relatives, community]
TASK	1. Recommend the best control source and why. 2. Advise on matching: whether to match, on what, and the ratio (1:1, 1:2), with the trade-offs. 3. Explain how to ascertain exposure equally in cases and controls. 4. Identify the key biases (selection, recall) and how to limit them. 5. Name the analysis that fits (e.g. conditional logistic regression if matched).
RULES	Warn me against matching on the exposure or its consequences, because overmatching hides the association I am looking for.
OUTPUT	Headed recommendations, then a bias table.

PRO TIP Controls should come from the same population that would have produced a case, had they developed the disease.

TRY IT WITH Cases: young adults with ischaemic stroke; exposure: hyperhomocysteinaemia.

→ **NEXT #160** Bias and confounding audit of my design

166 Follow-up plan that keeps loss to follow-up low

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When your thesis follows patients over time.

PERSONA	You are a cohort-study coordinator experienced in keeping follow-up rates high in Indian settings.
MY CONTEXT	Follow-up duration and visits: [brief] Patient population and where they live: [brief] Outcome ascertainment: [visit / phone / records]
TASK	1. Design the follow-up schedule with windows (e.g. day 30 ± 3). 2. List strategies to reduce loss to follow-up: contact details, alternate contacts, reminders, phone follow-up, travel support. 3. Define what counts as lost to follow-up and how to record reasons. 4. Explain how missing outcomes will be handled in the analysis.
RULES	Keep all contact details in a secure linking log separate from study data, because follow-up data creates privacy risk.
OUTPUT	A schedule table, a strategy list, then the definitions and analysis note.

PRO TIP Collect two phone numbers at enrolment, including a family member's. It halves loss to follow-up.

TRY IT WITH 6-month follow-up of patients after first acute coronary syndrome, rural catchment.

→ **NEXT #177** Data management plan for my thesis

167 Tighten a cross-sectional study design

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT / MBBS STUDENT

USE IT When your thesis is a cross-sectional study.

PERSONA	You are a methodologist who knows cross-sectional studies are easy to design and easy to get wrong.
MY CONTEXT	Topic and population: [brief] What I want to estimate or associate: [brief] Data sources: [interview / examination / tests / records]
TASK	1. Clarify whether the study is descriptive (prevalence) or analytical (association). 2. Define the sampling frame and time frame. 3. Explain how to avoid bias from prevalent cases (survivors, long-duration cases). 4. Explain what the design cannot conclude (temporality) and how to phrase results. 5. Recommend the effect measure (prevalence, prevalence ratio, odds ratio).
RULES	Make the temporality limitation explicit, because examiners ask "how do you know the exposure came first?".
OUTPUT	Five headed answers, then a one-paragraph design description.

PRO TIP For common outcomes, a prevalence ratio is often more honest than an odds ratio.

TRY IT WITH Prevalence of metabolic syndrome among people living with HIV on antiretroviral therapy.

→ **NEXT #163** Choose and describe my sampling method

168 Design a diagnostic accuracy study (STARD)

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When your thesis evaluates a test.

PERSONA	You are a diagnostic-research methodologist who applies STARD 2015 and QUADAS-2 when designing studies.
MY CONTEXT	Index test and reference standard: [brief] Target condition and population: [brief] Who performs and reads each test: [brief]
TASK	1. Define the population to match clinical use (spectrum), and consecutive enrolment. 2. Define index test conduct, the positivity threshold (pre-specified), and blinding of readers to the reference standard. 3. Define reference standard conduct, timing relative to the index test, and blinding. 4. Explain how to handle indeterminate results and patients without a reference standard. 5. Outline the 2 × 2 table and the accuracy measures with CIs.
RULES	Pre-specify thresholds and blinding, because post-hoc thresholds and unblinded reading inflate accuracy.
OUTPUT	Five headed sections, then the 2 × 2 layout.

PRO TIP Partial verification (only test-positives get the reference standard) is the most common fatal flaw. Avoid it.

TRY IT WITH Bedside lung ultrasound versus CT chest for pneumothorax in blunt chest trauma.

→ **NEXT #205** Sample size for a diagnostic accuracy study

169 Record-review (retrospective) methodology

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When your study uses existing case records.

PERSONA	You are a health-records researcher who has designed chart-review studies that survived peer review.
MY CONTEXT	Records and years: [brief] Variables to abstract: [list] Who will abstract: [brief]
TASK	1. Define case identification (codes, registers, keywords) and the period. 2. Create an abstraction protocol: a variable dictionary, where in the file each item is found, and rules for conflicting entries. 3. Plan abstractor training and a reliability check (e.g. double abstraction of 10% of records with kappa). 4. Explain how missing data will be recorded and handled. 5. Write the ethics note (waiver of consent, confidentiality).
RULES	Define missing versus normal explicitly, because "not documented" is not the same as "absent" and confusing them biases results.
OUTPUT	Headed protocol sections, then the ethics note.

PRO TIP Pilot on 20 files before committing to the full list of variables.

TRY IT WITH Outcomes of snakebite envenomation in a district hospital, 2019–2024.

→ **NEXT #177** Data management plan for my thesis

170 Identify confounders and draw my causal diagram

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When your study examines an association and you must decide what to adjust for.

PERSONA	You are an epidemiologist who teaches causal diagrams (DAGs) to clinicians in plain language.
MY CONTEXT	Exposure: [brief] Outcome: [brief] Other variables I collect: [list]
TASK	1. Classify each variable as confounder, mediator, collider, effect modifier or irrelevant, with a one-line reason. 2. Describe the causal diagram in text (arrows between variables), so I can draw it in dagitty or PowerPoint. 3. Recommend the minimal adjustment set. 4. Warn me about variables I must not adjust for (mediators and colliders).
RULES	Base classifications on clinical reasoning and state the assumption behind each, because adjusting for a mediator removes the effect I want to measure.
OUTPUT	A classification table, the diagram as a list of arrows, then the adjustment set.

PRO TIP A DAG in your methods chapter is a powerful answer to "why did you adjust for these variables?".

TRY IT WITH Exposure: smoking; outcome: post-operative pulmonary complications; variables: age, COPD, BMI, FEV1, type of surgery.

→ **NEXT #160** Bias and confounding audit of my design

171 Inter-observer agreement and quality-control plan

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When your measurements involve judgement: imaging, scales, clinical signs.

PERSONA	You are a measurement methodologist who designs reliability checks for clinical studies.
MY CONTEXT	Measurements needing judgement: [list] Observers available: [brief] Equipment needing calibration: [list]
TASK	1. Design an inter-observer (and, if relevant, intra-observer) agreement sub-study: sample, timing, blinding. 2. Recommend the statistic (Cohen's kappa, weighted kappa, ICC, Bland-Altman) for each measurement type. 3. Describe training and standardisation before data collection. 4. Describe equipment calibration and documentation. 5. Write the quality-control paragraph for my methods.
RULES	Match the agreement statistic to the variable type, because kappa for continuous data or correlation for agreement are common errors.
OUTPUT	Numbered plan, a statistic-choice table, then the paragraph.

PRO TIP Twenty to thirty paired readings are often enough for a thesis agreement sub-study. Confirm with your statistician.

TRY IT WITH Echocardiographic ejection fraction and a delirium scale, each measured by two residents.

→ **NEXT #158** Operational definitions for every variable

172 Translate and validate a questionnaire

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When your instrument must be used in Hindi or another Indian language.

PERSONA	You are a cross-cultural adaptation specialist who follows established translation and back-translation guidelines.
MY CONTEXT	Instrument: [name] Original language and target language: [brief] Is a validated version already available?: [yes / no / unsure] Licence status of the instrument: [free / licensed / unsure]
TASK	1. Tell me what to check first (existing validated translations, licensing, author permission). 2. Describe the adaptation steps: forward translation, synthesis, back-translation, expert committee, cognitive pre-testing. 3. Describe the psychometric checks for my sample (internal consistency, test-retest, construct validity). 4. Estimate the time this adds to my thesis.
RULES	Do not claim a validated translation exists; write [SEARCH FOR VALIDATED VERSION], because using an unvalidated translation undermines the whole outcome.
OUTPUT	A pre-check list, the step-by-step process, the psychometric plan, and the time estimate.

PRO TIP If a validated version exists, use it. Translation and validation can become a thesis of its own.

TRY IT WITH Adapting a sleep-quality questionnaire into Hindi for an OPD study.

→ **NEXT #158** Operational definitions for every variable

173 Study procedure as a patient journey

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When writing the "study procedure" section and designing your flowchart.

PERSONA	You are a clinical research coordinator who writes study procedures as clear, step-by-step patient journeys.
MY CONTEXT	Design: [design] Everything that happens to a participant, in rough order: [PASTE NOTES HERE] Time points: [list]
TASK	1. Write the procedure as numbered steps from screening to final follow-up, stating who does what, when and where. 2. Mark the points where consent, randomisation (if any), measurements and outcome assessment occur. 3. Describe a flowchart in text (boxes and arrows) that I can draw. 4. Flag any step that is ambiguous or unrealistic in my setting.
RULES	Use only the steps I described; where a necessary step is missing, write [ADD STEP], because the procedure must describe what actually happens.
OUTPUT	The numbered procedure, the flowchart description, then the flags.

PRO TIP Put the flowchart in your protocol. Committees understand a diagram faster than a page of text.

TRY IT WITH Screening ICU patients, consent from relatives, baseline labs, day-3 and day-7 scores, 28-day outcome by phone.

→ **NEXT #157** Write the full methodology chapter

174 Blinding when full blinding is impossible

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When your intervention cannot be hidden from patients or clinicians.

PERSONA	You are a trials methodologist who finds practical ways to reduce bias in open-label studies.
MY CONTEXT	Intervention and comparator: [brief] Outcome and who measures it: [brief] Staff available: [brief]
TASK	1. Identify who could still be blinded: outcome assessors, data analysts, adjudication committee. 2. Suggest objective outcomes or blinded assessment methods. 3. Describe how to document and report blinding (and its failure). 4. Write the blinding paragraph for my methods.
RULES	Do not describe a study as double-blind unless both participants and assessors are actually blinded; state exactly who was blinded, because "single-blind" and "double-blind" are ambiguous labels.
OUTPUT	Options table (who, how, feasibility), then the paragraph.

PRO TIP Blinding the statistician costs nothing. Code the groups as A and B until analysis is complete.

TRY IT WITH Early physiotherapy versus usual care in ventilated patients; outcome: ICU-acquired weakness.

→ **NEXT #164** Randomisation, allocation concealment and blinding

175 Before-after and quasi-experimental studies done right

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT / FACULTY

USE IT When you compare outcomes before and after a change in practice, without randomisation.

PERSONA	You are a health-services researcher experienced in quasi-experimental and interrupted time-series designs.
MY CONTEXT	The change or intervention and when it started: [brief] Outcome and how often it is measured: [brief] Data available before and after: [periods, numbers]
TASK	1. Explain the threats to validity (secular trends, regression to the mean, co-interventions, seasonal effects). 2. Recommend the strongest feasible design: simple before-after, a before-after with a control group, or interrupted time series. 3. Describe the data and analysis for the recommended design. 4. Write the methods paragraph.
RULES	Tell me clearly if my data cannot support an effect claim, because simple before-after studies usually cannot.
OUTPUT	A threats table, the recommendation, the analysis outline, then the paragraph.

PRO TIP Monthly data points (8 or more before and after) make an interrupted time series possible and far more convincing.

TRY IT WITH Introducing a sepsis bundle in the emergency department and its effect on mortality.

→ **NEXT #160** Bias and confounding audit of my design

176 Plan a pilot study for my thesis

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT Before main data collection, when tools, procedures or recruitment are untested.

PERSONA	You are a methodologist who designs pilot studies to test feasibility, not to estimate effects.
MY CONTEXT	Main study design and procedures: [brief] Uncertainties I want to test: [e.g. recruitment rate, proforma usability, test turnaround]
TASK	1. Define pilot objectives in feasibility terms (recruitment, retention, completeness, timing, acceptability). 2. Suggest a pilot size and duration with reasoning. 3. Set progression criteria (go, amend, stop) for each objective. 4. Explain whether pilot patients can be included in the main analysis.
RULES	Do not use pilot data to test hypotheses or report effects, because pilots are not powered for that and examiners know it.
OUTPUT	A pilot objectives table with progression criteria, then the size, duration and inclusion notes.

PRO TIP Report your pilot briefly in the methods. It strengthens your feasibility argument.

TRY IT WITH Pilot of a home-based pulmonary rehabilitation programme for 10 COPD patients.

→ **NEXT #159** Case record form (proforma) with a data-entry codebook

177 Data management plan for my thesis

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT Before data collection starts; many IECs now ask for it.

PERSONA	You are a research data manager who designs simple, secure data workflows for single-investigator studies.
MY CONTEXT	Data sources and format: [proforma / Google Forms / Excel / REDCap] Number of participants and variables: [brief] Who enters and who can access the data: [brief] Devices and storage available: [brief]
TASK	1. Describe data entry (timing, double entry or verification of a sample). 2. Describe cleaning checks and a correction log. 3. Describe storage, backup (3-2-1 rule), password protection and access control. 4. Describe de-identification and the separation of the linking log. 5. Describe retention after the thesis and data sharing.
RULES	Keep identifiable data off personal messaging apps and unencrypted drives, and say so explicitly, because that is where most thesis data breaches happen.
OUTPUT	A 5-section plan, then a one-paragraph summary for my IEC form.

PRO TIP Lock and date-stamp your final dataset before analysis. Analyse only the locked copy.

TRY IT WITH 150-patient cohort using a paper proforma and double entry into Excel.

→ **NEXT #196** Pre-specified statistical analysis plan

178 Laboratory sample handling and pre-analytical control

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When your study depends on blood, urine, fluid or tissue measurements.

PERSONA	You are a clinical pathologist who knows that most laboratory errors happen before the sample reaches the analyser.
MY CONTEXT	Samples and tests: [list] Where and when samples are collected: [brief] Lab processing and storage available: [brief]
TASK	1. Describe collection (tube type, volume, time of day, fasting, patient preparation). 2. Describe transport time, temperature and labelling (study ID only). 3. Describe processing, storage (temperature, freeze-thaw limits) and batch analysis if applicable. 4. Name the quality-control measures (internal QC, external QA participation). 5. Write the sample-handling paragraph for my methods.
RULES	Flag any test with known pre-analytical sensitivity (e.g. haemolysis, delayed separation) and mark specifics as [CONFIRM WITH LAB SOP], because local procedures override general advice.
OUTPUT	A step table, then the paragraph.

PRO TIP Agree the protocol with the lab in writing before you start. It prevents lost samples later.

TRY IT WITH Serum procalcitonin and lactate in suspected sepsis, emergency department collection.

→ **NEXT #158** Operational definitions for every variable

179 Standardised imaging acquisition and reading protocol

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT (RADIOLOGY AND CLINICAL)

USE IT When imaging findings are an exposure or outcome in your study.

PERSONA	You are a radiologist experienced in research imaging protocols and structured reporting.
MY CONTEXT	Modality and body part: [brief] Findings to be recorded: [list] Who acquires and who reads the images: [brief]
TASK	1. Standardise acquisition: equipment, settings, patient positioning and timing. 2. Define each finding with explicit criteria, and a structured reporting form. 3. Plan blinding of readers to clinical data and outcomes. 4. Plan a reliability check between two readers. 5. Write the imaging-methods paragraph.
RULES	Use named, recognised scoring systems where they exist and mark any you are unsure of as [VERIFY SCORE AND REFERENCE], because informal scoring is not reproducible.
OUTPUT	Protocol sections, a structured reporting-form outline, then the paragraph.

PRO TIP Store images with the study ID so re-reading is possible if an examiner or reviewer asks.

TRY IT WITH Lung ultrasound scoring in ARDS by two residents.

→ **NEXT #171** Inter-observer agreement and quality-control plan

180 Which reporting guideline applies to my study?

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT / MBBS STUDENT

USE IT At protocol stage, so you design and later write the study to the right checklist.

PERSONA	You are an EQUATOR Network-trained editor who matches studies to reporting guidelines.
MY CONTEXT	Design and topic: [brief] Planned outputs: [thesis / journal article / conference abstract]
TASK	1. Name the main reporting guideline (e.g. CONSORT, STROBE, STARD, TRIPOD, PRISMA, CARE, SQUIRE, COREQ or SRQR) and any extension that fits. 2. Explain how using it at protocol stage improves the study. 3. List the 5 checklist items residents most often fail to report for this design. 4. Say where to download the checklist.
RULES	Name only established guidelines and extensions; if you are unsure an extension exists, write [CHECK EQUATOR WEBSITE], because journals require the correct checklist at submission.
OUTPUT	The guideline and extension, the benefit note, the 5 common gaps, then the source.

PRO TIP Most journals require the completed checklist at submission. Fill it as you write, not afterwards.

TRY IT WITH A prognostic model predicting ICU mortality from admission variables.

→ **NEXT #157** Write the full methodology chapter

181 Describe the intervention so others can replicate it (TIDieR)

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT When your study delivers a treatment, exercise, education or other intervention.

PERSONA	You are a trial reporting expert who uses the TIDieR checklist to describe interventions completely.
MY CONTEXT	Intervention details as I have them: [PASTE NOTES HERE] Comparator or control: [brief]
TASK	1. Organise the description under TIDieR items: name, why, materials, procedures, who provides, how, where, when and how much, tailoring, modifications, and how adherence is assessed. 2. Flag items I have not described. 3. Write the intervention paragraph for my methods. 4. Describe the comparator to the same level of detail.
RULES	Use only details I gave; mark gaps as [DESCRIBE], because an intervention that cannot be replicated cannot be evaluated.
OUTPUT	A TIDieR table, the flags, then the paragraphs.

PRO TIP "Usual care" must also be described. Reviewers always ask what it involved.

TRY IT WITH A structured inhaler-technique education session for asthma patients in OPD.

→ **NEXT #173** Study procedure as a patient journey

182 Designing studies in patients who cannot consent

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT (ICU, EMERGENCY, PAEDIATRICS, PSYCHIATRY)

USE IT When your participants are unconscious, critically ill, children, or otherwise unable to consent for themselves.

PERSONA	You are an ethics-committee member and critical-care researcher experienced with consent from legally acceptable representatives.
MY CONTEXT	Population and why they cannot consent: [brief] Time window between eligibility and the first study procedure: [e.g. within 1 hour of admission] Risk level of the study: [observational / minimal-risk intervention / greater than minimal risk]
TASK	1. Describe consent options: legally acceptable representative (LAR), assent for children, and re-consent when the patient regains capacity. 2. Explain when deferred or waived consent may be considered, and what the IEC will need to see. 3. Adjust the recruitment procedure so that consent never delays clinical care. 4. Draft the consent section of my methods.
RULES	Refer to the current ICMR National Ethical Guidelines for specifics and mark every rule you state as [VERIFY IN ICMR GUIDELINES], because consent provisions for vulnerable participants are strictly interpreted by IECs.
OUTPUT	An options table (situation, consent route, IEC requirement), the recruitment adjustment, then the consent paragraph.

PRO TIP Discuss the consent route with your IEC member-secretary before submission. It prevents a full revision cycle.

TRY IT WITH Observational study of delirium in mechanically ventilated adults.

→ **NEXT #157** Write the full methodology chapter

183 Convert my protocol into thesis methods

STANDARD

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

USE IT After data collection, when writing the final thesis.

PERSONA	You are an academic editor who converts approved protocols into accurate thesis methodology chapters.
MY CONTEXT	Approved protocol methods: [PASTE PROTOCOL METHODS HERE] What actually happened differently (deviations): [list, or "none"] Final numbers (screened, enrolled, analysed): [numbers]
TASK	1. Convert the text to past tense and describe what was done, not what was planned. 2. Insert the deviations in the right places, with reasons. 3. Add the actual study period and final numbers. 4. List any protocol element that should now be removed because it didn't happen.
RULES	Report deviations openly, because hidden deviations discovered by an examiner are far more damaging than declared ones. Do not change numbers I gave.
OUTPUT	The converted methods text, then a deviation table (planned, actual, reason).

PRO TIP Examiners often compare the approved synopsis with the thesis. Make the differences explicit.

TRY IT WITH Protocol planned 120 patients over 18 months; 104 enrolled over 20 months due to COVID disruption.

→ **NEXT #157** Write the full methodology chapter

QUICK PROMPTS · 1.6

184 Name my study design correctly

QUICK

1.6 STUDY DESIGN & METHODOLOGY · MBBS STUDENT / PG RESIDENT

Act as an epidemiology examiner for postgraduate theses. From this description of what I will do, [DESCRIBE STUDY], name the exact study design (e.g. prospective cohort, hospital-based case-control), give the label to use in my title, and explain in 2 lines how you decided.

TRY IT WITH I will enrol all dengue patients this year and record who develops shock by day 7.

185 Prospective or retrospective: what's the difference for me?

QUICK

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

Act as a research methods tutor. Explain the difference between prospective and retrospective data collection using my study, [BRIEF STUDY DESCRIPTION], as the example. Give 3 advantages and 3 disadvantages of each for my topic, and tell me which fits better.

TRY IT WITH Outcomes of patients admitted with acute pancreatitis.

186 Strengths and weaknesses of my design

QUICK

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

Act as a viva examiner for MD/MS students. List 5 strengths and 5 weaknesses of a [DESIGN] for my topic [TOPIC], specific to my setting rather than textbook-general, and give a one-line answer to "why didn't you do an RCT?".

TRY IT WITH Hospital-based case-control study of risk factors for neonatal sepsis.

187 Operationally define one variable

QUICK

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

Act as a clinical research methodologist. Write a bedside-ready operational definition for [VARIABLE] in my study on [TOPIC]: what is measured, instrument, unit, cut-off or categories, time point and who measures it. Mark any criterion needing a citation as [CITE].

TRY IT WITH Variable: "controlled hypertension" in a diabetes clinic study.

188 Draw a simple random sample in Excel

QUICK

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT / MBBS STUDENT

Act as a biostatistics tutor. Give step-by-step instructions to draw a simple random sample of [n] from a list of [N] participants in Excel or Google Sheets using RAND() or RANDBETWEEN(), including how to freeze the values and document the process for my methods.

TRY IT WITH Selecting 150 households from a village list of 1,200.

189 Allocation concealment in my trial

QUICK

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

Act as a trials methodologist. Explain the practical difference between randomisation and allocation concealment using my trial, [BRIEF TRIAL DESCRIPTION], and describe an affordable, practical way to conceal allocation in an Indian government hospital setting in under 120 plain words.

TRY IT WITH Two dressing types for diabetic foot ulcers in a surgery ward.

190 Can I apply this criterion at the bedside?

QUICK

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

Act as a research coordinator. Review these criteria, [PASTE CRITERIA], and mark each as testable at screening or not. For each untestable one, rewrite it with a named test, threshold and time window so any resident would apply it identically.

TRY IT WITH "Patients with poor nutritional status", "patients unlikely to survive".

191 List the likely confounders

QUICK

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

Act as an epidemiologist. For the association between [EXPOSURE] and [OUTCOME] in [POPULATION], list the 6 most likely confounders, explaining each one's link to both exposure and outcome in one line, and mark which ones are usually available in routine hospital records.

TRY IT WITH Exposure: anaemia; outcome: post-operative complications after hip fracture surgery.

192 Cohort or case-control? Decide now

QUICK

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT / MBBS STUDENT

Act as a methodology tutor. For my question, [QUESTION], compare a cohort and a case-control design in a 4-row table (time, cost, outcome rarity, bias) and recommend one, with the reason, for a 2-year thesis with about [N] eligible patients per month.

TRY IT WITH Does prior statin use affect severity of community-acquired pneumonia?

193 Could observer bias affect my results?

QUICK

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

Act as a bias reviewer. In my study, [BRIEF DESCRIPTION OF WHO MEASURES WHAT], tell me whether observer, Hawthorne or recall bias is likely, which direction each could push the results, and one practical fix for each. Keep it under 120 words.

TRY IT WITH I assess patients' pain scores myself, knowing which analgesic they received.

194 Write the "study setting" paragraph

QUICK

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

Act as a methods editor. Write a 70–90 word study-setting paragraph from these details, [HOSPITAL TYPE, DEPARTMENT, BEDS, CATCHMENT, PATIENT LOAD], in formal past tense for my thesis. Keep every number exactly as I gave it and mark missing essentials as [ADD].

TRY IT WITH 1,200-bed government tertiary hospital, 40-bed medicine department, north Indian catchment.

195 My design in one line for the title page

QUICK

1.6 STUDY DESIGN & METHODOLOGY · PG RESIDENT

Act as a thesis committee member. Turn my design details, [DESIGN, SETTING, TIME FRAME], into a one-line subtitle for my thesis title page, e.g. "A prospective observational study at a tertiary care centre in north India". Give 3 options, and mark the one best suited to an Indian university format.

TRY IT WITH Hospital-based prospective cohort, 18 months, AIIMS-level tertiary centre.

1.7

Sample size & statistics

Deciding your test after seeing your data is how p-hacking happens by accident. Plan the analysis before data collection, and verify every number AI gives you.

45

prompts

6

Master

26

Standard

13

Quick

HOW TO USE THIS SECTION

Use it before data collection to fix the sample size and analysis plan, and again when you analyse.

Start with the Master prompts (#196–201) for the full workflow; use the Standard prompts (#202–227) for one task at a time; reach for the Quick prompts (#228–240) when you have two minutes. Tap any title below to jump to it, and follow → NEXT to move to the next step.

#196 Pre-specified statistical analysis plan

#197 Sample size with every assumption shown

#198 Run my analysis on the real dataset, safely

#199 Multivariable regression: plan, run and interpret

#200 Interpret my software output correctly

#201 Statistics viva drill for my thesis defence

#202 Sample size to estimate a proportion (prevalence)

#203 Sample size to compare two means

#204 Sample size to compare two proportions

#205 Sample size for a diagnostic accuracy study

#206 Sample size for a correlation study

#207 Sample size for a case-control study

#208 Is my sample big enough for regression?

#209 Which statistical test? A decision table

#210 Is my data normally distributed? Decide sensibly

#211 Choose and report the right non-parametric test

#212 Chi-square, Fisher's exact or something else?

#213 Before-after or matched data: paired analysis

#214 Outcomes measured three or more times

#215 Correlation or agreement? Methods comparison done right

#216 ROC curve, AUC and the best cut-off

#217 Time-to-event analysis: Kaplan-Meier and Cox

#218 Handle missing data honestly

#219 Data-cleaning checklist before analysis

#220 Step-by-step SPSS instructions for my analysis

#221 R code for my thesis analysis

#222 Free point-and-click analysis in Jamovi

#223 Effect sizes and confidence intervals for every result

#224 Multiple comparisons: correct or not?

#225 Subgroup and sensitivity analyses that hold up

#226 Report p values and CIs the way journals want

#227 Quick analysis inside Google Sheets with AI add-ons

#228 Classify my variables correctly

#229 Mean or median for this variable?

#230 What does my p value actually mean?

#231 Interpret my confidence interval

#232 Odds ratio or relative risk for my study?

#233 Explain statistical power with my study

#234 My result isn't significant: now what?

#235 Should I test baseline differences in my trial?

#236 Interpret my kappa value

#237 Which assumptions does my test need?

#238 Sanity-check an AI-generated sample size

#239 Explain any statistics term for the viva

#240 Build Table 1 from my variables

Pre-specified statistical analysis plan

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT

Write the plan before data collection. Run the analysis after the database is locked.

WHY

A pre-specified statistical analysis plan protects your integrity and your viva. It forces every test to be chosen for a reason, before the data can tempt you. It also becomes the analysis section of your thesis.

COPY THE PROMPT BELOW

PERSONA	You are a biostatistician with two decades of experience in clinical research who routinely explains statistical reasoning to clinicians. You are rigorous about assumptions and openly critical of common misuse, such as testing for normality in tiny samples or reporting p values without effect sizes.
MY CONTEXT	Design: [prospective cohort / RCT / cross-sectional / diagnostic accuracy / before-after] Total sample size, and per group if applicable: [N; n1, n2] Primary objective: [state] Secondary objectives: [list] Software: [SPSS / R / Stata / GraphPad / Jamovi / Python] My statistical comfort level: [beginner / intermediate / advanced] My variables, each with name, type and role (example format: "FEV1 % predicted, continuous, baseline and 12 weeks, primary outcome"): [LIST ALL VARIABLES]
TASK	1. Variable dictionary: type, expected distribution, unit, and how to summarise it in Table 1. 2. Data cleaning: implausible-value checks, range limits, duplicate detection, a documented correction log. 3. Missing data: how to quantify it, likely mechanisms, the threshold at which complete-case analysis becomes unsafe, and a practical alternative at my sample size. 4. Assumption checking for each planned comparison, and what to do when an assumption fails. 5. Test selection: a table mapping each objective to the exact test, the reason, and the non-parametric alternative. 6. Effect measures with confidence intervals for every comparison. 7. Adjusted analysis: covariates and their justification, how many my events or sample can support (events per variable), and a multicollinearity check. 8. Subgroup and sensitivity analyses, labelled exploratory. 9. Multiplicity: whether correction is needed, and the method. 10. Runnable, commented syntax in my software for every analysis. 11. Reporting templates: decimals, p values (including very small ones) and CI placement. 12. Table 1, the main results table, and two figures that best show the primary finding. 13. The five statistical mistakes most likely in this design, and how to avoid each.
RULES	Use clearly marked placeholders instead of any numeric result, because this is a plan, not an analysis. If my sample is too small for something I asked for, say so and tell me what I can legitimately claim instead. If my design cannot support a causal claim, give the correct associational phrasing. Explain every recommendation in plain clinical language first, then in statistical terms.
OUTPUT	Thirteen numbered sections, with tables for the variable dictionary and the test map, and code blocks for syntax.
TWEAK	End with the 5 criticisms a reviewer would make of this plan, and how to pre-empt each.

How to get more out of it

- Lock and date this plan before you unblind or begin analysis. Attach it to your thesis appendix.
- Run the syntax on 10 dummy rows first. Errors are easier to fix before the real data arrives.
- Never let the model interpret your actual results without checking the raw output yourself.
- → **NEXT #198** Run my analysis on the real dataset, safely

Sample size with every assumption shown

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT

While writing the protocol, before IEC submission.

WHY

The sample size is the most-questioned number in any synopsis. AI calculators make arithmetic errors, and the inputs matter more than the formula. This prompt shows every assumption and step so you can defend it and verify it in OpenEpi or G*Power.

COPY THE PROMPT BELOW

PERSONA

You are a biostatistician who has calculated and defended sample sizes for hundreds of postgraduate protocols. You show every formula, every input and every step, and you say where each input must come from.

MY CONTEXT

Design: [cross-sectional / case-control / cohort / RCT / diagnostic accuracy / correlation]
 Primary outcome and its type: [proportion / mean / correlation / sensitivity / time-to-event]
 Primary objective or hypothesis: [paste]
 Expected value(s) from literature or pilot data, with source: [e.g. prevalence 18% (Author, year); means 42 ± 9 vs 36 ± 9]
 Precision or effect size I want to detect: [e.g. absolute margin of error 5%; difference of 6 units]
 Alpha, power and sidedness: [e.g. 0.05, 80%, two-sided]
 Expected dropout or non-response: [e.g. 10%]
 Group ratio if comparing groups: [e.g. 1:1]
 Eligible patients per month and months available: [N, X]

TASK

1. Name the correct formula for my design and outcome, and explain why it fits.
2. Define every term in the formula.
3. Substitute my inputs and show the step-by-step calculation.
4. Apply the dropout adjustment and round up correctly.
5. Check feasibility: the months needed at my recruitment rate.
6. Show a sensitivity table: how the sample size changes if the key assumption is 20% better or worse.
7. Tell me exactly which inputs to enter in OpenEpi or G*Power to verify the result.
8. Write the sample-size paragraph for my protocol, including the source of each input.

RULES

Mark every input I did not supply as [INSERT FROM PUBLISHED SOURCE — VERIFY], because an assumption without a source is the first thing a committee rejects. Show your arithmetic so errors are visible. Do not choose an effect size to make the sample fit my patient numbers; if the study isn't feasible, say so.

OUTPUT

The formula and terms, the worked calculation, the feasibility check, the sensitivity table, the verification instructions, then the paragraph.

TWEAK

End with the 3 committee questions about this sample size and short answers.

How to get more out of it

- › Never submit the AI number alone. Verify it in OpenEpi or G*Power and report the verified figure.
- › The expected value must come from a real study similar to yours, cited in your protocol.
- › If the number is unfeasible, change the design or outcome. Don't massage the effect size.
- › → NEXT #202 Sample size to estimate a proportion (prevalence)

Run my analysis on the real dataset, safely

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT

After your dataset is cleaned and locked, in ChatGPT's data analysis mode, Julius AI or a similar tool that runs code.

WHY

AI data tools can run a full analysis in minutes, and can also silently mis-code a variable or pick the wrong test. This prompt makes the tool follow your pre-specified plan, show its code and checks, and flag anything uncertain, so you can verify every number.

COPY THE PROMPT BELOW

PERSONA	You are a careful biostatistician analysing a clinical thesis dataset. You follow the pre-specified plan exactly, show every step and every assumption check, and never proceed on a guess about coding.
MY CONTEXT	Uploaded file: [file name] with [N] rows and [K] columns Codebook (variable, meaning, codes, units): [PASTE CODEBOOK HERE] My pre-specified analysis plan: [PASTE SAP OR THE KEY TESTS HERE] Primary outcome and exposure: [brief]
TASK	1. Load the data and report rows, columns, variable types, missing values per variable, and any values outside the codebook's valid range. Stop and ask me before fixing anything. 2. Produce Table 1 (baseline characteristics) by group, with the summary statistic appropriate to each variable. 3. Run the primary analysis exactly as planned: show the assumption checks, the test used, the effect estimate with 95% CI and the exact p value. 4. Run the secondary analyses in the plan, labelled as such. 5. Produce the figures specified in the plan. 6. Show all code used, so I can re-run it. 7. List every decision you had to make that the plan did not specify.
RULES	Do not recode, drop or impute any data without asking me first, because silent data changes are the most dangerous AI analysis error. Follow my plan; if a planned test is inappropriate for the data, stop and explain rather than switching tests. Report numbers exactly as computed, without rounding before the final table.
OUTPUT	A data-check report, then Table 1, then results for each objective (test, estimate, CI, p), then the figures, the code and the decision log.
TWEAK	Finish with the 3 numbers I should verify by hand or in a second tool before I trust the analysis.

How to get more out of it

- › Upload de-identified data only: study IDs, no names, no hospital numbers.
- › Re-run the key analysis in SPSS, Jamovi or R. If the numbers differ, find out why before writing.
- › Save the code and outputs with the date. They are your audit trail.
- › → **NEXT #200** Interpret my software output correctly

Multivariable regression: plan, run and interpret

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT / FACULTY

USE IT

When your objective involves adjusting for confounders or finding independent predictors.

WHY

Regression is where theses most often overreach: too many variables for the events, stepwise selection, and adjusted odds ratios interpreted as causes. This prompt keeps the model honest and interpretable.

COPY THE PROMPT BELOW

PERSONA	You are a biostatistician who teaches regression modelling to clinicians and reviews models for medical journals. You care about events per variable, sensible variable selection and plain-language interpretation.
MY CONTEXT	Outcome and its type: [binary / continuous / time-to-event / count] Exposure of interest, if any: [variable] Candidate covariates, with why each matters: [list] Sample size and number of outcome events: [N, events] Purpose of the model: [adjust for confounding / identify predictors / build a prediction score] Software: [SPSS / R / Stata / Jamovi]
TASK	1. Choose the right model (linear, logistic, Cox, Poisson or negative binomial) and justify it. 2. Calculate the number of parameters my events can support, and prioritise covariates accordingly. 3. Recommend how to select variables for my purpose (clinical reasoning or a causal diagram, not automatic stepwise), and how to handle continuous variables. 4. List the assumption checks and diagnostics for my model (linearity, proportional hazards, multicollinearity, influential points, goodness of fit). 5. Give runnable, commented syntax. 6. Show how to present results: a table of crude and adjusted estimates with 95% CIs, and a plain-language interpretation of each. 7. Explain what the model can and cannot conclude.
RULES	Warn me clearly if I have too few events for my covariates, because an overfitted model produces unstable estimates that examiners and reviewers will question. Interpret adjusted estimates as associations unless the design is randomised. Use placeholders instead of invented results.
OUTPUT	Numbered sections, a covariate-priority table, code blocks, and a results-table template.
TWEAK	Finish with the 3 regression mistakes a viva examiner looks for, and how my plan avoids them.

How to get more out of it

- › Report crude and adjusted estimates side by side. Readers need to see what adjustment changed.
- › Avoid "independent predictor" language unless your purpose really is prediction, and validate if it is.
- › If the events are few, a simpler model with fewer, well-chosen covariates is more credible.
- › → **NEXT #200** Interpret my software output correctly

Interpret my software output correctly

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT

When SPSS, R, Jamovi or Stata output is in front of you and you need to understand and report it.

WHY

Statistical output is full of numbers you don't need, and a few you absolutely do. Misreading which p value or estimate matters is common and costly. This prompt identifies the right numbers, explains them in clinical language and writes the result sentence.

COPY THE PROMPT BELOW

PERSONA	You are a biostatistics tutor who explains software output line by line to clinicians and writes results exactly as journals expect.
MY CONTEXT	Software and test run: [e.g. SPSS independent-samples t-test] My objective for this analysis: [paste] Variable names and what they mean: [brief] Sample size I expected in this analysis: [n per group] What I expected to find, before running it: [brief] The output (copy as text or upload a screenshot): [PASTE OUTPUT HERE]
TASK	1. Identify which parts of the output answer my objective, and which to ignore. 2. Check that the right test was run for my variables, and that its assumptions appear to be met (e.g. Levene's test, expected counts). 3. Extract the key numbers: group summaries, effect estimate, 95% CI, test statistic, degrees of freedom and exact p value. 4. Explain the result in plain clinical language, then in statistical language. 5. Write the result sentence for my results section in journal style. 6. Flag anything suspicious: impossible values, suspiciously large effects, wrong n, or a mismatch with my expectations. 7. Say whether the result answers my objective fully, or whether another analysis is needed.
RULES	Use only the numbers in the output I pasted, reproduced exactly, because transcription errors in results are indistinguishable from fabrication. If the output is from the wrong test, say so and name the right one rather than interpreting it.
OUTPUT	A short "what to read" guide, the extracted-numbers table, the two explanations, the result sentence, then the flags.
TWEAK	End with the one follow-up analysis or check this output suggests I should run.

How to get more out of it

- › Always check that n in the output matches the n you expect. Missing data silently drops cases.
- › Paste the output as text where possible. Screenshots can be misread.
- › Keep a results log: output file, date, and the sentence you wrote from it.
- › → NEXT #226 Report p values and CIs the way journals want

Statistics viva drill for my thesis defence

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT (FINAL YEAR)

USE IT

In the weeks before your thesis defence or final viva.

WHY

Examiners love statistics questions because most residents can't explain why they used a test. A drill based on your own thesis, one question at a time, with feedback, turns "my statistician did it" into confident answers.

COPY THE PROMPT BELOW

PERSONA	You are a strict but fair biostatistics examiner on an MD/MS thesis viva panel. You ask questions one at a time, starting simple and escalating, and you give frank feedback after each answer.
MY CONTEXT	Thesis title and design: [paste] Sample size and how it was calculated: [brief] Tests used for each objective: [list] Key results (estimates, CIs, p values): [paste] Limitations I already know about: [list] My weakest statistics topics, in my own view: [list]
TASK	1. Ask me one statistics question at a time about my own thesis, starting with basics (variable types, why this test) and escalating (assumptions, power, confounding, multiplicity, clinical versus statistical significance). 2. After each of my answers: score it out of 10, give the ideal answer in 3–5 points, and name the gap in my reasoning. 3. Include at least 2 "trap" questions examiners commonly use (e.g. "what does $p = 0.04$ mean?", "was your study powered for this?"). 4. After 12 questions, summarise my weak areas and give a 3-day revision plan targeted at them, with one resource type for each (textbook chapter, video, practice set).
RULES	Ask only one question per turn and wait for my answer, because the value is in answering under pressure. Base questions on my actual thesis details; where you need a detail I have not given, ask for it instead of assuming.
OUTPUT	Conversational: one question per turn, feedback in a short structured block, then the summary and plan at the end.
TWEAK	Finish with the 5 statistical statements from my thesis I should be able to explain word for word.

How to get more out of it

- › Answer aloud before typing. The viva is spoken, not written.
- › Re-run the drill after revising your weak areas and compare scores.
- › Ask a friend to read the questions to you for a realistic rehearsal.
- › → **NEXT #239** Explain any statistics term for the viva

202 Sample size to estimate a proportion (prevalence)

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT / MBBS STUDENT

USE IT For descriptive studies estimating a prevalence or proportion with a chosen precision.

PERSONA	You are a biostatistician who teaches precision-based sample-size calculation.
MY CONTEXT	Expected proportion (with source): [e.g. 30% (Author, year)] Desired precision: [absolute margin, e.g. ±5%, or relative, e.g. 20% of p] Confidence level: [95%] Population size, if finite and small: [N or "large"] Expected non-response: [e.g. 10%]
TASK	1. Give the formula $n = Z^2 \cdot p(1-p)/d^2$ and define each term. 2. Calculate step by step with my inputs, and apply the finite-population correction if relevant. 3. Adjust for non-response and round up. 4. Show how n changes with d = 3%, 5% and 7%. 5. Write the protocol sentence and list the OpenEpi inputs for verification.
RULES	Make clear whether the precision is absolute or relative, because confusing them changes n several-fold. Mark any unsourced input as [VERIFY FROM PUBLISHED SOURCE].
OUTPUT	The formula, the working, a small sensitivity table, then the sentence and verification inputs.

PRO TIP If no prior estimate exists, $p = 50\%$ gives the maximum (most conservative) sample size. Justify it in the protocol.

TRY IT WITH Prevalence of depression in haemodialysis patients, expected 40%, ±7% absolute precision.

→ **NEXT #197** Sample size with every assumption shown

203 Sample size to compare two means

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When your primary outcome is continuous and you compare two groups.

PERSONA	You are a biostatistician who explains power-based sample sizes clearly.
MY CONTEXT	Expected mean and SD in each group (with source): [e.g. 45 ± 10 vs 38 ± 10] Minimum clinically important difference: [value and why] Alpha, power and sidedness: [0.05, 80%, two-sided] Group ratio and dropout: [1:1, 10%]
TASK	1. State the formula for comparing two independent means and define each term. 2. Calculate step by step, adjust for dropout and round up. 3. Show the effect of a smaller difference or larger SD in a sensitivity table. 4. Give the G*Power settings to verify. 5. Write the protocol sentence.
RULES	Base the difference on clinical importance, not on what makes the sample small, because committees ask "why this difference?". Mark unsourced SDs as [VERIFY FROM PUBLISHED SOURCE].
OUTPUT	Formula, working, sensitivity table, verification settings, sentence.

PRO TIP If you measure the outcome before and after in the same patients, ask for the paired calculation instead. It usually needs fewer patients.

TRY IT WITH Mean post-operative pain score (VAS) 4.5 ± 1.8 vs 3.5 ± 1.8 with two analgesic regimens.

→ **NEXT #197** Sample size with every assumption shown

204 Sample size to compare two proportions

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When your primary outcome is binary and you compare two groups (trial or cohort).

PERSONA	You are a biostatistician who calculates sample sizes for trials and cohorts.
MY CONTEXT	Expected proportion in each group (with source): [e.g. 30% vs 15%] Alpha, power and sidedness: [0.05, 80%, two-sided] Group ratio and dropout: [e.g. 1:1, 10%] Continuity correction wanted?: [yes / no / unsure]
TASK	1. State the formula for two independent proportions and define each term. 2. Calculate step by step, with and without continuity correction if relevant. 3. Adjust for dropout and round up. 4. Show a sensitivity table for smaller differences. 5. Give the OpenEpi or G*Power inputs and the protocol sentence.
RULES	Express the expected difference as both absolute and relative change, because committees look for a clinically meaningful difference. Mark unsourced proportions as [VERIFY FROM PUBLISHED SOURCE].
OUTPUT	Formula, working, sensitivity table, verification inputs, sentence.

PRO TIP Small absolute differences in rare outcomes need very large samples. Check feasibility before you commit.

TRY IT WITH Surgical-site infection 12% with standard care versus 5% with a new protocol.

→ **NEXT #197** Sample size with every assumption shown

205 Sample size for a diagnostic accuracy study

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When your primary objective is the sensitivity and/or specificity of a test.

PERSONA	You are a diagnostic-research statistician familiar with Buderer's method.
MY CONTEXT	Expected sensitivity and specificity (with source): [e.g. Se 85%, Sp 80%] Desired precision (half-width of 95% CI): [e.g. ±7%] Prevalence of the target condition in my study population: [e.g. 30%] Dropout or indeterminate results: [e.g. 5%]
TASK	1. Calculate the number of diseased participants needed for sensitivity and the number of non-diseased needed for specificity. 2. Convert each to a total sample size using the prevalence. 3. Take the larger total, adjust for losses and round up. 4. Explain why prevalence matters so much. 5. Write the protocol sentence and the verification approach.
RULES	Show both calculations, because many theses calculate only for sensitivity and are underpowered for specificity. Mark unsourced inputs as [VERIFY FROM PUBLISHED SOURCE].
OUTPUT	Both calculations, the total, the prevalence explanation, then the sentence.

PRO TIP If prevalence in your population is low, consider enriching with known cases, and report it as a limitation.

TRY IT WITH Lung ultrasound for pneumonia: expected Se 90%, Sp 85%, prevalence 40%, precision ±7%.

→ **NEXT #168** Design a diagnostic accuracy study (STARD)

206 Sample size for a correlation study

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When your primary objective is a correlation between two continuous variables.

PERSONA	You are a biostatistician who calculates sample sizes for correlation and agreement studies.
MY CONTEXT	Expected correlation coefficient (with source): [e.g. $r = 0.35$] Alpha, power and sidedness: [0.05, 80%, two-sided] Dropout: [e.g. 10%]
TASK	1. Explain the Fisher z-transformation approach in plain language. 2. Calculate the sample size step by step. 3. Adjust for dropout and round up. 4. Show n for $r = 0.2, 0.3, 0.4$ and 0.5 . 5. Warn if my real question is agreement (e.g. two methods measuring the same thing), which needs a different approach.
RULES	Mark unsourced expected correlations as [VERIFY FROM PUBLISHED SOURCE], because correlation strength drives the sample size dramatically.
OUTPUT	Explanation, working, sensitivity table, then the agreement warning if relevant.

PRO TIP A statistically significant weak correlation ($r = 0.2$) is rarely clinically useful. Plan for a meaningful r .

TRY IT WITH Correlation between serum ferritin and HbA1c in type 2 diabetes, expected $r = 0.3$.

→ **NEXT #215** Correlation or agreement? Methods comparison done right

207 Sample size for a case-control study

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When your design compares cases and controls for an exposure.

PERSONA	You are an epidemiologist who calculates case-control sample sizes, including matched designs.
MY CONTEXT	Expected exposure prevalence in controls (with source): [e.g. 20%] Odds ratio worth detecting: [e.g. 2.0] Controls per case: [1 / 2 / 3] Matched or unmatched: [brief] Alpha, power: [0.05, 80%]
TASK	1. Calculate the number of cases and controls step by step for an unmatched design. 2. Explain how matching and more controls per case change the requirement. 3. Show how n changes for OR = 1.5, 2.0 and 2.5. 4. Give OpenEpi inputs to verify. 5. Write the protocol sentence.
RULES	Explain why going beyond 4 controls per case gives little additional power. Mark unsourced inputs as [VERIFY FROM PUBLISHED SOURCE].
OUTPUT	Working, a sensitivity table, verification inputs, then the sentence.

PRO TIP If cases are scarce, adding controls per case is the cheapest way to gain power.

TRY IT WITH Exposure: indoor biomass fuel use; outcome: COPD in women; OR 2.0; control exposure 30%.

→ **NEXT #165** Choosing and matching controls in a case-control study

208 Is my sample big enough for regression?

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When you plan a multivariable model and need to know how many variables it can support.

PERSONA	You are a biostatistician who reviews regression plans for overfitting.
MY CONTEXT	Outcome type and expected number of outcome events (or N for continuous outcomes): [brief] Candidate predictors, and how many parameters each uses (categorical variables use several): [list] Purpose: [adjustment / prediction]
TASK	1. Count the parameters my candidate list uses. 2. Apply events-per-variable reasoning and explain why the 10-per-variable rule is only a rough guide. 3. Tell me the maximum parameters my study can support. 4. Suggest how to prioritise or combine variables. 5. Mention formal approaches for prediction models, and mark their details [CONFIRM WITH STATISTICIAN].
RULES	Count events, not total patients, for binary and time-to-event outcomes, because that is the most common error in thesis regression plans.
OUTPUT	A parameter count table, the calculation, then the recommendations.

PRO TIP A model with 3 well-chosen variables is more credible than one with 12 stepwise ones.

TRY IT WITH 30-day mortality with 45 deaths among 300 patients, 11 candidate predictors.

→ **NEXT #199** Multivariable regression: plan, run and interpret

209 Which statistical test? A decision table

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT / MBBS STUDENT

USE IT When you are unsure which test fits each objective.

PERSONA	You are a biostatistics teacher known for clear, correct test-selection advice.
MY CONTEXT	My objectives: [PASTE OBJECTIVES HERE] For each: outcome variable and type, comparison groups (number, independent or paired), sample size: [brief]
TASK	1. For each objective, name the parametric test, the non-parametric alternative, and when to use which. 2. Name the effect measure to report with each test. 3. Name the assumptions to check. 4. Put everything in one decision table.
RULES	Base the parametric versus non-parametric choice on the variable's distribution and sample size, not on a normality test alone, because normality tests are unreliable in small and very large samples.
OUTPUT	A decision table (objective, outcome type, groups, test, alternative, effect measure, assumptions).

PRO TIP Paste this table into your analysis plan and quote it in your viva when asked "why this test?".

TRY IT WITH Objectives comparing CRP (skewed) between 3 groups and mortality (yes/no) between 2 groups.

→ **NEXT #196** Pre-specified statistical analysis plan

210 Is my data normally distributed? Decide sensibly

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT Before choosing between parametric and non-parametric methods.

PERSONA	You are a biostatistician who teaches sensible distribution checks rather than blind reliance on normality tests.
MY CONTEXT	Variable and what it measures: [brief] Sample size per group: [n] Summary statistics (mean, SD, median, IQR, min, max): [paste] Histogram or Q-Q plot description, or the Shapiro-Wilk result: [paste]
TASK	1. Assess the distribution from the summaries (e.g. SD large relative to the mean for a positive-only variable suggests skew) and the plot. 2. Explain what the normality test does and does not tell me at my sample size. 3. Recommend how to summarise the variable (mean $\pm$ SD or median with IQR) and which test family to use. 4. Mention transformation (e.g. log) as an option, with its trade-offs.
RULES	Combine clinical knowledge of the variable with the data, because a normality test alone is underpowered in small samples and oversensitive in large ones.
OUTPUT	The assessment, the explanation, the recommendation, then the transformation note.

PRO TIP Hospital-stay length, CRP and most lab counts are usually skewed. Expect medians.

TRY IT WITH ICU length of stay: mean 6.2, SD 5.8, median 4, IQR 3–8, n = 90.

→ **NEXT #211** Choose and report the right non-parametric test

211 Choose and report the right non-parametric test

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When your data is skewed or ordinal and parametric tests don't fit.

PERSONA	You are a biostatistician who explains non-parametric methods in plain language.
MY CONTEXT	Outcome and its type: [skewed continuous / ordinal] Groups and whether independent or paired: [brief] Sample sizes: [n per group]
TASK	1. Name the right test (Mann-Whitney U, Wilcoxon signed-rank, Kruskal-Wallis, Friedman) and why. 2. Explain what it actually tests in plain language. 3. Tell me which summary and effect measure to report (median, IQR, Hodges-Lehmann estimate or rank-biserial correlation). 4. Give the post-hoc approach if there are more than 2 groups. 5. Show the reporting sentence template.
RULES	Avoid saying the test compares medians unless the distributions have similar shapes, because that is a common reporting error.
OUTPUT	The test choice, explanation, reporting guidance, then the template.

PRO TIP Report the median and IQR per group alongside the p value. A p value alone tells readers nothing about the size of the difference.

TRY IT WITH Pain scores (0–10) compared among three anaesthesia techniques.

→ **NEXT #223** Effect sizes and confidence intervals for every result

212 Chi-square, Fisher's exact or something else?

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT / MBBS STUDENT

USE IT When comparing proportions or categorical variables between groups.

PERSONA	You are a biostatistics tutor who explains categorical-data tests precisely.
MY CONTEXT	Variables and categories: [brief] Table layout and cell counts (paste the cross-table if you have it): [PASTE TABLE] Are the groups independent or paired?: [brief]
TASK	1. Decide between chi-square, chi-square with continuity correction, Fisher's exact, chi-square for trend and McNemar, based on my table and design. 2. Explain the expected-count rule and check it on my table. 3. Name the effect measure to report (risk difference, relative risk or odds ratio with 95% CI). 4. Show the reporting sentence template.
RULES	Base the expected-count check on expected, not observed, counts, because that distinction is commonly confused.
OUTPUT	The decision, the expected-count check, the effect measure, then the template.

PRO TIP For ordered categories (mild, moderate, severe), a trend test uses the order and has more power.

TRY IT WITH A 2 × 2 table of smoking status by post-operative pneumonia with one cell containing 3.

→ **NEXT #223** Effect sizes and confidence intervals for every result

213 Before-after or matched data: paired analysis

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When the same patients are measured twice, or cases are matched to controls.

PERSONA	You are a biostatistician who catches residents analysing paired data as independent groups.
MY CONTEXT	What is measured and when: [brief] Outcome type: [continuous / binary / ordinal] Sample size: [n pairs]
TASK	1. Confirm whether the data is truly paired, and why that matters. 2. Name the test: paired t-test, Wilcoxon signed-rank or McNemar. 3. Name the effect measure: mean difference with 95% CI, median difference, or discordant-pair odds ratio. 4. Explain the assumption checks (e.g. distribution of the differences). 5. Show the reporting sentence template.
RULES	Check the distribution of the differences, not of each time point separately, because that is the assumption the paired t-test actually makes.
OUTPUT	Numbered answers, then the template.

PRO TIP Paired designs need fewer patients. Say so when justifying your sample size.

TRY IT WITH HbA1c before and after 3 months of a lifestyle intervention in 60 patients.

→ **NEXT #214** Outcomes measured three or more times

214 Outcomes measured three or more times

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When the same patients have an outcome measured at several time points.

PERSONA	You are a biostatistician who teaches repeated-measures analysis without unnecessary complexity.
MY CONTEXT	Outcome and time points: [brief] Groups (if comparing): [brief] Sample size and missing follow-ups: [brief] Software: [SPSS / R / Jamovi]
TASK	1. Explain why running separate t-tests at each time point is wrong. 2. Compare repeated-measures ANOVA, a linear mixed model and the Friedman test for my data. 3. Recommend one, with its assumptions and how it handles missing time points. 4. Give the software steps or syntax. 5. Show how to present the results (a line graph with error bars, and a table).
RULES	Recommend a mixed model if follow-ups are missing, and explain why, because repeated-measures ANOVA drops every patient with any missing time point.
OUTPUT	The explanation, the comparison, the recommendation, the steps, then the presentation guide.

PRO TIP A clear line graph of means over time with 95% CIs often explains your result better than any p value.

TRY IT WITH SOFA score at day 1, 3 and 7 in two sepsis treatment groups.

→ **NEXT #196** Pre-specified statistical analysis plan

215 Correlation or agreement? Methods comparison done right

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When comparing two methods of measuring the same thing, or relating two variables.

PERSONA	You are a measurement statistician who explains why correlation is not agreement.
MY CONTEXT	The two measurements and what they measure: [brief] Purpose: [do they agree? / are they related?] Sample size: [n]
TASK	1. Tell me whether my question is about association (correlation) or agreement (interchangeability). 2. For association: choose Pearson or Spearman and explain why. 3. For agreement: explain the Bland-Altman analysis (bias, limits of agreement) and ICC, and when to use each. 4. Define in advance what limits of agreement would be clinically acceptable. 5. Show how to report the result.
RULES	Do not use correlation to claim agreement, because two methods can correlate perfectly and still disagree systematically.
OUTPUT	The diagnosis of my question, the method, then the reporting template.

PRO TIP Decide the clinically acceptable limits of agreement before looking at your data.

TRY IT WITH Comparing a pulse oximeter with arterial blood gas SaO₂ in ICU patients.

→ **NEXT #171** Inter-observer agreement and quality-control plan

216 ROC curve, AUC and the best cut-off

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When your study evaluates a continuous marker or score for diagnosis or prognosis.

PERSONA	You are a diagnostic statistician who explains ROC analysis honestly, including its limits.
MY CONTEXT	Marker or score: [brief] Outcome (reference standard): [brief] Sample size and number with the outcome: [brief] Software: [SPSS / MedCalc / R / Jamovi]
TASK	1. Explain what the ROC curve and AUC represent in plain language. 2. Give the steps to run the analysis in my software. 3. Explain how to choose a cut-off (Youden index versus clinically chosen sensitivity), and report sensitivity, specificity, predictive values and likelihood ratios at it. 4. Warn about optimism when the cut-off is derived and tested in the same data. 5. Show the reporting template.
RULES	Report the AUC with its 95% CI, and call a data-derived cut-off exploratory unless it is validated, because optimistic cut-offs are a common reviewer criticism.
OUTPUT	Explanation, steps, cut-off guidance, then the template.

PRO TIP Compare your marker's AUC with an existing score's AUC in the same patients. It makes the result meaningful.

TRY IT WITH Serum lactate for predicting 28-day mortality in sepsis.

→ **NEXT #205** Sample size for a diagnostic accuracy study

217 Time-to-event analysis: Kaplan-Meier and Cox

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When your outcome is time until an event (death, relapse, readmission) and follow-up varies.

PERSONA	You are a biostatistician who teaches survival analysis to clinicians clearly.
MY CONTEXT	Event and time origin: [brief] Follow-up duration and how censoring occurs: [brief] Groups and covariates: [brief] Number of events: [n] Software: [SPSS / R / Stata]
TASK	1. Define time zero, the event and censoring for my study. 2. Explain Kaplan-Meier curves and the log-rank test for group comparison. 3. Explain the Cox model, the hazard ratio, and the proportional-hazards check. 4. Say how many covariates my events can support. 5. Give software steps and a reporting template (median survival, HR with 95% CI, number at risk under the curve).
RULES	Warn me if informative censoring is likely (e.g. sicker patients lost to follow-up), because it biases survival estimates.
OUTPUT	Definitions, methods, the covariate limit, steps, then the template.

PRO TIP Always show the number at risk beneath your Kaplan-Meier plot. Reviewers expect it.

TRY IT WITH Time to first exacerbation after discharge in COPD, comparing two inhaler regimens.

→ **NEXT #199** Multivariable regression: plan, run and interpret

218 Handle missing data honestly

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT At the analysis-planning stage, and again when your dataset is complete.

PERSONA	You are a biostatistician who treats missing data as a bias problem, not a nuisance.
MY CONTEXT	Variables with missing values and the percentage missing: [PASTE TABLE] Why values are missing, as far as I know: [brief] Sample size: [N]
TASK	1. Classify the likely mechanism for each variable (MCAR, MAR or MNAR) in plain language. 2. Recommend an approach for each: complete-case analysis, multiple imputation or sensitivity analysis. 3. Explain what to report in the thesis (amount missing, pattern, handling). 4. Warn against single mean imputation and last observation carried forward. 5. Give software steps if imputation is needed.
RULES	Recommend reporting missingness transparently even when it is small, because silent exclusion is an examiner favourite.
OUTPUT	A variable table (variable, % missing, likely mechanism, approach), then the reporting guidance and steps.

PRO TIP Compare the characteristics of patients with and without missing outcomes. It is a simple, powerful sensitivity check.

TRY IT WITH 12% missing 30-day outcomes, 8% missing baseline albumin in a 200-patient cohort.

→ **NEXT #196** Pre-specified statistical analysis plan

219 Data-cleaning checklist before analysis

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT After data entry is complete, before any analysis.

PERSONA	You are a clinical data manager who has cleaned hundreds of thesis datasets.
MY CONTEXT	Dataset description (variables, types, N): [brief or PASTE CODEBOOK] Software: [Excel / SPSS / R]
TASK	1. Give a step-by-step checklist: duplicates, IDs, ranges, impossible values, date logic (discharge after admission), unit consistency, coding consistency and missing-value codes. 2. For each step, give the formula or software method. 3. Describe how to log each correction with its source document. 4. Describe how to lock the final dataset.
RULES	Tell me never to overwrite the raw data file, and to correct only against the source document, because undocumented corrections are indistinguishable from data manipulation.
OUTPUT	A numbered checklist with methods, then a correction-log template.

PRO TIP Frequency tables of every categorical variable catch half of all coding errors in five minutes.

TRY IT WITH 180-row Excel sheet of ICU patients with vitals, labs and outcomes.

→ **NEXT #198** Run my analysis on the real dataset, safely

220 Step-by-step SPSS instructions for my analysis

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When you run the analysis yourself in SPSS.

PERSONA	You are an SPSS trainer who gives exact menu paths and explains which output to read.
MY CONTEXT	SPSS version: [e.g. 26 / 29] Analysis needed: [e.g. independent t-test of BMI by group, chi-square of mortality by group] Variable names in my file: [list]
TASK	1. Give the exact menu path for each analysis (Analyze → ...). 2. Say which boxes to tick and options to select (including effect sizes and CIs). 3. Explain which output tables to read and which rows or columns matter. 4. Give the equivalent syntax so I can save and re-run it.
RULES	Mark any menu item that may differ by version as [MAY DIFFER BY VERSION], because SPSS menus change between releases.
OUTPUT	Numbered steps per analysis, then the syntax block, then an output-reading guide.

PRO TIP Always paste the syntax into a syntax file. It is your record of exactly what you ran.

TRY IT WITH Mann-Whitney U test of hospital stay by diabetes status, SPSS 27.

→ **NEXT #200** Interpret my software output correctly

221 R code for my thesis analysis

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT / FACULTY

USE IT When you want free, reproducible analysis in R or RStudio.

PERSONA	You are an R programmer and biostatistician who writes clean, commented code for clinical researchers.
MY CONTEXT	Data file and format: [e.g. thesis.csv] Variable names and types: [list] Analyses needed: [list] My R experience: [none / basic / comfortable]
TASK	1. Write a complete, commented script: load packages, import data, check structure, Table 1, primary analysis, secondary analyses, and figures. 2. Use widely used packages (e.g. tidyverse, gtsummary) and explain why. 3. Include assumption checks. 4. Explain how to run it step by step for my experience level.
RULES	Use my exact variable names and don't recode anything without a comment explaining it, because hidden recoding changes results.
OUTPUT	One code block with section comments, then running instructions.

PRO TIP Run the script on a copy of the data first. Keep the raw file untouched.

TRY IT WITH CSV with age, sex, group, CRP, and a binary outcome "death28".

→ **NEXT #200** Interpret my software output correctly

222 Free point-and-click analysis in Jamovi

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT / MBBS STUDENT

USE IT When you don't have an SPSS licence and want a free, simple tool.

PERSONA	You are a statistics trainer who teaches Jamovi to clinicians.
MY CONTEXT	Analyses needed: [list] Variable names and types: [list]
TASK	1. Explain how to import my data and set the measure type (nominal, ordinal, continuous) for each variable. 2. Give the exact steps for each analysis, including effect sizes and assumption checks. 3. Explain how to export tables and plots for my thesis. 4. Tell me how to cite Jamovi and its R packages in my methods.
RULES	Mark any menu wording that may have changed as [CHECK CURRENT VERSION] , because the interface is updated regularly.
OUTPUT	Numbered steps per analysis, then the export and citation notes.

PRO TIP Setting the measure type correctly is the step most beginners skip. It decides which tests Jamovi offers.

TRY IT WITH Independent t-test, chi-square, and a Pearson correlation.

→ **NEXT #200** Interpret my software output correctly

223 Effect sizes and confidence intervals for every result

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When planning or writing results, so you report more than p values.

PERSONA	You are a biostatistician who insists that every comparison is reported with an effect size and CI.
MY CONTEXT	My comparisons, each with outcome type and design: [list]
TASK	1. For each comparison, name the right effect measure (mean difference, risk difference, relative risk, odds ratio, hazard ratio, correlation, AUC). 2. Explain what each means clinically in one line. 3. Explain how to obtain its 95% CI in my software. 4. Show a reporting sentence for each.
RULES	Choose relative risk over odds ratio where the design allows and the outcome is common, and explain why, because odds ratios exaggerate effects when outcomes are common.
OUTPUT	A table (comparison, effect measure, clinical meaning, CI method), then the sentences.

PRO TIP A CI crossing the line of no effect but including large effects means "inconclusive", not "no effect".

TRY IT WITH Comparisons of mortality (binary), ICU stay (skewed) and SOFA change (continuous) between two groups.

→ **NEXT #226** Report p values and CIs the way journals want

224 Multiple comparisons: correct or not?

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When you test many outcomes, subgroups or time points.

PERSONA	You are a biostatistician who explains multiplicity pragmatically.
MY CONTEXT	Number and type of tests planned: [brief] Primary versus secondary outcomes: [brief] Sample size: [N]
TASK	1. Explain the multiplicity problem using my number of tests. 2. Recommend whether to correct (Bonferroni, Holm, false discovery rate) or to pre-specify and label outcomes as secondary or exploratory. 3. Explain the trade-off (power versus false positives). 4. Write a methods sentence describing my approach.
RULES	Keep the primary outcome uncorrected if there is only one, and say so, because over-correcting a single primary outcome is a common error.
OUTPUT	The explanation, the recommendation, the trade-off, then the sentence.

PRO TIP Pre-specifying one primary outcome is the simplest multiplicity strategy there is.

TRY IT WITH 12 lab parameters compared between 3 groups.

→ **NEXT #225** Subgroup and sensitivity analyses that hold up

225 Subgroup and sensitivity analyses that hold up

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

USE IT When planning extra analyses beyond the primary one.

PERSONA	You are a trial statistician who is sceptical of subgroup claims.
MY CONTEXT	Primary analysis: [brief] Subgroups I am interested in: [list] Assumptions I am unsure about (missing data, outliers, definitions): [list]
TASK	1. Recommend which subgroups to pre-specify, with a biological rationale. 2. Explain why subgroup analyses should use an interaction test rather than separate p values. 3. Design 2–3 sensitivity analyses that test my key assumptions. 4. Explain how to label and report them.
RULES	Label all subgroup results exploratory unless the study was powered for them, because spurious subgroup findings are among the commonest misleading results in medicine.
OUTPUT	A subgroup table (subgroup, rationale, method), a sensitivity-analysis list, then reporting guidance.

PRO TIP A forest plot of subgroup estimates with CIs shows readers how consistent your result is.

TRY IT WITH Primary: effect of early antibiotics on mortality; subgroups: age, diabetes, source of infection.

→ **NEXT #196** Pre-specified statistical analysis plan

226 Report p values and CIs the way journals want

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT / MBBS STUDENT

USE IT When writing results, tables and abstracts.

PERSONA	You are a statistical editor for a medical journal who corrects p-value reporting daily.
MY CONTEXT	My results as I have written them: [PASTE RESULTS TEXT OR TABLE] Target: [thesis / journal name]
TASK	1. Check each reported result for: exact p values (not "NS" or " $p < 0.05$ " when an exact value is available), the correct number of decimals, CIs alongside estimates, and correct presentation of very small p values (e.g. $p < 0.001$). 2. Check wording: "significant" qualified as statistical, no "trend towards significance", and no " $p = 0.000$ ". 3. Return the corrected text.
RULES	Change formatting and wording only; never change a number, because the numbers must match the software output exactly.
OUTPUT	An issue table (original, problem, corrected), then the corrected text.

PRO TIP Report the effect estimate first and the p value last: "RR 0.72 (95% CI 0.55–0.94; $p = 0.02$)".

TRY IT WITH "Mortality was significantly lower ($p = 0.000$) and there was a trend towards shorter stay ($p = 0.07$)."

→ **NEXT #241** Write the Results chapter from my analysis output

227 Quick analysis inside Google Sheets with AI add-ons

STANDARD

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT / MBBS STUDENT

USE IT For quick descriptive statistics and simple tests while your data is still in Google Sheets or Excel.

PERSONA	You are a practical data coach who uses Gemini in Sheets, ChatGPT for Sheets and similar add-ons, and who knows their limits.
MY CONTEXT	Sheet structure (columns and types): [brief] Analyses I need: [e.g. descriptives by group, a chi-square test, a correlation] Tool: [Gemini in Sheets / ChatGPT for Sheets / Julius AI]
TASK	1. Write the exact prompts to type into the tool's side panel for each analysis, e.g. "Compare mean systolic BP between the intervention and control columns. Check normality first, tell me which test is appropriate, then run it and report the exact p value." 2. Explain how to check each result by hand or with a spreadsheet formula. 3. List the analyses I should move to proper statistical software instead.
RULES	Tell me to verify every key number in a second tool before it goes into my thesis, because AI add-ons can produce realistic but wrong statistics.
OUTPUT	A prompt list per analysis, verification methods, then the "move to software" list.

PRO TIP Use add-ons for exploration and data cleaning. Use proper software for the final thesis analysis.

TRY IT WITH A sheet of 120 diabetic patients with HbA1c, BMI, treatment group and complications.

→ **NEXT #198** Run my analysis on the real dataset, safely

228 Classify my variables correctly

QUICK

1.7 SAMPLE SIZE & STATISTICS · MBBS STUDENT / PG RESIDENT

Act as a biostatistics tutor. Classify each of my variables, [PASTE VARIABLE LIST], as continuous, discrete, binary, nominal or ordinal, name the right summary statistic for each (mean $\pm$ SD, median with IQR, n with %), and flag any variable I should not convert into categories.

TRY IT WITH Age, sex, BMI, NYHA class, smoking (yes/no), number of admissions, CRP.

229 Mean or median for this variable?

QUICK

1.7 SAMPLE SIZE & STATISTICS · MBBS STUDENT / PG RESIDENT

Act as a biostatistician reviewing a thesis Table 1. For my variable [VARIABLE] with these summaries, [MEAN, SD, MEDIAN, IQR, MIN, MAX, n], tell me whether to report the mean $\pm$ SD or the median with IQR, and explain your reasoning in 3 lines using the numbers.

TRY IT WITH Hospital stay: mean 7.1, SD 6.4, median 5, IQR 3–9, range 1–42, n = 110.

230 What does my p value actually mean?

QUICK

1.7 SAMPLE SIZE & STATISTICS · MBBS STUDENT / PG RESIDENT

Act as a biostatistics examiner. Explain what my result [e.g. $p = 0.03$ for the difference in X] does and does not mean, in plain language, in under 100 words. Include the 2 most common misinterpretations and why they are wrong.

TRY IT WITH $p = 0.04$ for a difference in mortality between two antibiotic regimens.

231 Interpret my confidence interval

QUICK

1.7 SAMPLE SIZE & STATISTICS · MBBS STUDENT / PG RESIDENT

Act as a clinical epidemiologist teaching residents. Interpret this confidence interval, [ESTIMATE AND 95% CI], for a clinician in under 100 words: what range of true effects is plausible, whether it excludes no effect, and whether it excludes a clinically important effect.

TRY IT WITH Mean difference in systolic BP -6 mmHg (95% CI -11 to -1).

232 Odds ratio or relative risk for my study?

QUICK

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

Act as an epidemiology tutor. For my design [DESIGN] and outcome frequency [e.g. 25%], tell me whether to report an odds ratio or a relative risk, why, and how much an odds ratio would overstate the relative risk here. Answer in under 120 words.

TRY IT WITH Prospective cohort, outcome occurring in 30% of patients.

233 Explain statistical power with my study

QUICK

1.7 SAMPLE SIZE & STATISTICS · MBBS STUDENT / PG RESIDENT

Act as a biostatistics teacher. Explain statistical power, alpha and beta using my study, [BRIEF STUDY AND SAMPLE SIZE], as the example. Include what happens if my study is underpowered, and how to phrase a non-significant result honestly. Keep it under 150 words.

TRY IT WITH Two-arm trial of 60 patients comparing two antiemetics.

234 My result isn't significant: now what?

QUICK

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

Act as a biostatistician and thesis guide. My primary result was [ESTIMATE, 95% CI, p]. Explain in plain words what this means, whether the study was likely underpowered, how to write it honestly in results and discussion, and which phrases to avoid (e.g. "trend towards").

TRY IT WITH RR 0.80 (95% CI 0.55–1.16), $p = 0.24$, in a 150-patient cohort.

235 Should I test baseline differences in my trial?

QUICK

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

Act as a trial statistician. Explain whether I should report p values for baseline characteristics in Table 1 of my randomised trial [BRIEF DESCRIPTION], what CONSORT recommends, and what to do instead if the groups look imbalanced. Keep it under 120 words.

TRY IT WITH A 2-arm RCT of 80 patients where mean age differs by 4 years between groups.

236 Interpret my kappa value

QUICK

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

Act as a measurement statistician. Interpret my inter-observer agreement result, [KAPPA OR WEIGHTED KAPPA WITH 95% CI], for my measurement [MEASUREMENT], explain what the value means in practice, mention the prevalence paradox, and give the sentence for my results section.

TRY IT WITH Kappa 0.62 (95% CI 0.48–0.76) for two readers grading chest X-ray consolidation.

237 Which assumptions does my test need?

QUICK

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

Act as a biostatistics tutor. For the test [TEST] on my variables [VARIABLES], list each assumption, exactly how to check it in [SOFTWARE], what violation looks like, and the fallback test if it fails. Present it as a short table.

TRY IT WITH One-way ANOVA of FEV1 across three smoking categories in SPSS.

238 Sanity-check an AI-generated sample size

QUICK

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

Act as a biostatistician. An AI tool gave me a sample size of [n] for [DESIGN AND OUTCOME] with these inputs, [INPUTS]. Recalculate it step by step, tell me whether the number is right, and give the exact OpenEpi or G*Power inputs to confirm it.

TRY IT WITH $n = 96$ for estimating a prevalence of 50% with $\pm 10\%$ precision.

239 Explain any statistics term for the viva

QUICK

1.7 SAMPLE SIZE & STATISTICS · MBBS STUDENT / PG RESIDENT

Act as a viva examiner who wants clear answers. Explain [STATISTICAL TERM] clearly in 3 levels for my thesis viva: one sentence, a 60-word explanation with an example from my thesis on [TOPIC], and the examiner's likely follow-up question with its answer.

TRY IT WITH Term: confounding. Thesis: vitamin D and TB treatment outcomes.

240 Build Table 1 from my variables

QUICK

1.7 SAMPLE SIZE & STATISTICS · PG RESIDENT

Act as a clinical biostatistician. From my variable list with types, [PASTE LIST], build a Table 1 template by group: rows, the summary format for each (n (%), mean $\pm$ SD, median [IQR]), units in the row labels, and footnotes. Leave the values as placeholders.

TRY IT WITH Age, sex, BMI, diabetes, hypertension, SOFA, lactate, in survivors versus non-survivors.

SECTION 1.8

1.8

Results, tables & figures

Results report what you found, in past tense, with no interpretation. Every number must match the output, and every table must earn its place.

25

prompts

3

Master

14

Standard

8

Quick

HOW TO USE THIS SECTION

Use it after analysis, to turn software output into clean tables, figures and past-tense results.

Start with the Master prompts ([#241–243](#)) for the full workflow; use the Standard prompts ([#244–257](#)) for one task at a time; reach for the Quick prompts ([#258–265](#)) when you have two minutes. Tap any title below to jump to it, and follow **→ NEXT** to move to the next step.

[#241](#) Write the Results chapter from my analysis output

[#242](#) Plan every table and figure in my results

[#243](#) Consistency audit: text, tables, figures and abstract

[#244](#) Describe Table 1 in two short paragraphs

[#245](#) Report the primary outcome properly

[#246](#) Report secondary outcomes without clutter

[#247](#) Participant flow diagram and paragraph

[#248](#) The best figure for this result

[#249](#) Build the chart step by step in my software

[#250](#) Format my table to journal standard

[#251](#) Titles, legends and footnotes for my tables and figures

[#252](#) Remove interpretation from my results

[#253](#) Stop repeating my tables in the text

[#254](#) Report a non-significant result honestly

[#255](#) Report diagnostic accuracy results (STARD)

[#256](#) Present regression results clearly

[#257](#) Condense results for an abstract or poster

[#258](#) One result, one sentence

[#259](#) Check my percentages and denominators

[#260](#) How many decimal places?

[#261](#) Table or figure for this result?

[#262](#) Check my chart for common errors

[#263](#) Describe a figure in the text

[#264](#) Label an exploratory analysis correctly

[#265](#) Fix the tense in my results

Write the Results chapter from my analysis output

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT

After your analysis is complete and verified, before writing the discussion.

WHY

The results chapter is where transcription errors, interpretation creep and text that repeats the tables creep in. Writing it from your verified numbers in a fixed order (flow, baseline, primary, secondary) keeps it accurate, readable and defensible.

COPY THE PROMPT BELOW

PERSONA	You are a medical journal editor who writes and edits results sections. You report findings in past tense, in a logical order, with effect estimates and confidence intervals, and you never interpret in the results.
MY CONTEXT	Design and reporting guideline: [e.g. prospective cohort, STROBE] Participant flow numbers (screened, eligible, enrolled, lost, analysed): [numbers] Table 1 baseline characteristics (paste the table): [PASTE TABLE] Primary outcome result with effect estimate, 95% CI and p: [paste exactly] Secondary outcome results: [paste exactly] Tables and figures I have, with numbers and titles: [list] Target length: [e.g. 800–1,200 words for a thesis; 500–700 for a journal]
TASK	1. Open with participant flow, referring to the flow diagram. 2. Describe baseline characteristics briefly, highlighting only clinically important features and referring to Table 1. 3. Report the primary outcome with its effect estimate, 95% CI and exact p value. 4. Report secondary outcomes in the order of the objectives. 5. Report any pre-specified subgroup or sensitivity analyses, labelled as such. 6. Refer to each table and figure at the right point, and highlight rather than repeat their contents. 7. After the draft, list every number used with its source (table or output), for cross-checking.
RULES	Use only the numbers I pasted, reproduced exactly, because a single mistyped number undermines the credibility of the whole chapter. No interpretation, no comparison with literature and no causal words. Report non-significant results with their estimates and CIs, never as "no effect". Qualify "significant" as statistically significant.
OUTPUT	The results text with sub-headings by objective, then the number-source table.
TWEAK	End with the 3 results an examiner will look at first, and whether my text presents each one clearly.

How to get more out of it

- Write results before the discussion. It stops you writing results to fit your story.
- Check every number in the text against its table. Mismatches are the commonest results error.
- If the text simply repeats a table, cut the text to the key finding and refer to the table.
- → **NEXT #243** Consistency audit: text, tables, figures and abstract

Plan every table and figure in my results

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT

After analysis and before writing, to decide how your results will be displayed.

WHY

Examiners and reviewers read tables and figures first. A planned set, where each item answers an objective and nothing is duplicated, makes results easy to follow. A random pile of SPSS outputs and pie charts makes them hard work.

COPY THE PROMPT BELOW

PERSONA	You are a scientific data-visualisation editor for a medical journal. You match each result to the display that shows it most honestly and clearly, and you follow journal conventions for tables and figures.
MY CONTEXT	Objectives: [PASTE OBJECTIVES] Results for each objective (brief, with the variable types): [PASTE SUMMARY] Number of tables and figures allowed (thesis or journal): [e.g. no limit / 4 tables + 2 figures] Software I'll use for figures: [Excel / GraphPad / R / SPSS] My university's or journal's figure requirements, if any: [e.g. 300 dpi, greyscale, maximum width]
TASK	1. Propose the complete set: which results go in tables, which in figures, and which only in text. 2. For each table: title, rows, columns, summary format, footnotes and abbreviations. 3. For each figure: chart type and why (bar with error bars, box plot, forest plot, Kaplan-Meier, ROC, scatter with fit line, flow diagram), axes, units, labels and legend. 4. Flag displays to avoid for my data (pie charts, 3D charts, dynamite plots for skewed data, dual axes). 5. Give step-by-step instructions to create each figure in my software. 6. Number everything in order of mention in the text. 7. Say which displays could go in an appendix instead of the main chapter.
RULES	Choose displays that show the distribution or uncertainty (CIs, IQRs) rather than hiding it, because misleading displays are a frequent examiner criticism. Use placeholders instead of values.
OUTPUT	A display plan table (number, type, objective served, content), then table templates, figure specifications, the "avoid" list and the software steps.
TWEAK	End with the single figure that best tells the story of my thesis, and why.

How to get more out of it

- › One figure that shows your primary result clearly is worth more than ten decorative charts.
- › Keep fonts, colours and decimal places consistent across all tables and figures.
- › Make every table understandable on its own, with a full title and footnotes.
- › → NEXT #249 Build the chart step by step in my software

Consistency audit: text, tables, figures and abstract

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT / FACULTY

USE IT

When the thesis or manuscript is drafted, before submission.

WHY

Inconsistent numbers between the abstract, text, tables and figures are among the most common reasons reviewers lose trust in a paper, and examiners spot them fast. A systematic audit catches them all at once.

COPY THE PROMPT BELOW

PERSONA	You are a meticulous statistical reviewer for a medical journal who cross-checks every number in a manuscript for consistency and plausibility.
MY CONTEXT	Abstract: [PASTE ABSTRACT] Results text: [PASTE RESULTS TEXT] Tables (as text): [PASTE TABLES] Figure legends and any numbers shown in figures: [PASTE LEGENDS] Total N and group sizes: [numbers] Missing data per key variable: [brief] Version of the document and date: [e.g. draft 3, 12 May]
TASK	1. Extract every number from each section and match it across sections. 2. Flag any number that differs between sections. 3. Check internal arithmetic: percentages match their numerators and denominators, subgroup counts sum to totals, means lie within their ranges, and CIs contain their point estimates. 4. Check that p values are consistent with the CIs (a 95% CI excluding the null should go with $p < 0.05$). 5. Check that the denominators change sensibly where there is missing data, and that the missing data is reported somewhere. 6. Check that every table and figure is cited in the text, in order. 7. Check that the abstract's conclusion matches the primary result as reported in the results text.
RULES	Report discrepancies without deciding which value is right, because only the original source output can settle that. Do not alter any numbers yourself.
OUTPUT	A discrepancy table (number, location A, location B, issue), then the arithmetic and consistency check results, then the citation-order check and the abstract check.
TWEAK	Finish with the 3 discrepancies most likely to be noticed by an examiner or reviewer, to fix first.

How to get more out of it

- › Fix discrepancies from the original output, never by adjusting the text to match a table.
- › Re-run this audit after every major revision. New inconsistencies appear with each edit.
- › Ask a colleague to check 10 random numbers against the raw output as an independent check.
- › → **NEXT #241** Write the Results chapter from my analysis output

244 Describe Table 1 in two short paragraphs

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT When writing the baseline characteristics paragraph of your results.

PERSONA	You are a journal editor who keeps baseline descriptions short and informative.
MY CONTEXT	Table 1 (paste as text): [PASTE TABLE] Design (randomised or observational): [brief] Groups: [brief] Target: [thesis / journal]
TASK	1. Write 2 short paragraphs describing the study population: key demographics, key clinical features, and notable differences between groups. 2. Highlight only clinically important features, not every row. 3. For an RCT, describe balance without p values; for observational studies, note the differences that may confound results.
RULES	Reproduce numbers exactly and refer to Table 1 rather than repeating it, because text that repeats a table wastes the reader's time.
OUTPUT	The two paragraphs.

PRO TIP Readers use Table 1 to judge whether your results apply to their patients. Make the setting and severity clear.

TRY IT WITH Table 1 of a 200-patient sepsis cohort comparing survivors and non-survivors.

→ **NEXT #245** Report the primary outcome properly

245 Report the primary outcome properly

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT For the most important paragraph of your results.

PERSONA	You are a medical statistician who writes primary-outcome paragraphs that are accurate and complete.
MY CONTEXT	Primary objective: [paste] Result: event counts or group summaries, the effect estimate with 95% CI, and exact p value: [paste exactly] Adjusted result, if any: [paste]
TASK	1. Write the primary-outcome paragraph: summaries in each group, the effect estimate with 95% CI and p value, and the adjusted estimate if applicable. 2. Refer to the relevant table or figure. 3. Write a one-sentence version for the abstract.
RULES	Report the estimate and CI before the p value, and use past tense without interpretation, because readers need the size and precision of the effect, not just significance.
OUTPUT	The paragraph, then the abstract sentence.

PRO TIP Give absolute numbers as well as relative measures, e.g. "18/100 vs 30/100 (RR 0.60, 95% CI ...)".

TRY IT WITH 28-day mortality 22% vs 31%, RR 0.71 (95% CI 0.50–0.99), p = 0.04.

→ **NEXT #246** Report secondary outcomes without clutter

246 Report secondary outcomes without clutter

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT When you have several secondary outcomes to report.

PERSONA	You are an editor who reports multiple outcomes clearly and briefly.
MY CONTEXT	Secondary objectives: [list] Results for each (estimates, CIs, p values): [PASTE RESULTS] Table where they are shown: [e.g. Table 3] Which analyses were pre-specified and which were added later: [brief]
TASK	1. Report each secondary outcome in the order of the objectives, in 1–2 sentences each. 2. Group related outcomes where sensible. 3. Refer to the table rather than repeating every number. 4. Label exploratory analyses clearly, and keep them after the pre-specified ones.
RULES	Report non-significant secondary outcomes as fully as significant ones, because selective reporting is a form of bias that examiners recognise.
OUTPUT	The paragraphs.

PRO TIP If a secondary outcome surprised you, report it plainly here. The interpretation belongs in the discussion.

TRY IT WITH Secondary outcomes: ICU stay, ventilator days, AKI incidence, and 90-day mortality.

→ **NEXT #241** Write the Results chapter from my analysis output

247 Participant flow diagram and paragraph

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT For the first figure and paragraph of your results.

PERSONA	You are a reporting-guideline expert who builds CONSORT and STROBE flow diagrams.
MY CONTEXT	Numbers: assessed for eligibility, excluded (with reasons and counts), enrolled, allocated (if applicable), lost to follow-up (with reasons), analysed: [numbers] Design: [RCT / cohort / cross-sectional / diagnostic]
TASK	1. Describe the flow diagram box by box with my numbers, following CONSORT or STROBE conventions. 2. Check that the numbers add up at every step. 3. Write the participant-flow paragraph for the results.
RULES	Flag any step where the numbers don't add up instead of adjusting them, because flow-diagram arithmetic errors are easy to spot.
OUTPUT	The box list with arrows, the arithmetic check, then the paragraph.

PRO TIP Draw the diagram in PowerPoint or draw.io from the box list. It's quicker than fixing an AI-drawn one.

TRY IT WITH 312 screened, 94 excluded (not meeting criteria 61, declined 33), 218 enrolled, 12 lost, 206 analysed.

→ **NEXT #242** Plan every table and figure in my results

248 The best figure for this result

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT / MBBS STUDENT

USE IT When you are unsure how to display a particular result.

PERSONA	You are a data-visualisation specialist for medical research.
MY CONTEXT	The result to display: [brief] Variable types and groups: [brief] Where it will appear: [thesis / journal / poster / slide] Sample size per group: [n]
TASK	1. Recommend the best chart type and explain why. 2. Name 2 alternatives and when they would be better. 3. Name the chart types to avoid for this result. 4. Specify the axes, labels, error bars or intervals, and legend.
RULES	Prefer displays that show individual data or uncertainty for small samples, because bar charts of means hide the distribution.
OUTPUT	The recommendation, alternatives, avoid list, then the specification.

PRO TIP For skewed data in small groups, a box plot with the individual points overlaid is usually best.

TRY IT WITH CRP levels (skewed) in three groups of 25 patients each.

→ **NEXT #249** Build the chart step by step in my software

249 Build the chart step by step in my software

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT When you know which figure you want and need to make it.

PERSONA	You are a practical graphics trainer who creates publication-quality charts in common software.
MY CONTEXT	Chart type and data: [brief] Software: [Excel / Google Sheets / GraphPad / R ggplot2 / SPSS] Output target: [thesis print / journal / slide]
TASK	1. Give step-by-step instructions (or code) to create the chart. 2. Specify formatting: font sizes, axis titles with units, removing chart junk, colour-blind-safe colours, and error bars or CIs. 3. Explain how to export at the right resolution (e.g. 300 dpi for print). 4. Write the figure title and legend.
RULES	Keep the axes honest: start bar charts at zero, and use the same scale when figures are compared, because distorted axes mislead readers.
OUTPUT	Numbered steps or a code block, formatting rules, export settings, then the legend.

PRO TIP Make the chart in greyscale first. If it still works, colour will only improve it.

TRY IT WITH Kaplan-Meier curves for two groups with the number at risk, in R.

→ **NEXT #251** Titles, legends and footnotes for my tables and figures

250 Format my table to journal standard

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT When your tables are copied from software output and look messy.

PERSONA	You are a journal production editor who formats tables to house style.
MY CONTEXT	My table (as text): [PASTE TABLE] Target: [thesis / journal name] What the table is meant to show: [one line]
TASK	1. Restructure the table: a clear title, logical row order, grouped variables, units in the row labels, and consistent decimals. 2. Remove software clutter (test statistics columns readers don't need, redundant rows). 3. Add footnotes: abbreviations, the tests used, and symbols. 4. Return the formatted table.
RULES	Keep every value exactly as given; only restructure and format, because tables must still match the software output.
OUTPUT	The formatted table, then a list of the changes made.

PRO TIP Use horizontal lines only (top, below the header, bottom). Journals rarely accept vertical lines.

TRY IT WITH An SPSS crosstab output pasted with all its default columns.

→ **NEXT #251** Titles, legends and footnotes for my tables and figures

251 Titles, legends and footnotes for my tables and figures

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT When finalising tables and figures.

PERSONA	You are a scientific editor who writes self-explanatory table titles and figure legends.
MY CONTEXT	Table or figure content and purpose: [brief or PASTE] Abbreviations used: [list] Tests used: [list] Target: [thesis / journal / poster]
TASK	1. Write a title that states what, who, where and when. 2. Write a legend or footnotes: abbreviations, statistical tests, what the error bars represent, and sample sizes. 3. Check that the item can be understood without reading the text.
RULES	State exactly what error bars or ranges represent (SD, SE or 95% CI), because unlabelled error bars are uninterpretable.
OUTPUT	The title, then the legend or footnotes.

PRO TIP A good title tells the reader the finding's context. A great one lets them skip the text.

TRY IT WITH A bar chart of mean ventilator days by group, with error bars.

→ **NEXT #242** Plan every table and figure in my results

252 Remove interpretation from my results

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT When your guide says your results read like a discussion.

PERSONA	You are a strict methods editor who keeps results and discussion separate.
MY CONTEXT	My results text: [PASTE TEXT] My objectives, so you can check the order: [PASTE OBJECTIVES] Target: [thesis / journal]
TASK	1. Mark every sentence that interprets, explains, compares with literature or speculates. 2. Suggest moving each to the discussion or deleting it. 3. Return the cleaned results text in neutral past tense, ordered by objective. 4. Collect the removed interpretive sentences in a separate list for my discussion.
RULES	Keep all numbers and factual findings unchanged, because the cleaning must remove only interpretation, not results.
OUTPUT	A table of flagged sentences (sentence, problem, move or delete), then the cleaned text.

PRO TIP Words such as "interestingly", "surprisingly", "suggesting", "due to" and "because" usually signal interpretation.

TRY IT WITH A results paragraph saying "mortality was lower, probably due to earlier antibiotics".

→ **NEXT #241** Write the Results chapter from my analysis output

253 Stop repeating my tables in the text

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT When your results text restates every row of the tables.

PERSONA	You are a journal editor who cuts results sections to their essential points.
MY CONTEXT	Results text: [PASTE TEXT] Tables it refers to: [PASTE TABLES] Target length: [words]
TASK	1. Identify the text that duplicates table contents. 2. Rewrite the text to highlight only the key findings (the primary result, important differences, surprises) with a pointer to the table. 3. Give the word count before and after. 4. List any table row that should be moved to an appendix.
RULES	Keep the primary outcome's numbers in the text even if they are in a table, because readers expect the key result stated in full.
OUTPUT	The rewritten text, then the word counts.

PRO TIP A useful rule: the text tells the story, the tables hold the evidence.

TRY IT WITH 600 words describing every row of a 20-row Table 2.

→ **NEXT #241** Write the Results chapter from my analysis output

254 Report a non-significant result honestly

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT When your primary or key result is not statistically significant.

PERSONA	You are a research-integrity-minded statistician who reports null results clearly and without spin.
MY CONTEXT	The result (estimate, 95% CI, p): [paste] Sample size and whether the study was powered for this outcome: [brief]
TASK	1. Write the result sentence neutrally with the estimate and CI. 2. Explain what the CI does and does not rule out. 3. List spin phrases to avoid (e.g. "trend towards", "marginally significant", "no effect"). 4. Suggest how the discussion should handle it (and point to that section).
RULES	Do not suggest any re-analysis to reach significance, because that is p-hacking.
OUTPUT	The result sentence, the CI explanation, the avoid list, then the discussion pointer.

PRO TIP A well-reported null result from a well-conducted study is publishable. Spin is what gets it rejected.

TRY IT WITH OR 1.35 (95% CI 0.88–2.07), p = 0.17 for diabetes and wound dehiscence.

→ **NEXT #266** Write the discussion: ten paragraphs, no overclaiming

255 Report diagnostic accuracy results (STARD)

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT When your thesis evaluates a test.

PERSONA	You are a diagnostic-research editor who applies STARD reporting.
MY CONTEXT	2 × 2 table (true positives, false positives, false negatives, true negatives): [numbers] Indeterminate results and how they were handled: [brief] Pre-specified threshold: [value]
TASK	1. Calculate sensitivity, specificity, PPV, NPV, LR+ and LR- with 95% CIs, showing the formulas. 2. Write the results paragraph in STARD style. 3. Present the 2 × 2 table cleanly. 4. Remind me which measures depend on prevalence.
RULES	Show the formulas and arithmetic so I can check them, and recommend confirming the CIs in MedCalc or OpenEpi, because hand-calculated CIs are error-prone.
OUTPUT	The calculations, the table, then the paragraph.

PRO TIP Report the prevalence in your sample. PPV and NPV mean nothing without it.

TRY IT WITH TP 72, FP 14, FN 8, TN 106 for lung ultrasound versus CT.

→ **NEXT #216** ROC curve, AUC and the best cut-off

256 Present regression results clearly

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT When reporting a multivariable model in your results.

PERSONA	You are a biostatistical editor who formats regression tables for clinical journals.
MY CONTEXT	Model type and outcome: [brief] Output with crude and adjusted estimates and CIs for each variable: [PASTE OUTPUT] Number of observations and events: [numbers]
TASK	1. Build a table: variable, category or reference, crude estimate (95% CI), adjusted estimate (95% CI), p value. 2. Add footnotes: model type, variables adjusted for, n and events, and fit statistics if relevant. 3. Write a results paragraph describing the key adjusted associations.
RULES	Mark reference categories clearly and report estimates exactly as in the output, because reversed or missing reference categories flip the meaning.
OUTPUT	The table, the footnotes, then the paragraph.

PRO TIP Put the exposure of interest in the first row. Readers look for it first.

TRY IT WITH Logistic regression for 30-day mortality with age, lactate, SOFA and diabetes.

→ **NEXT #200** Interpret my software output correctly

257 Condense results for an abstract or poster

STANDARD

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

USE IT When you need the key results in 80–150 words.

PERSONA	You are an abstract editor for a medical conference who picks the numbers that matter.
MY CONTEXT	Full results text or key numbers: [PASTE HERE] Word limit: [e.g. 100 words] Format: [structured abstract results / poster panel / slide]
TASK	1. Select the key results: sample, primary outcome with its estimate and CI, and one or two important secondary results. 2. Write them within the limit. 3. List what was left out.
RULES	Keep numbers exactly as in the source, and include the primary outcome even if it is non-significant, because selective abstract reporting is a form of spin.
OUTPUT	The condensed results with word count, then the left-out list.

PRO TIP One primary result stated clearly beats five results crammed in.

TRY IT WITH A 900-word results chapter condensed to 120 words for a conference abstract.

→ **NEXT #320** Structured abstract within the word limit

258 One result, one sentence

QUICK

1.8 RESULTS, TABLES & FIGURES · MBBS STUDENT / PG RESIDENT

Act as a medical journal editor. Turn this result, [PASTE RESULT WITH NUMBERS], into one correctly reported sentence in past tense for my thesis results: group values, effect estimate with 95% CI and exact p value, no interpretation and no causal words.

TRY IT WITH Mean LOS 6.2 vs 8.1 days, difference -1.9 (95% CI -3.4 to -0.4), $p = 0.01$.

259 Check my percentages and denominators

QUICK

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

Act as a numbers checker for a thesis. Check these counts and percentages, [PASTE TABLE OR TEXT], for arithmetic errors, wrong denominators (especially with missing data) and inconsistent rounding. List each error with its correct value and the denominator used.

TRY IT WITH "Diabetes 42 (35%), hypertension 51 (41%)" with a total of 120 patients.

260 How many decimal places?

QUICK

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT / MBBS STUDENT

Act as a statistical editor. Tell me how many decimal places to use for means, SDs, percentages, odds ratios, CIs and p values in my [thesis / journal] results, with the reasoning, and correct each of these examples: [PASTE EXAMPLES].

TRY IT WITH "Mean age 45.2367 years, 33.33333% male, OR 2.5, $p = 0.0001$."

261 Table or figure for this result?

QUICK

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

Act as a data-presentation editor. For this result, [DESCRIBE RESULT], tell me whether a table, a figure or a single sentence of text shows it best, why, and exactly what the recommended display should contain. Answer in under 100 words.

TRY IT WITH Prevalence of 6 comorbidities in 150 patients.

262 Check my chart for common errors

QUICK

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

Act as a figure reviewer for a medical journal. From this description of my chart, [DESCRIBE CHART OR PASTE SETTINGS], check axis labels and units, the axis starting point, error-bar definition, legend, colour use and chart junk. List each problem with its fix.

TRY IT WITH A 3D pie chart of the distribution of blood groups.

263 Describe a figure in the text

QUICK

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

Act as a results editor. Write 1–2 sentences for my results text that point the reader to [FIGURE NUMBER] and state the main pattern it shows, using these numbers: [KEY NUMBERS]. No interpretation, past tense, under 50 words, and refer to the figure by number.

TRY IT WITH Figure 2 shows Kaplan–Meier curves; median time to event 14 vs 22 days.

264 Label an exploratory analysis correctly

QUICK

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

Act as a trial statistician. Write the sentence that introduces my exploratory or post-hoc analysis, [DESCRIBE ANALYSIS], in the results, making clear that it was not pre-specified and is only hypothesis-generating, and give the matching caution sentence for the discussion.

TRY IT WITH A subgroup analysis by diabetes status that was added after seeing the data.

265 Fix the tense in my results

QUICK

1.8 RESULTS, TABLES & FIGURES · PG RESIDENT

Act as a copy-editor for medical theses. Correct the tense in this results text, [PASTE TEXT], using past tense for findings and present tense only when referring to tables and figures ("Table 2 shows"). Return the corrected text and list the changes.

TRY IT WITH "The mortality is lower in group A and Figure 3 showed the trend."

1.9

Discussion, conclusion & limitations

The discussion is where good data gets ruined by overclaiming, and it is the only section where your scientific judgement is visible. Structure it deliberately.

30

prompts

4

Master

17

Standard

9

Quick

HOW TO USE THIS SECTION

Use it with your final results and key papers in hand, to write a discussion that doesn't overclaim.

Start with the Master prompts ([#266–269](#)) for the full workflow; use the Standard prompts ([#270–286](#)) for one task at a time; reach for the Quick prompts ([#287–295](#)) when you have two minutes. Tap any title below to jump to it, and follow → **NEXT** to move to the next step.

[#266](#) Write the discussion: ten paragraphs, no overclaiming

[#267](#) Limitations that are honest, ranked and specific

[#268](#) Hostile-reviewer attack on my discussion

[#269](#) Conclusion, recommendations and summary chapter

[#270](#) The first paragraph of the discussion

[#271](#) Compare one finding with the literature

[#272](#) Why my result differs from previous studies

[#273](#) The mechanism paragraph, clearly labelled as speculation

[#274](#) Clinical implications without overreach

[#275](#) Strengths without inflation

[#276](#) Future research that someone could actually do

[#277](#) A conclusion that doesn't exceed the data

[#278](#) Scan my discussion for overclaiming

[#279](#) Who do my results apply to?

[#280](#) Discuss a surprising result

[#281](#) Discuss a non-significant primary result

[#282](#) Check the structure of my discussion

[#283](#) The "Summary" chapter of an Indian thesis

[#284](#) What my findings mean for Indian practice

[#285](#) Cut my discussion to the word limit

[#286](#) Practical recommendations from my thesis

[#287](#) Rewrite "further studies are needed"

[#288](#) Calibrate my hedging

[#289](#) One take-home message

[#290](#) Qualify every "significant"

[#291](#) Which way does this bias push my result?

[#292](#) Swap causal words for correct ones

[#293](#) One comparison sentence with a study

[#294](#) Is this really a strength?

[#295](#) Do my conclusions match my objectives?

Write the discussion: ten paragraphs, no overclaiming

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT

After results are final. This is the section examiners and reviewers read most closely.

WHY

The discussion is where good data gets ruined by overclaiming, and it is the only section where your scientific judgement is visible. A fixed structure (answer, compare, explain, implications, strengths, limitations, future, conclusion) keeps it honest and complete.

COPY THE PROMPT BELOW

PERSONA	You are a journal editor and senior academic who has written and revised hundreds of discussion sections and rejected many more. You are ruthless about over-interpretation, unsupported causal language, and limitation sections that hide the real weaknesses.
MY CONTEXT	Design: [state] · Sample size: [state] Primary objective: [state] Primary result with effect estimate and CI: [paste exactly as computed] Secondary results: [paste, with numbers] Findings that surprised me or contradicted my hypothesis: [state] Findings that were not significant: [state] Known limitations: [list] Setting: [Indian tertiary / rural / private, etc.] · Target: [thesis / journal name] I have uploaded the [N] most relevant published papers. Use only these, plus anything you retrieve through live search with a verifiable PMID or DOI.
TASK	Draft a discussion of about [800–1,200] words: 1. Paragraph 1: answer the research question in 1–3 sentences with the principal finding and its numbers. No literature. 2. Paragraphs 2–4: for each key comparison, state what the prior study found, whether mine agrees or differs, and the likely reason (population, setting, definitions, measurement, sample size, follow-up). End every comparison with an interpretive sentence. 3. Paragraph 5: the most plausible mechanism, marked clearly where speculative. 4. Paragraph 6: clinical implications: who could act on this and how; say plainly if it is not yet practice-changing. 5. Paragraph 7: strengths, without inflation. 6. Paragraph 8: limitations in order of threat to the conclusion, each with the likely direction of bias. 7. Paragraph 9: specific, doable future research, not "further studies are needed". 8. Paragraph 10: a 2–3 sentence conclusion that does not exceed the data.
RULES	Use associational language unless my design is randomised; "caused", "led to", "resulted in" and "prevented" are banned otherwise, because causal overreach is the most common reviewer criticism. Qualify every "significant" as statistical or clinical. Never call a non-significant result a "trend". Where a comparison needs a source I have not supplied, write [CITATION NEEDED — FIND SOURCE]. Introduce no result that is not in my results section.
OUTPUT	The discussion in paragraphs with no sub-headings unless my target requires them, within the word range.
TWEAK	Then list separately the three sentences a hostile reviewer would attack first, and how to rewrite each one defensively.

How to get more out of it

- › Give the model your actual numbers. A discussion written around vague findings reads exactly as vague as it was written.
- › The hostile-reviewer list is worth more than the draft. Act on it before submission.
- › If the draft sounds more confident than you feel about your own data, the draft is wrong, not your instinct.
- › → **NEXT #267** Limitations that are honest, ranked and specific

Limitations that are honest, ranked and specific

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT

When writing the limitations paragraph, and before your viva.

WHY

Examiners judge your scientific maturity by your limitations. A generic list ("small sample, single centre") signals you didn't think. Limitations ranked by threat, each with the direction of bias and what you did to reduce it, signal that you did.

COPY THE PROMPT BELOW

PERSONA	You are a clinical epidemiologist and journal reviewer who evaluates limitation sections for honesty and insight. You expect each limitation to state how it could have changed the result.
MY CONTEXT	Design and setting: [paste] How participants were selected and how many declined or were lost: [brief with numbers] How the exposure and outcome were measured, and by whom: [brief] Confounders measured and not measured: [lists] Sample size, and whether the study was powered for each reported outcome: [brief] Primary result: [estimate with CI] Limitations I have thought of: [list]
TASK	1. Identify all important limitations under selection, information (measurement), confounding, sample size or precision, and generalisability. 2. Rank them by how much they threaten my main conclusion. 3. For each, state the likely direction of bias (towards or away from the null, or unpredictable) and whether it could explain my result entirely. 4. For each, state what I did in the design or analysis to reduce it, if anything. 5. Write the limitations paragraph (180–250 words) in ranked order, honest but not self-defeating. 6. Separately, list 2 limitations I should acknowledge in the viva even if they are not in the thesis text.
RULES	Be specific to my study, because generic limitations suggest you haven't understood your own design. Don't invent limitations my design doesn't have, and don't hide real ones behind softer wording. Keep numbers exactly as I gave them.
OUTPUT	A ranked limitation table (limitation, type, direction of bias, mitigation, impact), then the paragraph, then the viva list.
TWEAK	End with a one-line statement of how confident a reader should be in my primary conclusion, given these limitations.

How to get more out of it

- › Start with the limitation that matters most, not the most comfortable one.
- › "Single-centre" is only a limitation if you explain why your centre might differ from others.
- › Pair each limitation with what the next study should do differently. It leads naturally into future research.
- › → **NEXT #268** Hostile-reviewer attack on my discussion

Hostile-reviewer attack on my discussion

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT / FACULTY

USE IT

When your discussion is drafted, before your guide or a journal sees it.

WHY

The kindest thing you can do for your discussion is let someone attack it first. A structured hostile review finds overclaims, missing comparisons and weak logic, and it hands you defensive rewrites you can use on paper and at the viva.

COPY THE PROMPT BELOW

PERSONA	You are Reviewer 2: a senior, sceptical peer reviewer for a high-impact clinical journal. You are fair but relentless, and you target over-interpretation, selective comparisons and conclusions that outrun the data.
MY CONTEXT	Design, sample size and primary result: [brief with numbers] Target: [thesis / journal name and its readership] My results section: [PASTE RESULTS] My discussion: [PASTE DISCUSSION]
TASK	1. Write a reviewer report on the discussion with major comments (threats to validity or interpretation) and minor comments (wording, clarity, missing references), each quoting the text it refers to. 2. Identify every sentence that claims more than the results show, and every causal claim unsupported by the design. 3. Identify comparisons with literature that seem selective, where only agreeing studies are cited, and say what kind of study is missing. 4. Identify results reported in the discussion that do not appear in the results. 5. Identify important limitations the discussion leaves out or understates. 6. For each major comment, write a defensive rewrite of the relevant text and a one-line response I could use in a reply to reviewers or at the viva.
RULES	Quote the exact text for every criticism, because unquoted criticism cannot be acted on. Rewrites must not add new data or references; use [CITE] slots where a reference is needed. Keep my numbers unchanged.
OUTPUT	The reviewer report (major, then minor), then a table (quoted sentence, problem, rewrite, one-line response).
TWEAK	End with an overall verdict (accept, minor revision, major revision) with the single change that would most improve the discussion.

How to get more out of it

- › Run this before your guide reads the draft. Their time is better spent on judgement than on catching overclaims.
- › Keep the responses. They are ready-made answers for viva questions about your interpretation.
- › Run it again after revising. The major comments should turn into minor ones.
- › → **NEXT #278** Scan my discussion for overclaiming

Conclusion, recommendations and summary chapter

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT

For the final chapters of your thesis: summary, conclusion and recommendations.

WHY

Many Indian theses end with separate summary, conclusion and recommendations chapters, and examiners read them first when time is short. They must match the objectives exactly and claim no more than the data shows.

COPY THE PROMPT BELOW

PERSONA	You are a senior examiner for MD/MS theses who reads the summary and conclusion first to judge whether the student understood their study. You insist that every conclusion maps to an objective and every recommendation is proportionate to the evidence.
MY CONTEXT	Thesis title: [paste] Objectives (numbered): [PASTE OBJECTIVES] Key results for each objective (with numbers): [PASTE RESULTS] Main limitations: [list] My university's required end chapters and word limits: [e.g. Summary 500 words, Conclusion, Recommendations] Design and setting: [brief]
TASK	1. Write the summary chapter: background (2 lines), aims, methods (design, setting, sample, key measurements), key results by objective with numbers, and the conclusion, within the word limit. 2. Write the conclusion: one numbered statement per objective, each answering it directly and in associational terms where appropriate. 3. Write recommendations in 3 groups: for clinical practice (only if justified), for the department or health system, and for future research, each specific and feasible. 4. Build a check table showing each objective, its conclusion statement and the result that supports it. 5. Flag any result that is important but not captured by any conclusion.
RULES	Every conclusion must answer an objective using a result I gave, because conclusions without matching results are a classic examiner criticism. Recommendations for practice must be proportionate: a single-centre observational study can suggest or support, not establish. Keep numbers exactly as supplied.
OUTPUT	The summary, the numbered conclusions, the grouped recommendations, then the check table.
TWEAK	End with the one conclusion statement most likely to be challenged at the viva, and a safer wording for it.

How to get more out of it

- Read your conclusion and your objectives side by side. They should mirror each other.
- Future-research recommendations should name the design, population and outcome.
- Keep the summary free of any number that doesn't appear in your results chapter.
- → **NEXT #283** The "Summary" chapter of an Indian thesis

270 The first paragraph of the discussion

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT To open your discussion.

PERSONA	You are a journal editor who wants the discussion to open with the answer, not with background.
MY CONTEXT	Primary objective: [paste] Primary result with numbers: [paste] Most important secondary result: [paste] Design: [brief]
TASK	1. Write the opening paragraph (60–100 words) stating the answer to the research question with the key numbers. 2. Add one sentence on the most important secondary finding. 3. Give an alternative version for a journal article of 50 words or fewer.
RULES	No literature, no background and no interpretation in this paragraph, and use associational language for observational designs, because the first sentence is read as your main claim.
OUTPUT	The two versions.

PRO TIP If you can't state your main finding in one sentence, your results section needs work first.

TRY IT WITH Primary: in-hospital mortality 18% vs 27% between early and late mobilisation groups (RR 0.67, 95% CI 0.45–0.99).

→ **NEXT #271** Compare one finding with the literature

271 Compare one finding with the literature

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT For each key comparison paragraph in your discussion.

PERSONA	You are a clinical academic who writes comparison paragraphs that interpret rather than list.
MY CONTEXT	My finding with numbers: [paste] Comparable studies I have read, with design, setting, population and result: [PASTE DETAILS OR UPLOAD PAPERS]
TASK	1. Compare my finding with each study: agree, partly agree or differ. 2. For differences, give the most likely explanations (population, setting, definitions, measurement, timing, sample size). 3. Write a paragraph of 120–180 words that compares and interprets, ending with what the comparison means.
RULES	Use only the studies I supplied, citing each with its reference number, because comparisons with unseen or invented studies are academic misconduct. Present disagreeing studies too, not only supporting ones.
OUTPUT	A comparison table, then the paragraph.

PRO TIP One paragraph per key finding is a good discipline. Don't mix three findings in one paragraph.

TRY IT WITH My finding: prevalence of sarcopenia 38% in COPD; three studies reported 15%, 25% and 40%.

→ **NEXT #272** Why my result differs from previous studies

272 Why my result differs from previous studies

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT When your result disagrees with published findings and you must explain it.

PERSONA	You are an epidemiologist skilled at explaining discrepancies between studies without special pleading.
MY CONTEXT	My study: design, population, setting, definitions, sample size and result: [brief] The study or studies I differ from, with the same details: [brief]
TASK	1. Compare the studies systematically on each methodological dimension. 2. Rank the plausible explanations for the difference. 3. Consider chance (sample size, CI overlap) explicitly. 4. Write a discussion paragraph presenting the difference and the most likely explanations, without dismissing either result.
RULES	Don't claim my study is "better" without a specific methodological reason, and consider that my own study might be the outlier, because examiners respect balanced reasoning.
OUTPUT	The comparison table, the ranked explanations, then the paragraph.

PRO TIP Overlapping confidence intervals often mean the "difference" isn't really a difference. Check that first.

TRY IT WITH My study found no association between vitamin D and TB outcome; two earlier studies found one.

→ **NEXT #271** Compare one finding with the literature

273 The mechanism paragraph, clearly labelled as speculation

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT When your discussion needs a biological or clinical explanation for your finding.

PERSONA	You are a clinician-scientist who explains mechanisms carefully and labels speculation clearly.
MY CONTEXT	My finding: [paste] Relevant physiology or mechanisms I know of, with sources: [PASTE NOTES WITH CITATIONS]
TASK	1. Suggest the 2–3 most plausible mechanisms for my finding. 2. For each, say how strong the supporting evidence is (established, plausible, speculative). 3. Write a paragraph of 100–150 words presenting them with hedged language. 4. Suggest what study could test the leading mechanism.
RULES	Mark every mechanistic claim needing support as [CITE], and phrase speculation as speculation ("may", "could", "one possible explanation"), because mechanisms presented as fact are a common overreach.
OUTPUT	The mechanism list with evidence strength, then the paragraph, then the testing suggestion.

PRO TIP Keep this paragraph short. Examiners value an honest "the mechanism is unclear" over a confident guess.

TRY IT WITH Finding: low serum magnesium associated with higher ICU mortality.

→ **NEXT #274** Clinical implications without overreach

274 Clinical implications without overreach

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT For the implications paragraph of your discussion.

PERSONA	You are a guideline-panel member who judges how far a single study can change practice.
MY CONTEXT	Main findings: [paste] Design, sample size and setting: [brief] Who could use the findings (clinicians, administrators, programmes): [list]
TASK	1. State what the findings could mean for practice, and for whom. 2. Distinguish what can be acted on now, what needs confirmation, and what is hypothesis-generating. 3. Write the implications paragraph (100–150 words). 4. Flag any implication that overreaches the design.
RULES	Use proportionate verbs (suggests, supports, may inform) for single-centre observational results, because "should be adopted" claims are rarely justified by one study.
OUTPUT	The three-level classification, the paragraph, then the overreach flags.

PRO TIP Name the specific clinical decision your findings touch, e.g. "whether to repeat lactate at 6 hours".

TRY IT WITH Finding: a bedside frailty score predicted post-operative delirium in elderly surgical patients.

→ **NEXT #275** Strengths without inflation

275 Strengths without inflation

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT For the strengths paragraph of your discussion.

PERSONA	You are a sceptical reviewer who dislikes inflated strength statements.
MY CONTEXT	Design features: [e.g. prospective, consecutive recruitment, blinded outcome assessment, validated tools, low loss to follow-up] Anything unique about my setting or population: [brief]
TASK	1. Identify genuine methodological strengths and explain why each matters for validity. 2. Remove claims that are not real strengths (e.g. "first study in India" unless verified, or "large sample" when it isn't). 3. Write the strengths paragraph (80–120 words).
RULES	Tie each strength to a bias it reduces, because a strength without a reason is just a claim. Do not claim novelty unless I confirm it has been checked; write [VERIFY NOVELTY] instead.
OUTPUT	A strengths table (feature, bias reduced), the removed claims, then the paragraph.

PRO TIP "Prospective, consecutive recruitment with complete follow-up" is a real strength. Say it plainly.

TRY IT WITH A prospective cohort of 180 patients with 97% follow-up and blinded outcome assessment.

→ **NEXT #267** Limitations that are honest, ranked and specific

276 Future research that someone could actually do

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT Instead of writing "further studies are needed".

PERSONA	You are a research funder who only backs specific, feasible next studies.
MY CONTEXT	My findings and limitations: [brief] Open questions my study raised: [list] My setting and the resources a follow-up study could realistically use: [brief]
TASK	1. Propose 3–4 specific future studies, each with design, population, setting, outcome and why it follows from my findings. 2. Rank them by importance and feasibility, and say which one could be a thesis for a junior resident in my department. 3. Write the future-research paragraph (80–120 words).
RULES	Make each proposal specific enough to become someone's thesis topic, because vague calls for research add nothing.
OUTPUT	A proposal table, then the paragraph.

PRO TIP Link each future study to one of your limitations. It shows you understand what your study couldn't answer.

TRY IT WITH My cross-sectional study found an association between sleep apnoea and poor asthma control.

→ **NEXT #277** A conclusion that doesn't exceed the data

277 A conclusion that doesn't exceed the data

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT For the final paragraph of your discussion or abstract.

PERSONA	You are a journal editor who rewrites overreaching conclusions every day.
MY CONTEXT	Primary objective: [paste] Primary result: [paste] Design and setting: [brief] My current conclusion: [paste] Target: [thesis / journal abstract / poster]
TASK	1. Check my conclusion against the result and design, and list every overreach. 2. Rewrite it in 2–3 sentences: what was found, in whom, and what it suggests. 3. Write a one-sentence version for the abstract.
RULES	Match the strength of language to the design and precision of the result, because the conclusion is the most-quoted sentence of any study.
OUTPUT	The overreach list, the rewritten conclusion, then the abstract sentence.

PRO TIP Your conclusion should answer your primary objective in the same words the objective used.

TRY IT WITH Current: "Early antibiotics save lives in sepsis." Design: retrospective cohort.

→ **NEXT #269** Conclusion, recommendations and summary chapter

278 Scan my discussion for overclaiming

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT As a final check on any discussion or conclusion.

PERSONA	You are a research-integrity editor who scans manuscripts for spin.
MY CONTEXT	My design: [design] My primary result: [estimate, CI, p] My discussion text: [PASTE DISCUSSION]
TASK	1. Flag causal language used with an observational design. 2. Flag spin: "trend towards", "marginally significant", emphasis on secondary results over a null primary result, and extrapolation beyond the population. 3. Flag unqualified "significant". 4. Give a corrected version of each flagged sentence.
RULES	Keep the meaning where it is justified and change only the overreaching wording, because the goal is accuracy, not timidity.
OUTPUT	A table (sentence, type of spin, corrected sentence), then a count by type.

PRO TIP Search your text for "caus", "effect", "impact", "prevent", "led to", "due to". Most overclaims hide there.

TRY IT WITH A discussion claiming a supplement "improved outcomes" based on a before-after study.

→ **NEXT #268** Hostile-reviewer attack on my discussion

279 Who do my results apply to?

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT For the generalisability part of your limitations or implications.

PERSONA	You are an epidemiologist who thinks carefully about external validity.
MY CONTEXT	Setting and how patients reached my hospital: [brief] Population characteristics (severity, age, comorbidity, socio-economic profile): [brief] Exclusions: [list] Proportion of screened patients who were enrolled: [numbers]
TASK	1. Describe the population my results apply to most directly. 2. Identify populations to whom they may not apply, and why. 3. Discuss referral bias for a tertiary centre, if relevant, and the likely direction of its effect on my estimates. 4. Write a generalisability paragraph (80–120 words).
RULES	Base the assessment on my actual setting and exclusions, because generic statements about "single-centre studies" add nothing.
OUTPUT	Headed notes, then the paragraph.

PRO TIP Tertiary-centre patients are usually sicker. Say whether that makes your estimates higher or lower than in district practice.

TRY IT WITH ICU cohort at a referral hospital receiving transfers from 6 districts.

→ **NEXT #267** Limitations that are honest, ranked and specific

280 Discuss a surprising result

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT When a result contradicts your hypothesis or expectations.

PERSONA	You are a thoughtful senior researcher who treats surprising results as information, not embarrassment.
MY CONTEXT	What I expected, and why: [brief] What I found, with numbers: [paste] Possible issues with my data or methods: [brief]
TASK	1. List the explanations in order: chance, bias, confounding, measurement error, and a genuine effect. 2. Evaluate each against my study details. 3. Suggest a check I can still run (e.g. a sensitivity analysis) if appropriate. 4. Write a paragraph (100–150 words) presenting the finding honestly.
RULES	Don't explain away the finding, and don't overstate it, because examiners respect a balanced treatment of surprises.
OUTPUT	The explanations table, the check, then the paragraph.

PRO TIP Surprising findings often make the best future-research questions. Say so.

TRY IT WITH Expected obesity to worsen ARDS outcomes; found lower mortality in obese patients.

→ **NEXT #266** Write the discussion: ten paragraphs, no overclaiming

281 Discuss a non-significant primary result

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT When your primary outcome was not statistically significant.

PERSONA	You are a statistician-editor who helps authors write honest, publishable null-result discussions.
MY CONTEXT	Primary result (estimate, 95% CI, p): [paste] Sample size and whether it met the calculated target: [brief] Expected effect used for the sample size: [brief]
TASK	1. Interpret the CI: what effect sizes are ruled out, and what remain plausible. 2. Discuss power: whether the study could have detected a clinically important effect. 3. Compare with the literature honestly. 4. Write the opening paragraph of the discussion for this result.
RULES	Don't call the result a trend or imply a hidden effect, and don't call it proof of no effect, because both misrepresent a null result.
OUTPUT	The CI interpretation, the power note, then the paragraph.

PRO TIP "Our study was unable to demonstrate..." is better than "there was no difference".

TRY IT WITH RR 0.85 (95% CI 0.62–1.17), p = 0.32, sample of 180 against a target of 200.

→ **NEXT #266** Write the discussion: ten paragraphs, no overclaiming

282 Check the structure of my discussion

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT When your discussion is drafted and feels disorganised.

PERSONA	You are a thesis-writing coach who maps discussions against a standard structure.
MY CONTEXT	My discussion: [PASTE DISCUSSION] My primary result: [paste] Target structure, if my guide or journal prescribes one: [brief]
TASK	1. Map each paragraph to a structural element: principal findings, comparison, explanation, mechanism, implications, strengths, limitations, future research or conclusion. 2. Identify missing, repeated or misplaced elements. 3. Suggest a reordering. 4. Check that the first paragraph states the principal finding and the last states a proportionate conclusion.
RULES	Suggest structural changes only, keeping my text as it is, because the content and voice must remain mine.
OUTPUT	A paragraph map table, the problems, then the suggested order.

PRO TIP If your discussion opens with background, move it. The first paragraph must answer the question.

TRY IT WITH A discussion that opens with 2 paragraphs of epidemiology before mentioning the results.

→ **NEXT #266** Write the discussion: ten paragraphs, no overclaiming

283 The "Summary" chapter of an Indian thesis

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT When your university requires a separate summary chapter.

PERSONA	You are an MD/MS thesis examiner who reads the summary first.
MY CONTEXT	Title and objectives: [paste] Methods in brief: [paste] Key results by objective, with numbers: [paste] Conclusion: [paste] Word or page limit: [e.g. 500 words]
TASK	1. Write the summary within the limit, with short paragraphs for background, aims, methods, results and conclusion. 2. Keep results in the order of the objectives, with the key numbers. 3. Check that every number appears in my results chapter.
RULES	Add no information that isn't elsewhere in the thesis, because the summary must be a faithful miniature of it.
OUTPUT	The summary with its word count, then the number check.

PRO TIP Write the summary last and read it aloud. If it doesn't flow, your thesis argument doesn't either.

TRY IT WITH Summary for a thesis on thyroid dysfunction in chronic kidney disease, 500 words.

→ **NEXT #269** Conclusion, recommendations and summary chapter

284 What my findings mean for Indian practice

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT / FACULTY

USE IT For a paragraph relating your results to Indian clinical practice, programmes or policy.

PERSONA	You are a public-health physician familiar with Indian health programmes and resource settings.
MY CONTEXT	My findings: [brief] Relevant national programmes, guidelines or practices I know of: [list with sources] Resource context of typical Indian hospitals for this condition: [brief]
TASK	1. Relate the findings to current Indian practice and resource realities. 2. Identify which national programme or guideline, if any, the findings are relevant to. 3. Discuss feasibility of acting on the findings in district and primary settings. 4. Write the paragraph (100–150 words).
RULES	Mention a programme or guideline only if I listed it or you mark it [VERIFY PROGRAMME / GUIDELINE], because misnamed programmes undermine credibility.
OUTPUT	Headed notes, then the paragraph.

PRO TIP Examiners in India value this paragraph. It shows you thought beyond your own hospital.

TRY IT WITH Finding: low-cost bedside ultrasound detected pneumonia as accurately as X-ray in children.

→ **NEXT #274** Clinical implications without overreach

285 Cut my discussion to the word limit

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT When your discussion is over the limit for your thesis or target journal.

PERSONA	You are a journal editor who cuts discussions without losing substance.
MY CONTEXT	My discussion: [PASTE DISCUSSION] Current and target word count: [e.g. 1,900 → 1,200] Parts I must keep: [list]
TASK	1. Identify repetition of results, background that belongs in the introduction, over-long literature summaries and redundant limitations. 2. Produce the trimmed discussion at or under the target. 3. List what was cut, so I can restore anything essential.
RULES	Keep all citations with their claims and all numbers unchanged, because trimming must not alter the evidence. Keep the standard structure intact.
OUTPUT	The trimmed discussion with its word count, then the cut list.

PRO TIP Repeated results are the easiest cut. The discussion should interpret results, not restate them.

TRY IT WITH A 2,000-word thesis discussion needing to become a 1,100-word journal discussion.

→ **NEXT #268** Hostile-reviewer attack on my discussion

286 Practical recommendations from my thesis

STANDARD

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

USE IT When your thesis format requires a recommendations section.

PERSONA	You are a health-policy adviser who writes recommendations that are proportionate to the evidence.
MY CONTEXT	Key findings: [brief] Design strength and limitations: [brief] Stakeholders: [clinicians, hospital, programme, researchers] My university's format for this section, if any: [brief]
TASK	1. Write 4–6 recommendations grouped by stakeholder. 2. Grade each by the strength of support from my study (strong, moderate or preliminary). 3. Make each specific and feasible, with who should act, how, and what it would cost in effort.
RULES	Base practice recommendations only on results my design can support, and phrase the rest as research recommendations, because examiners question recommendations that outrun the data.
OUTPUT	The grouped, graded recommendations.

PRO TIP One specific, feasible recommendation is worth more than five generic ones.

TRY IT WITH Thesis finding: 40% of diabetic patients had undiagnosed peripheral neuropathy in OPD screening.

→ **NEXT #269** Conclusion, recommendations and summary chapter

QUICK PROMPTS · 1.9

287 Rewrite "further studies are needed"

QUICK

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

Act as a journal editor. Replace the phrase "further studies are needed" in my discussion with one specific sentence naming the design, population, outcome and reason, based on my study's main limitation, [LIMITATION], and main finding, [FINDING]. Give 2 alternatives.

TRY IT WITH Limitation: single-centre, 6-week follow-up.
Finding: early mobilisation reduced delirium.

288 Calibrate my hedging

QUICK

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

Act as an academic editor. For each of these sentences, [PASTE SENTENCES], tell me whether the hedging is too strong, too weak or right for my design [DESIGN], and rewrite it at the right level. Keep claims supported and avoid both overclaiming and excessive qualifiers.

TRY IT WITH "It is possible that this may perhaps suggest..." and "This proves that..."

289 One take-home message

QUICK

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT / MBBS STUDENT

Act as a senior clinician giving a grand-rounds summary. From my findings, [PASTE KEY FINDINGS], write the single take-home message a busy clinician on the ward should remember, in one sentence of under 25 words, accurate to my design and not overstated.

TRY IT WITH Findings on the association between delayed antibiotic administration and mortality in sepsis.

290 Qualify every "significant"

QUICK

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

Act as a statistical copy-editor. In this text, [PASTE TEXT], find every use of "significant" or "significantly" and rewrite each to say whether it means statistically significant (with the p value or CI) or clinically important. Return the corrected text with changes highlighted.

TRY IT WITH A discussion with "significantly higher", "a significant finding" and "significant patients".

291 Which way does this bias push my result?

QUICK

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

Act as an epidemiologist examining my thesis. For this limitation, [LIMITATION], in my study of [EXPOSURE] and [OUTCOME], tell me whether it likely biases my estimate towards or away from the null, why, and give one sentence I can use in my limitations paragraph.

TRY IT WITH Limitation: non-differential misclassification of smoking status by self-report.

292 Swap causal words for correct ones

QUICK

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

Act as an epidemiology editor. Rewrite these sentences, [PASTE SENTENCES], replacing causal words (causes, leads to, results in, prevents, due to) with accurate associational language suited to my [observational] design, while keeping the meaning as strong as the evidence allows.

TRY IT WITH "Anaemia leads to poor wound healing" from a cross-sectional study.

293 One comparison sentence with a study

QUICK

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

Act as a thesis writing coach. Write one discussion sentence comparing my result, [MY RESULT WITH NUMBERS], with this study's result, [THEIR RESULT, DESIGN, SETTING, CITATION NUMBER], noting agreement or difference and one likely methodological reason. Keep it under 45 words.

TRY IT WITH My prevalence 34% vs their 22% in a community sample.

294 Is this really a strength?

QUICK

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

Act as a sceptical reviewer. For each strength I plan to claim, [PASTE LIST], tell me whether it is a genuine methodological strength, what bias it reduces, or why it isn't really a strength. Suggest an accurate wording for each one worth keeping.

TRY IT WITH "First study of its kind", "large sample of 90 patients", "prospective design".

295 Do my conclusions match my objectives?

QUICK

1.9 DISCUSSION, CONCLUSION & LIMITATIONS · PG RESIDENT

Act as a thesis examiner. Compare my objectives, [PASTE OBJECTIVES], with my conclusions, [PASTE CONCLUSIONS]. For each objective, say whether a conclusion answers it directly, partly or not at all, and suggest the missing or corrected conclusion statement for each gap.

TRY IT WITH Three objectives but only two conclusions, one of which introduces a new claim.

SECTION 1.10

1.10

Protocol, ethics & synopsis

The protocol is the document you will be held to for three years, and the IEC is the gate you must pass first. Decide everything in advance.

20

prompts

3

Master

11

Standard

6

Quick

HOW TO USE THIS SECTION

Use it before the IEC meeting and your synopsis submission.

Start with the Master prompts (**#296–298**) for the full workflow; use the Standard prompts (**#299–309**) for one task at a time; reach for the Quick prompts (**#310–315**) when you have two minutes. Tap any title below to jump to it, and follow → **NEXT** to move to the next step.

#296 Complete study protocol, IEC-ready

#297 IEC submission pack and pre-review

#298 Participant information sheet and consent form

#299 The ethics section of my protocol or thesis

#300 Respond to IEC queries

#301 CTRI registration, field by field

#302 University synopsis in the required format

#303 Month-by-month thesis timeline

#304 Realistic thesis budget

#305 Risk-benefit assessment and adverse-event plan

#306 Confidentiality and data-protection plan

#307 Request a waiver of consent

#308 Protocol amendment request to the IEC

#309 Funding and conflict-of-interest declarations

#310 Rewrite a consent paragraph in plain language

#311 What documents does my IEC need?

#312 Does my study involve vulnerable participants?

#313 Compensation and insurance: what do I write?

#314 Assent form for children

#315 Version control for my study documents

Complete study protocol, IEC-ready

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

USE IT

Before ethics committee submission and before university synopsis deadlines.

WHY

The protocol is the document you will be held to for the next three years. Ambiguity here becomes an unanswerable question at your viva. This prompt makes you decide everything in advance, in the format Indian IECs expect.

COPY THE PROMPT BELOW

PERSONA	You are a clinical trials methodologist and ethics committee member who has reviewed several hundred postgraduate protocols. You know exactly which sections get sent back for revision and why. You write in formal scientific register suitable for institutional submission.
MY CONTEXT	Title: [working title] Research question (PICOT): [paste] Design: [prospective observational cohort / single-arm interventional / randomised controlled / diagnostic accuracy / cross-sectional] Setting: [hospital, unit, catchment] Duration approved for the study: [X months] Expected eligible patients per month: [N] Primary outcome and how it is measured: [state] Secondary outcomes: [list] Intervention or exposure, fully described (dose, timing, device, operator): [describe] Comparator: [describe]
TASK	Draft a complete protocol in this order, to a standard that would pass institutional ethics review at an Indian medical college: 1. Title-page fields for institutional submission. 2. Structured synopsis of 300 words. 3. Introduction and rationale ending in an explicit gap statement, with [CITE] slots. 4. Aims and objectives: one primary, numbered secondaries, each measurable and time-bound. 5. Null and alternative hypotheses. 6. Methodology: design and justification; setting and period; source population; operationally defined inclusion criteria; exclusion criteria with reasons; sampling and how it will be implemented; sample size (formula, every term, inputs marked [INSERT FROM PUBLISHED SOURCE — VERIFY], attrition); randomisation, concealment and blinding if applicable; the study procedure as a patient journey; operational definitions of every variable; data collection tools and who collects what; quality control and inter-observer agreement. 7. Statistical analysis plan in outline, with tests named against objectives. 8. Ethical considerations: consent (and assent), confidentiality and data storage, risks and mitigation, benefits, adverse-event provisions, compensation, conflicts of interest, and withdrawal. 9. Informed consent document in plain language (about eighth-grade reading level), suitable for translation. 10. A month-by-month Gantt-style timeline from ethics approval to thesis submission. 11. A budget with realistic line items. 12. CTRI registration requirements: what, when, and which fields will be difficult. 13. Limitations to acknowledge up front. 14. A numbered placeholder reference list.
RULES	Use [CITE] and [VERIFY] tags instead of any invented reference, prevalence figure, effect size or guideline statement, because the IEC checks them. Flag every section where an ethics committee is likely to raise a query, and say how to phrase it to avoid one. State methodological weaknesses plainly rather than writing around them. Formal scientific language throughout.
OUTPUT	The full protocol with numbered headings, then a separate list of likely IEC queries with pre-emptive fixes.
TWEAK	Finish with a follow-up I should run: "Act as the ethics committee reviewer and list every objection you would raise to this protocol."

How to get more out of it

- The operational-definitions section is where most protocols are weak. Don't shorten it.
- Get the sample size confirmed by a statistician. AI sample-size output is a starting draft, never the final number.
- Keep one master version with a version number and date. Every committee copy must match it.
- → NEXT #297 IEC submission pack and pre-review

IEC submission pack and pre-review

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT / FACULTY

USE IT

When your protocol is final and you are preparing the ethics committee submission.

WHY

IEC rejections and revision cycles cost months. Most queries are predictable: consent wording, risk-benefit, vulnerable participants, data protection, compensation and inconsistency between documents. This prompt reviews your pack the way the committee will.

COPY THE PROMPT BELOW

PERSONA	You are the member-secretary of an institutional ethics committee at an Indian medical college, experienced with the ICMR National Ethical Guidelines and the committee's usual queries. You review submissions strictly but constructively.
MY CONTEXT	Study type and risk level (in my view): [observational / minimal risk / greater than minimal risk] Participants, including any vulnerable groups: [brief] Protocol summary: [PASTE SUMMARY] Consent documents (English version): [PASTE PIS AND CONSENT FORM] Data handling and storage plan: [brief] Funding and conflicts of interest: [brief] Documents I plan to submit: [list]
TASK	1. Classify the likely review category (exempt, expedited or full committee) with reasoning, marked [CONFIRM WITH YOUR IEC]. 2. Build a submission checklist of documents Indian IECs commonly require (application form, protocol, PIS and consent in English and local language, CRF, investigator CVs and GCP certificates, undertakings, budget, CTRI plan) and mark what I am missing. 3. Review the consent documents for completeness (purpose, procedures, duration, risks, benefits, alternatives, confidentiality, voluntariness, withdrawal, compensation, contacts) and readability. 4. Review the risk-benefit assessment and the adverse-event plan. 5. Check consistency of title, objectives, sample size and procedures across all documents. 6. Write the 10 queries this committee is most likely to raise, each with a model response and the document change that resolves it.
RULES	Refer to the current ICMR National Ethical Guidelines and mark every specific rule you cite as [VERIFY IN ICMR GUIDELINES], because committees check citations against the source. Don't guarantee approval or classification; the committee decides.
OUTPUT	The category note, the checklist table (document, required, status), the consent review, the risk-benefit note, the consistency table, then the 10 queries with responses.
TWEAK	End with the 3 changes that would most reduce the chance of a revision cycle.

How to get more out of it

- › Mismatched titles and sample sizes between documents are the commonest avoidable query.
- › Get the local-language consent form translated and back-translated. Committees check this.
- › Keep your GCP training certificate current. Many IECs require it from all investigators.
- › → **NEXT #300** Respond to IEC queries

Participant information sheet and consent form

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

USE IT

While preparing your IEC submission.

WHY

A consent form written in dense medical English is not informed consent. Committees reject forms that are too technical, incomplete or inconsistent with the protocol. A plain-language, complete, translation-ready document gets approved and actually informs patients.

COPY THE PROMPT BELOW

PERSONA	You are a research-ethics specialist who writes participant information sheets and consent forms for Indian hospital studies. You write at about an eighth-grade reading level, in short sentences, with no jargon.
MY CONTEXT	Study title (formal and a lay version): [paste] What participants will experience, step by step: [paste] Time required and number of visits: [brief] Samples, tests or procedures, including any extra blood volume: [brief] Risks and discomforts, and benefits (direct and indirect): [brief] Alternatives to participation: [brief] Data confidentiality arrangements: [brief] Compensation for participation or injury: [brief] Investigator and IEC contact details: [placeholders] Participants who need a legally acceptable representative, or assent (children): [brief] Local language(s) needed: [e.g. Hindi, Marathi]
TASK	1. Write the participant information sheet under simple headings: why this study, what will happen, how long, risks, benefits, alternatives, privacy, your rights, compensation, and who to contact. 2. Write the consent form with the standard statements to initial, plus signature, thumb-impression and witness sections, and an impartial-witness provision for participants who cannot read. 3. If relevant, write an LAR consent version and a short assent form for children. 4. Check readability: flag any sentence over 20 words and any medical term, with a simpler version. 5. Give translation and back-translation instructions for the local language versions.
RULES	Use only the procedures, risks and arrangements I described, because anything added or omitted makes the form inconsistent with the protocol. Mark every regulatory requirement you mention as [VERIFY IN ICMR GUIDELINES]. Never promise benefits or confidentiality beyond what the protocol provides.
OUTPUT	The information sheet, the consent form, the LAR and assent versions if needed, the readability flags table, then the translation instructions.
TWEAK	End with the 3 questions a patient is most likely to ask when you take consent, with simple answers.

How to get more out of it

- Read the form aloud to a non-medical relative. Every place they frown needs rewriting.
- Say "you may stop at any time without affecting your treatment" in every language version.
- Keep a signed copy for the participant. Many IECs require this.
- → **NEXT #310** Rewrite a consent paragraph in plain language

299 The ethics section of my protocol or thesis

STANDARD

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

USE IT When writing the ethical considerations section.

PERSONA	You are an ethics committee member at an Indian government medical college who writes clear, compliant ethics sections.
MY CONTEXT	I am conducting a [STUDY DESIGN] study on [TOPIC] at a government medical college in India. My participants are [e.g. adult COPD patients]. Data collection involves [e.g. blood tests, interviews, chart review, imaging].
TASK	Write the ethics section covering: 1. The type of informed consent required and how it will be obtained. 2. The data confidentiality and anonymisation plan. 3. The risk-benefit analysis for participants. 4. Vulnerable populations, if applicable. 5. Compliance with the ICMR National Ethical Guidelines. 6. A statement on conflicts of interest.
RULES	Write in formal academic language suitable for IEC submission, and mark the guideline edition and any specific provision as [VERIFY CURRENT ICMR EDITION], because guideline editions are updated and committees check them.
OUTPUT	Six headed paragraphs, about 300 words in total.

PRO TIP Mirror the committee's own application-form headings where you can. It makes the reviewer's job easier.

TRY IT WITH Cross-sectional study using blood tests and questionnaires in adults with type 2 diabetes.

→ **NEXT #297** IEC submission pack and pre-review

300 Respond to IEC queries

STANDARD

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

USE IT When the ethics committee returns your submission with queries.

PERSONA	You are an experienced investigator who writes clear, respectful, point-by-point responses to ethics committees.
MY CONTEXT	The committee's queries, verbatim: [PASTE QUERIES] Relevant sections of my protocol and consent form: [PASTE SECTIONS]
TASK	1. Draft a point-by-point response letter: each query quoted, then my response, then the exact change made with page and section. 2. Suggest the revised protocol or consent text for each change. 3. Flag any query I should discuss with my guide before responding. 4. Draft the covering letter.
RULES	Address every query fully, and never argue with a query without a clear, respectful reason, because unresolved queries delay approval. Propose only changes consistent with my study.
OUTPUT	The response table (query, response, change, location), the revised text, the flags, then the covering letter.

PRO TIP Highlight all changes in the revised documents and submit a clean copy as well.

TRY IT WITH Queries on consent for illiterate patients, blood volume per visit, and data storage.

→ **NEXT #308** Protocol amendment request to the IEC

301 CTRI registration, field by field

STANDARD

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT / FACULTY

USE IT When your study must be registered with the Clinical Trials Registry – India.

PERSONA	You are a research coordinator who has registered many studies on CTRI.
MY CONTEXT	Study type: [interventional / observational] Protocol summary: [PASTE SUMMARY] IEC approval status: [approved on date / pending] Sites and investigators: [brief]
TASK	1. Tell me whether my study needs CTRI registration and when (before enrolling the first participant), marked [VERIFY CURRENT CTRI AND ICMR REQUIREMENTS]. 2. Walk through the main registration fields and draft my entries: titles, sponsors, sites, conditions, intervention and comparator, inclusion and exclusion criteria, method of allocation and concealment, blinding, primary and secondary outcomes with time points, target sample size, phase, dates and brief summary. 3. Flag fields residents commonly get wrong. 4. Explain what happens after submission and how to update the record.
RULES	Keep every entry identical to my approved protocol, because discrepancies between the registry and the thesis are flagged by journals. Mark any field name you are unsure of as [CHECK CURRENT FORM].
OUTPUT	The requirement note, a field table with draft entries, the common-errors list, then the post-submission steps.

PRO TIP Journals increasingly require prospective registration for trials. Register before the first participant, not after.

TRY IT WITH Single-centre RCT of two physiotherapy protocols after knee replacement.

→ **NEXT #296** Complete study protocol, IEC-ready

302 University synopsis in the required format

STANDARD

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

USE IT When your university or college requires a thesis synopsis for approval.

PERSONA	You are a university thesis-cell member who reviews postgraduate synopses and knows the common format.
MY CONTEXT	My university and its synopsis format or headings, if known: [paste or "standard format"] Protocol details: [PASTE PROTOCOL OR KEY SECTIONS] Word or page limit: [e.g. 8 pages]
TASK	1. Arrange the content under the required headings (typically title, introduction, need for the study, aims and objectives, review of literature, materials and methods, and references). 2. Condense each section to fit the limit. 3. Check consistency with the protocol and IEC documents. 4. List the attachments the synopsis usually needs (guide's certificate, IEC approval, proforma).
RULES	Keep objectives, sample size and title identical to the IEC submission, because universities cross-check them. Mark headings I haven't confirmed with my university as [CHECK YOUR UNIVERSITY FORMAT].
OUTPUT	The formatted synopsis, then the consistency check and attachment list.

PRO TIP Ask a senior for an approved synopsis from your department. Matching the local format saves a revision.

TRY IT WITH MD Anaesthesiology synopsis for a prospective RCT on post-operative nausea.

→ **NEXT #85** Short background for a synopsis or protocol

303 Month-by-month thesis timeline

STANDARD

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

USE IT For your protocol, synopsis and personal planning.

PERSONA	You are a thesis project manager who builds realistic timelines with buffers.
MY CONTEXT	Course start and thesis submission deadline: [dates] Expected IEC approval date: [date] Recruitment rate and target: [N per month, total] Follow-up duration: [months] Known busy periods (exams, rotations, postings): [list]
TASK	1. Build a month-by-month timeline: approvals, registration, pilot, recruitment, follow-up, data entry, analysis, writing each chapter, guide review, corrections, binding and submission. 2. Add buffers where delays are likely. 3. Calculate whether recruitment fits, and flag it if it doesn't. 4. Present it as a Gantt-style table (tasks against months).
RULES	Build in at least one month of buffer before submission, and say if my deadline is unrealistic, because optimistic timelines are the commonest reason for late theses.
OUTPUT	The Gantt-style table, then the feasibility check and buffer notes.

PRO TIP Start writing the introduction and ROL during recruitment. Waiting until data collection ends is the classic mistake.

TRY IT WITH MD course starting July 2025, submission due January 2028, recruitment 8 per month for 120 patients.

→ **NEXT #18** What could kill this topic? A risk register

304 Realistic thesis budget

STANDARD

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT / FACULTY

USE IT For your protocol, IEC form or a small grant application.

PERSONA	You are a research finance officer who builds realistic budgets for investigator-initiated studies.
MY CONTEXT	Tests, consumables and procedures per participant, beyond routine care: [list] Sample size: [N] Other costs: [printing, stationery, software, travel, translation, statistician, publication] Funding source: [self / department / grant]
TASK	1. Build a line-item budget: per-participant costs times N, plus fixed costs. 2. Mark who pays each item (hospital, investigator, grant). 3. Add a contingency line. 4. Flag any cost that would fall on patients, and how to avoid it.
RULES	Use placeholder unit costs marked [INSERT LOCAL COST] rather than inventing prices, because costs vary widely between institutions. Participants must not bear research-only costs, and the IEC will check this.
OUTPUT	The budget table, then the patient-cost flags.

PRO TIP Many state and national bodies offer small grants for postgraduate theses. Check your university's research cell.

TRY IT WITH 100 patients, each needing one extra serum biomarker test and a 6-month follow-up phone call.

→ **NEXT #296** Complete study protocol, IEC-ready

305 Risk-benefit assessment and adverse-event plan

STANDARD

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

USE IT When your study involves any procedure or intervention beyond routine care.

PERSONA	You are a clinical research safety officer who writes proportionate risk-benefit and adverse-event sections.
MY CONTEXT	Study procedures beyond routine care: [list] Participant population: [brief] Intervention, if any, and its known side effects: [brief]
TASK	1. List the risks of each study-specific procedure, with likelihood and severity. 2. Describe mitigation for each. 3. Describe direct and indirect benefits realistically. 4. Write the adverse-event plan: definitions, recording, grading, reporting timelines to the IEC, and management. 5. Summarise the overall risk category with justification.
RULES	Mark reporting timelines and definitions as [VERIFY IN ICMR / IEC SOP], because reporting requirements differ by study type and institution. Don't list routine-care risks as study risks.
OUTPUT	A risk table, the benefits paragraph, the AE plan, then the summary.

PRO TIP If a procedure is part of routine care anyway, say so clearly. It lowers the perceived risk accurately.

TRY IT WITH Study adding one extra 5 mL blood draw and a 6-minute walk test for COPD patients.

→ **NEXT #297** IEC submission pack and pre-review

306 Confidentiality and data-protection plan

STANDARD

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

USE IT For the confidentiality section of your IEC submission and protocol.

PERSONA	You are a health-data governance adviser familiar with Indian research-ethics requirements and data-protection principles.
MY CONTEXT	Data collected (identifiable and study data): [list] Where and how it will be stored: [brief] Who has access: [brief] Any data leaving the institution (cloud, AI tools, collaborators): [brief]
TASK	1. Describe de-identification: study IDs, a separate linking log, and restricted access. 2. Describe secure storage, backup, encryption and retention. 3. Describe the rules for using AI tools and cloud services with study data. 4. Describe what happens at study end (retention period, destruction). 5. Write the confidentiality paragraph for my protocol.
RULES	Recommend that no identifiable data ever be entered into consumer AI tools, and mark any legal requirement you mention (e.g. the Digital Personal Data Protection Act) as [VERIFY CURRENT LAW AND RULES], because legal obligations are specific and evolving.
OUTPUT	A 4-part plan, then the paragraph.

PRO TIP Keep the linking log (study ID → patient identity) on a different device from your study data.

TRY IT WITH Cohort of 150 patients with phone follow-up, data in Excel on a department laptop.

→ **NEXT #177** Data management plan for my thesis

307 Request a waiver of consent

STANDARD

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

USE IT For retrospective record-based studies or other minimal-risk research where consent is impracticable.

PERSONA	You are an IEC member who reviews waiver requests and knows what justifications committees accept.
MY CONTEXT	Study design and data source: [brief] Why obtaining consent is impracticable: [brief] Risk to participants: [brief] Confidentiality measures: [brief]
TASK	1. Assess whether my study plausibly qualifies for a waiver, with reasoning. 2. Write the waiver request: study description, justification of impracticability, minimal-risk argument, confidentiality safeguards, and how rights and welfare are protected. 3. List conditions the committee may impose.
RULES	Present the request honestly and mark the criteria you cite as [VERIFY IN ICMR GUIDELINES], because the committee alone decides eligibility for a waiver.
OUTPUT	The assessment, the request text, then the likely conditions.

PRO TIP Committees are far more comfortable granting waivers when data are anonymised at extraction.

TRY IT WITH Retrospective review of 5 years of ICU records on ventilator-associated pneumonia.

→ **NEXT #169** Record-review (retrospective) methodology

308 Protocol amendment request to the IEC

STANDARD

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

USE IT When you need to change the approved protocol (sample size, criteria, outcomes, procedures).

PERSONA	You are an experienced investigator who writes clear, well-justified amendment requests.
MY CONTEXT	Current approved protocol version and date: [version] Proposed changes and reasons: [list] Effect on participants, risks and consent: [brief]
TASK	1. Classify each change as minor or major, with reasoning, marked [CONFIRM WITH IEC SOP]. 2. Write the amendment request: a table of old text, new text and justification. 3. Identify documents that must be updated (protocol, consent form, CRF, CTRI record). 4. Note whether already-enrolled participants need re-consent.
RULES	Don't implement any change before approval unless it is needed to eliminate an immediate hazard, and say so, because unapproved changes are protocol violations.
OUTPUT	The classification, the amendment table, the documents list, then the re-consent note.

PRO TIP Update your CTRI record after approval. Journals compare the registry with the paper.

TRY IT WITH Increasing the sample size from 100 to 130 and adding a secondary outcome.

→ **NEXT #300** Respond to IEC queries

309 Funding and conflict-of-interest declarations

STANDARD

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT / FACULTY

USE IT When your study receives any funding, free materials or support, or when you have any industry relationship.

PERSONA	You are a research-integrity officer who drafts transparent declarations.
MY CONTEXT	Funding sources and amounts: [brief] Free equipment, kits or drugs received: [brief] Any personal or institutional relationships with relevant companies: [brief] Role of funders in design, analysis or publication: [brief]
TASK	1. Identify everything that must be declared, including non-financial interests. 2. Draft the declarations for the protocol, IEC form, thesis and future manuscript. 3. Recommend safeguards if a sponsor provides materials (e.g. no sponsor role in analysis or publication decisions).
RULES	Err on the side of disclosure, because an undeclared interest discovered later damages credibility far more than a declared one.
OUTPUT	The list of interests, the declarations for each document, then the safeguards.

PRO TIP Free diagnostic kits from a manufacturer count as support. Declare them.

TRY IT WITH A company provided free point-of-care test cartridges for my diagnostic accuracy study.

→ **NEXT #297** IEC submission pack and pre-review

310 Rewrite a consent paragraph in plain language

QUICK

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

Act as a patient-communication specialist. Rewrite this consent-form paragraph, [PASTE PARAGRAPH], at about an eighth-grade reading level for Indian patients, with sentences under 20 words, no jargon and no added promises. Keep every fact exactly, then list the medical terms you replaced.

TRY IT WITH A paragraph describing venepuncture risks and the storage of serum samples.

311 What documents does my IEC need?

QUICK

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

Act as an IEC office coordinator at an Indian medical college. List the documents typically required for a [STUDY TYPE] postgraduate thesis submission, what each must contain, and which are most often missing. Mark anything institution-specific as [CHECK YOUR IEC SOP].

TRY IT WITH Prospective observational thesis with blood sampling.

312 Does my study involve vulnerable participants?

QUICK

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

Act as an ethics committee member. For my study population, [DESCRIBE POPULATION], identify any vulnerable groups under research-ethics principles, the extra protections each needs, and how to describe them in my protocol. Mark specific requirements as [VERIFY IN ICMR GUIDELINES].

TRY IT WITH ICU patients, some unconscious, including pregnant women and adolescents.

313 Compensation and insurance: what do I write?

QUICK

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

Act as a research-ethics adviser. For my [observational / interventional] thesis on [TOPIC], explain what I should state about compensation for research-related injury and about payment for participation, and draft the wording for my protocol. Mark regulatory specifics as [VERIFY WITH IEC].

TRY IT WITH Interventional thesis testing a physiotherapy protocol after abdominal surgery.

314 Assent form for children

QUICK

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT (PAEDIATRICS)

Act as a paediatric research-ethics specialist. Write a short assent form for children aged [AGE RANGE] in my study, [BRIEF DESCRIPTION], using simple words, short sentences and a clear statement that they can say no. Mark age-band requirements as [VERIFY IN ICMR GUIDELINES].

TRY IT WITH Children aged 7–12 in a study requiring one finger-prick blood test.

315 Version control for my study documents

QUICK

1.10 PROTOCOL, ETHICS & SYNOPSIS · PG RESIDENT

Act as a research-documentation coordinator. Give me a simple version-control system for my protocol, consent forms, CRF and approvals: file naming, version numbers, dates, a change log and footer text, so every document the IEC sees matches my master copy.

TRY IT WITH Protocol v1.0, consent form v1.1, CRF v2.0 after an amendment.

1.11

Thesis to manuscript & publication

A thesis and a paper are different documents. Convert by cutting, comply with the journal's rules, and prepare for peer review before you submit.

35

prompts

4

Master

20

Standard

11

Quick

HOW TO USE THIS SECTION

Use it after thesis submission to convert it into a paper, choose a journal, submit and answer reviewers.

Start with the Master prompts ([#316–319](#)) for the full workflow; use the Standard prompts ([#320–339](#)) for one task at a time; reach for the Quick prompts ([#340–350](#)) when you have two minutes. Tap any title below to jump to it, and follow → **NEXT** to move to the next step.

[#316](#) Write the full manuscript for journal submission

[#317](#) Thesis → paper cut plan (and the second paper inside it)

[#318](#) Point-by-point response to peer reviewers

[#319](#) Pre-submission peer review of my manuscript

[#320](#) Structured abstract within the word limit

[#321](#) Five manuscript titles for my paper

[#322](#) Choose the right journal (and avoid predatory ones)

[#323](#) Check my manuscript against the author guidelines

[#324](#) Cover letter to the editor

[#325](#) Keywords that help my paper get found

[#326](#) Authorship order and CRediT contributions

[#327](#) Disclose AI use in my manuscript

[#328](#) Ethics, consent, funding, COI and data statements

[#329](#) Compress my thesis methods for a journal

[#330](#) Three-paragraph journal introduction

[#331](#) Condense my thesis discussion for a journal

[#332](#) Rejected: what now?

[#333](#) Suggest reviewers properly

[#334](#) Key points, highlights and "what this study adds"

[#335](#) Brief for a graphical abstract

[#336](#) Should I post a preprint?

[#337](#) Write a letter to the editor about a published paper

[#338](#) Check the galley proofs properly

[#339](#) Read and fix a similarity (plagiarism) report honestly

[#340](#) Running title under the limit

[#341](#) Word budget for each section

[#342](#) Cut my abstract to the limit

[#343](#) Summarise my paper for social media

[#344](#) Cut my references to the journal's limit

[#345](#) Write the acknowledgements

[#346](#) Polite status enquiry to the editor

[#347](#) Data-availability statement

[#348](#) Plain-language summary of my paper

[#349](#) Reply to one tough reviewer comment

[#350](#) Is splitting my thesis into two papers acceptable?

Write the full manuscript for journal submission

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

USE IT

When the thesis is done and you are converting it into a paper you can submit.

WHY

A thesis and a paper are different documents. This prompt handles the conversion, reporting-guideline compliance and the submission package in one structured pass, using only your real numbers.

COPY THE PROMPT BELOW

PERSONA	You are a senior academic author and manuscript editor who has published extensively in clinical journals and has served as a peer reviewer. You know reporting guidelines thoroughly and you write in clear, economical scientific English for an international readership.
MY CONTEXT	Design: [state] · Current full title: [paste] Setting and period: [state] Sample size and participant flow: [screened, eligible, enrolled, completed, analysed] Methods summary: [paste from protocol] All results with exact numbers, effect estimates and CIs: [paste] Tables and figures I already have: [list with titles] Target journal: [name] · Word, abstract, reference, table and figure limits: [paste from author guidelines] Reporting guideline that applies: [CONSORT / STROBE / STARD / PRISMA / CARE / SQUIRE / other]
TASK	1. Journal fit: assess scope, design and novelty; if it is a poor fit, say so and suggest what type of journal fits better. Do not quote impact factors or acceptance rates from memory. 2. Title: five options (descriptive, finding-stating, PICO-style, short, conservative), with the best one for this journal. 3. Structured abstract within the limit, with my actual numbers. 4. Six keywords, preferring MeSH terms. 5. Introduction: three paragraphs (known, unknown and why it matters, objective), with [CITE] slots. 6. Methods: past tense, replicable, following my reporting guideline, noting which checklist item each subsection satisfies, with ethics, consent and registration placeholders. 7. Results: past tense, no interpretation; flow, baseline (Table 1), primary, then secondary outcomes, each with its effect estimate and CI. 8. Discussion: principal finding, comparison, explanation, implications, strengths, ranked limitations, future research, conclusion. 9. Conclusion, funding, conflicts of interest, CRediT author contributions, acknowledgements and data-availability statement. 10. A reporting-guideline checklist table, flagging unaddressed items. 11. A cover letter under 300 words. 12. A pre-submission critical review: major and minor comments in journal format, and which to fix before submitting.
RULES	Every citation slot stays [CITE] unless I uploaded the paper or you verified it with a PMID or DOI, because fabricated references end in retraction. Don't alter, round differently or recompute any number I give you. Add no findings, comparisons or claims beyond my data. Flag any claim my design cannot support. Note that AI cannot be an author and remind me to check the journal's AI-disclosure policy.
OUTPUT	Twelve numbered parts, with the manuscript sections in submission order.
TWEAK	If time is short, do step 12 first and tell me the 3 fixes that most improve my chance of acceptance.

How to get more out of it

- › Do step 12 first if you are short on time. A pre-submission critical review saves months of revision.
- › Paste the actual author guidelines. Generic manuscript advice produces desk rejections.
- › Convert thesis to paper by cutting, not rewriting. Most theses contain two papers hiding inside one document.
- › → **NEXT #317** Thesis → paper cut plan (and the second paper inside it)

Thesis → paper cut plan (and the second paper inside it)

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

USE IT

Right after thesis submission, while the details are fresh.

WHY

Theses are long, comprehensive and repetitive; papers are short and focused. Most residents never publish because they try to shrink the whole thesis. This prompt finds the one or two papers inside it and tells you exactly what to cut.

COPY THE PROMPT BELOW

PERSONA	You are a senior clinical researcher who has turned dozens of MD/MS theses into published papers. You think in terms of "one paper, one message" and you cut ruthlessly.
MY CONTEXT	Thesis title and design: [paste] Objectives and key results for each: [PASTE OBJECTIVES AND RESULTS] Chapter word counts: [introduction, ROL, methods, results, discussion] Number of tables and figures in the thesis: [numbers] Target journal type and its word limit: [e.g. national indexed journal, 3,000 words] Co-authors and who will lead each paper: [brief]
TASK	1. Identify the main paper (primary objective) and any second paper (a distinct secondary objective, a sub-study or a methodological finding), each with a one-line message. 2. For the main paper, map what to keep, condense or drop from each thesis chapter, with target word counts. 3. Select the 3–5 tables or figures that carry the main message. 4. Explain what the ROL becomes (usually two sentences in the introduction and comparisons in the discussion). 5. Draft the paper's single-sentence message and a working title. 6. Flag duplicate-publication risks if two papers share data, and how to avoid salami slicing. 7. List any additional analysis the paper needs that the thesis didn't include, marked [NEW ANALYSIS — CONFIRM WITH GUIDE].
RULES	Each proposed paper must stand on its own question, because splitting one result into several papers is salami publication. Use only results I gave; don't invent new analyses.
OUTPUT	A papers table (paper, question, message, key tables), then a chapter-by-chapter cut plan with word targets, then the salami-slicing check.
TWEAK	End with a 4-week writing schedule to get the main paper to submission.

How to get more out of it

- › Write the paper's abstract first. If it doesn't hold together, the paper won't either.
- › Keep the thesis methods chapter open. You'll compress it, not rewrite it.
- › Discuss authorship with your guide before you start writing, not after.
- › → NEXT #316 Write the full manuscript for journal submission

Point-by-point response to peer reviewers

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

USE IT

When you receive a "major" or "minor revision" decision.

WHY

A revision is an invitation, not a rejection. Most revisions are accepted when every comment is answered respectfully and specifically, with changes clearly located. Defensive or vague responses are how revisions become rejections.

COPY THE PROMPT BELOW

PERSONA	You are a senior author who has responded to hundreds of peer reviews and sits on editorial boards. You are gracious, precise and firm where the science warrants it.
MY CONTEXT	Editor's decision letter: [PASTE LETTER] Reviewer comments, verbatim: [PASTE ALL COMMENTS] My manuscript (relevant sections): [PASTE OR UPLOAD] Comments I disagree with, and why: [brief] Revision deadline: [date] Analyses or data I can realistically add in that time: [brief]
TASK	1. Number every comment and classify it: requires new analysis, requires text change, requires clarification, or respectful disagreement. 2. For each, draft a response: thank the reviewer where appropriate, answer the point directly, and state exactly what changed and where (page and line). 3. Draft the revised manuscript text for each change. 4. For disagreements, write a courteous, evidence-based rebuttal, and suggest a partial concession where reasonable, such as adding a limitation sentence. 5. Flag comments that need new data or analysis I may not have, and how to handle each honestly (e.g. acknowledging it as a limitation). 6. Draft the cover letter to the editor summarising the main changes.
RULES	Answer every comment, including the minor ones, because an unanswered comment is the commonest reason for a second revision round. Don't promise analyses I cannot do, and don't invent new results or references; use [CITE] or [NEW ANALYSIS NEEDED] slots.
OUTPUT	A response table (comment number, verbatim comment, response, change and location), the revised text blocks, then the cover letter.
TWEAK	End with the 2 responses most likely to decide acceptance, and whether they are strong enough.

How to get more out of it

- › Reply in a table. Editors find it far easier to check.
- › Highlight changes in a tracked or coloured version of the manuscript, and submit a clean copy too.
- › If a reviewer misunderstood something, the text was probably unclear. Fix the text, not just the reply.
- › → NEXT #324 Cover letter to the editor

Pre-submission peer review of my manuscript

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

USE IT

One week before submission, when you think the manuscript is finished.

WHY

The fastest way to get published is to be your own harshest reviewer before the journal is. A structured pre-review, in journal format, catches the fatal flaws, the reporting gaps and the desk-rejection triggers while you can still fix them.

COPY THE PROMPT BELOW

PERSONA	You are an experienced peer reviewer and associate editor for clinical journals. You write structured reviews with major and minor comments, and you think about desk-rejection risk as an editor would.
MY CONTEXT	Target journal and its scope: [name, brief scope] Article type and word limit: [e.g. original article, 3,000 words] Reporting guideline: [e.g. STROBE] What I believe is the paper's main contribution: [one line] My biggest worry about the paper: [one line] My manuscript: [PASTE OR UPLOAD MANUSCRIPT]
TASK	1. Give an editor's first-read verdict: desk reject, send for review, or likely to be sent back for formatting, with reasons based on the title, abstract and cover-letter logic. 2. Write major comments on the question, design, analysis, interpretation and novelty, quoting the text. 3. Write minor comments on clarity, tables and figures, references and language. 4. Run a reporting-guideline check and list the items that are missing, with where each should go. 5. Check compliance with the author guidelines (word counts, abstract structure, reference style, declarations, figure formats). 6. Prioritise all issues: fix before submission, or can wait. 7. Say whether my stated main contribution comes across clearly in the title, abstract and conclusion.
RULES	Quote the manuscript for every comment, because specific comments can be acted on and vague ones can't. Don't rewrite the paper; identify and prioritise. Don't invent references or data.
OUTPUT	The verdict, major comments, minor comments, the guideline check, the compliance check, then a priority list.
TWEAK	End with the single weakness most likely to cause rejection, and the most efficient fix.

How to get more out of it

- › Ask a colleague outside your department to read the abstract and tell you the main finding. If they can't, rewrite it.
- › Fix every "before submission" item. Editors notice sloppy formatting within seconds.
- › Keep this review. Compare it with the real reviews when they arrive.
- › → **NEXT #322** Choose the right journal (and avoid predatory ones)

320 Structured abstract within the word limit

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

USE IT For journal submission, conference abstracts and the thesis abstract.

PERSONA	You are a journal abstract editor who writes precise, complete structured abstracts.
MY CONTEXT	Target journal or conference and its abstract format (headings, word limit): [paste] Design, setting and period: [brief] Participants and sample size: [brief] Primary outcome result with estimate, CI and p: [paste] Key secondary results: [paste] Conclusion: [paste]
TASK	1. Write the abstract with the required headings, within the limit. 2. Include the design in the Methods, the primary result with numbers in the Results, and a proportionate Conclusion. 3. Give the word count. 4. Check that every number matches the full text.
RULES	Use only the numbers I gave, and make sure the conclusion follows from the primary result, because abstracts are read far more than papers and spin there misleads the most readers.
OUTPUT	The abstract, its word count, then the number check.

PRO TIP Many readers only read the abstract. Make sure it states your primary result even if it was null.

TRY IT WITH 250-word structured abstract for a prospective cohort of 210 patients on AKI after cardiac surgery.

→ **NEXT #321** Five manuscript titles for my paper

321 Five manuscript titles for my paper

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

USE IT Before submission, or when your title feels flat.

PERSONA	You are a journal editor who knows which titles get read and cited.
MY CONTEXT	Main finding: [one line] Design and population: [brief] Target journal and any title limit: [name, characters] My current working title: [paste]
TASK	1. Write 5 titles: descriptive, finding-stating, question-form, short and punchy, and design-explicit (e.g. "...: a prospective cohort study"). 2. Check each for overclaiming, jargon, abbreviations and length. 3. Recommend one for my journal and explain why. 4. Suggest a running title.
RULES	Don't state causation in a title for an observational design, because editors flag it immediately.
OUTPUT	The 5 titles with character counts, the checks, the recommendation, then the running title.

PRO TIP Put the most important words at the start. Many databases truncate long titles.

TRY IT WITH Finding: frailty predicts 90-day mortality after emergency laparotomy in older Indian adults.

→ **NEXT #320** Structured abstract within the word limit

322 Choose the right journal (and avoid predatory ones)

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

USE IT Before writing the final version, so you write for a specific journal.

PERSONA	You are a publication adviser who matches manuscripts to journals and protects authors from predatory publishers.
MY CONTEXT	Main finding and design: [brief] Specialty: [specialty] Priorities: [indexing needed (e.g. PubMed, Scopus), speed, cost, open access, national or international readership] Budget for article processing charges: [amount or "none"]
TASK	1. Describe the types of journals that fit my paper (specialty or general, national or international). 2. Explain how to find candidate journals: the reference list of my own paper, journal-finder tools, and indexing databases. 3. Give a checklist for verifying legitimacy: indexing claims checked at the source, a real editorial board, transparent peer review and charges, DOAJ listing for open-access journals. 4. List red flags of predatory journals. 5. Help me build a shortlist table from the journals I find.
RULES	Don't recommend specific journals, impact factors or acceptance rates from memory; mark any journal named as [VERIFY INDEXING AND SCOPE], because journal metrics and indexing change and predatory journals mimic real ones.
OUTPUT	Journal types, finding methods, the legitimacy checklist, red flags, then a shortlist template.

PRO TIP If a journal emails you inviting a submission with rapid publication, treat it as a red flag until proven otherwise.

TRY IT WITH Prospective observational study in pulmonology from an Indian medical college, no APC budget.

→ **NEXT #323** Check my manuscript against the author guidelines

323 Check my manuscript against the author guidelines

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

USE IT Just before submission.

PERSONA	You are a journal's editorial assistant who checks submissions against the instructions for authors.
MY CONTEXT	The journal's author guidelines (relevant parts): [PASTE GUIDELINES] My manuscript: [PASTE OR UPLOAD]
TASK	1. Check article type, word counts (abstract and main text), heading structure, reference style and number, and table and figure limits and formats. 2. Check the required declarations: ethics, consent, registration, funding, conflicts of interest, data availability, AI disclosure and author contributions. 3. Check file requirements (title page, blinded manuscript, figures, checklist). 4. List every non-compliance with its fix.
RULES	Base every check on the guidelines I pasted, not on general assumptions, because journals differ in small details that cause technical returns.
OUTPUT	A compliance table (requirement, status, fix).

PRO TIP Technical returns cost weeks. Five minutes with this checklist saves them.

TRY IT WITH Original article for a national pulmonology journal with a 250-word abstract limit.

→ **NEXT #324** Cover letter to the editor

324 Cover letter to the editor

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

USE IT For every journal submission.

PERSONA	You are a senior author whose cover letters get manuscripts sent out for review.
MY CONTEXT	Journal and editor's name: [name] Title and article type: [paste] Main question and finding: [brief] Why it matters to this journal's readers: [brief] Required declarations (original work, not under consideration elsewhere, ethics, conflicts): [brief]
TASK	1. Write a cover letter of 300 words or fewer: the question, the finding, why it matters to this journal's readership, and the standard declarations. 2. Add a line on any related prior publication or preprint, if applicable. 3. Keep the tone professional and specific.
RULES	Don't exaggerate novelty or impact, because editors discount inflated letters. Don't name suggested reviewers without checking their independence.
OUTPUT	The letter.

PRO TIP Name the editor and the journal correctly. Generic "Dear Sir/Madam" letters signal mass submission.

TRY IT WITH Cover letter for a diagnostic accuracy study of lung ultrasound in paediatric pneumonia.

→ **NEXT #319** Pre-submission peer review of my manuscript

325 Keywords that help my paper get found

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

USE IT At submission, when the journal asks for keywords.

PERSONA	You are a medical librarian who chooses keywords for discoverability.
MY CONTEXT	Title and abstract: [PASTE] Number of keywords allowed: [e.g. 6] Target journal: [name]
TASK	1. Suggest keywords, preferring MeSH terms where they exist, plus 1–2 free-text terms readers would search. 2. Avoid words already in the title where the journal recommends that, and avoid overly broad terms. 3. Explain each choice in one line. 4. Suggest 2 alternative keywords for indexing services and search engines that don't use MeSH.
RULES	Mark any MeSH term you are unsure of as [CHECK IN MeSH BROWSER], because invalid terms reduce discoverability.
OUTPUT	A keyword list with the reason for each.

PRO TIP Think about how a clinician would search for your paper, not how you would describe it.

TRY IT WITH Abstract of a study on sarcopenia and outcomes in elderly COPD patients.

→ **NEXT #320** Structured abstract within the word limit

326 Authorship order and CRediT contributions

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

USE IT Before writing the paper, and again at submission.

PERSONA	You are a research-integrity adviser who applies ICMJE authorship criteria fairly.
MY CONTEXT	People involved and what each did (conception, data collection, analysis, writing, supervision, funding): [list] Department customs about authorship order: [brief]
TASK	1. Apply the ICMJE criteria to each person: author, or acknowledgement. 2. Suggest an order based on contribution, with reasoning. 3. Write the CRediT contribution statement. 4. Draft the acknowledgements for contributors who don't qualify as authors. 5. Suggest how to discuss disagreements respectfully.
RULES	Apply the criteria as written, and flag gift or ghost authorship if my list suggests it, because authorship disputes and misconduct findings can follow a paper for years.
OUTPUT	An authorship table, the order, the CRediT statement, then the acknowledgements.

PRO TIP Agree authorship in writing before you start writing. It prevents the most painful disputes.

TRY IT WITH Resident, guide, co-guide, statistician, a colleague who helped recruit, and the head of department.

→ **NEXT #328** Ethics, consent, funding, COI and data statements

327 Disclose AI use in my manuscript

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

USE IT When you used any AI tool while preparing the thesis or manuscript.

PERSONA	You are a publication-ethics editor familiar with ICMJE, COPE and journal policies on generative AI.
MY CONTEXT	AI tools used and for what (language editing, literature search, code, drafting): [list] Target journal's AI policy (paste if available): [PASTE]
TASK	1. Classify my uses by whether journals typically require disclosure. 2. Draft the disclosure statement in the journal's preferred location (methods, acknowledgements or a declaration). 3. Confirm the responsibilities that remain mine (accuracy, originality, citations). 4. Flag any use that may breach the journal's policy (e.g. AI-generated images or data).
RULES	Recommend full, honest disclosure of substantive AI use, and state that AI tools cannot be authors, because undisclosed AI use is increasingly treated as misconduct.
OUTPUT	The classification, the statement, the responsibilities, then the flags.

PRO TIP Keep a simple log of which AI tool you used for what and when. It makes disclosure easy.

TRY IT WITH Used Claude for language editing and ChatGPT for R code during the analysis.

→ **NEXT #328** Ethics, consent, funding, COI and data statements

328 Ethics, consent, funding, COI and data statements

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

USE IT For the declarations section at submission.

PERSONA	You are an editorial assistant who drafts complete, journal-ready declarations.
MY CONTEXT	IEC name, approval number and date: [details] Consent type: [written informed / waiver] Registration: [CTRI number or not applicable] Funding: [details or none] Conflicts of interest: [details or none] Data availability plan: [on request / repository / not shareable, with reason]
TASK	1. Draft each declaration: ethics approval, consent to participate, consent for publication (if applicable), registration, funding, competing interests, data availability and author contributions. 2. Flag anything missing that the journal is likely to require.
RULES	Use only the details I gave; mark missing essentials as [ADD DETAIL], because incomplete declarations cause technical returns.
OUTPUT	Headed declarations, then the missing-items list.

PRO TIP Keep a copy of your IEC approval letter handy. Many journals ask for it at submission.

TRY IT WITH IEC approval dated March 2025, CTRI registration, no funding, no conflicts.

→ **NEXT #323** Check my manuscript against the author guidelines

329 Compress my thesis methods for a journal

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

USE IT When turning the thesis methodology chapter into a 700–900 word methods section.

PERSONA	You are a manuscript editor who condenses methods without losing reproducibility.
MY CONTEXT	Thesis methodology chapter: [PASTE CHAPTER] Target word count: [e.g. 800] Reporting guideline: [e.g. STROBE]
TASK	1. Keep everything needed for replication and the reporting checklist. 2. Cut textbook explanations, repeated definitions and procedural detail that can go into a supplement. 3. Suggest what goes into supplementary material, and how to refer to it in the main text. 4. Return the compressed methods with the word count.
RULES	Don't change any definition, number or procedure, because the paper must describe exactly what the thesis did.
OUTPUT	The compressed methods, then the supplement list.

PRO TIP Cite your protocol or CTRI record for full details. It saves words legitimately.

TRY IT WITH A 4,000-word methods chapter from an MD thesis on AKI after cardiac surgery.

→ **NEXT #316** Write the full manuscript for journal submission

330 Three-paragraph journal introduction

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

USE IT When converting your thesis introduction and ROL into a journal introduction.

PERSONA	You are a journal editor who expects introductions of about 400 words that lead straight to the question.
MY CONTEXT	Thesis introduction and gap statement: [PASTE] Key references I will cite (with numbers): [list] Objective: [one line]
TASK	1. Write paragraph 1: the problem and why it matters, with burden figures from my sources. 2. Write paragraph 2: what is known and what is not, with the specific gap. 3. Write paragraph 3: the objective, stated precisely. 4. Keep it to about 350–450 words.
RULES	Use only my references and figures, keeping [CITE] slots where they are missing, because journal introductions are checked against the reference list.
OUTPUT	The three paragraphs with the word count.

PRO TIP The last sentence of the introduction should read like the question your abstract answers.

TRY IT WITH Converting a 2,500-word thesis introduction and ROL on dengue shock predictors.

→ **NEXT #316** Write the full manuscript for journal submission

331 Condense my thesis discussion for a journal

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

USE IT When your thesis discussion is far longer than the journal allows.

PERSONA	You are a journal editor who condenses discussions to the essential argument.
MY CONTEXT	Thesis discussion: [PASTE DISCUSSION] Target word count: [e.g. 1,000] The single main message of the paper: [one line]
TASK	1. Keep the structure: principal findings, key comparisons, explanation, implications, strengths and limitations, and conclusion. 2. Keep only the comparisons that matter for the main message. 3. Remove repetition of results and background. 4. Return the condensed discussion with the word count.
RULES	Keep citations with their claims and all numbers unchanged, and don't add new comparisons, because the paper must stay faithful to the thesis findings.
OUTPUT	The condensed discussion, then a list of the removed material.

PRO TIP A journal discussion usually cites 10–20 references, not the 60 in your thesis ROL.

TRY IT WITH A 3,500-word thesis discussion condensed to 1,100 words.

→ **NEXT #319** Pre-submission peer review of my manuscript

332 Rejected: what now?

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

USE IT When a journal rejects your manuscript.

PERSONA	You are a mentor who has turned many rejections into publications.
MY CONTEXT	Rejection type: [desk rejection / after review] The editor's and reviewers' comments: [PASTE] My next-choice journals: [list]
TASK	1. Interpret the rejection: scope, quality or priority? 2. Identify the comments worth addressing before resubmitting elsewhere. 3. Make a revision plan in order of priority. 4. List the reformatting needed for the next journal. 5. Advise whether an appeal is reasonable (rarely), with reasons.
RULES	Be honest about fundamental flaws if the reviews identify them, because resubmitting an unchanged flawed manuscript wastes months.
OUTPUT	The interpretation, a prioritised revision plan, the reformatting list, then the appeal advice.

PRO TIP Take reviewers' comments seriously even after rejection. The next journal may send it to the same reviewer.

TRY IT WITH Rejected after review for "limited novelty and single-centre design".

→ **NEXT #322** Choose the right journal (and avoid predatory ones)

333 Suggest reviewers properly

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

USE IT When the journal asks you to suggest reviewers.

PERSONA	You are an editor who knows how journals use and check suggested reviewers.
MY CONTEXT	Topic and methods of my paper: [brief] Authors of key papers I cited (from my reference list): [list] My co-authors' institutions and recent collaborators: [list]
TASK	1. Explain the criteria for a good suggested reviewer: expertise, independence and no recent collaboration. 2. Explain how to identify candidates from my reference list and PubMed. 3. Explain how to verify contact details from institutional pages. 4. Build a table of candidates from the authors I listed, with conflict checks.
RULES	Don't invent reviewer names or email addresses; work only from the authors I listed and tell me how to verify them, because fake or conflicted reviewer suggestions are treated as misconduct.
OUTPUT	The criteria, the method, then the candidate table.

PRO TIP Suggest reviewers from different countries and institutions. Editors trust diverse suggestions more.

TRY IT WITH Paper on high-flow nasal oxygen in COPD exacerbations.

→ **NEXT #324** Cover letter to the editor

334 Key points, highlights and "what this study adds"

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

USE IT When the journal asks for highlights, key messages or a "what is known / what this adds" box.

PERSONA	You are a journal editor who writes crisp summary boxes.
MY CONTEXT	Journal's required format (number of points, character limits): [paste] Main findings with numbers: [paste] Gap addressed: [brief] Design and setting: [brief]
TASK	1. Write the box in the journal's required format: what is already known, what this study adds, and how it might affect practice or research. 2. Keep each point within the character limit and in plain language. 3. Check that nothing overclaims beyond the results, and that every number matches the abstract.
RULES	Keep claims proportionate to the design, because highlights are often quoted out of context.
OUTPUT	The box, then the character counts.

PRO TIP "What this adds" should be something a reader couldn't have said before your paper.

TRY IT WITH Findings: 1 in 3 elderly surgical patients developed delirium, and frailty predicted it.

→ **NEXT #320** Structured abstract within the word limit

335 Brief for a graphical abstract

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

USE IT When the journal requests or encourages a graphical abstract.

PERSONA	You are a scientific illustrator who designs clear graphical abstracts.
MY CONTEXT	Question, design, population, main finding and conclusion: [brief] Journal's graphical-abstract specifications: [size, format] Tool I will use: [PowerPoint / Canva / BioRender]
TASK	1. Propose a left-to-right layout: population → method → key result → take-home. 2. Specify the icons, the text for each panel (under 15 words each) and the key numbers. 3. Suggest colours and fonts for readability. 4. List common mistakes to avoid.
RULES	Show the main result with its numbers accurately, and avoid visual exaggeration, because graphical abstracts are shared widely out of context.
OUTPUT	The panel-by-panel brief, then design notes and mistakes.

PRO TIP If it can't be understood in 10 seconds on a phone screen, simplify it.

TRY IT WITH RCT: early mobilisation vs usual care in ICU; outcome: delirium days.

→ **NEXT #334** Key points, highlights and "what this study adds"

336 Should I post a preprint?

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

USE IT Before submission, if you are considering medRxiv or a similar server.

PERSONA	You are a publication strategist who knows preprint servers and journal policies.
MY CONTEXT	Type of study and any clinical implications: [brief] Target journals: [list] My reasons for considering a preprint: [brief]
TASK	1. Explain the benefits and risks of a preprint for my study. 2. Explain how to check whether my target journals accept preprinted manuscripts. 3. Explain the screening that medical preprint servers apply and what is required (e.g. ethics and registration statements). 4. Explain how to link the preprint to the final paper.
RULES	Mark journal-specific policies as [CHECK JOURNAL POLICY], because preprint policies vary and change.
OUTPUT	Benefits and risks, the policy check, the requirements, then the linking step.

PRO TIP Clinical findings in preprints can be misreported by the media. Word the conclusion carefully.

TRY IT WITH A diagnostic accuracy study of a new point-of-care test for TB.

→ **NEXT #322** Choose the right journal (and avoid predatory ones)

337 Write a letter to the editor about a published paper

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

USE IT When you want to comment on a recently published article; it's often a first publication.

PERSONA	You are an experienced correspondent who writes concise, constructive letters that journals publish.
MY CONTEXT	The article I'm responding to (citation and key points): [PASTE] My comment: a methodological concern, an alternative interpretation, or relevant local data: [brief] Journal's letter limits (words, references, time window): [paste]
TASK	1. Structure the letter: reference to the article, the main point, supporting reasoning or data, and a constructive conclusion. 2. Write it within the limits. 3. Suggest a title.
RULES	Keep the tone collegial and evidence-based, and cite only sources I supply or [CITE], because letters that attack authors are rarely published.
OUTPUT	The title and the letter with its word count.

PRO TIP Submit quickly. Many journals only accept letters within a few weeks of publication.

TRY IT WITH Commenting on a trial that excluded patients with CKD, the majority in Indian practice.

→ **NEXT #324** Cover letter to the editor

338 Check the galley proofs properly

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

USE IT When the journal sends proofs, usually with 48–72 hours to respond.

PERSONA	You are a meticulous production editor who knows what goes wrong at the proof stage.
MY CONTEXT	Proof PDF text or sections to check: [PASTE OR UPLOAD] My accepted manuscript for comparison: [PASTE OR UPLOAD]
TASK	1. Give a checklist: author names, affiliations and order; corresponding author details; title; abstract numbers; tables and figures (correct versions, legends, symbols); units and special characters; references; and declarations. 2. Compare the proof text with my accepted manuscript and list the differences. 3. Answer each typesetter's query clearly.
RULES	Request only corrections of errors, not new content, because journals refuse substantive changes at proof stage.
OUTPUT	The checklist, a difference list, then responses to the queries.

PRO TIP Check symbols ($\pm$, $\geq$, μ) and minus signs in tables. Typesetting often breaks them.

TRY IT WITH Proofs for an original article with 4 tables and a Kaplan–Meier figure.

→ **NEXT #319** Pre-submission peer review of my manuscript

339 Read and fix a similarity (plagiarism) report honestly

STANDARD

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

USE IT When your university or journal returns a Turnitin, iThenticate or similar report.

PERSONA	You are a research-integrity officer who helps authors interpret similarity reports and fix genuine problems.
MY CONTEXT	Overall similarity percentage and the top matching sources: [PASTE SUMMARY] Highlighted passages with their matches: [PASTE PASSAGES] My university's or journal's threshold, if known: [e.g. 10%]
TASK	1. Separate acceptable matches (references, standard methods phrases, quotations with citation, my own registered protocol) from problematic ones (copied sentences without citation, close paraphrase). 2. For each problematic passage, explain the issue and how to rewrite it in my own words with the correct citation. 3. Advise how to report acceptable overlap, e.g. with my own thesis or preprint, and how to request exclusions properly.
RULES	Fix plagiarism by genuine rewriting and correct citation, never by tricks that hide text from the checker, because those are themselves misconduct. Mark institutional thresholds and rules as [CHECK YOUR UNIVERSITY / JOURNAL POLICY].
OUTPUT	A passage table (passage, match type, acceptable or not, fix), then the reporting advice.

PRO TIP Write methods and introduction in your own words from the start. Fixing overlap later takes far longer.

TRY IT WITH A 24% similarity report where most matches are from the methods and introduction.

→ **NEXT #82** Paraphrase and synthesise sources responsibly

340 Running title under the limit

QUICK

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

Act as a journal editor. Write 3 running titles for my paper, [FULL TITLE], each under [e.g. 50] characters including spaces, keeping the core topic and population recognisable and avoiding abbreviations the journal hasn't defined. Show the character count for each.

TRY IT WITH Frailty and 90-day mortality after emergency laparotomy in older Indian adults.

341 Word budget for each section

QUICK

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

Act as a manuscript editor. For a [ARTICLE TYPE] with a [WORD LIMIT] word limit, give a word budget for the introduction, methods, results and discussion, adjusted to my design, [DESIGN], and tell me which section usually runs over and how to control it.

TRY IT WITH Original article, 3,000 words, prospective cohort.

342 Cut my abstract to the limit

QUICK

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

Act as an abstract editor. Cut this abstract, [PASTE ABSTRACT], from [CURRENT] to [TARGET] words without removing the design, setting, sample size, primary result with its numbers, or conclusion. Show the new word count and list the phrases you removed.

TRY IT WITH A 310-word abstract that must fit a 250-word limit.

343 Summarise my paper for social media

QUICK

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

Act as a science communicator for doctors. Write a 3-post thread (each under 280 characters) summarising my published paper, [TITLE, KEY FINDING, DOI], accurately and without overclaiming, ending with the DOI link. Add one plain-language version for patients and families.

TRY IT WITH Paper on lung ultrasound for diagnosing pneumonia in children.

344 Cut my references to the journal's limit

QUICK

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

Act as a manuscript editor. My paper has [N] references and the limit is [LIMIT]. From this list with the claims each supports, [PASTE LIST], recommend which to remove or combine, keeping primary sources for key claims and the most recent evidence.

TRY IT WITH 62 references for a journal with a 40-reference limit.

345 Write the acknowledgements

QUICK

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

Act as an academic editor. Write a concise, professional acknowledgements paragraph thanking [LIST PEOPLE AND ROLES] who don't meet authorship criteria, plus all participants and any funding body, in under 80 words, warmly but formally, without claiming their endorsement of the conclusions.

TRY IT WITH Nursing staff, the lab technician, the statistician and all participants.

346 Polite status enquiry to the editor

QUICK

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

Act as a senior author. Write a short, polite email to the editorial office of [JOURNAL] asking about the status of my manuscript [ID, TITLE], submitted on [DATE], which has been "under review" for [DURATION]. Keep it under 80 words.

TRY IT WITH Manuscript under review for 4 months with no update.

347 Data-availability statement

QUICK

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT

Act as a research-data editor. Write a data-availability statement for my study, [BRIEF DESCRIPTION], that reflects my plan, [SHARING PLAN], respects participant confidentiality and IEC conditions, and meets typical journal requirements. Give one short and one detailed version, and say which the journal likely prefers.

TRY IT WITH De-identified data available on reasonable request, subject to IEC approval.

348 Plain-language summary of my paper

QUICK

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

Act as a patient-education writer. Write a 150-word plain-language summary of my paper, [PASTE ABSTRACT], for patients and the public: why the study was done, what was done, what was found and what it means, with no jargon and no overstatement.

TRY IT WITH Abstract of a study on inhaler technique errors in asthma patients.

349 Reply to one tough reviewer comment

QUICK

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

Act as a senior author. Draft a courteous, specific reply to this reviewer comment, [PASTE COMMENT], given my position, [AGREE / PARTLY AGREE / DISAGREE AND WHY], stating exactly what I changed and where, or why I respectfully did not. Keep it under 150 words.

TRY IT WITH "The single-centre design severely limits generalisability; the conclusions should be tempered."

350 Is splitting my thesis into two papers acceptable?

QUICK

1.11 THESIS TO MANUSCRIPT & PUBLICATION · PG RESIDENT / FACULTY

Act as a publication-ethics adviser. I plan two papers from one dataset: [PAPER 1 QUESTION] and [PAPER 2 QUESTION]. Tell me whether this is legitimate or salami slicing, why, and how to cross-reference the papers and disclose the shared dataset.

TRY IT WITH Paper 1 on mortality predictors; paper 2 on the same predictors for ICU length of stay.

1.12

Case reports & case series

Most case reports are rejected because the case isn't novel or the timeline is unclear, not because the writing is weak. Check publishability first, then write to CARE.

30

prompts

3

Master

18

Standard

9

Quick

HOW TO USE THIS SECTION

Use it when you see an unusual case: check whether it is publishable first, then write to CARE.

Start with the Master prompts ([#351–353](#)) for the full workflow; use the Standard prompts ([#354–371](#)) for one task at a time; reach for the Quick prompts ([#372–380](#)) when you have two minutes. Tap any title below to jump to it, and follow → NEXT to move to the next step.

[#351](#) From interesting case to CARE-compliant case report

[#352](#) Design and write a case series

[#353](#) Case-report discussion grounded in the literature

[#354](#) Is my case publishable?

[#355](#) Build the case timeline table

[#356](#) Consent for publication and de-identification

[#357](#) Select and prepare clinical images

[#358](#) Case report abstract

[#359](#) Case report titles that follow the CARE convention

[#360](#) Write the case presentation from my notes

[#361](#) Show how differentials were excluded

[#362](#) Learning points that readers remember

[#363](#) "Images in clinical medicine" submission

[#364](#) Journals for case reports: find and verify them

[#365](#) Patient-level table for a case series

[#366](#) Find previously reported similar cases

[#367](#) Patient-perspective section

[#368](#) Case report → conference poster

[#369](#) Critique my case report draft

[#370](#) Adverse drug reaction case report with causality assessment

[#371](#) Clinicopathological conference (CPC) write-up

[#372](#) One line on why my case is worth reading

[#373](#) Keywords to search for similar cases

[#374](#) Is my case report properly anonymised?

[#375](#) Cut my case report to the word limit

[#376](#) Legend for a clinical image

[#377](#) Turn my case into a teaching caption

[#378](#) Explain publication consent to my patient

[#379](#) Case report or case series?

[#380](#) The key sentence of my case-report cover letter

From interesting case to CARE-compliant case report

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT / FACULTY

USE IT

As soon as you suspect a case is unusual, while notes, images and consent are still easy to get.

WHY

Most case reports are rejected because the case isn't novel or the timeline is unclear, not because the writing is weak. This prompt checks publishability first, then drafts to the CARE guideline.

COPY THE PROMPT BELOW

PERSONA	You are the case-reports editor of an indexed [SPECIALTY] journal. You have handled hundreds of submissions and apply the CARE guidelines strictly. You are blunt about whether a case is actually publishable.
MY CONTEXT	Specialty and my level: [e.g. MD Respiratory Medicine, PG year 2] Setting: [tertiary / district / private] De-identified case summary (age band, sex, presentation, key investigations with dates as Day 0, Day 3..., treatment, outcome, follow-up): [PASTE] Why I think it is unusual: [state] Target journal and its case-report word limit: [name / limit] Patient consent for publication: [obtained / pending]
TASK	1. Publishability verdict: rare disease, rare presentation, new association, diagnostic lesson or management lesson? Score novelty out of 5 and justify it against my case. 2. Five title options that each contain the words "case report". 3. A CARE-structured draft: abstract (150 words or fewer), introduction, patient information, clinical findings, timeline table, diagnostic assessment (including the differentials you excluded and how), therapeutic intervention, follow-up and outcome, discussion, patient-perspective placeholder, and learning points. 4. A CARE checklist table showing where each item appears, flagging anything missing. 5. The consent and de-identification checklist I need before submission.
RULES	Use only the clinical details I gave; where something is missing, write [ASK ME: ...] instead of filling it in. Put [CITE] wherever literature is needed, and cite a paper only if I uploaded it or you can give a verified PMID or DOI, because fabricated references end in retraction. Keep every date relative (Day 0...) and remove anything identifying.
OUTPUT	Formal case-report register within my journal's word limit, then the two checklist tables.
TWEAK	Finish with the 3 comments a reviewer is most likely to make, and how to pre-empt each.

How to get more out of it

- If the verdict says "not novel", reframe it as a teaching image or a letter rather than forcing a full report.
- Build the timeline table first. Reviewers read it before the text.
- Upload 3–5 similar published cases so the discussion compares against real papers.
- → **NEXT #368** Case report → conference poster

Design and write a case series

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT / FACULTY

USE IT

When you have 3 or more patients with a shared unusual feature, diagnosis or treatment experience.

WHY

A case series is more than several case reports stapled together. Its value lies in a clear case definition, consistent data for every patient, and patterns across cases, presented honestly as descriptive evidence without a comparison group.

COPY THE PROMPT BELOW

PERSONA	You are a clinical researcher and journal editor who has published and reviewed many case series. You insist on a clear case definition, consecutive or complete inclusion where possible, and descriptive (not inferential) claims.
MY CONTEXT	Specialty and setting: [brief] The shared feature defining the series: [e.g. post-COVID mucormycosis with orbital involvement] Number of patients and the period: [N, dates] How the patients were identified (all consecutive cases, or selected): [brief] De-identified data for each patient (age band, sex, presentation, investigations, treatment, outcome, follow-up): [PASTE TABLE] Target journal and word limit: [name / limit] Ethics and consent status: [brief] Any patients excluded, and why: [brief]
TASK	1. Write a precise case definition and describe how cases were identified, including any selection. 2. Build a patient-level table (one row per patient) with consistent variables. 3. Summarise the patterns descriptively: counts, proportions, ranges and medians, and common and divergent features. 4. Draft the manuscript: abstract, introduction, methods (definition, identification, data sources, ethics), case descriptions or a summary table, results, discussion comparing with the literature, limitations, and conclusion. 5. State clearly what a case series can and cannot conclude.
RULES	Use only the data I gave. Avoid p values and causal claims, because a case series has no comparison group. Put [CITE] for literature unless I uploaded the paper or you give a verified PMID or DOI. Keep all details de-identified.
OUTPUT	The case definition, the patient table, the pattern summary, then the manuscript draft with headings.
TWEAK	End with the single most important pattern and the study design that would test it properly.

How to get more out of it

- › Include all eligible cases from the period, not only the interesting ones. Say how many were excluded, and why.
- › A clear, complete patient-level table is often the most-cited part of a case series.
- › Check whether your IEC requires approval for a retrospective case series. Many do.
- › → **NEXT #365** Patient-level table for a case series

Case-report discussion grounded in the literature

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT

When the case description is written and you need the discussion and learning points.

WHY

The discussion is where a case report earns its place; it shows what the case adds to what is already published. Comparing with real, similar cases rather than a textbook summary is what reviewers look for.

COPY THE PROMPT BELOW

PERSONA	You are a senior clinician-academic who writes case-report discussions that teach. You compare the case with published cases systematically and draw practical lessons without overgeneralising.
MY CONTEXT	Case summary (de-identified): [PASTE] What makes it unusual: [brief] Similar published cases I have found (uploaded PDFs or pasted abstracts with citations): [PASTE OR UPLOAD] Target word count for the discussion: [e.g. 500 words] My specialty and the journal's readership: [brief] The diagnostic or management decision that was hardest in my case: [brief]
TASK	1. Summarise the published cases in a comparison table: age, sex, presentation, diagnosis, treatment and outcome, next to my case. 2. Identify what is similar, what differs, and what my case adds. 3. Discuss the diagnostic reasoning: why it was difficult, which clues mattered, which differentials were excluded and how. 4. Discuss management choices and the outcome in the light of the literature, including what might be done differently. 5. Write 3–5 learning points, each practical, specific and supported by the case. 6. Write the discussion within the word count, starting with the key reason the case is notable. 7. Suggest 2 recent reviews or guidelines I should search for and cite, described by topic rather than by title.
RULES	Compare only with the cases I supplied, citing each, because inventing similar cases is fabrication. Don't generalise from one case to practice recommendations; phrase lessons as "clinicians should consider..." rather than "should treat..."
OUTPUT	The comparison table, the discussion text, then the learning points.
TWEAK	End with the one sentence that best captures why this case deserves publication.

How to get more out of it

- › Search PubMed with the "Case Reports" article-type filter to find similar cases. Read them in full.
- › If very similar cases have been published recently, emphasise what differs about yours.
- › Learning points are often what readers remember. Make them specific enough to act on.
- › → **NEXT #362** Learning points that readers remember

354 Is my case publishable?

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT / FACULTY

USE IT Before you invest time writing, the moment a case seems interesting.

PERSONA	You are a case-reports editor who rejects most submissions for lack of novelty and knows what gets accepted.
MY CONTEXT	De-identified case summary: [PASTE] What I think is unusual: [brief] Similar cases I have found so far: [brief or "not searched yet"]
TASK	1. Classify the case's potential contribution: rare disease, rare presentation of a common disease, new association, diagnostic lesson, therapeutic lesson, adverse drug reaction, or educational value only. 2. Score novelty and educational value out of 5 each. 3. Recommend the best format: full case report, clinical image, letter, or teaching case only. 4. List the information and images I must collect now, before they are lost.
RULES	Base novelty on the case itself; where it depends on the literature, write [SEARCH PUBMED CASE REPORTS FIRST], because "first reported case" claims are often wrong.
OUTPUT	The classification, the scores, the format recommendation, then the collection checklist.

PRO TIP Collect the images, reports and consent now. Most unpublished case reports die because the data went missing.

TRY IT WITH Young woman with recurrent pneumothorax found to have thoracic endometriosis.

→ **NEXT #351** From interesting case to CARE-compliant case report

355 Build the case timeline table

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT For every case report, before writing the narrative.

PERSONA	You are a meticulous clinical writer who builds CARE-style timelines.
MY CONTEXT	De-identified clinical events with their dates or intervals: [PASTE NOTES] Reference point (Day 0): [e.g. admission / symptom onset]
TASK	1. Convert all dates to relative days, weeks or months from Day 0. 2. Build a timeline table: time point, clinical event, investigations and key results, intervention, and response. 3. Flag gaps or inconsistencies in the sequence. 4. Suggest which time points need an image or a result shown.
RULES	Remove all actual calendar dates and identifying details, because timelines are a common source of accidental identification.
OUTPUT	The timeline table, then the flags and the image suggestions.

PRO TIP A clear timeline often lets you cut 200 words from the case narrative.

TRY IT WITH Notes on a patient with fever, jaundice and acute kidney injury later diagnosed with leptospirosis.

→ **NEXT #360** Write the case presentation from my notes

356 Consent for publication and de-identification

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT Before submitting any case report or clinical image.

PERSONA	You are a publication-ethics editor who ensures patient privacy in case reports.
MY CONTEXT	Case details and images I plan to publish: [brief] Patient status: [able to consent / deceased / minor / lacks capacity] Journal's consent requirements (paste if available): [PASTE]
TASK	1. Explain what consent for publication must cover and who can give it in my patient's situation. 2. List everything that must be removed or masked: names, initials, dates, hospital numbers, locations, rare occupations, faces, tattoos, and metadata in image files. 3. Explain what to do if consent cannot be obtained, marked [CHECK JOURNAL POLICY]. 4. Draft a plain-language explanation to give the patient.
RULES	Treat consent as required unless the journal explicitly states otherwise, because publishing without it breaches patient rights and journal policy. Note that eye bars do not adequately de-identify faces.
OUTPUT	The consent explanation, the de-identification checklist, the no-consent guidance, then the patient-facing text.

PRO TIP Use the journal's own consent form if it has one. Many journals insist on it.

TRY IT WITH A case report with a facial photograph of a rare dermatological condition.

→ **NEXT #357** Select and prepare clinical images

357 Select and prepare clinical images

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT When preparing images, radiographs or histology slides for a case report or poster.

PERSONA	You are a medical illustrator and journal image editor.
MY CONTEXT	Available images and what each shows: [list] Key finding each image must demonstrate: [brief] Journal requirements (resolution, format, size): [paste]
TASK	1. Recommend which images to include and in what order to tell the diagnostic story. 2. Explain how to crop, annotate (arrows, labels) and de-identify each image. 3. Explain file requirements (e.g. 300 dpi, TIFF or JPEG) and how to remove metadata. 4. Write a legend for each image.
RULES	Tell me to make only adjustments applied to the whole image (brightness, contrast) and never to alter findings, because image manipulation is misconduct.
OUTPUT	The image selection plan, the preparation steps, then the legends.

PRO TIP Remove patient details burned into radiographs and scan images before anything else.

TRY IT WITH Chest X-ray, CT and histology images for a case of pulmonary alveolar proteinosis.

→ **NEXT #356** Consent for publication and de-identification

358 Case report abstract

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT When your case report needs an abstract.

PERSONA	You are a case-reports editor who writes clear abstracts.
MY CONTEXT	Case report draft or summary: [PASTE] Journal's abstract format (structured or unstructured) and word limit: [paste] The single lesson the case teaches: [one line] Target readership: [e.g. general physicians]
TASK	1. Write the abstract in the journal's required format: background (why the case matters), case presentation (key features), conclusion (the lesson). 2. Keep it within the limit. 3. Suggest 3–5 keywords, preferring MeSH terms.
RULES	Use only details in the draft, and make the conclusion a lesson rather than a generalisation, because one case cannot establish a rule.
OUTPUT	The abstract with its word count, then the keywords.

PRO TIP Put the unusual feature in the first sentence. That is what makes an editor keep reading.

TRY IT WITH Case of Kikuchi disease presenting as fever of unknown origin in a young man.

→ **NEXT #359** Case report titles that follow the CARE convention

359 Case report titles that follow the CARE convention

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT When choosing your case report's title.

PERSONA	You are a journal editor who knows what titles attract readers and meet CARE conventions.
MY CONTEXT	Diagnosis and the unusual feature: [brief] Journal's title limits: [characters or words] Target readership: [e.g. general physicians / specialists]
TASK	1. Write 5 titles that include "case report" (or "a case report"). 2. Put the key diagnosis or phenomenon first. 3. Avoid overclaiming words such as "first", "unique" or "novel" unless verified. 4. Recommend one, with the reason.
RULES	Mark any "first report" claim as [VERIFY WITH LITERATURE SEARCH], because unverified priority claims are a common reason for reviewer criticism.
OUTPUT	Five titles with character counts, then the recommendation.

PRO TIP Titles that describe the lesson ("...mimicking...", "...presenting as...") are read more often.

TRY IT WITH Tuberculous pericarditis presenting as cardiac tamponade in a patient on dialysis.

→ **NEXT #358** Case report abstract

360 Write the case presentation from my notes

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT For the main narrative of a case report.

PERSONA	You are a clinical writer who produces concise, chronological case presentations.
MY CONTEXT	De-identified notes: history, examination, investigations with results, management, and course: [PASTE NOTES] Word limit for this section: [e.g. 400 words]
TASK	1. Write the case presentation chronologically: presenting complaint, relevant history, examination, initial investigations, diagnostic reasoning and further tests, treatment, and outcome. 2. Include only details relevant to the diagnosis or lesson. 3. Report laboratory values with units and reference ranges where helpful. 4. Point to the timeline table and images at the right places.
RULES	Use only facts in my notes; where a relevant detail is missing, write [ASK ME: ...] , because invented clinical details in a case report are fabrication.
OUTPUT	The case presentation with its word count, then any [ASK ME] items listed.

PRO TIP Negative findings that ruled out important differentials are relevant. Include them.

TRY IT WITH Notes on a 30-year-old with stroke found to have antiphospholipid syndrome.

→ **NEXT #361** Show how differentials were excluded

361 Show how differentials were excluded

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT For the diagnostic-assessment part of a case report.

PERSONA	You are a diagnostician who documents diagnostic reasoning transparently.
MY CONTEXT	Final diagnosis: [diagnosis] Differentials considered and the tests or findings used to exclude each: [PASTE NOTES] The investigation that finally confirmed the diagnosis: [brief]
TASK	1. Build a table: differential, why it was considered, the finding that excluded it, and the timing. 2. Identify any important differential that is not documented as excluded. 3. Write the diagnostic-assessment paragraph for the case report.
RULES	Use only the tests and findings I documented; flag any differential without a documented exclusion as [NOT EXCLUDED IN NOTES] , because reviewers will ask about it.
OUTPUT	The differential table, the missing-exclusion flags, then the paragraph.

PRO TIP Reviewers almost always ask "was X excluded?". This table answers it before they do.

TRY IT WITH Final diagnosis: adult-onset Still's disease; differentials: sepsis, lymphoma, SLE, TB.

→ **NEXT #353** Case-report discussion grounded in the literature

362 Learning points that readers remember

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT For the "learning points" or "key messages" box of a case report.

PERSONA	You are a medical educator who writes practical, memorable learning points.
MY CONTEXT	Case summary and the lesson I think it teaches: [brief] Journal's format (number of points, word limit): [paste] Intended readers: [e.g. residents, general practitioners]
TASK	1. Write 3–5 learning points, each one sentence. 2. Make each specific and actionable (e.g. "Consider X in patients with Y who do not respond to Z"). 3. Avoid generic points ("a high index of suspicion is needed"). 4. Check that each point follows from the case.
RULES	Phrase points as considerations, not rules, because a single case can't establish practice.
OUTPUT	The learning points, then a one-line check of each.

PRO TIP If a point could apply to any case report, it's too generic. Rewrite it.

TRY IT WITH Case of methaemoglobinaemia after dapsone use, initially misdiagnosed as cyanotic heart disease.

→ **NEXT #353** Case-report discussion grounded in the literature

363 "Images in clinical medicine" submission

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT / FACULTY

USE IT When a case has one striking image but may not justify a full case report.

PERSONA	You are an editor of a clinical-images section who selects images that teach at a glance.
MY CONTEXT	The image(s) and what they show: [describe] Brief clinical context: [2–3 lines] Journal's clinical-image format (word limit, number of images, references): [paste]
TASK	1. Judge whether the image is teaching quality: classic, striking and clearly showing the finding. 2. Write the text within the limit: a brief clinical context, what the image shows, the diagnosis, and the outcome. 3. Write the image legend with annotations described. 4. Suggest a title.
RULES	Use only the details I gave; de-identify fully and mark consent as required, because image sections have the same consent requirements as case reports.
OUTPUT	The judgement, the title, the text, then the legend.

PRO TIP Image sections are a quick first publication for residents. The image must be excellent.

TRY IT WITH Chest X-ray showing a classic "air crescent" sign in invasive aspergillosis.

→ **NEXT #356** Consent for publication and de-identification

364 Journals for case reports: find and verify them

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT When deciding where to submit a case report.

PERSONA	You are a publication adviser who knows the case-report publishing landscape and its predatory corners.
MY CONTEXT	Specialty and the type of case: [brief] Budget for publication charges: [amount or none] Indexing needed: [PubMed / Scopus / none required]
TASK	1. Describe the types of venues that publish case reports: specialty journals with case sections, dedicated case-report journals, and clinical-image sections. 2. Explain how to find candidates and verify their indexing and legitimacy. 3. List predatory red flags specific to case-report publishing. 4. Build a shortlist table from the journals I find.
RULES	Don't name journals, indexing status or charges from memory; mark anything named as [VERIFY], because case-report publishing has many predatory outlets.
OUTPUT	Venue types, finding and verification steps, red flags, then the shortlist template.

PRO TIP Check your own specialty society's journal first. Many accept case reports.

TRY IT WITH Unusual paediatric neurology case, no publication budget, PubMed indexing preferred.

→ **NEXT #351** From interesting case to CARE-compliant case report

365 Patient-level table for a case series

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT For the central table of your case series.

PERSONA	You are a clinical data editor who designs clear patient-level tables.
MY CONTEXT	Variables available for all patients: [list] Data for each patient (de-identified): [PASTE]
TASK	1. Choose the columns that show the key features consistently for every patient. 2. Build the table with one row per patient (Case 1, Case 2...). 3. Add a summary row or column (counts, ranges or medians). 4. Write the table title and footnotes.
RULES	Keep all data de-identified using age bands if needed, and write NA where data is unavailable rather than leaving blanks, because missing cells must be explained.
OUTPUT	The table, then the title and footnotes.

PRO TIP If the table is too wide, split it into clinical features and management/outcome tables.

TRY IT WITH Seven patients with scrub typhus meningoencephalitis.

→ **NEXT #352** Design and write a case series

366 Find previously reported similar cases

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT Before writing the discussion of a case report or series.

PERSONA	You are a medical librarian who searches for case reports efficiently.
MY CONTEXT	The key features of my case (diagnosis, unusual presentation, association): [brief]
TASK	1. Build a PubMed search string combining the key concepts with the "Case Reports" publication type. 2. Suggest Google Scholar queries for non-indexed reports. 3. Explain how to screen the results and extract comparison data (age, sex, presentation, treatment, outcome). 4. Give a comparison-table template.
RULES	Don't list case reports from memory; give me the search to run, because any list you produce without live search may contain fabricated cases.
OUTPUT	The search strings, the screening steps, then the table template.

PRO TIP Check the reference lists of the most recent similar case reports. They are often mini-reviews.

TRY IT WITH Case: Castleman disease presenting as a posterior mediastinal mass.

→ **NEXT #353** Case-report discussion grounded in the literature

367 Patient-perspective section

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT For the CARE "patient perspective" item, when the patient agrees to contribute.

PERSONA	You are a patient-centred clinician who helps patients describe their experience in their own words.
MY CONTEXT	Brief case summary: [brief] Notes from my conversation with the patient (their own words where possible): [PASTE NOTES] Language the patient spoke in: [language]
TASK	1. Suggest 5 gentle questions to ask the patient about their experience (symptoms, diagnosis journey, treatment and recovery). 2. From my notes, draft a short patient-perspective paragraph in the first person, faithful to their words. 3. Suggest how to have the patient review and approve the text.
RULES	Keep the patient's meaning and words; don't add emotions or statements they did not express, because this section must be genuinely theirs.
OUTPUT	The questions, the draft paragraph, then the approval steps.

PRO TIP Translate carefully and read the translation back to the patient before you use it.

TRY IT WITH A patient who went undiagnosed for 2 years with a rare autoimmune condition.

→ **NEXT #351** From interesting case to CARE-compliant case report

368 Case report → conference poster

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT When presenting your case at a conference as a poster or e-poster.

PERSONA	You are a conference poster designer who turns case reports into posters people stop to read.
MY CONTEXT	Case report text or summary: [PASTE] Key images: [list] Poster size and format: [e.g. A0 portrait / e-poster 16:9] Conference word limit, if any: [limit]
TASK	1. Plan the panels: title, introduction (2 lines), case presentation (as a timeline or bullet points), key images, diagnosis, discussion (3 points), learning points, and references. 2. Write the text for each panel within about 400–600 words in total. 3. Suggest the layout and reading flow, with images central. 4. Suggest a one-line "hook" for the top of the poster.
RULES	Use only details from my case report; keep images de-identified and consent in place, because posters are photographed and shared.
OUTPUT	A panel-by-panel plan with text, then layout notes.

PRO TIP People read posters from 2 metres away. If a panel needs a paragraph, it needs cutting.

TRY IT WITH Case report of a rare paediatric pulmonary condition with CT images.

→ **NEXT #382** Research poster from my manuscript or thesis

369 Critique my case report draft

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

USE IT When your case report is drafted, before submission.

PERSONA	You are a reviewer for a case-reports journal who applies CARE and the journal's standards.
MY CONTEXT	My draft: [PASTE DRAFT] Target journal and word limit: [brief]
TASK	1. Score the draft on novelty, clarity of the timeline, completeness of the diagnostic reasoning, quality of the literature comparison, strength of the learning points, and CARE compliance. 2. Quote the passage behind each score. 3. List the top 5 fixes. 4. Flag privacy risks and missing consent statements.
RULES	Base comments only on my text; quote rather than paraphrase, because vague feedback can't be acted on.
OUTPUT	A scoring table, the top fixes, then the privacy flags.

PRO TIP Run this again after revising. The timeline and diagnostic-reasoning scores usually rise the most.

TRY IT WITH A 1,400-word draft case report of paraquat poisoning with delayed lung fibrosis.

→ **NEXT #351** From interesting case to CARE-compliant case report

370 Adverse drug reaction case report with causality assessment

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT / FACULTY

USE IT When a case involves a suspected adverse drug reaction.

PERSONA	You are a clinical pharmacologist experienced in pharmacovigilance and ADR case reporting.
MY CONTEXT	Suspected drug(s), doses, start and stop dates (relative), reaction, timeline, dechallenge and rechallenge, and other possible causes: [PASTE DE-IDENTIFIED DETAILS]
TASK	1. Apply a causality assessment (e.g. WHO-UMC categories and the Naranjo scale), showing each criterion and its score. 2. Assess severity and seriousness, and whether the reaction is expected (labelled). 3. Draft the ADR case report following CARE and relevant ADR reporting guidance. 4. Remind me to report to the national pharmacovigilance programme, marked [VERIFY CURRENT REPORTING PROCESS].
RULES	Show the reasoning for each causality criterion from my details only, and mark unknown elements as [NOT DOCUMENTED], because causality scores are often challenged.
OUTPUT	The causality table, the severity assessment, the draft, then the reporting reminder.

PRO TIP Report to the pharmacovigilance programme even if you don't publish. Every report counts.

TRY IT WITH Suspected linezolid-induced lactic acidosis in a patient on TB treatment.

→ **NEXT #351** From interesting case to CARE-compliant case report

371 Clinicopathological conference (CPC) write-up

STANDARD

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT / FACULTY

USE IT When presenting or publishing a clinicopathological conference case, where the diagnosis is revealed at the end.

PERSONA	You are a senior physician and pathologist who has run CPCs for years, and knows how to build diagnostic suspense while staying fair to the reader.
MY CONTEXT	De-identified case details in chronological order (history, examination, investigations, course): [PASTE] Final diagnosis and how it was established (histopathology, autopsy or test): [brief] Audience: [residents / faculty / mixed]
TASK	1. Write the case protocol as it would be presented, revealing information in the order the clinicians received it, without giving away the diagnosis. 2. Write the discussant's differential-diagnosis reasoning, narrowing step by step to the most likely diagnosis. 3. Write the pathology or final-diagnosis section with key findings. 4. Add 3 teaching points and 2 questions for the audience.
RULES	Use only the details I gave, and keep the clinical clues fair, because a CPC that hides key data teaches nothing. Keep everything de-identified.
OUTPUT	The case protocol, the discussant's analysis, the final diagnosis, then the teaching points and questions.

PRO TIP Pick a case where the diagnosis could have been reached from the clinical data. Those make the best CPCs.

TRY IT WITH Fever, pancytopenia and splenomegaly in a young man, finally diagnosed as visceral leishmaniasis on marrow.

→ **NEXT #361** Show how differentials were excluded

372 One line on why my case is worth reading

QUICK

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

Act as a case-reports editor. From my case summary, [PASTE SUMMARY], write one clear sentence of under 30 words that states exactly what the case adds (a rare presentation, association, diagnostic lesson or management lesson), without unverified "first report" claims.

TRY IT WITH A young non-smoker with pulmonary Langerhans cell histiocytosis diagnosed on cryobiopsy.

373 Keywords to search for similar cases

QUICK

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

Act as a medical librarian. For my case, [KEY FEATURES], give me 8 useful search terms and synonyms (including older names and MeSH terms) and one ready-to-run PubMed search string limited to case reports, to find previously published similar cases.

TRY IT WITH Hypokalaemic periodic paralysis as the presenting feature of thyrotoxicosis.

374 Is my case report properly anonymised?

QUICK

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

Act as a privacy reviewer for a medical journal. Check this case text and image list, [PASTE TEXT AND IMAGE LIST], for anything that could identify the patient: dates, places, occupation, rare details, faces, tattoos and image metadata. List each risk with a fix.

TRY IT WITH A case mentioning the village name, the exact admission date and a distinctive tattoo.

375 Cut my case report to the word limit

QUICK

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

Act as a case-reports editor. Cut this case report, [PASTE TEXT], from [CURRENT] to [TARGET] words, keeping the timeline, key investigations, the excluded differentials and the learning points, and moving detail into the timeline table. Show the new word count.

TRY IT WITH A 1,600-word case report for a journal with a 1,000-word limit.

376 Legend for a clinical image

QUICK

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

Act as a journal image editor. Write a legend for this image, [DESCRIBE IMAGE AND FINDINGS], that states the modality or type, the view, what the arrows or labels point to, and the key finding, in under 50 words, with abbreviations defined.

TRY IT WITH Contrast CT showing a ring-enhancing lesion with a target sign in the right frontal lobe.

377 Turn my case into a teaching caption

QUICK

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT / FACULTY

Act as a medical educator. Turn my case, [BRIEF CASE], into a 60-word teaching caption for a department WhatsApp group or case-of-the-week slide, written for junior residents: the presentation, the key finding, the diagnosis and one learning point, fully de-identified.

TRY IT WITH Elderly man with hiccups for 2 weeks found to have lateral medullary infarct.

378 Explain publication consent to my patient

QUICK

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

Act as a compassionate clinician. Write a short, simple explanation (under 100 words) that I can say to my patient, [BRIEF CONTEXT], explaining why I want to publish their case, what will be shared, how their identity will be protected, and that they can refuse.

TRY IT WITH A patient who recovered from a rare poisoning, speaking Hindi.

379 Case report or case series?

QUICK

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

Act as a journal editor. I have [N] patients with [SHARED FEATURE]. Tell me whether to write a single case report, a case series or a clinical image, what each would need, and which journals typically prefer which, in under 120 words.

TRY IT WITH Three patients with the same rare drug reaction in 6 months.

380 The key sentence of my case-report cover letter

QUICK

1.12 CASE REPORTS & CASE SERIES · PG RESIDENT

Act as a senior author. Write the 2 key sentences of a cover letter to the editor for my case report, [TITLE AND ONE-LINE SUMMARY], explaining why the case matters to this journal's readers, [JOURNAL], without exaggerating novelty or making unverified priority claims.

TRY IT WITH Case of tuberculous mastitis mimicking breast carcinoma in a young woman.

SECTION 1.13

1.13

Conference abstracts, posters & oral papers

A conference presentation is often your research's first audience. Write to the exact limit, design for two metres away, and rehearse the questions.

20

prompts

3

Master

11

Standard

6

Quick

HOW TO USE THIS SECTION

Use it when an abstract deadline, a poster or a conference slot is coming up.

Start with the Master prompts (**#381–383**) for the full workflow; use the Standard prompts (**#384–394**) for one task at a time; reach for the Quick prompts (**#395–400**) when you have two minutes. Tap any title below to jump to it, and follow **→ NEXT** to move to the next step.

- #381** Conference abstract to the exact word limit
- #382** Research poster from my manuscript or thesis
- #383** Thesis → 8-minute oral paper, with slides and Q&A
- #384** Choose the right conference for my work
- #385** Score my abstract like a conference reviewer
- #386** Adapt my poster into an e-poster
- #387** Layout brief for Canva or PowerPoint
- #388** 90-second poster walk-through for judges
- #389** Slide plan for a short oral presentation
- #390** Prepare for questions after my talk

- #391** Prepare for an award-paper session
- #392** Application for a travel grant or bursary
- #393** From conference presentation to published paper
- #394** Record a presentation for a virtual or hybrid conference
- #395** An accurate, catchy abstract title
- #396** 25-word summary for the conference programme
- #397** One-page handout or QR page for my poster
- #398** Will my talk fit the time?
- #399** The closing take-home slide
- #400** Follow-up email after the conference

Conference abstract to the exact word limit

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT / FACULTY

USE IT

Before a conference abstract deadline, once your results (or interim results) are verified.

WHY

Conference abstracts are judged in minutes by reviewers scoring dozens of submissions. A clear question, a stated design, real numbers and a proportionate conclusion, within the exact limit and format, is what gets accepted, and often what gets you into the award session.

COPY THE PROMPT BELOW

PERSONA	You are a scientific committee member for national and international medical conferences who scores hundreds of abstracts each year. You know what reviewers reward and what gets rejected.
MY CONTEXT	Conference name and its abstract rules (word or character limit, headings, whether the title and authors count, category options): [PASTE RULES] Study design, setting and period: [brief] Sample size and participant flow: [numbers] Primary result with estimate, CI and p: [paste exactly] Key secondary results: [paste] Whether the data is final or interim: [final / interim with n so far]
TASK	1. Draft the abstract in the conference's exact format and headings, within the limit, stating the count used (words or characters, as the conference counts them). 2. Background: the problem and the gap in 2 sentences. Methods: design, setting, participants, key measurements and analysis. Results: the numbers that answer the objective. Conclusion: proportionate, and linked to practice or research. 3. Write 3 title options within any title limit. 4. Recommend the best category, and whether to opt for oral, poster or award consideration. 5. Score the draft against typical reviewer criteria (importance, methods, clarity of results, validity of the conclusion, presentation) and revise the weakest part.
RULES	Use only my numbers, reproduced exactly; label interim data as interim, because presenting interim results as final is misconduct. No references unless the conference allows them. No "results will be presented" placeholders, because reviewers reject abstracts without results.
OUTPUT	The final abstract with its count, the titles, the category advice, then the scoring table.
TWEAK	End with the one sentence a reviewer is most likely to criticise, and a safer rewrite.

How to get more out of it

- › Check how the conference counts: words or characters, and whether spaces, title and authors are included.
- › Submit a day early. Portals slow down or crash near deadlines.
- › Keep the abstract text. The journal version later must not contradict it.
- › → **NEXT #382** Research poster from my manuscript or thesis

Research poster from my manuscript or thesis

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT / FACULTY

USE IT

After abstract acceptance as a poster, 2–3 weeks before the conference.

WHY

Most posters are shrunken papers that nobody reads. A good poster is read from two metres away in under a minute: one clear question, one clear answer, figures instead of paragraphs. This prompt turns your paper into that.

COPY THE PROMPT BELOW

PERSONA	You are a scientific poster designer and conference judge who has evaluated hundreds of posters. You design for visual hierarchy, brevity and a clear take-home message.
MY CONTEXT	Accepted abstract: [PASTE ABSTRACT] Key results, tables and figures available: [list] Poster specifications (size, orientation, print or e-poster, template requirements): [paste] Design tool: [PowerPoint / Canva / other] Institution logo and branding requirements: [brief] Whether I will stand by the poster or it will be viewed alone: [brief]
TASK	1. Write a one-line take-home message to feature prominently. 2. Plan the panels: title, authors and affiliations, background (40 words), objective (1 line), methods (a small flow diagram or 4 bullets), results (2–3 figures or tables with one-line captions), conclusion (2–3 bullets), and references, acknowledgements and QR code. 3. Write the text for every panel, 500–800 words in total. 4. Specify the layout: columns, reading order, the central figure, font sizes (title, headings, body), colours and white space. 5. Convert any results table into a figure where that would be clearer. 6. Give step-by-step build instructions in my tool. 7. Write a 90-second walk-through script for when judges stop by.
RULES	Use only content and numbers from my abstract and results, keeping them identical to the abstract, because judges compare the poster with the abstract. Keep text sizes readable from 2 metres (body about 24–32 pt on an A0 poster; mark exact sizes [CHECK TEMPLATE]).
OUTPUT	The take-home message, a panel table with text, the layout specification, the build steps, then the walk-through script.
TWEAK	End with a checklist to run on the printed or displayed poster before the session.

How to get more out of it

- › If your poster has more than 800 words, it has too many.
- › Put your main result graph in the centre, at eye level.
- › A QR code linking to the abstract or full paper lets interested readers go deeper.
- › → **NEXT #388** 90-second poster walk-through for judges

Thesis → 8-minute oral paper, with slides and Q&A

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

USE IT

When your abstract is accepted for oral or award-paper presentation.

WHY

Oral paper slots are short and strictly timed, and chairs cut off speakers who overrun. A timed slide plan, a spoken script and rehearsed answers turn a nervous 12-minute ramble into a confident 8-minute presentation that wins awards.

COPY THE PROMPT BELOW

PERSONA	You are a conference presentation coach and award-session judge who has trained many residents for oral paper presentations.
MY CONTEXT	Abstract and key results: [PASTE] Time allowed for the talk and for discussion: [e.g. 8 + 2 minutes] Figures and tables available: [list] Award session?: [yes / no] My speaking pace and confidence: [brief] What the audience will mostly be: [specialists / mixed / students] Conference's slide requirements (template, aspect ratio, file format): [paste]
TASK	1. Plan 8–10 slides with a time budget per slide: title; background and gap; objective; methods (one visual); participant flow; primary result (one clear figure); secondary results; limitations; conclusion and take-home. 2. For each slide, give the title as a message ("Early mobilisation halved delirium days"), the content (maximum 3 bullets or 1 figure), and speaker notes. 3. Write a spoken script at about 130 words per minute that fits the time exactly. 4. Prepare the 8 most likely audience and judge questions with 30-second answers. 5. Give a rehearsal plan for the week before, including one rehearsal in front of colleagues who ask hard questions. 6. List the technical checks for the day (file format, fonts embedded, videos, backup on a pen drive and email).
RULES	Keep every number identical to my abstract, and don't add results that aren't in it, because judges check. Keep slides visual: no paragraphs, and at least 24 pt text.
OUTPUT	A slide table (slide, title, content, time, notes), the script with its word count, the Q&A table, then the rehearsal plan.
TWEAK	End with the opening 2 sentences and the closing 2 sentences to memorise word for word.

How to get more out of it

- Rehearse with a timer at least 5 times. Most residents overrun by 30–50%.
- Write slide titles as messages, not topics. The audience should get the story from the titles alone.
- Have 2–3 backup slides for predictable questions.
- → **NEXT #390** Prepare for questions after my talk

384 Choose the right conference for my work

STANDARD

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT / FACULTY

USE IT When deciding where to submit your abstract.

PERSONA	You are an academic mentor who guides residents on conference strategy.
MY CONTEXT	Specialty and the topic of my work: [brief] Stage of the work: [final results / interim / case report] Budget and travel constraints: [brief] Goals: [award, networking, first presentation, international exposure]
TASK	1. Describe the options: national specialty conferences, state or zonal meetings, international conferences and virtual meetings, and what each offers. 2. Explain how to find upcoming conferences and deadlines, and check that they are legitimate (predatory conferences exist too). 3. Explain the categories typically available (oral, poster, e-poster, award paper, case report sessions) and eligibility, e.g. age or membership requirements for awards. 4. Build a comparison table from the conferences I find.
RULES	Don't name specific conferences, dates or fees from memory; mark anything named as [VERIFY ON THE OFFICIAL SITE], because dates and rules change every year.
OUTPUT	The options overview, the finding and legitimacy steps, the categories note, then the comparison template.

PRO TIP Student and resident award sessions at national specialty conferences are among the best first stages for your work.

TRY IT WITH MD Pulmonary Medicine resident with final thesis results, limited travel budget.

→ **NEXT #381** Conference abstract to the exact word limit

385 Score my abstract like a conference reviewer

STANDARD

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

USE IT Before you submit a conference abstract.

PERSONA	You are a conference abstract reviewer who scores submissions on a standard rubric.
MY CONTEXT	My abstract: [PASTE ABSTRACT] Conference and category: [brief] Limit: [words or characters]
TASK	1. Score it out of 5 on: importance of the question, clarity of the objective, appropriateness of the methods, completeness of the results (with numbers), validity of the conclusion, and clarity of writing. 2. Quote the text behind each score. 3. Say whether it would likely be accepted, and in which format (oral or poster). 4. Give the 3 most important revisions.
RULES	Be honest about the likely outcome, and quote rather than paraphrase, because encouraging but vague feedback won't improve the abstract.
OUTPUT	A scoring table, the verdict, then the revisions.

PRO TIP Abstracts without numbers in the results are usually rejected. Check yours has them.

TRY IT WITH A 250-word abstract on ultrasound-guided versus landmark-guided central line insertion.

→ **NEXT #381** Conference abstract to the exact word limit

386 Adapt my poster into an e-poster

STANDARD

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

USE IT When the conference uses screen-displayed e-posters.

PERSONA	You are a digital poster designer who adapts print posters for screens.
MY CONTEXT	My poster content or file description: [PASTE OR DESCRIBE] E-poster specification (aspect ratio, single or multiple slides, resolution, file type, interactivity allowed): [paste]
TASK	1. Restructure the content for the format: a single 16:9 slide or a short slide set. 2. Reduce the text further for on-screen reading distance. 3. Suggest a reading flow and the use of any allowed interactivity (e.g. clickable panels). 4. Specify export settings.
RULES	Keep all numbers identical to the abstract; don't add new content, because e-posters are judged against the submitted abstract.
OUTPUT	The restructured layout plan, the reduced text, then the export settings.

PRO TIP Check your e-poster on a laptop from 1.5 metres before submitting. If you can't read it, neither can judges.

TRY IT WITH An A0 portrait poster converted to one 16:9 e-poster slide.

→ **NEXT #388** 90-second poster walk-through for judges

387 Layout brief for Canva or PowerPoint

STANDARD

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

USE IT When building the poster yourself.

PERSONA	You are a graphic designer who specialises in clean scientific posters.
MY CONTEXT	Poster size and orientation: [e.g. A0 portrait, 841 × 1189 mm] Panels and approximate text length for each: [list] Figures: [list] Tool: [Canva / PowerPoint]
TASK	1. Give the page set-up for my tool (custom size, margins, grid). 2. Suggest a 3-column grid and the placement of each panel. 3. Give a font system (one sans-serif family, sizes for title, headings, body and captions) and a colour palette of 2–3 colours plus neutrals that is colour-blind safe. 4. Describe alignment, spacing and how to place figures so they print sharply. 5. Give print-export settings.
RULES	Recommend readable, conservative choices, and mark exact template or institutional branding rules as [CHECK TEMPLATE], because conferences often mandate them.
OUTPUT	Numbered set-up and design steps, a font and colour table, then export settings.

PRO TIP Leave about 25% of the poster as white space. Crowded posters are skipped.

TRY IT WITH A0 landscape poster in PowerPoint with 3 figures and 1 table.

→ **NEXT #382** Research poster from my manuscript or thesis

388 90-second poster walk-through for judges

STANDARD

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

USE IT The week before your poster session.

PERSONA	You are a poster-session judge who knows what makes a presenter memorable.
MY CONTEXT	My poster content (panel text or summary): [PASTE] Time judges typically spend: [e.g. 2–3 minutes]
TASK	1. Write a 90-second script: the hook, the question, what I did, the key result (pointing to the figure), what it means, and one limitation. 2. Write a 30-second version for passers-by. 3. List 6 questions judges ask at posters, with short answers from my content. 4. Give body-language and positioning tips.
RULES	Use only what is on my poster, with numbers identical to it, because judges look at the poster while you talk.
OUTPUT	The two scripts with timing, the Q&A table, then the tips.

PRO TIP Start with the question, not the background. "We asked whether..." gets attention.

TRY IT WITH Poster on the prevalence of vitamin B12 deficiency in metformin-treated diabetics.

→ **NEXT #382** Research poster from my manuscript or thesis

389 Slide plan for a short oral presentation

STANDARD

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

USE IT When you need slides quickly for a 5–10 minute talk.

PERSONA	You are a presentation designer for medical conferences.
MY CONTEXT	Abstract and results: [PASTE] Time limit: [minutes] Slide template requirements: [brief] Audience: [specialists / mixed]
TASK	1. Recommend the number of slides (about one per minute). 2. For each slide, give a message title, a content suggestion (figure, table or a maximum of 3 bullets), and speaker notes. 3. Suggest the design: font sizes, contrast and one accent colour. 4. List slides to put in the backup section.
RULES	Keep numbers identical to the abstract and avoid text-heavy slides, because audiences read slides instead of listening.
OUTPUT	A slide table, the design notes, then the backup list.

PRO TIP One figure per results slide, with the key number called out in large text.

TRY IT WITH 7-minute oral paper on anaemia prevalence among adolescent girls in a rural block.

→ **NEXT #383** Thesis → 8-minute oral paper, with slides and Q&A

390 Prepare for questions after my talk

STANDARD

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

USE IT In the days before your oral or poster presentation.

PERSONA	You are a session chair who has asked and heard thousands of conference questions.
MY CONTEXT	My abstract and key slides: [PASTE] Known limitations: [list] Audience: [specialists / mixed / award judges]
TASK	1. List 12 likely questions: methodology, sample size, bias, generalisability, clinical relevance, "what next", and one hostile question. 2. Write a 30-second answer for each, grounded in my study. 3. Give a technique for questions I can't answer (acknowledge, offer a reasoned view, and follow up). 4. Suggest 2 backup slides that answer the most likely questions.
RULES	Base answers on my data only; where I lack the information, script an honest "we didn't measure that", because bluffing in front of experts is obvious.
OUTPUT	A Q&A table, the technique, then the backup-slide suggestions.

PRO TIP Repeat the question before answering. It buys you thinking time and helps the audience hear it.

TRY IT WITH Oral paper on a retrospective cohort of snakebite outcomes.

→ **NEXT #383** Thesis → 8-minute oral paper, with slides and Q&A

391 Prepare for an award-paper session

STANDARD

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

USE IT When your paper is selected for a best-paper or young-investigator award session.

PERSONA	You are an award-session judge at national medical conferences.
MY CONTEXT	Abstract and presentation plan: [PASTE] Award rules and judging criteria, if published: [paste] Time limits: [brief]
TASK	1. Map the typical judging criteria (originality, methodological rigour, clinical relevance, clarity, time management and handling of questions) to my presentation, and identify gaps. 2. Suggest changes to strengthen my weakest criterion. 3. Prepare answers to the judges' likely tough questions. 4. Give a checklist for the day: file formats, backup copies, arrival time, dress and timing.
RULES	Mark criteria that I haven't confirmed from the official rules as [CHECK AWARD RULES], because eligibility and criteria differ between conferences.
OUTPUT	A criteria-mapping table, the improvements, the Q&A, then the checklist.

PRO TIP Judges remember speakers who finish on time and answer questions honestly.

TRY IT WITH Award session for a prospective cohort on high-flow nasal oxygen in COPD.

→ **NEXT #390** Prepare for questions after my talk

392 Application for a travel grant or bursary

STANDARD

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

USE IT When applying for conference funding from your institution, a society or a foundation.

PERSONA	You are a grants officer who reviews travel-grant applications.
MY CONTEXT	Conference, dates and the type of presentation accepted: [brief] Estimated costs (registration, travel, stay): [amounts] Funding body and its criteria: [paste] My academic profile (programme, year, publications, presentations): [brief]
TASK	1. Write a concise application letter or statement: what I'm presenting, why it matters, what I'll gain, and how I'll share the learning. 2. Build a cost table with justification. 3. List the documents usually required (acceptance letter, abstract, guide's recommendation, CV).
RULES	Use only facts I provided and mark criteria I haven't confirmed as [CHECK FUNDER CRITERIA], because inaccurate applications are rejected.
OUTPUT	The statement, the cost table, then the document list.

PRO TIP Apply as soon as your abstract is accepted. Travel-grant deadlines are often earlier than you expect.

TRY IT WITH Oral presentation at an international respiratory conference, first-time applicant.

→ **NEXT #384** Choose the right conference for my work

393 From conference presentation to published paper

STANDARD

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT / FACULTY

USE IT After the conference, while feedback is fresh.

PERSONA	You are a publication mentor who helps residents turn presentations into papers.
MY CONTEXT	My abstract and presentation: [PASTE] Questions and feedback received at the conference: [brief] Whether the abstract was published in a journal supplement: [yes / no]
TASK	1. List the feedback that should shape the paper. 2. Explain how a published conference abstract affects journal submission (it is usually acceptable, but must be disclosed), marked [CHECK TARGET JOURNAL POLICY]. 3. Make a plan for writing the full paper within 6 weeks. 4. Draft the disclosure sentence about the prior presentation.
RULES	Keep results consistent with the presented abstract, or explain any differences (e.g. final versus interim data), because inconsistency raises questions about data integrity.
OUTPUT	The feedback list, the policy note, the 6-week plan, then the disclosure sentence.

PRO TIP Conference momentum fades in weeks. Start the paper the week you return.

TRY IT WITH Poster on inhaler technique errors that received strong interest at a national meeting.

→ **NEXT #316** Write the full manuscript for journal submission

394 Record a presentation for a virtual or hybrid conference

STANDARD

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT / FACULTY

USE IT When the conference asks for a pre-recorded talk or you present online.

PERSONA	You are a medical educator who produces clear, professional recorded talks on a modest budget.
MY CONTEXT	Talk length and format required (video file, live online, slide upload): [paste] Slides ready?: [yes / no] Equipment available: [laptop, phone, microphone, quiet room]
TASK	1. Give a set-up checklist: camera at eye level, lighting in front of me, a quiet room, microphone choice, and a plain background. 2. Explain how to record in PowerPoint, Zoom or similar tools, including slide-by-slide recording to make editing easier. 3. Adapt my script for the camera: shorter sentences, and a visual cue every 30 seconds. 4. Give export settings and the file-naming convention the conference requires. 5. Plan for live Q&A online: internet backup, and mute and unmute etiquette.
RULES	Mark platform-specific steps as [MAY DIFFER BY VERSION], because recording menus change often. Keep the talk within the time limit exactly, since recorded talks are often cut automatically.
OUTPUT	The set-up checklist, recording steps, script tips, export settings, then the Q&A plan.

PRO TIP Record one slide at a time. A mistake then costs 40 seconds, not the whole talk.

TRY IT WITH 10-minute pre-recorded oral paper for an international virtual respiratory conference.

→ **NEXT #383** Thesis → 8-minute oral paper, with slides and Q&A

QUICK PROMPTS · 1.13

395 An accurate, catchy abstract title

QUICK

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

Act as a conference abstract reviewer. Write 4 titles for my abstract, [ONE-LINE SUMMARY WITH DESIGN], under [LIMIT] characters: two descriptive and two that state the finding. None should overclaim or use undefined abbreviations. Mark the one most likely to attract reviewers.

TRY IT WITH Prospective cohort showing frailty predicts delirium after hip fracture surgery.

396 25-word summary for the conference programme

QUICK

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

Act as a conference programme editor. Summarise my accepted presentation, [PASTE ABSTRACT], in 25 words or fewer for delegates for the printed conference programme or app: what I studied, in whom, where, and the main finding, accurately, clearly and without jargon.

TRY IT WITH Abstract on the diagnostic accuracy of lung ultrasound in paediatric pneumonia.

397 One-page handout or QR page for my poster

QUICK

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

Act as a scientific communicator. Create the content for a one-page handout or QR-linked page for my poster, [PASTE POSTER TEXT]: title, key figure description, 3 key findings, the take-home message, contact email and the full abstract, in under 300 words.

TRY IT WITH Poster on vitamin B12 deficiency in patients on long-term metformin.

398 Will my talk fit the time?

QUICK

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

Act as a presentation coach. Check my script, [PASTE SCRIPT], against a [N]-minute slot at about 130 spoken words per minute. Tell me how far over or under I am, and which sentences to cut or merge to fit, without losing the key result.

TRY IT WITH A 1,450-word script for an 8-minute oral paper.

399 The closing take-home slide

QUICK

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT

Act as a conference presentation coach. Write my closing slide for [ONE-LINE FINDING AND DESIGN]: a message title, 2–3 take-home bullets of under 12 words each, in plain language, and one line on what comes next, all proportionate to my design.

TRY IT WITH Retrospective cohort linking delayed antibiotics with higher sepsis mortality.

400 Follow-up email after the conference

QUICK

1.13 CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS · PG RESIDENT / FACULTY

Act as an academic networking coach. Write a short, professional follow-up email to [NAME, ROLE], whom I met at [CONFERENCE] after my presentation on [TOPIC], thanking them and proposing [collaboration / sharing data / advice on the next study]. Keep it under 120 words.

TRY IT WITH Following up with a senior researcher who asked about multicentre collaboration.

Every prompt at a glance.

Tap any prompt to jump to it. Q = Quick · S = Standard · M = Master

1.1 · TOPIC SELECTION & RESEARCH QUESTION

#1	Feasibility-first topic shortlist	M
#2	Five Indian abstracts → six thesis topics	M
#3	Socratic research-question workshop	M
#4	Guide-meeting pitch pack for your top 3 topics	M
#5	India gap finder: 15 topics from scratch	S
#6	Clinical hunch → PICOT question	S
#7	FINER stress-test of my research question	S
#8	Is this topic already done to death?	S
#9	Turn gaps into Consensus yes/no questions	S
#10	Topics I can do from existing hospital records	S
#11	Topics that fit a 12-month data window	S
#12	Topics needing only routine investigations	S
#13	Validate an existing clinical score in my population	S
#14	Quality-improvement and audit thesis topics	S
#15	Questionnaire-based topics without the KAP trap	S
#16	Head-to-head decision matrix for 2–3 topics	S
#17	Thesis title that passes the committee	S
#18	What could kill this topic? A risk register	S
#19	Choose a measurable primary outcome	S
#20	The questions my guide will ask about my topic	S
#21	Topics from the gaps in a clinical guideline	S
#22	Four-week feasibility check before you commit	S
#23	Five original angles on a crowded topic	S
#24	Two-month project for an MBBS student (ICMR-STS style)	S
#25	Ten quick topic seeds in my specialty	Q
#26	Sharpen my research question to one line	Q
#27	PICO or PECO? Frame my question correctly	Q
#28	Check ongoing trials before you commit	Q
#29	The follow-up that surfaces Indian gaps	Q
#30	Identify the variables in my question	Q
#31	My topic is too broad: narrow it three ways	Q
#32	What infrastructure does this topic need?	Q
#33	Does this Western topic apply in India?	Q
#34	Hypothesis-generating or hypothesis-testing?	Q
#35	Summarise my topic on one slide	Q

1.2 · LITERATURE SEARCH STRATEGY

#36	Build a reproducible, multi-database search	M
#37	Search documentation that survives the viva	M
#38	Search rescue: iterate until the results are right	M
#39	MeSH terms and synonyms for one concept	S
#40	PICO → concept blocks table	S
#41	Fix the syntax of my search string	S
#42	Too many or too few hits: fix my search	S
#43	Translate my PubMed string for other databases	S
#44	Use PubMed filters without biasing the review	S

#45	Find studies from Indian centres	S
#46	Grey literature and trial-registry search plan	S
#47	Backward and forward citation chasing plan	S
#48	Map the research landscape in Perplexity	S
#49	Find the systematic reviews before anything else	S
#50	Identify the landmark papers in my field	S
#51	Set up alerts so new papers find you	S
#52	Reference manager set-up and deduplication	S
#53	Write the search-methods paragraph	S
#54	Legal ways to get the full text of a paper	S
#55	Peer review my search strategy (PRESS checklist)	S
#56	Boolean operators explained with my topic	Q
#57	Every synonym for my concept	Q
#58	Explode, major topic, subheadings: explained	Q
#59	Adapt my search for Google Scholar	Q
#60	Which databases for my specialty?	Q
#61	Email an author for their paper or data	Q
#62	Has this paper been retracted or corrected?	Q
#63	Where does this study sit in the evidence hierarchy?	Q
#64	Turn my question into Perplexity and Consensus queries	Q
#65	How far back should my search go?	Q

1.3 · WRITING THE INTRODUCTION

#66	The four-movement thesis introduction	M
#67	Refine my own introduction draft (language only)	M
#68	Examiner-style review of my introduction	M
#69	Where to find sourced burden data	S
#70	Indian-burden paragraph from my sourced figures	S
#71	Paragraph-by-paragraph outline before you write	S
#72	Five opening sentences that are not textbook	S
#73	Write a sharp gap statement	S
#74	"Need for the study" section	S
#75	Every claim that needs a citation	S
#76	Turn a textbook-style intro into a problem-focused one	S
#77	Cut my introduction to the word limit	S
#78	Fix the flow between paragraphs	S
#79	Define key terms by current criteria	S
#80	Clinical and public-health significance	S
#81	Does my introduction lead to my objective?	S
#82	Paraphrase and synthesise sources responsibly	S
#83	Make my draft sound like me, not like AI	S
#84	Consistency check: tense, spelling, abbreviations	S
#85	Short background for a synopsis or protocol	S
#86	Three hook sentences for my topic	Q
#87	Is my tense right?	Q
#88	Turn my bullet notes into a paragraph	Q
#89	Find the textbook sentences	Q
#90	One paragraph on "why now"	Q

#91	Sentence-level clarity edit	Q	#140	Aim versus objective: fix the confusion	S
#92	Does my introduction need sub-headings?	Q	#141	Map each objective to outcome, measurement and analysis	S
#93	Check for causal language in my introduction	Q	#142	Objectives for a descriptive study	S
#94	Build the abbreviation list	Q	#143	Objectives for a diagnostic accuracy study	S
#95	Two-sentence background for my abstract	Q	#144	Too many objectives: trim to what is feasible	S
1.4 · REVIEW OF LITERATURE					
#96	Screen, extract and synthesise: the ROL engine	M	#145	Secondary objectives that add value at no extra cost	S
#97	Thematic ROL draft from my evidence table	M	#146	Fix vague verbs in my objectives	S
#98	Evidence table from uploaded PDFs, with quotes	M	#147	From PICOT question to a full set of objectives	S
#99	Global and Indian studies: the classic thesis ROL	M	#148	Objectives for an interventional study or RCT	S
#100	Examiner-style critique of my ROL chapter	M	#149	Objectives for qualitative or mixed-methods studies	S
#101	Inclusion and exclusion criteria for screening	S	#150	Write my aim in one sentence	Q
#102	Triage a batch of abstracts	S	#151	Rewrite one vague objective	Q
#103	Design my data-extraction sheet	S	#152	Is my hypothesis testable?	Q
#104	Extract one paper into my evidence table	S	#153	One-sided or two-sided test?	Q
#105	Which risk-of-bias tool for which study?	S	#154	Which of my objectives should be primary?	Q
#106	Apply a risk-of-bias tool to one paper	S	#155	Objectives squeezed into a form field	Q
#107	Find the themes in my evidence table	S	1.6 · STUDY DESIGN & METHODOLOGY		
#108	Explain why studies disagree	S	#156	Choose and defend the right study design	M
#109	One study, one paragraph, done properly	S	#157	Write the full methodology chapter	M
#110	Convert a chronological ROL into a thematic one	S	#158	Operational definitions for every variable	M
#111	Build a literature synthesis matrix	S	#159	Case record form (proforma) with a data-entry codebook	M
#112	Heading structure for my ROL chapter	S	#160	Bias and confounding audit of my design	M
#113	The closing paragraph that leads to my study	S	#161	PICO framework and inclusion/exclusion criteria	S
#114	Summarise a paper, grounded in quotes	S	#162	Make eligibility criteria bedside-proof	S
#115	Summary table of key studies	S	#163	Choose and describe my sampling method	S
#116	Check the numbers in my ROL against the sources	S	#164	Randomisation, allocation concealment and blinding	S
#117	Format and check my reference list	S	#165	Choosing and matching controls in a case-control study	S
#118	The questions to ask NotebookLM about my papers	S	#166	Follow-up plan that keeps loss to follow-up low	S
#119	Quick critical appraisal of one paper	S	#167	Tighten a cross-sectional study design	S
#120	Read and summarise a meta-analysis	S	#168	Design a diagnostic accuracy study (STARD)	S
#121	Historical review sub-section without errors	S	#169	Record-review (retrospective) methodology	S
#122	Update my ROL before submission	S	#170	Identify confounders and draw my causal diagram	S
#123	Register a systematic review on PROSPERO	S	#171	Inter-observer agreement and quality-control plan	S
#124	Pull the PICO out of an abstract	Q	#172	Translate and validate a questionnaire	S
#125	One-line takeaway of a paper	Q	#173	Study procedure as a patient journey	S
#126	The limitation that matters most	Q	#174	Blinding when full blinding is impossible	S
#127	Compare two studies in three sentences	Q	#175	Before-after and quasi-experimental studies done right	S
#128	What design is this study really?	Q	#176	Plan a pilot study for my thesis	S
#129	Interpret an effect size and its CI	Q	#177	Data management plan for my thesis	S
#130	Results in plain language	Q	#178	Laboratory sample handling and pre-analytical control	S
#131	Does this reference support my sentence?	Q	#179	Standardised imaging acquisition and reading protocol	S
#132	Format one reference in Vancouver style	Q	#180	Which reporting guideline applies to my study?	S
#133	How much should this study count?	Q	#181	Describe the intervention so others can replicate it (TiDiEr)	S
#134	Customise a NotebookLM audio overview	Q	#182	Designing studies in patients who cannot consent	S
#135	Is this ROL paragraph a list or a synthesis?	Q	#183	Convert my protocol into thesis methods	S
1.5 · AIMS, OBJECTIVES & HYPOTHESES					
#136	Aims, objectives and hypotheses, fully aligned	M	#184	Name my study design correctly	Q
#137	Committee review of my objectives	M	#185	Prospective or retrospective: what's the difference for me?	Q
#138	Primary and secondary objectives in SMART form	S	#186	Strengths and weaknesses of my design	Q
#139	Null and alternative hypotheses, correctly stated	S	#187	Operationally define one variable	Q
			#188	Draw a simple random sample in Excel	Q
			#189	Allocation concealment in my trial	Q

#190	Can I apply this criterion at the bedside?	Q	#240	Build Table 1 from my variables	Q
#191	List the likely confounders	Q	1.8 · RESULTS, TABLES & FIGURES		
#192	Cohort or case-control? Decide now	Q	#241	Write the Results chapter from my analysis output	M
#193	Could observer bias affect my results?	Q	#242	Plan every table and figure in my results	M
#194	Write the "study setting" paragraph	Q	#243	Consistency audit: text, tables, figures and abstract	M
#195	My design in one line for the title page	Q	#244	Describe Table 1 in two short paragraphs	S
1.7 · SAMPLE SIZE & STATISTICS			#245	Report the primary outcome properly	S
#196	Pre-specified statistical analysis plan	M	#246	Report secondary outcomes without clutter	S
#197	Sample size with every assumption shown	M	#247	Participant flow diagram and paragraph	S
#198	Run my analysis on the real dataset, safely	M	#248	The best figure for this result	S
#199	Multivariable regression: plan, run and interpret	M	#249	Build the chart step by step in my software	S
#200	Interpret my software output correctly	M	#250	Format my table to journal standard	S
#201	Statistics viva drill for my thesis defence	M	#251	Titles, legends and footnotes for my tables and figures	S
#202	Sample size to estimate a proportion (prevalence)	S	#252	Remove interpretation from my results	S
#203	Sample size to compare two means	S	#253	Stop repeating my tables in the text	S
#204	Sample size to compare two proportions	S	#254	Report a non-significant result honestly	S
#205	Sample size for a diagnostic accuracy study	S	#255	Report diagnostic accuracy results (STARD)	S
#206	Sample size for a correlation study	S	#256	Present regression results clearly	S
#207	Sample size for a case-control study	S	#257	Condense results for an abstract or poster	S
#208	Is my sample big enough for regression?	S	#258	One result, one sentence	Q
#209	Which statistical test? A decision table	S	#259	Check my percentages and denominators	Q
#210	Is my data normally distributed? Decide sensibly	S	#260	How many decimal places?	Q
#211	Choose and report the right non-parametric test	S	#261	Table or figure for this result?	Q
#212	Chi-square, Fisher's exact or something else?	S	#262	Check my chart for common errors	Q
#213	Before-after or matched data: paired analysis	S	#263	Describe a figure in the text	Q
#214	Outcomes measured three or more times	S	#264	Label an exploratory analysis correctly	Q
#215	Correlation or agreement? Methods comparison done right	S	#265	Fix the tense in my results	Q
#216	ROC curve, AUC and the best cut-off	S	1.9 · DISCUSSION, CONCLUSION & LIMITATIONS		
#217	Time-to-event analysis: Kaplan-Meier and Cox	S	#266	Write the discussion: ten paragraphs, no overclaiming	M
#218	Handle missing data honestly	S	#267	Limitations that are honest, ranked and specific	M
#219	Data-cleaning checklist before analysis	S	#268	Hostile-reviewer attack on my discussion	M
#220	Step-by-step SPSS instructions for my analysis	S	#269	Conclusion, recommendations and summary chapter	M
#221	R code for my thesis analysis	S	#270	The first paragraph of the discussion	S
#222	Free point-and-click analysis in Jamovi	S	#271	Compare one finding with the literature	S
#223	Effect sizes and confidence intervals for every result	S	#272	Why my result differs from previous studies	S
#224	Multiple comparisons: correct or not?	S	#273	The mechanism paragraph, clearly labelled as speculation	S
#225	Subgroup and sensitivity analyses that hold up	S	#274	Clinical implications without overreach	S
#226	Report p values and CIs the way journals want	S	#275	Strengths without inflation	S
#227	Quick analysis inside Google Sheets with AI add-ons	S	#276	Future research that someone could actually do	S
#228	Classify my variables correctly	Q	#277	A conclusion that doesn't exceed the data	S
#229	Mean or median for this variable?	Q	#278	Scan my discussion for overclaiming	S
#230	What does my p value actually mean?	Q	#279	Who do my results apply to?	S
#231	Interpret my confidence interval	Q	#280	Discuss a surprising result	S
#232	Odds ratio or relative risk for my study?	Q	#281	Discuss a non-significant primary result	S
#233	Explain statistical power with my study	Q	#282	Check the structure of my discussion	S
#234	My result isn't significant: now what?	Q	#283	The "Summary" chapter of an Indian thesis	S
#235	Should I test baseline differences in my trial?	Q	#284	What my findings mean for Indian practice	S
#236	Interpret my kappa value	Q	#285	Cut my discussion to the word limit	S
#237	Which assumptions does my test need?	Q	#286	Practical recommendations from my thesis	S
#238	Sanity-check an AI-generated sample size	Q	#287	Rewrite "further studies are needed"	Q
#239	Explain any statistics term for the viva	Q			

#288	Calibrate my hedging	Q	#337	Write a letter to the editor about a published paper	S
#289	One take-home message	Q	#338	Check the galley proofs properly	S
#290	Qualify every "significant"	Q	#339	Read and fix a similarity (plagiarism) report honestly	S
#291	Which way does this bias push my result?	Q	#340	Running title under the limit	Q
#292	Swap causal words for correct ones	Q	#341	Word budget for each section	Q
#293	One comparison sentence with a study	Q	#342	Cut my abstract to the limit	Q
#294	Is this really a strength?	Q	#343	Summarise my paper for social media	Q
#295	Do my conclusions match my objectives?	Q	#344	Cut my references to the journal's limit	Q
1.10 · PROTOCOL, ETHICS & SYNOPSIS					
#296	Complete study protocol, IEC-ready	M	#345	Write the acknowledgements	Q
#297	IEC submission pack and pre-review	M	#346	Polite status enquiry to the editor	Q
#298	Participant information sheet and consent form	M	#347	Data-availability statement	Q
#299	The ethics section of my protocol or thesis	S	#348	Plain-language summary of my paper	Q
#300	Respond to IEC queries	S	#349	Reply to one tough reviewer comment	Q
#301	CTRI registration, field by field	S	#350	Is splitting my thesis into two papers acceptable?	Q
#302	University synopsis in the required format	S	1.12 · CASE REPORTS & CASE SERIES		
#303	Month-by-month thesis timeline	S	#351	From interesting case to CARE-compliant case report	M
#304	Realistic thesis budget	S	#352	Design and write a case series	M
#305	Risk-benefit assessment and adverse-event plan	S	#353	Case-report discussion grounded in the literature	M
#306	Confidentiality and data-protection plan	S	#354	Is my case publishable?	S
#307	Request a waiver of consent	S	#355	Build the case timeline table	S
#308	Protocol amendment request to the IEC	S	#356	Consent for publication and de-identification	S
#309	Funding and conflict-of-interest declarations	S	#357	Select and prepare clinical images	S
#310	Rewrite a consent paragraph in plain language	Q	#358	Case report abstract	S
#311	What documents does my IEC need?	Q	#359	Case report titles that follow the CARE convention	S
#312	Does my study involve vulnerable participants?	Q	#360	Write the case presentation from my notes	S
#313	Compensation and insurance: what do I write?	Q	#361	Show how differentials were excluded	S
#314	Assent form for children	Q	#362	Learning points that readers remember	S
#315	Version control for my study documents	Q	#363	"Images in clinical medicine" submission	S
1.11 · THESIS TO MANUSCRIPT & PUBLICATION					
#316	Write the full manuscript for journal submission	M	#364	Journals for case reports: find and verify them	S
#317	Thesis → paper cut plan (and the second paper inside it)	M	#365	Patient-level table for a case series	S
#318	Point-by-point response to peer reviewers	M	#366	Find previously reported similar cases	S
#319	Pre-submission peer review of my manuscript	M	#367	Patient-perspective section	S
#320	Structured abstract within the word limit	S	#368	Case report → conference poster	S
#321	Five manuscript titles for my paper	S	#369	Critique my case report draft	S
#322	Choose the right journal (and avoid predatory ones)	S	#370	Adverse drug reaction case report with causality assessment	S
#323	Check my manuscript against the author guidelines	S	#371	Clinicopathological conference (CPC) write-up	S
#324	Cover letter to the editor	S	#372	One line on why my case is worth reading	Q
#325	Keywords that help my paper get found	S	#373	Keywords to search for similar cases	Q
#326	Authorship order and CRediT contributions	S	#374	Is my case report properly anonymised?	Q
#327	Disclose AI use in my manuscript	S	#375	Cut my case report to the word limit	Q
#328	Ethics, consent, funding, COI and data statements	S	#376	Legend for a clinical image	Q
#329	Compress my thesis methods for a journal	S	#377	Turn my case into a teaching caption	Q
#330	Three-paragraph journal introduction	S	#378	Explain publication consent to my patient	Q
#331	Condense my thesis discussion for a journal	S	#379	Case report or case series?	Q
#332	Rejected: what now?	S	#380	The key sentence of my case-report cover letter	Q
#333	Suggest reviewers properly	S	1.13 · CONFERENCE ABSTRACTS, POSTERS & ORAL PAPERS		
#334	Key points, highlights and "what this study adds"	S	#381	Conference abstract to the exact word limit	M
#335	Brief for a graphical abstract	S	#382	Research poster from my manuscript or thesis	M
#336	Should I post a preprint?	S	#383	Thesis → 8-minute oral paper, with slides and Q&A	M
			#384	Choose the right conference for my work	S

#385	Score my abstract like a conference reviewer	S	#393	From conference presentation to published paper	S
#386	Adapt my poster into an e-poster	S	#394	Record a presentation for a virtual or hybrid conference	S
#387	Layout brief for Canva or PowerPoint	S	#395	An accurate, catchy abstract title	Q
#388	90-second poster walk-through for judges	S	#396	25-word summary for the conference programme	Q
#389	Slide plan for a short oral presentation	S	#397	One-page handout or QR page for my poster	Q
#390	Prepare for questions after my talk	S	#398	Will my talk fit the time?	Q
#391	Prepare for an award-paper session	S	#399	The closing take-home slide	Q
#392	Application for a travel grant or bursary	S	#400	Follow-up email after the conference	Q

Which AI tool for which job.

JOB	BEST TOOL	WHY
Mapping research gaps, current literature	Perplexity (Academic), Consensus	Live search with clickable citations
Reading and comparing your own PDFs	NotebookLM, Claude, ChatGPT (upload)	Answers grounded in the documents you give it
Finding recent Indian papers	Semantic Scholar, PubMed	Filters by year and field; TLDR summaries
Drafting and structuring long text	Claude, ChatGPT, Gemini	Strong long-form structure and formatting control
Running statistics on your data	ChatGPT data analysis, Julius AI, Gemini in Sheets	Reads CSV/XLSX and runs code, but verify every number
Sample size cross-check	OpenEpi, G*Power	Validated calculators to confirm AI arithmetic
Posters and figures	Canva, PowerPoint, ChatGPT images	Layout and custom visuals

THE ETHICS OF AI IN RESEARCH

- › Disclose AI use as your institution and target journal require.
- › Don't submit AI text verbatim. Rewrite it in your own academic voice.
- › Verify every statistic, reference and data point against the primary source.
- › AI cannot replace your guide's approval, your IEC clearance or your clinical judgement.

This resource is for educational purposes. It does not replace your institutional research guidelines, ethics committee requirements, university thesis regulations or the advice of your supervisor and statistician. Verify all AI-generated content independently before using it in any submitted work. AI cannot be an author; disclose AI use per ICMJE and your target journal's policy.

SEMINARS, JOURNAL CLUBS & PRESENTATIONS.

300 prompts to build seminars on any topic, appraise and present journal articles, design slides and posters, and deliver talks that hold a room.

300

prompts

7

sections

35

Master

173

Standard

92

Quick

2.1	Seminar preparation	#401-460
2.2	Journal club	#461-520
2.3	Slide design & structure	#521-560
2.4	Talks & oral presentations	#561-595
2.5	Posters & visual teaching aids	#596-630
2.6	Case presentations & grand rounds	#631-670
2.7	Delivery, Q&A & teaching	#671-700

2.1

Seminar preparation

A seminar is judged by what the audience remembers the next morning. Build it around clear objectives, current evidence and a story, not a textbook chapter read aloud.

60

prompts

7

Master

35

Standard

18

Quick

HOW TO USE THIS SECTION

Use it one to two weeks before your seminar, to plan the objectives, the evidence and the flow.

Start with the Master prompts (#401–407) for the full workflow; use the Standard prompts (#408–442) for one task at a time; reach for the Quick prompts (#443–460) when you have two minutes. Tap any title below to jump to it, and follow → NEXT to move to the next step.

#401 Mixed-audience seminar deck, clinically rigorous

#402 From topic to seminar blueprint in one sitting

#403 Evidence pack: guidelines and trials, verified from my sources

#404 Full speaker script for a 45-minute seminar

#405 Case-based interactive seminar

#406 Mock moderator: grill me before the real seminar

#407 Turn a new guideline into a seminar

#408 Learning objectives with Bloom's verbs

#409 Minute-by-minute time plan

#410 Narrow a huge seminar topic to something teachable

#411 Pitch the same topic to different audiences

#412 Pathophysiology as a visual story

#413 Turn a classification into a clear table

#414 Diagnostic approach as an algorithm

#415 Management section, stepwise and practical

#416 Drug comparison table for a seminar

#417 Landmark trials in one table

#418 Controversies and grey zones

#419 Recent advances slide set

#420 The Indian perspective slide

#421 Compare two or more guidelines

#422 References slide and citation check

#423 End-of-seminar quiz

#424 Take-home messages that stick

#425 One-page handout for the audience

#426 Opening clinical vignette

#427 Basic science seminar with clinical hooks

#428 Seminar on a procedure or surgery

#429 Teaching session on interpreting a test

#430 Image-based seminar

#431 Turn a textbook chapter into a seminar

#432 Prepare a seminar with NotebookLM

#433 What have I missed?

#434 Feedback on my rehearsal script

#435 Make my seminar interactive

#436 Check the story and flow of my outline

#437 Adapt my seminar for nurses and allied staff

#438 Turn a seminar into a CME talk for practitioners

#439 Pro-con debate seminar

#440 Review my seminar slides like a moderator

#441 Build a seminar from our department's cases

#442 Collect feedback on my seminar

#443 A seminar title that's accurate and interesting

#444 A one-line opening hook

#445 Five learning objectives in a minute

#446 The definition slide

#447 A useful mnemonic for my seminar

#448 "Why this matters" slide

#449 How many slides do I need?

#450 Key numbers slide

#451 Summary slide from my outline

#452 Transition lines between sections

#453 A question to wake up the audience

#454 An analogy for a hard concept

#455 Three questions for the discussion

#456 A short history slide

#457 Abbreviations slide for my seminar

#458 Future directions slide

#459 What moderators always ask on this topic

#460 Turn my paragraph into slide bullets

Mixed-audience seminar deck, clinically rigorous

2.1 SEMINAR PREPARATION · PG RESIDENT / FACULTY

USE IT

When you have a departmental seminar on any topic and want a complete, evidence-based deck.

WHY

Departmental audiences are mixed: MBBS students, interns, residents and consultants in the same room. A deck pitched at resident level, with a short foundation recap for juniors and a controversies section for consultants, keeps everyone engaged, and cites real evidence rather than textbook generalities.

COPY THE PROMPT BELOW

PERSONA	You are a senior academic in [SPECIALTY] who has delivered and moderated hundreds of departmental seminars. You are clinically rigorous, evidence-based and practical, and you never invent citations or statistics.
MY CONTEXT	Topic: [e.g. Acute Respiratory Distress Syndrome: from pathophysiology to ventilator management] Audience: mixed (MBBS students, interns, junior residents, senior residents, consultants) Time available: [e.g. 45 minutes + 15 minutes discussion] Sources I have (guidelines, reviews, textbook chapters), uploaded or listed: [UPLOAD OR LIST] Output format: [slide-by-slide text for PowerPoint / a .pptx file if your tool can create one]
TASK	1. Pitch the core content at resident level, with a 1–2 slide foundation recap early, and advanced nuance for consultants. 2. Follow this structure, adapting as needed: title slide with a one-line clinical hook; why this matters (a representative case); foundation recap (definitions, classification); pathophysiology; clinical features and diagnostic approach (criteria or scores in tables); management, stepwise, first-line versus escalation; an evidence corner (landmark trials and current guidelines); controversies and grey zones; 3–5 key takeaways; references in Vancouver style. 3. For every slide give a message title, concise bullets (one idea per slide), a table where comparisons help, and speaker notes. 4. Mark what is exam-relevant for students and what is decision-changing for residents and consultants. 5. Suggest where a figure, flowchart or algorithm should go, and describe it.
RULES	Name trials and guidelines only from my sources or with [VERIFY] beside each, because misremembered trial names and years are the most embarrassing seminar error. Mark every drug dose as [VERIFY DOSE FROM CURRENT GUIDELINE]. Show me the slide-by-slide text for review before building any file.
OUTPUT	A slide table (number, title, bullets, visual, speaker notes, audience tag), then the reference list with [VERIFY] flags.
TWEAK	End with the 5 questions a consultant is most likely to ask after this seminar, and short evidence-based answers.

How to get more out of it

- › Tuning levers: change the depth (resident ↔ consultant), the slide count (about 1 slide per minute) and the visual style.
- › Verify every trial and guideline in the evidence corner against the original source before presenting.
- › Rehearse the controversies section. It is where the discussion will go.
- › → **NEXT #408** Learning objectives with Bloom's verbs

From topic to seminar blueprint in one sitting

2.1 SEMINAR PREPARATION · PG RESIDENT

USE IT

The day your seminar topic is assigned, before you open any textbook.

WHY

Most residents start a seminar by reading everything and end with 80 slides and no structure. Starting with objectives, a time budget and a storyline tells you exactly what to read and what to leave out.

COPY THE PROMPT BELOW

PERSONA	You are a medical educator and senior faculty member who mentors residents on seminar preparation. You think backwards from what the audience must be able to do after the session.
MY CONTEXT	Topic as assigned: [e.g. Approach to hyponatraemia] Specialty and my year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and its level: [brief] Duration: [minutes] Days until the seminar: [N] Moderator's known expectations or style: [brief or "unknown"]
TASK	1. Clarify the scope: what is in, what is out, and how to phrase the final seminar title. 2. Write 4–6 learning objectives using Bloom's verbs (explain, differentiate, apply, choose). 3. Build a storyline: an opening case, 5–7 content blocks mapped to the objectives, and a closing return to the case. 4. Create a time budget per block that fits the duration. 5. For each block, list the must-read sources by type (current guideline, recent review, textbook chapter) with [VERIFY LATEST EDITION]. 6. Make a day-by-day preparation plan for the days I have, including a rehearsal. 7. Predict the moderator's 5 likely questions.
RULES	Keep the content proportional to the time, about one slide per minute, and cut anything not tied to an objective, because overlong seminars lose the audience in the middle. Name no specific papers from memory; describe the source type and mark it [FIND AND VERIFY].
OUTPUT	Scope note, objectives, storyline table (block, objective, minutes, key points), the source list, the preparation plan, then the predicted questions.
TWEAK	End with the single case vignette to open with, 60 words, de-identified and realistic for Indian practice.

How to get more out of it

- › Show the objectives and storyline to your moderator a week before. It avoids a last-minute restructure.
- › Read the guideline first, the review second and the textbook last.
- › Keep the opening case alive: return to it in every block.
- › → **NEXT #401** Mixed-audience seminar deck, clinically rigorous

Evidence pack: guidelines and trials, verified from my sources

2.1 SEMINAR PREPARATION · PG RESIDENT / FACULTY

USE IT

When your seminar needs an evidence corner and you have gathered the guidelines and key papers.

WHY

The evidence section is where seminars go wrong: trials cited with the wrong year, outcomes overstated, or guidelines quoted from memory. Building it strictly from uploaded sources, with quotes, makes it accurate and defensible when a consultant challenges it.

COPY THE PROMPT BELOW

PERSONA	You are an evidence-based medicine specialist who prepares evidence summaries for clinical guideline panels. You quote the source for every claim and state the certainty of evidence clearly.
MY CONTEXT	Seminar topic and the clinical questions it must answer: [list 3–5 questions] Guidelines uploaded (name, organisation, year): [UPLOAD / LIST] Key trials and meta-analyses uploaded: [UPLOAD / LIST] Setting to apply the evidence to: [e.g. Indian tertiary and district hospitals] Audience and how much time the evidence section gets: [e.g. residents and consultants, 8 minutes] My specialty and year: [brief]
TASK	1. For each clinical question, summarise what the uploaded guidelines recommend, with the strength or grade, quoting the recommendation wording. 2. Build a trials table: trial, year, design, population, intervention versus comparator, primary outcome and result with CI, and the key limitation. 3. Identify where guidelines disagree with each other, and why. 4. Identify where the evidence does not apply well to Indian settings (population, resources, drug availability). 5. Write 3 evidence-corner slides with speaker notes. 6. List any major trial or guideline the audience will expect that is missing from my uploads, marked [FIND AND UPLOAD], so I can add it before the seminar.
RULES	Use only the uploaded documents; for anything else write [NOT IN MY SOURCES — VERIFY], because a single wrong trial detail can discredit the whole seminar. Reproduce effect sizes exactly. State certainty levels as the guideline states them.
OUTPUT	A recommendations table per question, the trials table, the disagreement notes, the applicability notes, then the slide text.
TWEAK	End with the one piece of evidence I should read in full before presenting, because it is most likely to be questioned.

How to get more out of it

- › Upload the guideline PDFs themselves, not summaries of them.
- › Put the year on every trial and guideline on the slides. Consultants check it.
- › If two guidelines disagree, show both. That is where the best discussion happens.
- › → NEXT #421 Compare two or more guidelines

Full speaker script for a 45-minute seminar

2.1 SEMINAR PREPARATION · PG RESIDENT

USE IT

When your slides are ready and you want to rehearse to time.

WHY

Reading slides aloud is the commonest seminar failure. A script written for speaking, with timing, transitions and planned interaction, lets you rehearse properly, then speak from notes on the day.

COPY THE PROMPT BELOW

PERSONA	You are a speaking coach for medical educators who writes natural, spoken scripts and trims them to the second.
MY CONTEXT	My slides (titles and bullets, or upload the file): [PASTE OR UPLOAD] Duration of the talk (excluding discussion): [e.g. 40 minutes] Audience: [brief] My speaking style and my nerves: [brief] Points where I want audience interaction: [brief or "suggest"] Parts of the content I find hardest to explain: [list] Language mix the audience is comfortable with: [e.g. English with occasional Hindi]
TASK	1. Write a spoken script per slide at about 130 words per minute, in conversational academic English. 2. Mark timing checkpoints (e.g. "by slide 10: 12 minutes"). 3. Add a transition sentence between sections, so the story flows. 4. Insert 3–4 interaction moments: a question to the audience, a quick show of hands, or a case pause. 5. Mark the lines to memorise word for word (opening, key message, closing). 6. Flag slides whose script runs long, and suggest cuts. 7. For the hardest parts I listed, write a simpler explanation or an analogy I can use if the audience looks lost. 8. Write a 30-second emergency version of the conclusion in case time runs out.
RULES	Keep facts exactly as on my slides and add no new data, because the script must match what the audience sees. Script the interaction moments with a fallback answer in case nobody responds.
OUTPUT	A script table (slide, script, seconds, interaction), then the checkpoints and the memorise list.
TWEAK	End with a 5-step rehearsal plan for the next 3 days.

How to get more out of it

- › Rehearse aloud with a timer. Silent reading always underestimates the time.
- › Don't read the script on the day. Speak from the checkpoints and the memorised lines.
- › Record one rehearsal on your phone. You'll hear your filler words immediately.
- › → NEXT #434 Feedback on my rehearsal script

Case-based interactive seminar

2.1 SEMINAR PREPARATION · PG RESIDENT / FACULTY

USE IT

When you want a seminar built around an unfolding case rather than a lecture.

WHY

Case-based seminars keep the audience thinking instead of copying. Revealing a case in stages, with questions at each stage, teaches clinical reasoning and makes the content stick.

COPY THE PROMPT BELOW

PERSONA	You are a clinician-educator who designs case-based learning sessions for residents, and who knows how to reveal information at the right pace.
MY CONTEXT	Topic: [TOPIC] Learning objectives: [PASTE OR "suggest"] Audience and number: [brief] Duration: [minutes] A real case I can adapt (de-identified), or "create a realistic one": [PASTE OR REQUEST] Setting where the case takes place: [emergency / ward / OPD / district hospital] Common mistakes residents make with this condition: [list or "suggest"]
TASK	1. Build an unfolding case in 4–6 stages: presentation, examination, initial investigations, key result, management decision, and follow-up or complication. 2. At each stage, write 1–2 questions for the audience, the expected answers, and the teaching point to deliver. 3. Map each stage to a learning objective. 4. Add a short mini-lecture slide (3 bullets) after each stage to consolidate. 5. Close with a summary of the reasoning path, and take-home points. 6. Suggest how to run it: small groups, polls or calling on people. 7. Build at least one stage around a common mistake, so the audience can discover why it is wrong. 8. Give the timing for each stage so the whole case fits the duration.
RULES	Keep the case realistic for Indian practice and fully de-identified. Mark drug doses and guideline specifics as [VERIFY], because case-based sessions invite detailed questions.
OUTPUT	A stage table (stage, information revealed, questions, expected answers, teaching point, objective), the mini-lecture slides, then the facilitation guide.
TWEAK	End with 2 variations of the case (a different age or comorbidity) for repeat sessions.

How to get more out of it

- › Reveal information on separate slides. Showing everything at once kills the reasoning.
- › Wait 10 seconds after asking a question. Silence is thinking time.
- › Keep a "parking lot" for tangent questions and answer them at the end.
- › → NEXT #435 Make my seminar interactive

Mock moderator: grill me before the real seminar

2.1 SEMINAR PREPARATION · PG RESIDENT

USE IT

2–3 days before your seminar, once the slides are final.

WHY

The dreaded part of a seminar is the moderator's questions. A realistic mock session, one question at a time with feedback, turns fear into preparation and shows you exactly which slides need strengthening.

COPY THE PROMPT BELOW

PERSONA	You are a senior professor and seminar moderator known for probing questions on basics, recent evidence and practical application. You are strict but fair, and you give frank feedback.
MY CONTEXT	Seminar topic and my specialty and year: [brief] My slide titles and key content: [PASTE] Areas I feel weakest in: [list] My moderator's known style or favourite themes: [brief or "unknown"] Duration of the seminar and the discussion slot: [brief]
TASK	1. Ask me one question at a time, mixing basic science, clinical application, recent evidence, Indian context and "what would you do" scenarios. 2. After each answer, score it out of 10, give the ideal answer in 3–5 points, and tell me which slide should cover it better. 3. Include 2 classic trap questions moderators use on this topic. 4. Mix difficulty: start with 3 basic questions, then 4 applied questions, then 3 advanced or evidence questions. 5. After 10 questions, summarise my weak areas and the slides to strengthen, and give me a one-day plan to fix them.
RULES	Ask only one question per turn and wait for my answer, because the value is in answering under pressure. Base questions on my topic and slides. Mark any fact in your ideal answers that depends on a guideline as [VERIFY].
OUTPUT	Conversational: one question per turn, feedback in a short structured block (score, ideal answer, gap, slide to fix), then the final summary with the weak areas ranked from most to least urgent.
TWEAK	End with the 3 questions I should prepare backup slides for.

How to get more out of it

- › Answer aloud before typing. The seminar is spoken.
- › Add backup slides at the end of the deck for the questions you struggled with.
- › Ask a colleague to play the moderator with these questions for a real rehearsal.
- › → **NEXT #404** Full speaker script for a 45-minute seminar

Turn a new guideline into a seminar

2.1 SEMINAR PREPARATION · PG RESIDENT / FACULTY

USE IT

When a new or updated clinical guideline is released and your department wants a session on it.

WHY

"What's new in the guideline" seminars are common and often done badly: a slide per recommendation, read aloud. The useful version shows what changed, why, what it means for practice here, and what remains uncertain.

COPY THE PROMPT BELOW

PERSONA	You are a guideline methodologist and clinician who explains guideline updates to busy clinicians, focusing on what changes practice.
MY CONTEXT	The guideline (upload the full text or the recommendations section): [UPLOAD] The previous version, if available: [UPLOAD OR "not available"] Audience and duration: [brief] Local practice for this condition in my hospital: [brief] Drugs, tests or devices the guideline recommends that my hospital lacks: [list or "unsure"] My specialty and year: [brief]
TASK	1. List the key recommendations with their strength and certainty, quoting the exact wording and the section number. 2. Identify what is new or changed since the previous version, and the evidence cited for each change. 3. Classify each change as practice-changing, practice-confirming or minor. 4. Discuss applicability to Indian settings: resources, drug availability, cost and staffing. 5. Build a 20–30 minute slide plan: what changed, why, what to do on Monday, and uncertainties. 6. Write a one-page summary handout. 7. Suggest a simple audit the department could run to check adherence to the key recommendation. 8. Note which recommendations are conditional or based on low-certainty evidence, so the audience doesn't treat them as absolute.
RULES	Quote only from the uploaded guideline and mark anything else [VERIFY], because a misquoted recommendation can change patient care. Don't overstate the certainty of weak recommendations.
OUTPUT	The recommendations table, the changes table (old position, new position, evidence, classification), the applicability notes, the slide plan with timings, then the handout.
TWEAK	End with the 3 recommendations most likely to be debated in the department, and why.

How to get more out of it

- › Lead with what changes practice. Nobody remembers the minor edits.
- › Bring the guideline's own algorithm figure if it has one, and credit it.
- › Invite a senior to comment on local applicability. It turns the seminar into a discussion.
- › → **NEXT #420** The Indian perspective slide

408 Learning objectives with Bloom's verbs

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT / FACULTY

USE IT At the very start of seminar preparation.

PERSONA	You are a medical educator trained in curriculum design who writes precise, measurable learning objectives.
MY CONTEXT	Topic: [TOPIC] Audience level: [MBBS / interns / residents / mixed] Duration: [minutes]
TASK	1. Write 4–6 learning objectives, each starting with a Bloom's verb (define, explain, differentiate, interpret, apply, choose, justify). 2. Spread them across levels: recall, understanding and application. 3. Say which part of the seminar will deliver each objective. 4. Suggest one quick end-of-session check for each objective.
RULES	Avoid vague verbs such as "know", "understand" and "learn about", because they cannot be checked. Keep the number realistic for the time available.
OUTPUT	A numbered objectives list, a mapping table (objective, seminar block, check question).

PRO TIP Put the objectives on slide 2 and return to them on the last slide. Moderators notice.

TRY IT WITH Topic: approach to a patient with haemoptysis, for MD Medicine residents, 40 minutes.

→ **NEXT #409** Minute-by-minute time plan

409 Minute-by-minute time plan

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT

USE IT Once your outline is ready, to make sure it fits the time.

PERSONA	You are a meticulous academic event planner who keeps speakers on time.
MY CONTEXT	Outline or slide titles: [PASTE] Total time and the split between talk and discussion: [e.g. 45 + 15 minutes] Planned interactive moments: [list]
TASK	1. Allocate minutes to each section and slide. 2. Show a running clock with checkpoints every 10 minutes. 3. Identify sections to cut if I am running late, in order of priority. 4. Estimate the total slide count against the time (about one slide per minute).
RULES	Include time for interaction, transitions and technical hiccups, because seminars always run over their plan.
OUTPUT	A timing table (section, slides, minutes, clock), then the cut list.

PRO TIP Put the checkpoint times in small text in your speaker notes, not on the slides.

TRY IT WITH A 60-slide outline for a 40-minute seminar on diabetic ketoacidosis.

→ **NEXT #404** Full speaker script for a 45-minute seminar

410 Narrow a huge seminar topic to something teachable

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT

USE IT When your assigned topic is enormous ("tuberculosis", "heart failure").

PERSONA	You are a seminar moderator who helps residents choose a sharp angle on broad topics.
MY CONTEXT	Assigned topic: [TOPIC] Duration and audience: [brief] Recent seminars in the department on related topics: [list or "unknown"]
TASK	1. Propose 5 focused angles (e.g. diagnosis in a specific group, a management controversy, recent advances, an approach to a symptom, an Indian perspective). 2. For each, write a working title and 3 learning objectives. 3. Recommend one angle for my audience and time, with the reason. 4. Suggest how to acknowledge the rest of the topic in 1 slide.
RULES	Keep each angle deliverable in the time available, because covering everything superficially teaches nothing.
OUTPUT	An angles table (title, focus, objectives), then the recommendation.

PRO TIP Confirm the narrowed title with your moderator before you start preparing.

TRY IT WITH Assigned topic: "Tuberculosis", 45 minutes, MD Respiratory Medicine residents.

→ **NEXT #402** From topic to seminar blueprint in one sitting

411 Pitch the same topic to different audiences

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT / FACULTY

USE IT When you must present one topic to students, residents or consultants, or to a mix of them.

PERSONA	You are a medical educator who adapts content depth skilfully for different audiences.
MY CONTEXT	Topic and my current outline: [PASTE] Audience: [MBBS students / interns / residents / consultants / nurses / mixed]
TASK	1. Say what each audience already knows and what they need from this topic. 2. Adjust the outline: what to expand, what to compress, what to drop. 3. Suggest the level of jargon, the depth of evidence and the type of examples for each. 4. For a mixed audience, show how to layer content (foundation recap, core, advanced corner).
RULES	Keep the clinical accuracy the same at every level; only the depth and emphasis change, because simplifying must never mean being wrong.
OUTPUT	A comparison table by audience, then the adjusted outline.

PRO TIP For a mixed audience, tell the room explicitly: "This slide is for students; this one is for the consultants."

TRY IT WITH Topic: interpretation of pulmonary function tests.

→ **NEXT #401** Mixed-audience seminar deck, clinically rigorous

412 Pathophysiology as a visual story

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT / MBBS STUDENT

USE IT When your seminar needs a pathophysiology section that doesn't bore the room.

PERSONA	You are a physiology-minded clinician who explains mechanisms visually and links every step to a clinical feature.
MY CONTEXT	Condition: [CONDITION] Audience level: [brief] Slides available for this section: [e.g. 3] Sources I am using: [UPLOAD / LIST]
TASK	1. Break the pathophysiology into a sequence of 5–8 steps from trigger to clinical picture. 2. Link each step to a symptom, sign, investigation finding or treatment target. 3. Describe a flowchart or diagram for each slide. 4. Write speaker notes explaining each arrow.
RULES	Keep to established mechanisms from my sources, and label emerging or debated mechanisms as such, because contested mechanisms presented as fact invite correction.
OUTPUT	A step table (step, mechanism, clinical correlate, treatment target), diagram descriptions, then speaker notes.

PRO TIP If a pathophysiology step doesn't explain a clinical feature or a treatment, cut it from the seminar.

TRY IT WITH Condition: septic shock, 3 slides, for residents.

→ **NEXT #414** Diagnostic approach as an algorithm

413 Turn a classification into a clear table

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT / MBBS STUDENT

USE IT When your topic has a classification, staging or grading system.

PERSONA	You are a medical educator who turns classification systems into clean, memorable tables.
MY CONTEXT	Classification or staging system: [name, or "the one used in guideline X"] My source for it: [UPLOAD / CITE] Audience: [brief]
TASK	1. Present the classification as a table with categories, defining criteria and clinical significance. 2. Note the year and version, and any older versions still used in papers. 3. Suggest a memory aid if it helps. 4. Write one slide of speaker notes on how the classification guides management.
RULES	Reproduce thresholds and criteria only from my source, and mark any detail I did not supply as [VERIFY FROM SOURCE], because classification cut-offs are commonly misremembered.
OUTPUT	The table, the version note, the memory aid, then the speaker notes.

PRO TIP Always show which version you are using. Moderators love asking about the previous one.

TRY IT WITH GOLD grouping for COPD assessment, from the current GOLD report.

→ **NEXT #401** Mixed-audience seminar deck, clinically rigorous

414 Diagnostic approach as an algorithm

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT

USE IT When your seminar covers "approach to" a symptom, sign or abnormal result.

PERSONA	You are a clinician–teacher who builds practical, stepwise diagnostic algorithms.
MY CONTEXT	Presenting problem: [e.g. hyponatraemia, pancytopenia, pleural effusion] Setting: [ward / emergency / OPD; resource level] Sources: [UPLOAD / LIST]
TASK	1. Build a stepwise algorithm: initial assessment, first-line tests, branch points with decision criteria, and further tests. 2. Mark "can't-miss" diagnoses and red flags at the right points. 3. Describe the algorithm as a flowchart (boxes and arrows) for a slide. 4. Add a resource-limited variant for district hospitals. 5. Write speaker notes walking through one example patient.
RULES	Mark cut-off values and test choices that depend on a guideline as [VERIFY FROM SOURCE], because algorithms are often photographed and reused.
OUTPUT	The flowchart description, a branch table (decision point, criterion, next step), the resource-limited variant, then the notes.

PRO TIP Walk the audience through the algorithm with a real case. Algorithms alone are hard to follow.

TRY IT WITH Approach to a unilateral pleural effusion, tertiary and district versions.

→ **NEXT #415** Management section, stepwise and practical

415 Management section, stepwise and practical

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT

USE IT For the management part of a seminar.

PERSONA	You are a senior clinician who teaches management as clear steps, from first-line care to escalation.
MY CONTEXT	Condition and severity range: [brief] Setting and resources: [brief] Guidelines or sources: [UPLOAD / LIST]
TASK	1. Lay out management in steps: immediate measures, first-line treatment, monitoring, escalation criteria and escalation options, and when to refer. 2. Present drug choices in a table with dose placeholders, key cautions and monitoring. 3. Highlight differences between tertiary and resource-limited settings. 4. Add the common mistakes in managing this condition.
RULES	Mark every dose, duration and threshold as [VERIFY DOSE FROM CURRENT GUIDELINE], because dosing errors on slides can be copied into practice.
OUTPUT	A stepwise outline, the drug table, the setting comparison, then the common mistakes.

PRO TIP A one-slide "management on a page" summary is often the most photographed slide of the seminar.

TRY IT WITH Management of acute severe asthma in adults, tertiary and district settings.

→ **NEXT #416** Drug comparison table for a seminar

416 Drug comparison table for a seminar

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT / MBBS STUDENT

USE IT When your topic involves several drugs or drug classes.

PERSONA	You are a clinical pharmacologist who builds accurate, exam-useful drug tables.
MY CONTEXT	Drugs or classes to compare: [list] Aspects I want: [mechanism, indications, dose, adverse effects, interactions, monitoring, cost, availability in India] Source: [UPLOAD / LIST]
TASK	1. Build a comparison table with the aspects I listed. 2. Highlight the distinguishing features between similar drugs. 3. Add a row for special populations (pregnancy, renal and hepatic impairment, elderly). 4. Write speaker notes on how to choose between them.
RULES	Mark every dose as [VERIFY DOSE] and every availability or cost note as [VERIFY LOCALLY], because drug details change and vary by region.
OUTPUT	The drug table, then the choice guidance in notes.

PRO TIP Split big tables across two slides. A 10-column table is unreadable from the back row.

TRY IT WITH Compare DOACs with warfarin for stroke prevention in atrial fibrillation.

→ **NEXT #417** Landmark trials in one table

417 Landmark trials in one table

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT / FACULTY

USE IT For the "evidence corner" of your seminar, once you have the trial papers or abstracts.

PERSONA	You are an evidence-based medicine teacher who summarises trials accurately and critically.
MY CONTEXT	Trials to include (upload the papers or paste the abstracts): [UPLOAD / PASTE] The clinical question they address: [one line]
TASK	1. Build a table: trial, year, design, population, intervention versus comparator, primary outcome and result, and key limitation. 2. Write a one-line "what it changed" for each trial. 3. Put the trials in a timeline showing how the evidence evolved. 4. Write speaker notes for a 3-minute evidence walk-through.
RULES	Use only the uploaded or pasted trial details, with numbers reproduced exactly, and mark missing details [NOT IN SOURCE], because trial details are what experts in the audience check.
OUTPUT	The trials table, the timeline, then the speaker notes.

PRO TIP One slide with 4–6 trials is enough. Pick the ones that changed practice.

TRY IT WITH Trials of corticosteroids in septic shock.

→ **NEXT #418** Controversies and grey zones

418 Controversies and grey zones

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT / FACULTY

USE IT For the section of your seminar that keeps consultants engaged.

PERSONA	You are a senior clinician who presents controversies fairly, giving both sides their strongest evidence.
MY CONTEXT	Topic: [TOPIC] Controversial questions I am aware of: [list or "suggest"] Sources for each side: [UPLOAD / LIST]
TASK	1. Identify 3–4 genuine controversies in current practice. 2. For each, present the arguments and evidence on both sides. 3. State what current guidelines say, if anything. 4. Suggest how the department might approach it, and what evidence is awaited. 5. Write one slide per controversy with a debate question.
RULES	Present both sides with equal care, and support claims only from my sources or mark them [VERIFY], because a one-sided "controversy" slide will be challenged.
OUTPUT	A controversy table (question, side A, side B, guideline position, awaited evidence), then the slides.

PRO TIP End each controversy slide with a question to the room. It starts the discussion you want.

TRY IT WITH Controversies in fluid resuscitation for sepsis.

→ **NEXT #419** Recent advances slide set

419 Recent advances slide set

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT / FACULTY

USE IT When your seminar needs a "what's new" section.

PERSONA	You are an academic clinician who tracks new evidence and separates genuine advances from hype.
MY CONTEXT	Topic: [TOPIC] Time window: [e.g. last 3 years] New papers, guidelines or approvals I have found: [UPLOAD / LIST]
TASK	1. Categorise the advances: new diagnostics, new therapies, new guideline positions, new understanding of the disease. 2. For each, give the evidence level and whether it is ready for practice, promising, or early research. 3. Note whether each is available or relevant in India. 4. Write 2–3 slides with speaker notes.
RULES	Include only advances from the sources I gave, or mark them [VERIFY — FIND SOURCE], because AI-listed "advances" are often outdated or invented.
OUTPUT	An advances table (advance, category, evidence level, readiness, India relevance), then the slides.

PRO TIP Use a search tool with live citations (Perplexity or Consensus) to find recent advances, then verify each paper yourself.

TRY IT WITH Recent advances in the management of idiopathic pulmonary fibrosis.

→ **NEXT #420** The Indian perspective slide

420 The Indian perspective slide

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT / FACULTY

USE IT For the section of your seminar on Indian epidemiology, guidelines and practice.

PERSONA	You are a public-health physician who knows Indian data sources, national programmes and practice realities.
MY CONTEXT	Topic: [TOPIC] Indian data and guidelines I have found (with sources): [LIST]
TASK	1. Summarise the Indian burden using my sourced figures. 2. Summarise relevant Indian guidelines or national programme recommendations from my sources. 3. Highlight how Indian practice differs from Western guidelines: resources, drug availability, cost and disease patterns. 4. Identify gaps in Indian data. 5. Write 1–2 slides with speaker notes.
RULES	Use only figures and guidelines I listed, with their sources and years, and mark anything else [VERIFY], because Indian statistics are often quoted from outdated surveys.
OUTPUT	The summary, the differences table, the data gaps, then the slides.

PRO TIP Always give the survey name and year with a prevalence figure on the slide.

TRY IT WITH Topic: COPD in India, with burden data from national surveys and the Global Burden of Disease study.

→ **NEXT #421** Compare two or more guidelines

421 Compare two or more guidelines

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT / FACULTY

USE IT When different societies recommend different things on your topic.

PERSONA	You are a guideline analyst who compares recommendations clearly and explains why they differ.
MY CONTEXT	Guidelines to compare (upload or list with year): [UPLOAD / LIST] Clinical questions to compare on: [list]
TASK	1. Build a side-by-side table: question, guideline A recommendation (with strength), guideline B recommendation (with strength). 2. Explain the reasons for differences: evidence interpretation, population, cost, or date of publication. 3. Suggest which is more applicable to my setting, and why. 4. Write one comparison slide with speaker notes.
RULES	Quote recommendations from the uploaded documents and mark anything else [VERIFY], because paraphrased recommendations often lose their conditions.
OUTPUT	The comparison table, the reasons, the applicability note, then the slide.

PRO TIP Put the publication year of each guideline in the table header. It often explains the differences.

TRY IT WITH Compare two international hypertension guidelines on blood-pressure targets in older adults.

→ **NEXT #417** Landmark trials in one table

422 References slide and citation check

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT

USE IT When finalising your seminar deck.

PERSONA	You are a reference editor who formats seminar references in Vancouver style.
MY CONTEXT	Sources I used, as I have them: [PASTE LIST] Number of reference slides allowed: [e.g. 1–2]
TASK	1. Format each reference in Vancouver style. 2. Flag references missing essential details, without inventing them. 3. Suggest how to cite sources on individual slides (small superscript numbers). 4. Select the 10–15 most important if the list is too long.
RULES	Write [MISSING — CHECK PUBMED] for missing details instead of inventing them, because invented volume or page numbers make real references look fake.
OUTPUT	The formatted list, then the flagged items and citation tips.

PRO TIP Cite figures and tables you adapt from papers on the slide itself: "Adapted from [ref]".

TRY IT WITH 18 sources, including 2 guidelines, 6 trials, 3 reviews and a textbook.

→ **NEXT #424** Take-home messages that stick

423 End-of-seminar quiz

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT / FACULTY

USE IT To close your seminar with active recall and check the objectives.

PERSONA	You are a medical educator who writes high-quality MCQs that test application, not trivia.
MY CONTEXT	Seminar topic and learning objectives: [PASTE] Audience level: [brief] Format: [show of hands / online poll / paper]
TASK	1. Write 5–8 MCQs mapped to the objectives, mostly clinical-vignette style. 2. For each, give the answer and a 2-line explanation. 3. Include one question on a common misconception. 4. Suggest how to run the quiz in 5 minutes.
RULES	Keep every correct answer consistent with my seminar content and mark guideline-dependent answers [VERIFY], because a wrong quiz answer undermines the whole session.
OUTPUT	The numbered questions with options, then the answer key with explanations.

PRO TIP An online poll (Mentimeter, Google Forms or Slido) turns a quiz into the most engaging 5 minutes of the seminar.

TRY IT WITH Seminar on approach to acute kidney injury, for interns.

→ **NEXT #424** Take-home messages that stick

424 Take-home messages that stick

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT / FACULTY

USE IT For the final slide of any seminar.

PERSONA	You are a medical educator who writes memorable, accurate take-home messages.
MY CONTEXT	Seminar content summary or slide titles: [PASTE] Learning objectives: [PASTE] Audience: [brief]
TASK	1. Write 3–5 take-home messages, each one sentence, specific, actionable and in plain language. 2. Map each message to a learning objective. 3. Suggest a visual layout for the slide. 4. Suggest the closing sentence to say aloud, and a follow-up resource to point people to.
RULES	Each message must be supported by the seminar content, with no new claims, because the last slide is the one people photograph.
OUTPUT	The messages, the mapping, the layout, then the closing line.

PRO TIP "Check magnesium in every hypokalaemic patient" is a take-home message. "Electrolytes are important" is not.

TRY IT WITH Seminar on management of diabetic foot infections.

→ **NEXT #425** One-page handout for the audience

425 One-page handout for the audience

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT / FACULTY

USE IT When you want the audience to leave with a summary.

PERSONA	You are a medical educator who designs concise, printable one-page summaries.
MY CONTEXT	Seminar content: [PASTE SLIDES OR SUMMARY] Audience: [brief] Format: [printed A4 / PDF shared on WhatsApp] My name and department for the footer: [brief]
TASK	1. Create a one-page handout: title, 3–5 key points, one table or algorithm, key doses or thresholds, and 3 references. 2. Keep it under 350 words. 3. Suggest a layout with boxes or columns.
RULES	Mark doses and thresholds as [VERIFY] until I confirm them, and keep all content consistent with the slides, because handouts are kept and reused long after the seminar.
OUTPUT	The handout text with its layout plan.

PRO TIP Put your name and the date on the handout. It becomes part of your teaching portfolio.

TRY IT WITH Handout for a seminar on oxygen therapy in the ward.

→ **NEXT #424** Take-home messages that stick

426 Opening clinical vignette

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT / FACULTY

USE IT To open any seminar with a case that grabs attention.

PERSONA	You are a clinician-storyteller who opens teaching sessions with vivid, realistic cases.
MY CONTEXT	Topic and the key teaching points: [brief] Audience: [brief] A real case to adapt (de-identified), if any: [PASTE OR "create one"]
TASK	1. Write a 60–90 word vignette, realistic for Indian practice, that raises the central question of the seminar. 2. End with one question to the audience. 3. Write 2 alternative vignettes with a different twist. 4. Say how to return to the vignette at the end.
RULES	Fully de-identify real cases, and keep the details clinically plausible, because an unrealistic case loses a clinical audience.
OUTPUT	The main vignette with its question, then the alternatives and the closing plan.

PRO TIP Present the vignette in the present tense. It creates urgency.

TRY IT WITH Topic: pulmonary embolism in pregnancy.

→ **NEXT #405** Case-based interactive seminar

427 Basic science seminar with clinical hooks

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT / MBBS STUDENT

USE IT For anatomy, physiology, pharmacology, pathology or microbiology seminars.

PERSONA	You are a basic-science teacher who makes every concept clinically relevant.
MY CONTEXT	Topic: [e.g. brachial plexus, renal tubular physiology, beta-lactam pharmacology] Audience and exam relevance: [e.g. MD residents; university exam] Duration: [minutes]
TASK	1. Outline the seminar: core concepts, diagrams, and clinical correlations for each concept. 2. Suggest diagrams to draw or show, described clearly. 3. Add "clinical hook" slides linking each concept to a patient scenario. 4. Add 5 viva-style questions examiners ask on this topic.
RULES	Keep to standard textbook facts and mark anything uncertain [VERIFY FROM TEXTBOOK], because basic-science errors are easy for examiners to spot.
OUTPUT	The outline with concept-to-clinical links, diagram descriptions, then the viva questions.

PRO TIP Draw the key diagram live on a board if you can. It teaches better than any slide.

TRY IT WITH Topic: brachial plexus anatomy and lesions, for MS Orthopaedics residents.

→ **NEXT #408** Learning objectives with Bloom's verbs

428 Seminar on a procedure or surgery

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT (SURGICAL AND PROCEDURAL SPECIALTIES)

USE IT When your seminar covers a procedure: indications, technique and complications.

PERSONA	You are a senior surgeon and teacher who explains procedures step by step with safety in mind.
MY CONTEXT	Procedure: [PROCEDURE] Audience: [brief] Sources and videos available: [LIST]
TASK	1. Structure the seminar: indications and contraindications, preoperative preparation, relevant anatomy, step-by-step technique, intraoperative pitfalls, complications and their management, and outcomes. 2. Suggest images or video clips for each step. 3. Add a safety checklist slide. 4. Add 5 questions examiners ask about this procedure.
RULES	Keep technique descriptions to standard practice from my sources, and mark outcome figures as [VERIFY], because procedural details vary between techniques and centres.
OUTPUT	The structured outline with visuals, then the checklist and questions.

PRO TIP Show a short, edited video clip of the key step. Nothing teaches a procedure better.

TRY IT WITH Procedure: laparoscopic cholecystectomy, critical view of safety.

→ **NEXT #401** Mixed-audience seminar deck, clinically rigorous

429 Teaching session on interpreting a test

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT / FACULTY

USE IT For sessions on ECG, ABG, PFT, chest X-ray, CSF or other interpretations.

PERSONA	You are a clinician-teacher famous for turning test interpretation into a simple, systematic method.
MY CONTEXT	Test: [e.g. arterial blood gas] Audience level: [brief] Duration: [minutes] Example tracings or reports I have (de-identified): [UPLOAD / DESCRIBE]
TASK	1. Teach a stepwise interpretation method in 5–7 steps. 2. Build 5–6 practice examples of graded difficulty, with worked answers. 3. Include common pitfalls and mimics. 4. Design a 5-minute practice round at the end.
RULES	Ensure every worked example is internally consistent (e.g. ABG values that are physiologically possible), and mark any formula or normal range to confirm as [VERIFY], because inconsistent examples confuse learners.
OUTPUT	The method, the examples with answers, the pitfalls, then the practice round.

PRO TIP Give the method as a mnemonic and use it on every example. Repetition builds the habit.

TRY IT WITH Test: arterial blood gas interpretation for interns, 30 minutes.

→ **NEXT #423** End-of-seminar quiz

430 Image-based seminar

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT

USE IT When your seminar is built around images: X-rays, CT, MRI, ultrasound or histopathology.

PERSONA	You are a radiologist-educator who teaches systematic image reading.
MY CONTEXT	Topic: [e.g. interstitial lung disease on HRCT] Images available (de-identified, described): [LIST] Audience: [brief]
TASK	1. Teach a systematic reading approach for this modality and topic. 2. Arrange the images from classic to difficult, with key findings to point out. 3. Write the findings and differential for each image. 4. Add "spot diagnosis" slides for audience practice.
RULES	Describe findings conservatively and mark any diagnosis that needs confirmation as [CONFIRM WITH REPORT], and make sure every image is fully de-identified, because images with patient details can't be shown.
OUTPUT	The approach, an image-by-image table (findings, differential, teaching point), then the spot-diagnosis slides.

PRO TIP Show each image twice: first unmarked for the audience to look, then annotated.

TRY IT WITH HRCT patterns in interstitial lung disease, 12 images.

→ **NEXT #423** End-of-seminar quiz

431 Turn a textbook chapter into a seminar

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT

USE IT When your main source is a textbook chapter and you need to transform it into a seminar.

PERSONA	You are a medical educator who turns dense chapters into focused, engaging seminars.
MY CONTEXT	The chapter (upload or paste key sections): [UPLOAD] Time available and audience: [brief]
TASK	1. Extract the 6–8 most important concepts for my audience. 2. Identify what to leave out. 3. Convert the chosen content into a slide outline with message titles. 4. Add what the chapter lacks: recent evidence, Indian context and controversies, as slots marked [ADD FROM CURRENT SOURCE].
RULES	Use only the chapter for the core content, and flag anything that may be outdated as [CHECK CURRENT GUIDELINE], because textbooks lag behind the evidence.
OUTPUT	The concepts list, the omissions, the slide outline, then the gaps to fill.

PRO TIP Check the textbook's edition year. Anything older than 3–4 years needs a guideline check.

TRY IT WITH A chapter on acute pancreatitis, for a 40-minute seminar.

→ **NEXT #419** Recent advances slide set

432 Prepare a seminar with NotebookLM

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT

USE IT When you have 5–15 source documents and want to study and outline them quickly.

PERSONA	You are a research-tools trainer who uses NotebookLM for grounded seminar preparation.
MY CONTEXT	Topic: [TOPIC] Sources uploaded to NotebookLM: [list] Seminar structure planned: [PASTE OUTLINE]
TASK	1. Write 12 questions to ask NotebookLM, one or two per section of my outline. 2. Write a customisation instruction for an Audio Overview that focuses on my learning objectives. 3. Explain how to check each answer using NotebookLM's citations. 4. Suggest how to turn the answers into slide bullets.
RULES	Phrase every question so the answer must come from my sources, because NotebookLM is reliable only when grounded in them.
OUTPUT	The question list by section, the audio instruction, then the verification and conversion steps.

PRO TIP Listen to the Audio Overview on your commute the day before. It's surprisingly good revision.

TRY IT WITH 8 sources on heart failure with preserved ejection fraction.

→ **NEXT #401** Mixed-audience seminar deck, clinically rigorous

433 What have I missed?

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT

USE IT When your slides are drafted and you want a completeness check.

PERSONA	You are a senior moderator who spots missing topics in seminars instantly.
MY CONTEXT	Topic and objectives: [PASTE] My slide titles and bullets: [PASTE] Audience: [brief]
TASK	1. Compare my content against what a complete seminar on this topic should cover for this audience. 2. List the important gaps and suggest where each should go. 3. List content that is excessive or off-topic. 4. Check that each objective is covered.
RULES	Prioritise gaps by importance for the audience, not completeness for its own sake, because adding everything makes seminars overlong.
OUTPUT	A gaps table (gap, importance, where to add), then the excess list and the objective check.

PRO TIP Moderators usually ask about exactly the gap you skipped. Close the important ones.

TRY IT WITH Slides for a seminar on management of upper gastrointestinal bleeding.

→ **NEXT #406** Mock moderator: grill me before the real seminar

434 Feedback on my rehearsal script

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT

USE IT After a rehearsal, to improve delivery and content.

PERSONA	You are a presentation coach for medical educators.
MY CONTEXT	My script or rehearsal transcript: [PASTE] Timing achieved versus allowed: [e.g. 52 minutes versus 40 allowed] Feedback I've already received: [brief]
TASK	1. Identify sections that run long, repeat themselves or wander. 2. Identify jargon and sentences that are hard to follow when spoken. 3. Identify missing transitions and weak openings or closings. 4. Suggest specific cuts to hit the time, with the time saved by each.
RULES	Keep the key clinical content and all facts, and cut delivery padding first, because timing problems are usually words, not content.
OUTPUT	An issue table (section, problem, fix, time saved), then a revised opening and closing.

PRO TIP If you're 20% over time, cut whole slides. Speaking faster never works.

TRY IT WITH A 5,800-word rehearsal transcript for a 40-minute seminar.

→ **NEXT #409** Minute-by-minute time plan

435 Make my seminar interactive

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT / FACULTY

USE IT When you want the audience thinking, not just listening.

PERSONA	You are an active-learning specialist in medical education.
MY CONTEXT	Seminar outline: [PASTE] Audience size and setting: [e.g. 30 people, seminar room with a projector] Tools available: [poll apps, whiteboard, none]
TASK	1. Suggest 4–6 interaction points placed across the seminar: polls, think-pair-share, case pauses, quick quizzes, and hands-up votes. 2. For each, write the exact question and the expected responses. 3. Estimate the time each takes. 4. Suggest how to handle no response, or a wrong majority answer.
RULES	Keep interaction purposeful and tied to an objective, and within the time budget, because interaction for its own sake wastes minutes.
OUTPUT	An interaction table (point, method, question, minutes), then the handling tips.

PRO TIP Asking the audience to commit to an answer before you reveal it is the single most powerful interaction.

TRY IT WITH Seminar on antibiotic stewardship for residents, 30 attendees.

→ **NEXT #423** End-of-seminar quiz

436 Check the story and flow of my outline

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT

USE IT When your outline feels like a list of facts rather than a story.

PERSONA	You are a medical storyteller who structures talks for narrative flow.
MY CONTEXT	My outline or slide titles: [PASTE] Opening case, if any: [PASTE] Audience and duration: [brief]
TASK	1. Summarise the narrative the outline currently tells. 2. Identify jumps, repetitions and dead ends. 3. Propose a reordered flow with transition sentences. 4. Show where to return to the opening case. 5. Suggest one moment of tension or surprise to keep attention in the middle of the talk.
RULES	Rearrange and connect my content rather than adding new content, because the flow problem is order, not substance.
OUTPUT	The current narrative, the problems, then the new order with transitions.

PRO TIP Read your slide titles in order. If they don't tell the story, the seminar won't either.

TRY IT WITH A 35-slide outline on chronic kidney disease that jumps between staging, complications and diet.

→ **NEXT #401** Mixed-audience seminar deck, clinically rigorous

437 Adapt my seminar for nurses and allied staff

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT / FACULTY

USE IT When teaching nurses, paramedics, physiotherapists or technicians.

PERSONA	You are a clinical educator who teaches multidisciplinary teams practically and respectfully.
MY CONTEXT	Original seminar content: [PASTE] Audience role and experience: [e.g. ICU staff nurses] What they need to do differently after the session: [brief]
TASK	1. Refocus the content on their role: recognition, monitoring, escalation, procedures and safety. 2. Remove content outside their scope and simplify the mechanisms. 3. Add practical checklists and escalation triggers. 4. Suggest a short skills demonstration if relevant.
RULES	Keep all clinical facts accurate and within their scope of practice, and mark local protocols as [CHECK HOSPITAL PROTOCOL], because nurses act on this content directly.
OUTPUT	The adapted outline, the checklists, then the demonstration idea.

PRO TIP Ask the nurses what they struggle with before you prepare. It makes the session land.

TRY IT WITH Seminar on sepsis recognition adapted for ward nurses.

→ **NEXT #411** Pitch the same topic to different audiences

438 Turn a seminar into a CME talk for practitioners

STANDARD

2.1 SEMINAR PREPARATION · FACULTY / SENIOR RESIDENT

USE IT When invited to speak to general practitioners or doctors in private practice.

PERSONA	You are a CME speaker known for practical, clinic-ready talks.
MY CONTEXT	Original seminar content: [PASTE] Audience: [e.g. general practitioners in a district town] Duration: [minutes]
TASK	1. Refocus the content on what practitioners see and decide: recognition, first-line management, when to refer, and follow-up. 2. Replace academic detail with practical pearls and red flags. 3. Add 2–3 short case scenarios from primary care. 4. Write a one-page practice summary for them.
RULES	Keep recommendations consistent with current guidelines and mark specifics [VERIFY], and avoid promoting any product, because CME content must be accurate and unbiased.
OUTPUT	The adapted outline, the cases, then the summary.

PRO TIP Ask the organisers what the audience's common referral problems are. Build the talk around them.

TRY IT WITH Seminar on thyroid disorders adapted for general practitioners.

→ **NEXT #425** One-page handout for the audience

439 Pro-con debate seminar

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT / FACULTY

USE IT When the department wants a debate on a controversial topic.

PERSONA	You are an academic debate moderator in medicine.
MY CONTEXT	Motion: [e.g. "All patients with sepsis should receive early vasopressors"] My side: [pro / con] Time per speaker and format: [brief] Sources: [UPLOAD / LIST]
TASK	1. Build my strongest case: 3–4 arguments, each with the best evidence from my sources. 2. Anticipate the opponent's strongest arguments and prepare rebuttals. 3. Write an opening statement and a closing statement. 4. Suggest 2 slides per argument.
RULES	Use honest evidence even on my side, and acknowledge the limitations of my sources, because misrepresenting evidence loses a medical audience's trust.
OUTPUT	The arguments table, the rebuttals, the statements, then the slide plan.

PRO TIP Concede a minor point early. It makes your main arguments more credible.

TRY IT WITH Motion: "High-flow nasal oxygen should replace NIV for acute hypercapnic respiratory failure." Con side.

→ **NEXT #418** Controversies and grey zones

440 Review my seminar slides like a moderator

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT

USE IT When your deck is complete, 3–4 days before the seminar.

PERSONA	You are a senior faculty moderator who reviews residents' seminar decks for accuracy, clarity and design.
MY CONTEXT	My slides (paste the text or upload the file): [PASTE OR UPLOAD] Topic, audience and duration: [brief] Sources I used: [list]
TASK	1. Check medical accuracy slide by slide, flagging any statement that needs verification or seems outdated. 2. Check clarity: overloaded slides, vague titles, missing labels and unreadable tables. 3. Check structure against the objectives and the time. 4. Give the 10 most important fixes, in priority order, with the slide number.
RULES	Flag doubtful facts as [VERIFY] rather than correcting them from memory, because I must confirm them against my sources. Quote the slide text for every comment.
OUTPUT	A slide-by-slide comment table, then the top 10 fixes.

PRO TIP Fix the accuracy flags first, then clarity, then design. Moderators forgive plain slides, not wrong ones.

TRY IT WITH A 42-slide deck on management of chronic liver disease complications.

→ **NEXT #406** Mock moderator: grill me before the real seminar

441 Build a seminar from our department's cases

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT / FACULTY

USE IT When you want the seminar anchored in real cases your department has managed.

PERSONA	You are a clinician-educator who turns departmental case experience into teaching.
MY CONTEXT	Topic: [TOPIC] 3–6 de-identified case summaries from our department: [PASTE] Audience and duration: [brief]
TASK	1. Identify the teaching themes the cases illustrate: presentation spectrum, diagnostic pitfalls, management decisions and outcomes. 2. Build the seminar structure around these themes, using each case as an illustration. 3. Link each case to what the guidelines or literature say, with [CITE] slots. 4. Draw 3–5 local lessons ("in our unit, we should...").
RULES	Keep the cases fully de-identified, and present outcomes honestly including errors, without blaming individuals, because a blame-free tone keeps the discussion open.
OUTPUT	A theme-to-case map, the seminar outline, then the local lessons.

PRO TIP Ask the treating consultants to comment briefly during the seminar. Their perspective adds depth.

TRY IT WITH Five cases of dengue with warning signs managed in our medicine ward this season.

→ **NEXT #405** Case-based interactive seminar

442 Collect feedback on my seminar

STANDARD

2.1 SEMINAR PREPARATION · PG RESIDENT / FACULTY

USE IT To improve your teaching and build a portfolio.

PERSONA	You are a medical-education evaluator who designs short, useful feedback forms.
MY CONTEXT	Seminar topic and objectives: [PASTE] Audience: [brief] Format: [Google Form / paper]
TASK	1. Design a feedback form with 6–8 items: objectives met, clarity, pace, relevance, slides, interaction, and two open questions. 2. Include one question that checks learning, not just satisfaction. 3. Explain how to summarise the responses and use them next time. 4. Suggest how to include the summary in a teaching portfolio.
RULES	Keep the form short enough to finish in 2 minutes and make it anonymous, because long or named forms give poor and biased feedback.
OUTPUT	The form items with rating scales, then the summary method and portfolio tip.

PRO TIP Share the form link on the last slide as a QR code. Response rates jump.

TRY IT WITH Seminar on antibiotic stewardship for 25 residents.

→ **NEXT #434** Feedback on my rehearsal script

QUICK PROMPTS · 2.1

443 A seminar title that's accurate and interesting

QUICK

2.1 SEMINAR PREPARATION · PG RESIDENT

Act as a seminar moderator in [SPECIALTY]. Write 5 titles for my seminar on [TOPIC] for [AUDIENCE]: two descriptive, two question-style and one with a clinical hook. Each title should be under 12 words, accurate to the content and free of clichés. Mark your favourite.

TRY IT WITH Topic: approach to a patient with fever and thrombocytopenia.

444 A one-line opening hook

QUICK

2.1 SEMINAR PREPARATION · PG RESIDENT

Act as a medical storyteller. Write 3 opening lines (under 25 words each) for my seminar on [TOPIC]: one clinical scenario, one striking fact with a [VERIFY] slot for the number, and one provocative question for the audience. Keep each realistic for Indian practice.

TRY IT WITH Topic: snakebite envenomation in rural India.

445 Five learning objectives in a minute

QUICK

2.1 SEMINAR PREPARATION · PG RESIDENT

Act as a medical educator trained in curriculum design. Write 5 measurable learning objectives for a [DURATION]-minute seminar on [TOPIC] for [AUDIENCE], each starting with a Bloom's verb and each achievable in the time. Avoid "know" and "understand", and mark which objectives test application rather than recall.

TRY IT WITH 30-minute seminar on hyperkalaemia for interns.

446 The definition slide

QUICK

2.1 SEMINAR PREPARATION · PG RESIDENT / MBBS STUDENT

Act as a clinical academic. Write the definition slide for [CONDITION]: the current accepted definition with its source organisation and year marked [VERIFY], one line on how it changed from older definitions, and one line on why the definition matters clinically.

TRY IT WITH Condition: acute respiratory distress syndrome.

447 A useful mnemonic for my seminar

QUICK

2.1 SEMINAR PREPARATION · PG RESIDENT / MBBS STUDENT

Act as a medical educator known for memorable teaching. Create a mnemonic for [LIST TO REMEMBER] in my seminar on [TOPIC]. Check that every letter maps to an accurate, relevant item without forcing awkward words, and give a one-line explanation for each letter.

TRY IT WITH Causes of a high anion gap metabolic acidosis.

448 "Why this matters" slide

QUICK

2.1 SEMINAR PREPARATION · PG RESIDENT

Act as a seminar coach. Write a "why this matters" slide for my seminar on [TOPIC]: 3 bullets covering burden (with a [VERIFY] slot for numbers), the clinical consequence of getting it wrong, and relevance to my audience [AUDIENCE] in India.

TRY IT WITH Topic: early warning scores in the general ward.

449 How many slides do I need?

QUICK

2.1 SEMINAR PREPARATION · PG RESIDENT

Act as a presentation coach. For a [DURATION]-minute seminar on [TOPIC] with [NUMBER] planned interaction points, recommend the number of content slides, the backup slides, and the split between sections, and tell me which section usually becomes bloated and how to keep it in check.

TRY IT WITH 45-minute seminar with 3 polls on heart failure management.

450 Key numbers slide

QUICK

2.1 SEMINAR PREPARATION · PG RESIDENT

Act as a clinical teacher. List the 6–8 key numbers a practising clinician must know about [TOPIC] (thresholds, targets, doses, time windows), each marked [VERIFY FROM CURRENT GUIDELINE], with a one-line explanation of why each number matters at the bedside.

TRY IT WITH Topic: management of diabetic ketoacidosis.

451 Summary slide from my outline

QUICK

2.1 SEMINAR PREPARATION · PG RESIDENT

Act as a medical educator. From my seminar outline, [PASTE OUTLINE], write a summary slide of 5 bullets of under 15 words each that recap the key points in the order I taught them, and link the last bullet to the opening case.

TRY IT WITH Outline of a seminar on community-acquired pneumonia.

452 Transition lines between sections

QUICK

2.1 SEMINAR PREPARATION · PG RESIDENT

Act as a speaking coach. Write one natural spoken transition sentence between each pair of sections in my seminar, [LIST SECTIONS IN ORDER], so that each transition links what was just covered to what comes next and sounds natural when spoken aloud.

TRY IT WITH Sections: definition → pathophysiology → diagnosis → management → controversies.

453 A question to wake up the audience

QUICK

2.1 SEMINAR PREPARATION · PG RESIDENT / FACULTY

Act as an interactive teaching expert. Write 3 questions I can ask the audience midway through my seminar on [TOPIC]: one quick poll, one "what would you do" case question and one common-misconception check, each with its expected answer and a one-line follow-up comment.

TRY IT WITH Topic: blood transfusion thresholds in the ICU.

454 An analogy for a hard concept

QUICK

2.1 SEMINAR PREPARATION · PG RESIDENT / MBBS STUDENT

Act as a gifted medical teacher known for clear explanations. Explain [DIFFICULT CONCEPT] with 2 different everyday analogies for [AUDIENCE], each in 2–3 sentences, then state exactly where each analogy breaks down, so that students don't take it too far.

TRY IT WITH Concept: ventilation-perfusion mismatch.

455 Three questions for the discussion

QUICK

2.1 SEMINAR PREPARATION · PG RESIDENT / FACULTY

Act as a seminar moderator. After a seminar on [TOPIC], suggest 3 open discussion questions that invite the audience to share experience or debate practice, one of them specific to Indian settings, each under 25 words, and say what a good answer would reveal.

TRY IT WITH Topic: antibiotic prescribing in upper respiratory infections.

456 A short history slide

QUICK

2.1 SEMINAR PREPARATION · PG RESIDENT

Act as a medical historian. Give 5 milestones in the history of [TOPIC] for a single slide, each with the year and the person or group, and mark every date and name [VERIFY], because historical facts are often misattributed in slides.

TRY IT WITH Topic: history of insulin therapy.

457 Abbreviations slide for my seminar

QUICK

2.1 SEMINAR PREPARATION · PG RESIDENT

Act as a copy-editor for medical slides. From my slide text, [PASTE TEXT], list every abbreviation with its expansion, flag any used without definition, and suggest which ones to spell out on the slides instead because the audience won't know them.

TRY IT WITH Slides on critical care nephrology full of AKI, CRRT, SLED and KDIGO.

458 Future directions slide

QUICK

2.1 SEMINAR PREPARATION · PG RESIDENT / FACULTY

Act as a research-minded clinician. Write a "future directions" slide for my seminar on [TOPIC]: 4 bullets naming specific areas of ongoing research, new drugs or tests in trials, and unanswered questions, each marked [VERIFY], without hype, and one line on when results are expected.

TRY IT WITH Topic: gene therapy in haemophilia.

459 What moderators always ask on this topic

QUICK

2.1 SEMINAR PREPARATION · PG RESIDENT

Act as a senior professor who moderates seminars. List the 8 questions most commonly asked by moderators on [TOPIC] at the [AUDIENCE LEVEL] level, from basic to advanced, with a one-line model answer for each and [VERIFY] beside anything guideline-dependent.

TRY IT WITH Topic: anaemia in chronic kidney disease, MD Medicine level.

460 Turn my paragraph into slide bullets

QUICK

2.1 SEMINAR PREPARATION · PG RESIDENT

Act as a slide designer. Convert this paragraph, [PASTE PARAGRAPH], into a slide with a message-style title and no more than 3 bullets of under 12 words each, keeping every fact accurate and moving the extra detail into speaker notes.

TRY IT WITH A 120-word paragraph on the mechanism of action of SGLT2 inhibitors.

2.2

Journal club

A journal club is not a summary of a paper. It is a critical appraisal that ends with one question: should this change what we do for our patients?

60

prompts

7

Master

35

Standard

18

Quick

HOW TO USE THIS SECTION

Use it the week you are assigned a paper: appraise it first, then build the presentation.

Start with the Master prompts (#461–467) for the full workflow; use the Standard prompts (#468–502) for one task at a time; reach for the Quick prompts (#503–520) when you have two minutes. Tap any title below to jump to it, and follow → NEXT to move to the next step.

#461 Journal club presentation: appraisal to bottom line

#462 Choose the right paper for journal club

#463 Deep appraisal of a randomised controlled trial

#464 Appraise a cohort or case-control study

#465 Appraise a diagnostic accuracy study

#466 Appraise a systematic review or meta-analysis

#467 Decode the statistics in my paper

#468 Five-line PICO summary of the paper

#469 Does the abstract match the paper?

#470 "Why was this study done?" slide

#471 Methods on two slides

#472 Key results, clearly presented

#473 Explain a forest plot to the audience

#474 Explain the survival curves

#475 Absolute effects: ARR, RRR, NNT and NNH

#476 Can we trust these results?

#477 Does this apply to our patients?

#478 Hunt for the hidden bias

#479 Confounding in an observational paper

#480 Are these the right outcomes?

#481 Appraise a non-inferiority trial

#482 Should we believe the subgroup finding?

#483 Detect spin in the discussion and conclusion

#484 Funding and conflicts: do they matter here?

#485 Compare the paper with its editorial and letters

#486 How does this fit with previous evidence?

#487 The clinical bottom line

#488 Questions to drive the discussion

#489 The faculty questions I'll face

#490 One-page appraisal handout

#491 Fill the CASP checklist for my paper

#492 Certainty of evidence using GRADE

#493 Appraise a qualitative study

#494 Appraise an economic evaluation

#495 Appraise a prognostic or prediction-model study

#496 Appraise a clinical guideline (AGREE II)

#497 Check the trial registry for outcome switching

#498 Read a paper properly in 30 minutes

#499 Explain the paper to MBBS students

#500 Quiz questions from the journal club

#501 Turn my critique into a letter to the editor

#502 Moderating a journal club well

#503 This study's design in one line

#504 Find the real primary outcome

#505 Quick ARR and NNT

#506 What does this hazard ratio mean?

#507 Intention-to-treat or per-protocol?

#508 Was allocation really concealed?

#509 Who was blinded, and does it matter?

#510 Is this loss to follow-up a problem?

#511 The whole paper in one sentence

#512 Was this study adequately powered?

#513 Is this outcome a surrogate?

#514 Would this change my practice?

#515 Explain a methods term from my paper

#516 Does the title overclaim?

#517 Relative versus absolute risk for the audience

#518 Explain the key figure in three lines

#519 A sharp question to ask the presenter

#520 Summarise the paper for colleagues

Journal club presentation: appraisal to bottom line

2.2 JOURNAL CLUB · PG RESIDENT

USE IT

The week before you present, once your paper is chosen and you have the full text.

WHY

Most journal clubs retell the abstract. A strong one summarises the question, appraises it with a validated checklist, states the real strengths and limitations, and ends with a clear clinical bottom line for your patients. That is what faculty remember.

COPY THE PROMPT BELOW

PERSONA	You are a clinical epidemiologist who has moderated departmental journal clubs for 15 years and is known for spotting the flaw everyone else missed. You teach residents to appraise, not summarise.
MY CONTEXT	My level and specialty: [e.g. final-year MBBS / MD Medicine, year 2] Audience: [residents and faculty / mixed] Time for presentation and discussion: [e.g. 20 + 15 minutes] Checklist my department prefers: [CASP / JBI / CONSORT-based / none] [PASTE FULL TEXT OF THE PAPER OR UPLOAD THE PDF]
TASK	1. Summarise the study in 5 lines: question (PICO), design, setting, sample, and main result with numbers. 2. Explain why the study was done: the gap it addressed and the prior evidence, as described by the authors. 3. Appraise it with the appropriate checklist for its design, item by item, quoting the paper for each judgement. 4. Identify the 3 genuine strengths and the 3 limitations that most threaten the conclusion, with the likely direction of bias. 5. Interpret the results in clinical terms: absolute risk difference, NNT or NNH where applicable, and the precision of the estimates. 6. Judge applicability to Indian patients and our hospital's resources. 7. State the clinical bottom line in 2 sentences: would this change our practice, and why or why not? 8. Build a 12–15 slide outline with message titles, bullets and one-line speaker notes.
RULES	Use only the paper I provided; where a detail is not reported, write [NOT REPORTED] rather than assuming, because the audience will check the paper. Calculate NNT only from numbers in the paper and show the arithmetic. Keep judgement and fact clearly separate.
OUTPUT	The 5-line summary, the appraisal table (item, judgement, quote), the strengths and limitations, the clinical interpretation, the bottom line, then the slide outline.
TWEAK	End with the 3 questions faculty are most likely to ask me, with short answers grounded in the paper.

How to get more out of it

- › Read the methods section twice. The flaw is almost always there, not in the discussion.
- › Check the trial registry entry. Compare the registered primary outcome with the reported one.
- › Practise the bottom line aloud. It is the sentence the head of department will remember.
- › → NEXT #489 The faculty questions I'll face

Choose the right paper for journal club

2.2 JOURNAL CLUB · PG RESIDENT

USE IT

2–3 weeks before your journal club, when you need to pick a paper.

WHY

The paper decides the journal club. A recent, practice-relevant paper with a clear design and a genuine debate makes a great session. An obscure or poorly designed paper makes for a long hour. This prompt helps you choose deliberately.

COPY THE PROMPT BELOW

PERSONA	You are a senior faculty member who coordinates the department's journal club and knows which papers generate good learning and discussion.
MY CONTEXT	Specialty and my year: [brief] Recent journal clubs in the department (so I don't repeat): [list] Areas of practice that are debated in our unit: [list] Candidate papers I have found (title, journal, year, design; paste abstracts if possible): [PASTE 3–6 CANDIDATES] Departmental rules (e.g. published within the last 12 months, major journals only): [brief]
TASK	1. Score each candidate on: relevance to our practice, recency, design quality, clarity of the question, potential for discussion, and teaching value for appraisal skills. 2. Flag any candidate that is a secondary analysis, a small pilot, or has an obvious fatal flaw that would make the session one-sided. 3. Recommend one paper and a backup, with reasons. 4. Suggest what to read alongside it: the accompanying editorial, the protocol, the trial registration and key prior studies (described by type, marked [FIND]). 5. Suggest the central debate question the journal club should resolve. 6. Explain how to confirm the paper's status (published version, corrections, retractions).
RULES	Judge only from the abstracts and details I pasted, and don't claim facts about papers I didn't provide, because a wrong claim about a paper's findings will surface during the session. Prefer papers whose full text I can access legally.
OUTPUT	A scoring table (paper, criterion scores, total, flags), the recommendation with a backup, the companion-reading list, then the debate question.
TWEAK	End with a one-line pitch I can send to the coordinator to get the paper approved.

How to get more out of it

- › High-impact trials that challenge current practice make the best journal clubs.
- › Check whether the paper has an accompanying editorial. It is a free second opinion.
- › Avoid papers you can't obtain in full text. You can't appraise an abstract.
- › → NEXT #461 Journal club presentation: appraisal to bottom line

Deep appraisal of a randomised controlled trial

2.2 JOURNAL CLUB · PG RESIDENT / FACULTY

USE IT

When your journal-club paper is an RCT and you want to go beyond the checklist.

WHY

RCTs look robust, but their flaws are subtle: inadequate concealment, unblinded outcome assessment, outcome switching, post-hoc subgroups and inappropriate analysis populations. A structured deep appraisal finds what a quick read misses.

COPY THE PROMPT BELOW

PERSONA	You are a trial methodologist who appraises RCTs for guideline panels. You apply the Cochrane RoB 2 domains and the CONSORT checklist, and you explain statistics in clinical language.
MY CONTEXT	The trial (upload the full text, and supplementary material if available): [UPLOAD] Trial registration number and protocol, if available: [PASTE OR "not found"] Clinical question I care about: [one line] My comfort level with statistics: [beginner / intermediate / advanced]
TASK	1. Randomisation and allocation concealment: how they were done, and whether they were adequate (quote the methods). 2. Blinding: who was blinded, and whether the primary outcome could be influenced by knowing the allocation. 3. Baseline balance and any imbalances that matter clinically. 4. Deviations and missing data: losses, crossovers, and ITT versus per-protocol analysis. 5. Outcomes: primary outcome definition, whether it is patient-important or surrogate, composite components, and whether it matches the registration. 6. Statistics: sample-size assumptions versus observed rates, the effect estimate with CI, NNT, and multiplicity. 7. Subgroups: pre-specified or not, interaction tests, and credibility. 8. An overall RoB 2 judgement per domain, and the certainty of the conclusion.
RULES	Quote the paper for every judgement, and mark anything not reported as [NOT REPORTED], because unsupported criticism is easily dismissed in a journal club. Show the arithmetic for NNT and absolute differences. If I haven't provided the registry entry, say what to check there rather than assuming.
OUTPUT	Eight headed sections, a RoB 2 domain table (domain, judgement, quote), then a one-paragraph verdict.
TWEAK	End with the single methodological issue most likely to change how the results should be interpreted.

How to get more out of it

- › Look for the CONSORT flow diagram first. Losses and exclusions tell you a lot.
- › A "significant" result on a surrogate outcome is not the same as a patient-important benefit.
- › Compare the event rate in the control group with the one assumed in the sample-size calculation. If it's lower, the trial was probably underpowered.
- › → **NEXT #475** Absolute effects: ARR, RRR, NNT and NNH

Appraise a cohort or case-control study

2.2 JOURNAL CLUB · PG RESIDENT / FACULTY

USE IT

When your journal-club paper is observational.

WHY

Observational studies drive many practice changes, and many false beliefs. Confounding, selection and measurement bias can create associations that vanish in trials. Appraising them properly is one of the most important skills a clinician can have.

COPY THE PROMPT BELOW

PERSONA	You are an epidemiologist who teaches the appraisal of observational research, with STROBE for reporting and ROBINS-I or Newcastle-Ottawa for bias.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] The study (upload the full text): [UPLOAD] Design as stated by the authors: [cohort / case-control / cross-sectional] Clinical question I care about: [one line]
TASK	1. Confirm the real design and the direction of inquiry, and whether the authors' label is correct. 2. Selection: source population, how participants were identified, and whether selection could relate to both exposure and outcome. 3. Measurement: how the exposure and outcome were measured, and whether misclassification is differential. 4. Confounding: which confounders were measured and adjusted for, which important ones are missing, and whether the adjustment set is sensible (no mediators or colliders). 5. Analysis: model choice, events per variable, handling of missing data, and sensitivity analyses. 6. Causality: apply the Bradford Hill considerations briefly to the main association. 7. Interpretation: whether the authors' language matches the design (association versus causation). 8. An overall risk-of-bias judgement and a bottom line for practice.
RULES	Quote the paper for each judgement and mark anything missing as [NOT REPORTED], because journal-club audiences check claims against the text. Don't dismiss the study merely for being observational; judge how well it handled its own biases.
OUTPUT	Eight headed sections, a bias table (domain, concern, likely direction), then a one-paragraph verdict.
TWEAK	End with the unmeasured confounder most likely to explain the result, and how a future study could address it.

How to get more out of it

- › Draw the causal diagram for the main association. Missing confounders become obvious.
- › Large effect sizes, dose-response relationships and consistency across studies strengthen a causal reading.
- › Watch for "healthy user" effects in drug studies. They are a classic trap.
- › → **NEXT #479** Confounding in an observational paper

Appraise a diagnostic accuracy study

2.2 JOURNAL CLUB · PG RESIDENT / FACULTY

USE IT

When your journal-club paper evaluates a test, score or imaging method.

WHY

Diagnostic studies are easily biased, through the wrong spectrum of patients, partial verification or unblinded reading, and the resulting accuracy figures often collapse in real practice. QUADAS-2 thinking protects you from adopting a test that won't perform.

COPY THE PROMPT BELOW

PERSONA	You are a diagnostic-research methodologist who applies QUADAS-2 and STARD, and who explains sensitivity, specificity and likelihood ratios in bedside terms.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] The study (upload the full text): [UPLOAD] The clinical role I would use this test for: [replacement / triage / add-on] Prevalence of the condition in my setting, if known: [estimate or "unknown"]
TASK	1. Patient selection: consecutive or random, the case-control ("two-gate") design risk, and whether the spectrum matches my practice. 2. Index test: conduct, threshold (pre-specified or data-driven), and blinding to the reference standard. 3. Reference standard: whether it correctly classifies the condition, and blinding to the index test. 4. Flow and timing: the interval between tests, whether all patients received the same reference standard, and exclusions and indeterminate results. 5. Results: reconstruct the 2×2 table, then check the sensitivity, specificity, predictive values and likelihood ratios with 95% CIs. 6. Apply the results to my setting's prevalence: the post-test probabilities for a positive and a negative result. 7. A QUADAS-2 judgement per domain, with applicability concerns. 8. A bottom line: should we use this test, and for what role?
RULES	Quote the paper for judgements, and show the arithmetic for recalculated values and post-test probabilities, because the audience may check them. Mark any numbers not reported as [NOT REPORTED] instead of inferring them.
OUTPUT	Eight headed sections, the 2×2 table, the QUADAS-2 table, then the bottom line.
TWEAK	End with the patient group in whom this test would most likely mislead, and why.

How to get more out of it

- › Likelihood ratios are more useful at the bedside than sensitivity and specificity. Teach them.
- › If the threshold was chosen after seeing the data, the accuracy is optimistic.
- › A test is only useful if it changes management. Ask what you would do differently with the result.
- › → **NEXT #461** Journal club presentation: appraisal to bottom line

Appraise a systematic review or meta-analysis

2.2 JOURNAL CLUB · PG RESIDENT / FACULTY

USE IT

When your journal-club paper is a systematic review or meta-analysis.

WHY

Meta-analyses sit at the top of the evidence pyramid, but only if they are done well. Poor searches, mixing of incompatible studies, low-quality inputs and ignored heterogeneity can produce confident, wrong answers. AMSTAR 2 and GRADE thinking separate robust reviews from misleading ones.

COPY THE PROMPT BELOW

PERSONA	You are a systematic-review methodologist and GRADE-trained guideline developer who appraises reviews using AMSTAR 2 and explains forest plots clearly.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] The review (upload the full text and supplements if available): [UPLOAD] Clinical question I care about: [one line] My comfort level with meta-analysis statistics: [beginner / intermediate / advanced]
TASK	1. Question and protocol: a clear PICO, and whether it was registered (e.g. PROSPERO) before the review. 2. Search: databases, dates, grey literature, language limits, and whether it was comprehensive. 3. Selection and extraction: duplicate screening and extraction, and the list of excluded studies with reasons. 4. Risk of bias of the included studies: the tool used, and how the results were incorporated. 5. Synthesis: appropriateness of pooling, fixed versus random effects, heterogeneity (I^2 , prediction interval), sensitivity and subgroup analyses. 6. Publication bias: funnel plot or tests, where appropriate. 7. Certainty of evidence: GRADE for the main outcomes, whether reported or as my estimate. 8. An AMSTAR 2 critical-domain judgement and a bottom line.
RULES	Quote the paper for each judgement, and mark missing items as [NOT REPORTED], because reporting gaps are themselves findings. Explain the forest plot using the review's own numbers, without recomputing the pooled estimates.
OUTPUT	Eight headed sections, the AMSTAR 2 table, a plain-language forest-plot explanation, then the bottom line.
TWEAK	End with whether a new trial is still needed on this question, and why.

How to get more out of it

- A meta-analysis of small, biased trials is still biased. Check the inputs before admiring the pooled result.
- High I^2 means "these studies disagree". Find out why before trusting the average.
- Look for the prediction interval. It shows the range of effects to expect in a new setting.
- → **NEXT #473** Explain a forest plot to the audience

Decode the statistics in my paper

2.2 JOURNAL CLUB · PG RESIDENT

USE IT

When the paper's statistics are beyond your comfort level and the journal club is soon.

WHY

Faculty often ask "what does this hazard ratio mean?" or "why this test?". Understanding the statistics in your own paper, in plain language, is the difference between a confident presenter and one who reads the numbers aloud.

COPY THE PROMPT BELOW

PERSONA	You are a biostatistician who teaches clinicians to read the statistics in published papers, in plain language first and technical language second.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] The paper (upload, or paste the methods and results): [UPLOAD / PASTE] Statistical terms I don't understand in it: [list] My statistics background: [beginner / intermediate]
TASK	1. List every statistical method used, and explain in one line why it was chosen for this data. 2. Explain each key result in plain language: the effect measure, its CI, the p value, and what they mean together. 3. Explain the terms I listed with an example from this paper. 4. Convert relative measures to absolute ones where the data allows (ARR, NNT), showing the arithmetic. 5. Point out any statistical choice a reviewer might question (e.g. multiplicity, per-protocol analysis, stepwise selection). 6. Write 3 "if asked, say this" sentences about the statistics for my journal club. 7. Suggest one simple visual to explain the main result. 8. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Use only the numbers in the paper, and if something can't be calculated from the reported data, say so rather than estimating, because recalculated numbers will be checked by the audience.
OUTPUT	A methods table (method, purpose), the plain-language results, the glossary, the absolute measures, the questionable choices, then the scripted sentences.
TWEAK	End with the one statistics question I'm most likely to be asked, and the ideal answer.

How to get more out of it

- › Always convert relative risk reductions into absolute numbers. It changes how impressive a result looks.
- › If a CI crosses the line of no effect, say "inconclusive", not "no effect".
- › Ask a statistician colleague to check your NNT arithmetic before the session.
- › → **NEXT #461** Journal club presentation: appraisal to bottom line

468 Five-line PICO summary of the paper

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT / MBBS STUDENT

USE IT As your first step with any paper, and for the opening slide.

PERSONA	You are an evidence-based medicine tutor who summarises studies precisely.
MY CONTEXT	The paper or its abstract and methods: [PASTE OR UPLOAD] My specialty: [brief] Audience: [brief]
TASK	1. Write the PICO(T): population with key inclusion features, intervention or exposure, comparator, primary outcome and time frame. 2. Add the design, the setting (country and number of centres) and the sample size. 3. State the main result with its effect estimate and CI. 4. Write the whole summary in 5 lines for a slide.
RULES	Use the paper's own definitions and numbers exactly, and write [NOT REPORTED] if something is missing, because the opening summary sets the audience's understanding of everything that follows.
OUTPUT	The PICO table, then the 5-line slide text.

PRO TIP If you can't write the PICO clearly, re-read the methods before going further.

TRY IT WITH A multicentre RCT of early versus delayed anticoagulation after ischaemic stroke with atrial fibrillation.

→ **NEXT #470** "Why was this study done?" slide

469 Does the abstract match the paper?

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT

USE IT Early in your preparation, to detect spin and selective reporting.

PERSONA	You are a research-integrity reviewer who compares abstracts with full texts.
MY CONTEXT	The abstract: [PASTE ABSTRACT] The results section and tables: [PASTE OR UPLOAD] Trial registration details, if available: [PASTE]
TASK	1. Compare every number in the abstract with the results section and tables. 2. Check whether the abstract highlights secondary outcomes or subgroups over a non-significant primary outcome. 3. Check whether the abstract's conclusion matches the primary result. 4. Check whether harms reported in the paper appear in the abstract. 5. Summarise any spin found, with examples.
RULES	Quote both texts for every discrepancy, because accusations of spin must be demonstrated, not asserted.
OUTPUT	A comparison table (item, abstract, full text, match), then the spin summary.

PRO TIP Spin in abstracts is common. Finding it makes a memorable journal-club point.

TRY IT WITH A trial with a neutral primary outcome but an abstract emphasising a positive subgroup.

→ **NEXT #483** Detect spin in the discussion and conclusion

470 "Why was this study done?" slide

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT

USE IT For the introduction section of your journal club.

PERSONA	You are a journal-club presenter who sets the context in 2 slides.
MY CONTEXT	The paper's introduction: [PASTE] What I know about prior evidence and guidelines (with sources): [LIST] Audience: [brief]
TASK	1. Summarise the clinical problem and the knowledge gap as the authors describe it. 2. Summarise the key prior evidence the study builds on, citing the paper's references. 3. State the hypothesis the study tested. 4. Write 1–2 slides with speaker notes.
RULES	Represent the authors' rationale accurately, and mark any claim of mine that isn't from the paper or my sources as [VERIFY], because context slides are often where misquotation happens.
OUTPUT	The summary, then the slide text and notes.

PRO TIP One slide on "what we believed before this study" makes the results slide far more meaningful.

TRY IT WITH A trial testing conservative oxygen targets in mechanically ventilated patients.

→ **NEXT #471** Methods on two slides

471 Methods on two slides

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT

USE IT For the methods section of your journal-club presentation.

PERSONA	You are an experienced presenter who condenses methods without losing the critical details.
MY CONTEXT	The paper's methods section: [PASTE OR UPLOAD] Design: [design]
TASK	1. Extract the critical methods: design, setting, participants (key inclusion and exclusion criteria), intervention and comparator details, outcomes and their measurement, sample size, randomisation and blinding (for trials), and the analysis population. 2. Organise them into 2 slides, one with a flow diagram description. 3. Mark the details that matter for the appraisal, for me to emphasise.
RULES	Report the methods exactly as the paper states them, and flag missing essentials as [NOT REPORTED], because methods slides are where the audience looks for flaws.
OUTPUT	The extracted methods table, the 2-slide text, then the emphasis list.

PRO TIP Show the CONSORT or STROBE flow diagram from the paper. It answers half the methods questions.

TRY IT WITH A pragmatic cluster-randomised trial of a sepsis protocol in emergency departments.

→ **NEXT #472** Key results, clearly presented

472 Key results, clearly presented

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT

USE IT For the results section of your journal-club presentation.

PERSONA	You are a data presenter who shows results honestly and clearly.
MY CONTEXT	The paper's results section, tables and figures: [PASTE OR UPLOAD] Primary outcome: [state]
TASK	1. Present the primary outcome with the event rates in each group, the effect estimate, the CI and the p value. 2. Present the 2–3 most important secondary outcomes and the main harms. 3. Calculate absolute risk differences and NNT or NNH where possible, showing the arithmetic. 4. Suggest which figure from the paper to show, and how to annotate it. 5. Write 2–3 results slides with speaker notes.
RULES	Reproduce numbers exactly, and keep the primary outcome first even if it is non-significant, because leading with a secondary outcome is spin.
OUTPUT	A results table, the absolute measures with arithmetic, then the slides and notes.

PRO TIP Put the primary result in large font on its own slide. It's the most important slide you'll show.

TRY IT WITH A trial reporting 28-day mortality, ventilator-free days and adverse events.

→ **NEXT #475** Absolute effects: ARR, RRR, NNT and NNH

473 Explain a forest plot to the audience

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT

USE IT When your paper has a forest plot (meta-analysis or subgroups).

PERSONA	You are a statistics teacher who explains forest plots in plain language.
MY CONTEXT	A description of the forest plot, or the numbers in it: [PASTE OR UPLOAD] Outcome and effect measure: [e.g. risk ratio for mortality]
TASK	1. Explain the anatomy: study rows, squares (weight), lines (CIs), the diamond (pooled estimate) and the line of no effect. 2. Interpret this specific plot: the direction, size and precision of the pooled effect, and the consistency across studies. 3. Explain the heterogeneity statistics shown. 4. Write a 1-minute spoken explanation for the journal club.
RULES	Use only the plot's reported numbers and don't recompute the pooled estimate, because recalculations from a figure are unreliable.
OUTPUT	The anatomy explanation, the interpretation, then the spoken script.

PRO TIP Point at the diamond first, then at the studies that disagree with it.

TRY IT WITH A forest plot of 9 RCTs of corticosteroids in community-acquired pneumonia.

→ **NEXT #466** Appraise a systematic review or meta-analysis

474 Explain the survival curves

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT

USE IT When your paper reports Kaplan–Meier curves and hazard ratios.

PERSONA	You are a biostatistician who explains time-to-event results clearly.
MY CONTEXT	A description of the Kaplan–Meier figure, and the reported HR, CI, p value and numbers at risk: [PASTE]
TASK	1. Explain the axes, the censoring marks and the numbers at risk. 2. Explain when the curves separate, and what that implies clinically. 3. Explain the hazard ratio in plain language, and whether proportional hazards seem reasonable from the curves. 4. Point out the uncertainty in the tail of the curves, where few patients remain at risk. 5. Write a 1-minute spoken explanation.
RULES	Base the interpretation only on the reported figure and numbers, because survival figures are easy to over-read.
OUTPUT	The explanation points, then the script.

PRO TIP Crossing curves are a warning sign that a single hazard ratio may mislead.

TRY IT WITH Overall survival curves in an oncology trial with an HR of 0.78.

→ **NEXT #467** Decode the statistics in my paper

475 Absolute effects: ARR, RRR, NNT and NNH

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT / MBBS STUDENT

USE IT For every trial or cohort result you present.

PERSONA	You are an EBM tutor who converts relative results into absolute terms clinicians can use.
MY CONTEXT	Event counts or rates in each group, for benefits and harms: [PASTE NUMBERS FROM THE PAPER] Follow-up duration: [time]
TASK	1. Calculate the control event rate, the experimental event rate, ARR, RRR and NNT, with a CI for the NNT where possible. 2. Do the same for the main harm (ARI and NNH). 3. Interpret the benefit-harm balance in plain language. 4. Show how the NNT changes for a patient with a higher or lower baseline risk.
RULES	Show every formula and step, state the time frame of the NNT, and use only the paper's numbers, because NNT without a time frame is meaningless.
OUTPUT	A calculation table, then the plain-language interpretation.

PRO TIP "Treat 25 patients for one year to prevent one death" lands better than "a 20% relative risk reduction".

TRY IT WITH Mortality 18% vs 22% over 90 days; major bleeding 4% vs 2%.

→ **NEXT #487** The clinical bottom line

476 Can we trust these results?

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT

USE IT For the core appraisal section of your journal club.

PERSONA	You are a methodologist who judges internal validity systematically.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The paper (methods and results): [PASTE OR UPLOAD] Design: [design]
TASK	1. Assess selection bias, performance bias, detection bias, attrition bias and reporting bias. 2. For each, give a judgement (low, some concern, high) and a quote from the paper. 3. Say whether any bias could plausibly explain the entire result. 4. Summarise the overall internal validity in 3 sentences.
RULES	Quote the paper for each judgement, and mark unreported items [NOT REPORTED], because judgements without evidence won't survive faculty questioning.
OUTPUT	A bias table (bias, judgement, quote), then the summary.

PRO TIP Attrition above 20%, or unequal between groups, is a common and underrated threat.

TRY IT WITH An open-label RCT with a subjective primary outcome and 18% loss to follow-up.

→ **NEXT #477** Does this apply to our patients?

477 Does this apply to our patients?

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT / FACULTY

USE IT For the applicability section of your journal club.

PERSONA	You are a clinician-researcher who asks whether trial results transfer to real Indian practice.
MY CONTEXT	The paper's population, setting, intervention delivery and exclusions: [PASTE] Our patients and setting: [e.g. tertiary government hospital, high TB and malnutrition burden, limited ICU beds]
TASK	1. Compare the study population with our patients: age, severity, comorbidities, ethnicity and socio-economic factors. 2. Compare the setting and resources: staffing, monitoring, drug and device availability, and cost. 3. Identify exclusions that remove patients we commonly see. 4. Judge whether the effect would likely be larger, smaller or similar here. 5. Write the applicability slide.
RULES	Base the comparison on the paper's reported characteristics and my description, and mark assumptions as [ASSUMPTION], because applicability judgements are opinion and should be labelled as such.
OUTPUT	A comparison table (factor, study, our setting, impact), the judgement, then the slide.

PRO TIP The exclusion criteria often reveal who the results don't apply to. Read them carefully.

TRY IT WITH A European ICU trial applied to a busy Indian government ICU.

→ **NEXT #487** The clinical bottom line

478 Hunt for the hidden bias

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT

USE IT When the paper looks clean and you want to find what others will miss.

PERSONA	You are a sceptical peer reviewer with a gift for spotting subtle bias.
MY CONTEXT	The full paper: [PASTE OR UPLOAD]
TASK	1. Look for subtle issues: run-in periods, enrichment, early stopping for benefit, composite outcomes driven by soft components, per-protocol-only analyses, changes to outcomes or sample size, imbalance in co-interventions, and industry involvement in analysis. 2. For each issue found, quote the paper and explain its likely effect. 3. Rank the issues by how much they could change the interpretation. 4. Suggest how to present the key issue in the journal club without being unfair.
RULES	Report only issues supported by the text, and say "none found" for categories that are clean, because inventing problems is as bad as missing them.
OUTPUT	An issue table (issue, quote, likely effect, rank), then the presentation suggestion.

PRO TIP Trials stopped early for benefit often overestimate the effect. Check the stopping rule.

TRY IT WITH An industry-funded trial stopped early after an interim analysis showed benefit.

→ **NEXT #476** Can we trust these results?

479 Confounding in an observational paper

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT

USE IT When appraising a cohort or case-control study's adjustment strategy.

PERSONA	You are an epidemiologist who evaluates confounding control critically.
MY CONTEXT	Exposure, outcome and the variables adjusted for (from the methods and tables): [PASTE]
TASK	1. List the important confounders for this association from clinical reasoning. 2. Check which were measured and adjusted for, and which are missing. 3. Check for adjustment for mediators or colliders. 4. Judge whether residual confounding could explain the result, considering the effect size. 5. Suggest what a future study should measure.
RULES	Explain each confounder's link to both exposure and outcome, because a variable is only a confounder if it relates to both.
OUTPUT	A confounder table (variable, link to exposure, link to outcome, adjusted yes or no), then the judgement.

PRO TIP An E-value (if you know it) quickly shows how strong an unmeasured confounder would need to be.

TRY IT WITH A cohort linking proton-pump inhibitor use with dementia.

→ **NEXT #464** Appraise a cohort or case-control study

480 Are these the right outcomes?

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT

USE IT When appraising any trial or cohort study.

PERSONA	You are a patient-centred outcomes researcher.
MY CONTEXT	The primary and secondary outcomes, with their definitions: [PASTE] The registered outcomes, if available: [PASTE]
TASK	1. Classify each outcome as patient-important or surrogate, and hard or soft. 2. For composite outcomes, check whether the components have similar importance and frequency, and which component drives the result. 3. Compare the reported outcomes with the registered ones and note any switching. 4. Judge whether the outcomes justify the authors' conclusion.
RULES	Base conclusions on the definitions in the paper, because outcome definitions often differ from their labels.
OUTPUT	An outcome table (outcome, type, importance, registered yes or no), then the composite analysis and judgement.

PRO TIP A composite driven by "hospitalisation" rather than "death" deserves a closer look.

TRY IT WITH A heart-failure trial with a composite of cardiovascular death and heart-failure hospitalisation.

→ **NEXT #497** Check the trial registry for outcome switching

481 Appraise a non-inferiority trial

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT / FACULTY

USE IT When your paper claims a new treatment is "not worse" than the standard.

PERSONA	You are a trial statistician who specialises in non-inferiority and equivalence designs.
MY CONTEXT	The margin, its justification, the analysis populations (ITT and per-protocol) and the results with CIs: [PASTE]
TASK	1. Explain the non-inferiority logic and the margin in plain language. 2. Judge whether the margin is clinically acceptable and justified by prior evidence. 3. Check the results in both the ITT and per-protocol populations against the margin. 4. Assess whether the trial's conduct (adherence, cross-over) could bias towards non-inferiority. 5. Draw a simple CI-versus-margin diagram description for a slide.
RULES	Use the reported margin and CIs exactly, because the conclusion depends entirely on where the CI sits relative to the margin.
OUTPUT	The explanation, the margin judgement, the results check, then the diagram description.

PRO TIP Poor adherence makes treatments look similar, which favours a false non-inferiority result.

TRY IT WITH A trial showing oral antibiotics are non-inferior to IV antibiotics for bone infection.

→ **NEXT #463** Deep appraisal of a randomised controlled trial

482 Should we believe the subgroup finding?

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT

USE IT When a paper highlights a subgroup effect.

PERSONA	You are a methodologist who applies established credibility criteria to subgroup claims.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The subgroup analyses (forest plot or table) and how the authors describe them: [PASTE]
TASK	1. Check whether the subgroup was pre-specified, with a hypothesised direction. 2. Check the test for interaction, not separate p values. 3. Check the number of subgroups tested and the plausibility of the effect. 4. Check consistency with other studies. 5. Give an overall credibility judgement.
RULES	Treat subgroup findings as hypothesis-generating unless the credibility criteria are met, because most subgroup "effects" do not replicate.
OUTPUT	A credibility checklist table, then the judgement.

PRO TIP "The effect was significant in men but not in women" is not evidence of a difference. Ask for the interaction test.

TRY IT WITH A trial claiming benefit only in patients under 65 years.

→ **NEXT #483** Detect spin in the discussion and conclusion

483 Detect spin in the discussion and conclusion

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT

USE IT When the authors' conclusion seems more positive than their results.

PERSONA	You are a research-integrity researcher who studies spin in medical publications.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] The primary result: [paste] The abstract conclusion, the discussion and the conclusion: [PASTE]
TASK	1. Identify spin strategies: focusing on secondary outcomes, emphasising within-group changes, using "trend" language, claiming equivalence from a non-significant result, and extrapolating beyond the population. 2. Quote each example. 3. Rewrite the conclusion to match the primary result. 4. Suggest one slide showing the original and the rewritten conclusion side by side.
RULES	Quote the text exactly for each instance, because spin must be shown, not asserted.
OUTPUT	A spin table (strategy, quote), then the rewritten conclusion.

PRO TIP The rewritten, spin-free conclusion makes an excellent final journal-club slide.

TRY IT WITH A trial with a non-significant primary outcome whose conclusion says the drug "shows promise".

→ **NEXT #487** The clinical bottom line

484 Funding and conflicts: do they matter here?

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT

USE IT When appraising who funded the study and who controlled it.

PERSONA	You are a research-ethics specialist who assesses the impact of funding and conflicts on research.
MY CONTEXT	The funding and conflict-of-interest statements and the role of the sponsor: [PASTE]
TASK	1. Summarise the funding and the sponsor's role in design, conduct, analysis and writing. 2. Summarise the authors' declared conflicts. 3. Identify aspects of the study where sponsor influence could matter (comparator choice, dose, outcomes, early stopping). 4. Give a balanced judgement: a concern to note, not a reason to dismiss good methods.
RULES	Base the assessment only on the declarations, and avoid implying misconduct, because industry funding alone does not invalidate a study.
OUTPUT	A summary table, the areas of potential influence, then the balanced judgement.

PRO TIP Check whether the comparator was given at an optimal dose. Weak comparators are a classic issue.

TRY IT WITH A pharmaceutical-sponsored trial comparing a new drug with a low dose of the standard drug.

→ **NEXT #478** Hunt for the hidden bias

485 Compare the paper with its editorial and letters

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT / FACULTY

USE IT When the paper has an accompanying editorial or published correspondence.

PERSONA	You are an academic who synthesises expert commentary on major papers.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The editorial and any letters or author replies: [PASTE OR UPLOAD] My own appraisal points: [LIST]
TASK	1. Summarise the editorialist's main points of praise and concern. 2. Summarise the key criticisms in the letters and the authors' responses. 3. Compare them with my own appraisal: agreements and new points. 4. Write one slide summarising expert commentary.
RULES	Attribute every point to its source accurately, and quote where possible, because misattributing expert opinion is embarrassing in front of faculty.
OUTPUT	The summaries, the comparison, then the slide.

PRO TIP Letters to the editor often contain the sharpest critique of a paper, and they are free to read.

TRY IT WITH A landmark trial published with an editorial and 6 letters of correspondence.

→ **NEXT #461** Journal club presentation: appraisal to bottom line

486 How does this fit with previous evidence?

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT / FACULTY

USE IT For the "context" part of your appraisal.

PERSONA	You are an evidence synthesiser who places new studies within the existing body of evidence.
MY CONTEXT	The paper's main finding: [paste] Prior key studies or reviews (uploaded or with details): [UPLOAD / LIST] Current guideline position: [paste with source]
TASK	1. Compare the new finding with the prior evidence: consistent, conflicting or new territory. 2. Explain any differences (population, design, intervention, outcome). 3. Judge whether this study should shift the weight of evidence. 4. Write one "where this fits" slide.
RULES	Use only the studies and guidelines I provided, marking others [VERIFY], because comparisons with misremembered studies undermine the appraisal.
OUTPUT	A comparison table, the judgement, then the slide.

PRO TIP A single trial rarely overturns a body of evidence. Say so if that's the case.

TRY IT WITH A new trial finding no benefit of a therapy that earlier small trials supported.

→ **NEXT #487** The clinical bottom line

487 The clinical bottom line

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT / FACULTY

USE IT For the final slide of your journal club.

PERSONA	You are a senior clinician who turns appraisals into clear, honest practice decisions.
MY CONTEXT	My appraisal summary (validity, results, applicability): [PASTE] Current practice in our unit: [brief]
TASK	1. State whether the results are valid, important and applicable, in one line each. 2. State the bottom line in 2 sentences: change practice, consider in selected patients, or don't change practice. 3. Name the patients in whom it would or wouldn't apply. 4. Suggest what would need to happen before a local change (protocol, cost, training).
RULES	Match the strength of the recommendation to the strength of the evidence, because the bottom line is what the department will act on.
OUTPUT	The three one-liners, the bottom line, the patient groups, then the next steps.

PRO TIP End with a question to the head of department: "Should we change our protocol?". It starts the real discussion.

TRY IT WITH Appraisal of a trial of restrictive versus liberal transfusion in upper gastrointestinal bleeding.

→ **NEXT #488** Questions to drive the discussion

488 Questions to drive the discussion

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT

USE IT To prepare the discussion part of your journal club.

PERSONA	You are a journal-club moderator who knows how to start productive debates.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The paper and my appraisal summary: [PASTE] Audience: [brief]
TASK	1. Write 6 discussion questions: 2 on methodology, 2 on interpretation, and 2 on local practice. 2. For each, write the key points a good answer would contain. 3. Suggest the order to ask them in, and when to invite senior faculty. 4. Suggest a closing question that brings the discussion back to our own patients.
RULES	Make the questions open-ended, not yes or no, because closed questions end discussions.
OUTPUT	The questions with answer points, then the facilitation order.

PRO TIP Ask the most junior person first. It keeps seniors from closing the discussion early.

TRY IT WITH A trial on early versus delayed laparoscopic cholecystectomy.

→ **NEXT #489** The faculty questions I'll face

489 The faculty questions I'll face

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT

USE IT 2–3 days before your journal club.

PERSONA	You are a senior professor famous for difficult journal-club questions.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The paper: [PASTE OR UPLOAD] My slides or appraisal summary: [PASTE]
TASK	1. List 12 questions faculty are likely to ask, from basic design questions to deep methodology and practice implications. 2. For each, give a model answer grounded in the paper. 3. Mark the 3 hardest questions and suggest backup slides for them.
RULES	Ground every answer in the paper, and where the paper doesn't say, write "the paper does not report this" as the honest answer, because bluffing is obvious to faculty.
OUTPUT	A question-and-answer table, then the backup-slide suggestions.

PRO TIP "The paper doesn't report that, but I would expect..." is a perfectly good answer.

TRY IT WITH An RCT on tranexamic acid in gastrointestinal bleeding.

→ **NEXT #461** Journal club presentation: appraisal to bottom line

490 One-page appraisal handout

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT

USE IT To circulate before or after the journal club.

PERSONA	You are a medical educator who designs compact evidence summaries.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] My appraisal: [PASTE] The paper's citation: [paste]
TASK	1. Create a one-page summary: citation, PICO, design, key results with absolute measures, key strengths, key limitations, applicability and the bottom line. 2. Keep it under 300 words, laid out in boxes. 3. Add 2 questions for the reader to consider. 4. Add a one-line verdict and the date of the journal club, for the department archive.
RULES	Keep every number identical to the paper, because handouts are kept and quoted later.
OUTPUT	The handout text with layout guidance.

PRO TIP A folder of these one-pagers becomes your department's evidence library over time.

TRY IT WITH Appraisal of a trial of balanced crystalloids versus saline in critically ill adults.

→ **NEXT #487** The clinical bottom line

491 Fill the CASP checklist for my paper

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT / MBBS STUDENT

USE IT When your department requires a completed appraisal checklist.

PERSONA	You are an EBM tutor familiar with the CASP checklists for each study design.
MY CONTEXT	Study design: [RCT / cohort / case-control / diagnostic / systematic review / qualitative] The paper: [PASTE OR UPLOAD]
TASK	1. Identify the correct CASP checklist for the design. 2. Answer each checklist question with yes, no or can't tell, and a quote from the paper. 3. Summarise the overall appraisal in 3 sentences.
RULES	Base every answer on the paper's text and mark "can't tell" where it isn't reported, because the checklist is only valuable if it is honest.
OUTPUT	The checklist table (question, answer, quote), then the summary.

PRO TIP Check the current CASP checklist wording on the CASP website, as the questions are occasionally revised.

TRY IT WITH A prospective cohort on air pollution and childhood asthma exacerbations.

→ **NEXT #461** Journal club presentation: appraisal to bottom line

492 Certainty of evidence using GRADE

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT / FACULTY

USE IT When you want to rate how confident we can be in the paper's main finding.

PERSONA	You are a GRADE-trained guideline methodologist.
MY CONTEXT	The paper's design and main outcome result: [PASTE] Its main limitations from my appraisal: [LIST]
TASK	1. Start from the design's baseline certainty. 2. Assess each downgrading domain: risk of bias, inconsistency, indirectness, imprecision and publication bias. 3. Assess the upgrading domains for observational studies (large effect, dose-response, plausible confounding). 4. Give the final certainty (high, moderate, low or very low) with reasoning. 5. Write a GRADE-style sentence ("...probably reduces...").
RULES	Apply GRADE to a single study cautiously, and state that GRADE is designed for bodies of evidence, because rating one study alone overstates precision.
OUTPUT	A GRADE table (domain, judgement, reason), then the final rating and the sentence.

PRO TIP GRADE-style wording, such as "probably" or "may", is an excellent way to phrase your bottom line.

TRY IT WITH A single large RCT with some risk of bias and a wide CI.

→ **NEXT #487** The clinical bottom line

493 Appraise a qualitative study

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT / FACULTY

USE IT When your journal-club paper uses interviews or focus groups.

PERSONA	You are a qualitative health researcher familiar with COREQ and the CASP qualitative checklist.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The paper: [PASTE OR UPLOAD]
TASK	1. Assess the fit of a qualitative approach for the question. 2. Assess sampling, data collection, the researchers' reflexivity, analysis method, saturation and rigour (e.g. member checking, triangulation). 3. Assess whether the findings are supported by quotes. 4. Discuss transferability to our setting. 5. Summarise the value of the findings for practice.
RULES	Judge qualitative research by its own standards of rigour, not by sample size or statistics, because applying quantitative criteria misjudges it.
OUTPUT	Headed appraisal sections, then the summary.

PRO TIP Qualitative studies explain the "why" behind poor adherence and patient behaviour. They make excellent journal clubs.

TRY IT WITH Interviews with patients about reasons for stopping TB treatment.

→ **NEXT #461** Journal club presentation: appraisal to bottom line

494 Appraise an economic evaluation

STANDARD

2.2 JOURNAL CLUB · FACULTY / SENIOR RESIDENT

USE IT When your paper is a cost-effectiveness or cost-utility analysis.

PERSONA	You are a health economist who explains economic evaluations to clinicians.
MY CONTEXT	The paper: [PASTE OR UPLOAD] Our health system context: [e.g. Indian public sector, PMJAY]
TASK	1. Identify the type of evaluation, the perspective, the comparators and the time horizon. 2. Check the cost and outcome sources, discounting, and QALY or DALY calculations. 3. Explain the ICER and whether it is below the stated threshold. 4. Assess the sensitivity analyses and the uncertainty. 5. Judge transferability to Indian costs and willingness-to-pay thresholds.
RULES	Report the paper's own numbers and thresholds, and mark any Indian threshold as [VERIFY], because thresholds vary and are debated.
OUTPUT	Headed sections, then a transferability judgement.

PRO TIP An intervention can be cost-effective in one country and unaffordable in another. Emphasise that.

TRY IT WITH Cost-effectiveness of a new diagnostic test for TB in a high-burden setting.

→ **NEXT #477** Does this apply to our patients?

495 Appraise a prognostic or prediction-model study

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT / FACULTY

USE IT When your paper develops or validates a risk score or prediction model.

PERSONA	You are a clinical prediction-model methodologist familiar with TRIPOD and PROBAST.
MY CONTEXT	The paper: [PASTE OR UPLOAD] Intended use of the model: [brief]
TASK	1. Assess the participants, predictors, outcome definition and timing. 2. Assess the sample size and events per variable, the handling of missing data and continuous predictors, and the variable-selection method. 3. Assess performance: discrimination (C-statistic), calibration and clinical utility. 4. Check for internal and external validation. 5. Judge readiness for clinical use.
RULES	Distinguish development from validation clearly, because a model tested only on its own development data will look better than it is.
OUTPUT	A PROBAST-style table by domain, then the readiness judgement.

PRO TIP Calibration matters as much as the C-statistic, and it is often not reported. Look for it.

TRY IT WITH A new score predicting in-hospital mortality in dengue.

→ **NEXT #461** Journal club presentation: appraisal to bottom line

496 Appraise a clinical guideline (AGREE II)

STANDARD

2.2 JOURNAL CLUB · FACULTY / SENIOR RESIDENT

USE IT When the journal-club document is a clinical practice guideline.

PERSONA	You are a guideline appraiser trained in AGREE II.
MY CONTEXT	The guideline (upload or paste the methods and recommendations): [UPLOAD]
TASK	1. Appraise the AGREE II domains: scope and purpose, stakeholder involvement, rigour of development, clarity of presentation, applicability and editorial independence. 2. Give a domain score or judgement with evidence from the document. 3. Highlight the recommendations with weak evidence but strong wording. 4. Give an overall recommendation for use.
RULES	Base scores on what the document reports, and mark missing methodological details as [NOT REPORTED] , because good guidelines document their methods transparently.
OUTPUT	An AGREE II domain table, the flagged recommendations, then the overall judgement.

PRO TIP Look for how conflicts of interest among panel members were managed. It's often revealing.

TRY IT WITH A national guideline on management of hypertension.

→ **NEXT #461** Journal club presentation: appraisal to bottom line

497 Check the trial registry for outcome switching

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT

USE IT For any RCT, before you present it.

PERSONA	You are a research-transparency analyst.
MY CONTEXT	Registration number and the registered primary and secondary outcomes, sample size and dates: [PASTE FROM THE REGISTRY] What the paper reports: [PASTE]
TASK	1. Compare the registered and reported primary outcomes: definition, measure and time point. 2. Compare the secondary outcomes, and note any added or dropped. 3. Compare the planned and actual sample sizes and dates (was it registered prospectively?). 4. Check whether the paper explains any changes. 5. Judge the impact on credibility.
RULES	Report only differences that are verifiable from the two sources, because an outcome switch is a serious accusation.
OUTPUT	A comparison table (item, registered, reported, explained yes or no), then the judgement.

PRO TIP Many trials change outcomes legitimately with an explanation. Note whether one is given.

TRY IT WITH A trial registered on ClinicalTrials.gov whose paper reports a different primary time point.

→ **NEXT #480** Are these the right outcomes?

498 Read a paper properly in 30 minutes

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT / MBBS STUDENT

USE IT When you are short of time but must appraise a paper.

PERSONA	You are an efficient academic reader who teaches a structured reading method.
MY CONTEXT	The paper: [PASTE OR UPLOAD] My purpose: [journal club / clinical question / thesis]
TASK	1. Give a 30-minute reading plan: title and abstract (2 min), figures and tables (5 min), methods (10 min), results (8 min), discussion and limitations (5 min). 2. Give the 3 questions to answer at each stage. 3. Walk through the paper with me using this plan, answering those questions. 4. Summarise the key appraisal points found.
RULES	Answer the stage questions only from the paper, and note what is missing, because a fast read must still be accurate.
OUTPUT	The plan, the stage-by-stage answers, then the key points.

PRO TIP Look at the figures and tables before reading the text. They show you what the paper really found.

TRY IT WITH A 12-page cohort study on vitamin D and mortality.

→ **NEXT #461** Journal club presentation: appraisal to bottom line

499 Explain the paper to MBBS students

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT / FACULTY

USE IT When you want to use a journal-club paper to teach undergraduates.

PERSONA	You are an EBM teacher for medical students.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] The paper and my appraisal: [PASTE] Student level: [e.g. final-year MBBS] Time: [minutes]
TASK	1. Explain the study's question and why it matters in simple terms. 2. Teach 2–3 EBM concepts using this paper (e.g. randomisation, NNT, bias). 3. Create 5 questions to check understanding. 4. Write a 10-minute teaching plan.
RULES	Keep the numbers accurate and the concepts simple but correct, because students will remember what you say as fact.
OUTPUT	The explanation, the concept list with examples, the questions, then the plan.

PRO TIP One real paper teaches EBM better than any lecture on EBM.

TRY IT WITH A trial of azithromycin for COPD exacerbations, for final-year MBBS students.

→ **NEXT #500** Quiz questions from the journal club

500 Quiz questions from the journal club

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT / FACULTY

USE IT At the end of the journal club, or for department learning records.

PERSONA	You are an assessment writer who creates EBM-focused MCQs.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] The paper, my appraisal and the key teaching points: [PASTE]
TASK	1. Write 6 MCQs testing appraisal skills and results interpretation using this paper. 2. Include one question on NNT, one on bias and one on applicability. 3. Include one clinical-vignette question that asks how the result applies to a specific patient. 4. Give the answers with brief explanations, citing the relevant part of the paper.
RULES	Base every question on the paper's actual numbers and methods, because inaccurate questions teach the wrong lesson.
OUTPUT	The numbered MCQs, then the answer key.

PRO TIP Run the quiz through a live poll. It shows immediately what the group has understood.

TRY IT WITH A trial comparing two insulin regimens in hospitalised patients.

→ **NEXT #461** Journal club presentation: appraisal to bottom line

501 Turn my critique into a letter to the editor

STANDARD

2.2 JOURNAL CLUB · PG RESIDENT / FACULTY

USE IT When your appraisal found a significant issue that hasn't been raised.

PERSONA	You are an experienced correspondent who writes publishable letters to journals.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The paper's citation and publication date: [paste] My main critique point and the supporting evidence: [PASTE] The journal's letter limits and time window: [paste]
TASK	1. Check whether the issue has already been raised in published correspondence (tell me how to check). 2. Draft a concise, collegial letter within the limits. 3. Suggest a title.
RULES	Keep the tone constructive and cite only the paper and sources I provide, because letters that attack authors are rarely published.
OUTPUT	The checking steps, the title, then the letter with its word count.

PRO TIP Submit quickly. Most journals accept letters only within a few weeks of publication.

TRY IT WITH A trial whose exclusion criteria removed most patients with CKD, the majority in Indian practice.

→ **NEXT #485** Compare the paper with its editorial and letters

502 Moderating a journal club well

STANDARD

2.2 JOURNAL CLUB · FACULTY / SENIOR RESIDENT

USE IT When you are the faculty moderator or coordinator of a journal club.

PERSONA	You are a medical educator who has run a successful journal-club programme for years and trains new moderators.
MY CONTEXT	Format of our journal club (frequency, duration, presenters): [brief] The paper for this session and the presenter's level: [brief] Common problems in our sessions: [e.g. silence, seniors dominating, abstract-reading presentations]
TASK	1. Give a pre-session checklist: paper approval, the appraisal checklist, and the presenter's rehearsal. 2. Give a moderation script: opening, time-keeping, drawing out quieter participants, and managing dominant voices. 3. List 5 teaching points to weave in about appraisal methods. 4. Suggest how to close with a bottom line and an action item. 5. Suggest how to give the presenter constructive feedback afterwards.
RULES	Keep the focus on learning appraisal skills rather than catching the presenter out, because a psychologically safe session produces far better discussion.
OUTPUT	The checklist, the script, the teaching points, the closing, then the feedback structure.

PRO TIP Keep a log of each journal club's bottom line. Over a year it becomes the department's own evidence summary.

TRY IT WITH Monthly journal club for 15 residents where the discussion usually goes silent.

→ **NEXT #488** Questions to drive the discussion

QUICK PROMPTS · 2.2

503 This study's design in one line

QUICK

2.2 JOURNAL CLUB · MBBS STUDENT / PG RESIDENT

Act as an epidemiology tutor. From this methods section, [PASTE METHODS], state the study's true design in one line (e.g. "a multicentre, double-blind, placebo-controlled RCT"), and flag it if the authors' own label differs. Add one line on what this design can conclude.

TRY IT WITH A "prospective study" that analysed an existing registry.

504 Find the real primary outcome

QUICK

2.2 JOURNAL CLUB · PG RESIDENT

Act as a trial methodologist. From this paper, [PASTE METHODS AND RESULTS], identify the pre-specified primary outcome, its exact definition and time point, and whether the result for it was positive, negative or inconclusive. Answer in 4 lines, quoting the paper.

TRY IT WITH A trial reporting many outcomes in its abstract.

505 Quick ARR and NNT

QUICK

2.2 JOURNAL CLUB · MBBS STUDENT / PG RESIDENT

Act as an EBM tutor. With control event rate [CER] and experimental event rate [EER] over [TIME], calculate the ARR, RRR and NNT, showing the formulas and arithmetic step by step, and write one plain-language sentence I can say at journal club.

TRY IT WITH Mortality 24% in the control group vs 19% in the treatment group at 28 days.

506 What does this hazard ratio mean?

QUICK

2.2 JOURNAL CLUB · PG RESIDENT

Act as a biostatistician explaining to clinicians. Explain the hazard ratio [HR, 95% CI, p] for [OUTCOME] in this paper in plain language, including what the CI tells us, how an HR differs from a relative risk, and how to phrase it on a slide.

TRY IT WITH HR 0.82 (95% CI 0.70–0.96) for death from any cause.

507 Intention-to-treat or per-protocol?

QUICK

2.2 JOURNAL CLUB · PG RESIDENT

Act as a trial methodologist. Explain whether this trial's main analysis, [DESCRIBE ANALYSIS POPULATION], is intention-to-treat, modified ITT or per-protocol, why it matters for bias, and which analysis should drive the conclusion here. Keep it under 120 words. Flag anything you are unsure of so I can check it.

TRY IT WITH A trial that excluded patients who never received the study drug.

508 Was allocation really concealed?

QUICK

2.2 JOURNAL CLUB · PG RESIDENT

Act as a risk-of-bias assessor. From this description, [PASTE RANDOMISATION TEXT], judge whether allocation concealment was adequate, inadequate or unclear under RoB 2 thinking, quote the key phrase, and explain in 2 lines why it matters for the result. Flag anything you are unsure of so I can check it.

TRY IT WITH "Patients were randomised using sealed envelopes."

509 Who was blinded, and does it matter?

QUICK

2.2 JOURNAL CLUB · PG RESIDENT

Act as a trial appraiser. From this methods text, [PASTE TEXT], list who was blinded (participants, clinicians, outcome assessors, analysts), and judge whether the lack of blinding could have influenced the primary outcome [OUTCOME], given how subjective it is. Flag anything you are unsure of so I can check it.

TRY IT WITH An open-label trial with a pain-score primary outcome.

510 Is this loss to follow-up a problem?

QUICK

2.2 JOURNAL CLUB · PG RESIDENT

Act as a clinical epidemiologist. Losses to follow-up were [NUMBERS PER GROUP] out of [TOTALS]. Tell me whether this threatens validity, considering the size, the difference between groups and the event rate, and what sensitivity analysis the authors should have done.

TRY IT WITH 12% lost in the intervention arm vs 5% in the control arm.

511 The whole paper in one sentence

QUICK

2.2 JOURNAL CLUB · PG RESIDENT / MBBS STUDENT

Act as a senior academic. Summarise this paper, [PASTE ABSTRACT], in one sentence of under 35 words stating the design, population, intervention or exposure, and the main result with its key number, using associational language if the study is observational.

TRY IT WITH A cohort study of statin use and outcomes in sepsis.

512 Was this study adequately powered?

QUICK

2.2 JOURNAL CLUB · PG RESIDENT

Act as a biostatistician. Compare this trial's sample-size assumptions, [ASSUMED EVENT RATES AND EFFECT], with the observed results, [OBSERVED RATES]. Tell me whether the study was likely underpowered, and how that affects the interpretation of a non-significant result. Flag anything you are unsure of so I can check it.

TRY IT WITH Assumed control mortality 30%, observed 18%, with a non-significant result.

513 Is this outcome a surrogate?

QUICK

2.2 JOURNAL CLUB · PG RESIDENT

Act as an outcomes researcher. Tell me whether [OUTCOME] in this study is a surrogate or a patient-important outcome, whether it has been validated as a surrogate for [HARD OUTCOME], and what a clinician should conclude from a benefit on it. Keep it under 120 words.

TRY IT WITH HbA1c reduction as the primary outcome in a diabetes drug trial.

514 Would this change my practice?

QUICK

2.2 JOURNAL CLUB · PG RESIDENT / FACULTY

Act as a pragmatic senior clinician. Given this paper's result, [RESULT], quality, [APPRAISAL SUMMARY], and our setting, [SETTING], tell me in 4 lines whether this should change our practice now, in selected patients, or not yet, and why. Flag anything you are unsure of so I can check it.

TRY IT WITH A single-centre RCT showing benefit of a new sedation protocol.

515 Explain a methods term from my paper

QUICK

2.2 JOURNAL CLUB · MBBS STUDENT / PG RESIDENT

Act as an EBM tutor. Explain the term [TERM], as used in my paper [CONTEXT], in plain language in under 80 words, with an example from the paper, and say why it matters for judging the study's reliability. Flag anything you are unsure of so I can check it.

TRY IT WITH Term: "fragility index" in a trial of tranexamic acid.

516 Does the title overclaim?

QUICK

2.2 JOURNAL CLUB · PG RESIDENT

Act as a research-integrity reviewer. Compare this paper's title, [TITLE], with its design [DESIGN] and primary result [RESULT], tell me whether the title overclaims or implies causation inappropriately, and suggest an accurate alternative title. Flag anything you are unsure of so I can check it.

TRY IT WITH A cross-sectional study titled "Vitamin D deficiency causes severe COVID-19".

517 Relative versus absolute risk for the audience

QUICK

2.2 JOURNAL CLUB · MBBS STUDENT / PG RESIDENT

Act as a medical educator. Using this paper's numbers, [EVENT RATES], explain the difference between relative and absolute risk reduction to a mixed audience in 3 sentences, and show why the relative figure sounds more impressive. Flag anything you are unsure of so I can check it.

TRY IT WITH A drug that reduces stroke from 2% to 1% over 5 years.

518 Explain the key figure in three lines

QUICK

2.2 JOURNAL CLUB · PG RESIDENT

Act as a data-presentation coach. From this description of the paper's key figure, [DESCRIBE FIGURE], write 3 spoken lines explaining what the axes show, what the main pattern is, and what it means clinically, without overstating certainty. Flag anything you are unsure of so I can check it.

TRY IT WITH A figure showing cumulative incidence of infection in two groups over 90 days.

519 A sharp question to ask the presenter

QUICK

2.2 JOURNAL CLUB · PG RESIDENT / FACULTY

Act as an experienced journal-club participant. From this paper summary, [PASTE SUMMARY], suggest 3 sharp but fair questions to ask the presenter that test understanding of the methods and applicability, each under 25 words. Flag anything you are unsure of so I can check it.

TRY IT WITH Summary of a trial of early goal-directed therapy.

520 Summarise the paper for colleagues

QUICK

2.2 JOURNAL CLUB · PG RESIDENT / FACULTY

Act as a medical science communicator. Summarise this paper, [TITLE, DESIGN, KEY RESULT, APPRAISAL VERDICT], in a 4-line message for the department WhatsApp group: what was studied, what was found (with the absolute effect), the main limitation and our bottom line.

TRY IT WITH A trial of balanced crystalloids versus saline in the ICU.

2.3

Slide design & structure

Slides support the speaker; they are not the speech. One message per slide, visuals over paragraphs, and text readable from the back row.

40

prompts

4

Master

23

Standard

13

Quick

HOW TO USE THIS SECTION

Use it once your content is ready, to turn it into clear slides that are readable from the back row.

Start with the Master prompts ([#521–524](#)) for the full workflow; use the Standard prompts ([#525–547](#)) for one task at a time; reach for the Quick prompts ([#548–560](#)) when you have two minutes. Tap any title below to jump to it, and follow → **NEXT** to move to the next step.

[#521](#) Redesign my existing deck, slide by slide

[#522](#) A personal slide design system for all my talks

[#523](#) Build a deck with AI tools, safely

[#524](#) Turn data and tables into clear slide visuals

[#525](#) Rewrite slide titles as messages

[#526](#) Turn a text-heavy slide into a visual

[#527](#) Simplify a table for a slide

[#528](#) Build a flowchart slide

[#529](#) Adapt a guideline algorithm for a slide

[#530](#) Two conditions side by side

[#531](#) A timeline slide

[#532](#) Clinical image slide with annotation

[#533](#) Icons and simple graphics for concepts

[#534](#) Colours that work for everyone

[#535](#) Fonts and sizes that are readable

[#536](#) When to use animation, and when not to

[#537](#) Cut my deck to fit the time

[#538](#) Write speaker notes for my slides

[#539](#) Separate the slides from the handout

[#540](#) A design brief for building slides in Canva

[#541](#) A prompt for AI slide generators

[#542](#) Choose the right chart for a slide

[#543](#) Citing sources on slides

[#544](#) Title and opening slide design

[#545](#) Section divider slides that guide the audience

[#546](#) Slides for online teaching and webinars

[#547](#) Using images and figures legally

[#548](#) Turn a slide title into a message

[#549](#) Check a slide for clutter

[#550](#) Paragraph to one clean slide

[#551](#) A three-colour palette for my deck

[#552](#) Font sizes for my room

[#553](#) Icon ideas for my concepts

[#554](#) Alt text for my slide images

[#555](#) A caption for a figure on a slide

[#556](#) Should these bullets be a diagram?

[#557](#) Design my final slide

[#558](#) QR code slide content

[#559](#) Should this table be a chart?

[#560](#) Does my slide pass the 6 × 6 rule?

Redesign my existing deck, slide by slide

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

USE IT

When your deck is text-heavy, cluttered or inconsistent, and the talk is in a few days.

WHY

Most medical decks are written like documents: full sentences, tiny fonts, six colours and tables pasted from papers. A slide-by-slide redesign into message titles, one idea per slide and simple visuals transforms the talk, without changing a single fact.

COPY THE PROMPT BELOW

PERSONA	You are a presentation designer who specialises in medical and scientific talks. You apply the assertion-evidence approach (a message title plus visual evidence), and you are strict about readability and consistency.
MY CONTEXT	My deck (paste the slide text, or upload the .pptx or PDF): [PASTE OR UPLOAD] Talk type, audience and duration: [e.g. departmental seminar, residents, 40 minutes] Venue and screen: [e.g. seminar room projector / hall / online] Template or branding rules: [institution template or "free"] Tool I will edit in: [PowerPoint / Google Slides / Keynote / Canva]
TASK	1. Diagnose the deck as a whole: text density, font sizes, colour consistency, visual hierarchy and slide count against the time. 2. For every slide, give: the problem, a message title (a full-sentence takeaway), the redesigned content (maximum 3 bullets, or a visual), and what moves to the speaker notes. 3. Identify slides to merge, split or delete. 4. Suggest a visual for each text-heavy slide: flowchart, table-to-chart, icon grid, timeline or annotated image. 5. Propose a consistent style: 2 fonts at most, a 3-colour palette plus neutrals, fixed title positions and a grid. 6. Give the edit steps in my tool (master slides, layout reuse, alignment tools).
RULES	Keep every fact, number and reference exactly as in my deck, because the redesign must change the presentation, not the content. Mark anything that looks factually doubtful as [CHECK] rather than correcting it. Keep body text at 24 pt or more and titles at 32 pt or more for a projector.
OUTPUT	The overall diagnosis, a slide-by-slide table (slide, problem, new title, new content, notes), the merge and delete list, the style guide, then the edit steps.
TWEAK	End with the 3 slides whose redesign will make the biggest difference, and before-and-after text for each.

How to get more out of it

- › Redesign the master slide first. Every slide improves at once.
- › Read only the slide titles in order. If they tell the story, the redesign has worked.
- › Keep the old deck. If time runs out, present the 10 most important redesigned slides.
- › → NEXT #525 Rewrite slide titles as messages

A personal slide design system for all my talks

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

USE IT

When you present often and want every deck to look consistent and professional without starting from scratch.

WHY

Doctors present dozens of times a year: seminars, journal clubs, case presentations, CMEs. A simple, reusable design system (template, fonts, colours, layouts, chart style) saves hours, looks professional, and builds a recognisable teaching identity.

COPY THE PROMPT BELOW

PERSONA	You are a brand and presentation designer who builds simple design systems for clinicians and academics, focusing on readability, accessibility and speed of use.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] Tool: [PowerPoint / Google Slides / Keynote / Canva] Institution branding requirements: [logo, colours, or "none"] Types of talks I give: [e.g. seminars, journal clubs, case presentations, CMEs] Style I like, or examples I admire: [brief or "suggest"] Screen types: [projector / large display / online]
TASK	1. Define the typography: one heading font and one body font (widely available), with sizes for titles, body, captions and references. 2. Define the colour palette: a background, text, one primary accent, one secondary accent and a highlight colour for key numbers, checked for colour-blind safety and projector contrast. 3. Define 8 reusable layouts: title, section divider, message plus single visual, two-column comparison, table, flowchart, image with annotation, and take-home. 4. Define chart and table styles: minimal grid lines, direct labels, consistent decimals. 5. Define rules for icons, images and references on slides. 6. Give the step-by-step set-up of the master template in my tool.
RULES	Recommend only widely available fonts and free resources, and mark institutional requirements as [CHECK INSTITUTION TEMPLATE], because unavailable fonts break when you present on another computer. Keep contrast high enough for poor projectors.
OUTPUT	A style-guide table (element, specification), the layout descriptions, the chart rules, then the set-up steps.
TWEAK	End with a one-page checklist to run on any deck before presenting.

How to get more out of it

- › Embed fonts or use system fonts. Hospital computers rarely have your fancy fonts.
- › Save the template as a theme file so every new deck starts from it.
- › Dark text on a light background is the safest choice for bright seminar rooms.
- › → NEXT #535 Fonts and sizes that are readable

Build a deck with AI tools, safely

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

USE IT

When you want to use AI (Claude, ChatGPT, Copilot in PowerPoint, Gamma, Canva) to create a first draft of a deck quickly.

WHY

AI slide tools produce beautiful decks in minutes, with invented statistics, outdated guidelines and generic content. Used properly, with your verified content as the input and a strict structure, they save hours without putting errors in front of an audience.

COPY THE PROMPT BELOW

PERSONA	You are a medical educator and AI-tools trainer who builds presentation workflows that are fast and accurate.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] What I most want to get out of this: [one line] Topic, audience and duration: [brief] AI tool I will use: [Claude / ChatGPT / Copilot in PowerPoint / Gamma / Canva Magic Design / other] My verified content (outline, notes, key sources, numbers): [PASTE] Design preferences or template: [brief]
TASK	1. Write the exact prompt to paste into my tool to generate the deck from my content: the structure, one message per slide, speaker notes, citations kept as given, and the design constraints. 2. Add instructions telling the tool to use only my content, to mark any gaps as [ADD CONTENT], and to create no statistics or references. 3. Give a step-by-step workflow: generate, review the text, fix the design, verify facts, add references, rehearse. 4. Give a verification checklist for AI-generated slides: numbers, drug doses, guideline names and years, image rights, and citations. 5. List the tool-specific limitations and workarounds for the tool I chose, marked [MAY DIFFER BY VERSION].
RULES	Every fact in the final deck must come from my verified content, because AI slide tools confidently produce plausible but wrong medical details. Show me the slide-by-slide text for approval before any file is built.
OUTPUT	The ready-to-paste prompt in a code block, the workflow, the verification checklist, then the tool notes.
TWEAK	End with the 5 most common errors AI tools introduce into medical slides, and how to catch each one.

How to get more out of it

- › Give the AI your outline and sources, not just a topic. Topic-only prompts produce generic decks.
- › Always check drug doses and guideline years on AI-made slides. They are the most frequent errors.
- › Declare AI assistance in your teaching portfolio where your institution expects it.
- › → **NEXT #521** Redesign my existing deck, slide by slide

Turn data and tables into clear slide visuals

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

USE IT

When your talk includes results, statistics or comparisons that currently sit in dense tables.

WHY

A 12-column table on a slide is unreadable from the third row of the audience. Choosing the right chart, stripping the clutter and highlighting the one number that matters turns data into a message people remember.

COPY THE PROMPT BELOW

PERSONA	You are a data-visualisation specialist for medical talks who makes charts readable from the back of a lecture hall.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] What I most want to get out of this: [one line] The data or tables (paste): [PASTE DATA / TABLES] The message each table should convey: [list, or "suggest"] Tool: [PowerPoint / Excel / Google Sheets / Canva] Audience and room size: [brief]
TASK	1. For each table, identify the single message it supports. 2. Recommend the best visual for that message (bar, dot plot, line, forest-style plot, simple highlighted table, or a single big number), with the reason. 3. Specify the design: sorted order, direct labels instead of legends, minimal grid lines, one highlight colour and a message title. 4. Say what to remove: extra decimals, unnecessary columns, legends that could be direct labels, and 3D effects. 5. Give step-by-step build instructions in my tool. 6. Write the message title and a one-line speaker note for each visual. 7. Flag any table that should stay a table because readers need the exact values. 7. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Keep every number exactly as in my data, and keep the axes honest (bars start at zero), because distorted charts mislead the audience and damage your credibility.
OUTPUT	A table (original table, message, recommended visual, design notes), the build steps, then the titles and notes.
TWEAK	End with the one chart that best tells the main story of the talk.

How to get more out of it

- If only one number matters, show only that number, in very large font, with context.
- Sort bars by value, not alphabetically, unless the order has meaning.
- Use colour to highlight, not to decorate. Grey everything except the key bar.
- → NEXT #542 Choose the right chart for a slide

525 Rewrite slide titles as messages

STANDARD

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

USE IT When your slide titles are topic labels ("Pathophysiology", "Results").

PERSONA	You are a scientific presentation coach who teaches the assertion-evidence method.
MY CONTEXT	My slide titles and a one-line summary of each slide's content: [PASTE] Talk type and audience: [brief]
TASK	1. Rewrite each title as a full-sentence message stating the slide's takeaway (under 15 words). 2. Keep the titles consistent in tone and tense. 3. Read the new titles in order and check that they tell the story of the talk. 4. Flag slides with no clear message, which need rethinking.
RULES	Base each message title only on the content I described, without overstating it, because a message title is a claim the audience will remember.
OUTPUT	A before-and-after table, then the story check and the flags.

PRO TIP "Early antibiotics were linked to lower mortality" teaches. "Results" does not.

TRY IT WITH Titles such as "Introduction", "Epidemiology", "Diagnosis", "Treatment", "Evidence", "Conclusion".

→ **NEXT #526** Turn a text-heavy slide into a visual

526 Turn a text-heavy slide into a visual

STANDARD

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

USE IT For any slide with more than about 40 words.

PERSONA	You are an information designer for medical education.
MY CONTEXT	The slide text: [PASTE SLIDE TEXT] Purpose of the slide: [brief] Tool: [PowerPoint / Google Slides / Canva]
TASK	1. Identify the structure hidden in the text: sequence, comparison, hierarchy, cycle, cause-effect or list. 2. Recommend the matching visual (flowchart, two-column table, tree, cycle diagram, icon grid). 3. Describe the visual precisely: boxes, arrows, labels (under 5 words each) and layout. 4. Move the remaining detail to speaker notes. 5. Give the build steps in my tool (e.g. SmartArt or shapes).
RULES	Keep all the information either on the visual or in the notes, and lose nothing, because simplifying the slide must not delete teaching content.
OUTPUT	The structure diagnosis, the visual specification, the notes text, then the build steps.

PRO TIP If you can't find a structure in the text, the slide probably tries to say two things. Split it.

TRY IT WITH A 90-word slide listing the causes, features and management of hyperkalaemia.

→ **NEXT #528** Build a flowchart slide

527 Simplify a table for a slide

STANDARD

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

USE IT When you want to show a table from a paper or guideline on a slide.

PERSONA	You are a slide designer who makes tables readable from the back row.
MY CONTEXT	The original table: [PASTE TABLE] What the audience needs to take from it: [one line]
TASK	1. Keep only the rows and columns needed for the message. 2. Reduce the decimals, abbreviate consistently and add units in the headers. 3. Highlight the key cell or row. 4. Split it into two slides if necessary. 5. Write the source line for the bottom of the slide.
RULES	Don't alter any retained value, and cite the source as "Adapted from [ref]" when you modify a published table, because an altered table without attribution is a copyright and integrity problem.
OUTPUT	The simplified table, the highlight instruction, then the source line.

PRO TIP A table with more than 5 rows and 4 columns usually needs splitting or a chart.

TRY IT WITH A 10 × 8 table of antibiotic dosing across renal-function categories.

→ **NEXT #524** Turn data and tables into clear slide visuals

528 Build a flowchart slide

STANDARD

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

USE IT For processes, pathways and approaches.

PERSONA	You are a clinical pathway designer who draws clean flowcharts.
MY CONTEXT	The process or approach in text: [PASTE] Tool: [PowerPoint / Google Slides / draw.io / Canva]
TASK	1. Convert the process into steps and decision points. 2. Use standard shapes: rounded boxes for start and end, rectangles for actions, diamonds for decisions. 3. Keep the labels under 6 words, with yes and no on the decision branches. 4. Arrange it top-to-bottom or left-to-right, with no crossing arrows. 5. Give the build steps and the alignment tips.
RULES	Keep the clinical logic exactly as described, and mark any branch that seems incomplete as [CHECK BRANCH], because incomplete algorithms can mislead practice.
OUTPUT	A step and decision list, a layout description, then the build steps.

PRO TIP If the flowchart needs more than 12 boxes, split it into two slides.

TRY IT WITH Approach to a patient with a low platelet count.

→ **NEXT #529** Adapt a guideline algorithm for a slide

529 Adapt a guideline algorithm for a slide

STANDARD

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

USE IT When you want to show a published algorithm clearly.

PERSONA	You are a medical illustrator who adapts complex guideline algorithms for teaching.
MY CONTEXT	The algorithm (upload the figure or describe it): [UPLOAD / DESCRIBE] Its source (guideline and year): [cite] Audience: [brief]
TASK	1. Identify the core pathway most relevant to my audience. 2. Simplify it: keep the key decisions and collapse rare branches into a single "see guideline" box. 3. Describe the redesigned algorithm with fewer, larger boxes. 4. Add a source credit line. 5. Suggest building it up over 2–3 clicks with animation, if that helps comprehension.
RULES	Preserve the guideline's decision thresholds exactly, and mark every one as [VERIFY FROM SOURCE], because a simplified algorithm must still be correct.
OUTPUT	The simplified algorithm description, the credit line, then the animation plan.

PRO TIP Tell the audience you have simplified it, and where to find the full version.

TRY IT WITH A guideline algorithm for managing acute exacerbations of COPD.

→ **NEXT #528** Build a flowchart slide

530 Two conditions side by side

STANDARD

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / MBBS STUDENT

USE IT When teaching differences between two similar conditions, drugs or tests.

PERSONA	You are a clinical teacher who builds memorable comparison slides.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] The two items to compare: [A and B] Aspects to compare: [list, or "suggest"] Source: [cite]
TASK	1. Choose 5–7 comparison aspects that best discriminate between them. 2. Build a two-column table with short entries. 3. Highlight the 1–2 features that best distinguish them. 4. Write speaker notes with a clinical example.
RULES	Keep entries accurate to my source and mark uncertain items as [VERIFY], because comparison slides are often photographed for exam revision.
OUTPUT	The comparison table, the highlight, then the notes.

PRO TIP Put the most discriminating feature first. That's the one students will remember.

TRY IT WITH Transudate versus exudate pleural effusion.

→ **NEXT #527** Simplify a table for a slide

531 A timeline slide

STANDARD

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

USE IT For history, disease course, treatment phases or a patient journey.

PERSONA	You are a visual storyteller for medical talks.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Events with dates or time points: [PASTE] Purpose: [history / clinical course / treatment plan]
TASK	1. Select the 5–8 most important points. 2. Lay them out on a horizontal or vertical line, with labels of under 8 words. 3. Mark the key turning point visually. 4. Give the build steps.
RULES	Keep the dates as I gave them and mark historical dates as [VERIFY], because timeline dates are frequently wrong in medical slides.
OUTPUT	The selected points, the layout description, then the steps.

PRO TIP For a patient's course, use relative time (Day 0, Day 3), which also protects privacy.

TRY IT WITH The clinical course of a dengue patient from fever onset to recovery.

→ **NEXT #526** Turn a text-heavy slide into a visual

532 Clinical image slide with annotation

STANDARD

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

USE IT When showing an X-ray, CT, ECG, clinical photo or histology slide.

PERSONA	You are a radiology and pathology educator who annotates images for teaching.
MY CONTEXT	The image and its key findings: [DESCRIBE] Audience: [brief] Source or ownership, and consent: [brief]
TASK	1. Suggest the layout: a large image with a minimal title. 2. Plan the annotations: arrows or circles with short labels, revealed one click at a time. 3. Plan a "before" (unmarked) and an "after" (annotated) version. 4. Write the speaker notes, describing the findings systematically. 5. Write the credit or consent line.
RULES	Ensure the image is de-identified (names, dates, hospital numbers removed) and used with the right permission, because images are the commonest privacy breach in medical talks.
OUTPUT	The layout, the annotation plan, the notes, then the credit line.

PRO TIP Let the audience look at the unmarked image for 10 seconds before showing the arrows.

TRY IT WITH A chest X-ray showing a left-sided tension pneumothorax.

→ **NEXT #547** Using images and figures legally

533 Icons and simple graphics for concepts

STANDARD

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

USE IT When a slide explains abstract concepts that could be shown visually.

PERSONA	You are a graphic designer for health education.
MY CONTEXT	The concepts on the slide: [list] Tool: [PowerPoint (built-in icons) / Canva / Noun Project / other] Style of the deck: [brief]
TASK	1. Suggest a simple icon for each concept, with search keywords for my tool. 2. Suggest a layout (icon grid, process row, or hub and spoke). 3. Keep the icons in one consistent style and colour. 4. Pair each icon with a label of under 4 words.
RULES	Suggest only icons from sources that are free to use with attribution where needed, and mark licence checks as [CHECK LICENCE], because many online icons are not free for public use.
OUTPUT	A concept-icon table with keywords, then the layout and style notes.

PRO TIP Use icons from one family only. Mixed styles look amateur.

TRY IT WITH Concepts: risk factors for stroke (hypertension, diabetes, smoking, atrial fibrillation, dyslipidaemia).

→ **NEXT #526** Turn a text-heavy slide into a visual

534 Colours that work for everyone

STANDARD

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

USE IT When choosing a colour scheme, or checking an existing deck.

PERSONA	You are an accessibility specialist for visual presentations.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] My current colours (names or hex codes): [list] Background: [light / dark] Venue: [projector / screen / online]
TASK	1. Check the contrast between text and background, and flag failures. 2. Check for colour-blind problems (red-green pairs), and suggest alternatives. 3. Recommend a palette: text, background, primary accent, secondary accent and highlight. 4. Recommend redundant cues (labels, patterns) for charts.
RULES	Recommend combinations that remain readable on a dim projector, because seminar-room projectors wash out low-contrast colours.
OUTPUT	The contrast check, the colour-blind check, the palette table, then the chart advice.

PRO TIP Never use red versus green alone to show good versus bad. Add labels or symbols.

TRY IT WITH A deck with light-blue text on a white background and red/green charts.

→ **NEXT #535** Fonts and sizes that are readable

535 Fonts and sizes that are readable

STANDARD

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

USE IT When setting up a deck, or fixing one that's hard to read.

PERSONA	You are a typography specialist for presentations.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Current fonts and sizes: [list] Room size and screen: [brief]
TASK	1. Recommend fonts: one sans-serif for headings and body, widely available. 2. Recommend sizes: titles, body, table text, captions and references. 3. Recommend line spacing, the maximum number of lines per slide, and when to use bold. 4. Flag the problems in my current setup.
RULES	Recommend sizes that are readable from the back of my room, and system fonts or embedding, because fonts often fail on shared hospital computers.
OUTPUT	The font recommendation, a size table, the spacing rules, then the problems found.

PRO TIP Test your slide from the back of the room, or zoom out to 50% on your laptop.

TRY IT WITH A deck using a 14 pt script font for body text in a 100-seat hall.

→ **NEXT #522** A personal slide design system for all my talks

536 When to use animation, and when not to

STANDARD

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

USE IT When deciding on builds and transitions for your deck.

PERSONA	You are a presentation coach who uses animation only to support understanding.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Slides where I'm considering animation: [DESCRIBE]
TASK	1. For each slide, say whether animation helps (e.g. building an algorithm step by step, or revealing a case) or distracts. 2. Recommend simple builds only (appear or fade), in a consistent style. 3. Recommend a transition style for the whole deck. 4. Warn about technical risks (PDF export, other computers).
RULES	Recommend animation only where it controls the pace of information, because decorative animation distracts a clinical audience.
OUTPUT	A slide-by-slide recommendation table, then the style and risk notes.

PRO TIP Always keep a PDF backup. Animations disappear in PDF, so check the key slides still make sense.

TRY IT WITH An unfolding case presentation and a complex management algorithm.

→ **NEXT #537** Cut my deck to fit the time

537 Cut my deck to fit the time

STANDARD

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT

USE IT When you have far more slides than minutes.

PERSONA	You are a ruthless but fair presentation editor.
MY CONTEXT	Slide titles with a one-line content summary for each: [PASTE] Time available: [minutes] The core message and objectives: [brief]
TASK	1. Rank the slides by importance to the core message and objectives. 2. Identify slides to delete, merge or move to the backup section. 3. Produce the final slide list, fitting about one slide per minute. 4. Check that the story still flows after the cuts.
RULES	Protect the slides that deliver the objectives and the key evidence, and cut background and repetition first, because audiences forget background but remember messages.
OUTPUT	A ranking table, the cut list, then the final sequence.

PRO TIP Move cut slides to a backup section after your conclusion. They're useful for questions.

TRY IT WITH 68 slides for a 30-minute seminar on anaemia.

→ **NEXT #525** Rewrite slide titles as messages

538 Write speaker notes for my slides

STANDARD

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

USE IT When your slides are ready but you haven't planned what to say.

PERSONA	You are a speaking coach who writes concise, speakable notes.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Slide text (all slides): [PASTE] Talk duration: [minutes] My knowledge sources for the details: [LIST]
TASK	1. Write notes for each slide: the key point to say, supporting detail, and one transition line. 2. Keep each slide's notes under 80 words. 3. Mark the time checkpoints. 4. Add a "likely question" note on the key slides.
RULES	Add detail only from my listed sources, and mark anything I need to verify as [VERIFY], because notes often become what you say as fact.
OUTPUT	A slide-by-slide notes table.

PRO TIP Notes are prompts, not a script. Bullet fragments are easier to glance at than sentences.

TRY IT WITH A 25-slide deck on the management of acute stroke.

→ **NEXT #404** Full speaker script for a 45-minute seminar

539 Separate the slides from the handout

STANDARD

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

USE IT When you're tempted to put everything on the slides "so they have it".

PERSONA	You are a medical educator who designs slides for listening and handouts for reading.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] My current deck (text-heavy): [PASTE] Audience: [brief]
TASK	1. Identify the content that belongs on slides (messages and visuals) and in a handout (details, tables, references). 2. Produce the trimmed slide content. 3. Produce the handout content (1–2 pages). 4. Suggest how to share the handout (printed, PDF, QR code), and when to tell the audience about it.
RULES	Keep all the information across slides and handout combined, because the goal is to relocate detail, not lose it.
OUTPUT	The trimmed slide list, then the handout text.

PRO TIP Tell the audience at the start that a handout exists. They'll stop photographing every slide.

TRY IT WITH A 40-slide deck on antibiotic selection packed with dosing tables.

→ **NEXT #425** One-page handout for the audience

540 A design brief for building slides in Canva

STANDARD

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

USE IT When you'll build or polish the deck in Canva.

PERSONA	You are a Canva power user who designs clean medical presentations.
MY CONTEXT	Slide outline with content: [PASTE] Style preference: [e.g. minimal, dark, light] Branding: [institution colours / Parabox / none]
TASK	1. Recommend search terms for suitable Canva presentation templates. 2. Specify the brand kit set-up (fonts, colours). 3. Map each slide in my outline to a Canva layout type. 4. Give tips on using Canva elements, charts and Magic features, with fact-checking. 5. Give the export settings (PPTX or PDF) for presenting on a hospital computer.
RULES	Mark feature names that may have changed as [MAY DIFFER IN CURRENT CANVA], and remind me to replace any AI-generated text with my verified content, because Canva's AI writing tools can invent facts.
OUTPUT	The template search terms, the brand kit, a slide-to-layout map, the tips, then the export settings.

PRO TIP Export to PPTX and open it once on the presentation computer. Canva fonts sometimes don't carry over.

TRY IT WITH A 20-slide CME talk on thyroid disorders in pregnancy.

→ **NEXT #522** A personal slide design system for all my talks

541 A prompt for AI slide generators

STANDARD

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

USE IT When using Gamma, Copilot in PowerPoint, Canva Magic Design or a similar tool.

PERSONA	You are a prompt engineer for AI presentation tools.
MY CONTEXT	Topic, audience and duration: [brief] My verified content outline: [PASTE] Tool: [name]
TASK	1. Write a ready-to-paste prompt that specifies: the number of slides, one message per slide, message titles, a maximum of 3 bullets, speaker notes, my design preference, and "use only the content provided". 2. Add a line telling the tool to leave [ADD CONTENT] where information is missing instead of inventing it. 3. Suggest 3 follow-up prompts to refine the output.
RULES	Keep the prompt within typical tool input limits, and mark tool-specific syntax as [CHECK TOOL], because these tools change frequently.
OUTPUT	The main prompt in a code block, then the follow-up prompts.

PRO TIP Paste your outline, not just the topic. The quality of the output follows the quality of the input.

TRY IT WITH 15-slide seminar on management of heart failure with reduced ejection fraction.

→ **NEXT #523** Build a deck with AI tools, safely

542 Choose the right chart for a slide

STANDARD

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

USE IT When you have numbers to show and don't know which chart to use.

PERSONA	You are a data-visualisation teacher for clinicians.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The data and the comparison I want to show: [DESCRIBE] Audience: [brief]
TASK	1. Recommend the chart type and explain why. 2. Name the chart types to avoid (pie charts with many slices, 3D charts, dual axes) and why. 3. Specify the axes, labels, ordering and highlight colour. 4. Write the message title and a one-line speaker note explaining what to look at first.
RULES	Keep the representation honest (proportional areas, zero-based bars), because misleading charts undermine trust.
OUTPUT	The recommendation, the avoid list, the specification, then the title.

PRO TIP A dot plot is often clearer than a bar chart for comparing a few values.

TRY IT WITH Prevalence of five risk factors in our cohort versus national data.

→ **NEXT #524** Turn data and tables into clear slide visuals

543 Citing sources on slides

STANDARD

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

USE IT When finalising slides that use published data, figures or guidelines.

PERSONA	You are an academic presenter who cites cleanly without cluttering slides.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Sources used in my deck: [LIST] Slides that use figures or tables from papers: [list]
TASK	1. Recommend a citation style for slides (short author, journal, year at the bottom, or numbered superscripts). 2. Format the short citations for each source. 3. Write "Adapted from" credit lines for reused figures. 4. Format a references slide in Vancouver style.
RULES	Don't invent missing bibliographic details; mark them [CHECK], because incomplete references are better than fabricated ones.
OUTPUT	The citation style guide, the short citations, the credit lines, then the references slide.

PRO TIP A small source line under each evidence slide lets the audience judge the evidence immediately.

TRY IT WITH 12 sources, including 3 figures reproduced from guidelines.

→ **NEXT #547** Using images and figures legally

544 Title and opening slide design

STANDARD

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

USE IT When designing your first slide.

PERSONA	You are a presentation designer who makes strong first impressions.
MY CONTEXT	Talk title, my name, designation and institution: [brief] Event and date: [brief] Branding requirements: [brief]
TASK	1. Suggest 3 title-slide layouts: minimal text, an image background, and a split layout. 2. Recommend what to include and what to leave out. 3. Suggest a strong visual or a one-line hook. 4. Recommend a second slide (objectives or opening case).
RULES	Keep the text minimal and readable, and use only images I have rights to, because the title slide sits on screen longest.
OUTPUT	The 3 layouts, the content list, the hook, then the second-slide recommendation.

PRO TIP Put your name on the title slide once. Your institution's logo only needs to appear once, too.

TRY IT WITH Seminar on "Sepsis in 2026: what's changed?" by a second-year resident.

→ **NEXT #545** Section divider slides that guide the audience

545 Section divider slides that guide the audience

STANDARD

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

USE IT For talks longer than 20 minutes with distinct sections.

PERSONA	You are a presentation designer focused on audience navigation.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] My talk outline with its sections: [PASTE]
TASK	1. Design a consistent section-divider slide (a large section number, a title and a progress indicator). 2. Write a divider title and a one-line "what's next" for each section. 3. Suggest a simple agenda slide that the dividers echo, and how long to pause on each divider when speaking.
RULES	Keep dividers brief and consistent, because too many dividers slow the talk.
OUTPUT	The divider design, the divider texts, then the agenda slide.

PRO TIP A small progress bar or "Section 2 of 4" helps audiences stay with you in long talks.

TRY IT WITH A 50-minute seminar with 5 sections on chronic kidney disease.

→ **NEXT #522** A personal slide design system for all my talks

546 Slides for online teaching and webinars

STANDARD

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

USE IT When presenting on Zoom, Teams, Google Meet or a recorded platform.

PERSONA	You are an online medical educator.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] My deck: [PASTE OR DESCRIBE] Platform and audience devices (many on phones?): [brief]
TASK	1. Adjust the slides for small screens: larger fonts, fewer words, high contrast. 2. Plan interaction for online delivery: polls, chat questions, annotation. 3. Suggest pacing changes (more frequent visual changes). 4. Give a technical checklist: screen-sharing mode, notes on a second screen, a backup link.
RULES	Keep every key message readable on a phone screen, because many online attendees join from mobile phones.
OUTPUT	The adjustment list, the interaction plan, the pacing tips, then the checklist.

PRO TIP Change something on screen at least every 60–90 seconds to keep online attention.

TRY IT WITH A 45-minute webinar on insulin titration for general practitioners.

→ **NEXT #535** Fonts and sizes that are readable

547 Using images and figures legally

STANDARD

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

USE IT Before using images, figures or videos from the internet, papers or textbooks.

PERSONA	You are an academic copyright adviser.
MY CONTEXT	Images and figures I want to use and their sources: [LIST] Where the talk will be shown or shared: [internal teaching / conference / recorded online / published slides]
TASK	1. For each item, explain the likely permission status (open licence, publisher permission needed, fair-dealing educational use, or not allowed), marked [CHECK LICENCE]. 2. Suggest legal alternatives: open-licence sources, redrawing the figure, or using my own images. 3. Write the credit lines. 4. Explain patient-image consent requirements.
RULES	Don't give legal guarantees; recommend checking each licence, because rules differ between sources and uses.
OUTPUT	An item table (item, source, likely status, action), the alternatives, then the credit lines.

PRO TIP Redrawing a figure in your own style with "Adapted from" is usually the simplest safe option.

TRY IT WITH Figures from a textbook, a guideline PDF and images found through a search engine.

→ **NEXT #532** Clinical image slide with annotation

QUICK PROMPTS · 2.3

548 Turn a slide title into a message

QUICK

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT

Act as a presentation coach. Rewrite this topic-style slide title, [TITLE], into a full-sentence message title of under 15 words, based on the slide content [ONE-LINE CONTENT SUMMARY]. Give 3 options and mark the clearest one. Keep it accurate, without overclaiming.

TRY IT WITH Title: "Outcomes". Content: early mobilisation reduced delirium days in ventilated patients.

549 Check a slide for clutter

QUICK

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT

Act as a slide designer. Review this slide, [PASTE SLIDE TEXT AND DESCRIBE THE VISUALS], for clutter: word count, number of ideas, fonts, colours and decorative elements. Give a verdict and the 3 most useful fixes, keeping all the facts.

TRY IT WITH A slide with 8 bullets, 2 logos, a clip-art image and 3 font colours.

550 Paragraph to one clean slide

QUICK

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT

Act as an information designer. Convert this paragraph, [PASTE PARAGRAPH], into one slide: a message title, and either 3 short bullets or a described simple diagram, with the remaining detail moved into speaker notes. Keep every fact accurate. Flag anything you are unsure of so I can check it.

TRY IT WITH A paragraph on the stages of acute kidney injury and their criteria.

551 A three-colour palette for my deck

QUICK

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

Act as a presentation designer. Suggest a 3-colour palette plus neutrals (with hex codes) for a [TOPIC] talk on a [LIGHT / DARK] background, checked for projector contrast and colour-blind safety, and say which colour to use for what. Flag anything you are unsure of so I can check it.

TRY IT WITH A paediatrics seminar on a light background.

552 Font sizes for my room

QUICK

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT

Act as a typography expert for presentations. For a room of [NUMBER] people with a [PROJECTOR / LARGE SCREEN / ONLINE] set-up, give the minimum font sizes for titles, body text, table text and references, and one font pairing available on most computers.

TRY IT WITH A 60-seat seminar hall with an old projector.

553 Icon ideas for my concepts

QUICK

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT

Act as a graphic designer for health education. Suggest one simple icon for each of these concepts, [LIST CONCEPTS], with search keywords for PowerPoint or Canva icon libraries, and a one-line tip on keeping the icons in a consistent style.

TRY IT WITH Concepts: fever, cough, weight loss, night sweats, haemoptysis.

554 Alt text for my slide images

QUICK

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

Act as an accessibility specialist. Write concise alt text (under 25 words each) for these slide images, [DESCRIBE EACH IMAGE], describing what matters for learning, so that my slides work for screen readers and shared PDFs. Flag anything you are unsure of so I can check it.

TRY IT WITH A chest X-ray with consolidation, a flowchart and a bar chart of mortality.

555 A caption for a figure on a slide

QUICK

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT

Act as a scientific editor. Write a short figure caption for my slide showing [DESCRIBE FIGURE], stating what is shown, the key finding in under 15 words, and the source line "Adapted from [SOURCE]", ready to place under the figure.

TRY IT WITH A Kaplan-Meier curve from a published trial.

556 Should these bullets be a diagram?

QUICK

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT

Act as an information designer. Look at these bullets, [PASTE BULLETS], and tell me whether they hide a sequence, comparison, cycle or hierarchy. If they do, describe the diagram that should replace them, and if not, suggest how to trim them.

TRY IT WITH Bullets describing the steps of the coagulation cascade.

557 Design my final slide

QUICK

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

Act as a presentation coach. Design my final slide for a talk on [TOPIC]: 3 take-home messages of under 12 words each, a simple layout description, and my contact details or a QR code, instead of a plain "Thank you" slide. Keep it memorable and accurate.

TRY IT WITH A seminar on stroke thrombolysis in district hospitals.

558 QR code slide content

QUICK

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT / FACULTY

Act as a medical educator. Suggest what my QR code should link to after my talk on [TOPIC] (handout, references, feedback form or full slides), the one-line call to action to put beside it, and where on the slide it should sit to be scannable.

TRY IT WITH A CME talk on hypertension management for general practitioners.

559 Should this table be a chart?

QUICK

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT

Act as a data-visualisation specialist. Look at this table, [PASTE TABLE], and tell me whether a chart would communicate its main message better, which chart type, what to highlight, and which values to leave out, in under 100 words. Flag anything you are unsure of so I can check it.

TRY IT WITH A table of mortality rates across 5 age groups in two treatment arms.

560 Does my slide pass the 6 × 6 rule?

QUICK

2.3 SLIDE DESIGN & STRUCTURE · PG RESIDENT

Act as a presentation coach. Check this slide, [PASTE SLIDE TEXT], against the guideline of about 6 lines with about 6 words each. Count the lines and words, give a verdict, and rewrite it to fit while keeping the key facts.

TRY IT WITH A slide with 9 bullets of 15 words each on insulin types.

2.4

Talks & oral presentations

CMEs, invited lectures, community talks and panel discussions: every talk needs one big idea, a story to carry it, and a close the audience remembers.

35

prompts

4

Master

20

Standard

11

Quick

HOW TO USE THIS SECTION

Use it for CMEs, invited lectures, community talks and panel discussions.

Start with the Master prompts (#561–564) for the full workflow; use the Standard prompts (#565–584) for one task at a time; reach for the Quick prompts (#585–595) when you have two minutes. Tap any title below to jump to it, and follow → NEXT to move to the next step.

#561 Invited CME talk or guest lecture, start to finish

#562 Build any talk around a story

#563 Health talk for patients or the community

#564 Prepare for a panel discussion (as panellist or moderator)

#565 The first 90 seconds of my talk

#566 A closing the audience remembers

#567 A 5-minute lightning talk

#568 A PechaKucha-style talk (20 slides × 20 seconds)

#569 Motivational or career talk for students

#570 Deliver a talk in Hindi or a regional language

#571 Using patient stories ethically in talks

#572 Health talk for school students

#573 Chairing a scientific session

#574 Introduce a guest speaker

#575 A vote of thanks that isn't boring

#576 Host a medical webinar

#577 Script for a short educational video

#578 Rehearsal plan for an important talk

#579 Manage stage fright before a talk

#580 Shrink or stretch a talk to a new time slot

#581 Turn a key paper into a teaching talk

#582 Proposal and abstract for an invited talk

#583 A short speaker bio

#584 Learn from feedback on my talk

#585 An opening question that hooks the audience

#586 Organise my talk into three parts

#587 My talk's big idea in one sentence

#588 Five questions about my audience

#589 A safe icebreaker for a medical audience

#590 Tell a story in 60 seconds

#591 Recap before moving on

#592 A clear call to action

#593 A title for my invited talk

#594 Clean filler words out of my script

#595 A better way to open the Q&A

Invited CME talk or guest lecture, start to finish

2.4 TALKS & ORAL PRESENTATIONS · FACULTY / SENIOR RESIDENT

USE IT

When you are invited to speak at a CME, society meeting or another institution.

WHY

Invited talks are judged on relevance and clarity, not on how much you know. A talk built around the audience's real decisions, with current evidence, practical pearls and a memorable close, earns re-invitations. A textbook recital doesn't.

COPY THE PROMPT BELOW

PERSONA	You are a sought-after CME speaker and senior clinician who designs practical, evidence-based talks for practising doctors. You are known for clarity, good pacing and useful take-home points.
MY CONTEXT	Talk title as given by the organisers: [paste] Audience (specialty, practice setting, experience): [e.g. general practitioners and physicians in private practice] Duration and format (talk plus Q&A, panel, workshop): [brief] Organiser's brief or expectations: [paste or "none"] My sources (guidelines, key trials, reviews): [UPLOAD / LIST] Venue and technology: [hall, projector, online streaming]
TASK	1. Identify the audience's 3–5 real clinical decisions on this topic and the common mistakes they make. 2. Define the one big idea of the talk and 3 supporting messages. 3. Build the structure: an opening hook (a case or striking question), a section per message with evidence and a practical pearl, a mini-case for application, and a close. 4. Write a slide plan (about 1 slide per minute) with message titles and visuals. 5. Include an evidence summary from my sources and a "what's new" slide. 6. Write the first 90 seconds and the last 60 seconds word for word. 7. Prepare 8 likely audience questions with short answers. 8. Suggest a one-page take-home handout.
RULES	Base all recommendations on my sources and mark specifics such as doses and targets [VERIFY FROM CURRENT GUIDELINE], because practising doctors act on CME content the next day. Keep the talk free of brand promotion and declare any conflicts of interest on slide 2.
OUTPUT	The audience-decision analysis, the big idea and messages, the structure with timings, the slide plan, the scripted opening and close, the Q&A table, then the handout outline.
TWEAK	End with the single practical change the audience should make on Monday morning, phrased as one sentence.

How to get more out of it

- › Ask the organiser who attended last year and what they asked. It tells you the audience's real needs.
- › A 60-second case in the middle of the talk re-captures attention.
- › Finish 2–3 minutes early. Organisers remember speakers who respect time.
- › → NEXT #562 Build any talk around a story

Build any talk around a story

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT / FACULTY

USE IT

When your talk has good content but no shape.

WHY

Audiences remember stories, not lists. A talk that follows a simple arc (a hook, a problem, a journey through the evidence, a resolution and a call to action) holds attention and is remembered long after the slides are forgotten.

COPY THE PROMPT BELOW

PERSONA	You are a medical storyteller and speaking coach who helps clinicians turn information into memorable narratives, without sacrificing accuracy.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] What I most want to get out of this: [one line] Topic and audience: [brief] Duration: [minutes] My key content points (rough list): [PASTE] A patient story or clinical moment that inspired this talk (de-identified): [PASTE OR "none"] The one thing I want the audience to do or believe afterwards: [one line]
TASK	1. Write the one big idea in a single sentence. 2. Build the arc: the hook (first 60 seconds), the problem (why it matters), the journey (3 evidence-backed messages, each with an example), the turning point (a surprising fact or case twist), the resolution (what to do), and the call to action. 3. Map my content points onto the arc, and flag points that don't serve the story. 4. Place the patient story strategically, opening with it and returning to it at the end. 5. Suggest a recurring image, question or phrase that ties the talk together. 6. Write the transitions between the sections. 7. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Keep patient stories fully de-identified and respectful, and keep every fact accurate, because a story that distorts the evidence damages credibility. Cut content that doesn't serve the big idea, however interesting it is.
OUTPUT	The big idea, the arc table (section, content, minutes, story element), the cut list, then the transitions.
TWEAK	End with 3 alternative opening lines and 3 alternative closing lines to choose from.

How to get more out of it

- › If you can't state your big idea in one sentence, the audience won't be able to either.
- › Return to the opening story in your final minute. It closes the loop.
- › Rehearse the transitions. They are where talks lose their thread.
- › → NEXT #565 The first 90 seconds of my talk

Health talk for patients or the community

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT / FACULTY / MBBS INTERN

USE IT

For health camps, patient-education sessions, school or community talks, radio or local TV.

WHY

Talks to the public fail when doctors use jargon, lecture for too long or scare people without giving them clear actions. A community talk in simple language, with local examples, myths addressed respectfully and 3 clear actions, can change behaviour.

COPY THE PROMPT BELOW

PERSONA	You are a public-health physician and experienced community educator who speaks to rural and urban Indian audiences in simple language, with warmth and respect.
MY CONTEXT	Topic: [e.g. recognising stroke early, TB treatment adherence, diabetes foot care] Audience: [e.g. rural women's self-help group / factory workers / school parents] Language: [Hindi / Marathi / Tamil / English / mixed] Duration and setting: [e.g. 20 minutes, open-air health camp, no projector] Local beliefs or myths I know of: [list or "suggest common ones"] Resources to hand out: [pamphlets / none]
TASK	1. Choose the 3 key messages the audience must remember. 2. Write the talk in simple words in my chosen language, with local examples and analogies. 3. Address 3–5 common myths respectfully, with a correct and simple explanation for each. 4. Include a short demonstration or activity (e.g. FAST for stroke, a foot check). 5. Give 3 clear actions: what to do, where to go, and when to seek help. 6. Plan audience interaction: questions in their words, and repeating key messages aloud together. 7. Write a pamphlet summary of 100 words or fewer.
RULES	Avoid medical jargon, fear-based messaging and blaming language, and keep advice consistent with national programme guidance, marking specifics as [VERIFY WITH PROGRAMME GUIDELINES], because trust and accuracy matter most with public audiences. Give local helpline or facility details as placeholders only.
OUTPUT	The 3 messages, the talk script in my language, the myth-fact table, the activity, the actions, then the pamphlet text.
TWEAK	End with the 5 questions the audience is most likely to ask, with simple answers.

How to get more out of it

- Speak for no more than 15 minutes and leave the rest for questions. Community audiences learn most by asking.
- Use objects and demonstrations. A glucometer or a real inhaler teaches more than a slide.
- Ask a local health worker (ASHA or ANM) to co-present. It builds trust.
- → **NEXT #570** Deliver a talk in Hindi or a regional language

Prepare for a panel discussion (as panellist or moderator)

2.4 TALKS & ORAL PRESENTATIONS · FACULTY / SENIOR RESIDENT

USE IT

When you are on a panel, or moderating one, at a conference or CME.

WHY

Panels are either the liveliest session of a meeting or the dulllest. Good panellists prepare crisp positions and evidence, and listen; good moderators prepare questions that create real exchange, keep time and bring in the audience.

COPY THE PROMPT BELOW

PERSONA	You are an experienced conference moderator and panellist who has run lively, balanced medical panel discussions for years.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] What I most want to get out of this: [one line] Panel topic: [paste] My role: [panellist / moderator] Other panellists and their expertise: [list] Duration and format: [e.g. 45 minutes, 4 panellists, audience Q&A] My position or expertise on the topic: [brief] Sources: [UPLOAD / LIST]
TASK	If I am a panellist: 1. Define my 3 key positions, each with its best supporting evidence and a practical example. 2. Prepare a 90-second opening statement. 3. Anticipate the other panellists' views and prepare respectful responses. 4. Prepare answers to 6 likely audience questions. If I am the moderator: 5. Write the introduction of the topic and the panellists (60 seconds). 6. Write 8–10 questions that create contrast between the panellists and are directed at specific people. 7. Plan the time: openings, discussion, audience Q&A and a summary. 8. Prepare how to handle a dominant panellist, an off-topic answer, a heated disagreement and silence. 9. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Keep claims evidence-based and attribute positions fairly, marking evidence to confirm as [VERIFY], because panels are recorded and quoted. Declare any conflicts of interest.
OUTPUT	The section for my role: positions or questions, the scripts, the timing plan, then the handling strategies.
TWEAK	End with a 60-second closing summary template that captures agreement and disagreement.

How to get more out of it

- › As a panellist, answer in under 90 seconds. Short answers get follow-up questions; long ones get interrupted.
- › As a moderator, speak to the panellists before the session. It prevents surprises.
- › Invite audience questions early. Panels come alive when the room joins in.
- › → **NEXT #573** Chairing a scientific session

565 The first 90 seconds of my talk

STANDARD

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT / FACULTY

USE IT For any talk, when your current opening is "Good morning, today I will talk about..."

PERSONA	You are a speaking coach who specialises in strong openings for medical talks.
MY CONTEXT	Topic, audience and big idea: [brief] Material I could use (a case, a statistic with its source, a question, an image): [list] Duration of the whole talk: [minutes]
TASK	1. Write 3 alternative openings: a case story, a striking fact, and a direct question to the audience. 2. For each, include the transition into the talk's roadmap. 3. Recommend one for my audience, with the reason. 4. Mark the words to deliver slowly and the pauses.
RULES	Use only facts I supplied, with [VERIFY] for any number, and keep cases de-identified, because the opening sets your credibility.
OUTPUT	The 3 scripted openings (about 90 seconds each), the recommendation, then the delivery marks.

PRO TIP Memorise your opening word for word. Nerves are highest in the first minute.

TRY IT WITH A CME talk on deprescribing in elderly patients.

→ **NEXT #566** A closing the audience remembers

566 A closing the audience remembers

STANDARD

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT / FACULTY

USE IT For the last minute of any talk.

PERSONA	You are a speaking coach known for powerful, concise endings.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The big idea and 3 key messages: [paste] The opening I used: [paste] Audience: [brief]
TASK	1. Write a 60-second closing: a callback to the opening, the 3 messages in one sentence each, and a clear call to action. 2. Write a final one-line sentence to end on (not "thank you, any questions?"). 3. Suggest what should be on screen during the closing.
RULES	Introduce no new information in the closing, because new content at the end confuses the audience.
OUTPUT	The scripted closing, the final line, then the screen suggestion.

PRO TIP End on your strongest sentence, pause, then say "thank you". The pause is where it lands.

TRY IT WITH A talk on early recognition of sepsis in the wards.

→ **NEXT #562** Build any talk around a story

567 A 5-minute lightning talk

STANDARD

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT / FACULTY

USE IT For short-format sessions: lightning talks, flash presentations and 5-minute updates.

PERSONA	You are a coach for short-format scientific talks.
MY CONTEXT	Topic and my key message: [brief] Supporting content: [PASTE] Time: [e.g. 5 minutes] · slide limit: [e.g. 5 slides]
TASK	1. Reduce the content to one message and 2 supporting points. 2. Plan the slides (1 per minute), each with a message title and one visual. 3. Write a script of about 650 words for 5 minutes. 4. Mark where to cut if the time is shortened.
RULES	Keep one message only, and cut everything else, because a lightning talk with three messages delivers none.
OUTPUT	The message, the slide plan, the script with timing marks, then the cut points.

PRO TIP Rehearse with a strict timer. Lightning sessions often cut the microphone at the time limit.

TRY IT WITH A 5-minute update on a new antimicrobial stewardship policy for residents.

→ **NEXT #565** The first 90 seconds of my talk

568 A PechaKucha-style talk (20 slides × 20 seconds)

STANDARD

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT / FACULTY

USE IT When a session uses auto-advancing slides.

PERSONA	You are an expert in constrained presentation formats.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Topic and message: [brief] Content I want to include: [PASTE]
TASK	1. Plan 20 slides, each with one striking image or visual and at most 5 words. 2. Write a 20-second script (about 45 words) for each slide. 3. Group the slides into 4–5 story sections. 4. Mark slides where timing is risky, and give simpler scripts for them.
RULES	Keep each slide's script within 20 seconds, because slides advance automatically.
OUTPUT	A 20-row table (slide, visual, words on slide, script), then the risk notes.

PRO TIP Rehearse with the slides on auto-advance. It's a completely different experience from clicking.

TRY IT WITH "The life of an ICU nurse during a pandemic".

→ **NEXT #567** A 5-minute lightning talk

569 Motivational or career talk for students

STANDARD

2.4 TALKS & ORAL PRESENTATIONS · FACULTY / SENIOR RESIDENT

USE IT When invited to speak to MBBS students or interns about careers, exams or life in medicine.

PERSONA	You are a mentor who gives honest, practical and inspiring talks to medical students.
MY CONTEXT	Audience and what they're worried about: [e.g. final-year MBBS, NEET-PG pressure] My own journey and lessons (brief): [PASTE] Duration: [minutes]
TASK	1. Identify 3 messages that would genuinely help this audience. 2. Structure the talk with 2–3 personal stories that illustrate them. 3. Include practical advice (specific actions, resources and habits). 4. Address one uncomfortable truth honestly (e.g. burnout, failure, uncertainty). 5. Plan a Q&A segment with likely questions.
RULES	Keep advice honest and balanced, avoid survivorship bias ("just work hard"), and mention mental-health support resources, because students need realism as well as inspiration.
OUTPUT	The messages, the structure with stories, the advice list, the uncomfortable-truth segment, then the Q&A.

PRO TIP One honest failure story is worth ten success stories.

TRY IT WITH A talk for final-year MBBS students on choosing a specialty and preparing for NEET-PG.

→ **NEXT #566** A closing the audience remembers

570 Deliver a talk in Hindi or a regional language

STANDARD

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT / FACULTY

USE IT When your audience is more comfortable in Hindi or a regional language than in English.

PERSONA	You are a bilingual medical educator who explains clinical topics fluently in Indian languages.
MY CONTEXT	Topic and audience: [brief] Language: [Hindi / Marathi / Bengali / Tamil / other] My English content or script: [PASTE]
TASK	1. Translate the key messages into natural spoken language, not literal translation. 2. Decide which medical terms to keep in English (commonly understood) and which to translate, with simple explanations. 3. Suggest local analogies and examples. 4. Write the slide text in the language, or bilingually, with a readable script.
RULES	Keep the medical meaning exact and have a native speaker review the final text, because mistranslated health messages cause real harm.
OUTPUT	The spoken script, a term table (English, local term, explanation), then the slide text.

PRO TIP Mixed "Hinglish" is often the most natural register for urban audiences. Match how the audience actually speaks.

TRY IT WITH A talk on diabetes and diet for a Hindi-speaking community audience.

→ **NEXT #563** Health talk for patients or the community

571 Using patient stories ethically in talks

STANDARD

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT / FACULTY

USE IT When you want to include real patient experiences in a talk.

PERSONA	You are a clinician-educator who uses patient stories powerfully and ethically.
MY CONTEXT	The story I want to use (brief, de-identified): [PASTE] Purpose in the talk: [brief] Audience and whether the talk will be recorded: [brief]
TASK	1. Check the story for identifiable details and suggest changes (age band, location, occupation, dates). 2. Advise whether consent is needed for this use, marked [CHECK INSTITUTIONAL POLICY]. 3. Rewrite the story to be concise (under 150 words), respectful and focused on the teaching point. 4. Suggest how to introduce it ("details have been changed to protect privacy").
RULES	Keep the patient's dignity central and never use stories for shock value, because audiences and patients deserve respect.
OUTPUT	The de-identification changes, the consent note, the rewritten story, then the introduction line.

PRO TIP Composite cases (merging details from several patients) are a safe option for teaching.

TRY IT WITH A young patient's story of delayed TB diagnosis, for a public-health talk.

→ **NEXT #562** Build any talk around a story

572 Health talk for school students

STANDARD

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT / MBBS INTERN / FACULTY

USE IT For school health programmes and awareness events.

PERSONA	You are a paediatric health educator who engages children and teenagers.
MY CONTEXT	Topic: [e.g. hand hygiene, tobacco, menstrual health, road safety, mental health] Age group and school type: [brief] Language and duration: [brief]
TASK	1. Choose 3 age-appropriate key messages. 2. Design the talk as an interactive session: a question, an activity, a story and a quiz. 3. Use simple words and relatable examples. 4. Include a take-home activity or pledge. 5. Plan how to handle sensitive questions.
RULES	Keep the content age-appropriate and evidence-based, and coordinate sensitive topics with teachers, because schools and parents expect careful handling.
OUTPUT	The messages, the session plan with timings, the activity, then the guidance for sensitive questions.

PRO TIP Children remember what they do, not what they hear. Build in an activity every 5 minutes.

TRY IT WITH Tobacco and vaping awareness for 13–15-year-olds, in English and Hindi.

→ **NEXT #563** Health talk for patients or the community

573 Chairing a scientific session

STANDARD

2.4 TALKS & ORAL PRESENTATIONS · FACULTY / SENIOR RESIDENT

USE IT When you are invited to chair a conference or CME session.

PERSONA	You are an experienced session chair who runs sessions smoothly and fairly.
MY CONTEXT	The session format, speakers, their titles and time slots: [PASTE] Co-chair, if any: [name] Technical set-up: [brief]
TASK	1. Write short introductions for each speaker (name, designation, one line of relevance). 2. Plan time-keeping: signals and how to stop an overrunning speaker politely. 3. Prepare 1–2 questions per talk in case the audience is silent. 4. Plan the Q&A: how to take and distribute questions fairly. 5. Write a 60-second closing summary template.
RULES	Keep the introductions brief and accurate (check them with the speakers), and keep the session on time, because chairs are judged on time and fairness.
OUTPUT	The introductions, the time-keeping plan, the backup questions, the Q&A plan, then the closing template.

PRO TIP Meet the speakers 10 minutes before the session and agree the time signals.

TRY IT WITH A session with 4 speakers on critical care nephrology, 15 minutes each.

→ **NEXT #574** Introduce a guest speaker

574 Introduce a guest speaker

STANDARD

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT / FACULTY

USE IT When you have to introduce a speaker at a CME, conference or departmental event.

PERSONA	You are an event host who writes warm, concise speaker introductions.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The speaker's CV or bio (paste): [PASTE] Their talk title: [paste] Event and audience: [brief]
TASK	1. Write a 45–60 second introduction: the speaker's name, current role, 2–3 relevant achievements, and why they are the right person for this topic. 2. End with the talk title and an invitation to the speaker. 3. Mark names and terms to check pronunciation.
RULES	Use only facts from the bio and confirm them with the speaker, because inaccurate introductions embarrass both of you.
OUTPUT	The introduction script with pronunciation notes.

PRO TIP Don't read the whole CV. Choose what connects the speaker to the topic.

TRY IT WITH Introducing a visiting professor speaking on interventional pulmonology.

→ **NEXT #573** Chairing a scientific session

575 A vote of thanks that isn't boring

STANDARD

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT / FACULTY

USE IT At the end of a CME, conference or departmental event.

PERSONA	You are an event host known for gracious, crisp votes of thanks.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] The event, speakers, chief guests, organisers, sponsors (if any) and volunteers: [LIST] Duration allowed: [e.g. 2 minutes]
TASK	1. Write a 90–120 second vote of thanks in a logical order (chief guest, speakers, organisers, volunteers, participants). 2. Include one specific line about each speaker's talk. 3. End warmly and briefly.
RULES	Include everyone who should be thanked, and double-check names and titles, because omissions are remembered longer than the speech.
OUTPUT	The script with names marked [CHECK SPELLING / TITLE].

PRO TIP Mentioning one specific point from each talk shows that you actually listened.

TRY IT WITH The vote of thanks at a state-level CME on diabetes with 5 speakers.

→ **NEXT #574** Introduce a guest speaker

576 Host a medical webinar

STANDARD

2.4 TALKS & ORAL PRESENTATIONS · FACULTY / SENIOR RESIDENT

USE IT When organising or hosting an online CME or teaching webinar.

PERSONA	You are an experienced online event host for medical education.
MY CONTEXT	Platform: [Zoom / Teams / YouTube Live / other] Topic, speakers and audience size: [brief] Duration: [minutes]
TASK	1. Give a pre-event checklist: tech rehearsal, roles (host, co-host, chat moderator), registration and reminders. 2. Write the run-of-show with timings: welcome, housekeeping, speaker introductions, talks, Q&A and close. 3. Write the host scripts for the welcome and housekeeping (mute, chat, polls, recording consent). 4. Plan engagement: polls and chat prompts. 5. Give a backup plan for technical failures.
RULES	Announce the recording and any certificate criteria clearly, and mark platform-specific steps [MAY DIFFER BY VERSION], because attendees need clarity and platforms change.
OUTPUT	The checklist, the run-of-show table, the scripts, the engagement plan, then the backup plan.

PRO TIP Assign a co-host to watch the chat. The host can't present and moderate at the same time.

TRY IT WITH A 60-minute webinar on insulin pumps with 2 speakers and 300 attendees.

→ **NEXT #546** Slides for online teaching and webinars

577 Script for a short educational video

STANDARD

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT / FACULTY

USE IT For YouTube, Instagram or institutional e-learning videos.

PERSONA	You are a medical educator who creates short, engaging educational videos.
MY CONTEXT	Topic and audience (students / doctors / public): [brief] Video length and platform: [e.g. 3 minutes, YouTube] Key content: [PASTE]
TASK	1. Write a hook in the first 5 seconds. 2. Structure the video: the hook, 3 key points and a recap with a call to action. 3. Write the spoken script at about 150 words per minute. 4. Suggest a visual or on-screen text for each segment. 5. Add a short disclaimer line appropriate to the audience.
RULES	Keep the medical content accurate and avoid sensational claims, marking specifics [VERIFY], because medical videos are shared widely and quoted as fact.
OUTPUT	The script with timestamps, a visual plan table, then the disclaimer.

PRO TIP Record in short segments. Editing is easier and mistakes cost seconds, not minutes.

TRY IT WITH A 3-minute YouTube video on how to use an inhaler correctly, for patients.

→ **NEXT #571** Using patient stories ethically in talks

578 Rehearsal plan for an important talk

STANDARD

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT / FACULTY

USE IT 5–7 days before a significant talk.

PERSONA	You are a speaking coach who prepares clinicians for high-stakes talks.
MY CONTEXT	The talk, its duration and its importance: [brief] Days available: [N] My main concerns (timing, nerves, questions, content gaps): [list]
TASK	1. Build a day-by-day rehearsal plan: a content run, a timed run, a recorded run, a run in front of colleagues, and a final light run. 2. Build in specific work on my concerns. 3. Give a checklist for the day before and the day of the talk. 4. Suggest how to collect useful feedback from colleagues.
RULES	Include at least 3 full timed rehearsals aloud, because silent practice consistently underestimates the time needed.
OUTPUT	The day-by-day plan, the concern-specific exercises, the checklists, then the feedback questions.

PRO TIP Rehearse standing up, with the clicker, as you will on the day.

TRY IT WITH A 20-minute invited talk at a national conference, 6 days away.

→ **NEXT #579** Manage stage fright before a talk

579 Manage stage fright before a talk

STANDARD

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT / MBBS STUDENT

USE IT When nerves are affecting your preparation or delivery.

PERSONA	You are a performance coach who helps doctors manage public-speaking anxiety.
MY CONTEXT	What happens to me when I speak (shaking, blanking, rushing): [brief] The upcoming talk: [brief]
TASK	1. Explain briefly why anxiety happens and why some of it helps. 2. Give preparation strategies: over-preparing the opening, rehearsal and visualisation. 3. Give strategies for the day: breathing techniques, arriving early, and speaking to audience members beforehand. 4. Give in-the-moment recovery techniques for blanking or rushing. 5. Give a debrief routine after the talk.
RULES	Keep the advice practical and evidence-informed, and suggest professional support if the anxiety is severe or persistent, because stage fright can overlap with anxiety disorders.
OUTPUT	Five headed sections of practical techniques.

PRO TIP If you blank, pause, look at your slide title and say it aloud. It restarts your memory.

TRY IT WITH A first-year resident who freezes during departmental presentations.

→ **NEXT #578** Rehearsal plan for an important talk

580 Shrink or stretch a talk to a new time slot

STANDARD

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT / FACULTY

USE IT When the organisers change your slot, or you reuse a talk for a different event.

PERSONA	You are a presentation editor who adapts talks to any time slot.
MY CONTEXT	My current talk outline with its timings: [PASTE] Original and new durations: [e.g. 45 → 20 minutes] New audience, if different: [brief]
TASK	1. Identify the core that must stay (the big idea, key messages and key evidence). 2. For a shorter slot: specify the sections to cut, merge or summarise, with minutes saved. 3. For a longer slot: suggest what to add (a case, interaction, Q&A, depth). 4. Produce the revised outline with timings.
RULES	Protect the big idea and the closing, because shortened talks often lose their conclusion first.
OUTPUT	The core list, the changes with minutes, then the revised outline.

PRO TIP Cut whole sections rather than trimming every section. It keeps the pace natural.

TRY IT WITH Reducing a 45-minute seminar on heart failure to a 15-minute CME talk.

→ **NEXT #562** Build any talk around a story

581 Turn a key paper into a teaching talk

STANDARD

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT / FACULTY

USE IT When you want to teach the lessons of an important paper, not appraise it formally.

PERSONA	You are a clinical teacher who makes landmark studies come alive.
MY CONTEXT	The paper (upload or paste the key sections): [UPLOAD / PASTE] Audience and duration: [brief]
TASK	1. Tell the story of the study: the question, why it mattered, what they did, what they found and what changed. 2. Extract 3 teaching lessons (clinical and methodological). 3. Build a 10–15 minute talk outline with visuals. 4. Link it to current practice and guidelines, with [VERIFY].
RULES	Keep the study details accurate to the paper, because audiences may look it up afterwards.
OUTPUT	The story summary, the lessons, the outline, then the practice link.

PRO TIP Start with what doctors did before the study. It makes the result feel important.

TRY IT WITH A landmark trial on low tidal volume ventilation in ARDS.

→ **NEXT #562** Build any talk around a story

582 Proposal and abstract for an invited talk

STANDARD

2.4 TALKS & ORAL PRESENTATIONS · FACULTY / SENIOR RESIDENT

USE IT When organisers ask for a talk abstract, title and learning objectives.

PERSONA	You are an experienced speaker who writes clear, attractive talk proposals.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Proposed topic and my angle: [brief] Audience: [brief] Required format (word limit, objectives, bio): [paste]
TASK	1. Write 3 talk titles. 2. Write a talk abstract within the limit: the problem, what the talk covers, and what attendees will take away. 3. Write 3 learning objectives with Bloom's verbs. 4. Suggest the format (lecture, case-based, interactive), and the audience level it suits best.
RULES	Promise only what the talk will deliver, because organisers remember over-promising speakers.
OUTPUT	The titles, the abstract, the objectives, then the format note.

PRO TIP Organisers choose talks that solve their audience's problems. Lead with the problem.

TRY IT WITH A proposal for a talk on AI tools in medical practice for a state medical association.

→ **NEXT #583** A short speaker bio

583 A short speaker bio

STANDARD

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT / FACULTY

USE IT When an event, journal or website asks for a speaker bio.

PERSONA	You are a professional bio writer for academic clinicians.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] My CV highlights (qualifications, current role, experience, interests, key achievements, social handles): [PASTE] Word limits needed: [e.g. 50 and 150 words] Tone: [formal / conversational]
TASK	1. Write a 50-word bio and a 150-word bio in the third person. 2. Lead with my current role and the relevance to the event. 3. Include 2–3 achievements, without listing everything. 4. Write a one-line version for the host to read aloud.
RULES	Use only facts from my CV, without inflation, because inaccurate bios are easily checked online.
OUTPUT	The three versions.

PRO TIP Keep one bio document updated. You'll be asked for it far more often than you expect.

TRY IT WITH A pulmonologist and co-founder of an AI health-education platform.

→ **NEXT #574** Introduce a guest speaker

584 Learn from feedback on my talk

STANDARD

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT / FACULTY

USE IT After a talk, when you have feedback forms, comments or a recording.

PERSONA	You are a speaking coach who turns feedback into specific improvements.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Feedback received (ratings and comments): [PASTE] My own impressions and the recording notes: [brief]
TASK	1. Summarise the feedback themes: content, clarity, delivery, slides and time. 2. Identify 3 strengths to keep. 3. Identify 3 specific improvements, with an exercise for each. 4. Suggest what to change in the talk before giving it again.
RULES	Weigh repeated comments over one-off comments, and don't dismiss critical feedback, because patterns reveal real issues.
OUTPUT	The theme summary, the strengths, the improvements with exercises, then the talk changes.

PRO TIP Watch the recording once with the sound off. Your body language will surprise you.

TRY IT WITH Feedback from a CME with comments on "too fast" and "excellent cases".

→ **NEXT #578** Rehearsal plan for an important talk

585 An opening question that hooks the audience

QUICK

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT / FACULTY

Act as a speaking coach. Write 3 opening questions for my talk on [TOPIC] to [AUDIENCE] that make people think about their own practice or experience in the first 30 seconds, each under 20 words, with the answer I'll reveal later in the talk.

TRY IT WITH A CME talk on antibiotic overuse in upper respiratory infections.

586 Organise my talk into three parts

QUICK

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT

Act as a presentation strategist. Organise my content, [PASTE KEY POINTS], into exactly 3 memorable parts with short, parallel titles, and write a one-sentence roadmap that previews the 3 parts at the start of the talk. Flag anything you are unsure of so I can check it.

TRY IT WITH Points on the recognition, treatment and prevention of heat stroke.

587 My talk's big idea in one sentence

QUICK

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT / FACULTY

Act as a speaking coach. From my talk content, [PASTE OUTLINE], write the single big idea in one sentence of under 20 words that the audience should remember a week later. Give 3 options and mark the strongest one. Flag anything you are unsure of so I can check it.

TRY IT WITH A talk on the importance of early mobilisation in hospitalised elderly patients.

588 Five questions about my audience

QUICK

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT / FACULTY

Act as a presentation strategist. Before I prepare my talk on [TOPIC] for [AUDIENCE], list the 5 questions I should answer about them (knowledge level, decisions, pain points, expectations, attention span) and suggest how to find the answers. Flag anything you are unsure of so I can check it.

TRY IT WITH A talk for general practitioners in a district town on thyroid disorders.

589 A safe icebreaker for a medical audience

QUICK

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT / FACULTY

Act as an experienced medical speaker. Suggest 3 light, respectful icebreakers for the start of my talk on [TOPIC] to [AUDIENCE] that relate to the topic, avoid jokes about patients, and take under 30 seconds each. Flag anything you are unsure of so I can check it.

TRY IT WITH An after-lunch CME talk on hypertension management.

590 Tell a story in 60 seconds

QUICK

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT / FACULTY

Act as a medical storyteller. Turn this de-identified clinical experience, [BRIEF CASE], into a 60-second spoken story (about 150 words) with a setting, a tension, a turning point and a lesson that leads into my talk on [TOPIC]. Flag anything you are unsure of so I can check it.

TRY IT WITH A patient whose sepsis was recognised late on a busy night shift.

591 Recap before moving on

QUICK

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT

Act as a speaking coach. Write one-sentence spoken recaps to use at the end of each section of my talk, [LIST SECTIONS AND KEY POINTS], so the audience can follow the structure, each under 20 words and linking to the next section.

TRY IT WITH Sections on the causes, diagnosis and management of iron-deficiency anaemia.

592 A clear call to action

QUICK

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT / FACULTY

Act as a public-health communicator. Write 3 specific calls to action to end my talk on [TOPIC] for [AUDIENCE], each an action they can take within a week, phrased positively and concretely, in under 20 words. Flag anything you are unsure of so I can check it.

TRY IT WITH A talk to ward nurses on hand hygiene compliance.

593 A title for my invited talk

QUICK

2.4 TALKS & ORAL PRESENTATIONS · FACULTY / SENIOR RESIDENT

Act as a conference programme editor. Write 5 titles for my invited talk on [TOPIC] for [AUDIENCE]: 2 descriptive, 2 with a hook, and 1 in question form, each under 12 words, accurate to the content, and attractive in a printed programme.

TRY IT WITH A talk on how AI tools can help practising doctors.

594 Clean filler words out of my script

QUICK

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT

Act as a speaking coach. Review this transcript of my rehearsal, [PASTE TRANSCRIPT], list my filler words and verbal habits with their counts, and suggest a replacement strategy for each, such as pausing, instead of simply deleting them. Flag anything you are unsure of so I can check it.

TRY IT WITH A transcript full of "basically", "actually" and "so, uh".

595 A better way to open the Q&A

QUICK

2.4 TALKS & ORAL PRESENTATIONS · PG RESIDENT / FACULTY

Act as a speaking coach. Instead of "any questions?", write 3 ways to open the Q&A after my talk on [TOPIC] that invite real discussion, including one prepared question I can ask myself if the audience is silent, with a short answer.

TRY IT WITH A talk on non-invasive ventilation in COPD.

2.5

Posters & visual teaching aids

Ward protocol posters, infographics, patient leaflets, pocket cards and AI-generated illustrations: visuals teach faster than words, when they are accurate and simple.

35

prompts

4

Master

20

Standard

11

Quick

HOW TO USE THIS SECTION

Use it for ward posters, infographics, patient leaflets, pocket cards and teaching visuals.

Start with the Master prompts (#596–599) for the full workflow; use the Standard prompts (#600–619) for one task at a time; reach for the Quick prompts (#620–630) when you have two minutes. Tap any title below to jump to it, and follow → NEXT to move to the next step.

#596 Clinical protocol poster for a ward or ICU

#597 Comic-style teaching infographic (AI image tools)

#598 Handwritten-style revision notes image

#599 Awareness poster series for a health day

#600 A medical infographic for teaching or social media

#601 A patient-education leaflet

#602 Pocket quick-reference card

#603 A concept map for a complex topic

#604 Plan a chalk-and-talk or whiteboard session

#605 Sketchnote summary of a topic

#606 Educational poster for a student competition

#607 Patient-facing poster for the OPD waiting area

#608 Pictorial medication schedule for patients

#609 Brief for a medical illustration

#610 Prompt an AI image tool for medical visuals

#611 Educational carousel for Instagram or LinkedIn

#612 Low-cost teaching model or demonstration

#613 Step-by-step procedure wall chart

#614 Adapt a poster into several Indian languages

#615 Critique my poster or infographic

#616 Print specifications for posters and charts

#617 Poster with QR codes to videos and resources

#618 Emergency algorithm card for the crash cart

#619 A visual mnemonic poster

#620 Headline for my poster

#621 Short slogans for a health campaign

#622 Icons for my visual aid

#623 Colours for a patient-facing poster

#624 Check the reading level of my text

#625 Pictogram instructions for low literacy

#626 Caption for my infographic post

#627 Poster size and font sizes

#628 Fix a garbled AI image

#629 Description for a visual post

#630 Call to action for an awareness poster

Clinical protocol poster for a ward or ICU

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / FACULTY / NURSING EDUCATOR

USE IT

When your unit adopts a protocol (sepsis bundle, hypoglycaemia, anaphylaxis, blood transfusion) and staff need it at a glance.

WHY

Protocols in files are not followed; protocols on the wall, clear enough to use at 3 a.m., are. A well-designed poster turns a guideline into action, provided every dose and threshold is verified and it matches your hospital's approved protocol.

COPY THE PROMPT BELOW

PERSONA	You are a clinical quality-improvement lead and medical illustrator who designs bedside protocol posters that nurses and doctors actually use in emergencies.
MY CONTEXT	Protocol topic: [e.g. management of hypoglycaemia in adult inpatients] Source: our hospital's approved protocol or guideline: [UPLOAD / PASTE] Users: [nurses / residents / all staff] Where it will hang, and the size: [e.g. nursing station, A3] Language(s): [English / Hindi / bilingual] Drugs and equipment available locally: [list]
TASK	1. Extract the steps that must be on the poster: recognition criteria, immediate actions, drug doses and routes, monitoring, escalation triggers and who to call. 2. Design a top-to-bottom flow: a trigger box, action steps, decision points (yes/no), and escalation. 3. Write the text for each box in under 12 words, using action verbs. 4. Specify the design: large fonts, a colour code (red for emergency actions), icons and plenty of white space. 5. Add a footer with the source protocol, version and date, the approving authority, and the review date. 6. Write a 5-minute staff briefing to launch the poster.
RULES	Take every dose, threshold and step only from the approved protocol I provided, and mark each [VERIFY AGAINST APPROVED PROTOCOL] until it is checked, because a wrong dose on a wall poster can harm patients. Match the local drug formulation and route. Get sign-off from the unit in-charge before printing.
OUTPUT	The content extraction, the flow description (box by box), the design specification, the footer, then the briefing script.
TWEAK	End with a checklist for testing the poster: can a new nurse follow it correctly in 60 seconds?

How to get more out of it

- › Test the poster with 3 staff members before printing. Watch where they hesitate.
- › Laminate it and put a review date on it. Outdated protocol posters are dangerous.
- › Put the poster where the action happens, not where there's space.
- › → **NEXT #602** Pocket quick-reference card

Comic-style teaching infographic (AI image tools)

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / FACULTY / MEDICAL EDUCATOR

USE IT

When you want a shareable, memorable 5-panel comic on a clinical topic, using ChatGPT image generation, Gemini or similar tools.

WHY

A comic with two recurring characters, one asking and one answering, turns five high-yield facts into a story students remember and share. The prompt fixes the layout, characters and style so the tool produces a consistent, legible single image, and you verify every fact first.

COPY THE PROMPT BELOW

PERSONA	You are a medical educator and visual designer who writes precise image-generation prompts for educational comics, and who checks every clinical fact before it goes into an image.
MY CONTEXT	Topic: [e.g. hypokalaemia: 5 things you must know] The 5 questions and answers, written and verified by me: [PASTE 5 Q&A PAIRS] Character A (the asker): [e.g. a curious junior doctor in a white coat with a stethoscope, holding a clipboard] Character B (the answerer): [e.g. a calm, confident senior physician in scrubs] Brand colours and handle: [e.g. black and orange, @yourhandle] Image tool: [ChatGPT image / Gemini / other]
TASK	1. Check each of my 5 Q&A pairs for accuracy and brevity (answers under 12 words), and flag anything to verify. 2. Assign a tone to character A (curious, puzzled, worried, sceptical, alarmed, thinking) and an emotion to character B (reassuring, serious, knowing smile, warning finger) for each panel. 3. Write the complete image prompt: a single vertical comic-style infographic (portrait, 4:5), the title bar "[TOPIC]: 5 Things You Must Know", a clean modern medical-comic style with bold outlines, flat cell-shaded colouring and a soft palette, the brand colours on the title bar and panel borders, and 5 stacked numbered panels, each with both characters, a question bubble and an answer bubble. 4. Specify that both characters look identical in every panel, the text is short and legible, nothing is cropped, and a footer bar shows my handle. 5. Write 2 fix-up prompts for common problems: garbled text in one panel, and characters that drift between panels.
RULES	Put only facts I have verified into the image, and flag doubtful ones as [VERIFY] before generating, because errors in viral images can't be corrected once shared. Keep each bubble's text short, because image tools garble long text.
OUTPUT	The accuracy check, the tone table, the full image prompt in a code block, then the fix-up prompts.
TWEAK	End with a caption of under 60 words for posting the comic, with 2 references to cite.

How to get more out of it

- › Fix one panel at a time: "Fix the spelling in panel 3 only, keep everything else."
- › Keep answers under 12 words. Longer text is where image tools make spelling errors.
- › Read the generated image letter by letter before posting. Image tools still misspell medical terms.
- › → **NEXT #600** A medical infographic for teaching or social media

Handwritten-style revision notes image

2.5 POSTERS & VISUAL TEACHING AIDS · MBBS STUDENT / PG RESIDENT / MEDICAL EDUCATOR

USE IT

When you want an attractive A4 handwritten-style revision sheet on a topic, generated with an AI image tool.

WHY

Students love revising from colour-coded handwritten notes, and a consistent design system (paper, handwriting, colour-coded boxes, highlights and sketches) makes every topic look like a page from the same notebook. The content must be verified first, because image tools reproduce errors beautifully.

COPY THE PROMPT BELOW

PERSONA	You are a medical educator and designer who creates handwritten-style study sheets with AI image tools, and who checks every fact before generating.
MY CONTEXT	Topic: [TOPIC NAME] The verified content outline (definition, anatomy or pathophysiology, key features, clinical presentation, tests, differentials, treatment, complications, key points): [PASTE] Diagrams I want (2–4), described: [list] Footer text: [e.g. "1 / 1 @yourhandle"] Image tool: [ChatGPT image / Gemini / other]
TASK	1. Check my outline for accuracy and trim it so it fits one A4 page. 2. Write the full image prompt with this design system: warm cream background (#FEFDF6) with faint blue ruled lines; neat, slightly imperfect dark-navy gel-pen handwriting; rounded section boxes with solid header bars rotating between blue #3B5FAC, green #1E7A3D, red #C0392B and purple #6C3483, with white bold uppercase headings; highlighting in yellow #FFF176 for mnemonics, blue #BBDEFB for definitions, green #C8E6C9 for treatment, pink #F8BBD9 for clinical features and orange #FFCC80 for warnings; bullets, red stars before key points, and underlined key words. 3. Specify 2–4 small pen-and-ink labelled sketches with captions. 4. Specify the section order from my outline, a full-width yellow "key distinction" warning banner, and a "key points" box at the bottom. 5. Add the footer, and the instruction that it must look hand-done, not typed or printed. 6. Write 2 fix-up prompts: overcrowded text, and a misspelled term.
RULES	Include only content I have verified, marking anything doubtful [VERIFY] before generating, and don't reproduce copyrighted atlas plates, only original sketches "in the style of" clean anatomical line drawings, because both errors and copyright problems multiply once images are shared.
OUTPUT	The accuracy check, the trimmed outline, the full image prompt in a code block, then the fix-up prompts.
TWEAK	End with a list of the 5 terms most likely to be misspelled in the image, to proofread first.

How to get more out of it

- › One topic per page. If the outline is long, split it into two sheets.
- › Proofread every word of the generated image. Handwriting styles hide spelling errors.
- › Keep the design system identical across topics. It makes a recognisable series.
- › → **NEXT #600** A medical infographic for teaching or social media

Awareness poster series for a health day

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / FACULTY / COMMUNITY MEDICINE

USE IT

For World TB Day, World Diabetes Day, Breastfeeding Week and similar campaigns in the hospital, community or social media.

WHY

Awareness posters often say too much, scare people, or repeat the same slogan everywhere. A small, coordinated series with one message per poster, local language, a clear action and consistent design reaches more people and changes behaviour.

COPY THE PROMPT BELOW

PERSONA	You are a public-health communication specialist who designs health-awareness campaigns for Indian audiences in hospitals, communities and online.
MY CONTEXT	Health day or theme: [e.g. World TB Day] Official theme for the year, if any: [paste or "check"] Audiences: [patients in OPD / community / students / social media followers] Languages: [list] Where the posters will be used: [OPD walls / health camp / Instagram / WhatsApp] Key programme information (free services, helplines, facilities) from official sources: [PASTE]
TASK	1. Define 4–5 key messages, one per poster, each with a single action (e.g. "Cough for more than 2 weeks? Get tested free at [facility]."). 2. For each poster, write the headline (under 8 words), 2 supporting lines, the call to action and a visual idea. 3. Define a consistent design: colours, fonts, logo placement, and the space for local information. 4. Adapt the messages to each language, simply, not literally. 5. Adapt the formats: A3 print, a 1:1 social post and a WhatsApp status. 6. Suggest how to launch and distribute the series, and how to measure reach.
RULES	Take programme details only from the official sources I provided, marking anything else [VERIFY WITH PROGRAMME], and avoid stigmatising images and words, because stigma keeps people from seeking care. Follow the official campaign's branding rules if it has any.
OUTPUT	A message table, the poster-by-poster text and visuals, the design system, the language versions, the format list, then the distribution plan.
TWEAK	End with the one poster that would work best alone if only one could be printed.

How to get more out of it

- Show people like your audience in the visuals. People respond to posters that look like them.
- Always include where to go and that services are free, if they are. That's the action people need.
- Test the headline with 3 people from the audience. If they can't repeat it back, simplify it.
- → **NEXT #601** A patient-education leaflet

600 A medical infographic for teaching or social media

STANDARD

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / FACULTY

USE IT When you want to summarise a topic visually for students, colleagues or the public.

PERSONA	You are a medical infographic designer who balances accuracy and visual simplicity.
MY CONTEXT	Topic and audience: [brief] Verified key facts or data to include: [PASTE] Format: [A4 portrait / 1:1 Instagram / 4:5 / slide] Tool: [Canva / PowerPoint / AI image tool]
TASK	1. Choose a structure: a process, a comparison, a list with icons, a statistics highlight, or a timeline. 2. Write a headline, 5–7 content blocks (under 15 words each) and a source line. 3. Suggest icons and a colour scheme. 4. Suggest the reading flow and hierarchy.
RULES	Use only the facts I provided, with sources cited on the infographic, and mark numbers [VERIFY], because infographics are shared without context.
OUTPUT	The structure, the text blocks, the design notes, then the source line.

PRO TIP One infographic, one message. If it needs a second message, make a second infographic.

TRY IT WITH "5 signs of sepsis every nurse should know", 4:5 for Instagram.

→ **NEXT #611** Educational carousel for Instagram or LinkedIn

601 A patient-education leaflet

STANDARD

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / FACULTY / NURSE EDUCATOR

USE IT For patients to take home after a diagnosis, procedure or discharge.

PERSONA	You are a patient-education specialist who writes clear leaflets at a low reading level.
MY CONTEXT	Topic: [e.g. living with heart failure / preparing for colonoscopy] Patient group and language: [brief] Our hospital's advice and contact details (placeholders are fine): [PASTE]
TASK	1. Structure the leaflet: what it is, why it matters, what you should do, warning signs to come back for, and who to contact. 2. Write it at about a sixth-to-eighth-grade reading level, in short sentences. 3. Suggest simple pictures or icons. 4. Make a version in [LANGUAGE] if requested. 5. Keep it to one or two sides of A4.
RULES	Keep all advice consistent with our hospital's protocols, marking specifics [VERIFY WITH TREATING TEAM], because patients follow leaflets literally.
OUTPUT	The leaflet text with headings, the picture suggestions, then the translated version if requested.

PRO TIP Test the leaflet on a patient's relative. If they can't tell you the warning signs afterwards, rewrite them.

TRY IT WITH A discharge leaflet on inhaler use and COPD warning signs, in Hindi and English.

→ **NEXT #608** Pictorial medication schedule for patients

602 Pocket quick-reference card

STANDARD

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / MBBS INTERN

USE IT For a pocket card or phone-wallpaper reference on something you need often at the bedside.

PERSONA	You are a clinical educator who designs dense but readable quick-reference cards.
MY CONTEXT	Topic: [e.g. ABG interpretation steps / common emergency drug doses / GCS] Verified content from source: [PASTE WITH SOURCE] Size: [ID card / half A5 / phone screen]
TASK	1. Select the essential information that fits. 2. Organise it into a clear layout (sections, tables, a mnemonic). 3. Suggest a front and back design. 4. Add the source and version date in small print.
RULES	Mark every dose and threshold [VERIFY FROM SOURCE] until checked, and include "check the local protocol" wording, because pocket cards get used for years after guidelines change.
OUTPUT	The card content by side, the layout, then the source line.

PRO TIP Put the date on the card, and replace it when the guideline updates.

TRY IT WITH An emergency drug-dose card for adult cardiac arrest.

→ **NEXT #596** Clinical protocol poster for a ward or ICU

603 A concept map for a complex topic

STANDARD

2.5 POSTERS & VISUAL TEACHING AIDS · MBBS STUDENT / PG RESIDENT / FACULTY

USE IT For teaching or revising topics with many connected ideas.

PERSONA	You are a medical educator who uses concept maps to show how knowledge connects.
MY CONTEXT	Topic: [TOPIC] Level: [brief] Key concepts I want included: [list or "suggest"]
TASK	1. Identify the central concept and 12–20 linked concepts. 2. Define the links with labelled arrows ("causes", "leads to", "treated by", "detected by"). 3. Organise them into clusters (e.g. mechanisms, features, tests, treatment). 4. Describe the layout for drawing by hand or in a tool (e.g. draw.io, Miro). 5. Suggest 3 questions to test understanding using the map.
RULES	Keep every link clinically accurate and mark any uncertain relationship [VERIFY], because a wrong arrow teaches a wrong mechanism.
OUTPUT	A concept list, a link table (from, label, to), the layout description, then the questions.

PRO TIP Ask students to build the map themselves first, then compare it with yours. That's where the learning happens.

TRY IT WITH Topic: heart failure (causes, mechanisms, features, investigations and drug classes).

→ **NEXT #604** Plan a chalk-and-talk or whiteboard session

604 Plan a chalk-and-talk or whiteboard session

STANDARD

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / FACULTY

USE IT For bedside teaching, tutorials or rooms without projectors.

PERSONA	You are a master chalk-and-talk teacher.
MY CONTEXT	Topic and duration: [brief] Audience and group size: [brief] Board size: [small whiteboard / large blackboard]
TASK	1. Plan the board layout in zones: title, a central diagram, key points, and a "parking lot" for questions. 2. Plan the build order: what to draw first, and when to add each element as you talk. 3. Specify the diagrams to draw, with simple step-by-step drawing instructions. 4. Suggest colour use (if markers are available). 5. Plan a final summary that the finished board shows.
RULES	Keep drawings simple enough to draw in under 2 minutes each, because long drawing loses the audience.
OUTPUT	The board-zone layout, the build sequence with timings, the drawing instructions, then the summary.

PRO TIP Practise the diagram twice beforehand. Confident drawing looks expert; hesitant drawing looks unprepared.

TRY IT WITH A 20-minute tutorial on the brachial plexus for MBBS students.

→ **NEXT #603** A concept map for a complex topic

605 Sketchnote summary of a topic

STANDARD

2.5 POSTERS & VISUAL TEACHING AIDS · MBBS STUDENT / PG RESIDENT

USE IT For visual notes during or after a lecture, or as a teaching handout.

PERSONA	You are a visual-thinking coach who teaches sketchnoting to medical students.
MY CONTEXT	Topic or lecture content: [PASTE] Audience: [self / students]
TASK	1. Identify the central idea and 5–7 supporting ideas. 2. Suggest a simple icon or doodle for each idea. 3. Suggest a layout (radial, linear path or grid) with arrows and short labels. 4. Suggest one memorable example to draw. 5. If an AI image tool will be used, write a prompt with a text fallback.
RULES	Keep the labels accurate and short, and check the key facts, because sketchnotes are revised from repeatedly.
OUTPUT	The ideas, the icons, the layout, then the image prompt.

PRO TIP Simple shapes and arrows are enough. Sketchnoting isn't about drawing talent.

TRY IT WITH A sketchnote of the renin-angiotensin-aldosterone system.

→ **NEXT #603** A concept map for a complex topic

606 Educational poster for a student competition

STANDARD

2.5 POSTERS & VISUAL TEACHING AIDS · MBBS STUDENT / PG RESIDENT

USE IT For poster competitions at college festivals, health days or academic events.

PERSONA	You are a judge of educational poster competitions.
MY CONTEXT	Theme and rules (size, format, judging criteria): [PASTE] My topic idea: [brief] Audience: [brief]
TASK	1. Sharpen the topic into one clear message that fits the theme. 2. Plan the poster: headline, 3–5 content blocks, visuals and a call to action. 3. Map the plan against the judging criteria (content, creativity, clarity, relevance). 4. Suggest how to present the poster to the judges in 90 seconds.
RULES	Keep all medical content accurate and cited, and follow the size and format rules exactly, because rule violations disqualify entries.
OUTPUT	The message, the poster plan, the criteria map, then the pitch.

PRO TIP Judges walk past dozens of posters. A striking headline and one strong visual decide who stops.

TRY IT WITH A theme on antimicrobial resistance for a college health festival.

→ **NEXT #600** A medical infographic for teaching or social media

607 Patient-facing poster for the OPD waiting area

STANDARD

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / FACULTY

USE IT For waiting areas, where patients have a few minutes and a real need.

PERSONA	You are a patient-communication designer for Indian hospitals.
MY CONTEXT	Message topic: [e.g. how to use an inhaler / diabetic foot care / when to come back] Patient languages and literacy: [brief] Poster size and location: [brief]
TASK	1. Choose one message with one action. 2. Write a headline (under 7 words) and 3–4 picture-supported steps. 3. Suggest pictograms for low-literacy readers. 4. Produce versions in the required languages. 5. Suggest placement (eye level, near the relevant clinic).
RULES	Keep advice consistent with our clinicians' practice, marking specifics [VERIFY WITH DEPARTMENT], and avoid frightening images, because patients should feel informed, not scared.
OUTPUT	The headline, the steps with pictograms, the language versions, then the placement.

PRO TIP Pictures of hands doing each step work better than any text for inhaler technique.

TRY IT WITH "How to use your inhaler" for a chest-clinic waiting area, in Hindi and English.

→ **NEXT #601** A patient-education leaflet

608 Pictorial medication schedule for patients

STANDARD

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / NURSE EDUCATOR / PHARMACIST

USE IT For patients who struggle with complex medication schedules, especially elderly or low-literacy patients.

PERSONA	You are a clinical pharmacist experienced in designing pictorial medication aids.
MY CONTEXT	The patient's medication list with timings (de-identified): [PASTE] Patient literacy and language: [brief]
TASK	1. Design a daily chart with columns for morning, afternoon, evening and night, and pictures of meal times (before or after food). 2. Represent each medicine with its colour and shape (to be filled in from the actual tablets) and its purpose in simple words. 3. Add key warnings (e.g. do not stop suddenly). 4. Add a tick box for each dose.
RULES	Copy the medication details exactly from the prescription and have the pharmacist or doctor check the chart before giving it to the patient, because a wrong chart causes medication errors.
OUTPUT	The chart layout with the content filled in, then the verification step.

PRO TIP Use actual photos of the patient's tablets if possible. Generic pill icons confuse people.

TRY IT WITH An elderly patient on 7 medicines, including insulin, with limited literacy.

→ **NEXT #601** A patient-education leaflet

609 Brief for a medical illustration

STANDARD

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / FACULTY

USE IT When you need an accurate diagram for a talk, poster or paper and will draw it or commission it.

PERSONA	You are a medical illustrator who writes precise illustration briefs.
MY CONTEXT	What the illustration must show: [brief] Where it will be used and the size: [brief] Style: [line drawing / colour / schematic] Reference sources (textbook, atlas, published figure): [LIST]
TASK	1. Describe the illustration's content precisely: structures, labels, orientation and view. 2. Specify the labels (terminology) and leader-line placement. 3. Specify the style, colours and level of detail. 4. List accuracy checkpoints for the finished illustration.
RULES	Base anatomical and pathological details on the named reference sources, and don't copy copyrighted figures directly, because accuracy and copyright both matter.
OUTPUT	The content description, the label list, the style specification, then the checklist.

PRO TIP Ask for a draft sketch before the final version. Anatomical errors are easier to fix early.

TRY IT WITH A labelled diagram of the bronchopulmonary segments for a seminar.

→ **NEXT #610** Prompt an AI image tool for medical visuals

610 Prompt an AI image tool for medical visuals

STANDARD

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / FACULTY

USE IT When generating illustrations for slides or posters with ChatGPT, Gemini or other image tools.

PERSONA	You are a prompt engineer for AI image generation in medical education.
MY CONTEXT	What I need (concept, diagram or scene): [brief] Where it will be used: [slide / poster / social post] Tool: [name]
TASK	1. Write a detailed image prompt: subject, style (flat vector, line drawing, 3D render), composition, colours, text to include (minimal) and aspect ratio. 2. Add instructions to avoid anatomical inaccuracies and to keep text minimal. 3. Suggest a two-step approach: generate a text-free image, then add labels in PowerPoint or Canva. 4. Write a checklist for reviewing the generated image.
RULES	Don't rely on AI images for precise anatomy or data, and label them "illustrative", because AI images frequently contain anatomical and text errors.
OUTPUT	The image prompt in a code block, the approach, then the review checklist.

PRO TIP Add labels yourself afterwards. AI-generated text inside images is the most error-prone part.

TRY IT WITH A friendly illustration of a doctor explaining an inhaler to an elderly patient, for a slide.

→ **NEXT #597** Comic-style teaching infographic (AI image tools)

611 Educational carousel for Instagram or LinkedIn

STANDARD

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / FACULTY

USE IT When sharing medical education on social media in a swipeable format.

PERSONA	You are a medical educator and social-media content strategist.
MY CONTEXT	Topic and audience: [e.g. residents / students / the public] Verified key points: [PASTE] Number of slides: [e.g. 7–10] Brand style: [brief]
TASK	1. Write a hook for slide 1 (under 10 words). 2. Write the slide-by-slide text: one idea per slide, under 30 words each. 3. Write the final slide: a summary and a call to action (save, share, follow). 4. Suggest visuals for each slide. 5. Write a caption with sources.
RULES	Use only verified content and cite the sources in the caption, and avoid sensational claims, because medical misinformation spreads quickly on social media.
OUTPUT	The slide-by-slide text table, the visual notes, then the caption.

PRO TIP A strong first slide decides everything. If it doesn't stop the scroll, the rest won't be seen.

TRY IT WITH "7 ECG patterns every intern must recognise", 9 slides.

→ **NEXT #600** A medical infographic for teaching or social media

612 Low-cost teaching model or demonstration

STANDARD

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / FACULTY

USE IT When a physical model would teach better than a slide.

PERSONA	You are a creative medical educator who builds low-cost teaching models.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The concept to demonstrate: [brief] Audience: [brief] Materials and budget available: [brief]
TASK	1. Suggest 2–3 model or demonstration ideas using cheap, available materials. 2. Give build instructions for the best one. 3. Write a demonstration script linking the model to the clinical concept. 4. Note the model's limitations (where the analogy breaks down).
RULES	Ensure the model is safe to handle and doesn't introduce misconceptions, and state its limitations explicitly, because models are remembered vividly, including their errors.
OUTPUT	The ideas, the build instructions, the script, then the limitations.

PRO TIP A balloon in a jar teaches lung compliance better than any animation.

TRY IT WITH Demonstrating pneumothorax and chest-drain function for nursing students.

→ **NEXT #604** Plan a chalk-and-talk or whiteboard session

613 Step-by-step procedure wall chart

STANDARD

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / NURSE EDUCATOR

USE IT For procedures done often on a ward: central-line care, catheterisation, nebulisation, NG-tube insertion.

PERSONA	You are a clinical skills educator who designs procedure charts.
MY CONTEXT	Procedure: [name] Our hospital's approved procedure steps: [PASTE / UPLOAD] Users: [brief]
TASK	1. Break the procedure into 8–12 numbered steps with action verbs. 2. Mark the critical safety steps (hand hygiene, verification, sterility) visually. 3. Suggest a picture or icon for each step. 4. Add a checklist box format for audits. 5. Add the source, version and review date.
RULES	Follow our approved procedure exactly and mark any step to confirm as [VERIFY WITH SOP], because procedure charts become the de facto standard on the ward.
OUTPUT	The numbered steps, the safety highlights, the picture list, then the footer.

PRO TIP Put the chart where the procedure is prepared, such as the dressing trolley or the clean utility room.

TRY IT WITH Steps for central-line dressing change in the ICU.

→ **NEXT #596** Clinical protocol poster for a ward or ICU

614 Adapt a poster into several Indian languages

STANDARD

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / FACULTY

USE IT When a poster or leaflet must reach patients who speak different languages.

PERSONA	You are a health-communication translator experienced in Indian languages.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The English poster text: [PASTE] Target languages: [list] Audience literacy: [brief]
TASK	1. Adapt (not literally translate) the text into each language, keeping it simple. 2. Keep familiar medical words (e.g. "sugar", "BP") where patients use them. 3. Check the text length fits the layout, and suggest cuts if not. 4. Recommend a review by a native speaker from the target community.
RULES	Keep the meaning exactly consistent across languages, because inconsistent translations confuse patients and staff.
OUTPUT	A text table by language, the layout notes, then the review recommendation.

PRO TIP Back-translate the key warning line into English to check it survived translation intact.

TRY IT WITH A "warning signs of dengue" poster in Hindi, Bengali and Odia.

→ **NEXT #607** Patient-facing poster for the OPD waiting area

615 Critique my poster or infographic

STANDARD

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / FACULTY

USE IT Before printing or posting any visual aid.

PERSONA	You are a senior medical educator and design critic.
MY CONTEXT	My poster or infographic text and a description (or upload the image): [PASTE / UPLOAD] Purpose and audience: [brief]
TASK	1. Check accuracy: flag any claim, number or dose to verify. 2. Check clarity: the single message, reading order and text amount. 3. Check the design: contrast, font size, hierarchy and clutter. 4. Check accessibility: language, literacy and colour-blind safety. 5. Give the top 5 fixes.
RULES	Flag doubtful facts as [VERIFY] rather than correcting them from memory, because I must confirm them from my sources.
OUTPUT	A review table by criterion, then the top 5 fixes.

PRO TIP Look at your poster from across the room for 5 seconds. What you remember is its real message.

TRY IT WITH An A3 poster on hand hygiene with 180 words and 6 colours.

→ **NEXT #600** A medical infographic for teaching or social media

616 Print specifications for posters and charts

STANDARD

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / FACULTY

USE IT Before sending posters, charts or leaflets to print.

PERSONA	You are a print-production specialist.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Item, size and quantity: [brief] Where it will be used (wall, bedside, outdoors, handout): [brief] File source: [Canva / PowerPoint / other]
TASK	1. Recommend the file format (PDF), resolution (300 dpi), colour mode and bleed. 2. Recommend the paper or material (lamination, vinyl, sunboard) for the use and durability needed. 3. Recommend minimum font sizes for the viewing distance. 4. Give a pre-print checklist (spelling, logos, contact details, version date).
RULES	Recommend wipeable, laminated materials for clinical areas, because paper posters in wards become infection-control problems.
OUTPUT	The specification table, then the checklist.

PRO TIP Print one proof copy and look at it on the actual wall before printing the full batch.

TRY IT WITH 20 laminated A3 sepsis-protocol posters for medical wards.

→ **NEXT #596** Clinical protocol poster for a ward or ICU

617 Poster with QR codes to videos and resources

STANDARD

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / FACULTY

USE IT When a poster should lead people to videos, forms or detailed resources.

PERSONA	You are a digital-health educator who connects print and online resources.
MY CONTEXT	Poster topic: [brief] Resources to link (videos, PDFs, forms): [list] Audience and their phones or data access: [brief]
TASK	1. Choose what the QR code should link to, and why. 2. Write the call to action beside the QR code. 3. Suggest QR code size and placement for scanning distance. 4. Suggest a stable link strategy (a link you control and can update). 5. Suggest how to track scans.
RULES	Link only to trustworthy, stable resources and test every code before printing, because broken or hijacked links undermine trust.
OUTPUT	The link choice, the call to action, the size and placement, the link strategy, then the tracking.

PRO TIP Use a URL you can redirect later. Printed posters outlive most links.

TRY IT WITH A poster in the physiotherapy department linking to home-exercise videos.

→ **NEXT #607** Patient-facing poster for the OPD waiting area

618 Emergency algorithm card for the crash cart

STANDARD

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / NURSE EDUCATOR

USE IT For emergencies where seconds matter: anaphylaxis, cardiac arrest, status epilepticus, hyperkalaemia, massive haemorrhage.

PERSONA	You are a resuscitation-training instructor who designs algorithm cards for crash carts and emergency rooms.
MY CONTEXT	Emergency: [name] Source: our hospital protocol or the current resuscitation guideline: [UPLOAD / PASTE] Drugs and equipment on our crash cart: [list] Card size: [A5 / A4, laminated]
TASK	1. Extract the time-critical steps in order, with drug, dose, route and repeat interval. 2. Design a one-page flow that can be read in under 10 seconds per step. 3. Colour-code the actions (e.g. red for immediate, amber for next, green for monitor). 4. Add a "call for help" box with placeholders for local numbers. 5. Add the source, version date and review date.
RULES	Take every dose only from the protocol I provided and mark each [VERIFY AGAINST PROTOCOL], because an error on an emergency card is repeated in every emergency until someone notices.
OUTPUT	The step list, the card layout description, the colour code, then the footer.

PRO TIP Run a mock drill with the card. Where people hesitate, the card needs simplifying.

TRY IT WITH An anaphylaxis card for the emergency department, adults.

→ **NEXT #602** Pocket quick-reference card

619 A visual mnemonic poster

STANDARD

2.5 POSTERS & VISUAL TEACHING AIDS · MBBS STUDENT / PG RESIDENT / FACULTY

USE IT When a list must be memorised and a picture could anchor it.

PERSONA	You are a medical educator skilled in visual memory techniques.
MY CONTEXT	The list to remember: [LIST] Audience: [brief] Format: [poster / slide / flashcard]
TASK	1. Create a mnemonic (acronym, phrase or story) where each element maps accurately to an item. 2. Design a visual scene or image that links the mnemonic to the items. 3. Write a caption explaining each element. 4. Suggest an image prompt or drawing instructions.
RULES	Check that every mnemonic element maps to the correct item, and keep it respectful, avoiding offensive humour, because memorable errors are the worst kind.
OUTPUT	The mnemonic, the scene description, the caption, then the image prompt.

PRO TIP Absurd, vivid images are easiest to remember, as long as they stay professional.

TRY IT WITH Causes of clubbing, as a visual poster for final-year MBBS students.

→ **NEXT #597** Comic-style teaching infographic (AI image tools)

620 Headline for my poster

QUICK

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / FACULTY

Act as a health-communication copywriter. Write 5 headlines of under 8 words for a poster on [TOPIC] for [AUDIENCE], each stating the benefit or action rather than the topic name, accurate and free of fear-based language. Mark your favourite. Flag anything you are unsure of so I can check it.

TRY IT WITH A poster encouraging TB testing for anyone with a cough of more than 2 weeks.

621 Short slogans for a health campaign

QUICK

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / FACULTY

Act as a public-health campaign writer. Write 6 slogans of under 7 words for [HEALTH DAY OR THEME], in English and [LANGUAGE], that are memorable, positive, non-stigmatising and linked to one clear action. Flag anything you are unsure of so I can check it.

TRY IT WITH World Diabetes Day, English and Hindi.

622 Icons for my visual aid

QUICK

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT

Act as a graphic designer for health materials. For these items on my poster, [LIST ITEMS], suggest one simple, culturally appropriate icon for each, with search keywords for Canva or PowerPoint, and keep them in a single consistent style. Flag anything you are unsure of so I can check it.

TRY IT WITH Items: wash hands, wear mask, cover cough, see a doctor, take full treatment.

623 Colours for a patient-facing poster

QUICK

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT

Act as a health-communication designer. Suggest a colour scheme with hex codes for a patient-facing poster on [TOPIC] that is calm, readable, high-contrast, colour-blind safe and not alarming, and say which colour to use for the call to action. Flag anything you are unsure of so I can check it.

TRY IT WITH A poster on diabetes foot care for an OPD waiting area.

624 Check the reading level of my text

QUICK

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / NURSE EDUCATOR

Act as a plain-language editor. Check this patient text, [PASTE TEXT], for reading level, long sentences and medical jargon, and rewrite it at about a sixth-grade level, keeping every instruction accurate. List the words you replaced. Flag anything you are unsure of so I can check it.

TRY IT WITH Discharge instructions after a heart attack, written by a resident.

625 Pictogram instructions for low literacy

QUICK

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / NURSE EDUCATOR

Act as a patient-education designer. Turn these instructions, [PASTE STEPS], into 4–6 pictogram panels, describing each picture and giving a caption of 5 words or fewer, so that patients with limited literacy can follow them. Flag anything you are unsure of so I can check it.

TRY IT WITH How to use an insulin pen.

626 Caption for my infographic post

QUICK

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / FACULTY

Act as a medical social-media writer. Write a caption of under 80 words for my infographic on [TOPIC] for [AUDIENCE], with a hook line, a one-line summary, 2 sources, a call to save or share, and a short "educational, not medical advice" note.

TRY IT WITH An infographic on the warning signs of stroke (BE-FAST).

627 Poster size and font sizes

QUICK

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT

Act as a print designer. For a [POSTER SIZE] poster viewed from [DISTANCE] metres in [LOCATION], recommend the minimum font sizes for the headline, body text and footnotes, the maximum word count, and the margins, in a short table. Flag anything you are unsure of so I can check it.

TRY IT WITH An A3 poster viewed from 2 metres in a ward corridor.

628 Fix a garbled AI image

QUICK

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / FACULTY

Act as an AI image-prompt expert. My generated image has these problems: [DESCRIBE PROBLEMS, e.g. misspelled words, a drifting character, cropped text]. Write targeted fix-up prompts that change only the faulty parts and keep everything else identical. Flag anything you are unsure of so I can check it.

TRY IT WITH Panel 3 says "hypokalmia" and character B's face has changed.

629 Description for a visual post

QUICK

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT

Act as an accessibility writer. Write alt text of under 40 words for my medical infographic or carousel image, [DESCRIBE IMAGE], so that screen-reader users get the key message, and suggest where to add it on Instagram or LinkedIn. Flag anything you are unsure of so I can check it.

TRY IT WITH A carousel slide showing the 5 steps of an ABG interpretation.

630 Call to action for an awareness poster

QUICK

2.5 POSTERS & VISUAL TEACHING AIDS · PG RESIDENT / FACULTY

Act as a public-health communicator. Write 4 calls to action for a poster on [TOPIC] for [AUDIENCE] that tell people exactly what to do, where and when, with placeholders for local facility names and helplines, in under 15 words each.

TRY IT WITH A poster on early cancer screening for women in a district town.

2.6

Case presentations & grand rounds

A good case presentation tells a clinical story in the order the evidence arrived, reasons aloud, and ends with what everyone in the room should learn.

40

prompts

5

Master

23

Standard

12

Quick

HOW TO USE THIS SECTION

Use it before case presentations, grand rounds and mortality meetings.

Start with the Master prompts (**#631–635**) for the full workflow; use the Standard prompts (**#636–658**) for one task at a time; reach for the Quick prompts (**#659–670**) when you have two minutes. Tap any title below to jump to it, and follow → **NEXT** to move to the next step.

#631 Departmental case presentation, start to finish

#632 Grand rounds presentation

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#649 A 10–12 slide case-presentation deck

#650 Case discussion on a management controversy

#651 Present a case with an ethical dilemma

#652 Present a rare diagnosis without losing the audience

#653 Learning from a delayed or missed diagnosis

#654 The questions the head of department will ask

#655 Root-cause analysis for an M&M case

#656 Present a follow-up update on a previous case

#657 Turn a case into a teaching session for MBBS students

#658 Link a case to a recent paper

#659 Opening one-liner for my case

#660 Summarise my case in three lines

#661 Which negatives matter?

#662 Why is this case worth presenting?

#663 Build the problem list

#664 Describe the vitals trend

#665 Present the lab results concisely

#666 Hospital course as a timeline

#667 The final diagnosis statement

#668 One learning point from my case

#669 SBAR script for a phone call

#670 A title for my case presentation

Departmental case presentation, start to finish

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT / MBBS INTERN

USE IT

When you present a case at a departmental case discussion or clinical meeting.

WHY

Case presentations are where residents show clinical reasoning. A structured presentation (a crisp history, focused examination, a summary, a justified differential, a rational work-up and a clear plan) shows your thinking and invites useful discussion. A disorganised one invites interruption.

COPY THE PROMPT BELOW

PERSONA	You are a senior physician and clinical teacher who has trained generations of residents in case presentation. You value clinical reasoning over exhaustive lists, and you expect every statement to be justified.
MY CONTEXT	Specialty and my level: [e.g. MD Medicine, year 1] De-identified case notes (history, examination, investigations with results and dates as Day 0, Day 2..., treatment and course): [PASTE NOTES] Time allowed: [e.g. 15 minutes + 15 minutes discussion] Format: [bedside / slides / verbal] The discussion point my unit wants: [diagnostic dilemma / management decision / teaching]
TASK	1. Write the opening one-liner (age band, sex, key background, presenting problem and duration). 2. Structure the history: presenting complaint, history of present illness in chronological order with pertinent positives and negatives, and relevant past, drug, family, social and personal history. 3. Present the examination findings: general, vitals and systemic, relevant findings first, with pertinent negatives. 4. Write a 3–4 line summary that frames the problem. 5. Give a provisional diagnosis and a ranked differential, each with supporting and opposing features. 6. Present investigations in a logical order, with results interpreted and trends shown. 7. Present the management and course, with the reasoning behind key decisions. 8. End with the discussion points or questions for the audience, and 3 learning points.
RULES	Use only facts from my notes, keep everything de-identified, and write [NOT DOCUMENTED] where a key detail is missing rather than inventing it, because the audience will ask about it. Reason step by step when justifying the differential. This is for learning, not a substitute for the treating team's judgement.
OUTPUT	The full presentation in sections, suitable to speak or to put on 10–12 slides, then the discussion points and learning points.
TWEAK	End with the 5 questions the chairperson is most likely to ask, and short, evidence-aware answers marked [VERIFY] where guideline-dependent.

How to get more out of it

- › Present in the order the information became available. It lets the audience reason along with you.
- › Pertinent negatives are as important as positives. They show you considered the differential.
- › Rehearse the summary until you can say it without notes. It's the part everyone remembers.
- › → NEXT #636 A three-line case summary

Grand rounds presentation

2.6 CASE PRESENTATIONS & GRAND ROUNDS · SENIOR RESIDENT / FACULTY

USE IT

When you present at hospital or departmental grand rounds.

WHY

Grand rounds combine a case with a broader teaching review for a mixed, often senior audience. The best ones use the case as a doorway into current evidence, practical controversies and system lessons, and they keep time.

COPY THE PROMPT BELOW

PERSONA	You are an academic physician who has presented and chaired grand rounds for many years. You design sessions that are clinically grounded, evidence-rich and relevant to the whole hospital.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] What I most want to get out of this: [one line] The case (de-identified summary): [PASTE] The broader topic the case illustrates: [e.g. drug-induced liver injury] Audience (departments, seniority): [brief] Duration: [e.g. 45 minutes] Sources for the review part (guidelines, key papers): [UPLOAD / LIST] Guest discussants, if any: [brief]
TASK	1. Plan the structure: case presentation (10 minutes), case discussion or discussant (10 minutes), topic review (15 minutes), take-home points and system lessons (5 minutes), and Q&A. 2. Condense the case into a presentation with a clear clinical question. 3. Build the topic review from my sources: epidemiology, mechanisms relevant to the case, diagnosis, management and current evidence, with slide titles. 4. Identify system or quality lessons for the hospital, if any. 5. Write briefing questions for the discussant. 6. Write the take-home points. 7. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Base the review on my sources and mark guideline or trial specifics [VERIFY], because grand-rounds audiences include experts in the topic. Keep the case de-identified and blame-free.
OUTPUT	The session plan with timings, the case slides, the review slide outline, the system lessons, the discussant questions, then the take-home points.
TWEAK	End with the question the most senior person in the room is most likely to ask, and how to handle it.

How to get more out of it

- › Invite a discussant from another department. It broadens the discussion and the audience.
- › Keep the review focused on what the case raised. Grand rounds aren't a textbook chapter.
- › Circulate the take-home points afterwards. They extend the reach beyond the room.
- › → **NEXT #649** A 10–12 slide case-presentation deck

Morbidity and mortality (M&M) presentation

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT / FACULTY

USE IT

When presenting an adverse outcome, complication or death at an M&M or death-audit meeting.

WHY

M&M meetings exist to learn and prevent recurrence, not to blame. A structured presentation (the facts, a timeline, the analysis of contributing factors using a systems lens, the evidence, and concrete actions) makes the meeting productive and psychologically safe.

COPY THE PROMPT BELOW

PERSONA	You are a patient-safety physician who runs M&M meetings using a systems approach and a just culture. You are factual, balanced and focused on learning.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] The case, de-identified, with a timeline of events: [PASTE] The outcome (complication or death): [brief] Local protocols relevant to the case: [list] Institutional M&M format, if any: [paste]
TASK	1. Present the case factually as a timeline, without judgemental language. 2. Identify the key decision points and what information was available at each one. 3. Analyse the contributing factors with a structured framework: patient factors, task and technology, individual, team, environment and organisation. 4. Classify the event (e.g. known complication, possible error, system failure, unavoidable) as a discussion point, not a verdict. 5. Summarise the relevant evidence or guideline standards, marked [VERIFY]. 6. Propose 3–5 specific, actionable system improvements with owners and timelines (as suggestions). 7. Write discussion questions for the meeting, including one on how a similar event could be detected earlier. 8. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Use neutral, non-blaming language and no names or identifiers, because psychological safety is what makes M&M work. Use only facts from my timeline, and mark unknowns [NOT KNOWN].
OUTPUT	The timeline, the decision-point table, the contributing-factor analysis, the classification, the evidence summary, the improvement proposals, then the discussion questions.
TWEAK	End with one-line "learning messages" suitable for circulating to the department.

How to get more out of it

- › Ask "what made sense to the team at that time?" rather than "who made the mistake?".
- › Track your action items. An M&M without follow-up changes nothing.
- › Keep the presentation confidential according to your institution's rules for quality-improvement meetings.
- › → **NEXT #655** Root-cause analysis for an M&M case

Short "interesting case" presentation with teaching

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT

USE IT

For 10-minute "interesting case" or "case of the week" slots.

WHY

Short case slots succeed when they are tightly structured: the hook, the key findings, the reasoning to diagnosis, and one or two strong teaching points. Presenters who try to include everything run over and teach nothing.

COPY THE PROMPT BELOW

PERSONA	You are a clinical teacher who coaches residents to present interesting cases in 10 minutes with maximum teaching value.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] What I most want to get out of this: [one line] The case (de-identified notes): [PASTE] What makes it interesting: [rare diagnosis / atypical presentation / diagnostic twist / management lesson] Time: [e.g. 10 minutes] Audience: [brief] The key image or result that cracked the case: [describe] What the treating team would do differently next time: [brief]
TASK	1. Write a hook for the first 30 seconds. 2. Select only the history, examination and investigation findings that drive the reasoning. 3. Present the case as an unfolding puzzle: reveal the information in stages and pause for audience guesses. 4. Reveal the diagnosis and exactly how it was confirmed (the test, finding or response to treatment). 5. Deliver 2 teaching points with brief evidence, marked [VERIFY]. 6. Add a 30-second "what we would do differently" reflection. 7. Write 3 audience questions to use at the pauses, with the expected answers. 8. Plan 6–8 slides with timings that fit the slot exactly. 7. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Keep it fully de-identified and include only facts from my notes, because interesting cases are often recognisable to colleagues.
OUTPUT	The hook, the selected findings, the staged reveal plan, the diagnosis reveal, the teaching points, then the slide plan.
TWEAK	End with one "take-home pearl" for the audience to remember.

How to get more out of it

- The pause before revealing the diagnosis is where the audience learns. Don't rush it.
- Show one key image or result, large, at the turning point.
- Finish with what you would do differently next time. It's the most useful sentence in the talk.
- → NEXT #647 Teaching points from a case

Presenting at a multidisciplinary team meeting or tumour board

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT / SENIOR RESIDENT

USE IT

When presenting a patient at an MDT, tumour board, or joint medical-surgical meeting.

WHY

MDTs make major decisions in minutes. A crisp, standardised presentation with the right information for each specialty (radiology, pathology, oncology, surgery) leads to better decisions and saves everyone's time.

COPY THE PROMPT BELOW

PERSONA	You are an MDT coordinator and senior clinician who has run tumour boards and multidisciplinary meetings for years.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] Meeting type: [tumour board / ILD MDT / complex-case MDT] Patient summary (de-identified): age band, sex, performance status, comorbidities: [PASTE] Relevant history, imaging, pathology and prior treatment: [PASTE] The specific question for the MDT: [e.g. surgery versus chemoradiotherapy?] Patient's preferences, if known: [brief]
TASK	1. Write a 60-second standardised presentation: the one-liner, the key history, performance status, and the question. 2. Organise the data by specialty: imaging findings for radiology, pathology findings and markers, and staging if applicable. 3. List the options to be discussed, with the relevant considerations for each. 4. Prepare a template for documenting the MDT outcome. 5. Flag missing information needed for a decision, and who should obtain it before the meeting. 6. Note any patient-specific factors (frailty, social support, finances) the MDT should weigh. 6. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	State staging and markers exactly as in the reports, marking any staging version as [VERIFY EDITION], and include the patient's preferences where known, because MDT decisions must be accurate and patient-centred.
OUTPUT	The spoken presentation, the specialty data sections, the options table, the missing-information list, then the documentation template.
TWEAK	End with the 3 questions the MDT is most likely to ask the presenter.

How to get more out of it

- › State the question for the MDT first, not last. It focuses everyone's attention.
- › Have the images and pathology slides ready to display. MDTs stall while people search for them.
- › Document the decision and the rationale before leaving the room.
- › → NEXT #636 A three-line case summary

636 A three-line case summary

STANDARD

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT / MBBS INTERN

USE IT For the summary statement after history and examination, or at the start of rounds.

PERSONA	You are a clinical teacher who values concise, reasoning-rich summaries.
MY CONTEXT	Audience or purpose: [brief] Case details (de-identified): [PASTE] Specialty: [brief]
TASK	1. Write a 3-line summary: the one-liner (age band, sex, key background), the key clinical syndrome using semantic qualifiers (acute or chronic, unilateral or bilateral, progressive), and the key positive and negative findings. 2. Write an alternative 1-line version for handovers. 3. Explain which words carry the diagnostic reasoning.
RULES	Use only facts from the case, and don't insert a diagnosis into the summary unless it is confirmed, because summaries shape everyone's thinking.
OUTPUT	The 3-line summary, the 1-line version, then the reasoning note.

PRO TIP Semantic qualifiers ("acute, painful, unilateral") turn a summary into a diagnostic tool.

TRY IT WITH Notes on a 45-year-old man with 3 weeks of fever, weight loss and a right-sided pleural effusion.

→ **NEXT #639** Justify a differential diagnosis for a case

637 Organise a messy history for presentation

STANDARD

2.6 CASE PRESENTATIONS & GRAND ROUNDS · MBBS STUDENT / PG RESIDENT

USE IT When your clerking notes are complete but disorganised.

PERSONA	You are a clinical examiner who knows what a well-presented history sounds like.
MY CONTEXT	My clerking notes (de-identified): [PASTE] Specialty and presenting problem: [brief]
TASK	1. Reorganise the notes into the standard format: presenting complaint, HPI (chronological), past, drug, allergy, family, personal and social history, and review of systems. 2. Move the pertinent positives and negatives into the HPI. 3. Flag the gaps in the history that the audience will ask about. 4. Write the spoken version (about 2 minutes).
RULES	Use only information in my notes, and list the gaps rather than filling them in, because invented history is a serious error.
OUTPUT	The organised history, the gap list, then the spoken version.

PRO TIP For each symptom, cover onset, duration, progression and associated features. Examiners listen for them.

TRY IT WITH Notes on an elderly woman with breathlessness, leg swelling and reduced urine output.

→ **NEXT #638** Present examination findings clearly

638 Present examination findings clearly

STANDARD

2.6 CASE PRESENTATIONS & GRAND ROUNDS · MBBS STUDENT / PG RESIDENT

USE IT When presenting examination findings on rounds or in a case discussion.

PERSONA	You are a clinical examiner who values focused, organised findings.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Examination findings (de-identified): [PASTE] Presenting problem: [brief]
TASK	1. Order the findings: general appearance, vitals, general examination, then the relevant system first and other systems briefly. 2. Highlight the findings that support or oppose the leading diagnoses. 3. Include pertinent negatives. 4. Write the spoken version (about 90 seconds).
RULES	Report the findings as documented, and don't add findings that weren't examined, because examiners test what you actually elicited.
OUTPUT	The organised findings, the highlight list, then the spoken version.

PRO TIP Say the relevant negatives aloud: "no clubbing, no lymphadenopathy". They show a targeted examination.

TRY IT WITH Findings in a patient with suspected infective endocarditis.

→ **NEXT #636** A three-line case summary

639 Justify a differential diagnosis for a case

STANDARD

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT / MBBS STUDENT

USE IT For the differential-diagnosis part of a case presentation.

PERSONA	You are a diagnostician who teaches structured differential reasoning.
MY CONTEXT	Case summary with key findings: [PASTE] My provisional diagnosis and differentials: [list]
TASK	1. Rank the differentials by likelihood, with points for and against each from the case. 2. Identify the can't-miss diagnoses and whether they have been excluded. 3. Suggest the discriminating investigation for the top 2. 4. Write the spoken justification (about 2 minutes).
RULES	Reason step by step and use only the case facts, marking guideline-based criteria [VERIFY], because the audience will challenge each point. This supports learning, not the treating team's decisions.
OUTPUT	A differential table (diagnosis, for, against, next test), the can't-miss check, then the spoken version.

PRO TIP A short, well-justified differential beats a long, unranked list.

TRY IT WITH A young woman with fever, arthralgia, rash and pancytopenia.

→ **NEXT #640** Present and justify the investigations

640 Present and justify the investigations

STANDARD

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT

USE IT When presenting what tests were done and why.

PERSONA	You are a senior clinician who expects every test to answer a question.
MY CONTEXT	Investigations done with results and dates: [PASTE] Differential diagnosis: [list]
TASK	1. Group the tests: confirming the diagnosis, excluding differentials, assessing severity, and baseline for treatment. 2. For each important test, state the question it answered. 3. Present the abnormal results with trends in a compact table. 4. Identify any test that wasn't needed, or one that is still pending.
RULES	Report the values with units and reference ranges as in my data, because the audience interprets the numbers.
OUTPUT	The grouped test table, the trend table, then the needed and pending notes.

PRO TIP Present trends, not single values: "creatinine 1.2 → 2.8 → 1.6 over 5 days".

TRY IT WITH Investigations in a patient with suspected tuberculous meningitis.

→ **NEXT #641** Present the management plan and its reasoning

641 Present the management plan and its reasoning

STANDARD

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT

USE IT For the management part of a case presentation or rounds.

PERSONA	You are a senior clinician who teaches management reasoning.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Diagnosis, current status and management given: [PASTE] Guidelines or protocols followed: [list]
TASK	1. Present the management by problem (a problem-based list). 2. For each key decision, state the reasoning and the evidence or guideline behind it, marked [VERIFY]. 3. State the monitoring plan and the escalation criteria. 4. State the discharge or follow-up plan.
RULES	Describe the management as documented and don't recommend changes to real patient care, because this is a teaching presentation.
OUTPUT	The problem-based plan, the reasoning notes, the monitoring and escalation, then the follow-up.

PRO TIP A problem list (e.g. "1. Septic shock 2. AKI 3. Hyperglycaemia") keeps management presentations organised.

TRY IT WITH Management of a patient with diabetic ketoacidosis and pneumonia.

→ **NEXT #649** A 10–12 slide case-presentation deck

642 ISBAR handover for a patient

STANDARD

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT / MBBS INTERN / NURSE

USE IT For shift handovers and inter-team transfers.

PERSONA	You are a patient-safety educator who teaches structured handover.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Patient details (de-identified), current status and pending issues: [PASTE] Receiving team: [brief]
TASK	1. Write the handover in ISBAR format: Identify, Situation, Background, Assessment, Recommendation. 2. Highlight pending results, tasks and "watch for" items. 3. Include escalation criteria (if X, then do Y). 4. Keep it speakable in 60–90 seconds, and add a one-line code-status or goals-of-care note if relevant.
RULES	Include only verified information, and state uncertainty explicitly, because handover errors are a major cause of patient harm.
OUTPUT	The ISBAR handover, then the pending-task checklist.

PRO TIP Ask the receiving doctor to read back the key tasks. Read-back catches errors.

TRY IT WITH Night handover of a post-operative patient with falling urine output.

→ **NEXT #645** Present a case to a senior on the phone

643 Ward-round presentation (SOAP)

STANDARD

2.6 CASE PRESENTATIONS & GRAND ROUNDS · MBBS INTERN / PG RESIDENT

USE IT For presenting your patients on daily ward rounds.

PERSONA	You are a consultant who likes concise, organised ward-round presentations.
MY CONTEXT	Patient's data for today (de-identified): overnight events, symptoms, vitals, examination, new results and current treatment: [PASTE]
TASK	1. Write the presentation in SOAP format: Subjective, Objective (vitals, examination, results with trends), Assessment by problem, and Plan by problem. 2. Keep it to about 60 seconds for a stable patient and 2 minutes for a complex one. 3. Put the questions for the consultant at the end.
RULES	Present the data exactly and flag any abnormal result that needs a decision, because rounds are where decisions happen.
OUTPUT	The SOAP presentation, then the questions for the consultant.

PRO TIP Start with overnight events and anything urgent. Consultants want to know what's changed first.

TRY IT WITH Day 3 of admission for community-acquired pneumonia with new-onset atrial fibrillation.

→ **NEXT #644** Systems-based ICU rounds presentation

644 Systems-based ICU rounds presentation

STANDARD

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT (CRITICAL CARE, ANAESTHESIA, MEDICINE)

USE IT For presenting critically ill patients on ICU rounds.

PERSONA	You are an intensivist who expects systematic, concise ICU presentations.
MY CONTEXT	Patient's ICU data (de-identified), system by system: [PASTE] Day of ICU stay and the admitting diagnosis: [brief]
TASK	1. Present in a systems format: neurological (sedation), respiratory (ventilator settings, ABG), cardiovascular (haemodynamics, vasopressors), renal (fluid balance), GI and nutrition, haematology, infection (cultures, antibiotic day), lines and tubes, and prophylaxis. 2. Close with a problem list and today's goals. 3. Include a daily checklist (e.g. sedation hold, ventilator weaning, lines, DVT and ulcer prophylaxis, feeding).
RULES	Report numbers and settings exactly as recorded, because ICU decisions depend on precise values.
OUTPUT	The systems presentation, the problem list with goals, then the checklist.

PRO TIP Always state the antibiotic day and the culture results. It's the most frequently asked ICU rounds question.

TRY IT WITH Day 4 of septic shock from a urinary source, ventilated, on noradrenaline.

→ **NEXT #643** Ward-round presentation (SOAP)

645 Present a case to a senior on the phone

STANDARD

2.6 CASE PRESENTATIONS & GRAND ROUNDS · MBBS INTERN / PG RESIDENT

USE IT When calling a senior or another department about a patient, especially at night.

PERSONA	You are a senior resident who receives dozens of phone referrals and knows what a good one sounds like.
MY CONTEXT	Patient details (de-identified), the concern and the relevant data: [PASTE] Who I'm calling and what I need: [advice / review / transfer]
TASK	1. Write the call in SBAR: Situation (who I am, the patient, the concern), Background, Assessment (vitals and key findings, and my impression), Recommendation or Request (exactly what I need). 2. Keep it under 60 seconds. 3. List the data to have ready before dialling (vitals, recent results, current drugs, allergies).
RULES	State my specific request clearly, and include the most worrying finding early, because vague calls lead to vague responses.
OUTPUT	The SBAR script, then the "have ready" checklist.

PRO TIP Start with the request if it's urgent: "I need you to see a patient with a falling GCS now."

TRY IT WITH Calling a surgical resident about a ward patient with suspected acute abdomen.

→ **NEXT #642** ISBAR handover for a patient

646 Case presentation built around images

STANDARD

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT

USE IT When your case hinges on imaging, ECGs, fundoscopy, skin findings or histology.

PERSONA	You are a clinician-educator who uses images to teach reasoning.
MY CONTEXT	The case summary and the key images (described or uploaded, de-identified): [PASTE / UPLOAD] Audience: [brief]
TASK	1. Place each image at the point in the story where it changed thinking. 2. For each, write "what do you see?" prompts and the systematic description. 3. Plan annotated versions with arrows and labels. 4. Link the image findings to the differential.
RULES	De-identify all images (names, dates, hospital numbers) and describe the findings conservatively, marking uncertain ones [CONFIRM WITH REPORT], because images carry hidden identifiers and over-reading misleads.
OUTPUT	An image placement table, the prompts and descriptions, the annotation plan, then the reasoning links.

PRO TIP Show the image first without comment. Let the audience look before you speak.

TRY IT WITH A case of miliary tuberculosis with serial chest X-rays and a CT.

→ **NEXT #649** A 10–12 slide case-presentation deck

647 Teaching points from a case

STANDARD

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT / FACULTY

USE IT To close any case presentation.

PERSONA	You are a clinical educator who distils cases into memorable lessons.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Case summary and outcome: [PASTE] What went well, and what was difficult: [brief]
TASK	1. Write 3–5 teaching points, each one sentence, specific and actionable. 2. Classify them: diagnostic, management, systems or communication. 3. Link each to evidence or a guideline where relevant, marked [VERIFY]. 4. Suggest one question to test the audience on each point.
RULES	Make every point follow from this case, not general textbook facts, because case-specific lessons stick.
OUTPUT	The teaching points with classification, the evidence links, then the test questions.

PRO TIP The best teaching point is often "what I would do differently".

TRY IT WITH A case of hyponatraemia that worsened with normal saline (SIADH).

→ **NEXT #649** A 10–12 slide case-presentation deck

648 A brief literature review for my case

STANDARD

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT

USE IT When your case discussion needs 2–3 slides on what the literature says.

PERSONA	You are an academic clinician who summarises literature quickly and accurately.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The case's key question (diagnosis, management or rarity): [brief] Sources I found (uploaded or with details): [UPLOAD / LIST]
TASK	1. Summarise what the sources say about the key question. 2. Compare the case with published cases or series, if available. 3. Summarise the current guideline position, marked [VERIFY]. 4. Write 2–3 slides with speaker notes.
RULES	Use only the sources I provided, and mark anything else [FIND SOURCE], because invented citations in case presentations are easily exposed.
OUTPUT	The summary, the comparison, the guideline note, then the slides.

PRO TIP A search limited to case reports in PubMed often finds similar cases quickly.

TRY IT WITH A case of isoniazid-induced pellagra.

→ **NEXT #647** Teaching points from a case

649 A 10–12 slide case-presentation deck

STANDARD

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT

USE IT When your case presentation needs slides.

PERSONA	You are a presentation coach for clinical meetings.
MY CONTEXT	My structured case presentation: [PASTE] Time: [minutes] Images and tables available: [list]
TASK	1. Plan 10–12 slides: title and one-liner; history; examination; summary; differential; investigations (with a trend table); imaging; diagnosis; management and course; discussion; teaching points. 2. Write a message title and brief content for each slide. 3. Suggest where the reveals and pauses should be. 4. Check that the timing fits.
RULES	Keep the slides de-identified, with no names, dates or hospital numbers, including in image metadata, because slides get shared.
OUTPUT	The slide table (slide, title, content, visual, time), then the reveal plan.

PRO TIP Use a lab-trend table rather than pasting lab reports. It's readable and shows the trajectory.

TRY IT WITH A case of severe falciparum malaria with AKI.

→ **NEXT #631** Departmental case presentation, start to finish

650 Case discussion on a management controversy

STANDARD

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT / FACULTY

USE IT When a case raises a genuine management dilemma worth debating.

PERSONA	You are a senior clinician who moderates balanced case-based debates.
MY CONTEXT	The case and the decision point: [PASTE] The options considered, and what was chosen: [brief] Sources for each option: [UPLOAD / LIST]
TASK	1. Frame the dilemma as a clear question. 2. Present the evidence for each option, with its strengths and weaknesses, marked [VERIFY]. 3. Consider patient factors, preferences and resources. 4. Explain what was done and why, and the outcome. 5. Write the discussion questions for the audience.
RULES	Present both sides fairly and don't judge the treating team with hindsight, because outcome bias distorts learning.
OUTPUT	The dilemma, the evidence table, the patient factors, the decision explanation, then the questions.

PRO TIP Ask the audience to vote before and after the discussion. Changed votes show learning.

TRY IT WITH Anticoagulation in a patient with a recent GI bleed and new pulmonary embolism.

→ **NEXT #648** A brief literature review for my case

651 Present a case with an ethical dilemma

STANDARD

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT / FACULTY

USE IT For cases involving consent, capacity, end-of-life care, resource allocation or confidentiality.

PERSONA	You are a clinical ethicist who facilitates case discussions using principled frameworks.
MY CONTEXT	The case (de-identified) and the ethical question: [PASTE] Relevant Indian law or guidelines I'm aware of: [list]
TASK	1. State the ethical question clearly. 2. Analyse it using a framework (e.g. the four principles, or the four-box method: medical indications, patient preferences, quality of life, contextual features). 3. Identify the stakeholders and their perspectives. 4. Outline the options with their ethical and legal considerations, marked [VERIFY LEGAL POSITION]. 5. Write the discussion questions.
RULES	Present the perspectives respectfully and don't give legal advice, and mark legal points for verification, because ethics cases involve real legal and emotional stakes.
OUTPUT	The question, the framework analysis, the stakeholders, the options table, then the questions.

PRO TIP Invite a nurse, social worker or ethicist to the discussion. Ethics cases benefit from several perspectives.

TRY IT WITH A family requesting withdrawal of ventilation for an elderly patient without a documented advance directive.

→ **NEXT #647** Teaching points from a case

652 Present a rare diagnosis without losing the audience

STANDARD

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT

USE IT When your case is a rare disease that most of the audience hasn't seen.

PERSONA	You are a clinical teacher who makes rare cases relevant to everyday practice.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The case (de-identified) and the rare diagnosis: [PASTE] Audience: [brief]
TASK	1. Frame the case around the common presentation that hid the rare diagnosis. 2. Explain the diagnostic clues that pointed to it. 3. Give a brief overview of the condition (5 bullets), marked [VERIFY]. 4. Draw lessons that apply to common practice (e.g. when to think beyond the common).
RULES	Keep the overview accurate and brief, because rare-disease details are easily misremembered.
OUTPUT	The framing, the clues, the overview, then the lessons.

PRO TIP "When the common diagnosis doesn't fit, look again" is the universal lesson of rare cases.

TRY IT WITH Adult-onset Still's disease presenting as fever of unknown origin.

→ **NEXT #634** Short "interesting case" presentation with teaching

653 Learning from a delayed or missed diagnosis

STANDARD

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT / FACULTY

USE IT For teaching sessions on diagnostic error, using a de-identified case.

PERSONA	You are a diagnostic-reasoning expert who teaches cognitive bias without blame.
MY CONTEXT	The case timeline and the eventual diagnosis: [PASTE] The points where the diagnosis could have been made: [brief]
TASK	1. Map the diagnostic journey on a timeline. 2. Identify the cognitive biases likely at play (anchoring, premature closure, availability, framing) at each point. 3. Identify the system factors (workload, handover, test delays). 4. Suggest de-biasing strategies and system fixes. 5. Write the teaching points.
RULES	Frame the analysis around thinking processes and systems, not individuals, and keep it de-identified, because blame shuts down learning.
OUTPUT	The timeline, the bias table, the system factors, the strategies, then the teaching points.

PRO TIP A "diagnostic time-out" (asking "what else could this be?") is a simple, teachable de-biasing tool.

TRY IT WITH A pulmonary embolism initially treated as pneumonia for 3 days.

→ **NEXT #633** Morbidity and mortality (M&M) presentation

654 The questions the head of department will ask

STANDARD

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT

USE IT The day before presenting a case to your unit head or at a departmental meeting.

PERSONA	You are a senior professor known for incisive case-discussion questions.
MY CONTEXT	My case presentation: [PASTE] My specialty and year: [brief]
TASK	1. List 12 likely questions: on the history, examination technique, differential, investigations, management choices, recent evidence, and "what would you do if...". 2. Give a model answer for each, grounded in the case, with [VERIFY] for guideline facts. 3. Mark the 3 hardest questions and suggest how to prepare for them.
RULES	Base questions on the actual case and my level, and admit uncertainty in the model answers where appropriate, because bluffing is easily detected.
OUTPUT	A question-and-answer table, then the preparation plan for the hardest three.

PRO TIP "I don't know, but I would find out by..." is a respectable answer. Guessing isn't.

TRY IT WITH A presentation of a young patient with a first episode of status epilepticus.

→ **NEXT #631** Departmental case presentation, start to finish

655 Root-cause analysis for an M&M case

STANDARD

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT / FACULTY

USE IT When preparing the analysis part of an M&M or quality-improvement meeting.

PERSONA	You are a patient-safety specialist trained in root-cause analysis.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The event timeline (de-identified): [PASTE] Initial thoughts on contributing factors: [brief]
TASK	1. Build a fishbone (Ishikawa) diagram description with categories: people, processes, equipment, environment, communication and policies. 2. Apply "5 whys" to the 2 most important contributing factors. 3. Distinguish root causes from contributing factors. 4. Propose SMART corrective actions.
RULES	Focus on systems and processes, not individual blame, and base the analysis on the facts in the timeline, because RCA aims at prevention, not punishment.
OUTPUT	The fishbone description, the 5-whys chains, the root-cause list, then the actions table.

PRO TIP If your "root cause" is "human error", ask "why" at least twice more.

TRY IT WITH A delayed administration of antibiotics to a septic patient in the emergency department.

→ **NEXT #633** Morbidity and mortality (M&M) presentation

656 Present a follow-up update on a previous case

STANDARD

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT

USE IT When the department wants to hear how a previously presented patient did.

PERSONA	You are a clinical teacher who closes the loop on cases.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The original presentation summary: [PASTE] The follow-up course and outcome: [PASTE]
TASK	1. Remind the audience of the case in 2 lines. 2. Summarise the course since then: treatment response, complications and the current status. 3. Say whether the outcome confirms or changes the original diagnosis or plan. 4. Draw updated lessons.
RULES	Keep it de-identified and factual, and acknowledge it plainly if the original reasoning was wrong, because honest follow-up is powerful teaching.
OUTPUT	The reminder, the course summary, the diagnostic check, then the updated lessons.

PRO TIP Follow-up presentations teach humility and the natural history of disease. Do more of them.

TRY IT WITH A 6-month follow-up of a patient with autoimmune encephalitis.

→ **NEXT #647** Teaching points from a case

657 Turn a case into a teaching session for MBBS students

STANDARD

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT / FACULTY

USE IT When teaching undergraduates using a real ward case.

PERSONA	You are a clinical tutor for MBBS students.
MY CONTEXT	The case (de-identified): [PASTE] Students' year and what they have learned so far: [brief] Time: [minutes]
TASK	1. Choose 3 learning objectives appropriate for the students' level. 2. Plan a Socratic session: questions at each stage of the case. 3. Include one clinical skill to practise (e.g. eliciting a sign) if possible. 4. Link the case to their exam topics (e.g. NEET-PG and university exams). 5. Write 5 check questions.
RULES	Keep the questions at the students' level and correct misconceptions gently, and obtain the patient's consent for bedside teaching, because respect for patients is itself a lesson.
OUTPUT	The objectives, the Socratic question plan, the skill station, the exam links, then the check questions.

PRO TIP Ask students to commit to an answer before you reveal it. Commitment drives learning.

TRY IT WITH A patient with mitral stenosis for third-year MBBS students.

→ **NEXT #647** Teaching points from a case

658 Link a case to a recent paper

STANDARD

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT / FACULTY

USE IT When a case presentation would be stronger with one recent, relevant study.

PERSONA	You are an academic clinician who connects bedside cases to current evidence.
MY CONTEXT	The case summary and its key clinical question: [PASTE] The recent paper I found (upload or paste the abstract): [UPLOAD / PASTE] Audience: [brief]
TASK	1. Summarise the paper in 3 lines (design, population, key result). 2. Explain how it applies, or doesn't apply, to this patient. 3. Note the paper's main limitation for this context. 4. Write one slide connecting the case and the paper.
RULES	Use only the paper I provided, and don't overstate how directly it applies to one patient, because case-to-trial reasoning is easy to overstretch.
OUTPUT	The summary, the applicability note, the limitation, then the slide text.

PRO TIP One well-chosen recent paper makes a case discussion feel current and evidence-based.

TRY IT WITH A patient with severe community-acquired pneumonia and a recent trial of corticosteroids.

→ **NEXT #648** A brief literature review for my case

QUICK PROMPTS · 2.6

659 Opening one-liner for my case

QUICK

2.6 CASE PRESENTATIONS & GRAND ROUNDS · MBBS INTERN / PG RESIDENT

Act as a clinical teacher. Write the opening one-liner for my case presentation from these details, [PASTE KEY DETAILS], stating the age band, sex, the key relevant background and the presenting problem with its duration, in under 30 words and without giving away the diagnosis.

TRY IT WITH A 58-year-old diabetic woman with 5 days of fever and right upper-quadrant pain.

660 Summarise my case in three lines

QUICK

2.6 CASE PRESENTATIONS & GRAND ROUNDS · MBBS INTERN / PG RESIDENT

Act as a senior physician. Summarise this case, [PASTE CASE NOTES], in 3 lines using semantic qualifiers (acute or chronic, unilateral or bilateral, progressive) and the key positives and negatives, without naming an unconfirmed diagnosis. Flag anything you are unsure of so I can check it.

TRY IT WITH Notes on a teenager with fever, sore throat, rash and joint pains.

661 Which negatives matter?

QUICK

2.6 CASE PRESENTATIONS & GRAND ROUNDS · MBBS STUDENT / PG RESIDENT

Act as a clinical examiner. For a patient presenting with [PRESENTATION] and a leading diagnosis of [DIAGNOSIS], list the 6–8 pertinent negatives I should mention in history and examination, explaining in one line which differential each one argues against. Flag anything you are unsure of so I can check it.

TRY IT WITH Presentation: acute chest pain. Leading diagnosis: aortic dissection.

662 Why is this case worth presenting?

QUICK

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT

Act as a case-meeting coordinator. From this case summary, [PASTE SUMMARY], tell me in 3 sentences what makes the case worth presenting (a rare diagnosis, an atypical presentation, a diagnostic twist or a management lesson) and what the audience should take away.

TRY IT WITH A patient with hypokalaemic paralysis who turned out to have thyrotoxicosis.

663 Build the problem list

QUICK

2.6 CASE PRESENTATIONS & GRAND ROUNDS · MBBS INTERN / PG RESIDENT

Act as a physician on rounds. From this case, [PASTE NOTES], create a prioritised, numbered problem list (active problems first, then chronic and background issues), with one line on the current status of each, and without adding problems not supported by the notes.

TRY IT WITH An elderly patient with pneumonia, AKI, hyponatraemia, diabetes and delirium.

664 Describe the vitals trend

QUICK

2.6 CASE PRESENTATIONS & GRAND ROUNDS · MBBS INTERN / PG RESIDENT

Act as a clinical teacher. Describe the trend in these vital signs, [PASTE VITALS WITH TIMES], in 2–3 spoken sentences for a case presentation, highlighting clinically important changes and any that should trigger escalation. Flag anything you are unsure of so I can check it.

TRY IT WITH HR 88 → 104 → 118, BP 124/80 → 106/68 → 92/58, and RR 18 → 24 over 12 hours.

665 Present the lab results concisely

QUICK

2.6 CASE PRESENTATIONS & GRAND ROUNDS · MBBS INTERN / PG RESIDENT

Act as a senior resident. Convert these lab results, [PASTE RESULTS WITH DATES], into a compact trend table with units, highlight the abnormal values, and write one spoken sentence summarising what the labs show overall. Flag anything you are unsure of so I can check it.

TRY IT WITH Five days of creatinine, potassium, sodium, haemoglobin and platelet counts.

666 Hospital course as a timeline

QUICK

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT

Act as a clinical documentation expert. Turn this hospital course, [PASTE NOTES], into a day-by-day timeline of key events, investigations, treatment changes and responses, using relative days (Day 0, Day 1...) and removing identifiers, for one slide. Flag anything you are unsure of so I can check it.

TRY IT WITH A 10-day admission for dengue with warning signs and plasma leakage.

667 The final diagnosis statement

QUICK

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT

Act as a senior physician. Write the complete final-diagnosis statement for this case, [PASTE DETAILS], in standard format: the primary diagnosis with its aetiology and severity, then its complications, then the comorbidities, as used in discharge summaries and case presentations.

TRY IT WITH A patient with community-acquired pneumonia complicated by septic shock and AKI, with diabetes.

668 One learning point from my case

QUICK

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT

Act as a clinical educator. From this case, [PASTE BRIEF CASE], write the single most valuable learning point in one sentence of under 25 words, specific and actionable, and one question to test the audience on it. Flag anything you are unsure of so I can check it.

TRY IT WITH A patient whose hypoglycaemia was caused by a sulphonylurea in renal failure.

669 SBAR script for a phone call

QUICK

2.6 CASE PRESENTATIONS & GRAND ROUNDS · MBBS INTERN / PG RESIDENT / NURSE

Act as a patient-safety trainer. Write an SBAR phone script of under 60 seconds for calling [WHO] about this patient, [PASTE DETAILS], putting the most worrying finding early and ending with a specific request. Flag anything you are unsure of so I can check it.

TRY IT WITH Calling the ICU resident about a ward patient with a falling GCS and rising respiratory rate.

670 A title for my case presentation

QUICK

2.6 CASE PRESENTATIONS & GRAND ROUNDS · PG RESIDENT

Act as a case-meeting coordinator. Write 4 titles for my case presentation, [ONE-LINE SUMMARY], that intrigue without giving away the diagnosis, such as "Fever that wouldn't settle", each under 10 words, plus one descriptive subtitle. Flag anything you are unsure of so I can check it.

TRY IT WITH A case of infective endocarditis presenting as a stroke in a young adult.

2.7

Delivery, Q&A & teaching

Content earns attention; delivery keeps it. Handle questions with honesty, teach with methods that work, and give feedback that helps people grow.

30

prompts

4

Master

17

Standard

9

Quick

HOW TO USE THIS SECTION

Use it in the last days before any talk, and whenever you teach or give feedback.

Start with the Master prompts ([#671–674](#)) for the full workflow; use the Standard prompts ([#675–691](#)) for one task at a time; reach for the Quick prompts ([#692–700](#)) when you have two minutes. Tap any title below to jump to it, and follow → **NEXT** to move to the next step.

[#671](#) Plan a small-group teaching session

[#672](#) Structured bedside teaching session

[#673](#) Handle any question after my talk

[#674](#) Improve my voice, pace and presence

[#675](#) The one-minute preceptor, applied

[#676](#) Teach learners to present with SNAPPS

[#677](#) Design a flipped-classroom session

[#678](#) Write a simulation scenario

[#679](#) Teach a clinical skill step by step

[#680](#) Give feedback that helps

[#681](#) Orientation teaching for new interns

[#682](#) Engage a large lecture of 100+ students

[#683](#) Handle difficult small-group dynamics

[#684](#) Keep an online class engaged

[#685](#) Check learning during teaching

[#686](#) Build a teaching portfolio

[#687](#) Respond to a hostile or aggressive question

[#688](#) Answer "I don't know" with confidence

[#689](#) Exercises for voice and pace

[#690](#) Body language and stage presence

[#691](#) Peer observation of teaching

[#692](#) Rephrase an audience question

[#693](#) Handle an off-topic question

[#694](#) Use pauses to sound confident

[#695](#) Fix wandering eye contact

[#696](#) Remind myself to slow down

[#697](#) Opening activity for a teaching session

[#698](#) An exit-ticket question

[#699](#) One sentence of specific feedback

[#700](#) Praise that reinforces the right behaviour

Plan a small-group teaching session

2.7 DELIVERY, Q&A & TEACHING · PG RESIDENT / FACULTY

USE IT

When you teach interns, students or juniors in a tutorial, skills lab or classroom.

WHY

Most clinical teaching is a mini-lecture with no plan. A structured session (clear objectives, activation of prior knowledge, active learning, practice with feedback and a quick check of learning) teaches far more in the same time, and it gives you evidence for your teaching portfolio.

COPY THE PROMPT BELOW

PERSONA	You are a medical educator trained in instructional design who designs short, active small-group sessions for clinical learners, and you are known for making every minute count.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] Topic: [TOPIC] Learners: [e.g. 8 final-year MBBS students / interns / first-year residents] What they already know: [brief] Time and setting: [e.g. 45 minutes, seminar room with a whiteboard] Resources available: [projector, mannequin, cases, handouts]
TASK	1. Write 3 learning objectives with Bloom's verbs. 2. Plan a warm-up that activates prior knowledge (a question, a case or a quick quiz). 3. Plan the core content as active learning: case-based discussion, think-pair-share, or worked examples, not a lecture. 4. Plan a practice activity with feedback (applying the concept to a new case or skill). 5. Plan a closing check of learning (an exit question or mini-quiz) and a summary. 6. Build a minute-by-minute lesson plan with the teacher's actions and the learners' actions. 7. Suggest adaptations for a stronger and a weaker group. 8. Suggest how to collect feedback on the session.
RULES	Keep teacher talk to under a third of the session time, because learners retain far more from doing than from listening. Keep all clinical content accurate and mark guideline specifics [VERIFY].
OUTPUT	The objectives, a lesson-plan table (time, activity, teacher, learners, materials), the adaptations, then the feedback method.
TWEAK	End with the single question that best tests whether the session achieved its main objective.

How to get more out of it

- › Start with a question, not a slide. It shows you what the group already knows.
- › Plan less content than you think. Active learning takes more time, and teaches more.
- › Keep your lesson plans. Over a year they become a reusable teaching library.
- › → **NEXT #675** The one-minute preceptor, applied

Structured bedside teaching session

2.7 DELIVERY, Q&A & TEACHING · PG RESIDENT / FACULTY

USE IT

When teaching at the bedside on rounds or in a dedicated teaching session.

WHY

The bedside is the richest place to teach clinical skills and reasoning, and the easiest place to embarrass a patient or bore a learner. A plan covering patient consent, learner roles, focused skills and a debrief away from the bed makes bedside teaching safe, respectful and effective.

COPY THE PROMPT BELOW

PERSONA	You are a senior clinical teacher known for respectful, high-yield bedside teaching, who uses frameworks such as SNAPPS and the one-minute preceptor.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] The patient's condition and key signs (de-identified): [brief] Learners and their level: [brief] Time: [e.g. 20 minutes] Skill or concept to teach: [e.g. examining for a pleural effusion]
TASK	1. Plan the pre-briefing away from the bed: objectives, roles and etiquette. 2. Plan patient consent and involvement, and how to talk about findings in front of the patient. 3. Plan the bedside part: a focused history or examination by one learner, others observing with a checklist, and the teacher demonstrating key signs. 4. Plan questions that develop reasoning without grilling learners in front of the patient. 5. Plan the debrief away from the bed: findings, reasoning, differential and teaching points. 6. Plan how to thank and reassure the patient. 7. Plan a follow-up learning task. 8. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Obtain the patient's consent, protect their dignity and privacy, and discuss sensitive matters away from the bed, because patients' trust in the teaching hospital depends on it. Keep the learners' mistakes for the debrief, not the bedside.
OUTPUT	A session plan with phases and timings, the question list, the observation checklist, then the follow-up task.
TWEAK	End with 3 phrases to use at the bedside that teach without frightening the patient.

How to get more out of it

- › Ask the patient's permission in front of the learners. It models respect.
- › Let one learner examine while others watch for specific things. Everyone stays engaged.
- › Five minutes of focused bedside teaching beats thirty minutes of standing around.
- › → **NEXT #676** Teach learners to present with SNAPPS

Handle any question after my talk

2.7 DELIVERY, Q&A & TEACHING · PG RESIDENT / FACULTY

USE IT

Before any talk, presentation or viva-style discussion where questions will come.

WHY

The Q&A is where credibility is won or lost. Speakers who listen fully, clarify, answer briefly, admit uncertainty honestly and handle hostile or off-topic questions calmly are remembered as experts. Preparation makes that possible.

COPY THE PROMPT BELOW

PERSONA	You are a communication coach who trains doctors to handle audience questions with confidence and honesty.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] What I most want to get out of this: [one line] My talk topic and key content: [PASTE SUMMARY] Audience and the likely tough questioners: [brief] Known weaknesses or controversies in my content: [list]
TASK	1. Teach a response framework: listen fully, pause, rephrase or clarify, answer briefly (under 60 seconds), and check that the question was answered. 2. Prepare answers to 10 likely questions, including 3 difficult ones. 3. Script responses for special situations: a question I can't answer, a hostile or aggressive question, an off-topic question, a long statement disguised as a question, and a question that exposes a real error in my talk. 4. Plan how to manage time when many people want to ask. 5. Plan how to end the Q&A gracefully, including a final sentence that restates my key message. 6. Suggest 2 backup slides that would answer the most likely difficult questions. 6. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Model honesty: admitting "I don't know" and offering to follow up, and acknowledging real errors, because audiences forgive uncertainty but not bluffing. Ground the answers in my content, marking facts to verify as [VERIFY].
OUTPUT	The framework, the Q&A table, the special-situation scripts, the time-management plan, then the closing line.
TWEAK	End with the 3 questions I most fear, and a calm 30-second answer to each.

How to get more out of it

- › Repeat or rephrase the question before answering. It helps the audience and buys you thinking time.
- › Answer the question asked, not the one you prepared.
- › A short answer followed by "does that answer your question?" is better than a long one.
- › → **NEXT #687** Respond to a hostile or aggressive question

Improve my voice, pace and presence

2.7 DELIVERY, Q&A & TEACHING · PG RESIDENT / FACULTY

USE IT

When feedback says you speak too fast, too softly, or read your slides, or when you want to present with more authority.

WHY

Delivery determines whether good content lands. Pace, pauses, vocal variety, eye contact and movement are skills that improve quickly with specific practice and feedback, and they transfer to rounds, vivas and patient communication.

COPY THE PROMPT BELOW

PERSONA	You are a speech and presentation coach who works with clinicians and uses recordings and targeted exercises.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] My delivery issues as I see them, and feedback I've received: [list] A transcript or description of a recent talk (optional): [PASTE] Upcoming talks: [brief] Time I can spend practising per week: [e.g. 2 hours]
TASK	1. Diagnose the likely causes of my issues (nerves, over-preparation of content, under-rehearsal, habits). 2. Give specific exercises for pace (target words per minute, marked pauses), voice (projection, variety), filler words, eye contact and gestures. 3. Give a practice plan over 2–3 weeks, with measurable targets. 4. Explain how to self-assess with recordings (what to watch for). 5. Give a 5-minute warm-up routine before a talk. 6. Give tips for specific settings: a large hall, a small room, online. 7. Suggest how to get useful feedback from one trusted colleague after each talk. 7. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Keep the exercises practical and achievable within my time, and focus on 2–3 issues at a time, because trying to fix everything at once fixes nothing.
OUTPUT	The diagnosis, an exercises table (issue, exercise, target), the practice plan, the self-assessment guide, the warm-up, then the setting tips.
TWEAK	End with one change I can make in my very next talk that will have the biggest effect.

How to get more out of it

- › Record yourself on your phone. It's uncomfortable, and it's the fastest way to improve.
- › Mark pauses in your notes with a "/" symbol. Planned pauses slow you down naturally.
- › Speak to one person at a time for a sentence or two. It fixes wandering eye contact.
- › → **NEXT #584** Learn from feedback on my talk

675 The one-minute preceptor, applied

STANDARD

2.7 DELIVERY, Q&A & TEACHING · PG RESIDENT / FACULTY

USE IT When a learner presents a case to you in a busy OPD or ward.

PERSONA	You are a clinical teacher who uses the one-minute preceptor model efficiently.
MY CONTEXT	A case a learner might present to me (de-identified): [PASTE] The learner's level: [brief]
TASK	1. Walk through the 5 microskills with scripts for this case: get a commitment, probe for supporting evidence, teach a general rule, reinforce what was done well, and correct mistakes. 2. Write the likely learner responses and my follow-ups. 3. Keep the whole interaction to about 2 minutes.
RULES	Keep the teaching rule accurate and generalisable, and give the correction privately and constructively, because the learner's confidence matters in busy clinics.
OUTPUT	A 5-step script table (microskill, my words, learner's likely answer, follow-up).

PRO TIP "What do you think is going on?" is the most powerful teaching question in a busy clinic.

TRY IT WITH An intern presenting a patient with a first episode of seizure in the OPD.

→ **NEXT #676** Teach learners to present with SNAPPS

676 Teach learners to present with SNAPPS

STANDARD

2.7 DELIVERY, Q&A & TEACHING · PG RESIDENT / FACULTY

USE IT When you want learners to drive their own case presentations and reasoning.

PERSONA	You are a clinical educator who uses the learner-centred SNAPPS model.
MY CONTEXT	The learner's level and a sample case (de-identified): [brief]
TASK	1. Explain the SNAPPS steps briefly: Summarise, Narrow the differential, Analyse it, Probe the preceptor, Plan management, Select a learning issue. 2. Model a SNAPPS presentation of the case. 3. Write the preceptor's responses at each step. 4. Give a pocket card for learners to use.
RULES	Keep the clinical content of the example accurate, and emphasise that probing the preceptor is encouraged, because SNAPPS works only when learners feel safe to ask.
OUTPUT	The explanation, the modelled presentation, the preceptor responses, then the pocket card.

PRO TIP The "select a learning issue" step turns every case into self-directed learning.

TRY IT WITH A second-year MBBS student presenting a child with fever and a rash.

→ **NEXT #675** The one-minute preceptor, applied

677 Design a flipped-classroom session

STANDARD

2.7 DELIVERY, Q&A & TEACHING · FACULTY / SENIOR RESIDENT

USE IT When learners can prepare in advance and you want class time for application.

PERSONA	You are a medical educator experienced in flipped and blended learning.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Topic and learners: [brief] Class time: [minutes] Resources for pre-work (videos, readings, notes): [list or "suggest"]
TASK	1. Design the pre-work: 20–30 minutes, with a short check quiz. 2. Design the class: a quick quiz review, application activities (cases, problem-solving), and a wrap-up. 3. Plan for learners who didn't do the pre-work. 4. Plan the assessment and feedback.
RULES	Keep the pre-work short and clearly linked to the class activity, because long pre-work doesn't get done.
OUTPUT	The pre-work plan, the class plan with timings, the contingency, then the assessment.

PRO TIP Start the class with a question that only makes sense if people did the pre-work.

TRY IT WITH A flipped session on ABG interpretation for interns.

→ **NEXT #671** Plan a small-group teaching session

678 Write a simulation scenario

STANDARD

2.7 DELIVERY, Q&A & TEACHING · FACULTY / SENIOR RESIDENT

USE IT For skills-lab or in-situ simulation training.

PERSONA	You are a simulation educator who writes realistic, safe scenarios with structured debriefs.
MY CONTEXT	Learning objectives: [list] Learners and their team composition: [brief] Equipment or mannequin available: [brief] Duration: [e.g. 10-minute scenario + 20-minute debrief]
TASK	1. Write the scenario: a patient brief, the initial state (vitals, findings) and the expected actions. 2. Build the progression: how vitals change with the right and wrong actions. 3. Specify the confederate roles and cues. 4. Write the debrief plan (e.g. reactions, analysis, summary) with key questions. 5. Add a checklist of critical actions.
RULES	Base the expected actions on current guidelines marked [VERIFY], and maintain psychological safety during the debrief, because simulation should build confidence, not fear.
OUTPUT	The scenario sheet, the progression table, the roles, the debrief guide, then the checklist.

PRO TIP The debrief is where the learning happens. Plan twice as much time for it as for the scenario.

TRY IT WITH Anaphylaxis in a ward patient after an IV antibiotic, for interns and nurses.

→ **NEXT #680** Give feedback that helps

679 Teach a clinical skill step by step

STANDARD

2.7 DELIVERY, Q&A & TEACHING · PG RESIDENT / FACULTY

USE IT When teaching a procedure or examination skill to juniors.

PERSONA	You are a clinical skills tutor who uses Peyton's four-step approach.
MY CONTEXT	The skill: [e.g. arterial blood gas sampling, lumbar puncture, CPR] Learners and their level: [brief] Equipment: [brief]
TASK	1. Break the skill into steps with the key safety points. 2. Plan Peyton's 4 steps: demonstrate at normal speed, deconstruct (demonstrate with explanation), comprehension (the learner talks you through it while you perform), and performance (the learner performs). 3. Write the common errors and how to correct them. 4. Create a competency checklist for assessment.
RULES	Follow the standard technique and local protocols, marking any variation [CHECK LOCAL SOP], because skills are learned as they are first taught.
OUTPUT	The step list, the 4-step plan, the error list, then the checklist.

PRO TIP Step 3, where the learner explains while you do, is the step most teachers skip, and it's the most powerful.

TRY IT WITH Teaching arterial blood gas sampling to interns on a mannequin arm.

→ **NEXT #680** Give feedback that helps

680 Give feedback that helps

STANDARD

2.7 DELIVERY, Q&A & TEACHING · PG RESIDENT / FACULTY

USE IT After observing a learner's presentation, procedure or patient interaction.

PERSONA	You are a medical educator trained in constructive feedback models (Pendleton, ask-tell-ask, the feedback sandwich done well).
MY CONTEXT	What I observed (behaviours, not personality): [brief] The learner's level and goals: [brief] Setting (private or public): [brief]
TASK	1. Choose the feedback model best suited to the situation. 2. Write a feedback script: start by asking for self-assessment, reinforce specific strengths, name 1–2 specific improvements with examples, and agree an action plan. 3. Suggest phrasing that is specific and behavioural. 4. Plan a follow-up.
RULES	Keep the feedback specific, behavioural and private, focus on 1–2 improvements, and avoid vague praise or criticism, because vague or public feedback doesn't change behaviour.
OUTPUT	The chosen model, the script, the phrasing examples, then the follow-up plan.

PRO TIP "Your summary was clear because you used semantic qualifiers" teaches. "Good job" doesn't.

TRY IT WITH Feedback to an intern whose case presentation was disorganised but whose examination was thorough.

→ **NEXT #675** The one-minute preceptor, applied

681 Orientation teaching for new interns

STANDARD

2.7 DELIVERY, Q&A & TEACHING · PG RESIDENT / FACULTY

USE IT When a new batch of interns joins your department.

PERSONA	You are a department educator who designs practical orientation sessions.
MY CONTEXT	Department and the interns' roles: [brief] Time available: [e.g. 2 hours] Common problems new interns face here: [list]
TASK	1. Plan the session: introductions, roles and expectations, key protocols, documentation standards, escalation pathways and common pitfalls. 2. Include hands-on elements (e.g. writing a note, an SBAR call). 3. Create a one-page survival guide. 4. Plan a check-in after one week.
RULES	Keep protocols consistent with the department's approved versions, marking them [VERIFY WITH DEPARTMENT], and emphasise when to call for help, because new interns are most at risk of not escalating.
OUTPUT	The session plan, the hands-on activities, the survival guide, then the check-in plan.

PRO TIP "When in doubt, call" should be said more than once. New interns worry about disturbing seniors.

TRY IT WITH Orientation for 20 interns joining the medicine department.

→ **NEXT #671** Plan a small-group teaching session

682 Engage a large lecture of 100+ students

STANDARD

2.7 DELIVERY, Q&A & TEACHING · FACULTY / SENIOR RESIDENT

USE IT When you must lecture to a large MBBS batch.

PERSONA	You are an award-winning lecturer who keeps large classes engaged.
MY CONTEXT	Topic and duration: [brief] Batch size and year: [brief] Technology available: [projector, microphone, poll app, none]
TASK	1. Chunk the lecture into 10–15 minute segments, with an activity between segments. 2. Suggest activities that work with large groups: polls, think-pair-share, a show of hands, one-minute papers. 3. Plan a clinical hook for each segment. 4. Plan the end-of-lecture summary and a check of learning.
RULES	Keep each segment focused on one objective, because attention drops sharply after about 15 minutes of continuous lecturing.
OUTPUT	A segment plan with activities and timings, then the summary and check.

PRO TIP A 2-minute "turn to your neighbour and explain..." resets attention in any size of room.

TRY IT WITH A 60-minute lecture on the pharmacology of antihypertensives for 150 second-year students.

→ **NEXT #671** Plan a small-group teaching session

683 Handle difficult small-group dynamics

STANDARD

2.7 DELIVERY, Q&A & TEACHING · FACULTY / SENIOR RESIDENT

USE IT When a tutorial group has a dominant talker, a silent group or disengaged learners.

PERSONA	You are an experienced small-group facilitator.
MY CONTEXT	The group problem: [e.g. one learner dominates; others are silent; phones out] Group size and level: [brief]
TASK	1. Diagnose the likely causes. 2. Suggest facilitation techniques: structured turn-taking, directed questions, pair work, roles and ground rules. 3. Script how to redirect a dominant learner respectfully. 4. Script how to draw in quiet learners without putting them on the spot. 5. Suggest how to set ground rules at the next session.
RULES	Keep the strategies respectful of every learner, because humiliation damages the whole group's learning.
OUTPUT	The diagnosis, the techniques, the scripts, then the ground rules.

PRO TIP "Let's hear from someone who hasn't spoken yet" works better than naming the quiet person.

TRY IT WITH A group of 10 residents where one senior answers every question.

→ **NEXT #671** Plan a small-group teaching session

684 Keep an online class engaged

STANDARD

2.7 DELIVERY, Q&A & TEACHING · FACULTY / SENIOR RESIDENT

USE IT When teaching over Zoom, Teams or Google Meet.

PERSONA	You are an online medical educator who keeps virtual learners active.
MY CONTEXT	Topic, duration and number of learners: [brief] Platform features available: [polls, breakout rooms, chat, whiteboard]
TASK	1. Plan interaction every 5–10 minutes: polls, chat questions, breakout cases and annotation. 2. Plan the camera and presence strategy (encouraging cameras, using names). 3. Plan breakout-room activities with clear instructions. 4. Plan a check of learning at the end. 5. Plan for technical problems.
RULES	Keep instructions for activities very clear and posted in the chat, because online learners can't ask neighbours what to do.
OUTPUT	An interaction plan with timings, the breakout design, the check, then the tech backup.

PRO TIP Open with a chat question everyone can answer in 10 seconds. It gets people participating from minute one.

TRY IT WITH A 60-minute online tutorial on ECG basics for 40 MBBS students.

→ **NEXT #546** Slides for online teaching and webinars

685 Check learning during teaching

STANDARD

2.7 DELIVERY, Q&A & TEACHING · PG RESIDENT / FACULTY

USE IT To know whether learners are following, before the end of the session.

PERSONA	You are an assessment-for-learning specialist.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Topic and objectives: [PASTE] Session format and group size: [brief]
TASK	1. Suggest 5 quick formative checks: hinge questions, a show of fingers (confidence 1–5), mini-whiteboards, polls and one-minute papers. 2. Write 3 hinge questions for my topic, with the misconception each wrong option reveals. 3. Explain how to adapt the teaching based on the responses.
RULES	Design the questions so that the wrong answers reveal specific misconceptions, because that is what makes a check useful.
OUTPUT	The checks list, the hinge questions with their distractor analysis, then the adaptation guide.

PRO TIP If more than a third of the group gets a hinge question wrong, re-teach before moving on.

TRY IT WITH Teaching acid-base disorders to interns.

→ **NEXT #698** An exit-ticket question

686 Build a teaching portfolio

STANDARD

2.7 DELIVERY, Q&A & TEACHING · PG RESIDENT / FACULTY

USE IT For academic promotion, job applications, or to track your growth as a teacher.

PERSONA	You are an academic development mentor for clinician-educators.
MY CONTEXT	My teaching activities so far (seminars, lectures, bedside, simulation, mentoring): [list] Feedback or evaluations I have: [brief] Purpose of the portfolio: [promotion / application / personal growth]
TASK	1. Suggest the portfolio structure: teaching philosophy, activities log, evidence of effectiveness (feedback, learner outcomes), materials developed, and reflections. 2. Draft a 250-word teaching philosophy from my activities. 3. Create a teaching-activity log template. 4. Suggest how to collect better evidence going forward.
RULES	Include only activities and feedback I actually have, because portfolios are verified during academic review.
OUTPUT	The structure, the philosophy draft, the log template, then the evidence plan.

PRO TIP Save feedback forms and a copy of every teaching material. Evidence is the hardest part to reconstruct.

TRY IT WITH A second-year resident with 10 seminars, weekly intern teaching and one simulation session.

→ **NEXT #691** Peer observation of teaching

687 Respond to a hostile or aggressive question

STANDARD

2.7 DELIVERY, Q&A & TEACHING · PG RESIDENT / FACULTY

USE IT When preparing for a talk where someone may challenge you aggressively.

PERSONA	You are a communication coach experienced in high-pressure academic settings.
MY CONTEXT	The likely hostile question or challenge: [paste] The relevant facts from my work: [brief]
TASK	1. Separate the emotion from the content of the question. 2. Write a calm response: acknowledge, clarify the substance, answer with evidence, and offer to continue afterwards. 3. Write what not to say. 4. Plan how to handle persistence (a second challenge).
RULES	Keep the response respectful and factual, and concede valid points, because the audience judges how you respond more than who is right.
OUTPUT	The analysis, the response script, the avoid list, then the persistence plan.

PRO TIP "That's a fair challenge" defuses most hostility, and it costs nothing.

TRY IT WITH "Your sample is tiny and your conclusions are meaningless. Why present this at all?"

→ **NEXT #673** Handle any question after my talk

688 Answer "I don't know" with confidence

STANDARD

2.7 DELIVERY, Q&A & TEACHING · PG RESIDENT / MBBS STUDENT

USE IT When preparing for Q&A, rounds or vivas where you may not know an answer.

PERSONA	You are a senior clinician-teacher who models intellectual honesty.
MY CONTEXT	The setting (talk, rounds, viva, patient conversation): [brief] Examples of questions I might not know: [list]
TASK	1. Write 4 honest responses for not knowing, adapted to each setting. 2. Show how to reason aloud from first principles when partly unsure. 3. Show how to offer to follow up, and actually do it. 4. Explain when a guess is acceptable and how to label it.
RULES	Never recommend bluffing, especially about patient care, because a confident wrong answer is dangerous in medicine.
OUTPUT	The responses by setting, a reasoning-aloud example, then the follow-up guidance.

PRO TIP "I'm not sure. Let me reason it through..." often earns more respect than a quick right answer.

TRY IT WITH A viva question about a rare drug interaction.

→ **NEXT #673** Handle any question after my talk

689 Exercises for voice and pace

STANDARD

2.7 DELIVERY, Q&A & TEACHING · PG RESIDENT / MBBS STUDENT

USE IT When you speak too fast, too softly or in a monotone.

PERSONA	You are a voice coach for professional speakers.
MY CONTEXT	My main voice issue: [fast / soft / monotone / accent clarity] Time available to practise daily: [minutes]
TASK	1. Give 5 daily exercises (breathing, projection, articulation, pausing and vocal variety). 2. Set a target pace (e.g. 130–150 words per minute) and explain how to measure it. 3. Explain how to mark scripts for pauses and emphasis. 4. Give a 2-week practice plan.
RULES	Keep the exercises safe for the voice (no strain) and practical in a small space, because busy residents practise in hostels and call rooms.
OUTPUT	The exercises table, the pace method, the script-marking guide, then the plan.

PRO TIP Record a 1-minute reading daily and time it. Pace problems become obvious and fixable.

TRY IT WITH A resident who speaks at over 180 words per minute when nervous.

→ **NEXT #674** Improve my voice, pace and presence

690 Body language and stage presence

STANDARD

2.7 DELIVERY, Q&A & TEACHING · PG RESIDENT / FACULTY

USE IT When you want to look more confident while presenting.

PERSONA	You are a presentation coach specialising in non-verbal communication.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] My habits (e.g. pacing, hands in pockets, facing the screen, fidgeting): [list] Venue: [podium / open stage / small room / online]
TASK	1. Explain the effect of each habit on the audience. 2. Give a replacement behaviour for each. 3. Give guidance on stance, gestures, movement with purpose and using the pointer. 4. Give online-specific guidance (camera framing, eye contact with the lens).
RULES	Keep the advice natural rather than theatrical, because forced body language looks less confident than none.
OUTPUT	A habit-to-fix table, the general guidance, then the online tips.

PRO TIP Face the audience, not the screen. Glance at your laptop instead of turning to the projector.

TRY IT WITH A presenter who reads from the screen with their back to the audience.

→ **NEXT #674** Improve my voice, pace and presence

691 Peer observation of teaching

STANDARD

2.7 DELIVERY, Q&A & TEACHING · FACULTY / SENIOR RESIDENT

USE IT When a colleague observes your teaching, or you observe theirs.

PERSONA	You are a faculty-development specialist who runs peer-observation programmes.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] The session type and the teacher's goals for feedback: [brief]
TASK	1. Create an observation form: objectives clarity, structure, engagement, questioning, use of time, clinical accuracy and learner participation. 2. Suggest how to agree the focus before the session. 3. Suggest how to record specific observations (time-stamped, behavioural). 4. Plan a supportive post-observation conversation.
RULES	Focus on observable behaviours and the teacher's chosen goals, because peer observation fails when it becomes judgemental.
OUTPUT	The observation form, the pre-meeting guide, then the conversation plan.

PRO TIP Observe for what the teacher asked you to look at. Unrequested critique rarely helps.

TRY IT WITH Peer observation of a resident's 30-minute bedside teaching session.

→ **NEXT #680** Give feedback that helps

QUICK PROMPTS · 2.7

692 Rephrase an audience question

QUICK

2.7 DELIVERY, Q&A & TEACHING · PG RESIDENT

Act as a speaking coach. Show me how to rephrase this audience question, [PASTE QUESTION], in one neutral sentence that confirms I understood it, buys thinking time and helps the rest of the audience hear it, before I answer. Flag anything you are unsure of so I can check it.

TRY IT WITH "Isn't this whole approach outdated now that the new guidelines have come out?"

693 Handle an off-topic question

QUICK

2.7 DELIVERY, Q&A & TEACHING · PG RESIDENT / FACULTY

Act as a session chair. Write a polite 2-sentence response to this off-topic question, [PASTE QUESTION], after my talk on [TOPIC], that acknowledges it, bridges back to the talk if possible, and offers to discuss it afterwards. Flag anything you are unsure of so I can check it.

TRY IT WITH A question about hospital salaries after a talk on sepsis management.

694 Use pauses to sound confident

QUICK

2.7 DELIVERY, Q&A & TEACHING · PG RESIDENT / MBBS STUDENT

Act as a speaking coach. Mark where to pause in this script, [PASTE SCRIPT], using "/" for a short pause and "//" for a long pause, before key numbers, after questions and before the conclusion, and explain in 2 lines why each long pause is there.

TRY IT WITH The opening and closing paragraphs of a seminar on stroke.

695 Fix wandering eye contact

QUICK

2.7 DELIVERY, Q&A & TEACHING · PG RESIDENT

Act as a presentation coach. Give me 5 practical techniques for better eye contact when presenting to [AUDIENCE SIZE] people in a [ROOM TYPE], including what to do when using notes and slides, and one exercise to practise before my next talk.

TRY IT WITH A 40-seat seminar room with the slides projected behind me.

696 Remind myself to slow down

QUICK

2.7 DELIVERY, Q&A & TEACHING · PG RESIDENT / MBBS STUDENT

Act as a speaking coach. I speak too fast when nervous. Suggest 5 cues I can build into my slides, notes and routine for a [DURATION]-minute talk that will naturally slow me down, such as breathing points and planned audience questions.

TRY IT WITH A 15-minute case presentation.

697 Opening activity for a teaching session

QUICK

2.7 DELIVERY, Q&A & TEACHING · PG RESIDENT / FACULTY

Act as a medical educator. Suggest 3 quick opening activities (under 3 minutes each) for a teaching session on [TOPIC] with [LEARNERS] that activate prior knowledge and show me what they already know. Flag anything you are unsure of so I can check it.

TRY IT WITH A tutorial on heart failure for final-year MBBS students.

698 An exit-ticket question

QUICK

2.7 DELIVERY, Q&A & TEACHING · PG RESIDENT / FACULTY

Act as an assessment-for-learning specialist. Write 3 exit-ticket questions for the end of my session on [TOPIC], each answerable in 1–2 minutes, that test the main objective [OBJECTIVE] and reveal the most common misconception. Flag anything you are unsure of so I can check it.

TRY IT WITH A session on interpreting chest X-rays for interns.

699 One sentence of specific feedback

QUICK

2.7 DELIVERY, Q&A & TEACHING · PG RESIDENT / FACULTY

Act as a clinical teacher. Turn this vague feedback, [VAGUE FEEDBACK], into one specific, behavioural sentence that names what the learner did, its effect, and exactly what to do next time, in a supportive tone. Flag anything you are unsure of so I can check it.

TRY IT WITH "Your presentation was a bit confusing."

700 Praise that reinforces the right behaviour

QUICK

2.7 DELIVERY, Q&A & TEACHING · PG RESIDENT / FACULTY

Act as a medical educator. Rewrite this generic praise, [GENERIC PRAISE], into specific praise for [LEARNER ACTION] that names the behaviour and why it mattered for the patient or the team, in one or two sentences. Flag anything you are unsure of so I can check it.

TRY IT WITH "Good job with that patient" after an intern calmly explained a diagnosis to a family.

Every prompt at a glance.

Tap any prompt to jump to it. Q = Quick · S = Standard · M = Master

2.1 · SEMINAR PREPARATION

#401	Mixed-audience seminar deck, clinically rigorous	M
#402	From topic to seminar blueprint in one sitting	M
#403	Evidence pack: guidelines and trials, verified from my sources	M
#404	Full speaker script for a 45-minute seminar	M
#405	Case-based interactive seminar	M
#406	Mock moderator: grill me before the real seminar	M
#407	Turn a new guideline into a seminar	M
#408	Learning objectives with Bloom's verbs	S
#409	Minute-by-minute time plan	S
#410	Narrow a huge seminar topic to something teachable	S
#411	Pitch the same topic to different audiences	S
#412	Pathophysiology as a visual story	S
#413	Turn a classification into a clear table	S
#414	Diagnostic approach as an algorithm	S
#415	Management section, stepwise and practical	S
#416	Drug comparison table for a seminar	S
#417	Landmark trials in one table	S
#418	Controversies and grey zones	S
#419	Recent advances slide set	S
#420	The Indian perspective slide	S
#421	Compare two or more guidelines	S
#422	References slide and citation check	S
#423	End-of-seminar quiz	S
#424	Take-home messages that stick	S
#425	One-page handout for the audience	S
#426	Opening clinical vignette	S
#427	Basic science seminar with clinical hooks	S
#428	Seminar on a procedure or surgery	S
#429	Teaching session on interpreting a test	S
#430	Image-based seminar	S
#431	Turn a textbook chapter into a seminar	S
#432	Prepare a seminar with NotebookLM	S
#433	What have I missed?	S
#434	Feedback on my rehearsal script	S
#435	Make my seminar interactive	S
#436	Check the story and flow of my outline	S
#437	Adapt my seminar for nurses and allied staff	S
#438	Turn a seminar into a CME talk for practitioners	S
#439	Pro-con debate seminar	S
#440	Review my seminar slides like a moderator	S
#441	Build a seminar from our department's cases	S
#442	Collect feedback on my seminar	S
#443	A seminar title that's accurate and interesting	Q
#444	A one-line opening hook	Q
#445	Five learning objectives in a minute	Q

#446	The definition slide	Q
#447	A useful mnemonic for my seminar	Q
#448	"Why this matters" slide	Q
#449	How many slides do I need?	Q
#450	Key numbers slide	Q
#451	Summary slide from my outline	Q
#452	Transition lines between sections	Q
#453	A question to wake up the audience	Q
#454	An analogy for a hard concept	Q
#455	Three questions for the discussion	Q
#456	A short history slide	Q
#457	Abbreviations slide for my seminar	Q
#458	Future directions slide	Q
#459	What moderators always ask on this topic	Q
#460	Turn my paragraph into slide bullets	Q

2.2 · JOURNAL CLUB

#461	Journal club presentation: appraisal to bottom line	M
#462	Choose the right paper for journal club	M
#463	Deep appraisal of a randomised controlled trial	M
#464	Appraise a cohort or case-control study	M
#465	Appraise a diagnostic accuracy study	M
#466	Appraise a systematic review or meta-analysis	M
#467	Decode the statistics in my paper	M
#468	Five-line PICO summary of the paper	S
#469	Does the abstract match the paper?	S
#470	"Why was this study done?" slide	S
#471	Methods on two slides	S
#472	Key results, clearly presented	S
#473	Explain a forest plot to the audience	S
#474	Explain the survival curves	S
#475	Absolute effects: ARR, RRR, NNT and NNH	S
#476	Can we trust these results?	S
#477	Does this apply to our patients?	S
#478	Hunt for the hidden bias	S
#479	Confounding in an observational paper	S
#480	Are these the right outcomes?	S
#481	Appraise a non-inferiority trial	S
#482	Should we believe the subgroup finding?	S
#483	Detect spin in the discussion and conclusion	S
#484	Funding and conflicts: do they matter here?	S
#485	Compare the paper with its editorial and letters	S
#486	How does this fit with previous evidence?	S
#487	The clinical bottom line	S
#488	Questions to drive the discussion	S
#489	The faculty questions I'll face	S
#490	One-page appraisal handout	S
#491	Fill the CASP checklist for my paper	S

#492	Certainty of evidence using GRADE	S	#542	Choose the right chart for a slide	S
#493	Appraise a qualitative study	S	#543	Citing sources on slides	S
#494	Appraise an economic evaluation	S	#544	Title and opening slide design	S
#495	Appraise a prognostic or prediction-model study	S	#545	Section divider slides that guide the audience	S
#496	Appraise a clinical guideline (AGREE II)	S	#546	Slides for online teaching and webinars	S
#497	Check the trial registry for outcome switching	S	#547	Using images and figures legally	S
#498	Read a paper properly in 30 minutes	S	#548	Turn a slide title into a message	Q
#499	Explain the paper to MBBS students	S	#549	Check a slide for clutter	Q
#500	Quiz questions from the journal club	S	#550	Paragraph to one clean slide	Q
#501	Turn my critique into a letter to the editor	S	#551	A three-colour palette for my deck	Q
#502	Moderating a journal club well	S	#552	Font sizes for my room	Q
#503	This study's design in one line	Q	#553	Icon ideas for my concepts	Q
#504	Find the real primary outcome	Q	#554	Alt text for my slide images	Q
#505	Quick ARR and NNT	Q	#555	A caption for a figure on a slide	Q
#506	What does this hazard ratio mean?	Q	#556	Should these bullets be a diagram?	Q
#507	Intention-to-treat or per-protocol?	Q	#557	Design my final slide	Q
#508	Was allocation really concealed?	Q	#558	QR code slide content	Q
#509	Who was blinded, and does it matter?	Q	#559	Should this table be a chart?	Q
#510	Is this loss to follow-up a problem?	Q	#560	Does my slide pass the 6 × 6 rule?	Q
#511	The whole paper in one sentence	Q	2.4 · TALKS & ORAL PRESENTATIONS		
#512	Was this study adequately powered?	Q	#561	Invited CME talk or guest lecture, start to finish	M
#513	Is this outcome a surrogate?	Q	#562	Build any talk around a story	M
#514	Would this change my practice?	Q	#563	Health talk for patients or the community	M
#515	Explain a methods term from my paper	Q	#564	Prepare for a panel discussion (as panellist or moderator)	M
#516	Does the title overclaim?	Q	#565	The first 90 seconds of my talk	S
#517	Relative versus absolute risk for the audience	Q	#566	A closing the audience remembers	S
#518	Explain the key figure in three lines	Q	#567	A 5-minute lightning talk	S
#519	A sharp question to ask the presenter	Q	#568	A PechaKucha-style talk (20 slides × 20 seconds)	S
#520	Summarise the paper for colleagues	Q	#569	Motivational or career talk for students	S
2.3 · SLIDE DESIGN & STRUCTURE			#570	Deliver a talk in Hindi or a regional language	S
#521	Redesign my existing deck, slide by slide	M	#571	Using patient stories ethically in talks	S
#522	A personal slide design system for all my talks	M	#572	Health talk for school students	S
#523	Build a deck with AI tools, safely	M	#573	Chairing a scientific session	S
#524	Turn data and tables into clear slide visuals	M	#574	Introduce a guest speaker	S
#525	Rewrite slide titles as messages	S	#575	A vote of thanks that isn't boring	S
#526	Turn a text-heavy slide into a visual	S	#576	Host a medical webinar	S
#527	Simplify a table for a slide	S	#577	Script for a short educational video	S
#528	Build a flowchart slide	S	#578	Rehearsal plan for an important talk	S
#529	Adapt a guideline algorithm for a slide	S	#579	Manage stage fright before a talk	S
#530	Two conditions side by side	S	#580	Shrink or stretch a talk to a new time slot	S
#531	A timeline slide	S	#581	Turn a key paper into a teaching talk	S
#532	Clinical image slide with annotation	S	#582	Proposal and abstract for an invited talk	S
#533	Icons and simple graphics for concepts	S	#583	A short speaker bio	S
#534	Colours that work for everyone	S	#584	Learn from feedback on my talk	S
#535	Fonts and sizes that are readable	S	#585	An opening question that hooks the audience	Q
#536	When to use animation, and when not to	S	#586	Organise my talk into three parts	Q
#537	Cut my deck to fit the time	S	#587	My talk's big idea in one sentence	Q
#538	Write speaker notes for my slides	S	#588	Five questions about my audience	Q
#539	Separate the slides from the handout	S	#589	A safe icebreaker for a medical audience	Q
#540	A design brief for building slides in Canva	S	#590	Tell a story in 60 seconds	Q
#541	A prompt for AI slide generators	S	#591	Recap before moving on	Q

#592	A clear call to action	Q	#640	Present and justify the investigations	S
#593	A title for my invited talk	Q	#641	Present the management plan and its reasoning	S
#594	Clean filler words out of my script	Q	#642	ISBAR handover for a patient	S
#595	A better way to open the Q&A	Q	#643	Ward-round presentation (SOAP)	S
2.5 · POSTERS & VISUAL TEACHING AIDS					
#596	Clinical protocol poster for a ward or ICU	M	#644	Systems-based ICU rounds presentation	S
#597	Comic-style teaching infographic (AI image tools)	M	#645	Present a case to a senior on the phone	S
#598	Handwritten-style revision notes image	M	#646	Case presentation built around images	S
#599	Awareness poster series for a health day	M	#647	Teaching points from a case	S
#600	A medical infographic for teaching or social media	S	#648	A brief literature review for my case	S
#601	A patient-education leaflet	S	#649	A 10–12 slide case-presentation deck	S
#602	Pocket quick-reference card	S	#650	Case discussion on a management controversy	S
#603	A concept map for a complex topic	S	#651	Present a case with an ethical dilemma	S
#604	Plan a chalk-and-talk or whiteboard session	S	#652	Present a rare diagnosis without losing the audience	S
#605	Sketchnote summary of a topic	S	#653	Learning from a delayed or missed diagnosis	S
#606	Educational poster for a student competition	S	#654	The questions the head of department will ask	S
#607	Patient-facing poster for the OPD waiting area	S	#655	Root-cause analysis for an M&M case	S
#608	Pictorial medication schedule for patients	S	#656	Present a follow-up update on a previous case	S
#609	Brief for a medical illustration	S	#657	Turn a case into a teaching session for MBBS students	S
#610	Prompt an AI image tool for medical visuals	S	#658	Link a case to a recent paper	S
#611	Educational carousel for Instagram or LinkedIn	S	#659	Opening one-liner for my case	Q
#612	Low-cost teaching model or demonstration	S	#660	Summarise my case in three lines	Q
#613	Step-by-step procedure wall chart	S	#661	Which negatives matter?	Q
#614	Adapt a poster into several Indian languages	S	#662	Why is this case worth presenting?	Q
#615	Critique my poster or infographic	S	#663	Build the problem list	Q
#616	Print specifications for posters and charts	S	#664	Describe the vitals trend	Q
#617	Poster with QR codes to videos and resources	S	#665	Present the lab results concisely	Q
#618	Emergency algorithm card for the crash cart	S	#666	Hospital course as a timeline	Q
#619	A visual mnemonic poster	S	#667	The final diagnosis statement	Q
#620	Headline for my poster	Q	#668	One learning point from my case	Q
#621	Short slogans for a health campaign	Q	#669	SBAR script for a phone call	Q
#622	Icons for my visual aid	Q	#670	A title for my case presentation	Q
#623	Colours for a patient-facing poster	Q	2.7 · DELIVERY, Q&A & TEACHING		
#624	Check the reading level of my text	Q	#671	Plan a small-group teaching session	M
#625	Pictogram instructions for low literacy	Q	#672	Structured bedside teaching session	M
#626	Caption for my infographic post	Q	#673	Handle any question after my talk	M
#627	Poster size and font sizes	Q	#674	Improve my voice, pace and presence	M
#628	Fix a garbled AI image	Q	#675	The one-minute preceptor, applied	S
#629	Description for a visual post	Q	#676	Teach learners to present with SNAPPS	S
#630	Call to action for an awareness poster	Q	#677	Design a flipped-classroom session	S
2.6 · CASE PRESENTATIONS & GRAND ROUNDS					
#631	Departmental case presentation, start to finish	M	#678	Write a simulation scenario	S
#632	Grand rounds presentation	M	#679	Teach a clinical skill step by step	S
#633	Morbidity and mortality (M&M) presentation	M	#680	Give feedback that helps	S
#634	Short "interesting case" presentation with teaching	M	#681	Orientation teaching for new interns	S
#635	Presenting at a multidisciplinary team meeting or tumour board	M	#682	Engage a large lecture of 100+ students	S
#636	A three-line case summary	S	#683	Handle difficult small-group dynamics	S
#637	Organise a messy history for presentation	S	#684	Keep an online class engaged	S
#638	Present examination findings clearly	S	#685	Check learning during teaching	S
#639	Justify a differential diagnosis for a case	S	#686	Build a teaching portfolio	S
			#687	Respond to a hostile or aggressive question	S
			#688	Answer "I don't know" with confidence	S
			#689	Exercises for voice and pace	S

#690	Body language and stage presence	S
#691	Peer observation of teaching	S
#692	Rephrase an audience question	Q
#693	Handle an off-topic question	Q
#694	Use pauses to sound confident	Q
#695	Fix wandering eye contact	Q

#696	Remind myself to slow down	Q
#697	Opening activity for a teaching session	Q
#698	An exit-ticket question	Q
#699	One sentence of specific feedback	Q
#700	Praise that reinforces the right behaviour	Q

Which AI tool for which job.

JOB	BEST TOOL	WHY
Studying sources for a seminar	NotebookLM, Claude, ChatGPT (upload)	Answers and audio overviews grounded in your PDFs
Current evidence and recent trials	Perplexity (Academic), Consensus	Live search with clickable citations
Outlines, scripts and speaker notes	Claude, ChatGPT, Gemini	Strong structure and plain-language rewriting
Slide decks	PowerPoint (Copilot), Gamma, Canva	Fast first drafts from your verified outline
Posters, infographics and leaflets	Canva, ChatGPT images, Gemini images	Layouts and illustrations; proofread every word
Polls and live quizzes	Mentimeter, Slido, Google Forms	Turn a lecture into an interactive session
Rehearsal feedback	Phone recording + AI transcript review	See your pace, filler words and timing

TEACHING WITH AI, RESPONSIBLY

- › Verify every drug dose, guideline year and trial detail before it goes on a slide.
- › Never put identifiable patient data or images into AI tools or slides.
- › Credit figures you adapt, and use only images you have the right to use.
- › AI drafts the deck. You own the content, the delivery and the answers.

This resource is for educational purposes. Clinical content in any seminar, poster or protocol must be verified against current guidelines and your institution's approved protocols. It does not replace the advice of your moderator, department or hospital policy. Verify all AI-generated content independently before presenting or distributing it.

STUDY, EXAM & VIVA PREP.

300 prompts to learn any topic, plan revision, practise MCQs for NEET-PG and UPSC CMS, write university answers, face the viva and reason through a differential diagnosis.

300

prompts

7

sections

37

Master

169

Standard

94

Quick

3.1	Learning any topic	#701-750
3.2	Study planning & revision	#751-780
3.3	MCQ practice & exam strategy	#781-830
3.4	Long & short answer writing	#831-860
3.5	Viva, OSCE & practical exams	#861-910
3.6	Clinical reasoning & differential diagnosis	#911-970
3.7	Memory aids & high-yield summaries	#971-1000

3.1

Learning any topic

Understanding beats memorising. Learn in layers, test yourself as you go, and connect every fact to a patient.

50

prompts

6

Master

28

Standard

16

Quick

HOW TO USE THIS SECTION

Use it the first time you study a topic, to understand it rather than memorise it.

Start with the Master prompts (#701–706) for the full workflow; use the Standard prompts (#707–734) for one task at a time; reach for the Quick prompts (#735–750) when you have two minutes. Tap any title below to jump to it, and follow → NEXT to move to the next step.

#701 Structured topic master for any subject

#702 A personal tutor that checks my understanding

#703 Build a custom tutor that answers only from my notes

#704 Study a chapter with NotebookLM

#705 Understand a mechanism from first principles

#706 One topic across all subjects: vertical integration

#707 Learn a topic in four layers

#708 Explain it back and find my gaps (Feynman technique)

#709 Learn a hard concept through analogies

#710 Surface features and deeper mechanisms (iceberg)

#711 Learn a process as a flowchart

#712 Compare options in a decision matrix

#713 Classifications as a tree

#714 A learning roadmap from my level to my goal

#715 Connect this topic to what I already know

#716 Learn through a worked example

#717 Common misconceptions in a topic

#718 Sort a topic into high- and low-yield

#719 Turn a textbook chapter into structured notes

#720 Clean up my lecture notes

#721 Notes from a video lecture transcript

#722 Learn a clinical guideline for exams

#723 Learn any drug with one framework

#724 A standard template for any disease

#725 Learn an anatomy topic that sticks

#726 Understand physiology graphs and curves

#727 Learn to read a type of clinical image

#728 Normal values and interpretation

#729 Learn a microorganism with one template

#730 Simplify a biochemistry pathway

#731 Learn national health programmes (community medicine)

#732 Make my own questions from my notes

#733 Design an interleaved practice session

#734 Diagnose my knowledge gaps

#735 Explain it like I'm a second-year student

#736 One-page summary of a topic

#737 Ten must-know facts

#738 Ask "why" five times

#739 Clear up two confusing terms

#740 A real-world example of a concept

#741 Why does this basic-science fact matter clinically?

#742 A quick 5-question quiz

#743 Define it, with an example

#744 Clear up one specific confusion

#745 The whole topic in 280 characters

#746 A diagram I can draw

#747 What should I study next?

#748 Is my understanding correct?

#749 Translate the jargon

#750 Three questions the examiner will ask

Structured topic master for any subject

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT

For deep revision of any topic in any subject, before university exams, NEET-PG or UPSC CMS.

WHY

Scattered notes from five sources waste revision time. One structured, exam-ready note per topic (definition to complications, with high-yield points and examiner favourites flagged) becomes your master copy for every revision cycle.

COPY THE PROMPT BELOW

PERSONA	You are an expert medical educator in [SUBJECT] who prepares students for university examinations, NEET-PG and UPSC CMS. You write structured, exam-ready notes, use tables wherever comparison helps, and flag what examiners love to ask.
MY CONTEXT	Audience and time available: [brief] What I most want to get out of this: [one line] Topic: [e.g. primary open-angle glaucoma / nephrotic syndrome / brachial plexus] My level: [MBBS year / intern / PG resident in SPECIALTY] Exam I'm preparing for: [university theory / NEET-PG / UPSC CMS / MD-MS-DNB final / USMLE / PLAB] Standard textbook I use: [name and edition] Time I can spend on this topic: [e.g. 2 hours]
TASK	1. Definition and classification. 2. Aetiology and risk factors. 3. Pathophysiology, with the key mechanisms as a flow. 4. Clinical features: symptoms, signs, and grading or staging scales where applicable. 5. Investigations, with interpretation pearls. 6. Management: medical, surgical and procedural, in a stepwise format. 7. Complications and their management. 8. High-yield exam points, previous-year question themes, and commonly asked viva questions with model answers. 9. A one-line "if you remember only one thing" summary.
RULES	Keep facts consistent with standard textbooks and current guidelines, and mark any drug dose, cut-off or guideline-dependent statement as [VERIFY FROM TEXTBOOK / GUIDELINE], because exam answers must match the standard source. Flag anything that is a favourite examiner question. Say so where textbooks disagree.
OUTPUT	Headed sections with bullets and tables, then the high-yield box and the viva questions with answers.
TWEAK	End with 5 active-recall questions I should answer tomorrow without looking at the notes.

How to get more out of it

- › Replace [TOPIC] with anything: retinal detachment, uveitis, heart failure, brachial plexus.
- › Cross-check the output against your standard textbook once, then revise only from this note.
- › Answer the 5 recall questions the next day. That's what turns reading into remembering.
- › → NEXT #702 A personal tutor that checks my understanding

A personal tutor that checks my understanding

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT

When a topic feels confusing and you want to understand it, not just read it.

WHY

Reading explanations feels like learning but often isn't. A tutor that explains in small parts, asks you a question after each part, and moves on only when you answer correctly forces real understanding, and shows you exactly where your gaps are.

COPY THE PROMPT BELOW

PERSONA	You are an expert, patient tutor in [SUBJECT] who teaches using the Socratic method. You explain simply, use real-world and clinical analogies, and never move ahead until the learner shows understanding.
MY CONTEXT	Topic: [TOPIC] My level and what I already know: [brief] Exam or purpose: [university exam / NEET-PG / UPSC CMS / residency exam / clinical work] Where I usually get confused: [brief or "not sure"]
TASK	1. Break the topic into 5–7 small, logical parts, and show me the list first. 2. Teach one part at a time in simple language, with an analogy and a clinical example. 3. After each part, ask me one question that tests understanding, not recall. 4. Wait for my answer. If it's correct, confirm it and add one deeper point; if it's wrong, explain the misconception differently and ask another question. 5. Move to the next part only when I answer correctly. 6. At the end, summarise the topic in 5 bullet points. 7. Give me 3 exam-style questions to test myself, with answers hidden until I attempt them.
RULES	Ask only one question at a time and wait for my reply, because the learning happens in my answering. Keep facts accurate to standard textbooks and mark guideline-dependent details [VERIFY]. Don't give me the answers to your check questions before I try.
OUTPUT	Conversational: one part and one question per turn, then the final summary and exam questions.
TWEAK	At the end, tell me which part I struggled with most and suggest one resource type to strengthen it.

How to get more out of it

- › Answer honestly, including "I don't know". The tutor adapts to your answers.
- › Ask "explain it differently" whenever an explanation doesn't click.
- › Revisit the part you struggled with after 2 days. Spacing makes it stick.
- › → **NEXT #707** Learn a topic in four layers

Build a custom tutor that answers only from my notes

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT

When setting up a Custom GPT (ChatGPT), a Claude Project or a Gemini Gem with your own notes and textbooks.

WHY

A general chatbot mixes your standard textbook with random internet content. A custom tutor instructed to answer only from your uploaded material, cite the file and page, and say "not in your notes" when it doesn't know gives exam-consistent answers you can trust and check.

COPY THE PROMPT BELOW

PERSONA	You are an expert at configuring AI study assistants for medical exams. You write system instructions that keep the assistant grounded in the uploaded files and in exam style.
MY CONTEXT	Tool: [ChatGPT Custom GPT / Claude Project / Gemini Gem] Exam: [NEET-PG / UPSC CMS / MD-MS-DNB final / university exam] Files I will upload (textbooks, notes, PYQs): [list] How I want to use it: [Q&A / viva drill / MCQ practice / revision summaries]
TASK	1. Write the complete system instructions for the tutor: answer ONLY from the uploaded files; if the answer isn't in the files, reply "Not in your uploaded notes. Verify from a textbook."; always cite the file name and page or section; use exam-style language; highlight high-yield facts and previous-year traps. 2. Add modes the tutor can switch between on command: "Explain", "Viva", "MCQ", "Summary" and "Compare", each with its rules. 3. Add instructions for handling conflicting information between files. 4. Write 5 starter prompts I can use daily. 5. Explain how to test the tutor on 5 questions whose answers I already know, to check its grounding.
RULES	Keep the instructions within typical length limits for the tool and mark tool-specific settings as [CHECK CURRENT SETTINGS], because platform features change often. Upload only material I'm allowed to use.
OUTPUT	The system instructions in a code block, the mode definitions, the conflict rule, the starter prompts, then the testing procedure.
TWEAK	End with the 3 warning signs that the tutor is answering from outside my files, and what to do.

How to get more out of it

- Test the tutor with a question you know isn't in your notes. It should say so.
- Keep one tutor per subject. Mixed subjects confuse retrieval.
- Upload your previous-year papers too, so it can link topics to past questions.
- → **NEXT #704** Study a chapter with NotebookLM

Study a chapter with NotebookLM

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT

When you have a chapter, notes or a set of PDFs and want grounded summaries, questions and audio revision.

WHY

NotebookLM answers only from the sources you upload and shows exactly where each answer came from. Used with the right questions and a customised Audio Overview, it turns a dense chapter into structured notes, self-tests and a podcast for your commute.

COPY THE PROMPT BELOW

PERSONA	You are a study coach who has built efficient NotebookLM workflows for medical exam preparation.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] Sources I'll upload: [chapter / notes / guideline PDFs] Topic and exam: [e.g. hepatitis B, NEET-PG] Time available: [e.g. 90 minutes]
TASK	1. Give a step-by-step workflow: upload, first-pass summary, structured questions, self-test, audio overview and review. 2. Write 12 questions to ask NotebookLM that cover definitions, classification, mechanisms, diagnosis, management, complications and exam traps. 3. Write the Audio Overview customisation: "Generate an Audio Overview of this chapter for a [EXAM] aspirant. Cover only high-yield points: definitions, classifications, diagnostic criteria, drug doses, side effects and the 5 most-asked previous-year themes. Skip history and rare exceptions." 4. Write prompts to generate a study guide, a comparison table and 15 self-test questions from the sources. 5. Explain how to check each answer using NotebookLM's citations. 6. Suggest how to organise notebooks (one per subject or system) and name sources so that citations stay easy to follow. 6. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Phrase everything so that answers come only from my sources, and tell me to verify drug doses against the source, because grounding is NotebookLM's strength only if I use it.
OUTPUT	The workflow, the question list, the audio instruction in a code block, the generation prompts, then the verification steps.
TWEAK	End with a 20-minute review routine for the next day using the same notebook.

How to get more out of it

- › Listen to the Audio Overview on your commute, then answer the self-test questions.
- › Click the citation numbers. They take you to the exact passage, and that is your verification.
- › One notebook per subject or system keeps the sources focused.
- › → **NEXT #703** Build a custom tutor that answers only from my notes

Understand a mechanism from first principles

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT

When you have memorised something but can't explain why it happens.

WHY

Mechanisms learned from first principles don't need to be memorised: you can reconstruct them in the exam hall. "Why" chains, from a clinical feature back to physiology, turn isolated facts into understanding that survives pressure and tricky questions.

COPY THE PROMPT BELOW

PERSONA	You are a physiology-minded clinician-teacher who explains every clinical fact by asking "why" until reaching basic principles.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] The fact or phenomenon I want to understand: [e.g. why hypokalaemia causes muscle weakness, why the JVP rises in right heart failure] My level: [brief] The basic science I'm comfortable with: [brief]
TASK	1. Start from the clinical fact and build a "why" chain of 4–6 steps down to the basic principle. 2. Explain each step in plain language with a simple analogy. 3. Draw the chain as a flow diagram in text, with arrows between the steps. 4. Show how the same principle explains 2–3 other clinical facts. 5. Give 3 "predict the effect" questions that test whether I can apply the principle. 6. Link the principle to one treatment and explain why the treatment works. 7. Point out one common exam question that this principle answers, and show the answer built from the chain. 7. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Keep each step physiologically accurate and mark any simplification as "simplified", because oversimplified mechanisms cause errors in harder questions. Mark anything textbook-dependent [VERIFY].
OUTPUT	The why-chain, the explanations, the flow diagram, the connected facts, the prediction questions, then the treatment link.
TWEAK	End with a one-sentence version of the principle I can say in a viva.

How to get more out of it

- › Try to answer each "why" yourself before reading the explanation.
- › One principle often explains ten facts. Find the principle and you save hours.
- › Draw the chain from memory the next day.
- › → NEXT #706 One topic across all subjects: vertical integration

One topic across all subjects: vertical integration

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT

When you want to see how a topic connects across anatomy, physiology, pathology, pharmacology, microbiology and clinical medicine.

WHY

Exams, especially NEET-PG and clinical vivas, increasingly test integration: a question about a disease may ask about its anatomy, mechanism or drug. Studying one topic vertically across subjects builds connected knowledge that is easier to recall.

COPY THE PROMPT BELOW

PERSONA	You are a medical educator who designs integrated, organ-system teaching across preclinical, paraclinical and clinical subjects.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] Topic or disease: [e.g. myocardial infarction / tuberculosis / diabetes mellitus] My year and exam: [brief] Subjects to cover: [all / list]
TASK	1. Build an integration map: for each subject (anatomy, physiology, biochemistry, pathology, microbiology, pharmacology, community medicine, forensic medicine if relevant, and clinical medicine or surgery), list the 3–5 high-yield points about this topic. 2. Show the links between subjects (e.g. coronary anatomy → infarct territory → ECG leads → complications). 3. Write 5 integrated questions that cross at least 2 subjects each, with answers. 4. Suggest a one-page visual map of the connections, with the disease in the centre. 5. Mark the facts most often asked in NEET-PG and UPSC CMS style questions, as recurring themes, not claimed statistics. 6. Suggest the order in which to revise the subjects for this topic, and how long each should take. 6. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Keep facts consistent with standard textbooks and mark doses and cut-offs [VERIFY], because integrated questions punish small factual errors.
OUTPUT	The integration table by subject, the link list, the integrated questions with answers, the map description, then the high-yield marks.
TWEAK	End with the 3 cross-subject links students most often miss.

How to get more out of it

- › Use this for the 20 most important diseases. It covers a surprising share of the syllabus.
- › Draw the map by hand. It sticks better than a printed one.
- › Revise the integrated questions before mock tests.
- › → NEXT #701 Structured topic master for any subject

707 Learn a topic in four layers

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT When a topic is new or feels overwhelming.

PERSONA	You are a teacher who builds understanding in layers, from simple to advanced.
MY CONTEXT	Topic: [TOPIC] My level: [brief] Exam: [brief]
TASK	Teach me the topic in 4 layers: 1. A one-sentence explanation. 2. A beginner explanation (about 100 words). 3. The underlying mechanism (about 150 words). 4. An advanced, exam-level explanation with nuances and exceptions. Define new terms as they appear, use the same clinical example throughout, and end each layer with one check question.
RULES	Keep each layer accurate, labelling simplifications as such, and mark guideline-dependent facts [VERIFY], because early simplifications must not become later errors.
OUTPUT	Four labelled layers, each with its check question.

PRO TIP Stop at the layer your exam needs. Most MBBS exams need layer 3; residency exams need layer 4.

TRY IT WITH Topic: the renin-angiotensin-aldosterone system.

→ **NEXT #708** Explain it back and find my gaps (Feynman technique)

708 Explain it back and find my gaps (Feynman technique)

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT After studying a topic, to test whether you really understand it.

PERSONA	You are a demanding but kind examiner who uses the Feynman technique.
MY CONTEXT	Topic: [TOPIC] My explanation in my own words (typed or voice-to-text): [PASTE MY EXPLANATION] My level: [brief]
TASK	1. Identify what I explained correctly. 2. Identify errors, gaps, vague phrases and jargon I used without understanding. 3. Ask me 3 follow-up questions targeting the gaps. 4. After I answer, give a corrected, simple explanation.
RULES	Be specific about each gap, quoting my words, and don't simply rewrite everything for me, because the goal is to find what I don't understand.
OUTPUT	The correct points, a gaps table (my words, problem), the follow-up questions, then (after my answers) the corrected explanation.

PRO TIP If you can't explain it to a 12-year-old, you don't understand it yet. That's the Feynman test.

TRY IT WITH My explanation of why loop diuretics cause hypokalaemia.

→ **NEXT #702** A personal tutor that checks my understanding

709 Learn a hard concept through analogies

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT When a concept won't stick through textbook explanations.

PERSONA	You are a gifted medical teacher known for memorable analogies.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Concept: [CONCEPT] My level: [brief]
TASK	1. Give 3 different everyday analogies for the concept. 2. For each, map the parts of the analogy to the parts of the concept. 3. State where each analogy breaks down. 4. Recommend the best analogy and give one exam question it helps answer, with a short model answer.
RULES	Make each analogy's limitations explicit, because an analogy taken too far teaches the wrong physiology.
OUTPUT	Three analogies with mapping tables and limitations, then the recommendation.

PRO TIP Draw your chosen analogy. Visual analogies stick best.

TRY IT WITH Concept: oxygen-haemoglobin dissociation curve shifts.

→ **NEXT #707** Learn a topic in four layers

710 Surface features and deeper mechanisms (iceberg)

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT When you know a disease's features but not why they occur.

PERSONA	You are a clinician-teacher who uses the iceberg model to connect signs to mechanisms.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Disease or syndrome: [name] My level: [brief]
TASK	1. Above the surface: list the visible symptoms, signs and test findings. 2. Just below: the patterns and physiological changes that produce them. 3. Deep below: the underlying mechanisms and causes. 4. Link each surface feature to its mechanism with an arrow. 5. Distinguish established mechanisms from hypotheses.
RULES	Don't invent mechanisms to fill the diagram; say where the mechanism is unclear, because false certainty is harder to unlearn.
OUTPUT	A three-level iceberg as a table, then the feature-to-mechanism links.

PRO TIP Examiners love "explain why..." questions. The deep layer answers them.

TRY IT WITH Disease: nephrotic syndrome.

→ **NEXT #705** Understand a mechanism from first principles

711 Learn a process as a flowchart

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT For pathways, algorithms, cascades and step-by-step processes.

PERSONA	You are a teacher who converts processes into clear flowcharts.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Process: [e.g. coagulation cascade, approach to anaemia, steps of an ABG] My level: [brief]
TASK	1. Draw the process as a text flowchart with a start, steps, decision points (yes/no) and outcomes. 2. Label every branch. 3. Add the clinical relevance at the key steps. 4. Give a numbered-outline version as a text fallback. 5. Give 3 questions that test the flowchart.
RULES	Keep the steps accurate to standard sources and mark guideline thresholds [VERIFY], because flowcharts are often reproduced in exam answers.
OUTPUT	The flowchart, the outline, then the questions.

PRO TIP Redraw the flowchart from memory. The step you forget is the one to revise.

TRY IT WITH Approach to a patient with metabolic acidosis.

→ **NEXT #713** Classifications as a tree

712 Compare options in a decision matrix

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT When you need to choose between treatments, tests or approaches, as exams often ask.

PERSONA	You are a clinical pharmacologist and teacher who builds comparison matrices.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The options to compare: [e.g. antiepileptic drugs, imaging modalities, antihypertensives in special populations] Criteria: [list or "suggest"] Exam context: [brief]
TASK	1. Build a decision matrix with the options as rows and the criteria as columns. 2. Assess each cell briefly, stating assumptions. 3. State which option fits which clinical situation. 4. Add 3 exam-style "which is best for..." questions with answers.
RULES	Don't invent scores; use qualitative assessments and mark doses and specifics [VERIFY], because a false precision score misleads.
OUTPUT	The matrix, the situation guide, then the questions.

PRO TIP "Drug of choice in pregnancy" and "drug to avoid in renal failure" columns answer many exam questions.

TRY IT WITH First-line antihypertensives in diabetes, CKD, pregnancy and heart failure.

→ **NEXT #723** Learn any drug with one framework

713 Classifications as a tree

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT For topics with complex classifications (drugs, tumours, anaemias, glomerular diseases).

PERSONA	You are a teacher who organises knowledge into hierarchical trees.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Topic: [TOPIC] Classification basis (if a standard one exists): [brief]
TASK	1. Organise the topic as a hierarchy: main concept → categories → subcategories → examples. 2. Use one consistent classification rule at each level. 3. Flag overlaps rather than forcing items into single branches. 4. Add one distinguishing feature per category. 5. Give a text-tree version for quick revision.
RULES	Follow the standard textbook classification and mark any that differs between sources [CHECK TEXTBOOK], because exams expect the standard version.
OUTPUT	The tree, the distinguishing features, then the text-tree.

PRO TIP Classifications are frequent 3-mark questions. Practise writing them from memory.

TRY IT WITH Classification of anaemias by red-cell indices.

→ **NEXT #711** Learn a process as a flowchart

714 A learning roadmap from my level to my goal

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT When starting a large subject or skill, such as ECGs, radiology or pharmacology.

PERSONA	You are a curriculum designer who builds realistic learning roadmaps.
MY CONTEXT	Subject or skill: [e.g. ECG interpretation] Current level: [brief] Goal: [e.g. confidently interpret common ECGs for the NEET-PG image-based questions] Time available: [weeks and hours per week]
TASK	1. Break the path into 4–6 stages with prerequisites. 2. For each stage: learning goals, resources by type, practice tasks and a measurable checkpoint. 3. Estimate the time per stage. 4. State whether the scope exceeds the available time, and what to cut if so.
RULES	Keep the plan realistic for my stated time, because over-ambitious roadmaps are abandoned by week 2.
OUTPUT	A stage table (stage, goals, practice, checkpoint, time), then the feasibility note.

PRO TIP A checkpoint you can test yourself on (e.g. "interpret 20 ECGs at 80% accuracy") beats "finish chapter 3".

TRY IT WITH Learning chest X-ray interpretation in 4 weeks, 5 hours a week.

→ **NEXT #751** A personalised day-by-day study plan

715 Connect this topic to what I already know

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT When a new topic feels isolated from everything else you've learned.

PERSONA	You are a teacher who builds knowledge by connecting new ideas to old ones.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] New topic: [TOPIC] Topics I already know well: [list]
TASK	1. Identify 5 links between the new topic and what I know (shared mechanisms, similar presentations, related drugs). 2. Explain each link in 2 sentences. 3. Identify one common confusion between the new topic and a related known topic. 4. Give 3 questions that require using both topics together.
RULES	Keep every link accurate, and flag superficial similarities that could mislead, because false connections cause exam errors.
OUTPUT	A link table, the confusion note, then the questions.

PRO TIP Knowledge sticks to knowledge. Always ask "what does this remind me of?"

TRY IT WITH New topic: SIADH. Known topics: diabetes insipidus and hyponatraemia.

→ **NEXT #713** Classifications as a tree

716 Learn through a worked example

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT For problem-solving topics: acid-base, dosing calculations, ECG axes, statistics, fluid prescriptions.

PERSONA	You are a tutor who teaches problem-solving through clear worked examples.
MY CONTEXT	Problem type: [e.g. ABG interpretation, paediatric fluid calculation] My level: [brief]
TASK	1. Show a fully worked example, step by step, explaining the reasoning at each step. 2. Show a second example with a twist. 3. Give me a third problem to solve alone, and wait for my answer. 4. Check my answer step by step and correct any errors.
RULES	Make every example internally consistent (numbers that are physiologically possible) and mark formulas and normal ranges [VERIFY], because inconsistent examples confuse learners.
OUTPUT	Two worked examples, a practice problem, then feedback after my attempt.

PRO TIP Cover the steps and try each one yourself before reading it.

TRY IT WITH Calculating maintenance IV fluids for a 22-kg child using the Holliday-Segar method.

→ **NEXT #708** Explain it back and find my gaps (Feynman technique)

717 Common misconceptions in a topic

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT Before exams, to avoid the traps most students fall into.

PERSONA	You are an experienced examiner who has marked thousands of answers and knows the common errors.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Topic: [TOPIC] Exam: [brief]
TASK	1. List 8 common misconceptions or confusions about this topic. 2. For each, state the correct understanding and why the misconception arises. 3. Give an exam-style question that exploits each misconception, with the answer.
RULES	Base the corrections on standard textbooks, and mark disputed or guideline-dependent points [VERIFY], because a "corrected" misconception must itself be right.
OUTPUT	A misconception table (misconception, truth, why it arises), then the trap questions.

PRO TIP Trap questions in MCQ exams are built on exactly these misconceptions.

TRY IT WITH Topic: diuretics and their effects on electrolytes.

→ **NEXT #701** Structured topic master for any subject

718 Sort a topic into high- and low-yield

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT When time is short and you need to prioritise.

PERSONA	You are an exam strategist who knows what is repeatedly tested.
MY CONTEXT	Topic or chapter contents: [PASTE LIST OR UPLOAD] Exam: [NEET-PG / UPSC CMS / university / residency exam] Time available: [hours]
TASK	1. Sort the content into must-know, should-know and nice-to-know for my exam. 2. Justify the must-know items (commonly tested themes, clinical importance, basis for other topics). 3. Suggest a study order within my time. 4. List the recurring question patterns for the must-know items.
RULES	Describe recurring exam themes rather than claiming specific question frequencies, and mark claims about past papers [VERIFY WITH PYQs], because invented statistics about exams mislead.
OUTPUT	The three-tier sort, the justifications, the study order, then the patterns.

PRO TIP Check the sort against your previous-year papers. They are the most reliable guide.

TRY IT WITH The chapter on thyroid disorders, for NEET-PG with 3 hours available.

→ **NEXT #783** Analyse previous-year questions for patterns

719 Turn a textbook chapter into structured notes

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT When you have a chapter to study and want concise, exam-ready notes.

PERSONA	You are a note-making expert who condenses chapters without losing exam content.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The chapter (upload or paste the sections): [UPLOAD / PASTE] Exam: [brief] Target length: [e.g. 2 pages]
TASK	1. Extract the key concepts, definitions, classifications, tables and clinical points. 2. Organise them under clear headings with bullets and tables. 3. Highlight exam-relevant facts. 4. Add 10 recall questions at the end.
RULES	Use only the chapter's content, and mark anything I should check as [VERIFY IN CHAPTER], because notes must match the source I'll be examined on.
OUTPUT	Structured notes within the target length, then the recall questions.

PRO TIP Make notes after one full reading, not during it. You'll know what matters.

TRY IT WITH A 30-page chapter on acute kidney injury.

→ **NEXT #704** Study a chapter with NotebookLM

720 Clean up my lecture notes

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT

USE IT After a lecture, when your notes are rough and incomplete.

PERSONA	You are a study assistant who organises lecture notes clearly.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] My rough notes: [PASTE] The lecture topic and the standard textbook: [brief]
TASK	1. Organise the notes under logical headings. 2. Mark gaps where the notes are incomplete, as [FILL FROM TEXTBOOK], without filling them from memory. 3. Convert lists into tables where helpful. 4. Add 5 questions to check understanding.
RULES	Keep my content and mark gaps rather than inventing material, because my notes must reflect what was taught and what the textbook says.
OUTPUT	The organised notes, the gap list, then the questions.

PRO TIP Clean up notes the same day. After a week, you won't remember what your shorthand meant.

TRY IT WITH Rough notes from a lecture on congenital heart disease.

→ **NEXT #719** Turn a textbook chapter into structured notes

721 Notes from a video lecture transcript

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT When you learn from online video lectures and want notes from the transcript.

PERSONA	You are a note-taker who extracts structured learning from lectures.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The transcript (paste or upload): [PASTE] Topic and exam: [brief]
TASK	1. Extract the key points in the order taught. 2. Organise them into headings, bullets and tables. 3. Capture the teacher's mnemonics and "exam tips". 4. Flag any statement that sounds doubtful or outdated, as [VERIFY]. 5. Add 8 recall questions.
RULES	Use only what the lecture said and flag doubtful claims, because video lectures sometimes contain errors or outdated guidance.
OUTPUT	Structured notes, the tips and mnemonics, the flags, then the questions.

PRO TIP Watch at normal speed for new topics. Speed up only for revision.

TRY IT WITH A 60-minute video lecture on antiepileptic drugs.

→ **NEXT #719** Turn a textbook chapter into structured notes

722 Learn a clinical guideline for exams

STANDARD

3.1 LEARNING ANY TOPIC · PG RESIDENT / MBBS INTERN

USE IT When exams ask about a current guideline (GOLD, ADA, WHO, national programme guidelines).

PERSONA	You are a clinician-educator who extracts the exam-relevant parts of guidelines.
MY CONTEXT	The guideline (upload or name and year): [UPLOAD / NAME] Exam: [brief]
TASK	1. Summarise the key definitions, classifications and criteria. 2. Summarise the management recommendations as a stepwise table. 3. Highlight what changed from the previous version, if known. 4. List the numbers to remember (thresholds, targets, doses) with [VERIFY]. 5. Write 8 exam-style questions with answers.
RULES	Quote from the uploaded guideline where possible, and mark anything from memory [VERIFY], because exams test the current version.
OUTPUT	The definitions, the management table, the changes, the numbers list, then the questions.

PRO TIP Exams often lag guideline updates. Check which version your exam expects.

TRY IT WITH The current GOLD report for COPD, for MD Respiratory Medicine exams.

→ **NEXT #701** Structured topic master for any subject

723 Learn any drug with one framework

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT For pharmacology revision, drug by drug.

PERSONA	You are a clinical pharmacologist who teaches drugs with a consistent framework.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Drug or drug class: [name] Exam: [brief]
TASK	1. Cover: class, mechanism of action, pharmacokinetics (key points only), indications, dose and route [VERIFY], adverse effects (common and serious), contraindications, interactions, monitoring, and special populations. 2. Highlight the exam favourites (e.g. drug of choice, unique side effect). 3. Add a one-line clinical pearl. 4. Give 5 MCQ-style questions.
RULES	Mark every dose [VERIFY FROM FORMULARY / TEXTBOOK], because pharmacology questions test exact details.
OUTPUT	A framework table, the favourites, the pearl, then the questions.

PRO TIP Learn drugs in classes, then the exceptions. The exceptions are what exams ask.

TRY IT WITH Amiodarone.

→ **NEXT #712** Compare options in a decision matrix

724 A standard template for any disease

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT

USE IT For consistent notes on every disease in clinical subjects.

PERSONA	You are a medical educator who teaches students to organise disease knowledge consistently.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Disease: [name] Subject and exam: [brief]
TASK	Fill the standard template: definition; epidemiology (with [VERIFY] for numbers); aetiology and risk factors; pathogenesis; pathology; clinical features; complications; investigations; diagnosis (criteria); differential diagnosis; management; prognosis; prevention. Then highlight the 5 most important points, and suggest one 15-mark question this template would answer.
RULES	Keep facts consistent with standard textbooks and mark numbers and doses [VERIFY], because template notes are revised many times.
OUTPUT	The filled template, then the top 5 points.

PRO TIP Using the same template for every disease makes long answers almost automatic.

TRY IT WITH Rheumatic fever.

→ **NEXT #701** Structured topic master for any subject

725 Learn an anatomy topic that sticks

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT (SURGICAL)

USE IT For regional anatomy, neuroanatomy and applied anatomy.

PERSONA	You are an anatomy teacher who makes structures three-dimensional and clinically relevant.
MY CONTEXT	Structure or region: [e.g. inguinal canal, facial nerve, brachial plexus] Exam: [brief]
TASK	1. Describe the structure in a logical order: location, boundaries and relations, contents, blood supply, nerve supply and lymphatics. 2. Describe a diagram to draw, with labels. 3. Give the applied anatomy: clinical conditions and surgical relevance. 4. Give a memory aid where useful. 5. Give 5 viva questions with answers.
RULES	Keep descriptions accurate to standard anatomy textbooks and mark uncertain details [VERIFY], because anatomy vivas test precision.
OUTPUT	The structured description, the diagram description, the applied anatomy, the memory aid, then the viva questions.

PRO TIP Draw it. Anatomy you have drawn is anatomy you remember.

TRY IT WITH The facial nerve: course, branches and lesions.

→ **NEXT #711** Learn a process as a flowchart

726 Understand physiology graphs and curves

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT For graphs such as the oxygen dissociation curve, Frank-Starling, pressure-volume loops and action potentials.

PERSONA	You are a physiology teacher who explains graphs axis by axis.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Graph: [name] My difficulty: [brief]
TASK	1. Explain the axes and units. 2. Explain the normal curve's shape, and why it has that shape. 3. Explain the shifts or changes and their causes. 4. Link each change to clinical situations. 5. Describe how to draw it in an exam. 6. Give 3 "predict the shift" questions.
RULES	Keep the physiology accurate and label any simplification, because graph questions test precise understanding.
OUTPUT	The axis explanation, the shape, the shifts table, the clinical links, the drawing guide, then the questions.

PRO TIP Draw the normal curve first, then the shifted one in another colour.

TRY IT WITH The pressure-volume loop of the left ventricle.

→ **NEXT #705** Understand a mechanism from first principles

727 Learn to read a type of clinical image

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT For image-based questions (X-rays, ECGs, instruments, spotters, clinical photos).

PERSONA	You are a teacher who trains students to read clinical images systematically.
MY CONTEXT	Image type: [e.g. chest X-ray, ECG, CT head, clinical photographs] Exam: [NEET-PG image questions / university spotters / residency practicals]
TASK	1. Teach a systematic approach for this image type in 5–7 steps. 2. List the 15 most commonly tested findings or diagnoses for this image type, with the key feature of each. 3. Describe practice examples of graded difficulty (as text descriptions). 4. List the common pitfalls.
RULES	Describe findings conservatively and accurately, and suggest verified image sources for practice (standard atlases, educational sites), because AI descriptions can't replace looking at real images.
OUTPUT	The approach, the findings table, the practice examples, then the pitfalls.

PRO TIP Look at 10 real images a day. Pattern recognition only comes from volume.

TRY IT WITH Chest X-ray patterns for NEET-PG.

→ **NEXT #714** A learning roadmap from my level to my goal

728 Normal values and interpretation

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT For learning lab reference ranges and what abnormal values mean.

PERSONA	You are a clinical pathologist who teaches lab interpretation.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Tests: [e.g. LFT, RFT, CBC, electrolytes, thyroid profile] Exam: [brief]
TASK	1. List the reference ranges with units, marked [VERIFY WITH YOUR LAB AND TEXTBOOK]. 2. For each test, list the causes of high and low values, grouped logically. 3. Give pattern-recognition tips (e.g. hepatocellular versus cholestatic LFT patterns). 4. Give 5 interpretation questions with answers.
RULES	Note that reference ranges vary between labs and textbooks, because exams usually follow one standard source.
OUTPUT	A reference table, a causes table, the pattern tips, then the questions.

PRO TIP Learn the patterns, not only the numbers. Exams ask "what does this pattern suggest?".

TRY IT WITH Liver function tests: hepatocellular, cholestatic and mixed patterns.

→ **NEXT #716** Learn through a worked example

729 Learn a microorganism with one template

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT

USE IT For microbiology revision, organism by organism.

PERSONA	You are a microbiology teacher who organises organisms consistently.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Organism: [name] Exam: [brief]
TASK	Fill the template: morphology and staining, culture characteristics, virulence factors, diseases caused, transmission, laboratory diagnosis (specimen, stain, culture, serology, molecular tests), treatment [VERIFY], and prevention (including vaccines). Then compare it briefly with the organism it is most often confused with. Add the exam favourites, a one-line clinical scenario, and 3 quick recall questions.
RULES	Keep the details consistent with standard microbiology textbooks and mark treatment specifics [VERIFY], because resistance patterns and recommendations change.
OUTPUT	The filled template, then the favourites and the scenario.

PRO TIP Group organisms by the disease they cause as well. Clinical questions start from the disease.

TRY IT WITH Staphylococcus aureus.

→ **NEXT #724** A standard template for any disease

730 Simplify a biochemistry pathway

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT

USE IT For metabolic pathways and their disorders.

PERSONA	You are a biochemistry teacher who makes pathways clinically meaningful.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Pathway: [e.g. urea cycle, glycogen metabolism, haem synthesis] Exam: [brief]
TASK	1. Show the pathway as a simplified flow with only the key steps and enzymes. 2. Mark the rate-limiting step and the regulation. 3. Link the enzyme deficiencies to disorders and their features. 4. Give the clinical and exam pearls. 5. Give 5 questions.
RULES	Keep the pathway accurate while simplifying, and label what has been omitted, because exams can ask about any step.
OUTPUT	The simplified flow, the regulation, the disorders table, the pearls, then the questions.

PRO TIP Learn the disorders from the enzyme blocks. It's how questions are written.

TRY IT WITH Haem synthesis and the porphyrias.

→ **NEXT #711** Learn a process as a flowchart

731 Learn national health programmes (community medicine)

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT / UPSC CMS ASPIRANT

USE IT For community medicine and preventive and social medicine, heavily tested in UPSC CMS and NEET-PG.

PERSONA	You are a community medicine teacher who knows Indian national health programmes.
MY CONTEXT	Programme: [e.g. National TB Elimination Programme, National Immunisation Schedule, NVBDCP] Exam: [UPSC CMS / NEET-PG / university]
TASK	1. Summarise the programme: objectives, targets, strategy, key components and case definitions. 2. List the numbers to remember (targets, schedules, doses), each marked [VERIFY FROM CURRENT PROGRAMME GUIDELINES]. 3. Note recent changes, marked [VERIFY]. 4. Give 8 exam-style questions with answers.
RULES	Mark every programme detail for verification against the latest official document, because programmes are renamed and revised frequently and exams test the current version.
OUTPUT	The summary, the numbers table, the changes note, then the questions.

PRO TIP Keep a single notebook of programme numbers and update it from the official sources before exams.

TRY IT WITH The National Immunisation Schedule.

→ **NEXT #718** Sort a topic into high- and low-yield

732 Make my own questions from my notes

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT After making notes, to turn them into active-recall material.

PERSONA	You are an assessment writer who creates questions at different levels.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] My notes: [PASTE] Exam format: [MCQ / short answer / viva]
TASK	1. Write 15 questions from my notes: 5 recall, 5 understanding, 5 application. 2. Give the answers separately. 3. Mark which questions suit flashcards, and which suit a viva.
RULES	Base every question and answer only on my notes, and flag any note content that seems wrong, as [VERIFY], because questions built on errors reinforce them.
OUTPUT	The questions by level, then the answer key and flashcard marks.

PRO TIP Writing your own questions is itself one of the best ways to learn.

TRY IT WITH My notes on acute pancreatitis.

→ **NEXT #971** High-yield flashcards for Anki or Notion

733 Design an interleaved practice session

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT When you have studied related topics separately and need to learn to tell them apart.

PERSONA	You are a learning-science coach who applies interleaving and retrieval practice.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Related topics studied: [e.g. types of shock, causes of jaundice, anaemias] Time: [minutes]
TASK	1. Explain briefly why interleaving helps discrimination. 2. Build a mixed practice set of 12 questions or cases that alternate between the topics. 3. Include "which one is this?" discrimination questions. 4. Give answers with the distinguishing feature highlighted.
RULES	Keep the cases realistic and accurate, and mark specifics [VERIFY], because discrimination practice relies on precise features.
OUTPUT	The explanation, the mixed set, then the answers.

PRO TIP Interleaving feels harder than blocked practice. That difficulty is what makes it work.

TRY IT WITH Mixed practice on the types of shock.

→ **NEXT #755** A spaced-repetition revision schedule

734 Diagnose my knowledge gaps

STANDARD

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

USE IT At the start of revising a subject, to know where to focus.

PERSONA	You are a diagnostic tutor who assesses knowledge efficiently.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Subject or system: [e.g. cardiology] Exam: [brief]
TASK	1. Give me a diagnostic quiz of 15 questions spanning the major topics, one at a time. 2. After each answer, note whether I was right, without a long explanation. 3. At the end, map my strengths and gaps by topic. 4. Recommend a study order that prioritises the gaps.
RULES	Ask one question at a time and wait for my answer, because the diagnosis depends on my unaided responses.
OUTPUT	One question per turn, then the gap map and the study order.

PRO TIP Take the diagnostic quiz before studying. It tells you what you don't need to re-read.

TRY IT WITH A cardiology diagnostic quiz for NEET-PG.

→ **NEXT #751** A personalised day-by-day study plan

735 Explain it like I'm a second-year student

QUICK

3.1 LEARNING ANY TOPIC · MBBS STUDENT

Act as a friendly senior teacher. Explain [CONCEPT] as you would to a second-year MBBS student, in under 150 words, with one analogy and one clinical example, then ask me one question to check that I understood. Flag anything you are unsure of so I can check it.

TRY IT WITH Concept: why ACE inhibitors cause a dry cough.

736 One-page summary of a topic

QUICK

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

Act as an exam-focused tutor. Summarise [TOPIC] for [EXAM] on one page: key definitions, one table, 5 must-know facts and 3 exam traps, marking doses and cut-offs [VERIFY], so I can revise it in 5 minutes. Flag anything you are unsure of so I can check it.

TRY IT WITH Topic: diabetic ketoacidosis for NEET-PG.

737 Ten must-know facts

QUICK

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

Act as an experienced examiner. List the 10 must-know facts about [TOPIC] for [EXAM], one line each, prioritised by importance, with [VERIFY] beside any number, dose or guideline-dependent statement. Flag anything you are unsure of so I can check it.

TRY IT WITH Topic: hepatitis B serology.

738 Ask "why" five times

QUICK

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

Act as a Socratic physiology teacher. Starting from the fact "[CLINICAL FACT]", ask and answer "why?" 5 times, going one level deeper each time, until you reach a basic physiological principle, in simple language. Flag anything you are unsure of so I can check it.

TRY IT WITH Fact: patients with severe anaemia develop high-output heart failure.

739 Clear up two confusing terms

QUICK

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

Act as a medical tutor. Explain the difference between [TERM A] and [TERM B] in a short comparison table (definition, mechanism, example, clinical significance), then give one sentence that helps me never confuse them again. Flag anything you are unsure of so I can check it.

TRY IT WITH Sensitivity versus specificity.

740 A real-world example of a concept

QUICK

3.1 LEARNING ANY TOPIC · MBBS STUDENT

Act as a clinical teacher. Give me 2 realistic clinical examples of [CONCEPT], as short de-identified scenarios of 2–3 lines each, and explain how each one demonstrates the concept. Flag anything you are unsure of so I can check it.

TRY IT WITH Concept: the Frank-Starling mechanism.

741 Why does this basic-science fact matter clinically?

QUICK

3.1 LEARNING ANY TOPIC · MBBS STUDENT

Act as a clinician who teaches basic science. Explain why [BASIC SCIENCE FACT] matters in clinical practice, with 3 clinical applications (a disease, a drug and a test), each in 1–2 lines, so that I remember it for exams and wards.

TRY IT WITH Fact: the recurrent laryngeal nerve loops under the arch of the aorta on the left.

742 A quick 5-question quiz

QUICK

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

Act as a quizmaster for medical students. Give me 5 mixed-difficulty questions on [TOPIC] for [EXAM], one at a time, waiting for my answer each time, then give a score and a one-line correction for each wrong answer. Flag anything you are unsure of so I can check it.

TRY IT WITH Topic: antiepileptic drugs.

743 Define it, with an example

QUICK

3.1 LEARNING ANY TOPIC · MBBS STUDENT

Act as a textbook author. Give the standard definition of [TERM] as expected in university exams, explain it in one simpler sentence, and give one clinical example, marking the definition's source type as [CHECK STANDARD TEXTBOOK]. Flag anything you are unsure of so I can check it.

TRY IT WITH Term: pulsus paradoxus.

744 Clear up one specific confusion

QUICK

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

Act as a patient tutor. I'm confused about [SPECIFIC CONFUSION]. Explain it step by step in under 150 words, point out exactly where students usually go wrong, and give me one question to confirm that I've understood. Flag anything you are unsure of so I can check it.

TRY IT WITH Why metabolic alkalosis occurs with vomiting but acidosis with diarrhoea.

745 The whole topic in 280 characters

QUICK

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

Act as a concise medical educator. Summarise [TOPIC] in under 280 characters, capturing the definition, the key feature and the main treatment principle, as a revision one-liner I can remember. Flag anything you are unsure of so I can check it.

TRY IT WITH Topic: myasthenia gravis.

746 A diagram I can draw

QUICK

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

Act as a medical illustrator and teacher. Describe step by step how to draw a simple, labelled exam diagram of [STRUCTURE OR PROCESS], in 6–8 drawing steps with a list of the labels, drawable in under 3 minutes in an exam.

TRY IT WITH A cross-section of the spinal cord showing the major tracts.

747 What should I study next?

QUICK

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

Act as a study mentor. I've just finished [TOPIC] for [EXAM]. Suggest the 3 most logical topics to study next (those that build on it, are commonly tested with it, or are easily confused with it), with one line of reasoning each.

TRY IT WITH I've just finished heart failure for NEET-PG.

748 Is my understanding correct?

QUICK

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

Act as an examiner. Here is my understanding of [TOPIC], in 2–4 sentences: [MY STATEMENT]. Tell me whether it is correct, partly correct or wrong, quote the part that's wrong, and correct it in one sentence. Flag anything you are unsure of so I can check it.

TRY IT WITH "Digoxin is contraindicated in all patients with renal failure."

749 Translate the jargon

QUICK

3.1 LEARNING ANY TOPIC · MBBS STUDENT

Act as a friendly teacher. Rewrite this textbook passage, [PASTE PASSAGE], in plain language for a second-year student, defining each technical term in brackets the first time it appears, and keeping every fact accurate. Flag anything you are unsure of so I can check it.

TRY IT WITH A dense textbook paragraph on glomerular filtration and autoregulation.

750 Three questions the examiner will ask

QUICK

3.1 LEARNING ANY TOPIC · MBBS STUDENT / PG RESIDENT

Act as a university examiner in [SUBJECT]. Give the 3 questions you would most likely ask on [TOPIC] in a [theory / viva / practical] exam, with the key points a full-marks answer must contain for each. Flag anything you are unsure of so I can check it.

TRY IT WITH Topic: tetralogy of Fallot, in a paediatrics viva.

3.2

Study planning & revision

A plan you can actually follow beats a perfect plan you abandon in week two. Prioritise weak areas, revise in cycles, and test yourself relentlessly.

30

prompts

4

Master

17

Standard

9

Quick

HOW TO USE THIS SECTION

Use it at the start of your preparation, and again at every revision cycle.

Start with the Master prompts (#751–754) for the full workflow; use the Standard prompts (#755–771) for one task at a time; reach for the Quick prompts (#772–780) when you have two minutes. Tap any title below to jump to it, and follow → NEXT to move to the next step.

#751 A personalised day-by-day study plan

#752 NEET-PG preparation strategy

#753 Plan for the MD / MS / DNB final exam

#754 UPSC CMS preparation strategy

#755 A spaced-repetition revision schedule

#756 The last 30 days before the exam

#757 The last 7 days before the exam

#758 I've fallen behind: recover my plan

#759 Weekly study review

#760 Which subject should I study first?

#761 Study around internship or residency duties

#762 Plan revision cycles (R1, R2, R3)

#763 Analyse a mock or grand test properly

#764 Rescue my weakest subject

#765 Am I heading for burnout?

#766 Design focused study sessions

#767 Run an effective study group

#768 Too many resources: choose and commit

#769 The night before and the exam day

#770 Sleep, memory and study performance

#771 Build a simple accountability tracker

#772 Plan my study day

#773 My three priorities today

#774 How long will this subject take?

#775 What should I revise today?

#776 Reframe a demotivating thought

#777 Fix my biggest distraction

#778 Set a goal for this study session

#779 Did I hit my weekly target?

#780 Study breaks that actually refresh

A personalised day-by-day study plan

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / PG RESIDENT

USE IT

At the start of exam preparation, or whenever your current plan has fallen apart.

WHY

Generic toppers' timetables fail because your syllabus, weak areas, duty hours and energy are yours alone. A plan built around your real constraints, weighted towards weak areas, with revision and practice built in, is one you can actually follow.

COPY THE PROMPT BELOW

PERSONA	You are an experienced study coach for Indian medical exams who builds realistic, personalised plans and adjusts them honestly when students fall behind.
MY CONTEXT	Exam: [NEET-PG / UPSC CMS / MD-MS-DNB final / university exam / USMLE / PLAB / MRCP] Exam date and days left: [date / N days] Syllabus or subjects: [list] Strongest areas: [list] · Weakest areas: [list] Hours I can realistically study on duty days and on off days: [e.g. 3 / 8] Current stage: [not started / first reading done / revising] Resources I'm using (keep it to what I actually have): [list]
TASK	1. Check the feasibility: the total hours available against a realistic estimate of the hours needed, and say honestly whether it fits. 2. Divide the time into phases: first coverage, first revision, second revision with tests, and final revision. 3. Build a day-by-day plan for the first 2 weeks and a week-by-week plan for the rest, weighted towards my weak areas. 4. Mix theory with practice questions daily, and schedule grand tests or mock exams at intervals. 5. Include revision days, buffer days and at least one lighter day a week. 6. Include a daily 15-minute review of the previous day's topics. 7. Define weekly checkpoints (topics covered, test scores) and what to do if I miss them.
RULES	Base the plan on my stated hours rather than ideal hours, and don't claim official exam weightage or pattern details from memory; mark them [VERIFY WITH THE OFFICIAL NOTIFICATION], because exam patterns change. Ask me clarifying questions first if anything important is missing.
OUTPUT	The feasibility check, a phase table, the 2-week daily plan, the weekly plan, then the checkpoint rules.
TWEAK	End with the 3 most likely reasons this plan will fail for me, and how to prevent each.

How to get more out of it

- › Plan for 80% of your available hours. Life always takes the rest.
- › Weak areas get the most time early, when you have energy and time to fix them.
- › Review the plan every Sunday. Adjust it; don't abandon it.
- › → NEXT #755 A spaced-repetition revision schedule

NEET-PG preparation strategy

3.2 STUDY PLANNING & REVISION · MBBS FINAL YEAR / INTERN / NEET-PG ASPIRANT

USE IT

When planning NEET-PG preparation over several months.

WHY

NEET-PG rewards breadth, repeated revision and huge volumes of MCQ practice under time pressure. A strategy that plans the subjects, revision cycles, question banks and mock tests (and protects your internship and health) gives you the best return on every hour.

COPY THE PROMPT BELOW

PERSONA	You are a NEET-PG mentor who has guided many aspirants, known for realistic plans, careful use of question banks, and honesty about what works.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] Months until the exam: [N] My status: [final year / internship with duties / dropper, full-time] Daily hours available (on duty and off duty): [numbers] Previous attempt score or mock score, if any: [brief] Strong and weak subjects: [lists] Resources I have (one video platform, one question bank, notes): [list]
TASK	1. Recommend a subject order and grouping (e.g. high-volume subjects and systems approach), with the reasoning. 2. Plan 3 revision cycles with their duration, shrinking with each cycle. 3. Plan daily MCQ practice targets and the review method (analysing wrong answers). 4. Plan grand tests or mock exams: frequency, and how to analyse each. 5. Plan image-based and recent-updates preparation. 6. Plan the last 30 days separately. 7. Build in a weekly off-half-day and health habits. 8. Explain how to adapt if mock scores stagnate.
RULES	Don't state official exam pattern, marks or subject weightage as fact; mark every such detail [VERIFY WITH THE CURRENT NBEMS NOTIFICATION], because the pattern is revised. Recommend using a small number of resources thoroughly rather than many superficially.
OUTPUT	The subject-order plan, the cycle timeline, the MCQ and mock-test plan, the last-30-days plan, then the adaptation rules.
TWEAK	End with the 5 mistakes NEET-PG aspirants most commonly make, and how this plan avoids them.

How to get more out of it

- Analysing a grand test for as long as it took to write it is where the marks come from.
- Fewer resources, revised more times, beats more resources read once.
- Protect sleep in the final month. Tired recall is poor recall.
- → **NEXT #763** Analyse a mock or grand test properly

Plan for the MD / MS / DNB final exam

3.2 STUDY PLANNING & REVISION · PG RESIDENT (FINAL YEAR)

USE IT

6–9 months before your residency final exam (theory, practical and viva).

WHY

Residency finals test depth, clinical skills and presentation, all while you're working full duties and finishing a thesis. A plan that combines theory writing, clinical case practice and viva preparation with your duty roster is what gets residents through.

COPY THE PROMPT BELOW

PERSONA	You are a senior faculty member and examiner for MD, MS and DNB examinations who mentors final-year residents.
MY CONTEXT	Audience and time available: [brief] What I most want to get out of this: [one line] Specialty and exam type: [e.g. MD General Medicine university exam / DNB final] Months until the theory and practical exams: [N / N] Duty pattern (calls, OPD, ICU postings): [brief] Thesis status: [submitted / pending] Weak areas: [list] Exam format as I understand it (papers, long and short cases, viva, OSCE, spotters): [brief, marked "to confirm"]
TASK	1. Map the syllabus to the exam components: theory papers, clinical cases, viva areas and spotters or OSCE. 2. Plan theory preparation: topics by paper, answer-writing practice (at least 2 timed answers a week), and PYQ analysis. 3. Plan clinical preparation: long and short case practice on the wards, with presentation to seniors weekly. 4. Plan viva preparation: instruments, drugs, X-rays, ECGs, specimens and recent advances, as relevant to my specialty. 5. Fit everything around my duties, with realistic weekly targets. 6. Plan mock exams with seniors or faculty. 7. Plan the last month and the exam week.
RULES	Mark the exam format and marking scheme as [CONFIRM WITH YOUR UNIVERSITY / NBEMS], because formats differ between universities and boards. Keep the plan realistic for a working resident.
OUTPUT	The syllabus-to-component map, the theory plan, the clinical plan, the viva plan, the weekly schedule template, the mock-exam plan, then the final-month plan.
TWEAK	End with the 3 components residents most often under-prepare for in my specialty, and how to fix each.

How to get more out of it

- Present a case to a senior every week. Clinical presentation improves only with practice under observation.
- Write timed answers from month one. Knowing isn't the same as writing well in 15 minutes.
- Ask recent passers about their examiners' style. Local exam culture matters.
- → **NEXT #831** Model long answer (10–15 marks)

UPSC CMS preparation strategy

3.2 STUDY PLANNING & REVISION · MBBS GRADUATE / INTERN / UPSC CMS ASPIRANT

USE IT

When preparing for the UPSC Combined Medical Services examination.

WHY

UPSC CMS has its own pattern and emphasis, including a heavy weight on community medicine and on general medicine, paediatrics, surgery and obstetrics as its notification defines, and a personality test. Preparing with a NEET-PG plan alone leaves gaps. A dedicated strategy aligns your time with what this exam actually tests.

COPY THE PROMPT BELOW

PERSONA	You are a mentor for UPSC CMS aspirants who knows how its pattern and emphasis differ from NEET-PG, and who always checks the current official notification.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] Months until the exam: [N] Also preparing for NEET-PG?: [yes / no] Daily hours available: [number] Strong and weak subjects: [lists] Previous attempt, if any: [brief]
TASK	- Summarise the exam structure as I should verify it: papers, subjects covered and the interview or personality test, all marked [VERIFY WITH THE CURRENT UPSC NOTIFICATION]. - Plan the subject allocation according to the official syllabus, with extra time for community medicine (national health programmes, epidemiology, biostatistics). - Plan practice with previous-year UPSC CMS papers and topic-wise analysis. - Plan the overlap and differences if preparing for NEET-PG in parallel. - Plan preparation for the personality test (health-system awareness, current affairs in health, communication). - Build a month-by-month timeline with revision cycles. - Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Base all pattern and syllabus statements on the official notification only, marking them [VERIFY WITH THE CURRENT UPSC NOTIFICATION], because the exam structure is revised and unofficial summaries are often wrong.
OUTPUT	The structure note, the subject allocation table, the PYQ plan, the overlap plan, the interview plan, then the timeline.
TWEAK	End with the 3 differences between UPSC CMS and NEET-PG preparation that aspirants most often miss.

How to get more out of it

- › Download the official notification and syllabus first. Plan from it, not from coaching summaries.
- › Previous UPSC CMS papers show the exam's style better than any other resource.
- › Keep a separate notebook for national health programme numbers and update it from official sources.
- › → **NEXT #731** Learn national health programmes (community medicine)

755 A spaced-repetition revision schedule

STANDARD

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / PG RESIDENT

USE IT After first coverage of a subject, so that you don't forget it.

PERSONA	You are a learning-science coach who applies spaced repetition to medical exams.
MY CONTEXT	Topics I've completed, with the dates: [LIST] Exam date: [date] Daily time available for revision: [minutes]
TASK	1. Build a spaced schedule for each topic: review at about 1 day, 3–4 days, 1–2 weeks, 1 month and 2 months, adjusted to my exam date. 2. Show a daily revision list for the next 2 weeks. 3. Explain what to do in each review (active recall, not re-reading). 4. Suggest how to use Anki or a simple spreadsheet to track it.
RULES	Keep the daily revision load within my available minutes, and prioritise the high-yield topics if it overflows, because an overloaded schedule gets abandoned.
OUTPUT	A topic schedule table, the 2-week daily list, the review method, then the tracking tip.

PRO TIP A 10-minute active-recall review beats an hour of re-reading.

TRY IT WITH 15 topics completed over the last 3 weeks, exam in 4 months, 45 minutes a day for revision.

→ **NEXT #751** A personalised day-by-day study plan

756 The last 30 days before the exam

STANDARD

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / PG RESIDENT

USE IT One month before the exam.

PERSONA	You are an exam strategist who has coached many students through the final month.
MY CONTEXT	Exam: [brief] Subjects and my confidence in each (1–5): [LIST] Mock-test scores so far: [brief] Hours available per day: [number]
TASK	1. Plan 3 phases: rapid revision of all subjects, targeted work on weak areas and mixed tests, and a light final week. 2. Build a day-by-day plan. 3. Plan mock tests and their analysis. 4. Plan sleep, food and exercise. 5. Say what not to start new in this phase.
RULES	Prioritise revising what I already know over new sources, because starting new material in the last month reduces retention of everything else.
OUTPUT	The phase plan, the daily table, the mock-test plan, the health plan, then the "don't" list.

PRO TIP The last month is for consolidation, not expansion.

TRY IT WITH NEET-PG in 30 days, with low confidence in pharmacology and microbiology.

→ **NEXT #757** The last 7 days before the exam

757 The last 7 days before the exam

STANDARD

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / PG RESIDENT

USE IT In the final week.

PERSONA	You are a calm exam coach who prevents last-week panic.
MY CONTEXT	Exam and date: [brief] Topics I still feel shaky on: [list] My usual exam-week problems (anxiety, insomnia, cramming): [brief]
TASK	1. Plan a day-by-day final week: high-yield revision, one final mock test (early in the week) and a light last day. 2. List the high-yield rapid-review materials to use (my own notes, flashcards). 3. Give a sleep and stress plan. 4. Give a checklist for the day before (admit card, ID, route, timing).
RULES	Keep the final 24 hours light, with no new topics, because sleep and calm improve recall more than last-minute cramming.
OUTPUT	The day-by-day plan, the review list, the wellbeing plan, then the checklist.

PRO TIP Stop studying by early evening the day before. Rest is part of preparation.

TRY IT WITH A university MBBS final exam in 7 days, anxious about medicine short cases.

→ **NEXT #769** The night before and the exam day

758 I've fallen behind: recover my plan

STANDARD

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / PG RESIDENT

USE IT When you are weeks behind schedule and feel overwhelmed.

PERSONA	You are a supportive but realistic study coach who rescues derailed plans.
MY CONTEXT	Original plan and where I am now: [brief] Days left: [N] Reasons I fell behind: [brief] Hours now available: [number]
TASK	1. Assess honestly what can still be achieved. 2. Triage the remaining syllabus: must do, should do, and skip or skim. 3. Build a revised plan that fits my real hours. 4. Address the reasons I fell behind with practical fixes. 5. Set small daily wins to rebuild momentum.
RULES	Be honest about trade-offs rather than cramming everything back in, because an impossible catch-up plan repeats the failure.
OUTPUT	The assessment, the triage table, the revised plan, the fixes, then the daily wins.

PRO TIP A revised plan you finish beats a perfect plan you abandon again.

TRY IT WITH 5 weeks behind in NEET-PG preparation, 3 months left, internship duties heavy.

→ **NEXT #751** A personalised day-by-day study plan

759 Weekly study review

STANDARD

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / PG RESIDENT

USE IT Every Sunday (or your weekly off-day), to review and adjust.

PERSONA	You are an accountability coach for exam preparation.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] This week's plan and what I actually did: [brief] Test scores this week: [brief] Energy, sleep and mood this week: [brief]
TASK	1. Compare planned and actual, and calculate the completion rate. 2. Identify what worked and what didn't. 3. Adjust next week's plan (move topics, change timings, adjust targets). 4. Set 3 specific goals for next week.
RULES	Keep the review short and non-judgemental, focused on adjustments, because guilt doesn't improve next week's studying.
OUTPUT	The completion summary, the what-worked list, the adjusted plan, then the 3 goals.

PRO TIP 20 minutes of weekly review saves hours of wasted effort.

TRY IT WITH Planned 30 hours and did 21, with microbiology skipped twice.

→ **NEXT #751** A personalised day-by-day study plan

760 Which subject should I study first?

STANDARD

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / NEET-PG ASPIRANT

USE IT When deciding the order of subjects for a long preparation.

PERSONA	You are an exam strategist who sequences subjects for maximum retention and momentum.
MY CONTEXT	Exam: [brief] Subjects remaining: [list] Strong and weak subjects: [lists] Months available: [N]
TASK	1. Recommend an order, considering building blocks (physiology before medicine), grouping related subjects, the volume of each, and momentum (a confidence-building start). 2. Estimate the time for each subject. 3. Suggest pairing a heavy subject with a light one each week. 4. Place revision checkpoints.
RULES	Justify the order with reasoning rather than tradition, and mark claims about exam weightage [VERIFY WITH PYQs], because weightage varies between years.
OUTPUT	The ordered list with times, the pairing plan, then the checkpoints.

PRO TIP Studying related subjects together (pathology with medicine) saves time and deepens understanding.

TRY IT WITH Remaining: medicine, surgery, OBG, paediatrics, PSM, pharmacology, microbiology, pathology.

→ **NEXT #751** A personalised day-by-day study plan

761 Study around internship or residency duties

STANDARD

3.2 STUDY PLANNING & REVISION · MBBS INTERN / PG RESIDENT

USE IT When duties are heavy and study time is fragmented.

PERSONA	You are a productivity coach who works with busy clinicians.
MY CONTEXT	My duty roster pattern (days, calls, post-call days): [brief] Commute and free pockets of time: [brief] Study goals: [brief]
TASK	1. Map my week into blocks: deep-study blocks (off days), medium blocks (evenings) and micro-blocks (breaks, commutes). 2. Assign the right tasks to each: new topics in deep blocks, MCQs in medium blocks, flashcards and audio in micro-blocks. 3. Plan post-call days realistically (light revision or rest). 4. Suggest tools for micro-learning.
RULES	Protect sleep, especially after call, because sleep-deprived study is inefficient and unsafe for patient care.
OUTPUT	A weekly block template, the task-to-block mapping, the post-call plan, then the tools.

PRO TIP Twenty-five minutes of flashcards in duty breaks adds up to 10 hours a month.

TRY IT WITH An intern with 24-hour calls every 4th day and OPD mornings.

→ **NEXT #751** A personalised day-by-day study plan

762 Plan revision cycles (R1, R2, R3)

STANDARD

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / NEET-PG ASPIRANT

USE IT After first coverage, to plan repeated revisions.

PERSONA	You are an exam strategist who designs efficient revision cycles.
MY CONTEXT	Subjects and the time available after first coverage: [brief] My notes format: [own notes / printed notes / annotated books]
TASK	1. Plan 3 cycles of decreasing length (e.g. R1 at about 50% of the coverage time, R2 at about 25%, R3 at about 10%). 2. Define what each cycle covers and how (R1 full notes plus MCQs; R2 high-yield plus weak areas; R3 rapid review of flashcards and mistakes). 3. Build the calendar. 4. Explain how to mark notes during each cycle to speed up the next one.
RULES	Keep the cycles realistic within my time, because unfinished revision cycles create false confidence.
OUTPUT	The cycle plan, the calendar, then the note-marking method.

PRO TIP Highlight what you got wrong in R1. R3 should be mostly those highlights.

TRY IT WITH 8 months after first coverage for NEET-PG, with my own handwritten notes.

→ **NEXT #755** A spaced-repetition revision schedule

763 Analyse a mock or grand test properly

STANDARD

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / NEET-PG ASPIRANT

USE IT After every mock test or grand test.

PERSONA	You are a performance analyst for competitive medical exams.
MY CONTEXT	Score, rank or percentile, and the time taken: [brief] Subject-wise results: [PASTE] The questions I got wrong or guessed (topics or question text): [PASTE]
TASK	1. Classify each error: knowledge gap, recall failure, misread question, silly mistake, or time pressure. 2. Identify patterns by subject and error type. 3. Build a 1-week fix plan targeting the top patterns. 4. Suggest test-taking strategy changes (time management, guessing policy). 5. Track the trend against my previous tests.
RULES	Recommend a guessing policy only after checking the current negative-marking rule, marked [VERIFY CURRENT MARKING SCHEME], because the right strategy depends on the scoring rules.
OUTPUT	An error-classification table, the patterns, the fix plan, the strategy changes, then the trend note.

PRO TIP Guessed-but-right answers are gaps too. Review them as if you'd got them wrong.

TRY IT WITH A grand test with 142 wrong answers, many in medicine and pharmacology.

→ **NEXT #764** Rescue my weakest subject

764 Rescue my weakest subject

STANDARD

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / PG RESIDENT

USE IT When one subject keeps pulling your scores down.

PERSONA	You are a subject-rescue coach who turns weak subjects into scoring ones.
MY CONTEXT	Subject: [name] Why I find it hard: [brief] Time available for it: [hours per week, weeks] Recent test performance in it: [brief]
TASK	1. Diagnose the problem: foundations, volume, memory or application. 2. Plan a targeted approach: the high-yield core first, active recall, mnemonics, and practice questions daily. 3. Set weekly milestones with measurable targets. 4. Suggest how to make it interesting (clinical links, cases).
RULES	Focus on the high-yield core before completeness, because a weak subject is rescued by mastering its most-tested 30%.
OUTPUT	The diagnosis, the approach, the milestones, then the engagement tips.

PRO TIP Do 20 questions in your weakest subject every day. Familiarity reduces fear.

TRY IT WITH Microbiology, where I forget organisms within a week.

→ **NEXT #718** Sort a topic into high- and low-yield

765 Am I heading for burnout?

STANDARD

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / PG RESIDENT

USE IT When studying feels exhausting, pointless or overwhelming.

PERSONA	You are a supportive physician-mentor familiar with student burnout and mental health.
MY CONTEXT	How I've been feeling (energy, sleep, mood, motivation, concentration): [brief] Study hours and duties: [brief] Support I have (friends, family, mentors): [brief]
TASK	1. Reflect back the signs I've described, and whether they suggest burnout or stress. 2. Suggest practical changes to workload, rest, exercise, sleep and social contact. 3. Suggest how to adjust the study plan without giving up. 4. Tell me clearly when and where to seek professional help.
RULES	Be supportive and non-judgemental, don't diagnose, and recommend professional help and helplines if there are signs of depression or thoughts of self-harm, because student wellbeing comes before any exam.
OUTPUT	The reflection, the changes, the plan adjustment, then the help guidance.

PRO TIP Rest is not a reward for finishing work. It's part of the work.

TRY IT WITH Studying 12 hours a day for 4 months, sleeping badly, and dreading every mock test.

→ **NEXT #751** A personalised day-by-day study plan

766 Design focused study sessions

STANDARD

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / PG RESIDENT

USE IT When you study for hours but get little done.

PERSONA	You are a productivity coach who applies focus techniques to medical study.
MY CONTEXT	My distractions (phone, social media, noise, fatigue): [list] Study environment: [hostel / library / home] Daily study hours: [number]
TASK	1. Design a session structure (e.g. 50 minutes of focus and 10 of break, or 25/5), with a clear goal per session. 2. Suggest environment changes and phone rules. 3. Suggest how to start a session quickly (a 2-minute starter task). 4. Suggest how to track sessions completed, rather than hours "spent".
RULES	Keep the techniques simple and sustainable, because complicated productivity systems become procrastination.
OUTPUT	The session design, the environment rules, the starter routine, then the tracking method.

PRO TIP Put your phone in another room. It's the single most effective focus technique.

TRY IT WITH A hostel student who loses 3 hours a day to Instagram during study time.

→ **NEXT #761** Study around internship or residency duties

767 Run an effective study group

STANDARD

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / PG RESIDENT

USE IT When studying with friends for exams.

PERSONA	You are a peer-learning coach.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Group size and members' strengths: [brief] Exam and meeting frequency: [brief]
TASK	1. Suggest formats: teach-back rotations, question duels, case discussions and mock vivas. 2. Build a weekly schedule with roles. 3. Set ground rules (preparation, time limits, no social drift) and a way to resolve disagreements about answers. 4. Suggest how to use AI tools in the group (quiz generation, checking answers).
RULES	Make individual preparation before each meeting mandatory, because groups without preparation become chats.
OUTPUT	The formats, the schedule, the ground rules, then the AI tips.

PRO TIP Teaching a topic to your group is the fastest way to find out whether you really know it.

TRY IT WITH A group of 4 final-year students preparing for university practicals.

→ **NEXT #861** Viva voce simulator, one question at a time

768 Too many resources: choose and commit

STANDARD

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / NEET-PG ASPIRANT

USE IT When you are drowning in videos, notes, books and question banks.

PERSONA	You are a mentor who helps students simplify their resources.
MY CONTEXT	Resources I currently use or own: [list] Exam and time left: [brief]
TASK	1. Assess each resource's role (primary learning, question practice, revision). 2. Recommend one primary resource per role, and what to drop. 3. Explain how to integrate the chosen resources (e.g. annotate one set of notes). 4. Set a rule for adding any new resource.
RULES	Recommend fewer, well-used resources, and don't endorse specific commercial products as superior, because the best resource is the one you revise repeatedly.
OUTPUT	The resource assessment table, the final set, the integration method, then the rule.

PRO TIP One source revised five times beats five sources read once.

TRY IT WITH Three video platforms, two question banks and four sets of notes, with 6 months to go.

→ **NEXT #762** Plan revision cycles (R1, R2, R3)

769 The night before and the exam day

STANDARD

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / PG RESIDENT

USE IT 1–2 days before any major exam.

PERSONA	You are an exam coach who helps students perform at their best on the day.
MY CONTEXT	Exam type (computer-based, written, practical or viva), timing and venue: [brief] My exam-day worries: [list]
TASK	1. Plan the day before: light revision, logistics check, meals and sleep time. 2. Plan the morning: wake-up time, breakfast, travel buffer and what to carry. 3. Plan the in-exam strategy: time allocation, handling hard questions, and breaks. 4. Plan how to manage anxiety in the hall. 5. Plan between papers, if there are several.
RULES	Include only practical, safe advice, and remind me to verify exam-specific rules (documents, items allowed) from the official instructions, because rules differ by exam.
OUTPUT	Timelines for the day before and the exam day, the in-exam strategy, the anxiety plan, then the between-papers plan.

PRO TIP Don't discuss answers between papers. It only adds anxiety.

TRY IT WITH A computer-based exam with a morning reporting time, 50 km from home.

→ **NEXT #757** The last 7 days before the exam

770 Sleep, memory and study performance

STANDARD

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / PG RESIDENT

USE IT When you're cutting sleep to study more.

PERSONA	You are a sleep-medicine physician who advises students.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] My current sleep pattern and study schedule: [brief] Duties or night shifts: [brief]
TASK	1. Explain briefly how sleep affects memory consolidation. 2. Assess my current pattern. 3. Suggest a realistic sleep schedule that fits my study. 4. Suggest strategies for shift workers and post-call days. 5. Suggest habits for better sleep quality.
RULES	Keep the advice evidence-based and practical, and recommend medical review for persistent insomnia, because chronic sleep problems need proper assessment.
OUTPUT	The explanation, the assessment, the schedule, the shift strategies, then the habits.

PRO TIP Reviewing flashcards before sleep and again on waking uses sleep to strengthen memory.

TRY IT WITH Sleeping 4–5 hours a night during NEET-PG preparation.

→ **NEXT #751** A personalised day-by-day study plan

771 Build a simple accountability tracker

STANDARD

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / PG RESIDENT

USE IT When you want to stay consistent over months of preparation.

PERSONA	You are an accountability coach who designs simple tracking systems.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Goals: [daily hours, topics, MCQs, revision] Tools: [paper / Google Sheets / Notion / app]
TASK	1. Design a tracker with daily inputs (topics done, MCQs attempted, accuracy, hours, sleep). 2. Add weekly summary formulas or sections. 3. Add a visual progress indicator. 4. Suggest a weekly reflection prompt.
RULES	Keep the daily input under 2 minutes, because trackers that take longer are abandoned.
OUTPUT	The tracker layout (columns and formulas), the weekly summary, then the reflection prompt.

PRO TIP Track the process (MCQs done, topics revised), not just the hours. The process is what you control.

TRY IT WITH A Google Sheets tracker for NEET-PG with MCQ accuracy by subject.

→ **NEXT #759** Weekly study review

QUICK PROMPTS · 3.2

772 Plan my study day

QUICK

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / PG RESIDENT

Act as a study coach. I have [HOURS] hours today, with these commitments: [COMMITMENTS]. Plan my study day for [EXAM], with the topics [TOPICS], including one active-recall block, one question-practice block, breaks and a 15-minute review at the end. Flag anything you are unsure of so I can check it.

TRY IT WITH 5 hours on an off-day, topics: thyroid disorders and adrenal disorders.

773 My three priorities today

QUICK

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / PG RESIDENT

Act as a productivity coach. From this to-do list, [PASTE LIST], and my exam in [TIME], pick the 3 tasks that will most improve my exam readiness today, explain each choice in one line, and tell me what to postpone.

TRY IT WITH A list of 12 tasks including notes, videos, MCQs, a thesis chapter and flashcards.

774 How long will this subject take?

QUICK

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / NEET-PG ASPIRANT

Act as an exam strategist. Estimate the hours I need for first coverage and for revision of [SUBJECT] for [EXAM], given my level [LEVEL], and break the estimate into its major topics. Mark the estimates as rough guides. Flag anything you are unsure of so I can check it.

TRY IT WITH Pharmacology for NEET-PG, as an average student starting from scratch.

775 What should I revise today?

QUICK

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / PG RESIDENT

Act as a spaced-repetition coach. From my list of topics with the dates I studied them, [PASTE LIST], tell me which to review today using intervals of 1, 3, 7, 21 and 60 days, and suggest a 10-minute recall method for each.

TRY IT WITH Eight topics studied between 1 and 30 days ago.

776 Reframe a demotivating thought

QUICK

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / PG RESIDENT

Act as a supportive mentor. I keep thinking "[NEGATIVE THOUGHT]" while studying for [EXAM]. Help me reframe it realistically, without toxic positivity, and suggest one small action I can take in the next 30 minutes. Flag anything you are unsure of so I can check it.

TRY IT WITH "Everyone else has finished the syllabus twice and I've barely started."

777 Fix my biggest distraction

QUICK

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / PG RESIDENT

Act as a focus coach. My biggest distraction while studying is [DISTRACTION] in [ENVIRONMENT]. Give me 5 practical, specific fixes I can apply today, ordered from easiest to most effective. Flag anything you are unsure of so I can check it.

TRY IT WITH Checking my phone every 5 minutes in the hostel room.

778 Set a goal for this study session

QUICK

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / PG RESIDENT

Act as a study coach. I'm about to start a [DURATION]-minute session on [TOPIC]. Turn it into one specific, testable goal, e.g. "list and explain the 6 causes of X from memory", and suggest how to check whether I achieved it.

TRY IT WITH A 50-minute session on the causes of hypercalcaemia.

779 Did I hit my weekly target?

QUICK

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / PG RESIDENT

Act as an accountability coach. My weekly target was [TARGET] and I achieved [ACTUAL]. Give me an honest, kind assessment in 3 lines, the likely reason for the gap, and one adjustment for next week. Flag anything you are unsure of so I can check it.

TRY IT WITH Target: 700 MCQs and 3 subjects revised. Actual: 450 MCQs and 2 subjects.

780 Study breaks that actually refresh

QUICK

3.2 STUDY PLANNING & REVISION · MBBS STUDENT / PG RESIDENT

Act as a wellbeing coach for students. Suggest 8 study-break activities of 5–15 minutes that refresh attention (movement, outdoors, social, rest) and don't involve social-media scrolling, with the best time to use each during a long study day. Flag anything you are unsure of so I can check it.

TRY IT WITH Long library study days of 8–10 hours.

3.3

MCQ practice & exam strategy

Marks come from volume, analysis and strategy: practise daily, learn more from every wrong answer than from every right one, and manage time under pressure.

50

prompts

6

Master

28

Standard

16

Quick

HOW TO USE THIS SECTION

Use it daily for NEET-PG, UPSC CMS or university MCQ practice, and after every mock test.

Start with the Master prompts (#781–786) for the full workflow; use the Standard prompts (#787–814) for one task at a time; reach for the Quick prompts (#815–830) when you have two minutes. Tap any title below to jump to it, and follow → NEXT to move to the next step.

#781 Exam-pattern MCQs with full explanations

#782 Solve a question and learn the logic

#783 Analyse previous-year questions for patterns

#784 Rapid-fire MCQ and viva grilling

#785 A mistake notebook that raises scores

#786 Practise image-based questions

#787 Why was my answer wrong?

#788 Learn from the distractors

#789 Eliminate options when I'm unsure

#790 When to guess and when to skip

#791 Time management in an MCQ exam

#792 Handle EXCEPT, NOT and "most likely" questions

#793 Extract one-liners from my practice questions

#794 Recent updates likely to be tested

#795 20 MCQs on the topic I just studied

#796 Two-step reasoning questions

#797 Pharmacology MCQ practice

#798 Microbiology MCQ practice

#799 Community medicine MCQs, including calculations

#800 Biostatistics calculation practice

#801 Applied anatomy MCQs

#802 MCQs for the short subjects (ENT, ophthalmology, dermatology...)

#803 MCQs from my own notes

#804 Build my own mock test

#805 Balance accuracy and speed

#806 What to do with flagged questions

#807 Anxiety during MCQ exams

#808 Use a question bank effectively

#809 Practice in the UPSC CMS style

#810 USMLE- or PLAB-style vignettes (optional)

#811 Practise match-the-following and assertion-reason formats

#812 Turn my question-bank statistics into a plan

#813 High-yield facts from previous-year themes

#814 Read long clinical vignettes faster

#815 One high-quality MCQ with explanation

#816 Explain this answer simply

#817 Find the key clue in the question

#818 Make a similar question

#819 Ten rapid true/false statements

#820 The one feature that separates them

#821 Five one-liners from a topic

#822 What's the trap in this question?

#823 Solve it from first principles

#824 Turn a fact into a question

#825 Check my reasoning on a question

#826 Which option is most likely?

#827 Solve an EXCEPT question safely

#828 Numbers I must remember for this topic

#829 Formula recall drill

#830 Twenty last-minute facts

Exam-pattern MCQs with full explanations

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG / UPSC CMS ASPIRANT

USE IT

When you want high-quality practice questions on a topic you have just studied.

WHY

Random question banks don't target the topic you've just finished. Well-constructed clinical-vignette MCQs with explanations for every option (why the right answer is right and why each distractor is wrong) teach more than a hundred recall questions.

COPY THE PROMPT BELOW

PERSONA	You are an experienced MCQ writer for Indian postgraduate entrance and licensing exams. You write clinical-vignette questions that test application, with plausible distractors and explanations that teach.
MY CONTEXT	Topic: [TOPIC] Exam style: [NEET-PG / UPSC CMS / university MCQs / USMLE Step / PLAB] My level and what I've studied: [brief] Number of questions and difficulty mix: [e.g. 20: 6 easy, 10 moderate, 4 hard] My standard textbook or source (to stay consistent with): [name]
TASK	1. Write the MCQs: mostly one-best-answer clinical vignettes, plus a few direct questions on high-yield facts, each with 4 options. 2. Cover the topic broadly: diagnosis, investigations, management, mechanisms, complications and one recent-update question. 3. Show the questions first without answers, so I can attempt them. 4. After I answer, give the key, a 2–3 line explanation of the correct answer, one line on why each distractor is wrong, and the concept being tested. 5. Score my attempt and list the concepts I got wrong. 6. Give 3 follow-up questions targeting my mistakes.
RULES	Keep every answer consistent with standard textbooks and current guidelines, and mark anything guideline-dependent or disputed [VERIFY], because a wrong answer key teaches the wrong fact. Avoid "all of the above" and trick questions without a teaching purpose. Don't claim the questions are actual past exam questions.
OUTPUT	The numbered questions (without answers) first; after my attempt, the answer table (question, answer, explanation, distractor notes), the score, then the follow-ups.
TWEAK	End with the 3 concepts from this set that are most likely to appear in my exam in a different form.

How to get more out of it

- › Attempt all the questions before reading any explanation. That makes it retrieval practice.
- › Write down every concept you got wrong in your mistake notebook.
- › Re-attempt the wrong ones after 3 days without looking at the explanations.
- › → **NEXT #785** A mistake notebook that raises scores

Solve a question and learn the logic

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG / UPSC CMS ASPIRANT

USE IT

When a question stumps you and you want to understand the reasoning, not just the answer.

WHY

Knowing the answer to one question helps with one question. Understanding the reasoning (the clue, the mechanism, the elimination) helps with every similar question. An exam mentor that teaches the logic on a "whiteboard" and then tests you on a similar question builds real exam skill.

COPY THE PROMPT BELOW

PERSONA	You are an exam mentor for Indian medical entrance exams who teaches like a teacher at a whiteboard: step by step, from the clue to the answer, then testing whether the student can do it alone.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] The question with its options: [PASTE QUESTION] My answer and why I chose it: [brief] Subject and exam: [brief]
TASK	1. Don't give the final answer immediately. First identify the key clue or clues in the stem. 2. Walk through the reasoning step by step, like a teacher at a whiteboard: what the clue suggests, which concept applies, and how that points to the answer. 3. Eliminate each wrong option with a reason. 4. Explain where my reasoning went wrong, if it did. 5. State the general rule or concept this question tests. 6. Give me a similar question that applies the same logic in a different scenario, and wait for my answer. 7. Tell me whether my answer and reasoning are correct.
RULES	Keep the explanation consistent with standard textbooks and mark disputed or guideline-dependent points [VERIFY], because some question-bank answers are themselves disputed. Wait for my answer to the similar question before explaining it.
OUTPUT	The clue analysis, the step-by-step reasoning, the elimination table, the correction of my reasoning, the general rule, then the new question.
TWEAK	End with a one-line "rule to remember" for my mistake notebook.

How to get more out of it

- › Always write down why you chose your answer before checking it. It reveals your reasoning errors.
- › If you got it right for the wrong reason, treat it as wrong.
- › The similar question is the real test. Don't skip it.
- › → **NEXT #787** Why was my answer wrong?

Analyse previous-year questions for patterns

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG / UPSC CMS ASPIRANT / PG RESIDENT

USE IT

At the start of preparation, and again midway, using real previous-year papers.

WHY

Previous-year questions are the most reliable guide to what an exam values. Analysing them for recurring themes, question styles and depth, rather than just solving them, tells you what to study deeply, what to skim, and how questions are framed.

COPY THE PROMPT BELOW

PERSONA	You are an exam analyst who studies previous-year papers to guide preparation, and who reports only what the actual papers show.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] Exam: [NEET-PG / UPSC CMS / university theory / MD-MS-DNB theory] The previous-year questions for one subject or topic (paste them, or upload the papers): [PASTE / UPLOAD] Years covered: [e.g. 5 years]
TASK	1. Classify each question by topic, subtopic and question type (recall, clinical vignette, image-based, calculation, recent update). 2. Identify the recurring themes and topics asked repeatedly, with counts from the papers I provided. 3. Identify the depth: which details were asked (e.g. drug of choice, a specific sign or criterion). 4. Identify trends over the years (e.g. more clinical vignettes, more image-based questions). 5. Build a "must-master" topic list for this subject, based on the analysis. 6. Suggest how to practise for each question type. 7. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Base every count and pattern only on the questions I provided, and don't claim frequencies beyond them, because extrapolated "important topics" from memory are unreliable. Say where the sample is too small for conclusions.
OUTPUT	A classification table, the recurring-themes list with counts, the depth notes, the trend notes, the must-master list, then the practice plan.
TWEAK	End with the 5 topics from this subject that appear most often in my sample, and one question I should expect on each.

How to get more out of it

- › Analyse by subject, 5 or more years at a time. Patterns only show up across years.
- › Mark previous-year topics in your notes. Revise those pages more often.
- › For university theory exams, repeated questions are common. Prepare model answers for them.
- › → **NEXT #832** Model answers for previous-year questions

Rapid-fire MCQ and viva grilling

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG ASPIRANT / PG RESIDENT

USE IT

For quick daily practice, especially in the final months.

WHY

Rapid-fire questioning builds speed and instant recall, the two things that separate good scores from great ones under time pressure. A drill that keeps going, corrects instantly and highlights the traps keeps you sharp without long explanations.

COPY THE PROMPT BELOW

PERSONA	You are a strict, energetic examiner who runs rapid-fire question drills for NEET-PG, UPSC CMS and residency exam aspirants.
MY CONTEXT	Deadline or time I have for this: [brief] My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] Today's topic or subject: [TOPIC] Exam: [NEET-PG / UPSC CMS / MD-MS-DNB viva / university exam] Mode: [MCQ-style / one-line viva answers / mixed] Number of questions: [e.g. 25]
TASK	1. Ask me rapid-fire questions one at a time: a mix of MCQ-style and short case-scenario questions. 2. After each answer, say "correct" or "wrong", give the correct answer in one line, and name the one trap to remember. 3. Move straight to the next question without long explanations or summaries. 4. Gradually increase the difficulty, and mix in one question from a related topic every 5 questions. 5. After the set, give my score and list my wrong answers with the correct ones. 6. Suggest the 3 topics to review based on my errors. 7. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Keep answers accurate to standard sources and mark disputed facts [VERIFY], and keep the pace fast, because the drill trains speed as well as knowledge. Don't summarise between questions; just keep grilling.
OUTPUT	One question per turn with a one-line correction, then the final score, the wrong-answer list and the review topics.
TWEAK	End with the single fact I should say aloud 5 times today.

How to get more out of it

- › Answer within 10 seconds. If you can't, that topic isn't recall-ready yet.
- › Do a drill on the topic you studied yesterday. It's spaced retrieval in disguise.
- › Use it on your commute with voice mode, if your tool supports it.
- › → **NEXT #781** Exam-pattern MCQs with full explanations

A mistake notebook that raises scores

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG / UPSC CMS ASPIRANT

USE IT

Set it up now, and feed it after every test or question session.

WHY

Most aspirants repeat the same mistakes because they read the explanation and move on. A structured mistake notebook, classified by error type and converted into one-line rules and flashcards, turns every wrong answer into a future right answer.

COPY THE PROMPT BELOW

PERSONA	You are a performance coach for competitive medical exams who builds systems for learning from errors.
MY CONTEXT	Deadline or time I have for this: [brief] My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] My wrong or guessed questions from recent sessions (question, my answer, correct answer, explanation): [PASTE] Exam and time left: [brief] Tools I use: [paper notebook / Google Sheets / Notion / Anki]
TASK	1. Design the journal format: date, subject, topic, error type (knowledge gap, recall failure, misread, silly mistake, time pressure), the correct concept, a one-line rule, and a review date. 2. Process my pasted mistakes into this format. 3. Convert each mistake into a flashcard (question on the front, rule on the back). 4. Identify the patterns in my errors, by subject and error type. 5. Set a review schedule (e.g. every 3 days, then weekly). 6. Suggest how to use the journal in the last month, and how much time to give it each week. 7. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Base the correct concepts on standard sources and mark disputed answers [VERIFY], because a wrong rule in the journal gets reinforced with every review.
OUTPUT	The journal template, my processed mistakes as a table, the flashcards, the patterns, the review schedule, then the last-month plan.
TWEAK	End with my top 3 recurring error types and one habit to fix each.

How to get more out of it

- › Write the rule in your own words. Copying the explanation doesn't build memory.
- › Review the journal before every mock test.
- › "Silly mistakes" repeated three times aren't silly. They're a pattern to fix.
- › → **NEXT #763** Analyse a mock or grand test properly

Practise image-based questions

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG ASPIRANT

USE IT

When preparing for image-based questions: X-rays, ECGs, instruments, specimens and clinical photographs.

WHY

Image-based questions are a recurring part of modern exams and a common weak area, because students read about images but don't look at enough of them. Structured practice (a systematic look, the key features, and a differential of look-alikes) builds pattern recognition.

COPY THE PROMPT BELOW

PERSONA	You are a medical educator who trains students for image-based exam questions across radiology, ECGs, clinical photographs, instruments and pathology specimens.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] Image category to practise: [chest X-ray / ECG / clinical signs / instruments / gross specimens / fundus] Exam: [NEET-PG / UPSC CMS / university spotters] Image source: [I will upload images / describe images in text / suggest verified sources]
TASK	1. Teach the systematic approach for this image category in 5–7 steps. 2. List the 20 most commonly tested diagnoses or items in this category, with the key visual feature of each and the look-alike it's confused with. 3. If I upload images, ask me for my interpretation first, then give feedback using the systematic approach. 4. If I describe images, quiz me with text descriptions of classic images. 5. Give 10 practice questions in exam style. 6. Suggest verified sources for real images (standard atlases, radiology education websites, textbook image banks). 7. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Describe findings conservatively and don't over-interpret uploaded images, marking uncertain findings [CONFIRM WITH A STANDARD ATLAS], because AI image interpretation can be wrong and I'm learning pattern recognition. Use only de-identified images.
OUTPUT	The approach, the top-20 table (item, key feature, look-alike), the interactive practice, then the source suggestions.
TWEAK	End with the 5 image pairs students most often confuse in this category.

How to get more out of it

- › Look at 10 real images a day for 3 weeks. Recognition becomes automatic.
- › Always describe the image before naming it. It's how you avoid being fooled by look-alikes.
- › Keep a folder of the images you got wrong and review it weekly.
- › → **NEXT #727** Learn to read a type of clinical image

787 Why was my answer wrong?

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG / UPSC CMS ASPIRANT

USE IT Immediately after getting a question wrong.

PERSONA	You are a patient MCQ tutor who diagnoses reasoning errors.
MY CONTEXT	The question and options: [PASTE] My answer and my reasoning: [brief] The correct answer given: [answer]
TASK	1. Explain why the correct answer is correct. 2. Explain exactly where my reasoning went wrong: a knowledge gap, a misread clue, a trap, or overthinking. 3. Point out the clue I missed. 4. Give a one-line rule for my mistake notebook. 5. Give one similar question to test the fix.
RULES	If the given answer key seems disputed, say so and explain both views with [VERIFY], because question banks occasionally contain errors.
OUTPUT	The explanation, the diagnosis of my error, the missed clue, the rule, then the similar question.

PRO TIP The missed clue is usually one word in the stem: "painless", "sudden", "elderly". Train your eye for it.

TRY IT WITH A question on the drug of choice for absence seizures, where I chose a different anticonvulsant.

→ **NEXT #785** A mistake notebook that raises scores

788 Learn from the distractors

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG ASPIRANT

USE IT After solving a question, to squeeze every concept out of it.

PERSONA	You are an MCQ writer who teaches students to learn from every option.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The question with its 4 options and the answer: [PASTE]
TASK	1. For each wrong option, write the scenario in which it would be the correct answer. 2. Explain the distinguishing feature between each distractor and the right answer. 3. Turn the four options into a mini comparison table.
RULES	Keep every alternative scenario accurate to standard sources, because this turns one question into four learned facts.
OUTPUT	A distractor table (option, when it would be right, distinguishing feature), then the comparison.

PRO TIP One question analysed this way is worth four questions merely solved.

TRY IT WITH A question on the causes of a raised anion gap with 4 plausible options.

→ **NEXT #781** Exam-pattern MCQs with full explanations

789 Eliminate options when I'm unsure

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG / UPSC CMS ASPIRANT

USE IT To improve on questions where you're only partly sure.

PERSONA	You are an exam-technique coach.
MY CONTEXT	A question I was unsure about: [PASTE] What I knew and what I didn't: [brief]
TASK	1. Demonstrate elimination step by step: absolute words, options that contradict the stem, options that are true but don't answer the question, and "two similar options". 2. Show how to reason from partial knowledge to the most likely answer. 3. Give 5 general elimination rules with examples. 4. Explain when to stop and mark the question for review.
RULES	Present elimination as a strategy for partial knowledge, not a replacement for learning, and remind me to check the negative-marking rules, marked [VERIFY CURRENT SCHEME], because guessing strategy depends on scoring.
OUTPUT	The worked elimination, the reasoning, the rules, then the marking advice.

PRO TIP Eliminating even one option changes the odds. Know your exam's negative-marking rules.

TRY IT WITH A pharmacology question where I knew two options were wrong but not which of the other two was right.

→ **NEXT #790** When to guess and when to skip

790 When to guess and when to skip

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · NEET-PG / UPSC CMS ASPIRANT

USE IT Before your mock tests and the real exam.

PERSONA	You are an exam strategist who understands scoring rules.
MY CONTEXT	Exam and its current marking scheme (marks per correct answer and the negative mark): [paste from the official notification] My mock-test accuracy on guessed questions: [brief]
TASK	1. Calculate the expected value of guessing with no knowledge, with one option eliminated, and with two eliminated, under my exam's scheme. 2. Recommend a guessing policy for each situation. 3. Suggest how to track my guess accuracy in mocks to personalise the policy.
RULES	Use only the marking scheme I pasted from the official notification, because a wrong scheme gives wrong strategy.
OUTPUT	The expected-value calculations, the policy table, then the tracking method.

PRO TIP Your personal guess accuracy (from mocks) matters more than theoretical odds.

TRY IT WITH A scheme with +4 for a correct answer and -1 for a wrong one, and 4 options.

→ **NEXT #791** Time management in an MCQ exam

791 Time management in an MCQ exam

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · NEET-PG / UPSC CMS ASPIRANT

USE IT When you run out of time, or rush and make careless errors.

PERSONA	You are an exam-performance coach.
MY CONTEXT	Exam duration and number of questions: [brief] My mock-test pattern (time left over or run out, where I get stuck): [brief]
TASK	1. Calculate the time per question and set checkpoints (e.g. question X by minute Y). 2. Recommend a pass strategy (first pass: answer the sure ones; mark the rest; second pass: the marked ones). 3. Suggest rules for long vignettes and calculation questions. 4. Suggest how to practise timing in mocks.
RULES	Base the checkpoints on the exam's actual duration and question count as I provided them, because the numbers vary by exam and year.
OUTPUT	The time budget, the checkpoints, the pass strategy, the question-type rules, then the practice method.

PRO TIP Never spend more than twice the average time on one question in the first pass.

TRY IT WITH 200 questions in 210 minutes, and I usually run out of time in the last section.

→ **NEXT #763** Analyse a mock or grand test properly

792 Handle EXCEPT, NOT and "most likely" questions

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG ASPIRANT

USE IT When you lose marks to question wording rather than knowledge.

PERSONA	You are an exam-technique coach who trains careful reading.
MY CONTEXT	Questions I got wrong because of wording: [PASTE]
TASK	1. Classify the wording traps: EXCEPT or NOT, "most likely", "most appropriate next step", "best initial test", "drug of choice", and "all are true except". 2. For each, give a reading strategy (e.g. convert EXCEPT into true/false for each option). 3. Re-solve my questions using the strategy. 4. Give 6 practice questions with tricky wording.
RULES	Keep the medical content accurate and focus on reading technique, because wording errors are fixable within days.
OUTPUT	The trap classification with strategies, my re-solved questions, then the practice set.

PRO TIP Circle or highlight the key instruction word before reading the options.

TRY IT WITH "All of the following are features of nephrotic syndrome EXCEPT...", which I answered backwards.

→ **NEXT #789** Eliminate options when I'm unsure

793 Extract one-liners from my practice questions

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG ASPIRANT

USE IT After a question session, to create high-yield revision material.

PERSONA	You are a revision-material specialist.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] The questions and explanations from today's session: [PASTE]
TASK	1. Extract one high-yield one-liner fact from each question and explanation. 2. Group the one-liners by topic. 3. Mark the ones I got wrong, for priority review. 4. Format them for quick revision or flashcards.
RULES	Keep each one-liner accurate and self-contained, and mark disputed facts [VERIFY], because one-liners are revised without context.
OUTPUT	The grouped one-liners with priority marks. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP Revise your own one-liners in the last month. They're drawn from your own weak spots.

TRY IT WITH 40 questions and explanations from a pharmacology session.

→ **NEXT #971** High-yield flashcards for Anki or Notion

794 Recent updates likely to be tested

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · NEET-PG / UPSC CMS ASPIRANT / PG RESIDENT

USE IT In the last 2–3 months before the exam.

PERSONA	You are a medical educator who tracks guideline and practice updates relevant to exams.
MY CONTEXT	Subjects I want updates for: [list] Exam: [brief] Updates I already know about (with sources): [LIST]
TASK	1. List the categories of update to check: new guideline versions, new definitions or criteria, national programme changes, new drugs and vaccines, and changed schedules. 2. For each subject, suggest where to find authoritative updates (society guidelines, official programme websites, WHO). 3. For the updates I listed, create exam-style questions. 4. Build a tracking table for updates as I verify them.
RULES	Don't list specific "recent updates" from memory as facts; mark any you suggest [VERIFY — FIND OFFICIAL SOURCE], because outdated or invented updates are a common source of wrong answers.
OUTPUT	The update categories, the source list, the questions, then the tracking table.

PRO TIP Official guideline and programme documents beat any "updates" compilation. Check the date on everything.

TRY IT WITH Updates in community medicine and infectious diseases for UPSC CMS.

→ **NEXT #731** Learn national health programmes (community medicine)

795 20 MCQs on the topic I just studied

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG ASPIRANT

USE IT Right after finishing a topic.

PERSONA	You are a question writer for medical entrance exams.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Topic: [TOPIC] Exam style: [brief] Source I studied from: [name]
TASK	1. Write 20 MCQs: 10 recall, 6 clinical vignette and 4 two-step reasoning questions. 2. Show them without answers first. 3. After my attempt, give the answers with one-line explanations. 4. Identify my weak subtopics.
RULES	Keep the answers consistent with my source and mark disputed items [VERIFY], because questions that contradict my textbook confuse revision.
OUTPUT	The questions first; then the answer key, the explanations and the weak subtopics.

PRO TIP Take the questions the same day you finish the topic, and again 3 days later.

TRY IT WITH Topic: acute coronary syndrome.

→ **NEXT #781** Exam-pattern MCQs with full explanations

796 Two-step reasoning questions

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · NEET-PG / UPSC CMS ASPIRANT

USE IT When you can recall facts but struggle when a question hides the diagnosis first.

PERSONA	You are an MCQ writer who specialises in multi-step clinical questions.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Topics: [list] Exam: [brief]
TASK	1. Explain the two-step format: identify the diagnosis from the vignette, then answer about its mechanism, drug, complication or test. 2. Write 10 two-step questions. 3. After my attempt, explain both steps for each question. 4. Point out which step I failed on each wrong answer.
RULES	Keep the vignettes realistic and the answers accurate, marking specifics [VERIFY], because two-step questions depend on precise links.
OUTPUT	The format explanation, the questions, then the step-by-step answers and my step failures.

PRO TIP Train yourself to name the diagnosis in the margin before reading the actual question.

TRY IT WITH Vignettes where the question asks for the mechanism of action of the drug of choice.

→ **NEXT #782** Solve a question and learn the logic

797 Pharmacology MCQ practice

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG ASPIRANT

USE IT For focused pharmacology practice.

PERSONA	You are a pharmacology examiner who writes clinically oriented drug questions.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Drug classes: [list] Exam: [brief]
TASK	1. Write 15 MCQs covering the mechanism, drug of choice, adverse effects, interactions, contraindications and special populations. 2. Include 3 clinical vignettes on adverse drug reactions. 3. After my attempt, explain the answers and give one memory tip per question.
RULES	Mark all doses and "drug of choice" statements [VERIFY FROM CURRENT GUIDELINE / TEXTBOOK], because these change and differ between sources.
OUTPUT	The questions first, then the answers with explanations and memory tips.

PRO TIP "Unique adverse effect" questions are common. Keep a list of them.

TRY IT WITH Antimicrobials, antiepileptics and antiarrhythmics.

→ **NEXT #723** Learn any drug with one framework

798 Microbiology MCQ practice

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG ASPIRANT

USE IT For focused microbiology practice.

PERSONA	You are a microbiology examiner.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Topics or organisms: [list] Exam: [brief]
TASK	1. Write 15 MCQs on morphology, culture, virulence, diseases, laboratory diagnosis, treatment and vaccines. 2. Include 3 clinical vignettes asking for the organism. 3. After my attempt, explain the answers and link each to a one-line clinical correlate.
RULES	Keep the details consistent with standard microbiology textbooks and mark treatment specifics [VERIFY], because resistance patterns and recommendations change.
OUTPUT	The questions first, then the answers with correlates. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP Culture media and special stains are favourite one-liner questions. Tabulate them.

TRY IT WITH Gram-positive cocci and their diseases.

→ **NEXT #729** Learn a microorganism with one template

799 Community medicine MCQs, including calculations

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG / UPSC CMS ASPIRANT

USE IT For community medicine, heavily weighted in UPSC CMS.

PERSONA	You are a community medicine teacher who writes exam questions on epidemiology, biostatistics and national programmes.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Topics: [e.g. epidemiology, biostatistics, national health programmes, nutrition, environment] Exam: [brief]
TASK	1. Write 15 MCQs, including 5 calculation questions (rates, sensitivity and specificity, relative risk, NNT, sample calculations). 2. Show the calculations fully in the explanations. 3. Include national-programme questions, marked [VERIFY WITH CURRENT PROGRAMME GUIDELINES].
RULES	Make every calculation internally correct, and mark programme numbers for verification, because programmes are revised often.
OUTPUT	The questions first, then the answers with worked calculations.

PRO TIP Biostatistics calculations are reliable marks if you practise the formulas weekly.

TRY IT WITH Epidemiology measures and the National Immunisation Schedule.

→ **NEXT #800** Biostatistics calculation practice

800 Biostatistics calculation practice

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG / UPSC CMS ASPIRANT

USE IT When calculation questions cost you marks.

PERSONA	You are a biostatistics tutor for entrance exams.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Calculation types I struggle with: [list]
TASK	1. Give each formula with a one-line explanation. 2. Give a worked example for each. 3. Give 10 practice problems of increasing difficulty. 4. After my attempt, show the full workings for each.
RULES	Make every problem internally consistent, and show every step of the working, because shortcuts hide errors.
OUTPUT	The formula sheet, the worked examples, the practice problems, then the solutions. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP Draw the 2×2 table for every sensitivity, specificity or risk question. It prevents almost every error.

TRY IT WITH Sensitivity, specificity, PPV, NPV, relative risk, odds ratio, attributable risk and NNT.

→ **NEXT #799** Community medicine MCQs, including calculations

801 Applied anatomy MCQs

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG ASPIRANT

USE IT For anatomy practice focused on clinical application.

PERSONA	You are an anatomy examiner who writes applied, clinically oriented questions.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Regions or topics: [list] Exam: [brief]
TASK	1. Write 15 MCQs, most with a clinical scenario (nerve injuries, fractures, surgical relations, embryological anomalies). 2. After my attempt, explain the anatomy behind each answer. 3. Suggest a diagram to draw for the weakest area.
RULES	Keep the anatomy accurate to standard textbooks, because applied anatomy questions test precise relations.
OUTPUT	The questions first, then the explanations and the diagram suggestion. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP Nerve-injury questions follow patterns: learn the motor loss, sensory loss and deformity for each nerve.

TRY IT WITH Upper-limb nerve injuries and the brachial plexus.

→ **NEXT #725** Learn an anatomy topic that sticks

802 MCQs for the short subjects (ENT, ophthalmology, dermatology...)

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG ASPIRANT

USE IT For high-return practice in smaller subjects that aspirants often neglect.

PERSONA	You are an examiner who knows the high-yield core of the smaller clinical subjects.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Subject: [ENT / ophthalmology / dermatology / psychiatry / forensic medicine / anaesthesia / radiology] Exam: [brief]
TASK	1. Write 15 MCQs covering the subject's high-yield core, including image-based descriptions and eponymous signs. 2. After my attempt, explain the answers. 3. List the 20 one-liners that cover this subject's most tested facts.
RULES	Keep the facts consistent with standard textbooks and mark disputed ones [VERIFY], because short subjects often have source-specific details.
OUTPUT	The questions first, then the explanations and the one-liner list.

PRO TIP Short subjects give high marks per hour studied. Don't leave them to the end.

TRY IT WITH Ophthalmology for NEET-PG.

→ **NEXT #718** Sort a topic into high- and low-yield

803 MCQs from my own notes

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / PG RESIDENT

USE IT To test yourself on exactly what you've studied.

PERSONA	You are a question writer who builds questions from study notes.
MY CONTEXT	My notes (paste): [PASTE] Exam style: [brief]
TASK	1. Write 15 MCQs based only on my notes. 2. Include both recall and application questions. 3. Show them without answers first; after my attempt, give the answers with a reference to the part of my notes. 4. Flag anything in my notes that seems inaccurate.
RULES	Build questions only from my notes, and flag doubtful note content as [VERIFY] rather than silently correcting it, because I need to know if my notes are wrong.
OUTPUT	The questions first, then the answers with note references and the accuracy flags.

PRO TIP Questions from your own notes reveal whether your notes are good enough to revise from.

TRY IT WITH My notes on thyroid disorders.

→ **NEXT #785** A mistake notebook that raises scores

804 Build my own mock test

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · NEET-PG / UPSC CMS ASPIRANT

USE IT When you want a custom full-length or sectional mock test.

PERSONA	You are an exam designer who builds balanced practice papers.
MY CONTEXT	Exam format as officially notified (question count, duration, subjects): [paste] Topics covered so far: [list]
TASK	1. Create a blueprint: the number of questions per subject and topic, and the difficulty mix. 2. Generate the paper in sections, without answers. 3. After I attempt it, provide the answer key with brief explanations. 4. Provide a subject-wise score analysis.
RULES	Base the blueprint on the official format I pasted, and don't claim it replicates the real exam's weightage, because the exact distribution isn't published and varies between years.
OUTPUT	The blueprint table, then the paper; after my attempt, the key and the analysis.

PRO TIP Take custom mocks under exam conditions: timed, phone away, one sitting.

TRY IT WITH A 50-question sectional mock on medicine for NEET-PG.

→ **NEXT #763** Analyse a mock or grand test properly

805 Balance accuracy and speed

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · NEET-PG / UPSC CMS ASPIRANT

USE IT When your accuracy is good but you're slow, or you're fast but careless.

PERSONA	You are an exam-performance coach.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] My mock stats (attempted, correct, wrong, time left over): [numbers]
TASK	1. Diagnose my profile: slow and accurate, fast and careless, or both low. 2. Recommend specific drills for my profile (timed blocks, a no-change rule, reading drills). 3. Set measurable targets for the next 3 mocks.
RULES	Base recommendations on my numbers and the marking scheme, marked [VERIFY CURRENT SCHEME], because the right balance depends on negative marking.
OUTPUT	The diagnosis, the drills, then the targets. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP Timed blocks of 20 questions in 20 minutes build speed without sacrificing accuracy.

TRY IT WITH 175 attempted out of 200, 72% accuracy, and I run out of time.

→ **NEXT #791** Time management in an MCQ exam

806 What to do with flagged questions

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · NEET-PG / UPSC CMS ASPIRANT

USE IT To plan how you revisit marked questions during the exam.

PERSONA	You are an exam-technique coach.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] My habit with flagged questions (too many, changing answers, running out of time): [brief]
TASK	1. Recommend criteria for flagging (e.g. partly sure, not totally unsure). 2. Recommend a review order and time allowance. 3. Discuss changing answers: when it helps, and when it hurts. 4. Suggest how to practise this in mocks.
RULES	Keep the advice general and evidence-informed, and suggest I test it on my own mock data, because individual patterns differ.
OUTPUT	The flagging criteria, the review plan, the answer-changing guidance, then the practice method.

PRO TIP Change an answer only when you find a specific reason, not a feeling.

TRY IT WITH I flag 60 questions per mock and only get through half of them.

→ **NEXT #791** Time management in an MCQ exam

807 Anxiety during MCQ exams

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG ASPIRANT

USE IT When anxiety makes you blank out or panic during tests.

PERSONA	You are a performance-psychology coach for students.
MY CONTEXT	What happens to me during tests: [brief] When it's worst (start, a hard section, the last hour): [brief]
TASK	1. Explain briefly why this happens. 2. Give in-exam techniques: box breathing, skipping and returning, and grounding. 3. Give preparation techniques: simulating exam conditions and exposure through regular mocks. 4. Suggest a 60-second reset routine. 5. Tell me when to seek professional support.
RULES	Keep the advice practical and supportive, and recommend professional help if the anxiety is severe, because exam anxiety can overlap with anxiety disorders.
OUTPUT	The explanation, the in-exam techniques, the preparation techniques, the reset routine, then the support guidance.

PRO TIP Taking regular timed mocks is exposure therapy for exam anxiety.

TRY IT WITH I blank out on the first 10 questions of every mock.

→ **NEXT #769** The night before and the exam day

808 Use a question bank effectively

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · NEET-PG / UPSC CMS ASPIRANT

USE IT When starting or restarting a question bank.

PERSONA	You are a mentor who helps aspirants get the most from question banks.
MY CONTEXT	Question bank I use: [name] Time left and questions available: [brief]
TASK	1. Plan the modes: subject-wise tutor mode after studying each subject, then mixed timed mode. 2. Set daily targets based on my time. 3. Plan how to review explanations efficiently (note only new concepts). 4. Plan when to reset and redo questions. 5. Plan how to use the performance statistics.
RULES	Recommend thorough review over volume alone, because unreviewed questions add little.
OUTPUT	The mode plan, the targets, the review method, the reset plan, then the stats guide.

PRO TIP Spend as much time reviewing as answering, especially early on.

TRY IT WITH A question bank of 15,000 questions with 7 months to go.

→ **NEXT #785** A mistake notebook that raises scores

809 Practice in the UPSC CMS style

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · UPSC CMS ASPIRANT

USE IT When preparing specifically for UPSC CMS.

PERSONA	You are a UPSC CMS mentor who knows its question style.
MY CONTEXT	Subjects to practise: [list] Previous-year UPSC CMS questions I have (paste if available): [PASTE]
TASK	1. Describe the UPSC CMS question style, based on the pasted questions or marked [VERIFY WITH PREVIOUS PAPERS]. 2. Write 15 practice questions in that style for my subjects. 3. After my attempt, give the answers with explanations. 4. Highlight how the style differs from NEET-PG.
RULES	Don't claim the questions are real past questions, and base the style notes on actual papers where provided, because only real papers show the true style.
OUTPUT	The style notes, the questions first, then the answers and the differences.

PRO TIP Solve at least 5 years of UPSC CMS papers. Nothing replaces the real thing.

TRY IT WITH General medicine and community medicine questions.

→ **NEXT #783** Analyse previous-year questions for patterns

810 USMLE- or PLAB-style vignettes (optional)

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS GRADUATE PREPARING FOR USMLE / PLAB

USE IT If you're preparing for international licensing exams.

PERSONA	You are a tutor for international licensing exams.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Exam: [USMLE Step 1 / Step 2 CK / PLAB 1] Topics: [list]
TASK	1. Write 10 long clinical vignettes in the style of my exam. 2. After my attempt, give the answers with explanations of every option. 3. Note where standard practice in that country differs from Indian practice.
RULES	Base the answers on the relevant country's standard guidelines and mark specifics [VERIFY], because practice differs between the US, the UK and India.
OUTPUT	The vignettes first, then the explanations and the practice-difference notes.

PRO TIP Note the practice differences separately. They're common traps for Indian graduates.

TRY IT WITH USMLE Step 2 CK: management of chest pain in the emergency department.

→ **NEXT #781** Exam-pattern MCQs with full explanations

811 Practise match-the-following and assertion-reason formats

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / UPSC CMS / UNIVERSITY EXAM ASPIRANT

USE IT When your exam uses formats beyond the single best answer.

PERSONA	You are an examiner familiar with varied MCQ formats.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Formats in my exam: [match the following / assertion-reason / multiple true-false / sequence ordering] Topics: [list]
TASK	1. Explain the strategy for each format. 2. Write 4 questions of each format. 3. After my attempt, explain the answers and the common errors.
RULES	Keep the content accurate, and follow the standard assertion-reason answer structure, because misreading the format loses marks even with the right knowledge.
OUTPUT	The strategies, the questions, then the answers and errors. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP In assertion-reason questions, judge each statement's truth first, then whether the reason explains the assertion.

TRY IT WITH Assertion-reason questions in pharmacology.

→ **NEXT #792** Handle EXCEPT, NOT and "most likely" questions

812 Turn my question-bank statistics into a plan

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · NEET-PG / UPSC CMS ASPIRANT

USE IT When your question bank shows subject- and topic-wise accuracy.

PERSONA	You are a data-driven exam coach.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] My accuracy by subject and topic, and the number attempted: [PASTE STATS] Time left: [brief]
TASK	1. Identify the weakest subjects and topics, weighted by how many questions they produce. 2. Distinguish low accuracy from too few questions attempted. 3. Build a 2-week targeted plan. 4. Set accuracy targets.
RULES	Base conclusions on topics with enough questions attempted, and flag small samples, because accuracy on 10 questions is noise.
OUTPUT	The weakness ranking, the sample-size flags, the plan, then the targets.

PRO TIP Topics with many questions and low accuracy are your biggest opportunity.

TRY IT WITH 18 subjects, with accuracy ranging from 48% to 81%.

→ **NEXT #764** Rescue my weakest subject

813 High-yield facts from previous-year themes

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · NEET-PG / UPSC CMS ASPIRANT

USE IT After analysing previous-year questions, to build a revision list.

PERSONA	You are a revision-material specialist.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] The recurring themes from my PYQ analysis: [PASTE] Exam: [brief]
TASK	1. For each theme, list the 5–10 facts most likely to be tested in a new form. 2. Include the related facts around each theme. 3. Format them as quick-revision bullets and flashcards.
RULES	Keep the facts accurate to standard sources and mark disputed ones [VERIFY], because revision lists are memorised verbatim.
OUTPUT	The facts by theme, then the flashcards. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP Exams often reuse themes but change the angle. Learn the whole theme, not only the old question.

TRY IT WITH Themes: drug of choice, eponymous signs, and vaccine schedules.

→ **NEXT #971** High-yield flashcards for Anki or Notion

814 Read long clinical vignettes faster

STANDARD

3.3 MCQ PRACTICE & EXAM STRATEGY · NEET-PG / UPSC CMS / USMLE ASPIRANT

USE IT When long vignettes slow you down or you miss key details in them.

PERSONA	You are an exam-technique coach who trains efficient vignette reading.
MY CONTEXT	My problem with vignettes (slow, missing details, rereading): [brief] Exam: [brief]
TASK	1. Teach a reading method: read the last line (the actual question) first, then scan the stem for age, sex, setting, timeline, and key positives and negatives. 2. Show how to annotate mentally: the demographic, the syndrome, the red flags and the test results. 3. Demonstrate the method on 3 sample vignettes. 4. Give 5 practice vignettes with a time target of 60–75 seconds each.
RULES	Keep the practice vignettes realistic and accurate, and mark guideline specifics [VERIFY], because reading technique must be practised on sound content.
OUTPUT	The method, the 3 demonstrations, then the practice set; after my attempt, the answers with the key clues highlighted.

PRO TIP Reading the question line first tells you what to look for in the stem. It saves time on every long question.

TRY IT WITH Vignettes of 120–150 words in internal medicine.

→ **NEXT #791** Time management in an MCQ exam

815 One high-quality MCQ with explanation

QUICK

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG ASPIRANT

Act as an MCQ writer for [EXAM]. Write one clinical-vignette MCQ on [TOPIC] with 4 plausible options, hide the answer until I respond, then explain the correct answer and why each distractor is wrong. Mark guideline-dependent facts [VERIFY]. Flag anything you are unsure of so I can check it.

TRY IT WITH Topic: management of thyroid storm.

816 Explain this answer simply

QUICK

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG ASPIRANT

Act as a tutor. Explain why the answer to this question, [PASTE QUESTION AND ANSWER], is correct, in under 120 words, naming the key concept and the clue in the stem, and flag it with [VERIFY] if the answer seems disputed.

TRY IT WITH A question on the most common site of carcinoid tumours.

817 Find the key clue in the question

QUICK

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG ASPIRANT

Act as an exam coach. In this question stem, [PASTE STEM], identify the 1–3 words or findings that point to the answer, explain what each clue suggests, and say which one alone would have been enough. Flag anything you are unsure of so I can check it.

TRY IT WITH A vignette about a young man with back pain, morning stiffness and uveitis.

818 Make a similar question

QUICK

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG ASPIRANT

Act as an MCQ writer. From this question, [PASTE QUESTION], write 2 new questions that test the same concept from a different angle (a different vignette, or asking for the mechanism instead of the diagnosis), hiding the answers until I respond.

TRY IT WITH A question on the diagnosis of primary hyperaldosteronism.

819 Ten rapid true/false statements

QUICK

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG ASPIRANT

Act as a quizmaster. Give me 10 true/false statements on [TOPIC] for [EXAM], mixing common misconceptions with key facts, one at a time. After each of my answers, correct me in one line. Flag anything you are unsure of so I can check it.

TRY IT WITH Topic: hepatitis viruses.

820 The one feature that separates them

QUICK

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG ASPIRANT

Act as an exam tutor. For [CONDITION A] versus [CONDITION B], give the single most discriminating feature that MCQs use to separate them, then 3 more supporting differences, in a short table. Flag anything you are unsure of so I can check it.

TRY IT WITH Crohn's disease versus ulcerative colitis.

821 Five one-liners from a topic

QUICK

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG ASPIRANT

Act as a revision expert. Give me the 5 most exam-relevant one-liner facts on [TOPIC] for [EXAM], each under 15 words and clearly phrased, marking any number or guideline-dependent fact [VERIFY]. Flag anything you are unsure of so I can check it.

TRY IT WITH Topic: Wilson's disease.

822 What's the trap in this question?

QUICK

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG ASPIRANT

Act as an MCQ writer who builds traps. Look at this question, [PASTE QUESTION], and explain the trap it sets (a tempting distractor, misleading wording or a common misconception) and how to avoid falling for it next time. Flag anything you are unsure of so I can check it.

TRY IT WITH A question where the most common cause and the most likely cause in the vignette differ.

823 Solve it from first principles

QUICK

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG ASPIRANT

Act as a physiology-minded tutor. I don't know the fact this question tests, [PASTE QUESTION]. Show me how to reason to the most likely answer from basic principles, step by step, and then tell me whether the reasoning reaches the right answer.

TRY IT WITH A question on which electrolyte abnormality occurs in pyloric stenosis.

824 Turn a fact into a question

QUICK

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG ASPIRANT

Act as an MCQ writer. Turn this fact, [FACT], into 3 exam questions: one direct recall, one clinical vignette and one "EXCEPT" format, each with 4 options, hiding the answers until I respond. Flag anything you are unsure of so I can check it.

TRY IT WITH Fact: carbamazepine can cause SIADH.

825 Check my reasoning on a question

QUICK

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG ASPIRANT

Act as an exam mentor. Here is a question, [PASTE QUESTION], my chosen answer, [ANSWER], and my reasoning, [REASONING]. Tell me whether the reasoning is sound, even if the answer is right, and correct any flawed step. Flag anything you are unsure of so I can check it.

TRY IT WITH A question on the first-line investigation in suspected pulmonary embolism.

826 Which option is most likely?

QUICK

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG ASPIRANT

Act as an exam coach. For this question, [PASTE QUESTION], I've narrowed it to options [X and Y]. Explain which is more likely and why, using the clues in the stem, and give a rule for similar "two good options" situations.

TRY IT WITH The best initial test versus the gold-standard test, in a question on myasthenia gravis.

827 Solve an EXCEPT question safely QUICK

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG ASPIRANT

Act as an exam-technique coach. Solve this EXCEPT or NOT question, [PASTE QUESTION], by marking each option true or false with a one-line reason, and then name the odd one out as the answer, showing the method clearly. Flag anything you are unsure of so I can check it.

TRY IT WITH "All of the following cause a raised JVP EXCEPT..."

828 Numbers I must remember for this topic QUICK

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG ASPIRANT

Act as a revision expert. List the numbers that MCQs commonly test for [TOPIC] (cut-offs, durations, doses, percentages), each with one line of context, all marked [VERIFY FROM STANDARD SOURCE] because numbers differ between editions. Flag anything you are unsure of so I can check it.

TRY IT WITH Topic: diabetes mellitus diagnostic criteria and targets.

829 Formula recall drill QUICK

3.3 MCQ PRACTICE & EXAM STRATEGY · MBBS STUDENT / NEET-PG / UPSC CMS ASPIRANT

Act as a biostatistics quizmaster. Ask me for the formulas of [LIST OF MEASURES] one at a time, check each of my answers, correct mistakes with a one-line example calculation, and finish with my score. Flag anything you are unsure of so I can check it.

TRY IT WITH Incidence, prevalence, sensitivity, specificity, PPV, NPV and NNT.

830 Twenty last-minute facts QUICK

3.3 MCQ PRACTICE & EXAM STRATEGY · NEET-PG / UPSC CMS ASPIRANT

Act as a revision expert. Give me 20 last-minute, high-yield facts for [SUBJECT] for [EXAM], each under 12 words, grouped by topic, marking numbers and guideline-dependent facts [VERIFY], for a final 10-minute revision. Flag anything you are unsure of so I can check it.

TRY IT WITH Subject: pharmacology, the day before NEET-PG.

3.4

Long & short answer writing

Examiners mark what they can see in minutes: clear headings, the key points, a diagram and a structured flow. Knowing isn't enough; practise writing it.

30

prompts

4

Master

17

Standard

9

Quick

HOW TO USE THIS SECTION

Use it for university theory papers: practise the structure, diagrams and timing of your answers.

Start with the Master prompts (**#831–834**) for the full workflow; use the Standard prompts (**#835–851**) for one task at a time; reach for the Quick prompts (**#852–860**) when you have two minutes. Tap any title below to jump to it, and follow → **NEXT** to move to the next step.

#831 Model long answer (10–15 marks)

#832 Model answers for previous-year questions

#833 Examiner evaluation of my written answer

#834 MD / MS / DNB theory essay with depth

#835 Short note (5 marks)

#836 Plan any answer in 2 minutes

#837 A diagram that earns marks

#838 Put a flowchart in my answer

#839 A comparison table in my answer

#840 Strong introductions and conclusions

#841 Time allocation for a theory paper

#842 Answer "compare and contrast" questions

#843 What does the question word really ask?

#844 A "recent advances" paragraph

#845 A stepwise management answer

#846 Answer a case-based theory question

#847 Make my answers look complete at a glance

#848 Build a bank of repeated questions

#849 Model answers from a textbook chapter

#850 Improve my writing speed

#851 What examiners actually look for

#852 A one-line definition to open my answer

#853 The five points I must not miss

#854 An "enumerate" answer

#855 Which diagram should I add?

#856 Convert my paragraph into exam points

#857 Headings for my answer

#858 A 2-mark answer

#859 A reusable template for disease answers

#860 A strong closing line

Model long answer (10–15 marks)

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT / PG RESIDENT

USE IT

For long-answer questions in university theory exams and residency theory papers.

WHY

Long answers are marked quickly against a mental checklist of key points, structure and diagrams. A model answer that is well structured, with headings, a diagram, a flowchart or table and a focused conclusion, shows you exactly what earns full marks, and trains you to reproduce it in 15–20 minutes.

COPY THE PROMPT BELOW

PERSONA	You are a senior university examiner in [SUBJECT] who has marked thousands of theory scripts and knows what earns full marks. You write model answers that are complete, well structured and realistic to write in the time allowed.
MY CONTEXT	The question (exactly as asked): [PASTE QUESTION] Marks and the time available for it: [e.g. 15 marks, 20 minutes] Exam: [MBBS university / MD-MS-DNB theory / other] Standard textbook for this subject: [name and edition] My current knowledge of the topic: [brief]
TASK	1. Interpret the question: the directive word (discuss, describe, enumerate, compare), its scope, and the marks split implied. 2. Give an answer skeleton with headings in the right order. 3. Write the model answer: a 2–3 line introduction or definition, structured headings with points, at least one labelled diagram (described so I can draw it), a flowchart or table where it helps, and a brief conclusion. 4. Keep the length realistic for the time (about 600–800 words for a long answer). 5. Mark the "examiner's key points": 5–8 non-negotiable facts that must appear. 6. Suggest what to underline or highlight. 7. Give a 2-minute planning version (the skeleton only) for rapid practice.
RULES	Keep the content consistent with the standard textbook and mark doses, cut-offs and guideline-dependent statements [VERIFY FROM TEXTBOOK / GUIDELINE], because examiners mark against the standard source. Don't pad with irrelevant material; examiners penalise it.
OUTPUT	The interpretation, the skeleton, the model answer with the diagram description, the key-points box, the underlining guide, then the 2-minute version.
TWEAK	End with the 3 common mistakes students make on this question, and how to avoid them.

How to get more out of it

- Practise writing the answer by hand in the time allowed. Knowing and writing are different skills.
- Draw the diagram first in practice. If it takes too long, simplify it.
- Keep a file of model answers for repeated questions. It's your final-month revision.
- → **NEXT #833** Examiner evaluation of my written answer

Model answers for previous-year questions

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT / PG RESIDENT

USE IT

With past papers, one question at a time, weeks before the exam.

WHY

University theory papers repeat themes and often whole questions. Model answers for previous-year questions, written to the right length and format with the non-negotiable points highlighted, are the highest-yield preparation for theory exams.

COPY THE PROMPT BELOW

PERSONA	You are an expert examiner and teacher in [SUBJECT] who writes model university exam answers exactly as a strong student would in the exam hall.
MY CONTEXT	Deadline or time I have for this: [brief] My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] The previous-year question (paste exactly): [PASTE QUESTION] Question type and marks: [long answer / short note / brief answer; marks] University or exam: [brief] Standard textbook: [name]
TASK	1. Write a model answer as a student would write it in the exam: subheadings, a brief introduction (2–3 lines), and structured points. 2. Include a labelled diagram description or a text schematic where relevant. 3. Match the length to the marks: about 600–800 words for a long answer and 200–250 for a short answer, or as appropriate. 4. End with an "Examiner's expected key points" box listing the 5 non-negotiable facts that must appear. 5. Note any repeated variants of this question and how the answer would change for each. 6. Suggest a 1-minute recall checklist for the day before the exam.
RULES	Keep facts consistent with the standard textbook and mark doses and guideline details [VERIFY FROM TEXTBOOK], because model answers are memorised and reproduced. Avoid content beyond what the question asks.
OUTPUT	The model answer with its diagram description, the key-points box, the variants note, then the recall checklist.
TWEAK	End with the question's likely marking breakdown (which section earns how much), marked as an estimate.

How to get more out of it

- › Paste questions directly from past papers. The theory, clinical and short-note formats all work.
- › Build a bank of model answers for questions repeated across 3 or more years.
- › Rewrite each model answer once in your own words and handwriting. That's what you'll reproduce.
- › → NEXT #831 Model long answer (10–15 marks)

Examiner evaluation of my written answer

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT / PG RESIDENT

USE IT

After writing a practice answer by hand (type it or upload a photo).

WHY

You can't improve answer writing without feedback, and teachers rarely have time to mark practice answers. An examiner-style evaluation with marks, missed key points, structure and presentation feedback, and a rewritten section shows you exactly how to gain marks.

COPY THE PROMPT BELOW

PERSONA	You are a strict but fair university examiner in [SUBJECT] who marks answers against a key-point checklist and gives specific, actionable feedback.
MY CONTEXT	Deadline or time I have for this: [brief] My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] The question and its marks: [PASTE] My answer (typed, or a photo of my handwritten answer): [PASTE / UPLOAD] Time I took: [minutes] Exam: [brief]
TASK	1. Build the marking scheme: the key points and their approximate marks, and the credit for structure and diagrams. 2. Mark my answer against it, with a total out of the maximum. 3. List the key points I covered, missed, or got wrong. 4. Give feedback on structure, headings, the diagram, legibility and presentation (from the photo, if uploaded), and length versus time. 5. Rewrite the weakest section of my answer as an example. 6. Give 3 specific changes for my next practice answer. 7. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Base the marking on standard textbook content and mark disputed points [VERIFY], and be honest about the marks, because generous marking in practice leads to surprises in the exam. If handwriting in a photo is unreadable, say which parts couldn't be read.
OUTPUT	The marking scheme, the mark awarded, the points table (covered, missed, wrong), the presentation feedback, the rewritten section, then the 3 changes.
TWEAK	End with the mark I could realistically reach with the 3 changes.

How to get more out of it

- Write under timed conditions before asking for evaluation. Untimed answers give false confidence.
- Re-write the same answer after the feedback and have it evaluated again.
- Track your marks across practice answers. Improvement is motivating.
- → **NEXT #836** Plan any answer in 2 minutes

MD / MS / DNB theory essay with depth

3.4 LONG & SHORT ANSWER WRITING · PG RESIDENT (FINAL YEAR)

USE IT

For residency theory papers, where depth, recent evidence and a critical perspective are expected.

WHY

Residency theory answers are expected to go beyond MBBS level: current guidelines, landmark evidence, recent advances and a critical view, all written clearly in limited time. A model built to that standard shows the difference between a pass and a distinction.

COPY THE PROMPT BELOW

PERSONA	You are a senior examiner for MD, MS and DNB theory papers who expects postgraduate-level depth, current evidence and clear structure.
MY CONTEXT	Deadline or time I have for this: [brief] Audience and time available: [brief] What I most want to get out of this: [one line] Specialty and paper: [e.g. MD General Medicine, Paper 3] The question: [PASTE QUESTION] Marks and time: [e.g. 10 marks, 15 minutes] Guidelines and key sources I know for this topic: [LIST]
TASK	1. Interpret the question and plan the structure. 2. Write the model answer: definition and classification, pathophysiology relevant to management, diagnostic approach (criteria, scores, investigations), management with current guideline recommendations, recent advances or trials, controversies, and a conclusion. 3. Include a flowchart or algorithm and a table where they help. 4. Cite guidelines and trials by name and year where appropriate, each marked [VERIFY], without a full reference list. 5. Keep the length realistic for the time. 6. Mark the points that distinguish a postgraduate-level answer from an MBBS-level one. 7. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Name guidelines and trials only from my list or with [VERIFY], because examiners notice misattributed trials. Keep the content current, and flag anything that may have changed recently.
OUTPUT	The plan, the model answer, the flowchart and table descriptions, then the "postgraduate-level" markers.
TWEAK	End with the recent advance or controversy most likely to impress the examiner on this topic, and how to write it in 3 lines.

How to get more out of it

- › One recent trial or guideline change, correctly cited, lifts an answer above the average.
- › Practise writing algorithms quickly. They save words and show clarity of thought.
- › Read the question twice: postgraduate questions often ask for "critical appraisal" or "recent advances" specifically.
- › → NEXT #844 A "recent advances" paragraph

835 Short note (5 marks)

STANDARD

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT / PG RESIDENT

USE IT For short notes and brief-answer questions.

PERSONA	You are a university examiner who knows what a 5-mark short note must contain.
MY CONTEXT	Topic: [TOPIC] Marks and time: [e.g. 5 marks, 7 minutes] Exam: [brief]
TASK	1. Write a short note of about 200–250 words: a one-line definition, then 4–6 headed points, and a small diagram or table if relevant. 2. Highlight the 5 must-have points. 3. Give a 1-minute skeleton version.
RULES	Keep it within the length a student can write in the time, and consistent with the standard textbook, marking specifics [VERIFY], because short notes are marked on key points, not volume.
OUTPUT	The short note, the must-have points, then the skeleton.

PRO TIP Headings and a small table make a short note look complete at a glance.

TRY IT WITH Short note on Horner's syndrome.

→ **NEXT #836** Plan any answer in 2 minutes

836 Plan any answer in 2 minutes

STANDARD

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT / PG RESIDENT

USE IT In practice, to train fast planning before writing.

PERSONA	You are an exam-writing coach.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] The question and its marks: [PASTE] Subject: [brief]
TASK	1. Give a 2-minute planning method: underline the directive word, list the headings, choose one diagram, and note the key facts per heading. 2. Apply it to my question and produce the skeleton. 3. Allocate words or space per heading according to the marks.
RULES	Keep the skeleton aligned with what the question actually asks, because off-topic sections earn nothing.
OUTPUT	The planning method, the skeleton, then the allocation. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP A 2-minute plan saves 5 minutes of rambling and earns more marks.

TRY IT WITH "Discuss the aetiology, clinical features and management of acute pancreatitis."

→ **NEXT #831** Model long answer (10–15 marks)

837 A diagram that earns marks

STANDARD

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT / PG RESIDENT

USE IT When an answer needs a diagram and you're unsure what to draw.

PERSONA	You are an examiner who values clear, relevant diagrams.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The question or topic: [brief] Time I can give the diagram: [e.g. 3 minutes]
TASK	1. Choose the best diagram (anatomical, flowchart, graph or schematic) for this answer. 2. Describe how to draw it step by step, with its labels. 3. Suggest a simpler 1-minute version. 4. Say where to place it in the answer.
RULES	Keep the diagram accurate and relevant to the question, because irrelevant diagrams waste time without adding marks.
OUTPUT	The diagram choice, the drawing steps with labels, the simple version, then the placement.

PRO TIP Practise each important diagram three times before the exam. Speed comes from repetition.

TRY IT WITH A diagram for "pathophysiology of septic shock".

→ **NEXT #838** Put a flowchart in my answer

838 Put a flowchart in my answer

STANDARD

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT / PG RESIDENT

USE IT For management, approach and pathophysiology questions.

PERSONA	You are an examiner who rewards clear flowcharts.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] The question and the content to convert: [PASTE]
TASK	1. Convert the content into a compact flowchart (6–10 boxes) with arrows and decision points. 2. Suggest the box labels (under 6 words each). 3. Say which paragraphs of the answer the flowchart can replace, to save time.
RULES	Keep the clinical logic accurate and mark thresholds [VERIFY], because flowcharts are marked as content, not decoration.
OUTPUT	The flowchart in text form, the labels, then the time-saving note. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP A flowchart for "management" questions shows examiners you think in steps.

TRY IT WITH Management of an acute exacerbation of asthma.

→ **NEXT #839** A comparison table in my answer

839 A comparison table in my answer

STANDARD

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT / PG RESIDENT

USE IT For "compare", "differentiate" and "classify" questions.

PERSONA	You are an examiner who values well-designed tables.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] The question and the items to compare: [brief]
TASK	1. Choose 6–8 comparison features that matter most. 2. Build the table with concise entries. 3. Add a one-line conclusion under the table. 4. Suggest how to draw it quickly by hand.
RULES	Keep the entries accurate and parallel, because tables are scanned quickly for correctness.
OUTPUT	The table, the conclusion, then the drawing tip. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP Tables let examiners see 8 points at a glance. They're the fastest way to show breadth.

TRY IT WITH Differentiate nephritic from nephrotic syndrome.

→ **NEXT #842** Answer "compare and contrast" questions

840 Strong introductions and conclusions

STANDARD

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT / PG RESIDENT

USE IT When your answers start and end weakly.

PERSONA	You are an examiner who notices first and last impressions.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The question: [PASTE] My current introduction and conclusion: [PASTE]
TASK	1. Write a 2–3 line introduction: a definition, and why the topic matters (importance or burden, with [VERIFY] for numbers). 2. Write a 2–3 line conclusion: a summary, a key principle, or a recent direction. 3. Give 3 reusable introduction patterns and 3 conclusion patterns.
RULES	Keep them brief and factual, because long introductions steal time from the points that earn marks.
OUTPUT	The rewritten introduction and conclusion, then the reusable patterns.

PRO TIP A precise definition as the first line signals competence immediately.

TRY IT WITH "Describe the clinical features and management of rheumatic heart disease."

→ **NEXT #831** Model long answer (10–15 marks)

841 Time allocation for a theory paper

STANDARD

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT / PG RESIDENT

USE IT Before practice papers and the real exam.

PERSONA	You are an exam strategist for theory papers.
MY CONTEXT	Paper duration and the question pattern (number, types, marks): [brief] My usual problem (running out of time, long first answers): [brief]
TASK	1. Calculate the time per question by marks, with reading and review time. 2. Recommend the order (e.g. strongest questions first, or in paper order). 3. Set checkpoints for the paper. 4. Plan what to do when time runs out (bullet-point the remaining answers).
RULES	Base the allocation on my paper's actual pattern, because patterns differ between universities and papers.
OUTPUT	The time table, the order recommendation, the checkpoints, then the emergency plan.

PRO TIP Never exceed a question's time by more than 20%. The next question's marks are easier to earn.

TRY IT WITH A 3-hour paper with 2 long answers of 15 marks and 10 short notes of 5 marks.

→ **NEXT #836** Plan any answer in 2 minutes

842 Answer "compare and contrast" questions

STANDARD

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT / PG RESIDENT

USE IT For comparison questions.

PERSONA	You are an examiner who marks comparison questions.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] The question: [PASTE] Marks: [number]
TASK	1. Write a brief introduction defining both items. 2. Build a comparison table (the core of the answer). 3. Add a short paragraph on the key distinguishing points and their clinical relevance. 4. Add a diagram if relevant.
RULES	Keep the comparisons parallel and accurate, and mark specifics [VERIFY], because comparison answers are checked point by point.
OUTPUT	The introduction, the table, the paragraph, then the diagram suggestion. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP Include "similarities" briefly. Many students forget, and examiners notice.

TRY IT WITH Compare type 1 and type 2 diabetes mellitus.

→ **NEXT #839** A comparison table in my answer

843 What does the question word really ask?

STANDARD

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT / PG RESIDENT

USE IT When you're unsure how to approach "discuss", "describe", "enumerate", "outline" or "critically evaluate".

PERSONA	You are an examiner who explains question directives.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] The directive words in my exam papers: [list] Example questions: [PASTE 2–3]
TASK	1. Define each directive word and what it expects: depth, format and length. 2. Show how the same topic changes with different directives. 3. Rewrite the approach to my example questions.
RULES	Keep the definitions consistent with common university conventions, and suggest checking local expectations, because institutions differ slightly.
OUTPUT	A directive table (word, meaning, expected format), the same-topic examples, then my question approaches. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP "Enumerate" means a list, not an essay. Answering the directive saves time and earns marks.

TRY IT WITH "Enumerate the causes of haemoptysis" versus "discuss the approach to haemoptysis".

→ **NEXT #836** Plan any answer in 2 minutes

844 A "recent advances" paragraph

STANDARD

3.4 LONG & SHORT ANSWER WRITING · PG RESIDENT / MBBS FINAL YEAR

USE IT When a question asks about recent advances, or an answer would benefit from one.

PERSONA	You are an academic clinician who writes accurate recent-advances sections.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Topic: [TOPIC] Recent developments I know of, with sources: [LIST] Exam: [brief]
TASK	1. Write a recent-advances paragraph or bullet list (5–7 points) from my sources. 2. Include the year and source for each, marked [VERIFY]. 3. Indicate which are established, and which are emerging.
RULES	Use only the developments I listed or mark new suggestions [VERIFY — FIND SOURCE], because "recent advances" copied from old notes are often outdated.
OUTPUT	The paragraph or list with sources and status.

PRO TIP Two correct, recent, well-cited points beat seven vague ones.

TRY IT WITH Recent advances in the management of heart failure with reduced ejection fraction.

→ **NEXT #834** MD / MS / DNB theory essay with depth

845 A stepwise management answer

STANDARD

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT / PG RESIDENT

USE IT For "discuss the management of..." questions.

PERSONA	You are an examiner who expects structured management answers.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The condition and the question wording: [PASTE] Marks: [number]
TASK	1. Structure the answer: general measures, specific treatment (first line, then second line), management of complications, monitoring, prevention and follow-up. 2. Add a management flowchart. 3. Mark drug doses and durations [VERIFY]. 4. Add special situations (pregnancy, renal failure, children) if relevant.
RULES	Keep the recommendations consistent with standard textbooks and current guidelines, and mark every dose [VERIFY], because dosing errors lose marks and credibility.
OUTPUT	The structured answer, then the flowchart description.

PRO TIP "General measures" first shows examiners you think of the whole patient.

TRY IT WITH Management of acute severe ulcerative colitis.

→ **NEXT #838** Put a flowchart in my answer

846 Answer a case-based theory question

STANDARD

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT / PG RESIDENT

USE IT When a theory question starts with a clinical scenario.

PERSONA	You are an examiner who writes case-based questions.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The question with its scenario and sub-questions: [PASTE] Marks: [number]
TASK	1. Identify the diagnosis from the scenario and justify it briefly. 2. Answer each sub-question in order, with headings matching the sub-questions. 3. Keep each part proportional to its marks. 4. Link the answers to the scenario details.
RULES	Reason step by step from the scenario, and keep facts accurate with [VERIFY] for specifics, because case-based questions reward reasoning.
OUTPUT	The diagnosis with its justification, then the headed answers to each part.

PRO TIP Quote the scenario's clues in your answer ("the history of X suggests..."). It shows your reasoning.

TRY IT WITH A scenario of a young woman with fever, a malar rash and arthralgia, with questions on diagnosis, investigations and management.

→ **NEXT #831** Model long answer (10–15 marks)

847 Make my answers look complete at a glance

STANDARD

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT / PG RESIDENT

USE IT When your content is good but your marks are average.

PERSONA	You are an examiner who knows how presentation affects marks.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] A photo or description of a recent answer I wrote: [UPLOAD / DESCRIBE]
TASK	1. Assess the presentation: headings, spacing, underlining, point format, diagrams, legibility and margins. 2. Give 8 specific presentation rules. 3. Show a before-and-after layout of one page.
RULES	Focus on presentation techniques allowed in the exam (pen colours, underlining), and remind me to check my university's rules, because some restrict colours or formats.
OUTPUT	The assessment, the rules, then the before-and-after layout. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP Headings underlined, points numbered and one diagram per page make an answer look complete before it is even read.

TRY IT WITH A photo of a handwritten 15-mark answer that is one continuous paragraph.

→ **NEXT #833** Examiner evaluation of my written answer

848 Build a bank of repeated questions

STANDARD

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT / PG RESIDENT

USE IT With past papers from 5–10 years.

PERSONA	You are an exam analyst for university theory papers.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Past questions for my subject (paste them): [PASTE] Years covered: [range]
TASK	1. Group the questions by topic and identify the repeats and near-repeats. 2. Rank the topics by how often they recur. 3. Create a list of model answers to prepare, in priority order. 4. Suggest a schedule to write them.
RULES	Base the counts only on the questions I pasted, because claims about "frequently asked" topics beyond my papers are guesses.
OUTPUT	The grouped table with counts, the priority list, then the schedule.

PRO TIP Questions repeated 3 or more times deserve a memorised model answer.

TRY IT WITH 8 years of MBBS pathology theory papers.

→ **NEXT #832** Model answers for previous-year questions

849 Model answers from a textbook chapter

STANDARD

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT / PG RESIDENT

USE IT After studying a chapter, to prepare answer-ready material.

PERSONA	You are an examiner who converts chapters into exam answers.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The chapter (upload or paste the key sections): [UPLOAD] Exam: [brief]
TASK	1. Identify the 3–5 most likely long and short questions from this chapter. 2. Write a model skeleton for each (headings, key points and a diagram). 3. Write one full model answer for the most likely long question.
RULES	Use only the chapter's content, and mark details to check [VERIFY IN CHAPTER], because answers must match the source.
OUTPUT	The likely questions, the skeletons, then the full model answer.

PRO TIP Skeletons are enough for revision. Write the full answers only for high-priority questions.

TRY IT WITH A chapter on tuberculosis in a medicine textbook.

→ **NEXT #836** Plan any answer in 2 minutes

850 Improve my writing speed

STANDARD

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT / PG RESIDENT

USE IT When you know the answers but can't finish the paper.

PERSONA	You are an exam coach who trains writing speed and efficiency.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] My current speed (e.g. words per 10 minutes) and my handwriting issues: [brief] Weeks until the exam: [N]
TASK	1. Give techniques: planning first, bullet points instead of prose, standard abbreviations (defined once), and quick diagrams. 2. Plan a weekly practice routine with timed writing. 3. Set speed targets with measurement. 4. Suggest ergonomic tips (pen, posture, breaks).
RULES	Keep the abbreviations to standard, recognisable ones, because examiners won't decode private shorthand.
OUTPUT	The techniques, the routine, the targets, then the ergonomics.

PRO TIP Timed practice twice a week for 6 weeks noticeably improves speed.

TRY IT WITH I write 250 words in 15 minutes and need 400 or more.

→ **NEXT #841** Time allocation for a theory paper

851 What examiners actually look for

STANDARD

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT / PG RESIDENT

USE IT To understand how theory answers are marked.

PERSONA	You are a senior examiner who explains marking honestly.
MY CONTEXT	Exam type and subject: [brief] My typical marks versus my expectations: [brief]
TASK	1. Explain how examiners typically mark: key points, structure, diagrams, relevance and presentation. 2. Explain what loses marks: irrelevance, missing key points, poor structure and illegibility. 3. Explain how the marks for a 10-mark answer might be split, as an illustrative estimate. 4. Give a checklist to run on every practice answer.
RULES	Present the marking insights as general practice, not official schemes, because marking varies between examiners and universities.
OUTPUT	The marking explanation, the mark-losers, the illustrative split, then the checklist.

PRO TIP Examiners often mark tired and fast. Make key points easy to find.

TRY IT WITH MBBS final-year medicine theory.

→ **NEXT #833** Examiner evaluation of my written answer

QUICK PROMPTS · 3.4

852 A one-line definition to open my answer

QUICK

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT / PG RESIDENT

Act as a university examiner. Give a precise, textbook-standard one-line definition of [TOPIC] to open a theory answer, plus one line on its clinical importance, marking the source type as [CHECK STANDARD TEXTBOOK]. Flag anything you are unsure of so I can check it.

TRY IT WITH Topic: acute respiratory distress syndrome.

853 The five points I must not miss

QUICK

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT / PG RESIDENT

Act as an experienced university examiner. For the question "[QUESTION]", list the 5 non-negotiable points a full-marks answer must contain, each in one line, in the order they should appear. Flag anything you are unsure of so I can check it.

TRY IT WITH "Describe the clinical features of hypothyroidism."

854 An "enumerate" answer

QUICK

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT

Act as an examiner. For "Enumerate [TOPIC]", give a numbered, grouped list (grouped by mechanism or category), with no explanations unless one word is needed, so I can write it in under 3 minutes, and mark anything uncertain [VERIFY]. Flag anything you are unsure of so I can check it.

TRY IT WITH "Enumerate the causes of splenomegaly."

855 Which diagram should I add?

QUICK

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT / PG RESIDENT

Act as an examiner. For my answer on [TOPIC], suggest the single most valuable diagram to add, describe it in 4–5 drawing steps with labels, and say where in the answer it should go. Flag anything you are unsure of so I can check it.

TRY IT WITH An answer on the pathogenesis of atherosclerosis.

856 Convert my paragraph into exam points

QUICK

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT / PG RESIDENT

Act as an exam-writing coach for university theory papers. Convert this paragraph from my answer, [PASTE PARAGRAPH], into a heading with 4–6 concise numbered points, keeping every fact and removing filler words. Flag anything you are unsure of so I can check it.

TRY IT WITH A 150-word paragraph on the complications of diabetes.

857 Headings for my answer

QUICK

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT / PG RESIDENT

Act as a university examiner. Give the heading structure (in order) for the question "[QUESTION]" worth [MARKS] marks, with a suggested word allocation for each heading and the place for one diagram. Flag anything you are unsure of so I can check it.

TRY IT WITH "Discuss the approach to a patient with pyrexia of unknown origin" (15 marks).

858 A 2-mark answer

QUICK

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT

Act as an examiner. Write a 2-mark answer for [QUESTION] in 40–60 words: the direct answer first, then one supporting point or example, with nothing extra. Mark any number [VERIFY]. Flag anything you are unsure of so I can check it.

TRY IT WITH "What is the Kussmaul sign?"

859 A reusable template for disease answers

QUICK

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT

Act as an exam coach. Give me a reusable heading template for long answers on any disease in [SUBJECT] (e.g. definition, aetiology, pathogenesis, clinical features, investigations, management, complications, prognosis), with one line on what to write under each heading. Flag anything you are unsure of so I can check it.

TRY IT WITH Subject: general surgery.

860 A strong closing line

QUICK

3.4 LONG & SHORT ANSWER WRITING · MBBS STUDENT / PG RESIDENT

Act as an examiner. Write 3 possible concluding lines (under 30 words each) for my answer on [TOPIC] (a summary principle, a clinical pearl, and a recent direction marked [VERIFY]), and say which suits a long answer best. Flag anything you are unsure of so I can check it.

TRY IT WITH An answer on the management of hypertension in pregnancy.

3.5

Viva, OSCE & practical exams

Vivas and practicals test what you can say, do and reason out under pressure. Practise aloud, one question at a time, with feedback.

50

prompts

6

Master

28

Standard

16

Quick

HOW TO USE THIS SECTION

Use it in the weeks before practicals, vivas and OSCEs, and always practise aloud.

Start with the Master prompts ([#861–866](#)) for the full workflow; use the Standard prompts ([#867–894](#)) for one task at a time; reach for the Quick prompts ([#895–910](#)) when you have two minutes. Tap any title below to jump to it, and follow → **NEXT** to move to the next step.

[#861](#) Viva voce simulator, one question at a time

[#862](#) Long case: presentation and examiner questions

[#863](#) Short cases: examination and spot findings

[#864](#) OSCE station simulator

[#865](#) Thesis defence viva preparation

[#866](#) Spotters and table viva: instruments, specimens, images, drugs

[#867](#) Opening questions examiners use on a topic

[#868](#) Structure any viva answer

[#869](#) Instrument viva: one instrument, fully prepared

[#870](#) Pathology or anatomy specimen viva

[#871](#) X-ray viva: describe, diagnose, discuss

[#872](#) ECG viva: systematic and confident

[#873](#) Pharmacology practical: drug and prescription viva

[#874](#) Charts and graphs viva (partograph, growth charts, epidemic curves)

[#875](#) Demonstrate a system examination properly

[#876](#) Elicit a specific clinical sign

[#877](#) History-taking OSCE with a simulated patient

[#878](#) Breaking bad news and difficult conversations

[#879](#) Counselling station (medication, lifestyle, procedure consent)

[#880](#) Procedural skills station

[#881](#) Data-interpretation station

[#882](#) Prescription-writing station

[#883](#) Viva etiquette and presence

[#884](#) When I blank out in a viva

[#885](#) Handle different examiner styles

[#886](#) DNB-style bedside case discussion

[#887](#) Table viva: rapid structured answers

[#888](#) Community medicine spotters (vaccines, models, charts)

[#889](#) Forensic medicine spotters (weapons, poisons, specimens)

[#890](#) Obstetrics and gynaecology practical preparation

[#891](#) Paediatric practical preparation

[#892](#) Surgery short case: examining a lump or swelling

[#893](#) The practical exam day

[#894](#) Last-minute viva revision list

[#895](#) A 60-second answer to "what is X?"

[#896](#) Classify it for the viva

[#897](#) Five viva questions on my topic

[#898](#) An instrument in one breath

[#899](#) How do I elicit this sign?

[#900](#) Describe this finding like an examiner expects

[#901](#) Where will the examiner go next?

[#902](#) "What would you do next?"

[#903](#) The differential they'll ask for

[#904](#) This topic's examiner-favourite question

[#905](#) Say "I don't know" well in a viva

[#906](#) OSCE station opening lines

[#907](#) Close an OSCE station well

[#908](#) Sound confident without overclaiming

[#909](#) Plain-language phrases for counselling stations

[#910](#) Describe a spot diagnosis

Viva voce simulator, one question at a time

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

USE IT

In the weeks before a university viva, a residency practical, or any oral examination.

WHY

Vivas punish silence and rambling, not just ignorance. A simulator that presents a clinical scenario, asks one question at a time, waits for your answer, gives feedback and escalates like a real examiner builds the calm, structured answering that examiners reward.

COPY THE PROMPT BELOW

PERSONA	You are a strict but fair university examiner in [SUBJECT] conducting a viva voce. You start from a clinical scenario, ask one question at a time, give frank feedback, and escalate from basic to advanced questions as a real examiner does.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Subject and topic: [e.g. ophthalmology: sudden painless loss of vision] Exam: [MBBS final practical / MD-MS-DNB practical / other] My level: [brief] Weak areas I want to be tested on: [list or "any"] Number of questions: [e.g. 10]
TASK	1. Start with a clinical scenario as you would present it in the exam hall. 2. Ask me one question at a time, covering signs, investigations, differentials, management decisions and applied basic science. 3. Wait for my answer each time. 4. After each answer, give feedback: what I got right, what I missed, and a model answer in 3–5 points. 5. Adjust the difficulty according to my answers, as a real examiner would. 6. Include one "why" question and one "what if" twist. 7. After the set, give a performance summary: score, strengths, weak areas and 3 topics to revise.
RULES	Ask only one question per turn and wait for my answer, because the value is in answering under pressure. Keep the model answers accurate to standard textbooks and mark guideline-dependent facts [VERIFY].
OUTPUT	Conversational: one question per turn with short structured feedback; then the final performance summary.
TWEAK	End with the 3 questions I answered worst, rephrased as I would hear them from a different examiner.

How to get more out of it

- Answer aloud before typing. The viva is spoken.
- Try topics such as "a patient with sudden painless loss of vision" or "diabetic retinopathy grading".
- Repeat the drill with a friend reading the questions for a realistic rehearsal.
- → **NEXT #868** Structure any viva answer

Long case: presentation and examiner questions

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS FINAL YEAR / PG RESIDENT

USE IT

For long-case practice before university or residency practical exams.

WHY

The long case is the centrepiece of most clinical practicals. Examiners judge your history, examination, summary, differential and management, and how you defend them. Structured practice with a realistic examiner discussion prepares you for both the presentation and the grilling.

COPY THE PROMPT BELOW

PERSONA	You are a senior clinical examiner for MBBS and MD/MS/DNB practical examinations who has examined long cases for 20 years. You expect a structured presentation and probe the reasoning behind every statement.
MY CONTEXT	Subject and exam: [e.g. MD General Medicine practical] The case I clerked (de-identified history and examination findings): [PASTE] Time allowed for presentation: [e.g. 10 minutes] My provisional diagnosis: [brief]
TASK	1. Review my history for completeness and organisation, and list the key questions I missed asking the patient. 2. Review my examination findings for completeness, and list the signs I should have looked for. 3. Help me write the presentation: the one-liner, a concise history, the positive and relevant negative findings, and a summary. 4. Build the diagnosis in the format examiners expect (anatomical, pathological, aetiological and functional, where relevant). 5. List the differentials with points for and against each from the case. 6. Plan the investigations and management with their reasoning. 7. Conduct an examiner discussion: ask me 8 questions one at a time about this case, and give feedback after each.
RULES	Use only the findings I clerked, and flag missing information as [NOT ELICITED] rather than assuming it, because examiners test what you actually found. Mark guideline-dependent management points [VERIFY]. This is exam practice, not a treatment plan for the patient.
OUTPUT	The history and examination review, the presentation script, the diagnosis statement, the differential table, the investigation and management plan, then the interactive examiner questions.
TWEAK	End with the 3 questions examiners most often use to test a candidate on this type of case.

How to get more out of it

- › Practise presenting in the time allowed with a timer. Examiners stop overrunning candidates.
- › Always state the diagnosis in the complete format your department expects.
- › Know why you ordered each investigation. "To rule out X" is a complete answer.
- › → NEXT #863 Short cases: examination and spot findings

Short cases: examination and spot findings

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS FINAL YEAR / PG RESIDENT

USE IT

For short-case practice, where examiners watch you examine and ask quick questions.

WHY

Short cases test examination technique, recognition of findings and quick reasoning, all within minutes. Knowing the examination sequence cold, what examiners look for, and the standard follow-up questions for common short cases is what gets you through.

COPY THE PROMPT BELOW

PERSONA	You are a clinical examiner who conducts short cases in medicine, surgery, paediatrics and other clinical subjects, and who watches technique as closely as answers.
MY CONTEXT	Deadline or time I have for this: [brief] My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] Subject and exam: [brief] The system or short-case type: [e.g. cardiovascular system / a neck swelling / hepatosplenomegaly / a peripheral nerve lesion] My weak points in examination technique: [brief]
TASK	1. Give the complete examination sequence for this short case, step by step, including inspection, palpation, percussion and auscultation, and the special tests. 2. List what examiners watch for (hand hygiene, introduction, consent, exposure, positioning, technique). 3. Give the 10 most common short cases in this category with their key findings. 4. For each, give the typical examiner questions and short model answers. 5. Give a "presenting your findings" template (in 60 seconds). 6. Run a practice round: describe a case's findings, and ask me for the diagnosis and the next questions. 7. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Keep the examination technique consistent with standard clinical-methods textbooks, and mark answers that depend on guidelines [VERIFY], because short cases test standard technique.
OUTPUT	The examination sequence, the examiner checklist, the common cases table, the Q&A, the presentation template, then the practice round.
TWEAK	End with the 3 technique errors examiners penalise most in this short case.

How to get more out of it

- › Practise the examination sequence on friends until you do it without thinking.
- › Say what you're doing as you examine, if the examiner allows it. It shows your system.
- › Present findings, then your conclusion: "These findings suggest..."
- › → **NEXT #875** Demonstrate a system examination properly

OSCE station simulator

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT / LICENSING-EXAM CANDIDATE

USE IT

For OSCE practice: history-taking, communication, examination, procedures and data interpretation stations.

WHY

OSCE stations are marked against checklists, often including items candidates forget under time pressure, such as introductions, consent, summarising and safety-netting. Simulated stations with the checklist revealed afterwards train both the content and the process.

COPY THE PROMPT BELOW

PERSONA	You are an OSCE examiner and simulated-patient trainer who designs realistic stations and marks strictly against checklists.
MY CONTEXT	Deadline or time I have for this: [brief] My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] Station type: [history-taking / communication / examination / procedure / data interpretation / counselling] Topic: [e.g. chest pain history / breaking bad news / inhaler counselling] Exam: [university OSCE / PLAB 2 / OSCE for MD/MS / other] Station duration: [e.g. 8 minutes]
TASK	1. Give the candidate instructions exactly as they would appear on the station door. 2. Play the simulated patient or examiner: respond realistically to my questions and actions, revealing information only when asked. 3. Keep track of time and tell me when it's time to close. 4. After I finish, reveal the marking checklist with the items I achieved and missed. 5. Give global feedback on communication, structure and empathy. 6. Give a model performance script for comparison. 7. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Stay in character during the station, and reveal the checklist only after I finish, because knowing the checklist in advance defeats the practice. Keep the clinical content accurate and mark guideline specifics [VERIFY].
OUTPUT	The station instructions, then an interactive role-play; afterwards the checklist table (item, achieved yes or no), the global feedback and the model script.
TWEAK	End with the 3 checklist items candidates most often miss in this type of station.

How to get more out of it

- › Always introduce yourself, confirm identity, take consent and summarise. These are easy marks.
- › Practise with a timer. Running out of time before the summary loses marks.
- › Rotate station types. Candidates often over-practise history-taking and neglect communication.
- › → **NEXT #878** Breaking bad news and difficult conversations

Thesis defence viva preparation

3.5 VIVA, OSCE & PRACTICAL EXAMS · PG RESIDENT (FINAL YEAR)

USE IT

Before your thesis defence or the thesis part of the practical exam.

WHY

Examiners use the thesis to judge whether you understand research: why this question, why this design, what the numbers mean and what the limitations are. Rehearsing the defence with realistic questions, one at a time, turns a nervous recitation into a confident discussion.

COPY THE PROMPT BELOW

PERSONA	You are an external examiner for MD/MS/DNB theses who conducts thesis defences, probing the rationale, the methods, the statistics and the interpretation.
MY CONTEXT	Deadline or time I have for this: [brief] My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] Thesis title and design: [paste] Objectives: [paste] Sample size and how it was calculated: [brief] Key results (estimates, CIs, p values): [paste] Limitations I already acknowledge: [list] My presentation slides, if any (titles): [PASTE]
TASK	1. Ask me questions one at a time across: the rationale and gap, the objectives, design choice, sample size, sampling, operational definitions, ethics, statistical tests, interpretation of results, limitations, clinical implications and future work. 2. After each answer, score it out of 10 and give an ideal answer grounded in my thesis details. 3. Include 3 challenging questions (e.g. "why not an RCT?", "is your result clinically significant?", "how would you redo this study?"). 4. After 12 questions, summarise my weak areas. 5. Give a 5-minute thesis presentation outline. 6. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Base the ideal answers on my thesis details, and where information is missing, ask me rather than assuming, because examiners test what I actually did. Ask one question at a time.
OUTPUT	Conversational questions with feedback, then the weakness summary and the presentation outline.
TWEAK	End with the 3 sentences about my thesis I should be able to say fluently: the question, the main finding and the main limitation.

How to get more out of it

- › Know your sample-size calculation and every test you used. They're asked in almost every defence.
- › Admit limitations before the examiner points them out. It shows maturity.
- › Rehearse the "how would you do it differently?" answer. It's almost always asked.
- › → **NEXT #861** Viva voce simulator, one question at a time

Spotters and table viva: instruments, specimens, images, drugs

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

USE IT

For spotter stations and table vivas in anatomy, pathology, pharmacology, community medicine, forensic medicine, surgery, OBG, anaesthesia and radiology practicals.

WHY

Spotters and table vivas reward fast, structured recall: identify it, describe it, and state its uses, and its complications or significance. A consistent answer framework for each category, practised on the commonly kept items, converts panic into points.

COPY THE PROMPT BELOW

PERSONA	You are a practical examiner who has set spotters and conducted table vivas for years, and who expects crisp, structured answers.
MY CONTEXT	Deadline or time I have for this: [brief] My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] Subject and practical exam: [e.g. MBBS final surgery practical / pharmacology practical / MD Anaesthesia] Spotter categories in my exam: [instruments / specimens / X-rays / drugs / charts / models / photographs] Items I'm weakest on: [list or "suggest the common ones"]
TASK	1. Give an answer framework for each category, e.g. instruments (identify, parts, uses, how to use, complications or precautions) and specimens (identify, gross features, diagnosis, differential, clinical correlation). 2. List the 20 most commonly kept items for my subject and category, with a 3–4 line model answer for each. 3. Give the follow-up questions examiners typically ask for each item. 4. Run a rapid practice round: describe an item, then ask me to identify it and answer. 5. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Keep the identifications and descriptions accurate to standard textbooks and practical manuals, and mark doses, uses and specifics [VERIFY], because spotter answers are marked on precise facts. Recommend handling the real items in the department wherever possible.
OUTPUT	The frameworks, a common-items table with model answers, the follow-up questions, then the practice round.
TWEAK	End with the 5 look-alike pairs students most often confuse in my spotter categories.

How to get more out of it

- Visit the department museum or practical lab and handle the items. Photos aren't enough.
- Answer in the same framework every time. Examiners hear the structure.
- Make a flashcard for each commonly kept item, with its image on the front.
- → **NEXT #869** Instrument viva: one instrument, fully prepared

867 Opening questions examiners use on a topic

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

USE IT When preparing a topic for a viva.

PERSONA	You are a university examiner who knows the standard opening questions.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Topic and subject: [brief] Exam: [brief]
TASK	1. List the 10 most common opening questions on this topic. 2. Give a 30-second model answer for each. 3. For each opener, give the likely follow-up question.
RULES	Keep the model answers accurate and concise, marking specifics [VERIFY], because viva openers set the examiner's impression.
OUTPUT	A table (opener, 30-second answer, likely follow-up). Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP A crisp answer to the first question often decides the tone of the whole viva.

TRY IT WITH Topic: jaundice, in a medicine viva.

→ **NEXT #861** Viva voce simulator, one question at a time

868 Structure any viva answer

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

USE IT When your viva answers come out disorganised.

PERSONA	You are a viva coach who teaches structured spoken answers.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Common question types in my subject (define, classify, causes, how to manage): [brief] A question I answered poorly, and my answer: [PASTE]
TASK	1. Give a spoken-answer structure for each question type (e.g. "causes": classify them first, then give examples). 2. Rewrite my poor answer using the structure. 3. Give a 3-step method for thinking before speaking (pause, classify, answer).
RULES	Keep the answers short and structured, because examiners interrupt long answers.
OUTPUT	The structures by question type, my rewritten answer, then the 3-step method.

PRO TIP "The causes can be classified as..." buys you 3 seconds and sounds organised.

TRY IT WITH "What are the causes of hepatomegaly?", which I answered as a random list.

→ **NEXT #861** Viva voce simulator, one question at a time

869 Instrument viva: one instrument, fully prepared

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT (SURGICAL, OBG, ANAESTHESIA)

USE IT For each instrument you may be asked about.

PERSONA	You are a surgical examiner who conducts instrument vivas.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Instrument: [name] Subject and exam: [brief]
TASK	Answer in the viva framework: identification, parts and features, uses, how to hold or use it, sterilisation, complications or precautions, and similar instruments with the differences. Then list 5 likely follow-up questions with short answers.
RULES	Keep the details accurate to standard surgical texts and practical manuals, marking uncertain details [VERIFY], because instrument vivas test specifics.
OUTPUT	The framework answer, then the follow-up questions. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP Always hold the instrument correctly while answering. Examiners watch.

TRY IT WITH Instrument: Allis tissue forceps.

→ **NEXT #866** Spotters and table viva: instruments, specimens, images, drugs

870 Pathology or anatomy specimen viva

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

USE IT For specimen stations in pathology, anatomy or surgery.

PERSONA	You are a pathology examiner who conducts specimen vivas.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Specimen: [name or description] Subject: [brief]
TASK	Answer in the framework: identification (organ and side), gross description (size, surfaces, cut section), diagnosis, differential diagnosis, microscopy (brief), complications, and clinical correlation. Then list 5 follow-up questions with short answers.
RULES	Describe gross features systematically and mark specifics [VERIFY], because examiners expect accurate gross descriptions.
OUTPUT	The framework answer, then the follow-up questions. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP Start with the organ and whether the specimen was cut or whole. It shows a systematic approach.

TRY IT WITH Specimen: gastric carcinoma (linitis plastica).

→ **NEXT #866** Spotters and table viva: instruments, specimens, images, drugs

871 X-ray viva: describe, diagnose, discuss

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

USE IT For X-ray stations in medicine, surgery, orthopaedics and paediatrics.

PERSONA	You are a clinical examiner who shows X-rays in vivas.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] X-ray type and the finding to practise (or upload a de-identified image): [DESCRIBE / UPLOAD] Subject: [brief]
TASK	1. Give a systematic description sequence for this X-ray type. 2. Model the spoken description and diagnosis. 3. Give the differentials and the next investigation. 4. Give 5 likely follow-up questions with answers.
RULES	Describe findings conservatively and mark uncertain interpretations [CONFIRM WITH ATLAS], because over-reading an X-ray in a viva invites trouble.
OUTPUT	The sequence, the spoken model, the differentials, then the follow-ups.

PRO TIP Always check the name, date, side marker and projection first. Examiners notice when you do.

TRY IT WITH A chest X-ray showing a right-sided pleural effusion.

→ **NEXT #786** Practise image-based questions

872 ECG viva: systematic and confident

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

USE IT For ECG stations and vivas.

PERSONA	You are a cardiology examiner who tests ECG interpretation.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] ECG finding to practise (describe, or upload a de-identified tracing): [DESCRIBE / UPLOAD]
TASK	1. Give the systematic interpretation sequence (rate, rhythm, axis, P wave, PR interval, QRS, ST segment, T wave, QT interval). 2. Model the spoken interpretation. 3. Give the diagnosis, the differentials and the immediate management principles [VERIFY]. 4. Give 5 follow-up questions with answers.
RULES	Keep the interpretation criteria accurate and mark thresholds [VERIFY], because ECG criteria are tested precisely.
OUTPUT	The sequence, the spoken model, the diagnosis and management, then the follow-ups.

PRO TIP State the rate and rhythm first, confidently. It sets the tone.

TRY IT WITH An ECG showing complete heart block.

→ **NEXT #861** Viva voce simulator, one question at a time

873 Pharmacology practical: drug and prescription viva

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT

USE IT For pharmacology practical exams and drug vivas.

PERSONA	You are a pharmacology practical examiner.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Drug or drug group: [name] Practical format (spotters, prescriptions, case-based): [brief]
TASK	1. Give a viva answer framework (class, mechanism, uses, dose and route [VERIFY], adverse effects, contraindications, interactions). 2. Give the 5 most common questions on this drug with short answers. 3. Show how the drug may appear in a prescription-writing or case question.
RULES	Mark every dose [VERIFY FROM FORMULARY / TEXTBOOK], because pharmacology practicals test exact details.
OUTPUT	The framework, the Q&A, then the practical-question example. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP Know the brand-generic distinction and why generic prescribing matters. It's a common viva theme.

TRY IT WITH Drug: metformin.

→ **NEXT #882** Prescription-writing station

874 Charts and graphs viva (partograph, growth charts, epidemic curves)

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

USE IT For practical exams with charts, graphs or records to interpret.

PERSONA	You are a practical examiner in OBG, paediatrics or community medicine.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Chart type: [e.g. partograph / WHO growth chart / epidemic curve / temperature chart] Exam: [brief]
TASK	1. Explain the chart's components and how to read them. 2. Give a systematic interpretation method. 3. Give 3 example scenarios with interpretations. 4. Give 5 common viva questions with answers.
RULES	Keep the chart conventions accurate to the current standard version, marked [VERIFY CURRENT VERSION], because charts are revised.
OUTPUT	The components, the method, the scenarios, then the questions.

PRO TIP Point to the chart as you speak. Examiners want to see that you can read it, not just recite it.

TRY IT WITH A partograph showing a prolonged active phase of labour.

→ **NEXT #866** Spotters and table viva: instruments, specimens, images, drugs

875 Demonstrate a system examination properly

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

USE IT Before practicals where you examine under observation.

PERSONA	You are a clinical-skills examiner who assesses examination technique.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] System: [e.g. respiratory / cardiovascular / abdomen / nervous system / a joint] Exam: [brief]
TASK	1. Give the complete, ordered examination sequence with the correct technique for each step. 2. Note the positioning, exposure and patient-comfort points. 3. List the common technique errors. 4. Give the 60-second findings-presentation template.
RULES	Follow standard clinical-methods textbook technique, and emphasise consent, privacy and hand hygiene, because examiners mark professionalism as well as skill.
OUTPUT	The sequence, the positioning notes, the errors, then the presentation template.

PRO TIP Practise the full sequence on a friend 10 times. Fluent technique looks competent.

TRY IT WITH Examination of the abdomen for hepatosplenomegaly.

→ **NEXT #863** Short cases: examination and spot findings

876 Elicit a specific clinical sign

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

USE IT When you may be asked to demonstrate a sign.

PERSONA	You are a clinical teacher who teaches bedside signs precisely.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Sign: [e.g. shifting dullness, Babinski sign, JVP, Murphy's sign]
TASK	1. Explain the step-by-step method to elicit it correctly. 2. Explain what a positive result looks like, and the pitfalls. 3. Explain its clinical significance and causes. 4. Give 3 viva questions about the sign with answers.
RULES	Describe the technique accurately and safely, because incorrect technique is marked down even when the answer is right.
OUTPUT	The method, the positive finding and pitfalls, the significance, then the questions.

PRO TIP Explain to the patient what you're about to do. It's part of the technique.

TRY IT WITH Sign: eliciting fluid thrill and shifting dullness.

→ **NEXT #875** Demonstrate a system examination properly

877 History-taking OSCE with a simulated patient

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PLAB / OSCE CANDIDATE

USE IT For practising history stations.

PERSONA	You are a simulated patient and OSCE examiner.
MY CONTEXT	Presenting complaint: [e.g. chest pain, headache, breathlessness] Station time: [minutes]
TASK	1. Give the candidate instructions. 2. Role-play the patient: answer only what I ask, realistically, with an appropriate emotional tone. 3. After I finish, give the checklist (introduction, open questions, the HPI using a structured tool, red flags, past, drug, family and social history, ideas, concerns and expectations, summary, and closure) with the items I missed. 4. Give feedback on my communication.
RULES	Stay in character until I finish, and reveal nothing unless asked, because real patients don't volunteer every detail.
OUTPUT	The instructions, an interactive role-play, then the checklist and feedback.

PRO TIP Ask about the patient's ideas, concerns and expectations. It's a checklist item many candidates miss.

TRY IT WITH Presenting complaint: a 45-year-old with chest pain for 2 hours.

→ **NEXT #864** OSCE station simulator

878 Breaking bad news and difficult conversations

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT / OSCE CANDIDATE

USE IT For communication stations and real practice.

PERSONA	You are a communication-skills examiner trained in structured frameworks such as SPIKES.
MY CONTEXT	Scenario: [e.g. breaking a cancer diagnosis, an angry relative, a medication error] Station time: [minutes]
TASK	1. Explain the relevant framework briefly (e.g. SPIKES: Setting, Perception, Invitation, Knowledge, Emotions, Strategy and summary). 2. Role-play the patient or relative realistically, including their emotions. 3. After the station, mark me against a checklist and give feedback on empathy, clarity and structure. 4. Give model phrases for the key moments.
RULES	Keep the role-play realistic and emotionally authentic, and the medical information accurate, because communication stations test both empathy and honesty.
OUTPUT	The framework, the interactive role-play, then the checklist, feedback and model phrases.

PRO TIP Silence after bad news is powerful. Don't rush to fill it.

TRY IT WITH Scenario: informing a patient's daughter that her father's biopsy shows lung cancer.

→ **NEXT #879** Counselling station (medication, lifestyle, procedure consent)

879 Counselling station (medication, lifestyle, procedure consent)

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT / OSCE CANDIDATE

USE IT For counselling and consent stations.

PERSONA	You are an OSCE examiner who marks counselling stations.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Counselling topic: [e.g. starting insulin, inhaler technique, consent for endoscopy, smoking cessation] Station time: [minutes]
TASK	1. Give the structure: introduction, explore the patient's understanding, explain in plain language, check understanding (teach-back), address concerns, and summarise and safety-net. 2. Role-play the patient with realistic questions and concerns. 3. Give checklist-based feedback afterwards. 4. Give the key facts I must convey accurately [VERIFY].
RULES	Keep all medical information accurate and in plain language, because counselling errors mislead patients.
OUTPUT	The structure, the interactive role-play, then the checklist feedback and key facts.

PRO TIP Use teach-back: "Can you tell me how you'll take this at home?". It's a high-value checklist item.

TRY IT WITH Counselling a newly diagnosed diabetic patient about starting insulin.

→ **NEXT #864** OSCE station simulator

880 Procedural skills station

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

USE IT For skills stations on mannequins: IV cannulation, catheterisation, suturing, CPR, ABG, NG tube.

PERSONA	You are a clinical-skills examiner.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Procedure: [name] Exam: [brief]
TASK	1. Give the step-by-step procedure, including preparation, consent, asepsis, the procedure itself, disposal and documentation. 2. Mark the critical steps whose omission fails the station. 3. Give the common errors. 4. Give a checklist for self-assessment.
RULES	Follow standard technique and current guidelines (e.g. resuscitation guidelines, marked [VERIFY CURRENT GUIDELINE]), because skills stations mark safety steps strictly.
OUTPUT	The steps, the critical steps, the errors, then the checklist. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP Say your safety checks aloud: "confirming patient identity", "checking the expiry date". Examiners can't mark what they don't see.

TRY IT WITH Procedure: urinary catheterisation in a male patient.

→ **NEXT #875** Demonstrate a system examination properly

881 Data-interpretation station

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT / OSCE CANDIDATE

USE IT For stations with lab results, ABGs, CSF, spirometry or other data to interpret.

PERSONA	You are an examiner who sets data-interpretation stations.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Data type: [ABG / LFT / CSF / PFT / thyroid profile / urine] Exam: [brief]
TASK	1. Give a systematic interpretation method for this data type. 2. Give 4 practice data sets of graded difficulty (physiologically consistent). 3. After my attempt, give model interpretations with the diagnosis and next steps.
RULES	Make every data set internally consistent and mark reference ranges [VERIFY], because inconsistent practice data teaches wrong patterns.
OUTPUT	The method, the data sets; after my attempt, the model interpretations.

PRO TIP State the pattern before the diagnosis: "this shows a cholestatic pattern, suggesting..."

TRY IT WITH Data type: CSF analysis in meningitis.

→ **NEXT #716** Learn through a worked example

882 Prescription-writing station

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / INTERN

USE IT For prescription-writing stations and real-world prescribing skill.

PERSONA	You are a clinical pharmacologist who marks prescription stations.
MY CONTEXT	Clinical scenario: [brief] Exam: [brief]
TASK	1. Show the components of a complete prescription: patient details, date, the drug in generic name and capitals, dose, route, frequency, duration, instructions, and the prescriber's signature and registration. 2. Write a model prescription for the scenario, with doses marked [VERIFY]. 3. List the common errors that fail the station. 4. Give counselling points to accompany it.
RULES	Mark every dose [VERIFY FROM FORMULARY], and follow the rational-prescribing principles and the prescription format my exam requires, because prescription errors are marked as safety failures.
OUTPUT	The components, the model prescription, the errors, then the counselling points.

PRO TIP Write drug names in generic form and capital letters. Many stations specifically check this.

TRY IT WITH A prescription for an adult with uncomplicated community-acquired pneumonia in OPD.

→ **NEXT #873** Pharmacology practical: drug and prescription viva

883 Viva etiquette and presence

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

USE IT The week before your viva.

PERSONA	You are a senior examiner who knows how candidates come across.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] My viva concerns (nervous, talk too much, go blank, argue): [list]
TASK	1. Give etiquette guidance: greeting, posture, eye contact, dress, and listening fully before answering. 2. Show how to handle an interruption, a correction or a challenge from the examiner. 3. Show how to steer towards topics I know (without evading). 4. Give phrases for thinking time.
RULES	Keep the advice respectful and honest, because examiners dislike bluffing more than not knowing.
OUTPUT	The etiquette list, the handling strategies, the steering tips, then the phrases.

PRO TIP Listen to the whole question. Half of viva mistakes are answers to questions that weren't asked.

TRY IT WITH I get defensive when an examiner corrects me.

→ **NEXT #884** When I blank out in a viva

884 When I blank out in a viva

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

USE IT To prepare a recovery plan for going blank.

PERSONA	You are a performance coach for oral examinations.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] What happens when I blank: [brief]
TASK	1. Give an immediate recovery routine (pause, breathe, restate the question, start from a definition or classification). 2. Give phrases to buy time honestly. 3. Show how to reason aloud from first principles. 4. Show how to recover the next answer after a bad one.
RULES	Recommend honest strategies only, with no bluffing, because examiners recognise and penalise it.
OUTPUT	The routine, the phrases, the reasoning example, then the recovery tips. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP "Let me think about this systematically..." followed by a classification recovers most blanks.

TRY IT WITH I blanked when asked for the causes of clubbing.

→ **NEXT #883** Viva etiquette and presence

885 Handle different examiner styles

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

USE IT When preparing mentally for the viva.

PERSONA	You are an experienced examiner who has seen every style of examiner and candidate.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Examiner styles I'm worried about: [e.g. rapid-fire, silent, friendly-then-tough, a subject expert]
TASK	1. Describe each style and what they are looking for. 2. Give a strategy for each. 3. Give a phrase or approach that works with each.
RULES	Keep the strategies professional and respectful, because the goal is to show knowledge clearly, not to manipulate the examiner.
OUTPUT	A style table (style, what they want, strategy, useful phrase). Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP A silent examiner isn't a hostile one. Keep answering clearly and briefly.

TRY IT WITH A rapid-fire internal examiner and a silent external examiner.

→ **NEXT #883** Viva etiquette and presence

886 DNB-style bedside case discussion

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · PG RESIDENT (DNB / RESIDENCY PRACTICAL)

USE IT For practical exams with discussion at the bedside with examiners.

PERSONA	You are a DNB practical examiner who conducts bedside discussions.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The case type: [brief] My findings: [PASTE]
TASK	1. Help me prepare a crisp bedside presentation (in 3 minutes). 2. Anticipate the bedside questions: signs to demonstrate, differentials and immediate management. 3. Conduct a mock discussion, one question at a time, with feedback. 4. List the patient-etiquette points examiners notice at the bedside.
RULES	Use only my findings and keep the answers evidence-aware with [VERIFY], because bedside discussions test clinical judgement in front of the patient.
OUTPUT	The presentation, the anticipated questions, an interactive mock discussion, then the etiquette points.

PRO TIP Thank the patient and cover them after the examination. Examiners notice.

TRY IT WITH A patient with mitral stenosis and atrial fibrillation.

→ **NEXT #862** Long case: presentation and examiner questions

887 Table viva: rapid structured answers

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

USE IT For table vivas in pharmacology, community medicine, forensic medicine or anaesthesia.

PERSONA	You are a table-viva examiner.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Subject: [brief] Topics likely on the table: [list or "suggest"]
TASK	1. List the 20 most likely table-viva topics for my subject. 2. Give a 3-line model answer for each. 3. Give the follow-up chains for the top 5. 4. Run a rapid practice set, one question at a time.
RULES	Keep the answers accurate and mark specifics [VERIFY], because table vivas test quick recall of precise facts.
OUTPUT	The topic list with answers, the follow-up chains, then the practice set. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP Table vivas move fast. Answer in short, complete sentences.

TRY IT WITH A community medicine table viva: vaccines, disinfectants and water purification.

→ **NEXT #866** Spotters and table viva: instruments, specimens, images, drugs

888 Community medicine spotters (vaccines, models, charts)

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT

USE IT For community medicine practical spotters.

PERSONA	You are a community medicine practical examiner.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Spotter types in my exam: [vaccines, insecticides, charts, models, water-testing kits, contraceptives]
TASK	1. List the common spotters in each category with an identification clue. 2. Give a model answer for each (identification, uses, key facts, national-programme link), with programme specifics marked [VERIFY WITH CURRENT PROGRAMME GUIDELINES]. 3. Give 3 follow-up questions per category.
RULES	Mark vaccine schedules, doses and programme details for verification, because national schedules are updated.
OUTPUT	A category table with model answers, then the follow-ups. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP Link every spotter to a national programme. It's the answer examiners want to hear.

TRY IT WITH Spotters: the vaccine carrier, ORS packet and oral contraceptive pills.

→ **NEXT #731** Learn national health programmes (community medicine)

889 Forensic medicine spotters (weapons, poisons, specimens)

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT

USE IT For forensic medicine practicals.

PERSONA	You are a forensic medicine practical examiner.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Spotter types: [weapons / poisons / bones / specimens / photographs]
TASK	1. List the common spotters with identification features. 2. Give model answers (identification, medicolegal importance, and signs, symptoms and management for poisons [VERIFY]). 3. Give the common viva questions.
RULES	Keep the medicolegal content accurate to standard forensic textbooks and current Indian law, marked [VERIFY LEGAL SECTION], because laws and their section numbers have changed.
OUTPUT	A spotter table, the model answers, then the viva questions. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP Always state the medicolegal importance. It's what makes a forensic answer complete.

TRY IT WITH Spotters: dhatura seeds, a single-edged knife, and the femur for age estimation.

→ **NEXT #866** Spotters and table viva: instruments, specimens, images, drugs

890 Obstetrics and gynaecology practical preparation

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS FINAL YEAR / PG RESIDENT

USE IT For OBG practicals: case presentations, instruments, partograph, specimens and drills.

PERSONA	You are an OBG practical examiner.
MY CONTEXT	Components of my OBG practical: [brief] Weak areas: [list]
TASK	1. Give the obstetric history and examination format (including gravida, parity, abortion and living-children notation, and the obstetric grips). 2. List the commonly kept instruments and specimens with model answers. 3. Give the partograph interpretation method. 4. List the drills and emergencies asked (e.g. PPH, eclampsia) with management steps [VERIFY]. 5. Give the common viva questions.
RULES	Keep management protocols consistent with current national and standard guidelines, marked [VERIFY], because obstetric emergencies are tested precisely.
OUTPUT	The formats, the instruments and specimens table, the partograph method, the emergency drills, then the viva questions.

PRO TIP Present the obstetric history in the standard order every time. Examiners listen for it.

TRY IT WITH A case presentation of a primigravida at 36 weeks with pre-eclampsia.

→ **NEXT #874** Charts and graphs viva (partograph, growth charts, epidemic curves)

891 Paediatric practical preparation

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS FINAL YEAR / PG RESIDENT

USE IT For paediatric practicals: case presentations, growth charts, development, nutrition and vaccines.

PERSONA	You are a paediatrics practical examiner.
MY CONTEXT	Components of my paediatric practical: [brief] Weak areas: [list]
TASK	1. Give the paediatric history format (including birth, developmental, immunisation and dietary history). 2. Explain growth-chart plotting and interpretation. 3. Explain developmental milestone assessment, with the key milestones by age [VERIFY]. 4. Give nutritional assessment and the common deficiency signs. 5. Give the common viva questions and spotters.
RULES	Keep the milestone ages and schedules consistent with standard textbooks and current guidance, marked [VERIFY], because paediatric numbers are tested exactly.
OUTPUT	The history format, the growth-chart method, the milestones table, the nutrition guide, then the viva questions.

PRO TIP Plot the growth chart during the exam. Examiners expect to see it done.

TRY IT WITH A child with severe acute malnutrition, as a long case.

→ **NEXT #874** Charts and graphs viva (partograph, growth charts, epidemic curves)

892 Surgery short case: examining a lump or swelling

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS FINAL YEAR / PG RESIDENT (SURGERY)

USE IT For surgery short cases involving lumps, ulcers, swellings and hernias.

PERSONA	You are a surgery practical examiner.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Case type: [lump / ulcer / hernia / thyroid swelling / breast lump / scrotal swelling]
TASK	1. Give the systematic examination (site, size, shape, surface, edge, consistency, tenderness, temperature, mobility, fixity, transillumination, regional lymph nodes and the special tests). 2. Give the presentation template. 3. Give the differentials, investigations and management principles. 4. Give the common examiner questions with answers.
RULES	Follow standard clinical-surgery textbook technique, because surgical short cases are marked on systematic description.
OUTPUT	The examination method, the template, the differentials and management, then the Q&A.

PRO TIP Describe before you diagnose. "A 4 × 3 cm, smooth, firm, non-tender swelling..." earns marks before the diagnosis.

TRY IT WITH Case type: thyroid swelling.

→ **NEXT #863** Short cases: examination and spot findings

893 The practical exam day

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

USE IT The day before and the morning of practical exams.

PERSONA	You are an experienced practical examiner and mentor.
MY CONTEXT	Exam components and timings: [brief] My worries: [list]
TASK	1. Give a checklist of what to carry (stethoscope, torch, tendon hammer, measuring tape, pens, watch, ID). 2. Give guidance on dress and professional presentation. 3. Plan the day: case allotment, clerking time management, presenting and the vivas. 4. Give a patient-interaction etiquette reminder. 5. Suggest a brief review routine for between components.
RULES	Keep the advice practical, and remind me to check the exam-specific rules about what I can carry, because rules vary.
OUTPUT	The checklist, the dress note, the day plan, the etiquette reminder, then the review routine.

PRO TIP Time your clerking with a watch. Many candidates run out of time for the examination.

TRY IT WITH A clinical practical with a long case, 2 short cases and a table viva.

→ **NEXT #894** Last-minute viva revision list

894 Last-minute viva revision list

STANDARD

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

USE IT The night before the viva.

PERSONA	You are a viva coach who prepares focused last-minute lists.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Subject and topics likely in my viva: [brief] My weak spots: [list]
TASK	1. Give the 30 most asked viva one-liners for my subject. 2. Give 10 "definitions you must say perfectly". 3. Give 5 classifications to recite. 4. Give a 30-minute revision routine.
RULES	Keep every fact accurate and mark numbers [VERIFY], because last-minute lists are memorised word for word.
OUTPUT	The one-liners, the definitions, the classifications, then the routine. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP Stop revising 1–2 hours before sleeping. A rested mind answers better than a crammed one.

TRY IT WITH A pathology viva the next morning.

→ **NEXT #861** Viva voce simulator, one question at a time

895 A 60-second answer to "what is X?"

QUICK

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

Act as a viva examiner. Give me a model spoken answer (about 150 words, 60 seconds) to "What is [TOPIC]?" that covers the definition, classification in one line, the key feature and the clinical significance, in the order an examiner expects.

TRY IT WITH "What is cor pulmonale?"

896 Classify it for the viva

QUICK

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

Act as an examiner. Give the standard classification of [TOPIC] as expected in a viva, as a short, recitable list with one example each, and mark any classification that differs between textbooks [CHECK TEXTBOOK]. Flag anything you are unsure of so I can check it.

TRY IT WITH Classification of shock.

897 Five viva questions on my topic

QUICK

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

Act as a university examiner in [SUBJECT]. Ask me 5 viva questions on [TOPIC] one at a time, from basic to advanced, wait for each answer, and give one-line feedback before the next question. Flag anything you are unsure of so I can check it.

TRY IT WITH Topic: nephrotic syndrome in children.

898 An instrument in one breath

QUICK

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT

Act as a surgical examiner. For [INSTRUMENT], give a one-breath viva answer (identification, 2 uses and one precaution) of under 40 words, then the most likely follow-up question with its answer. Flag anything you are unsure of so I can check it.

TRY IT WITH A Foley's catheter.

899 How do I elicit this sign?

QUICK

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT

Act as a clinical-methods teacher. Give the step-by-step method to elicit [SIGN] in 5–6 short steps, what a positive result looks like, and one common error to avoid, suitable for demonstrating in a practical exam. Flag anything you are unsure of so I can check it.

TRY IT WITH Sign: Kernig's sign.

900 Describe this finding like an examiner expects

QUICK

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

Act as a clinical examiner. Model how to describe [FINDING] aloud in a short case, using the systematic terms examiners expect, in 2–3 sentences, then give the most likely diagnosis and one differential. Flag anything you are unsure of so I can check it.

TRY IT WITH Finding: a pulsatile swelling in the epigastrium.

901 Where will the examiner go next?

QUICK

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

Act as a viva examiner. After I answer "[FIRST QUESTION]" on [TOPIC], show the chain of 5 follow-up questions you would ask, each with a one-line model answer, going deeper each time. Flag anything you are unsure of so I can check it.

TRY IT WITH First question: "What are the causes of anaemia?"

902 "What would you do next?"

QUICK

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

Act as a clinical examiner. For this scenario, [BRIEF SCENARIO], model the answer to "What would you do next?" with an immediate action, the investigation and the reasoning, in under 80 words, marking guideline specifics [VERIFY]. Flag anything you are unsure of so I can check it.

TRY IT WITH A patient with a GCS of 8 after a head injury, in the emergency room.

903 The differential they'll ask for

QUICK

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

Act as a viva examiner. For a case of [CONDITION], give the top 4 differentials an examiner expects, each with the one feature that distinguishes it, in a short table I can recite. Flag anything you are unsure of so I can check it.

TRY IT WITH A case of a neck swelling that moves with swallowing.

904 This topic's examiner-favourite question

QUICK

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

Act as a senior examiner in [SUBJECT]. Tell me the one question on [TOPIC] that examiners love to ask, why they ask it, and a crisp model answer of under 80 words, marking specifics [VERIFY]. Flag anything you are unsure of so I can check it.

TRY IT WITH Topic: digoxin toxicity.

905 Say "I don't know" well in a viva

QUICK

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

Act as a viva coach. Give me 4 honest ways to respond in a viva when I don't know the answer to [TYPE OF QUESTION], including how to reason aloud from what I do know and how to invite the next question gracefully.

TRY IT WITH A question about a rare syndrome I've never read about.

906 OSCE station opening lines

QUICK

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / OSCE CANDIDATE

Act as an OSCE examiner. Write the ideal first 30 seconds for a [STATION TYPE] station, covering the introduction, identity check, explanation of purpose and consent, as a script I can adapt to any patient. Flag anything you are unsure of so I can check it.

TRY IT WITH Station type: a history-taking station for abdominal pain.

907 Close an OSCE station well

QUICK

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / OSCE CANDIDATE

Act as an OSCE examiner. Write the ideal final 45 seconds for a [STATION TYPE] station: a summary, checking understanding, inviting questions, safety-netting and thanking the patient, as a reusable script. Flag anything you are unsure of so I can check it.

TRY IT WITH Station type: counselling a patient about a new asthma inhaler.

908 Sound confident without overclaiming

QUICK

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

Act as a viva coach. Rewrite these hesitant answers, [PASTE MY ANSWERS], into confident but accurate phrasing, removing fillers and "I think maybe" hedges while keeping honest uncertainty where the evidence genuinely is uncertain. Flag anything you are unsure of so I can check it.

TRY IT WITH "I think maybe it could be something like nephrotic syndrome, probably?"

909 Plain-language phrases for counselling stations

QUICK

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / OSCE CANDIDATE

Act as a communication-skills trainer. Give me 8 plain-language phrases for a counselling station on [TOPIC] (explaining the condition, checking understanding, addressing worries and safety-netting), avoiding jargon and keeping each phrase under 20 words. Flag anything you are unsure of so I can check it.

TRY IT WITH Topic: explaining hypertension and why lifelong medication matters.

910 Describe a spot diagnosis

QUICK

3.5 VIVA, OSCE & PRACTICAL EXAMS · MBBS STUDENT / PG RESIDENT

Act as a practical examiner. For the spot diagnosis [CONDITION OR SIGN], give the description I should say aloud before naming it (the key visible features in order), then the diagnosis and 2 follow-up facts examiners commonly ask. Flag anything you are unsure of so I can check it.

TRY IT WITH Spot diagnosis: a butterfly rash on the face.

3.6

Clinical reasoning & differential diagnosis

Good diagnosticians don't know more diagnoses; they reason better. Represent the problem, rank the possibilities, never miss the dangerous ones, and let tests change your mind. (For learning and exam practice, not a substitute for clinical judgement.)

60

prompts

7

Master

34

Standard

19

Quick

HOW TO USE THIS SECTION

Use it to practise diagnostic reasoning on cases and exam vignettes, never as a substitute for bedside judgement.

Start with the Master prompts (#911–917) for the full workflow; use the Standard prompts (#918–951) for one task at a time; reach for the Quick prompts (#952–970) when you have two minutes. Tap any title below to jump to it, and follow → NEXT to move to the next step.

#911 Prioritised differential with can't-miss diagnoses

#912 Build illness scripts for diseases

#913 Interactive patient case simulator

#914 Approach to any presenting symptom

#915 Debrief my reasoning for cognitive bias

#916 Choose and interpret tests like an expert (Bayesian reasoning)

#917 Reason through a management decision

#918 Write a problem representation

#919 Practise semantic qualifiers

#920 Generate causes systematically (VINDICATE)

#921 Localise the lesion (neurology)

#922 Estimate a pretest probability

#923 Use likelihood ratios at the bedside

#924 Red flags for a presenting symptom

#925 Can't-miss diagnoses for a presentation

#926 The next best investigation

#927 Interpret a lab pattern in clinical context

#928 Reason through an acid-base problem

#929 Interpret an ECG in clinical context

#930 Interpret imaging in clinical context

#931 Approach to fever, including PUO

#932 Approach to chest pain

#933 Approach to breathlessness

#934 Approach to altered sensorium

#935 Approach to acute abdominal pain

#936 Approach to anaemia

#937 Approach to jaundice

#938 Approach to acute kidney injury

#939 Approach to hyponatraemia

#940 Age-specific differential in children

#941 Differential diagnosis in pregnancy

#942 Atypical presentations in the elderly

#943 Tropical infection differential (Indian setting)

#944 Could a drug be causing this?

#945 Recognise the syndrome from scattered findings

#946 Discriminate between two close diagnoses

#947 Apply a clinical decision rule

#948 A diagnostic time-out before I close the case

#949 Turn a ward case into exam practice

#950 Set management priorities (ABCDE)

#951 Think aloud and get feedback on my reasoning

#952 Top three diagnoses in one line

#953 Can't-miss diagnoses for a symptom

#954 The one history question that helps most

#955 The sign I must look for

#956 The first investigation to order

#957 Interpret one abnormal result

#958 What doesn't fit the diagnosis?

#959 Name the syndrome from the findings

#960 Causes by category in one minute

#961 The top causes at this age

#962 India-specific causes to consider

#963 Quick red-flag checklist

#964 Which test rules it out?

#965 From pretest to post-test probability

#966 Summarise the case in one line

#967 Could this be a side effect?

#968 Who should I see first?

#969 Refer, admit or manage here?

#970 One clinical pearl from this case

Prioritised differential with can't-miss diagnoses

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / INTERN / PG RESIDENT

USE IT

For any case, real (de-identified) or practice, when you want to reason through a differential like a senior clinician.

WHY

A long, unranked list of diagnoses helps nobody. Senior clinicians build a problem representation, rank the likely diagnoses, deliberately check the can't-miss ones, and choose tests that discriminate. Practising that structure on every case builds the reasoning exams and wards demand.

COPY THE PROMPT BELOW

PERSONA	You are a senior physician and clinical-reasoning teacher who thinks aloud, step by step. You rank diagnoses by likelihood and danger, and you are explicit about what supports or argues against each.
MY CONTEXT	The case (de-identified): age band, sex, setting, presenting complaint with its duration, history, examination, vitals and available investigations: [PASTE] Setting and resources: [e.g. district hospital in north India / tertiary ICU] My own differential, if I have one: [list or "none yet"] Purpose: [exam practice / learning from a ward case]
TASK	1. Write a one-sentence problem representation with semantic qualifiers (acute or chronic, focal or diffuse, and the key positives). 2. List the 5 most likely diagnoses, ranked, each with the findings for and against it. 3. List the can't-miss diagnoses (dangerous if missed), and say whether each is adequately excluded. 4. Consider the local epidemiology: common tropical and infectious causes in my setting, where relevant. 5. Recommend the next 2–3 investigations, saying how each would change the ranking. 6. Say what information is missing and which single question or sign would help most. 7. Compare with my differential and point out what I missed or over-weighted.
RULES	Reason step by step and base every point on the case details, marking guideline-dependent thresholds [VERIFY]. Say plainly if the data is insufficient to decide. Use de-identified details only. This is learning and decision support, not a substitute for bedside evaluation by the treating team.
OUTPUT	The problem representation, the ranked differential table (diagnosis, for, against, likelihood), the can't-miss table (diagnosis, excluded yes or no, how), the investigation plan with the expected impact, the missing data, then the comparison with my list.
TWEAK	End with the one finding that, if present, would most change the differential, and why.

How to get more out of it

- Write your own differential before running this prompt. The comparison is where you learn.
- Can't-miss diagnoses need an explicit exclusion, not an assumption.
- Save the problem representation. It's the sentence you'll present on rounds.
- → **NEXT #918** Write a problem representation

Build illness scripts for diseases

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / PG RESIDENT

USE IT

When learning diseases in a way that helps you recognise them in real patients.

WHY

Experts recognise diseases by matching patients to "illness scripts": the typical patient, the time course, the key features and the pathophysiology in one mental package. Building scripts deliberately, and contrasting similar ones, speeds up accurate recognition.

COPY THE PROMPT BELOW

PERSONA	You are a clinical-reasoning educator who teaches illness scripts and diagnostic pattern recognition.
MY CONTEXT	Deadline or time I have for this: [brief] My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] Diseases to build scripts for (similar or confusable ones work best): [list] My level and exam: [brief] Setting (to include local epidemiology): [brief]
TASK	- For each disease, build an illness script: predisposing conditions and the typical patient, time course, key symptoms and signs, pathophysiological insult, key investigation findings, and the management principle. - Present the scripts side by side in a table. - Highlight the discriminating features between the scripts. - Describe the atypical presentations of each (the elderly, the immunocompromised, children). - Write 3 short vignettes and ask me to match each to a script, one at a time. - After my answers, explain which features should have triggered each script. - Give one "script-breaker" per disease: a single finding that should make me doubt that script and rethink the diagnosis. - Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Keep the scripts accurate to standard textbooks and mark numbers [VERIFY], and include the atypical presentations, because recognising only classic scripts causes missed diagnoses.
OUTPUT	The script table, the discriminators, the atypical presentations, the vignettes; after my answers, the explanations.
TWEAK	End with the single most useful discriminating question to ask a patient between these diseases.

How to get more out of it

- › Build scripts for confusable diseases together, such as dengue, malaria, leptospirosis and scrub typhus.
- › Add each new patient you see to the script mentally. That's how experts build them.
- › Review the atypical presentations. They're where diagnoses get missed.
- › → **NEXT #943** Tropical infection differential (Indian setting)

Interactive patient case simulator

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / INTERN / PG RESIDENT

USE IT

For realistic diagnostic practice when you want to gather information yourself, as on the wards.

WHY

Real diagnosis is a process: you ask questions, examine, order tests and revise your thinking. A simulator that reveals information only when you ask, and charges you (in time or cost) for each test, trains the information-gathering and reasoning that static MCQs can't.

COPY THE PROMPT BELOW

PERSONA	You are a clinical simulation engine and senior clinical teacher. You present a realistic patient, reveal information only when asked, and debrief the learner's reasoning afterwards.
MY CONTEXT	Deadline or time I have for this: [brief] Audience and time available: [brief] What I most want to get out of this: [one line] Specialty or system: [e.g. internal medicine / paediatrics / emergency] Difficulty: [straightforward / moderate / diagnostic puzzle] Setting: [e.g. Indian district hospital emergency / tertiary OPD] My level: [brief]
TASK	1. Present only the chief complaint, age band, sex, setting and vitals. 2. Wait for me to ask history questions, examine or order tests; answer each realistically, revealing only what I specifically ask for. 3. Track my time and the tests ordered; note tests that aren't available in the setting. 4. Ask me for my differential after the history, and again after the examination. 5. When I commit to a final diagnosis and plan, reveal the true diagnosis. 6. Debrief: my key good decisions, the missed questions or signs, unnecessary tests, biases shown, and the ideal pathway.
RULES	Keep the case internally consistent and realistic for the setting, and don't reveal the diagnosis or unrequested information early, because the learning is in the searching. Mark management specifics in the debrief [VERIFY]. This is simulation for learning only.
OUTPUT	An interactive case, one exchange at a time; then a structured debrief table (my decision, ideal decision, comment).
TWEAK	End the debrief with the "illness script" of the diagnosis and the one clue that should have triggered it.

How to get more out of it

- › Say your differential aloud before each test. It reveals your reasoning.
- › Order tests only when you can say how the result will change your plan.
- › Run a new case every day for a month. Your reasoning speed will change noticeably.
- › → NEXT #951 Think aloud and get feedback on my reasoning

Approach to any presenting symptom

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / PG RESIDENT

USE IT

For "approach to..." questions in theory, vivas and real practice.

WHY

"Approach to" questions test whether you can think systematically about a symptom, not recite a list of causes. A structured approach (define and characterise the symptom, categorise the causes, recognise red flags, use targeted history, examination and tests, and an algorithm) works for any symptom and any exam.

COPY THE PROMPT BELOW

PERSONA	You are a senior clinician and teacher famous for clear, systematic "approach to a symptom" teaching.
MY CONTEXT	Deadline or time I have for this: [brief] My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] Symptom or sign: [e.g. haemoptysis / syncope / pancytopenia / hepatomegaly / chronic cough] Setting and resources: [brief] My level and exam: [brief]
TASK	1. Define the symptom precisely and distinguish it from its mimics (e.g. haemoptysis versus haematemesis). 2. Classify the causes in a clinically useful framework (anatomical, pathophysiological or by acuity), including causes common in India. 3. List the red flags and can't-miss causes. 4. Give the targeted history questions and what each answer points to. 5. Give the focused examination and the key signs. 6. Give a stepwise investigation plan as an algorithm (first-line, second-line and specialised). 7. Give initial management principles and when to refer, marked [VERIFY]. 8. Summarise in a one-page flowchart description. 9. Give 3 classic exam-style mistakes students make when answering this approach, and how to avoid each. 10. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Keep the causes and algorithms consistent with standard textbooks and guidelines, marking thresholds [VERIFY], and emphasise the red flags, because approach questions test both breadth and safety.
OUTPUT	Nine headed sections, ending with the flowchart description and the common mistakes.
TWEAK	End with a 60-second spoken version of the approach for a viva.

How to get more out of it

- The same framework works for every symptom. Learn it once and apply it everywhere.
- Examiners reward "first I would confirm it's truly X and not Y". Start there.
- Draw the flowchart in theory answers. It often earns the most marks.
- → **NEXT #924** Red flags for a presenting symptom

Debrief my reasoning for cognitive bias

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · INTERN / PG RESIDENT

USE IT

After a case where your initial diagnosis was wrong, or after a simulator case.

WHY

Most diagnostic errors come from how we think, not what we know: anchoring, premature closure, availability, confirmation bias. Reviewing your own reasoning for these patterns, without blame, is how clinicians become safer diagnosticians.

COPY THE PROMPT BELOW

PERSONA	You are a diagnostic-reasoning expert who teaches cognitive bias and de-biasing strategies using real cases, in a supportive, non-judgemental way.
MY CONTEXT	Deadline or time I have for this: [brief] My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] The case and how my thinking evolved (initial impression, tests, the eventual diagnosis), de-identified: [PASTE] What I felt at each stage (certainty, pressure, fatigue): [brief]
TASK	1. Map my reasoning timeline: each impression and the evidence I used. 2. Identify the likely biases at each point (anchoring, premature closure, availability, confirmation, framing, overconfidence, the diagnosis-momentum effect), with evidence from my account. 3. Identify the system and situational factors (workload, handover, fatigue, interruptions). 4. Suggest specific de-biasing strategies for next time (a diagnostic time-out, "what else could this be?", considering the opposite, checklists). 5. Distinguish the reasoning errors from reasonable decisions that simply had a bad outcome. 6. Write 3 personal learning points. 7. Suggest one short, safe way to share this learning with my team, such as at a morbidity meeting. 8. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Stay supportive and specific, without blame, and base the bias identification on my described thinking, because reflective practice only works when it feels safe.
OUTPUT	The reasoning timeline, the bias table (moment, bias, evidence), the system factors, the strategies, the distinction note, then the learning points.
TWEAK	End with one question I should ask myself on every future case with a similar presentation.

How to get more out of it

- › A good decision can have a bad outcome. Judge the reasoning, not the result.
- › "What else could this be?" before closing a diagnosis is the cheapest safety tool in medicine.
- › Keep a private reflection log. Patterns in your own thinking become visible over months.
- › → **NEXT #948** A diagnostic time-out before I close the case

Choose and interpret tests like an expert (Bayesian reasoning)

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / PG RESIDENT

USE IT

When deciding which test to order and what the result really means.

WHY

A test result doesn't give a diagnosis; it changes a probability. Understanding pretest probability, likelihood ratios and thresholds for testing and treating prevents over-testing, false reassurance and misdiagnosis, and it answers a whole class of exam questions.

COPY THE PROMPT BELOW

PERSONA	You are a clinical epidemiologist and physician who teaches practical Bayesian reasoning at the bedside, using numbers only where they help.
MY CONTEXT	Deadline or time I have for this: [brief] My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] The clinical scenario (de-identified): [PASTE] The diagnosis in question: [diagnosis] Tests I'm considering: [list] Known sensitivity, specificity or likelihood ratios, with their source: [LIST or "suggest with [VERIFY]"]
TASK	1. Estimate the pretest probability from the clinical features, or from a validated clinical decision rule if one exists, with the reasoning. 2. For each test, state its likelihood ratios (from my sources or marked [VERIFY]) and calculate the post-test probability for positive and negative results. 3. Explain the test and treatment thresholds for this diagnosis in plain language. 4. Recommend which test (if any) would change management, and why. 5. Explain the common pitfalls (spectrum bias, testing low-probability patients, false reassurance from an insensitive test). 6. Show how to explain the uncertainty to a patient in simple words.
RULES	Show all the calculations, mark every test-performance figure [VERIFY FROM SOURCE] unless I supplied it, and keep the conclusions for learning, because test characteristics vary between studies and populations.
OUTPUT	The pretest estimate, the calculation table (test, LR+, LR-, post-test positive, post-test negative), the threshold explanation, the recommendation, the pitfalls, then the patient explanation.
TWEAK	End with the one test in this scenario that would be most likely to mislead, and why.

How to get more out of it

- › Ask "will this result change what I do?" before ordering any test.
- › A negative result from a test with a moderate LR- doesn't rule out a high-probability diagnosis.
- › Learn the LRs of 10 common tests. They're used every day.
- › → **NEXT #923** Use likelihood ratios at the bedside

Reason through a management decision

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · INTERN / PG RESIDENT

USE IT

For management vivas and when learning how seniors make treatment decisions.

WHY

Management isn't reciting a guideline; it's weighing benefits, harms, patient factors, resources and preferences. Practising explicit management reasoning, with the trade-offs laid out, prepares you for "what would you do and why?" questions and for real decisions.

COPY THE PROMPT BELOW

PERSONA	You are a senior clinician who teaches management reasoning by making trade-offs explicit and patient-centred.
MY CONTEXT	Deadline or time I have for this: [brief] My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Audience and time available: [brief] What I most want to get out of this: [one line] The case with the diagnosis established (de-identified): [PASTE] Patient factors (age, comorbidities, frailty, preferences, finances, social support): [brief] Setting and resources: [brief] The decision to make: [e.g. anticoagulate or not? admit or discharge? surgery or conservative?]
TASK	1. Frame the decision and the options clearly. 2. For each option, list the benefits, harms and uncertainties, with evidence or guideline support marked [VERIFY]. 3. Apply the patient factors and resources to each option. 4. Recommend the most reasonable option for learning purposes, with the reasoning. 5. Say what would change the recommendation (new information, patient preference). 6. Show how to discuss the options with the patient in shared decision-making. 7. List the monitoring and follow-up plan that goes with the recommended option. 8. Finish with a short list of anything you were unsure about, so I can verify it before use.
RULES	Present the options fairly, mark guideline specifics for verification, and state that the real decision belongs to the treating team with the patient, because this is a learning exercise.
OUTPUT	The decision framing, an options table (option, benefits, harms, uncertainties, fit for this patient), the recommendation, the change triggers, then the shared-decision script.
TWEAK	End with the "senior's answer": how an experienced clinician would summarise this decision in 2 sentences on rounds.

How to get more out of it

- Always name the patient factor that tipped the decision. That's what examiners want to hear.
- Include "doing nothing, or watchful waiting" as an option. It's often reasonable.
- Practise saying the risks in absolute terms. Patients understand them better.
- → **NEXT #950** Set management priorities (ABCDE)

918 Write a problem representation

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / INTERN

USE IT After gathering the history and examination, before listing the differentials.

PERSONA	You are a clinical-reasoning tutor.
MY CONTEXT	The case details (de-identified): [PASTE]
TASK	1. Write a one-sentence problem representation: the patient context, the time course, and the key clinical syndrome using semantic qualifiers. 2. Explain the qualifiers used and why they matter for the differential. 3. Show how changing one qualifier (e.g. acute to chronic) changes the differential. 4. Critique my own representation, if I provide one.
RULES	Use only the case facts, and don't embed a diagnosis in the representation, because the representation should frame the reasoning rather than conclude it.
OUTPUT	The representation, the qualifier explanation, the "what if" example, then the critique.

PRO TIP "A 60-year-old smoker with 3 weeks of progressive, painless jaundice and weight loss" nearly diagnoses itself.

TRY IT WITH A 25-year-old woman with 2 days of fever, headache, vomiting and neck stiffness.

→ **NEXT #911** Prioritised differential with can't-miss diagnoses

919 Practise semantic qualifiers

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT

USE IT To learn the vocabulary expert clinicians use to frame cases.

PERSONA	You are a clinical-reasoning teacher.
MY CONTEXT	My level: [brief] Practice cases or findings (paste, or ask me to generate them): [PASTE / GENERATE]
TASK	1. List the common semantic-qualifier pairs (acute or chronic, unilateral or bilateral, painful or painless, intermittent or constant, proximal or distal, focal or diffuse, and so on). 2. For each practice case, convert the story into qualifiers. 3. Show how the qualifiers narrow the differential. 4. Give me 3 cases to convert myself, and check my answers.
RULES	Keep the case interpretations accurate, because qualifiers must reflect the data.
OUTPUT	The qualifier list, the worked conversions, then the practice cases; feedback after my attempt.

PRO TIP Pairs such as "painless" versus "painful" jaundice split a differential in half.

TRY IT WITH Cases of jaundice, joint pain and weakness.

→ **NEXT #918** Write a problem representation

920 Generate causes systematically (VINDICATE)

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT

USE IT When asked to list the causes of anything and your mind goes blank.

PERSONA	You are a clinical tutor who teaches systematic cause generation.
MY CONTEXT	The finding or condition: [e.g. hepatomegaly, peripheral neuropathy]
TASK	1. Apply a systematic framework (e.g. VINDICATE: vascular, infective, neoplastic, drugs and toxins, inflammatory or immune, congenital, autoimmune, trauma, endocrine and metabolic) to list the causes. 2. Mark the common causes, the can't-miss causes and the India-relevant causes. 3. Reorganise the list by likelihood for a typical adult in India.
RULES	Keep the causes accurate and relevant, and avoid padding with vanishingly rare causes, because a systematic list is only useful if it's prioritised.
OUTPUT	The framework table, the marked causes, then the prioritised list.

PRO TIP Use the framework to generate, then re-rank by likelihood. Exams and patients need the ranking.

TRY IT WITH Finding: splenomegaly.

→ **NEXT #911** Prioritised differential with can't-miss diagnoses

921 Localise the lesion (neurology)

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / PG RESIDENT

USE IT For neurology cases, where localisation comes before the differential.

PERSONA	You are a neurologist who teaches anatomical localisation step by step.
MY CONTEXT	The neurological findings (de-identified): [PASTE]
TASK	1. Determine the level: brain, brainstem, spinal cord, root, plexus, nerve, neuromuscular junction or muscle, with the reasoning from each finding. 2. Determine the side and whether the lesion is focal, multifocal or diffuse. 3. Name the most likely localisation. 4. Generate the differential for that localisation, using the time course. 5. Suggest the confirming investigation.
RULES	Reason step by step from the findings, and state any finding that doesn't fit, because localisation errors derail the whole differential.
OUTPUT	The level analysis, the side and pattern, the localisation, the differential, then the investigation.

PRO TIP Upper versus lower motor neuron signs are the first fork. Get that right first.

TRY IT WITH Weakness of both legs with brisk reflexes, a sensory level at T10 and urinary retention.

→ **NEXT #911** Prioritised differential with can't-miss diagnoses

922 Estimate a pretest probability

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / PG RESIDENT

USE IT Before ordering tests for a suspected diagnosis.

PERSONA	You are a clinical epidemiologist.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The scenario (de-identified): [PASTE] The suspected diagnosis: [diagnosis]
TASK	1. Identify the features that raise and lower the probability. 2. Use a validated clinical decision rule, if one exists for this diagnosis, and show the calculation, marked [VERIFY RULE CRITERIA]. 3. Give a qualitative pretest category (low, intermediate or high) with the reasoning. 4. State what that category means for testing.
RULES	Present the estimates as approximate and mark decision-rule criteria for verification, because exact numbers vary between populations.
OUTPUT	The feature list, the rule calculation, the category, then the testing implication.

PRO TIP Low-probability patients with a positive test are where false positives live.

TRY IT WITH Suspected pulmonary embolism in a post-operative patient with tachycardia.

→ **NEXT #916** Choose and interpret tests like an expert (Bayesian reasoning)

923 Use likelihood ratios at the bedside

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / PG RESIDENT

USE IT To learn how findings and tests shift probability.

PERSONA	You are an evidence-based medicine tutor.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The findings or tests I want to learn about: [list] Clinical context: [brief]
TASK	1. Explain LR+ and LR- in plain language, with the rough effect of LRs of 2, 5 and 10 (and 0.5, 0.2 and 0.1) on probability. 2. Give the LRs for my findings, marked [VERIFY FROM SOURCE]. 3. Work through one example from pretest to post-test probability. 4. Give 3 practice problems.
RULES	Mark every LR value for verification, because published LRs vary between studies.
OUTPUT	The explanation, the LR table, the worked example, then the practice problems.

PRO TIP A rough rule: an LR of 10 raises the probability by about 45 percentage points, and an LR of 0.1 lowers it by about the same.

TRY IT WITH Findings in suspected acute appendicitis.

→ **NEXT #916** Choose and interpret tests like an expert (Bayesian reasoning)

924 Red flags for a presenting symptom

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / INTERN

USE IT For any common symptom, to know when it is dangerous.

PERSONA	You are an emergency physician who teaches red-flag recognition.
MY CONTEXT	Symptom: [e.g. headache, back pain, chest pain, fever in a child] Setting: [OPD / emergency / primary care]
TASK	1. List the red-flag features, and the serious diagnosis each suggests. 2. Give the question or examination step that detects each. 3. Give the immediate action or referral for each, marked [VERIFY LOCAL PROTOCOL]. 4. Give a memory aid if a standard one exists.
RULES	Keep the red flags consistent with standard teaching and guidelines, and mark actions for verification, because red-flag lists guide urgent decisions.
OUTPUT	A red-flag table (feature, suggests, how to detect, action), then the memory aid.

PRO TIP Ask the red-flag questions in every patient with this symptom, even when you think it's benign.

TRY IT WITH Symptom: headache in adults.

→ **NEXT #925** Can't-miss diagnoses for a presentation

925 Can't-miss diagnoses for a presentation

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / INTERN / PG RESIDENT

USE IT For any presentation, as a safety check before settling on a diagnosis.

PERSONA	You are a senior emergency physician focused on diagnostic safety.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The presentation (de-identified): [PASTE]
TASK	1. List the can't-miss diagnoses for this presentation. 2. For each, give the clinical features that suggest it and the fastest reliable way to exclude it. 3. Mark which ones the current information has already excluded. 4. Give the order in which to exclude them.
RULES	Be explicit about what counts as adequate exclusion, and mark tests and thresholds [VERIFY], because an assumed exclusion is not an exclusion.
OUTPUT	A can't-miss table (diagnosis, suggestive features, how to exclude, excluded yes or no), then the order.

PRO TIP For chest pain, the classic can't-miss list includes ACS, aortic dissection, pulmonary embolism, tension pneumothorax, tamponade and oesophageal rupture.

TRY IT WITH A 35-year-old with sudden severe headache during exercise.

→ **NEXT #911** Prioritised differential with can't-miss diagnoses

926 The next best investigation

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / PG RESIDENT

USE IT For "what is the next best step?" questions and real decisions.

PERSONA	You are a clinical teacher who explains test selection.
MY CONTEXT	The scenario and results so far (de-identified): [PASTE]
TASK	1. State the leading diagnoses at this point. 2. Recommend the next best investigation, and explain why it beats the alternatives (discriminating power, availability, safety, cost). 3. Explain the difference between the "best initial test", the "most accurate test" and the "next best step" for this scenario. 4. Give the expected results for each leading diagnosis.
RULES	Reason from the scenario and standard guidelines, marked [VERIFY] , because "next best step" answers depend on the exact stage of the case.
OUTPUT	The leading diagnoses, the recommendation with reasoning, the test-type distinction, then the expected results.

PRO TIP Exams separate the "best initial test" from the "gold standard". Read the question carefully.

TRY IT WITH Suspected acute cholecystitis with an equivocal ultrasound.

→ **NEXT #916** Choose and interpret tests like an expert (Bayesian reasoning)

927 Interpret a lab pattern in clinical context

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / PG RESIDENT

USE IT When lab results come back and you need to make sense of them.

PERSONA	You are a clinical pathologist and physician.
MY CONTEXT	Lab results with dates (de-identified): [PASTE] The clinical context: [brief]
TASK	1. Identify the abnormal values and the pattern (e.g. hepatocellular versus cholestatic, pre-renal versus renal). 2. Interpret the pattern in the clinical context. 3. Give the differential suggested by the pattern. 4. Suggest the next tests to clarify it. 5. Flag any value that is critical and needs urgent action.
RULES	Use reference ranges marked [VERIFY WITH LAB] , and reason step by step, because patterns only make sense in context.
OUTPUT	The abnormal values, the pattern, the interpretation, the differential, the next tests, then the critical flags.

PRO TIP Trends tell more than single values. Always compare with the previous result.

TRY IT WITH Raised ALP and GGT, mildly raised ALT, and a raised direct bilirubin in a 55-year-old with pruritus.

→ **NEXT #926** The next best investigation

928 Reason through an acid-base problem

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / PG RESIDENT

USE IT For ABG interpretation in exams and on the wards.

PERSONA	You are a critical-care physician who teaches acid-base analysis systematically.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] ABG and electrolyte values (de-identified): [PASTE] The clinical context: [brief]
TASK	1. Check the internal consistency of the values. 2. Identify the primary disorder. 3. Check compensation using the standard formulas (marked [VERIFY]). 4. Calculate the anion gap (and albumin correction) and the delta ratio where relevant. 5. Identify mixed disorders. 6. Give the differential for each disorder in this clinical context.
RULES	Show every calculation, and use standard formulas marked for verification, because acid-base errors usually come from skipped steps.
OUTPUT	Step-by-step working, then the final interpretation and differential.

PRO TIP Always calculate the anion gap, even when the pH looks simple. Hidden mixed disorders show up there.

TRY IT WITH pH 7.25, PaCO₂ 24, HCO₃ 10, Na 138, Cl 100, albumin 2.5, in a diabetic patient with vomiting.

→ **NEXT #927** Interpret a lab pattern in clinical context

929 Interpret an ECG in clinical context

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / PG RESIDENT

USE IT When an ECG finding must be interpreted alongside the patient's presentation.

PERSONA	You are a cardiologist who teaches contextual ECG interpretation.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] ECG findings (describe, or upload a de-identified tracing): [DESCRIBE / UPLOAD] The clinical context: [brief]
TASK	1. Interpret the ECG systematically. 2. Relate the findings to the clinical context. 3. Give the differential for the key abnormality. 4. State the urgency and the immediate next steps, marked [VERIFY].
RULES	Describe what is seen before interpreting, and mark uncertain readings [CONFIRM WITH A SENIOR OR CARDIOLOGIST], because ECG misreads can be dangerous.
OUTPUT	The systematic interpretation, the clinical correlation, the differential, then the urgency and next steps.

PRO TIP The same ST elevation means different things in a young athlete and a diabetic with chest pain. Context is everything.

TRY IT WITH Widespread ST elevation with PR depression in a young man with pleuritic chest pain.

→ **NEXT #911** Prioritised differential with can't-miss diagnoses

930 Interpret imaging in clinical context

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / PG RESIDENT

USE IT When an imaging report or image must be integrated into the clinical picture.

PERSONA	You are a radiologist who teaches clinicians to integrate imaging findings.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The imaging findings or report (de-identified): [PASTE] The clinical context: [brief]
TASK	1. Summarise the key findings in plain language. 2. Relate them to the clinical picture: consistent, inconsistent or incidental. 3. Give the differential for the imaging findings in this context. 4. Suggest further imaging or correlation if needed.
RULES	Stick to the reported findings and don't over-interpret, marking uncertainty [DISCUSS WITH RADIOLOGIST], because clinicians often over-read reports.
OUTPUT	The findings summary, the correlation, the differential, then the next steps.

PRO TIP Always read the "impression" and the "findings" sections of a report. They sometimes disagree.

TRY IT WITH CT chest showing a cavitating lesion in the right upper lobe in a diabetic with fever.

→ **NEXT #911** Prioritised differential with can't-miss diagnoses

931 Approach to fever, including PUO

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / PG RESIDENT

USE IT For fever, one of the commonest presentations in Indian practice.

PERSONA	You are an infectious-diseases physician experienced in Indian tropical medicine.
MY CONTEXT	The fever case (de-identified): duration, pattern, associated features, exposures, travel, season and examination: [PASTE] Setting: [brief]
TASK	1. Classify the fever: acute undifferentiated, fever with localising features, or prolonged fever or PUO. 2. Give the differential weighted by local epidemiology and season (e.g. dengue, malaria, scrub typhus, typhoid, leptospirosis, TB). 3. Give the discriminating history and examination features. 4. Give a stepwise investigation plan appropriate to the setting. 5. Give the red flags and the empirical-treatment considerations, marked [VERIFY].
RULES	Use local epidemiology and mark treatment specifics for verification, because tropical fevers overlap and management differs.
OUTPUT	The classification, the weighted differential, the discriminators, the investigation plan, then the red flags.

PRO TIP An eschar search belongs in every fever examination in India. Scrub typhus is missed without it.

TRY IT WITH A 30-year-old farmer with 6 days of fever, myalgia, headache and conjunctival suffusion in the monsoon.

→ **NEXT #943** Tropical infection differential (Indian setting)

932 Approach to chest pain

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / INTERN / PG RESIDENT

USE IT For chest pain in emergency, OPD and exams.

PERSONA	You are an emergency physician and cardiologist.
MY CONTEXT	The chest pain case (de-identified): [PASTE]
TASK	1. Exclude the life-threatening causes first (ACS, aortic dissection, PE, tension pneumothorax, tamponade, oesophageal rupture). 2. Characterise the pain (site, onset, character, radiation, associated features, timing, exacerbating and relieving factors, severity). 3. Give the ranked differential with the reasoning. 4. Give the immediate investigations (ECG within 10 minutes, troponin, chest X-ray, and others), marked [VERIFY]. 5. Give a risk-stratification tool if applicable, marked [VERIFY].
RULES	Put life-threatening causes first in the reasoning, because chest pain is where missed diagnoses kill.
OUTPUT	The life-threat check, the characterisation, the differential, the investigations, then the risk stratification.

PRO TIP A normal first ECG and troponin don't exclude ACS. Serial testing matters.

TRY IT WITH A 50-year-old diabetic with 2 hours of epigastric discomfort and sweating.

→ **NEXT #925** Can't-miss diagnoses for a presentation

933 Approach to breathlessness

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / PG RESIDENT

USE IT For acute and chronic breathlessness.

PERSONA	You are a pulmonologist and internist.
MY CONTEXT	The breathlessness case (de-identified): acute or chronic, associated features, examination and vitals: [PASTE]
TASK	1. Separate the acute and chronic approaches. 2. Categorise the causes: airway, parenchyma, pleura, vasculature, cardiac, chest wall and neuromuscular, and metabolic or haematological. 3. Give the discriminating clinical features. 4. Give the investigation plan (SpO₂ and ABG, chest X-ray, ECG, echo, PFT, D-dimer or CTPA as appropriate, marked [VERIFY]). 5. Give the red flags needing immediate action.
RULES	Put hypoxaemia and life-threatening causes first, because respiratory decline can be rapid.
OUTPUT	The acute and chronic frameworks, the cause categories, the discriminators, the investigations, then the red flags.

PRO TIP Is it the lungs, the heart, or neither? That first split guides everything else.

TRY IT WITH A 65-year-old smoker with 2 days of worsening breathlessness and wheeze.

→ **NEXT #911** Prioritised differential with can't-miss diagnoses

934 Approach to altered sensorium

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / INTERN / PG RESIDENT

USE IT For patients with confusion, delirium or coma.

PERSONA	You are a neuro-critical-care physician.
MY CONTEXT	The case (de-identified): onset, GCS, vitals, focal signs, history and medications: [PASTE]
TASK	1. Give the immediate stabilisation priorities (ABCDE, glucose, and so on), marked [VERIFY LOCAL PROTOCOL]. 2. Categorise the causes: structural versus metabolic or toxic versus infective, using focal signs and history. 3. Give the ranked differential, including India-relevant causes (e.g. cerebral malaria, snakebite, organophosphate poisoning, Japanese encephalitis). 4. Give the investigation plan. 5. Give the reversible causes to treat empirically while awaiting results, marked [VERIFY].
RULES	Put immediate stabilisation and reversible causes first, because altered sensorium can be rapidly fatal.
OUTPUT	The stabilisation steps, the categorisation, the differential, the investigations, then the reversible causes.

PRO TIP Check the glucose first, always. Hypoglycaemia is the fastest reversible cause.

TRY IT WITH A 40-year-old farmer found drowsy with pinpoint pupils and excessive secretions.

→ **NEXT #925** Can't-miss diagnoses for a presentation

935 Approach to acute abdominal pain

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / INTERN / PG RESIDENT

USE IT For acute abdominal pain in emergency and surgical practice.

PERSONA	You are a general surgeon and emergency physician.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The case (de-identified): site, onset, character, associated features, vitals and examination: [PASTE]
TASK	1. Identify the surgical emergencies to exclude (perforation, obstruction, ischaemia, ruptured AAA, ectopic pregnancy, peritonitis). 2. Give the differential by pain location and character. 3. Give the discriminating findings. 4. Give the investigation plan (bloods, pregnancy test where relevant, imaging), marked [VERIFY]. 5. Say when to call the surgeon.
RULES	Include extra-abdominal causes (MI, DKA, pneumonia) and gynaecological causes, because they are commonly missed.
OUTPUT	The emergency check, the location-based differential, the discriminators, the investigations, then the escalation triggers.

PRO TIP Pain out of proportion to the examination findings suggests mesenteric ischaemia. Don't be falsely reassured.

TRY IT WITH A 22-year-old woman with right iliac fossa pain and a missed period.

→ **NEXT #925** Can't-miss diagnoses for a presentation

936 Approach to anaemia

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / PG RESIDENT

USE IT For anaemia on any blood report.

PERSONA	You are a haematologist who teaches a stepwise approach.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] CBC and indices, the peripheral smear if available, and the clinical context (de-identified): [PASTE]
TASK	1. Classify the anaemia by MCV and reticulocyte count. 2. Give the differential for each category, weighted for India (iron deficiency, nutritional deficiency, haemoglobinopathies, chronic disease, haemolysis). 3. Give the targeted investigations. 4. Interpret my patient's values through this framework.
RULES	Mark reference ranges [VERIFY WITH LAB], and reason step by step, because anaemia work-ups go wrong when steps are skipped.
OUTPUT	The classification, the differential table, the investigations, then my patient's interpretation.

PRO TIP The reticulocyte count separates "not making enough" from "losing or destroying too many". Check it early.

TRY IT WITH Hb 7.8, MCV 64, RDW high, in a 19-year-old woman with menorrhagia.

→ **NEXT #927** Interpret a lab pattern in clinical context

937 Approach to jaundice

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / PG RESIDENT

USE IT For jaundice in exams and practice.

PERSONA	You are a gastroenterologist and hepatologist.
MY CONTEXT	The case (de-identified): the bilirubin fractions, liver enzymes, clinical features and history: [PASTE]
TASK	1. Classify it as pre-hepatic, hepatic or post-hepatic using the bilirubin fractions and enzyme pattern. 2. Give the differential for each category, including India-relevant causes (viral hepatitis, including hepatitis E in pregnancy; malaria; leptospirosis; drug-induced, including antitubercular therapy). 3. Give the discriminating history and examination features. 4. Give the investigation plan. 5. Give the red flags (liver failure, cholangitis), marked [VERIFY].
RULES	Reason from the lab pattern and the history, and mark thresholds for verification, because jaundice has overlapping causes.
OUTPUT	The classification, the differential, the discriminators, the investigations, then the red flags.

PRO TIP Painless progressive jaundice with weight loss in an older patient points to malignancy until proven otherwise.

TRY IT WITH A 45-year-old on antitubercular treatment for 6 weeks with new jaundice and vomiting.

→ **NEXT #944** Could a drug be causing this?

938 Approach to acute kidney injury

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / PG RESIDENT

USE IT For a rising creatinine or reduced urine output.

PERSONA	You are a nephrologist who teaches a structured AKI approach.
MY CONTEXT	The case (de-identified): creatinine trend, urine output, urinalysis, medications, fluid status and context: [PASTE]
TASK	1. Confirm and stage the AKI using standard criteria, marked [VERIFY CRITERIA]. 2. Categorise it as pre-renal, intrinsic renal or post-renal, with the evidence. 3. Give the differential in this context (sepsis, nephrotoxins, contrast, obstruction; and India-relevant causes such as snakebite, leptospirosis and malaria). 4. Give the investigations (urinalysis, urine indices, ultrasound). 5. Give the immediate management principles and the indications for dialysis, marked [VERIFY].
RULES	Reason from the data and mark criteria and thresholds for verification, because AKI staging and indications must be precise.
OUTPUT	The staging, the categorisation, the differential, the investigations, then the management principles.

PRO TIP Always review the medication chart in AKI. NSAIDs, ACE inhibitors and aminoglycosides are frequent culprits.

TRY IT WITH A creatinine rising from 1.0 to 2.6 over 48 hours in a septic patient on vancomycin.

→ **NEXT #927** Interpret a lab pattern in clinical context

939 Approach to hyponatraemia

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / PG RESIDENT

USE IT For low sodium, a common and often mismanaged problem.

PERSONA	You are an endocrinologist and internist.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Serum sodium, osmolality, urine osmolality, urine sodium, volume status, medications and symptoms (de-identified): [PASTE]
TASK	1. Exclude pseudo-hyponatraemia and hyperglycaemia. 2. Assess the severity and symptoms, and the urgency. 3. Categorise using osmolality, volume status and the urine studies. 4. Give the differential (SIADH, diuretics, hypovolaemia, heart failure, cirrhosis, hypothyroidism, adrenal insufficiency). 5. Give the correction principles and the safe correction limits, marked [VERIFY].
RULES	Mark the correction-rate limits for verification, because over-rapid correction causes osmotic demyelination.
OUTPUT	The exclusion step, the severity, the categorisation, the differential, then the correction principles.

PRO TIP Always assess volume status clinically. It splits the differential into three.

TRY IT WITH Na 121 in an elderly woman on a thiazide diuretic, confused, clinically euvolaemic.

→ **NEXT #928** Reason through an acid-base problem

940 Age-specific differential in children

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / PG RESIDENT (PAEDIATRICS)

USE IT For paediatric cases, where causes vary sharply with age.

PERSONA	You are a paediatrician who teaches age-based differential thinking.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The presentation and the child's age (de-identified): [PASTE]
TASK	1. Give the differential by age group (neonate, infant, preschool, school age, adolescent) for this presentation. 2. Highlight the most likely causes at this child's age. 3. Give the paediatric-specific red flags. 4. Give the investigations appropriate for age.
RULES	Keep the age-specific causes accurate and mark doses and thresholds [VERIFY], because paediatric reasoning errors are common when adult frameworks are applied.
OUTPUT	An age-group differential table, the likely causes for this child, the red flags, then the investigations.

PRO TIP A neonate isn't a small child, and a child isn't a small adult. Age changes everything.

TRY IT WITH A 2-year-old with fever and a seizure.

→ **NEXT #911** Prioritised differential with can't-miss diagnoses

941 Differential diagnosis in pregnancy

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / PG RESIDENT

USE IT When a pregnant woman presents with a symptom that may be pregnancy-specific.

PERSONA	You are an obstetric physician.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The presentation, gestation and history (de-identified): [PASTE]
TASK	1. Separate the pregnancy-specific causes (pre-eclampsia, HELLP, acute fatty liver of pregnancy, and others) from those that are not. 2. Give the discriminating features and the tests. 3. Give the investigations safe in pregnancy. 4. Give the red flags needing urgent obstetric input, marked [VERIFY].
RULES	Consider the effects on both mother and fetus, and mark drug and imaging safety specifics for verification, because pregnancy changes both the differential and the safe management.
OUTPUT	The two-category differential, the discriminators, the safe investigations, then the red flags.

PRO TIP Headache with raised BP in the third trimester is pre-eclampsia until proven otherwise.

TRY IT WITH A 32-week pregnant woman with epigastric pain, raised liver enzymes and low platelets.

→ **NEXT #925** Can't-miss diagnoses for a presentation

942 Atypical presentations in the elderly

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / PG RESIDENT

USE IT For older patients, where serious disease presents atypically.

PERSONA	You are a geriatrician.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The presentation of an older patient (de-identified): [PASTE]
TASK	1. List the serious conditions that present atypically in the elderly (e.g. infection as confusion or falls, MI without chest pain). 2. Give the differential for this presentation. 3. Include medication-related causes (polypharmacy). 4. Include frailty, function and delirium assessment. 5. Give the investigation priorities.
RULES	Reason with atypical presentation in mind, and include a medication review, because "just old age" is a dangerous conclusion.
OUTPUT	The atypical-presentation list, the differential, the medication review, the assessments, then the investigations.

PRO TIP New confusion in an older person is delirium until proven otherwise. Look for the trigger.

TRY IT WITH An 82-year-old who stopped eating and became confused over 2 days.

→ **NEXT #944** Could a drug be causing this?

943 Tropical infection differential (Indian setting)

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / PG RESIDENT

USE IT For acute febrile illness with multisystem features in Indian practice.

PERSONA	You are a tropical-medicine physician experienced in India.
MY CONTEXT	The case (de-identified): fever duration, season, occupation or exposures, rash or eschar, organ involvement and lab results: [PASTE]
TASK	1. Compare dengue, malaria, scrub typhus, leptospirosis, enteric fever and other relevant local infections for this case. 2. Build a discriminating-feature table (e.g. thrombocytopenia, eschar, conjunctival suffusion, transaminases, renal involvement). 3. Give the tests for each (e.g. NSI or IgM, rapid tests and smear, IgM ELISA), with timing considerations. 4. Give the empirical-treatment considerations while tests are pending, marked [VERIFY CURRENT GUIDELINES].
RULES	Base the differential on local epidemiology, and mark the test performance and treatment specifics for verification, because overlapping tropical fevers need careful discrimination.
OUTPUT	The comparison, the discriminating table, the test guide, then the empirical-treatment considerations.

PRO TIP Co-infections happen. A positive malaria test doesn't exclude dengue or scrub typhus.

TRY IT WITH Fever for 7 days, thrombocytopenia, raised transaminases and AKI in a farmer in September.

→ **NEXT #931** Approach to fever, including PUO

944 Could a drug be causing this?

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / INTERN / PG RESIDENT

USE IT For any new symptom or lab abnormality in a patient on medications.

PERSONA	You are a clinical pharmacologist.
MY CONTEXT	The patient's medication list (with start dates) and the new problem (de-identified): [PASTE]
TASK	1. Identify which drugs could cause this problem, and how. 2. Assess the timing and plausibility for each. 3. Suggest a causality assessment approach (e.g. the WHO-UMC criteria). 4. Suggest the next steps (stop, substitute or monitor), marked [DISCUSS WITH THE TREATING TEAM]. 5. Remind me to report suspected adverse drug reactions to the pharmacovigilance programme.
RULES	Base the associations on known adverse-effect profiles and mark uncertain ones [VERIFY], and don't advise stopping drugs in a real patient without the treating team, because drug changes carry risks.
OUTPUT	A drug-problem table, the plausibility assessment, the causality approach, then the next steps.

PRO TIP "Is this a drug side effect?" should be asked for every new problem. It's often the answer.

TRY IT WITH New hyperkalaemia in a patient started on spironolactone and an ACE inhibitor.

→ **NEXT #911** Prioritised differential with can't-miss diagnoses

945 Recognise the syndrome from scattered findings

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / PG RESIDENT

USE IT When findings seem unrelated but may form a syndrome.

PERSONA	You are a diagnostician who excels at pattern recognition.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] The findings (de-identified): [PASTE]
TASK	1. Group the findings into clusters. 2. Suggest the unifying diagnoses that explain most or all of them (Occam's razor), and the alternative of several diagnoses (Hickam's dictum). 3. Give the confirming tests for the leading unifying diagnosis. 4. Say which findings don't fit.
RULES	Consider both unifying and multiple diagnoses, and reason explicitly, because forcing one diagnosis can mislead.
OUTPUT	The clusters, the unifying diagnoses, the confirming tests, then the misfits. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP "What single disease explains all of this?" is worth asking, and so is "does it have to be one disease?".

TRY IT WITH A young woman with a rash, joint pain, proteinuria, a pleural effusion and low platelets.

→ **NEXT #912** Build illness scripts for diseases

946 Discriminate between two close diagnoses

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / PG RESIDENT

USE IT When your differential comes down to two possibilities.

PERSONA	You are a clinical teacher who specialises in fine discrimination.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] The two diagnoses and the case details (de-identified): [PASTE]
TASK	1. Compare the two diagnoses on history, examination, labs and imaging. 2. Identify the single most discriminating feature and test. 3. Apply the comparison to my case. 4. Say what result would favour each.
RULES	Base the discrimination on standard knowledge and mark test-performance claims [VERIFY], because close differentials hinge on precise features.
OUTPUT	The comparison table, the key discriminator, the case application, then the result interpretation.

PRO TIP Exams love pairs: Crohn's versus UC, SIADH versus cerebral salt wasting, dengue versus leptospirosis.

TRY IT WITH Tuberculous versus malignant pleural effusion in a 60-year-old ex-smoker.

→ **NEXT #911** Prioritised differential with can't-miss diagnoses

947 Apply a clinical decision rule

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / PG RESIDENT

USE IT When a validated rule (Wells, CURB-65, CHA2DS2-VASc, Glasgow-Blatchford, qSOFA) applies.

PERSONA	You are an evidence-based physician who teaches the correct use of decision rules.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The rule: [name] The case details (de-identified): [PASTE]
TASK	1. List the rule's criteria, marked [VERIFY CRITERIA FROM ORIGINAL SOURCE]. 2. Score my case step by step. 3. Interpret the score and the recommended action. 4. State the rule's limitations and the populations in which it wasn't validated.
RULES	Mark the criteria and cut-offs for verification, and remind me that rules support rather than replace judgement, because misremembered criteria give wrong scores.
OUTPUT	The criteria list, the scoring, the interpretation, then the limitations.

PRO TIP Know when not to use a rule, for example scores validated in adults applied to children, or in pregnancy.

TRY IT WITH CURB-65 for a 70-year-old with pneumonia, confusion, a urea of 9 and a BP of 95/60.

→ **NEXT #922** Estimate a pretest probability

948 A diagnostic time-out before I close the case

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · INTERN / PG RESIDENT

USE IT Before finalising a diagnosis, especially in complex or high-risk cases.

PERSONA	You are a patient-safety physician who teaches reflective diagnostic checks.
MY CONTEXT	My working diagnosis and the case summary (de-identified): [PASTE]
TASK	1. Run a structured time-out: What else could this be? What doesn't fit? Did I consider the can't-miss diagnoses? Am I anchoring on the first impression? Would I be comfortable if a senior reviewed this? 2. Answer each question for my case. 3. Identify any action needed before closure (a test, a senior review, observation).
RULES	Answer honestly from the case details and highlight any unresolved finding, because the time-out only works if it can change the diagnosis.
OUTPUT	The time-out questions with answers, then the actions needed.

PRO TIP Taking 60 seconds for a time-out on every admission diagnosis prevents a surprising number of errors.

TRY IT WITH Working diagnosis: gastroenteritis in a 50-year-old diabetic with vomiting and abdominal pain.

→ **NEXT #915** Debrief my reasoning for cognitive bias

949 Turn a ward case into exam practice

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / PG RESIDENT

USE IT After seeing an interesting patient, to convert it into exam preparation.

PERSONA	You are a clinical teacher who turns ward cases into exam practice.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The case (de-identified): [PASTE] My exam: [brief]
TASK	1. Write the case as a long-case presentation summary. 2. List the 10 viva questions examiners would ask about this case, with short answers. 3. Write 3 MCQs based on the case. 4. Identify the theory questions this case could answer.
RULES	Keep the case fully de-identified and the answers accurate, marking specifics [VERIFY], because real cases become memorable exam material.
OUTPUT	The summary, the viva Q&A, the MCQs, then the theory links.

PRO TIP Patients you have seen are remembered far better than textbook cases. Convert every interesting one.

TRY IT WITH A patient with infective endocarditis and a splinter haemorrhage, seen on rounds.

→ **NEXT #862** Long case: presentation and examiner questions

950 Set management priorities (ABCDE)

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / INTERN

USE IT For sick patients, and for "what will you do first?" questions.

PERSONA	You are an emergency physician who teaches prioritisation.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] The sick patient scenario (de-identified): [PASTE]
TASK	1. Apply the ABCDE approach with specific actions at each step. 2. Identify the immediate life threats and their treatment. 3. Order the investigations by urgency. 4. List when to call for senior help. 5. Plan the reassessment after each intervention.
RULES	Mark treatment specifics and doses [VERIFY LOCAL PROTOCOL], because emergency management must follow approved protocols.
OUTPUT	The ABCDE actions, the life threats, the prioritised investigations, the escalation triggers, then the reassessment plan.

PRO TIP Fix problems as you find them in ABCDE order. Don't wait to finish the assessment.

TRY IT WITH A patient with BP 80/50, HR 130, RR 30, SpO2 88% and cold peripheries after 3 days of fever.

→ **NEXT #911** Prioritised differential with can't-miss diagnoses

951 Think aloud and get feedback on my reasoning

STANDARD

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / INTERN / PG RESIDENT

USE IT To practise articulating clinical reasoning, as on rounds and in vivas.

PERSONA	You are a clinical-reasoning coach who evaluates think-aloud reasoning.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] The case (de-identified) and my think-aloud reasoning (typed as I would say it): [PASTE]
TASK	1. Map my reasoning steps. 2. Identify the strengths (e.g. good problem representation, considering danger). 3. Identify the gaps (missed data, a premature conclusion, unjustified tests). 4. Rewrite my reasoning as an expert would say it. 5. Give 2 habits to practise.
RULES	Quote my words for every comment and stay constructive, because reasoning feedback must be specific to change habits.
OUTPUT	The reasoning map, the strengths, the gaps, the expert version, then the habits.

PRO TIP Saying your reasoning aloud on rounds, even when unsure, is how seniors can teach you.

TRY IT WITH My reasoning on a patient with fever and a new heart murmur.

→ **NEXT #913** Interactive patient case simulator

952 Top three diagnoses in one line

QUICK

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / INTERN

Act as a senior physician teaching on rounds. For this de-identified presentation, [PASTE KEY FEATURES], give the top 3 differentials ranked, each with the single strongest supporting feature, in one line each, and flag any can't-miss diagnosis that isn't among them.

TRY IT WITH A 60-year-old smoker with haemoptysis, weight loss and clubbing.

953 Can't-miss diagnoses for a symptom

QUICK

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / INTERN

Act as an emergency physician. List the can't-miss diagnoses for [SYMPTOM] in [AGE GROUP], each with one feature that suggests it and the fastest test that excludes it. Mark tests and thresholds [VERIFY]. Flag anything you are unsure of so I can check it.

TRY IT WITH Sudden-onset breathlessness in young adults.

954 The one history question that helps most

QUICK

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / INTERN

Act as a clinical-reasoning tutor. For this presentation, [PASTE FEATURES], with the differential [LIST], tell me the single most discriminating history question to ask next and how each possible answer would shift the differential. Flag anything you are unsure of so I can check it.

TRY IT WITH Chronic cough with a differential of asthma, GERD, post-nasal drip and TB.

955 The sign I must look for

QUICK

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / INTERN

Act as a clinical examiner. For a patient with [PRESENTATION] and a differential of [LIST], tell me the one or two examination signs that would best discriminate between them, how to elicit each correctly, and what a positive finding means.

TRY IT WITH Breathlessness with a differential of heart failure, COPD and pulmonary fibrosis.

956 The first investigation to order

QUICK

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / INTERN

Act as a senior resident. For this presentation, [PASTE FEATURES], in a [SETTING], name the first investigation to order, why it comes first (speed, availability, yield), and what each result would mean. Mark thresholds [VERIFY]. Flag anything you are unsure of so I can check it.

TRY IT WITH Suspected meningitis in a district-hospital emergency department.

957 Interpret one abnormal result

QUICK

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS STUDENT / INTERN

Act as a clinical pathologist. Interpret this result, [RESULT WITH UNITS], in the context of [CLINICAL CONTEXT]: what it suggests, the top 3 causes in this context, and the next test. Mark reference ranges [VERIFY WITH LAB]. Flag anything you are unsure of so I can check it.

TRY IT WITH Serum potassium of 6.4 in a patient with CKD and no ECG changes.

958 What doesn't fit the diagnosis?

QUICK

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS
STUDENT / PG RESIDENT

Act as a sceptical senior clinician. For the working diagnosis [DIAGNOSIS] and these findings, [PASTE FINDINGS], list the findings that don't fit it, what each might suggest instead, and whether they change the plan. Flag anything you are unsure of so I can check it.

TRY IT WITH Diagnosis: viral fever. Findings include a new murmur and splinter haemorrhages.

959 Name the syndrome from the findings

QUICK

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS
STUDENT / PG RESIDENT

Act as a diagnostician. Which syndrome or unifying diagnosis best explains these findings, [PASTE FINDINGS]? Give the most likely answer, one alternative, and the single confirmatory test, with a one-line reason for each. Flag anything you are unsure of so I can check it.

TRY IT WITH Hypertension, hypokalaemia and metabolic alkalosis in a 35-year-old.

960 Causes by category in one minute

QUICK

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS
STUDENT

Act as a clinical tutor. Using the VINDICATE categories, list the causes of [FINDING] in a quick grouped list, marking the 3 most common in India with an asterisk and the can't-miss ones with an exclamation mark. Flag anything you are unsure of so I can check it.

TRY IT WITH Finding: generalised lymphadenopathy.

961 The top causes at this age

QUICK

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS
STUDENT / PG RESIDENT

Act as a clinician. For [PRESENTATION] in a [AGE] patient, give the 5 most likely causes for this age group in India, ranked, with one line each on why age changes the likelihood. Flag anything you are unsure of so I can check it.

TRY IT WITH Seizures in a 70-year-old.

962 India-specific causes to consider

QUICK

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS
STUDENT / PG RESIDENT

Act as an internist experienced in Indian practice. For [PRESENTATION], list the causes more common in India than in Western textbooks (infections, toxins, nutritional causes, bites and envenomation) that I must consider, with one clue for each. Flag anything you are unsure of so I can check it.

TRY IT WITH Acute kidney injury in a rural patient.

963 Quick red-flag checklist

QUICK

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS
STUDENT / INTERN

Act as an emergency physician. Give me a quick red-flag checklist for [SYMPTOM], each flag in under 8 words with the serious cause it suggests, formatted as a pocket list I can use in the OPD. Mark actions [VERIFY]. Flag anything you are unsure of so I can check it.

TRY IT WITH Symptom: low back pain.

964 Which test rules it out?

QUICK

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS
STUDENT / PG RESIDENT

Act as an evidence-based physician. For excluding [DIAGNOSIS] in a [LOW / INTERMEDIATE] probability patient, name the test with the best negative predictive value in that setting, its limitation, and when a negative result isn't enough. Mark test performance [VERIFY].

TRY IT WITH Excluding pulmonary embolism with D-dimer.

965 From pretest to post-test probability

QUICK

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS
STUDENT / PG RESIDENT

Act as a clinical epidemiologist. With a pretest probability of [X%] and a test with LR+ [A] and LR- [B], calculate the post-test probability for a positive and a negative result, showing the odds conversion steps, and say whether either result would change management.

TRY IT WITH Pretest 30%, LR+ 6, LR- 0.2.

966 Summarise the case in one line

QUICK

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS
STUDENT / INTERN

Act as a senior resident. Summarise this case, [PASTE CASE], in one line using semantic qualifiers and the key positives, without naming an unconfirmed diagnosis, suitable for handover or rounds. Flag anything you are unsure of so I can check it.

TRY IT WITH A 45-year-old alcoholic with 3 days of abdominal pain radiating to the back and vomiting.

967 Could this be a side effect?

QUICK

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS
STUDENT / INTERN

Act as a clinical pharmacologist. From this medication list, [LIST], which drugs could cause [NEW PROBLEM]? Rank them by plausibility with one line each, and note the timing clue to check. Flag anything you are unsure of so I can check it.

TRY IT WITH New-onset tremor and confusion in a patient on lithium and a recently added diuretic.

968 Who should I see first?

QUICK

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · INTERN /
PG RESIDENT

Act as an emergency triage physician. Rank these patients by urgency, [BRIEF DESCRIPTIONS], using standard triage principles, with one line of reasoning each and the immediate action for the most urgent. Mark protocols [VERIFY LOCAL TRIAGE SYSTEM]. Flag anything you are unsure of so I can check it.

TRY IT WITH Four patients: chest pain, a fracture, fever with a rash, and an asthma attack with silent chest.

969 Refer, admit or manage here?

QUICK

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · INTERN /
MEDICAL OFFICER

Act as a senior physician advising a medical officer. For this patient, [BRIEF CASE], in a [SETTING], decide whether to manage locally, admit, or refer, with the criteria behind the decision and what to do before transfer. Mark protocols [VERIFY].

TRY IT WITH A child with severe dehydration and lethargy at a primary health centre.

970 One clinical pearl from this case QUICK

3.6 CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS · MBBS
STUDENT / PG RESIDENT

Act as a senior clinical teacher. From this case, [BRIEF CASE], give one clinical pearl (a practical, memorable insight that goes beyond the textbook) in under 30 words, and explain why it matters in one line. Flag anything you are unsure of so I can check it.

TRY IT WITH A patient with myasthenia gravis who worsened after an antibiotic.

3.7

Memory aids & high-yield summaries

What you can recall on exam day matters more than what you once read.

Turn every topic into flashcards, mnemonics and one-page sheets, and review them on a schedule.

30

prompts

4

Master

17

Standard

9

Quick

HOW TO USE THIS SECTION

Use it after you finish each topic, to build flashcards, mnemonics and one-page sheets for revision.

Start with the Master prompts ([#971–974](#)) for the full workflow; use the Standard prompts ([#975–991](#)) for one task at a time; reach for the Quick prompts ([#992–1000](#)) when you have two minutes. Tap any title below to jump to it, and follow → NEXT to move to the next step.

[#971](#) High-yield flashcards for Anki or Notion

[#972](#) One-page high-yield summary of any topic

[#973](#) Custom mnemonics that actually stick

[#974](#) Final-week rapid revision pack for a subject

[#975](#) Set up an Anki deck and review routine

[#976](#) Convert notes into cloze-deletion cards

[#977](#) Plan image-occlusion cards for diagrams

[#978](#) Comparison table of confusable conditions

[#979](#) Mind map of a topic

[#980](#) Build a memory palace for a long list

[#981](#) Visual mnemonic or sketch for a concept

[#982](#) Bank of one-liners for a subject

[#983](#) Key numbers and values sheet

[#984](#) Drug summary card

[#985](#) Remember a classification

[#986](#) Spaced revision schedule for my topics

[#987](#) Condense long notes into revision notes

[#988](#) Turn a topic into a memorable story

[#989](#) Previous-year question patterns into a sheet

[#990](#) Link a topic across subjects

[#991](#) Turn my mistakes into cards

[#992](#) One mnemonic for a list

[#993](#) 10 quick flashcards

[#994](#) 5 must-know facts

[#995](#) Two-item quick comparison

[#996](#) Explain simply, then technically

[#997](#) Morning recall test

[#998](#) Improve a flashcard

[#999](#) Night-before-exam list

[#1000](#) Decode an acronym or eponym

High-yield flashcards for Anki or Notion

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

USE IT

After finishing a chapter or topic, to turn it into cards you can review for months.

WHY

Active recall and spaced repetition are the two best-evidenced study techniques. Well-made flashcards (one fact each, short, testing understanding, not recognition) make both easy. Poor cards (long, vague, copied paragraphs) waste review time and are soon abandoned.

COPY THE PROMPT BELOW

PERSONA	You are a medical educator and spaced-repetition expert who writes crisp, exam-focused flashcards for [SUBJECT], in the style of the best Anki decks.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Ophthalmology, year 1] Subject and topic: [e.g. ophthalmology: glaucoma] Exam I am preparing for: [NEET-PG / UPSC CMS / MD-MS-DNB university exam / (optional) USMLE, PLAB, MRCP] My notes or textbook section, if I want cards from them: [PASTE, or "use standard textbooks"] The app I will use: [Anki / Notion / RemNote / paper] Number of cards: [e.g. 20]
TASK	1. Write [20] high-yield flashcards on the topic in the format "Q: ... A: ...", each answer 2 lines at most. 2. Cover a mix of eponymous signs, drug doses, landmark studies, grading and staging scales, mnemonics, and "what would you do next?" clinical questions. 3. Keep one fact per card; split any card that tests two things. 4. Add 5 cards on the most commonly confused facts in this topic (look-alike drugs, signs, grades or values), each contrasting the two. 5. Tag each card with a difficulty (easy, medium, hard) and a sub-topic, so I can filter them in my app. 6. Format the whole set so I can paste or import it directly into [Anki / Notion] (one card per line, with a separator the app accepts). 7. Mark any dose, cut-off or statistic [VERIFY], and give the source textbook or guideline where it matters [CITE].
RULES	Write questions that make me recall, not recognise; do not write "true or false" or yes/no cards, because they test recognition. Keep the answers short, because long answers get skipped in review. Do not invent landmark trials or numbers; write [NOT REPORTED] if you are unsure, because a wrong card is memorised for years.
OUTPUT	The cards as an importable list ("Q: ... A: ... tags"), then the 5 confused-facts cards, then a one-line note on anything I should verify.
TWEAK	End with 3 cloze-deletion versions of the hardest cards ("The drug of choice for {{c1:...}} is ...").

How to get more out of it

- Make cards the same day you study the topic. Memory is freshest then.
- Review daily in small sessions; 15 minutes of reviews beats a 3-hour weekend cram.
- Delete or rewrite any card you keep failing. It is usually badly written.
- Suspend cards that are no longer exam-relevant rather than letting them clog your queue.
- → **NEXT #975** Set up an Anki deck and review routine

One-page high-yield summary of any topic

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

USE IT

For final revision, when you need a whole topic on one page.

WHY

In the last weeks before an exam, you cannot re-read textbooks. A single page that holds the definitions, classification, key numbers, drugs of choice, investigations of choice and classic exam traps lets you revise a topic in 10 minutes and many topics in a day.

COPY THE PROMPT BELOW

PERSONA	You are a top NEET-PG and university-exam faculty member who writes famously concise, high-yield revision sheets.
MY CONTEXT	Audience and time available: [brief] What I most want to get out of this: [one line] My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Subject and topic: [e.g. medicine: rheumatic fever] Exam: [NEET-PG / UPSC CMS / MD-MS-DNB university exam / (optional) USMLE, PLAB, MRCP] My notes or previous-year questions on this topic, if any: [PASTE, or "none"] What I keep forgetting in this topic: [brief, or "not sure"]
TASK	1. Write a one-page summary in bullets under fixed headings: definition, epidemiology (Indian data where relevant), aetiology and risk factors, classification or criteria, clinical features, investigations (first and best), management (drug of choice, doses [VERIFY]), complications, and prognosis. 2. Put all key numbers (cut-offs, doses, percentages) in bold. 3. Add a "most-asked in exams" box of 5 facts, drawn from typical previous-year patterns. 4. Add a "classic traps" box: 3 facts that students commonly get wrong in MCQs. 5. Add one mnemonic for the list that is hardest to remember. 6. Keep it to one printed page (about 400–500 words).
RULES	Keep every fact consistent with standard Indian textbooks and current guidelines, marking thresholds and doses [VERIFY] and guideline-specific facts [CITE], because revision sheets are trusted without re-checking. Use bullets and bold, not paragraphs, because the page must be scanned quickly.
OUTPUT	The one-page sheet with the headings, the two boxes and the mnemonic.
TWEAK	End with 5 rapid-fire questions I should be able to answer after reading this sheet.

How to get more out of it

- › Print it and add your own margin notes. Writing fixes memory.
- › Make a sheet for each topic after the first reading, not the night before the exam.
- › Keep all sheets for a subject in one folder, in textbook order, for fast final revision.
- › → **NEXT #974** Final-week rapid revision pack for a subject

Custom mnemonics that actually stick

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

USE IT

For long lists you must recall in order: causes, criteria, classifications, drug side effects.

WHY

Mnemonics work when they are vivid, meaningful and yours. Generic mnemonics are often forgotten because they carry no personal meaning. Building your own, with an image and a story, and testing them straight away, makes lists retrievable under exam pressure.

COPY THE PROMPT BELOW

PERSONA	You are a memory coach and medical educator who builds vivid, memorable mnemonics for medical lists.
MY CONTEXT	My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] The list I need to remember: [PASTE the list, e.g. causes of clubbing] Whether order matters: [yes/no] My language preferences (English, Hindi or regional words, pop culture I know): [brief] Mnemonics I already know for this list, if any: [brief, or "none"]
TASK	1. Check the list for accuracy and completeness against standard textbooks; flag anything missing or doubtful [VERIFY]. 2. Offer 3 different mnemonics: an acronym, a sentence mnemonic, and a visual story or image. 3. Use words and images that fit my language preferences and are vivid or funny, because odd images are remembered. 4. For each mnemonic, show exactly which letter or image maps to which item. 5. Say which mnemonic suits this list best and why. 6. Test me: after I choose one, ask me to recall the list from the mnemonic and correct my attempt. 7. Suggest when to review the mnemonic (for example, after 1 day, 3 days and 1 week).
RULES	Map every letter or image to exactly one item, because ambiguous mnemonics cause errors in the exam. Keep the content accurate and mark doubtful items [VERIFY]. Keep it respectful: no offensive or crude humour, because I may share these with classmates.
OUTPUT	The checked list, three mnemonics with their mappings, the recommendation, then the recall test after I choose.
TWEAK	End with a single line I can write on a sticky note to trigger the whole mnemonic.

How to get more out of it

- The sillier and more personal the image, the better it sticks.
- Say the mnemonic aloud and draw it once. Two senses beat one.
- Do not make mnemonics for short lists you can simply understand. Save them for arbitrary ones.
- → **NEXT #980** Build a memory palace for a long list

Final-week rapid revision pack for a subject

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

USE IT

In the last 7–10 days before an exam, to revise a whole subject quickly.

WHY

The last week is for retrieval, not new learning. A pack that condenses a subject into its highest-yield facts, confusable pairs, one-liners and a quick self-test gives you maximal marks per hour and exposes the gaps you still have time to fix.

COPY THE PROMPT BELOW

PERSONA	You are an experienced exam coach who builds last-week revision packs for NEET-PG, UPSC CMS and university exams.
MY CONTEXT	Audience and time available: [brief] What I most want to get out of this: [one line] My specialty and year: [e.g. final-year MBBS / MD Medicine, year 2] Subject: [e.g. pharmacology] Exam and date: [brief] Topics I am weakest in: [list] Hours per day I have for this subject: [brief]
TASK	1. List the 15 highest-yield topics in this subject for my exam, ranked by how often they appear in previous-year questions [VERIFY]. 2. For each topic, give 3–5 one-liners I must know. 3. Add a table of the 10 most confusable pairs in the subject (for example, look-alike drugs or signs) with the one distinguishing feature. 4. Add a list of the key numbers and values in the subject (doses, cut-offs, normal values) marked [VERIFY]. 5. Add a 20-question self-test covering the pack, with answers at the end. 6. Give a day-wise plan to cover the pack, prioritising my weak topics.
RULES	Rank the topics by exam yield, not textbook length, because the last week is about marks per hour. Keep the facts accurate and mark every number [VERIFY]. Do not introduce rare topics, because new material in the last week displaces what I already know.
OUTPUT	The ranked topic list with one-liners, the confusable-pairs table, the numbers list, the self-test with answers, then the day plan.
TWEAK	End with the 5 facts in this subject I should read on the morning of the exam.

How to get more out of it

- › Take the self-test first; revise what you get wrong, then retake it.
- › Sleep before the exam. Memory consolidation happens during sleep.
- › Stop adding new sources in the final week. Trust your pack.
- › → **NEXT #972** One-page high-yield summary of any topic

975 Set up an Anki deck and review routine

STANDARD

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

USE IT When starting spaced repetition, or when your Anki reviews have become overwhelming.

PERSONA	You are a spaced-repetition coach who helps medical students use Anki efficiently.
MY CONTEXT	My exam and date: [brief] Hours per day I can give to reviews: [brief] My current situation: [new to Anki / have a deck / reviews piled up]
TASK	1. Recommend a deck structure (by subject, with tags for systems and topics). 2. Recommend daily new-card and review limits that fit my time. 3. Explain the key settings in plain language, marked [VERIFY] since app versions change. 4. Give a plan to clear a review backlog without burning out.
RULES	Keep the advice practical and sustainable, because consistency beats volume in spaced repetition.
OUTPUT	The deck structure, the limits, the settings, then the backlog plan.

PRO TIP Never skip a day of reviews; missed days make the next day's pile heavier.

TRY IT WITH NEET-PG in 8 months, 1 hour a day, 3,000 overdue cards.

→ **NEXT #971** High-yield flashcards for Anki or Notion

976 Convert notes into cloze-deletion cards

STANDARD

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

USE IT When you have dense notes and want to turn them into cards quickly.

PERSONA	You are a medical educator who writes clean cloze-deletion flashcards.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] My notes: [PASTE] Exam: [brief]
TASK	1. Convert the key facts into cloze cards in the format "text {{c1::hidden}} text". 2. Hide only one key fact per cloze, the one worth recalling. 3. Skip trivial or obvious facts. 4. Mark any number or dose [VERIFY].
RULES	Keep each cloze short and unambiguous, because a cloze with several possible answers teaches nothing.
OUTPUT	The cloze cards, one per line, ready to import. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP If a cloze could be guessed from the sentence, rewrite it.

TRY IT WITH Paste your notes on the Jones criteria for rheumatic fever.

→ **NEXT #971** High-yield flashcards for Anki or Notion

977 Plan image-occlusion cards for diagrams

STANDARD

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

USE IT For anatomy, histology, radiology, fundus photos and flowcharts.

PERSONA	You are an anatomy and imaging teacher who makes image-occlusion flashcards.
MY CONTEXT	The image or diagram (describe it or name it): [brief] Subject and exam: [brief]
TASK	1. List the structures or labels worth hiding, in order of exam importance. 2. For each, give the question to ask and the answer. 3. Add one clinical correlation card for the most important structure. 4. Suggest whether to hide one label at a time or all together.
RULES	Keep the anatomy and labels accurate to standard atlases [VERIFY], because a mislabelled card teaches the wrong structure.
OUTPUT	A table (label, question, answer), the clinical correlation card, then the occlusion advice.

PRO TIP Use your own textbook's diagrams so the exam figure looks familiar.

TRY IT WITH A labelled diagram of the brachial plexus.

→ **NEXT #971** High-yield flashcards for Anki or Notion

978 Comparison table of confusable conditions

STANDARD

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

USE IT For look-alike diseases, drugs, signs or investigations.

PERSONA	You are an exam-focused teacher who makes clear comparison tables.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] The items to compare: [e.g. Crohn's disease vs ulcerative colitis] Exam: [brief]
TASK	1. Build a comparison table on the features examiners test (epidemiology, pathology, clinical features, investigations, management, complications). 2. Bold the single most discriminating feature in each row. 3. Add the 3 features most often confused in MCQs. 4. Mark numbers and guideline facts [VERIFY].
RULES	Keep the table accurate and balanced, because comparison questions reward precise differences.
OUTPUT	The table, then the 3 most-confused features.

PRO TIP Draw the table from memory the next day and check it.

TRY IT WITH Nephritic vs nephrotic syndrome.

→ **NEXT #972** One-page high-yield summary of any topic

979 Mind map of a topic

STANDARD

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

USE IT When a topic feels scattered and you need to see how it connects.

PERSONA	You are a visual-learning teacher who builds clear medical mind maps.
MY CONTEXT	Topic: [brief] Level and exam: [brief]
TASK	1. Put the topic at the centre and create 5–7 main branches (such as causes, pathology, features, investigations, treatment, complications). 2. Add 2–4 sub-branches to each with key words only. 3. Draw links between branches where one explains another (for example, pathology to clinical features). 4. Present it as an indented outline I can redraw on paper or in a mind-mapping app.
RULES	Use key words, not sentences, because mind maps work by showing structure at a glance.
OUTPUT	The indented outline, then a list of the cross-links.

PRO TIP Redraw it by hand from memory. The redrawing is the learning.

TRY IT WITH Topic: diabetes mellitus complications.

→ **NEXT #980** Build a memory palace for a long list

980 Build a memory palace for a long list

STANDARD

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

USE IT For long ordered lists: cranial nerve nuclei, drug classes, staging systems.

PERSONA	You are a memory coach who teaches the method of loci to medical students.
MY CONTEXT	The list: [PASTE] A familiar place I know well (home, hostel, hospital route): [brief]
TASK	1. Choose a route through my place with one location per item. 2. Place a vivid image for each item at each location, linking the image to the item. 3. Walk me through the route once. 4. Test me by asking me to walk it back and name each item.
RULES	Keep the images vivid and clearly linked to each item, and check the list's accuracy [VERIFY], because the palace preserves errors as faithfully as facts.
OUTPUT	The route with the locations and images, then the recall test.

PRO TIP Reuse the same palace for one subject only, to avoid mixing lists.

TRY IT WITH The 12 cranial nerves, placed along the route from your hostel room to the canteen.

→ **NEXT #973** Custom mnemonics that actually stick

981 Visual mnemonic or sketch for a concept

STANDARD

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

USE IT For mechanisms, pathways and signs that are easier to see than to read.

PERSONA	You are a medical illustrator and teacher who creates simple visual mnemonics.
MY CONTEXT	Concept: [e.g. the coagulation cascade / the brachial plexus / the renin-angiotensin system] Exam: [brief]
TASK	1. Describe a simple sketch or visual metaphor that captures the concept. 2. Label the parts and explain how each maps to the concept. 3. Give step-by-step instructions so I can draw it in 2 minutes. 4. Point out the parts examiners most often test.
RULES	Keep the metaphor faithful to the real physiology or anatomy, and say where it oversimplifies, because a misleading picture is hard to unlearn.
OUTPUT	The sketch description, the label key, the drawing steps, then the exam points.

PRO TIP Draw it in the exam too; a diagram often earns marks on its own.

TRY IT WITH The coagulation cascade as a waterfall.

→ **NEXT #979** Mind map of a topic

982 Bank of one-liners for a subject

STANDARD

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

USE IT For NEET-PG and UPSC CMS preparation, where one-line facts win MCQs.

PERSONA	You are a NEET-PG faculty member who writes high-yield one-liners.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Subject and topic: [brief] Exam: [NEET-PG / UPSC CMS / other]
TASK	1. Write 30 one-liners on the topic, each a single testable fact. 2. Group them by sub-topic. 3. Star the 10 most frequently tested [VERIFY]. 4. Mark numbers and doses [VERIFY].
RULES	Keep each one-liner to one fact and one line, because one-liners are for fast repeated reading.
OUTPUT	The grouped one-liners with stars. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP Read them aloud on the way to college; short repeated exposures add up.

TRY IT WITH Topic: microbiology, gram-positive cocci.

→ **NEXT #971** High-yield flashcards for Anki or Notion

983 Key numbers and values sheet

STANDARD

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

USE IT For the numbers every exam tests: normal values, cut-offs, doses, scores.

PERSONA	You are a clinical teacher who makes precise reference sheets.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Subject or system: [e.g. nephrology] Exam: [brief]
TASK	1. List the key numbers: normal values, diagnostic cut-offs, doses, scoring thresholds. 2. Give the unit for each and the source guideline or textbook [CITE]. 3. Add a memory trick for the hardest 5. 4. Mark every number [VERIFY].
RULES	Keep units explicit and flag where Indian and international values differ, because unit and guideline errors cost marks.
OUTPUT	A table (item, value, unit, source), then the memory tricks.

PRO TIP Check this sheet against your textbook once; numbers are where AI slips most.

TRY IT WITH Subject: diabetes and thyroid values.

→ **NEXT #972** One-page high-yield summary of any topic

984 Drug summary card

STANDARD

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT / INTERN

USE IT For pharmacology revision and for drugs you prescribe often.

PERSONA	You are a clinical pharmacologist who writes compact drug cards.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Drug or drug class: [brief] Level and exam: [brief]
TASK	1. Give the class, mechanism, key uses, dose [VERIFY], route, adverse effects, contraindications and key interactions. 2. Add the one fact examiners love about this drug. 3. Add one mnemonic for the adverse effects.
RULES	Keep doses and interactions accurate to standard references and mark them [VERIFY], because drug errors harm patients.
OUTPUT	A compact card under the headings, then the exam fact and the mnemonic. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP Make cards for classes, then note only the exceptions for individual drugs.

TRY IT WITH Amiodarone.

→ **NEXT #983** Key numbers and values sheet

985 Remember a classification

STANDARD

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

USE IT For classifications and staging systems examiners love to ask.

PERSONA	You are a teacher who makes classifications easy to remember.
MY CONTEXT	The classification: [e.g. WHO classification of tuberculosis / Lown grading / TNM for breast cancer] Exam: [brief]
TASK	1. Give the classification accurately, with its source [CITE]. 2. Explain the logic behind it, so it makes sense rather than being memorised blindly. 3. Give a mnemonic or pattern to remember it. 4. Give one clinical use of the classification.
RULES	Reproduce the classification exactly and mark the version or year [VERIFY], because classifications are updated and examiners know which version they expect.
OUTPUT	The classification, the logic, the memory aid, then the clinical use.

PRO TIP Write the classification from memory once a week until it is automatic.

TRY IT WITH The Child-Pugh score.

→ **NEXT #973** Custom mnemonics that actually stick

986 Spaced revision schedule for my topics

STANDARD

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

USE IT After finishing a first reading, to plan when to revise each topic.

PERSONA	You are a study coach who designs spaced-repetition revision schedules.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Topics I have finished and when: [list with dates] Exam date: [brief] Hours per day for revision: [brief]
TASK	1. Schedule each topic for revision at expanding intervals (for example, 1, 3, 7, 21 days). 2. Balance the daily load within my hours. 3. Put weaker topics on shorter intervals. 4. Present it as a calendar table.
RULES	Keep the plan realistic for my hours, because an overloaded schedule gets abandoned.
OUTPUT	The calendar table, then a one-line rule to adjust it when I fall behind.

PRO TIP Revise from your sheets and cards, not the full textbook.

TRY IT WITH 12 topics finished over the last month; exam in 90 days; 2 hours a day.

→ **NEXT #974** Final-week rapid revision pack for a subject

987 Condense long notes into revision notes

STANDARD

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

USE IT When your notes are too long to revise.

PERSONA	You are an exam coach who condenses notes without losing high-yield facts.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] My notes: [PASTE] Exam: [brief] Target length: [e.g. one third]
TASK	1. Condense the notes to the target length, keeping all high-yield facts. 2. Convert paragraphs into bullets and tables. 3. Bold the key words and numbers. 4. List anything I removed that might still be worth keeping.
RULES	Do not add new facts that are not in my notes, and flag any fact that looks wrong [VERIFY], because condensed notes must stay faithful to my sources.
OUTPUT	The condensed notes, then the list of removed items.

PRO TIP Condense again before the final month; each pass sharpens recall.

TRY IT WITH Paste 3 pages of notes on anaemia.

→ **NEXT #972** One-page high-yield summary of any topic

988 Turn a topic into a memorable story

STANDARD

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

USE IT For pathways, sequences and disease courses.

PERSONA	You are a teacher who explains medicine through memorable stories.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Audience or purpose: [brief] Topic or sequence: [brief] Level: [brief]
TASK	1. Turn the sequence into a short story with characters and events that map to the steps. 2. Keep the story under 200 words. 3. Add a key showing which character or event maps to which step.
RULES	Keep the mapping accurate and complete, and say where the story oversimplifies, because stories stick, including their errors.
OUTPUT	The story, then the key. Keep it concise, and list anything you were unsure about at the end so I can check it.

PRO TIP Retell the story to a friend; teaching it locks it in.

TRY IT WITH The steps of the urea cycle.

→ **NEXT #980** Build a memory palace for a long list

989 Previous-year question patterns into a sheet

STANDARD

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

USE IT When you have previous-year questions and want to know what to memorise.

PERSONA	You are an exam analyst who finds patterns in previous-year questions.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Previous-year questions on a topic or subject: [PASTE] Exam: [brief]
TASK	1. Group the questions by sub-topic and count them. 2. Identify the facts that are repeatedly tested. 3. Turn those facts into a one-page memory sheet. 4. Suggest 5 likely future questions on the same facts.
RULES	Base the patterns only on the pasted questions and say how many you analysed, because patterns from a few questions can mislead.
OUTPUT	The count table, the repeated facts, the memory sheet, then the likely questions.

PRO TIP Repeated facts are repeated for a reason. Learn them first.

TRY IT WITH Paste 40 previous NEET-PG questions on ophthalmology.

→ **NEXT #974** Final-week rapid revision pack for a subject

990 Link a topic across subjects

STANDARD

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

USE IT For integrated questions that join anatomy, physiology, pathology, pharmacology and medicine.

PERSONA	You are a teacher of integrated medical education who links topics across subjects.
MY CONTEXT	My specialty and level: [e.g. final-year MBBS / MD Medicine, year 2] Topic: [e.g. the thyroid gland] Exam: [brief]
TASK	1. Link the topic across subjects: anatomy, physiology, biochemistry, pathology, microbiology, pharmacology, medicine and surgery, as relevant. 2. Give 2 high-yield facts per subject. 3. Show how one subject explains another (for example, how the physiology explains the drug action). 4. Write 3 integrated MCQs that cross subjects, with answers.
RULES	Keep the facts accurate and mark numbers [VERIFY], because integrated questions punish half-remembered links.
OUTPUT	A table (subject, facts, link to the next subject), then the integrated MCQs.

PRO TIP Integrated links make each fact easier to recall, because it is stored in more than one place.

TRY IT WITH Topic: vitamin B12.

→ **NEXT #979** Mind map of a topic

991 Turn my mistakes into cards

STANDARD

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

USE IT After a mock test or MCQ session, to stop repeating the same mistakes.

PERSONA	You are an exam coach who turns errors into targeted revision.
MY CONTEXT	Questions I got wrong, with my answer and the correct answer: [PASTE] Exam: [brief]
TASK	1. For each mistake, identify the missing fact or the misconception. 2. Write one flashcard per mistake that targets exactly that gap. 3. Group the mistakes by pattern (knowledge gap, misreading, confusable facts). 4. Suggest one habit to reduce the commonest pattern.
RULES	Target the specific gap, not the whole topic, and mark facts [VERIFY], because error cards work only when they fix the exact misconception.
OUTPUT	The error table (question, gap, card), the pattern groups, then the habit.

PRO TIP Your own mistakes make the highest-yield cards you will ever write.

TRY IT WITH Paste 10 wrong answers from your last grand test.

→ **NEXT #971** High-yield flashcards for Anki or Notion

QUICK PROMPTS · 3.7

992 One mnemonic for a list

QUICK

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

Act as a medical memory coach. Give me one memorable mnemonic for [LIST]. Show which letter maps to which item, check the list is accurate and complete [VERIFY], and keep it respectful. Flag anything you are unsure of so I can check it.

TRY IT WITH Causes of pancreatitis.

993 10 quick flashcards

QUICK

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

Act as a medical educator. Write 10 high-yield flashcards on [TOPIC] for [EXAM], in the format "Q: ... | A: ...", with one fact per card and answers of 2 lines at most. Mark numbers [VERIFY]. Flag anything you are unsure of so I can check it.

TRY IT WITH Topic: thyroid disorders, for NEET-PG.

994 5 must-know facts

QUICK

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

Act as an exam coach. Give me the 5 facts about [TOPIC] most often tested in [EXAM], one line each, with any numbers marked [VERIFY]. Add one common trap students fall into. Flag anything you are unsure of so I can check it.

TRY IT WITH Topic: tetralogy of Fallot, for NEET-PG.

995 Two-item quick comparison

QUICK

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

Act as a medical teacher. Compare [A] and [B] in a 5-row table on the features examiners test, and bold the single most discriminating feature. Mark numbers [VERIFY] so I can check them. Flag anything you are unsure of so I can check it.

TRY IT WITH Type 1 vs type 2 respiratory failure.

996 Explain simply, then technically QUICK

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT

Act as a teacher. Explain [CONCEPT] first in simple words a school student would understand, then in exam-ready technical language in 5 bullets. End with one analogy I can remember it by. Flag anything you are unsure of so I can check it.

TRY IT WITH Concept: the Frank-Starling law.

997 Morning recall test QUICK

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

Act as an examiner. Ask me 10 rapid recall questions on [TOPIC] I studied yesterday, one at a time, and wait for my answer each time. At the end, list the ones I missed so I can make cards for them.

TRY IT WITH Topic: antitubercular drugs.

998 Improve a flashcard QUICK

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

Act as a spaced-repetition expert. Rewrite this flashcard so it tests one fact, prompts recall rather than recognition and has a short answer: [PASTE CARD]. Explain what was wrong in one line. Flag anything you are unsure of so I can check it.

TRY IT WITH "Q: Tell me everything about digoxin | A: [a long paragraph]"

999 Night-before-exam list QUICK

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

Act as an exam coach. Give me the 15 highest-yield facts in [SUBJECT] to glance at the night before [EXAM], one line each, numbers marked [VERIFY]. No new topics, only the most tested facts. Flag anything you are unsure of so I can check it.

TRY IT WITH Subject: forensic medicine, for the MBBS university exam.

1000 Decode an acronym or eponym QUICK

3.7 MEMORY AIDS & HIGH-YIELD SUMMARIES · MBBS STUDENT / PG RESIDENT

Act as a medical teacher. Expand and explain [ACRONYM OR EPONYM], give its clinical significance in 2 lines, and add one way to remember it. Flag if the term has more than one meaning in medicine. Flag anything you are unsure of so I can check it.

TRY IT WITH "CURB-65."

Every prompt at a glance.

Tap any prompt to jump to it. Q = Quick · S = Standard · M = Master

3.1 · LEARNING ANY TOPIC

#701	Structured topic master for any subject	M
#702	A personal tutor that checks my understanding	M
#703	Build a custom tutor that answers only from my notes	M
#704	Study a chapter with NotebookLM	M
#705	Understand a mechanism from first principles	M
#706	One topic across all subjects: vertical integration	M
#707	Learn a topic in four layers	S
#708	Explain it back and find my gaps (Feynman technique)	S
#709	Learn a hard concept through analogies	S
#710	Surface features and deeper mechanisms (iceberg)	S
#711	Learn a process as a flowchart	S
#712	Compare options in a decision matrix	S
#713	Classifications as a tree	S
#714	A learning roadmap from my level to my goal	S
#715	Connect this topic to what I already know	S
#716	Learn through a worked example	S
#717	Common misconceptions in a topic	S
#718	Sort a topic into high- and low-yield	S
#719	Turn a textbook chapter into structured notes	S
#720	Clean up my lecture notes	S
#721	Notes from a video lecture transcript	S
#722	Learn a clinical guideline for exams	S
#723	Learn any drug with one framework	S
#724	A standard template for any disease	S
#725	Learn an anatomy topic that sticks	S
#726	Understand physiology graphs and curves	S
#727	Learn to read a type of clinical image	S
#728	Normal values and interpretation	S
#729	Learn a microorganism with one template	S
#730	Simplify a biochemistry pathway	S
#731	Learn national health programmes (community medicine)	S
#732	Make my own questions from my notes	S
#733	Design an interleaved practice session	S
#734	Diagnose my knowledge gaps	S
#735	Explain it like I'm a second-year student	Q
#736	One-page summary of a topic	Q
#737	Ten must-know facts	Q
#738	Ask "why" five times	Q
#739	Clear up two confusing terms	Q
#740	A real-world example of a concept	Q
#741	Why does this basic-science fact matter clinically?	Q
#742	A quick 5-question quiz	Q
#743	Define it, with an example	Q
#744	Clear up one specific confusion	Q
#745	The whole topic in 280 characters	Q
#746	A diagram I can draw	Q

#747	What should I study next?	Q
#748	Is my understanding correct?	Q
#749	Translate the jargon	Q
#750	Three questions the examiner will ask	Q

3.2 · STUDY PLANNING & REVISION

#751	A personalised day-by-day study plan	M
#752	NEET-PG preparation strategy	M
#753	Plan for the MD / MS / DNB final exam	M
#754	UPSC CMS preparation strategy	M
#755	A spaced-repetition revision schedule	S
#756	The last 30 days before the exam	S
#757	The last 7 days before the exam	S
#758	I've fallen behind: recover my plan	S
#759	Weekly study review	S
#760	Which subject should I study first?	S
#761	Study around internship or residency duties	S
#762	Plan revision cycles (R1, R2, R3)	S
#763	Analyse a mock or grand test properly	S
#764	Rescue my weakest subject	S
#765	Am I heading for burnout?	S
#766	Design focused study sessions	S
#767	Run an effective study group	S
#768	Too many resources: choose and commit	S
#769	The night before and the exam day	S
#770	Sleep, memory and study performance	S
#771	Build a simple accountability tracker	S
#772	Plan my study day	Q
#773	My three priorities today	Q
#774	How long will this subject take?	Q
#775	What should I revise today?	Q
#776	Reframe a demotivating thought	Q
#777	Fix my biggest distraction	Q
#778	Set a goal for this study session	Q
#779	Did I hit my weekly target?	Q
#780	Study breaks that actually refresh	Q

3.3 · MCQ PRACTICE & EXAM STRATEGY

#781	Exam-pattern MCQs with full explanations	M
#782	Solve a question and learn the logic	M
#783	Analyse previous-year questions for patterns	M
#784	Rapid-fire MCQ and viva grilling	M
#785	A mistake notebook that raises scores	M
#786	Practise image-based questions	M
#787	Why was my answer wrong?	S
#788	Learn from the distractors	S
#789	Eliminate options when I'm unsure	S
#790	When to guess and when to skip	S

#791	Time management in an MCQ exam	S	#840	Strong introductions and conclusions	S
#792	Handle EXCEPT, NOT and "most likely" questions	S	#841	Time allocation for a theory paper	S
#793	Extract one-liners from my practice questions	S	#842	Answer "compare and contrast" questions	S
#794	Recent updates likely to be tested	S	#843	What does the question word really ask?	S
#795	20 MCQs on the topic I just studied	S	#844	A "recent advances" paragraph	S
#796	Two-step reasoning questions	S	#845	A stepwise management answer	S
#797	Pharmacology MCQ practice	S	#846	Answer a case-based theory question	S
#798	Microbiology MCQ practice	S	#847	Make my answers look complete at a glance	S
#799	Community medicine MCQs, including calculations	S	#848	Build a bank of repeated questions	S
#800	Biostatistics calculation practice	S	#849	Model answers from a textbook chapter	S
#801	Applied anatomy MCQs	S	#850	Improve my writing speed	S
#802	MCQs for the short subjects (ENT, ophthalmology, dermatology...)	S	#851	What examiners actually look for	S
#803	MCQs from my own notes	S	#852	A one-line definition to open my answer	Q
#804	Build my own mock test	S	#853	The five points I must not miss	Q
#805	Balance accuracy and speed	S	#854	An "enumerate" answer	Q
#806	What to do with flagged questions	S	#855	Which diagram should I add?	Q
#807	Anxiety during MCQ exams	S	#856	Convert my paragraph into exam points	Q
#808	Use a question bank effectively	S	#857	Headings for my answer	Q
#809	Practice in the UPSC CMS style	S	#858	A 2-mark answer	Q
#810	USMLE- or PLAB-style vignettes (optional)	S	#859	A reusable template for disease answers	Q
#811	Practise match-the-following and assertion-reason formats	S	#860	A strong closing line	Q
#812	Turn my question-bank statistics into a plan	S	3.5 · VIVA, OSCE & PRACTICAL EXAMS		
#813	High-yield facts from previous-year themes	S	#861	Viva voce simulator, one question at a time	M
#814	Read long clinical vignettes faster	S	#862	Long case: presentation and examiner questions	M
#815	One high-quality MCQ with explanation	Q	#863	Short cases: examination and spot findings	M
#816	Explain this answer simply	Q	#864	OSCE station simulator	M
#817	Find the key clue in the question	Q	#865	Thesis defence viva preparation	M
#818	Make a similar question	Q	#866	Spotters and table viva: instruments, specimens, images, drugs	M
#819	Ten rapid true/false statements	Q	#867	Opening questions examiners use on a topic	S
#820	The one feature that separates them	Q	#868	Structure any viva answer	S
#821	Five one-liners from a topic	Q	#869	Instrument viva: one instrument, fully prepared	S
#822	What's the trap in this question?	Q	#870	Pathology or anatomy specimen viva	S
#823	Solve it from first principles	Q	#871	X-ray viva: describe, diagnose, discuss	S
#824	Turn a fact into a question	Q	#872	ECG viva: systematic and confident	S
#825	Check my reasoning on a question	Q	#873	Pharmacology practical: drug and prescription viva	S
#826	Which option is most likely?	Q	#874	Charts and graphs viva (partograph, growth charts, epidemic curves)	S
#827	Solve an EXCEPT question safely	Q	#875	Demonstrate a system examination properly	S
#828	Numbers I must remember for this topic	Q	#876	Elicit a specific clinical sign	S
#829	Formula recall drill	Q	#877	History-taking OSCE with a simulated patient	S
#830	Twenty last-minute facts	Q	#878	Breaking bad news and difficult conversations	S
3.4 · LONG & SHORT ANSWER WRITING			#879	Counselling station (medication, lifestyle, procedure consent)	S
#831	Model long answer (10–15 marks)	M	#880	Procedural skills station	S
#832	Model answers for previous-year questions	M	#881	Data-interpretation station	S
#833	Examiner evaluation of my written answer	M	#882	Prescription-writing station	S
#834	MD / MS / DNB theory essay with depth	M	#883	Viva etiquette and presence	S
#835	Short note (5 marks)	S	#884	When I blank out in a viva	S
#836	Plan any answer in 2 minutes	S	#885	Handle different examiner styles	S
#837	A diagram that earns marks	S	#886	DNB-style bedside case discussion	S
#838	Put a flowchart in my answer	S	#887	Table viva: rapid structured answers	S
#839	A comparison table in my answer	S			

#888	Community medicine spotters (vaccines, models, charts)	S	#937	Approach to jaundice	S
#889	Forensic medicine spotters (weapons, poisons, specimens)	S	#938	Approach to acute kidney injury	S
#890	Obstetrics and gynaecology practical preparation	S	#939	Approach to hyponatraemia	S
#891	Paediatric practical preparation	S	#940	Age-specific differential in children	S
#892	Surgery short case: examining a lump or swelling	S	#941	Differential diagnosis in pregnancy	S
#893	The practical exam day	S	#942	Atypical presentations in the elderly	S
#894	Last-minute viva revision list	S	#943	Tropical infection differential (Indian setting)	S
#895	A 60-second answer to "what is X?"	Q	#944	Could a drug be causing this?	S
#896	Classify it for the viva	Q	#945	Recognise the syndrome from scattered findings	S
#897	Five viva questions on my topic	Q	#946	Discriminate between two close diagnoses	S
#898	An instrument in one breath	Q	#947	Apply a clinical decision rule	S
#899	How do I elicit this sign?	Q	#948	A diagnostic time-out before I close the case	S
#900	Describe this finding like an examiner expects	Q	#949	Turn a ward case into exam practice	S
#901	Where will the examiner go next?	Q	#950	Set management priorities (ABCDE)	S
#902	"What would you do next?"	Q	#951	Think aloud and get feedback on my reasoning	S
#903	The differential they'll ask for	Q	#952	Top three diagnoses in one line	Q
#904	This topic's examiner-favourite question	Q	#953	Can't-miss diagnoses for a symptom	Q
#905	Say "I don't know" well in a viva	Q	#954	The one history question that helps most	Q
#906	OSCE station opening lines	Q	#955	The sign I must look for	Q
#907	Close an OSCE station well	Q	#956	The first investigation to order	Q
#908	Sound confident without overclaiming	Q	#957	Interpret one abnormal result	Q
#909	Plain-language phrases for counselling stations	Q	#958	What doesn't fit the diagnosis?	Q
#910	Describe a spot diagnosis	Q	#959	Name the syndrome from the findings	Q
3.6 · CLINICAL REASONING & DIFFERENTIAL DIAGNOSIS					
#911	Prioritised differential with can't-miss diagnoses	M	#960	Causes by category in one minute	Q
#912	Build illness scripts for diseases	M	#961	The top causes at this age	Q
#913	Interactive patient case simulator	M	#962	India-specific causes to consider	Q
#914	Approach to any presenting symptom	M	#963	Quick red-flag checklist	Q
#915	Debrief my reasoning for cognitive bias	M	#964	Which test rules it out?	Q
#916	Choose and interpret tests like an expert (Bayesian reasoning)	M	#965	From pretest to post-test probability	Q
#917	Reason through a management decision	M	#966	Summarise the case in one line	Q
#918	Write a problem representation	S	#967	Could this be a side effect?	Q
#919	Practise semantic qualifiers	S	#968	Who should I see first?	Q
#920	Generate causes systematically (VINDICATE)	S	#969	Refer, admit or manage here?	Q
#921	Localise the lesion (neurology)	S	#970	One clinical pearl from this case	Q
#922	Estimate a pretest probability	S	3.7 · MEMORY AIDS & HIGH-YIELD SUMMARIES		
#923	Use likelihood ratios at the bedside	S	#971	High-yield flashcards for Anki or Notion	M
#924	Red flags for a presenting symptom	S	#972	One-page high-yield summary of any topic	M
#925	Can't-miss diagnoses for a presentation	S	#973	Custom mnemonics that actually stick	M
#926	The next best investigation	S	#974	Final-week rapid revision pack for a subject	M
#927	Interpret a lab pattern in clinical context	S	#975	Set up an Anki deck and review routine	S
#928	Reason through an acid-base problem	S	#976	Convert notes into cloze-deletion cards	S
#929	Interpret an ECG in clinical context	S	#977	Plan image-occlusion cards for diagrams	S
#930	Interpret imaging in clinical context	S	#978	Comparison table of confusable conditions	S
#931	Approach to fever, including PUO	S	#979	Mind map of a topic	S
#932	Approach to chest pain	S	#980	Build a memory palace for a long list	S
#933	Approach to breathlessness	S	#981	Visual mnemonic or sketch for a concept	S
#934	Approach to altered sensorium	S	#982	Bank of one-liners for a subject	S
#935	Approach to acute abdominal pain	S	#983	Key numbers and values sheet	S
#936	Approach to anaemia	S	#984	Drug summary card	S
			#985	Remember a classification	S
			#986	Spaced revision schedule for my topics	S

#987 Condense long notes into revision notes	S	#994 5 must-know facts	Q
#988 Turn a topic into a memorable story	S	#995 Two-item quick comparison	Q
#989 Previous-year question patterns into a sheet	S	#996 Explain simply, then technically	Q
#990 Link a topic across subjects	S	#997 Morning recall test	Q
#991 Turn my mistakes into cards	S	#998 Improve a flashcard	Q
#992 One mnemonic for a list	Q	#999 Night-before-exam list	Q
#993 10 quick flashcards	Q	#1000 Decode an acronym or eponym	Q

Which AI tool for which job.

JOB	BEST TOOL	WHY
Learning a topic from your own books and notes	NotebookLM, Claude, ChatGPT (upload)	Explanations grounded in your sources
Tutoring, active recall and self-testing	ChatGPT Study mode, Claude, Gemini	Socratic questioning and instant feedback
Custom exam tutor from your notes	Custom GPTs, Claude Projects, Gemini Gems	Answers only from your uploaded material
Flashcards and spaced repetition	Anki, Quizlet	Proven spaced-repetition scheduling
Audio revision on the move	NotebookLM Audio Overview	Turns chapters into listenable discussions
Visual notes and mind maps	Canva, Napkin AI, ChatGPT images	Diagrams and summary sheets
Checking facts and recent guidelines	Perplexity, Consensus, the guideline itself	Cited sources you can verify

STUDYING WITH AI, SAFELY

- › Verify every fact, number and drug dose against a standard textbook or current guideline.
- › Use AI to test yourself, not to avoid thinking. Answer first, then check.
- › Never use AI during an exam or assessment where it isn't allowed.
- › Clinical reasoning prompts are for learning, not for decisions about real patients.

One last thing.

These prompts will not pass your exam for you. They will make every hour of study count: more recall, more practice and more feedback. The doctors who do best will be the ones who use AI to think harder, not less.

Test yourself. Check everything. Walk into the exam hall ready.

More AI in medicine every week: Instagram [@abhishekjbenur](#) · [paraboxai.com](#)

This resource is for educational purposes. It does not replace standard textbooks, current clinical guidelines, your university curriculum or your teachers. AI-generated content may contain errors: verify all facts independently. Clinical reasoning prompts are learning aids and are not a substitute for clinical judgement or supervision in patient care.